

Prepladder Version X Qbank

Subject: Biochemistry

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Carbohydrates

1. A 55-year-old patient presents with symptoms suggestive of a metabolic disorder. Biochemical tests are performed to analyze protein structures and their carbohydrate components. The medical team needs to differentiate between glycoproteins and proteoglycans to better understand the patient's condition. Which of the following statements is correct?

- A. Glycoproteins contain long mucopolysaccharide chains
 - B. Proteoglycans have long mucopolysaccharide chains and high carbohydrate content.
 - C. Glycoproteins have long mucopolysaccharide chains and higher carbohydrate content
 - D. Proteoglycans contain short oligosaccharide chains and lower carbohydrate content
-

2. A 15-year-old girl is brought to the clinic by her parents because she is more fatigued than usual and frequently thirsty. A urinalysis reveals positive results for Benedict's test, Barfoed's test, and Glucose Oxidase Peroxidase Test are also positive. Which of the following is the most likely diagnosis?

- A. Diabetes Mellitus
 - B. Essential Fructosuria
 - C. Lactose due to pregnancy and lactation
 - D. Classical Galactosemia
-

3. A laboratory technician performs an Osazone test on a patient's urine sample and observes the formation of characteristic crystals. The results show the presence of fluffy, hedgehog-shaped crystals. Which sugar is most likely present in the urine sample?

- A. Glucose

- B. Lactose
 - C. Maltose
 - D. Sucrose
-

4. What is the reagent used in Barfoed's test to detect monosaccharides?

- A. Sodium carbonate, copper sulfate, and sodium citrate
 - B. Cupric acetate in acetic acid solution
 - C. Nitric acid
 - D. Phenolphthalein blue
-

5. A new mother comes to the pediatric OPD with her infant as she notices dark staining in diapers after the baby urinates. What is the possible enzyme defect here?

- A. Fumarylacetoacetase hydrolase
 - B. Homogentisic acid oxidase
 - C. Cystathionine β -synthase
 - D. Phenylalanine hydroxylase
-

6. A urine sample of a diabetes patient was tested to see the presence of reducing sugars in the laboratory. Initial color of solution is blue. What would be the color of the solution after the test?

- A. Blue
- B. Orange
- C. Colourless

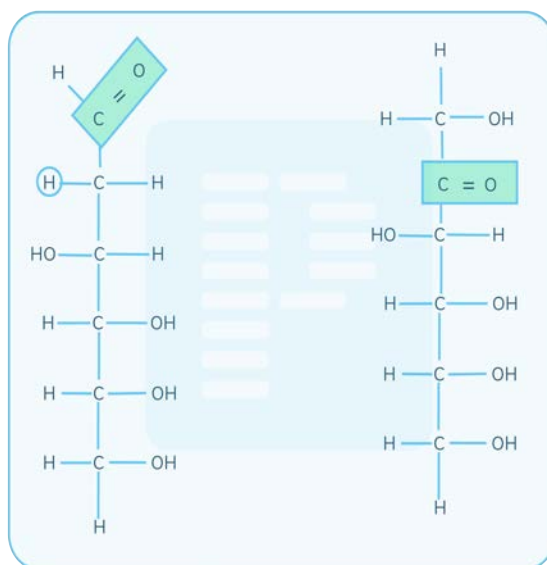
7. A 55-year-old female patient presents with symptoms of increased thirst, frequent urination, and weight loss. The physician suspects diabetes mellitus and decides to test the patient's urine for reducing sugars using Benedict's and Fehling's solutions. Below are images of the test results. Image A: Benedict's Test Results Image B: Fehling's Test Results Based on the images provided, which of the following statements is correct?





- A. Both tests indicate the absence of reducing sugars in the urine sample.
 - B. Both tests indicate the presence of reducing sugars in the urine sample.
 - C. Benedict's test indicates the presence of reducing sugars, while Fehling's test indicates their absence.
 - D. Benedict's test indicates the absence of reducing sugars, while Fehling's test indicates their presence.
-

8. Which type of isomerism is exhibited by these two compounds?



- A. Functional Isomerism
- B. Epimerism
- C. Enantiomerism
- D. Anomerism

9. Xylose and arabinose are examples of?

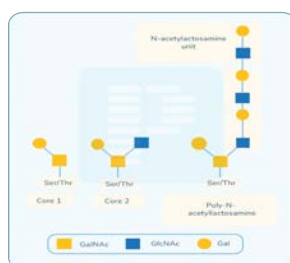
- A. Optical isomerism
- B. Diastereoisomerism
- C. Enantiomerism
- D. Epimerism

10. Which type of isomerism involves two molecules that have the same molecular formula and structural formula but differ in the spatial arrangement of their groups?

- A. Optical Isomerism

- B. Functional Isomerism
- C. Stereoisomerism
- D. All of the above

11. Identify the type of linkage in the given Structure.



- A. O-linked glycoproteins
- B. N- N-linked glycoproteins
- C. GPI anchored Glycoproteins
- D. None of the above

12. A 45-year-old patient presents with symptoms of fatigue, abdominal pain, and dark-coloured urine, especially noticed in the morning. Laboratory tests reveal elevated levels of hemoglobinuria. Which of the following tests is most appropriate for confirming a diagnosis of Paroxysmal Nocturnal Hemoglobinuria (PNH)?

- A. Bone Marrow Biopsy

- B. Flow Cytometry
 - C. Serum Ferritin Test
 - D. Haemoglobin Electrophoresis
-

13. Which category of carbohydrates consists of molecules that cannot be hydrolyzed into simpler sugars?

- A. Monosaccharides
 - B. Disaccharides
 - C. Oligosaccharides
 - D. Polysaccharides
-

14. Which of the following statements is a distinguishing feature between Hunter syndrome and Hurler syndrome?

- A. Hunter syndrome is associated with corneal clouding; Hurler is not.
 - B. Hurler syndrome is due to a deficiency in iduronate-2-sulfatase; Hunter syndrome is due to a deficiency in alpha-L-iduronidase.
 - C. Both syndromes result in the accumulation of heparan sulfate and dermatan sulfate.
 - D. Hunter syndrome is X-linked recessive; Hurler syndrome is autosomal recessive.
-

15. A 4-year-old child presents with developmental delay, behavioural problems, and progressive mental deterioration. Physical examination reveals coarse facial features and hepatosplenomegaly, his ocular findings are normal. The paediatrician suspects a type of mucopolysaccharidosis (MPS). Based on these clinical features, which kind of MPS is most likely?

- A. MPS I (Hurler syndrome)

- B. MPS IX (Natowicz syndrome)
 - C. MPS III (Sanfilippo syndrome)
 - D. MPS IV (Morquio syndrome)
-

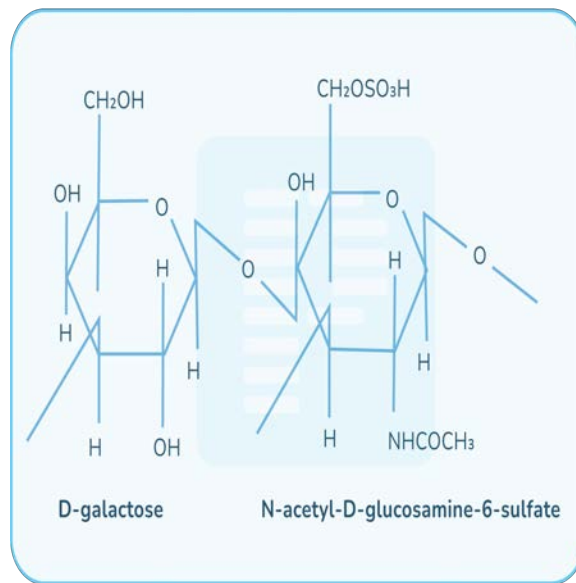
16. A patient presents with symptoms indicative of joint issues, including pain and reduced mobility. Examination reveals that the patient's synovial fluid is lacking a key mucopolysaccharide responsible for lubrication. Which mucopolysaccharide is most likely deficient in this patient's synovial fluid?

- A. Heparin
 - B. Chondroitin Sulfate
 - C. Dermatan Sulfate
 - D. Hyaluronic Acid
-

17. A patient is diagnosed with a rare genetic disorder affecting the structural integrity of their skin and blood vessels, resulting in hyperextensible skin and increased susceptibility to vascular damage. Which mucopolysaccharide is present in the skin and blood vessels, most likely involved in this condition?

- A. Hyaluronic Acid
 - B. Heparin
 - C. Chondroitin Sulfate
 - D. Dermatan Sulfate
-

18. Analyse the structure of the mucopolysaccharide shown below: Which mucopolysaccharide is represented in this image?



- A. Heparan sulfate
- B. Keratan sulfate
- C. Chondroitin sulfate
- D. Dermatan sulfate

19. Mucopolysaccharides are heteropolysaccharides containing uronic acid and amino sugars. Which among the following mucopolysaccharides contradicts the above statement?

- A. Keratan sulfate
- B. Chondroitin sulfate
- C. Dermatan sulfate
- D. Heparin

20. Which of the following is accurate about dietary fibres?

- A. Insoluble fibre is fermented in the large intestine
 - B. Dietary fibre provides a significant source of energy
 - C. Soluble fibre helps lower LDL cholesterol
 - D. Soluble fibre promotes bowel movements
-

21. Which of the following pairs correctly distinguishes between homoglycans and heteroglycans?

- A. Homoglycans - repeated units of the same monosaccharide; Heteroglycans - two or more different types of monosaccharide units.
 - B. Homoglycans - used for structural purposes; Heteroglycans - energy storage.
 - C. Homoglycans- typically soluble in water; Heteroglycans - insoluble.
 - D. Homoglycans - only in plant cells; Heteroglycans - only in animal cells.
-

22. A patient comes with complaints of abdominal pain, bloating, flatulence and frothy stools. It is noticed that these symptoms arise only when he consumes milk or other dairy products. Which of the following is the IOC in this situation to diagnose him?

- A. Lactose tolerance test
 - B. Hydrogen breath test
 - C. Stool osmolarity test
 - D. Serum calcium test
-

23. Which of the following is non-reducing sugar?

- A. Maltose

- B. Glucose
 - C. Sucrose
 - D. Fructose
-

24. A 28-year-old patient with a history of gastrointestinal issues is undergoing metabolic testing. The clinical team is investigating the role of different types of disaccharides in digestion and absorption. Which of the following are the constituent monosaccharides of the disaccharide lactose?

- A. Galactose and glucose
 - B. Glucose and Fructose
 - C. Galactose and Sucrose
 - D. Maltose and Glucose
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	2
Question 4	2
Question 5	2
Question 6	2
Question 7	2
Question 8	1
Question 9	4
Question 10	3
Question 11	1
Question 12	2
Question 13	1
Question 14	4
Question 15	3
Question 16	4
Question 17	4
Question 18	2

Question 19	1
Question 20	3
Question 21	1
Question 22	2
Question 23	3
Question 24	1

Solution for Question 1:

Correct Answer: B) Proteoglycans have long mucopolysaccharide chains and are characterized by their high carbohydrate content.

Explanation : Proteoglycans contain long mucopolysaccharide chains, are negatively charged due to sulfated or acetylated groups, and have higher carbohydrate content compared to glycoproteins, which have shorter oligosaccharide chains and are less negatively charged.

Property	Glycoproteins	Proteoglycans
Carbohydrates content	Low	High
Basic difference	Oligosaccharide chain	Mucopolysaccharide chain
Charge	Not negatively charged	Negatively charged sulfated or acetylated
Location	Plasma proteins Cellular receptors – except albumin Receptors made up of simple proteins can not be heterogenous, as simple proteins are made up of just 2- aminoacids which can be linked by peptide linkages. Lack of heterogeneity causes lack of specificity	ECM - extracellular matrix

Solution for Question 2:

Correct Answer: A) Diabetes Mellitus

Explanation:

The most probable diagnosis, considering the symptoms and test results, is Diabetes Mellitus, indicated by the presence of glucose as a reducing monosaccharide in the urine. The positive results in both Benedict's and Barfoed's tests support this conclusion. It is further verified by a positive Glucose Oxidase Peroxidase Test (GOD-POD).

Algorithm for detecting sugars in urine:

Test	Purpose	Interpretation
Benedict's Test	Detects reducing substances in urine.	Positive indicates presence of reducing sugars, but not specific for monosaccharides.
Barfoed's Test	Differentiates between reducing monosaccharides and other sugars.	Positive indicates reducing monosaccharides (e.g., glucose, fructose, galactose).
Glucose Oxidase Peroxidase Test (GOD-POD)	Specifically tests for glucose presence.	Positive confirms glucose, supporting diagnosis of diabetes mellitus.
Seliwanoff Test	Distinguishes ketoses like fructose from aldoses like glucose.	Positive indicates fructose; observed in essential fructosuria if combined with other symptoms.
Mucic Acid Test	Identifies galactose in the urine.	Positive confirms galactose, supporting a diagnosis of classical galactosemia.
Bial's Test	Identifies pentoses in the urine.	Used for diagnosing conditions like essential pentosuria.

Other options:

Essential Fructosuria (Option B) would show a negative Glucose Oxidase Peroxidase Test test.

Lactose due to pregnancy and lactation (Option C) would show a negative Barfoed's test, and generally would have history and features of pregnancy.

Classical Galactosemia (Option D) would also show a negative Glucose Oxidase Peroxidase Test test.

Solution for Question 3:

Correct Answer: B) Lactose

Explanation: The Osazone test is used to differentiate reducing sugars based on the unique shapes and formation times of crystals. Lactose forms Hedgehog or “Pincushion with pins” or “flower of touch-me-not plant” shaped Lactosazones, with an excess of Phenylhydrazine when kept at boiling temperature.

Sugar	Crystal Shape	Notes
Lactose (Option B)	Hedgehog or “Pincushion with pins” or “flower of touch-me-not plant”	Indicates lactose presence, as seen in galactosemia.
Glucose (Option A) Fructose	Needle-shaped crystals arranged like a broom	Often associated with conditions like Diabetes Mellitus.
Maltose (Option C)	Sunflower-shaped or Petal-shaped	Takes longer time to form, associated with menopausal status in women.
Sucrose (Option D)	No crystals	Sucrose is a non-reducing sugar and does not form crystals in the Osazone test.

Solution for Question 4:

Correct answer: B) Cupric acetate in acetic acid solution

Explanation:

- Barfoed’s test is used to distinguish between monosaccharides and disaccharides.
- The reagent used is cupric acetate in an acetic acid solution.
- It is weakly acidic and is reduced by only monosaccharides.
- A false positive result can occur with prolonged boiling of disaccharides.

- Monosaccharides react with this reagent within 5 minutes to give a brick-red precipitate of copper (I) oxide .
 - Disaccharides take a longer time to react because the aldehyde function is masked in the acetal linkage and gives an orange-red color.
-

Solution for Question 5:

Correct answer: B) Homogentisic acid oxidase

Explanation:

- The scenario here points to Alkaptonuria which is an autosomal recessive condition.
- Caused due to deficiency of homogentisic acid oxidase leading to excretion of homogentisic acid through urine.
- Investigations :

Urine becomes darker on standing : blackening is accelerated on exposure to sunlight and oxygen. In a test tube urine starts blackening from the top layer.

Ferric chloride test: positive (urine)

Benedict's test: strongly positive, screening test. Can be a differential diagnosis for reducing sugars in urine.

Enzyme	Disease
Option A) Fumarylacetoacetate hydrolase	Tyrosinemia type I
Option B) Homogentisic acid oxidase	Alkaptonuria
Option C) Cystathionine β -synthase	Homocystinuria
Option D) Phenylalanine hydroxylase	Phenylketonuria

Solution for Question 6:

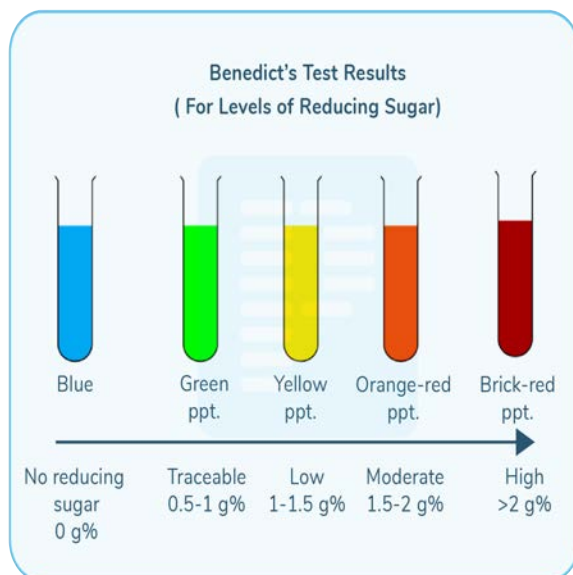
Correct answer: B) Orange

Explanation:

- The presence of reducing sugars is detected by Benedict's test which is a blue solution that turns its color in the range from green to yellow to orange to red according to the concentration of the reducing sugars.
- A positive Benedict's test : change in color from blue to green / yellow / orange / red

A negative Benedict's test : no change in color.

- The reagent here is Benedict's solution which contains sodium carbonate, copper sulphate, and sodium citrate . The copper sulfate gives it a blue color.
- In an alkaline medium, sugars form enediol that in turn reduces the cupric ions in the reagent. This changes the color of the reagent.
- Any sugar with a free aldehyde or keto group will reduce the Benedict's reagent and hence this is not a reliable test for detecting glucose in urine.
- Disaccharides like sucrose don't react with the reagent as such but only after hydrolysis of sucrose does it give a positive test result. This is a specific test for detecting sucrose.
- Interpretation of Benedict's test:



Solution for Question 7:

Correct Answer : B) Both tests indicate the presence of reducing sugars in the urine sample.

Explanation :

Image A (Benedict's Test Results):	The color changes from blue to green, yellow, and red, indicating the presence of reducing substances in the urine sample.
Image B (Fehling's Test Results)	The color changes from blue to brick-red, indicating the presence of reducing substances in the urine sample.

Therefore, both tests confirm the presence of reducing sugars, which may suggest a condition like diabetes mellitus in the patient.

Solution for Question 8:

Correct Answer: A) Functional Isomerism

Explanation:

The given structures are that of Glucose and Fructose , which are examples of functional isomers . They have the same molecular formula but differ in their functional groups, with glucose being an aldehyde (aldohexose) and fructose being a ketone (ketohexose).

Explanation of Isomerism Types:

Type of Isomerism	Description	Examples
Functional Isomerism	Compounds with the same molecular formula but different structural formulas, specifically due to different functional groups.	Glucose (an aldehyde) and Fructose (a ketone)
Epimerism (Option B)	Compounds differing in configuration around just one chiral carbon atom.	D-Glucose and D-Mannose are epimers because they differ at the C-2 carbon.
Enantiomerism (Option C)	Compounds that are non-superimposable mirror images, differing at all chiral centers.	L-Glucose and D-Glucose are enantiomers, being mirror images of each other
Anomerism (Option D)	Compounds differing at the anomeric carbon formed during the ring closure of carbohydrates.	α -D-Glucose and β -D-Glucose differ in configuration at the anomeric carbon (C-1)

Solution for Question 9:

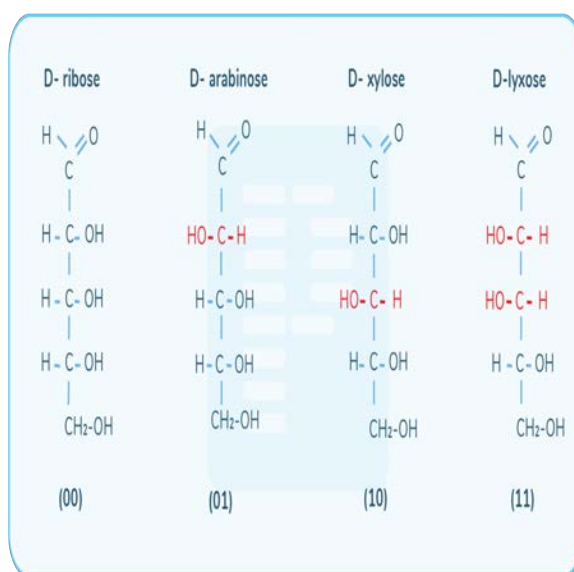
Correct answer : D) Epimerism

Explanation:

Epimerism refers to a type of diastereoisomerism in which two sugars differ in configuration at only one chiral center. Xylose and arabinose are both aldopentoses (five-carbon sugars) and differ only in the configuration around one specific carbon atom, specifically at the C-3 position.

In Ribose, the first and the last carbon atoms are symmetric, and the fourth carbon atom is the penultimate carbon atom. Therefore, the 2nd and 3rd carbon atom differences can provide the epimers of ribose. They are Arabinose and Xylose.

Epimerism is defined as the Isomerism at any carbon atom apart from the functional and penultimate carbon atom.



Other options:

Optical Isomerism(OPTION A) : Two molecules with same molecular formula, but differ in the way they turn the plain polarised lights.

- The orientation of the —H and —OH groups around the carbon atom adjacent to the terminal alcohol carbon (carbon 5 in glucose) determines whether the sugar belongs to the d or l series.

- The isomer that turns the light to the right side is called dextrorotatory (d/ +)
- The isomer that turns the light to the right side is called Levorotatory (l / -)

Diastereomers(OPTION B): The molecule should differ in spatial orientation with respect to one or more asymmetric carbon atoms other than the penultimate carbon Atom.

Enantiomerism(OPTION C) : Exact mirror image of the carbon atoms. Another name for enantiomerism is Racemism.

Solution for Question 10:

Correct answer : C) Stereoisomerism

Explanation:

Stereoisomerism: A broad category of isomerism where molecules have the same molecular formula and connectivity of atoms but differ in the spatial arrangement of their atoms.

Optical Isomerism: (Option A)

- Here the two molecules have the same molecular formula, but they differ in the way they turn the plane polarized light.
- The orientation of the —H and —OH groups around the carbon atom adjacent to the terminal alcohol carbon (carbon 5 in glucose) determines whether the sugar belongs to the d or l series.
- The isomer that turns the light to the right side is called Dextrorotatory (d/ +)
- The isomer that turns the light to the right side is called Levorotatory (l / -)
- Example: Lactic acid, which exists as two enantiomers: L-lactic acid and D-lactic acid.

Functional Isomerism: (Option B)

- Two molecules have the same molecular formula but different functional groups.
- Example: Ethanol (an alcohol) and dimethyl ether (an ether) both have the formula C_2H_6O but different functional groups and properties.

Solution for Question 11:

Correct answer- a) O-linked Glycoproteins

Explanation

O-linked glycoproteins (post-translation modification)	Carbohydrate chains are attached to the oxygen atom of serine or threonine residues. Mucins: Major components of mucus, providing lubrication and protection to epithelial surfaces. Glycoproteins in collagen
N-linked glycoproteins (Option B)	Carbohydrate chains are attached to the nitrogen atom of asparagine residues. Immunoglobulins (IgG) & Erythropoietin
GPI anchored Glycoproteins (Option C)	Carbohydrates are not attached to any amino acid Anchor consists of a lipid tail (phosphatidylinositol) and a glycan chain, which covalently links the protein to the membrane.

Solution for Question 12:

Correct Answer : B) Flow Cytometry

Explanation:

Flow cytometry is used to detect the absence or reduced expression of surface proteins CD55 and CD59 on red blood cells. These proteins are typically absent or deficient in PNH due to the genetic mutation affecting their GPI anchors.

Paroxysmal Nocturnal Hemoglobinuria (PNH):

A rare, acquired blood disorder characterized by the destruction of red blood cells due to a genetic mutation affecting cell membrane protection.

Cause:

- Mutation in the PIGA gene.
- Defective synthesis of glycosylphosphatidylinositol (GPI) anchors, which are crucial for attaching protective proteins to red blood cells.

Key Proteins Involved:

- CD55 (Decay Accelerating Factor) : Prevents complement system activation on the red blood cell surface.
- CD59 (Membrane Inhibitor of Reactive Lysis) inhibits the formation of the complement system's membrane attack complex.

Bone Marrow Biopsy(Option A) : This test is used to diagnose various bone marrow disorders but is not specific for PNH.

Serum Ferritin Test(Option C) : This test measures iron storage levels and is not specific to PNH. It is more useful for diagnosing iron-related disorders.

Haemoglobin Electrophoresis(Option D) : This test separates different types of haemoglobin but does not diagnose PNH. It is used to diagnose hemoglobinopathies such as sickle cell disease or thalassemia.

Solution for Question 13:

Correct answer: A) Monosaccharides

Explanation:

Monosaccharides are simple carbohydrates made up of single carbohydrate units and hence cannot be hydrolysed.

Carbohydrates are divided into simple and complex carbohydrates.

Simple carbohydrates can further be classified as follows:

Type	Feature
Monosaccharides	Single carbohydrate unit in them Ex: glucose, fructose
Disaccharides	Two monosaccharide units Ex: sucrose, maltose, isomaltose, trehalose, lactose
Oligosaccharides	3- 10 monosaccharide units Ex: Maltotriose
Polysaccharides	More than 10 monosaccharide units Ex: starch, cellulose, glycogen

Solution for Question 14:

Correct Answer: D) Hunter syndrome is X-linked recessive; Hurler syndrome is autosomal recessive.

Explanation:

Hunter syndrome arises due to a lack of the enzyme iduronate-2-sulfatase and follows an X-linked recessive inheritance pattern. In contrast, Hurler syndrome is due to a deficiency in the enzyme alpha-L-iduronidase and is inherited in an autosomal recessive manner (Option B) .

Both disorders result in the buildup of heparan sulfate and dermatan sulfate in tissues (Option C) . However, corneal clouding, which is observed in Hurler syndrome, is not typically seen in Hunter syndrome (Option A) .

Differentiating Features of Hurler and Hunter Syndromes

Feature	Hurler Syndrome (MPS I)	Hunter Syndrome (MPS II)
Deficient Enzyme	L-Iduronidase	Iduronate sulfatase
Inheritance (Option D)	Autosomal recessive	X-linked recessive
Mental Retardation	Severe (MR+++)	Mild to moderate (MR+)
Corneal Opacity	Present (Corneal opacity++)	Absent (No corneal clouding)

Solution for Question 15:

Correct Answer: C) MPS III (Sanfilippo syndrome)

Explanation:

MPS III, also known as Sanfilippo syndrome, is the most common type of mucopolysaccharidosis. It is characterized by severe central nervous system involvement leading to progressive mental deterioration, behavioral problems, and developmental delay.

Clinical Features and Investigations of Mucopolysaccharidoses

Disease Name	Abbreviation	Enzyme Defective	GAG(s) Affected	Symptoms
Hurler, Scheie, Hurler-Scheie syndrome (Option A)	MPS I	α -L-Iduronidase	Dermatan sulfate, heparan sulfate	Mental retardation, coarse facial features, hepatosplenomegaly, cloudy cornea
Hunter syndrome	MPS II	Iduronate sulfatase	Dermatan sulfate, heparan sulfate	Mental retardation
Sanfilippo syndrome A (Option C)	MPS IIIA	Heparan sulfate-N-sulfatase	Heparan sulfate	Delay in development, motor dysfunction
Sanfilippo syndrome B	MPS IIIB	α -N-Acetylglucosaminidase	Heparan sulfate	As MPS IIIA
Sanfilippo syndrome C	MPS IIIC	α -Glucosaminide N-acetyltransferase	Heparan sulfate	As MPS IIIA
Sanfilippo syndrome D	MPS IIID	N-Acetylglucosamine 6-sulfatase	Heparan sulfate	As MPS IIIA
Morquio syndrome A (Option D)	MPS IVA	Galactosamine 6-sulfatase	Keratan sulfate, chondroitin 6-sulfate	Skeletal dysplasia, short stature
Morquio syndrome B	MPS IVB	β -Galactosidase	Keratan sulfate	Skeletal dysplasia, short stature
Maroteaux-Lamy syndrome	MPS VI	N-Acetylgalactosamine 4-sulfatase	Dermatan sulfate	Curvature of the spine, short stature, skeletal dysplasia, cardiac defects
Sly syndrome	MPS VII	β -Glucuronidase	Dermatan sulfate, heparan sulfate, chondroitin 4-sulfate, chondroitin 6-sulfate	Skeletal dysplasia, short stature, hepatosplenomegaly, cloudy cornea
Natowicz syndrome (Option B)	MPS IX	Hyaluronidase	Hyaluronic acid	Joint pain, short stature

Solution for Question 16:

Correct Answer: D) Hyaluronic Acid

Explanation:

Hyaluronic acid is predominantly found in connective tissues, tendons, synovial fluid, and the vitreous humor of the eye. It serves as a crucial lubricant in joint cavities, helping to maintain joint health and mobility. A deficiency in hyaluronic acid can lead to joint pain and reduced mobility due to insufficient lubrication in the synovial fluid.

Functions of Various Mucopolysaccharides

Mucopolysaccharide	Functions
Hyaluronic Acid (Option D)	Serves as a lubricant in joint cavities and contributes to the structure of connective tissues
Heparin (Option A)	Acts as an anticoagulant by activating antithrombin III, which inactivates thrombin and other clotting factors
Chondroitin Sulfate (Option B)	Provides structural support to connective tissues and contributes to their resilience and elasticity
Dermatan Sulfate (Option C)	Contributes to the structure of the skin, blood vessels, and heart valves

Solution for Question 17:

Correct Answer: D) Dermatan Sulfate

Explanation:

Dermatan sulfate contributes to the structure of the skin, blood vessels, and heart valves. A deficiency or dysfunction in dermatan sulfate can lead to issues with the structural integrity of the skin and blood vessels, resulting in symptoms such as hyperextensible skin and increased vascular fragility.

Locations of Various Mucopolysaccharides:

Mucopolysaccharide	Primary Locations
Hyaluronic Acid (Option A)	Connective tissues, tendons, synovial fluid, vitreous humor

Heparin (Option B)	Liver, lungs, spleen, monocytes
Chondroitin Sulfate (Option C)	Cartilage, bone, tendons, cornea, skin
Keratan Sulfate	Cornea, cartilage
Dermatan Sulfate (Option D)	Skin, blood vessels, heart valves

Solution for Question 18:

Correct Answer: B) Keratan sulfate

Explanation:

The image depicts Keratan sulfate, identifiable by its unique composition of galactose and N-acetylglucosamine, along with possible sulfate groups. It is unique among glycosaminoglycans (GAGs) because it lacks uronic acid. It is primarily found in the cornea and cartilage, contributing to their structural properties.

Identification Tips:

- Look for the presence of galactose as one of the primary sugars in the disaccharide unit.
- N-acetylglucosamine can be identified by the presence of the N-acetyl group (CH_3CONH).
- Sulfate groups may be attached to either of the sugar moieties, contributing to the molecule's negative charge.

Mucopolysaccharide (MPS)	Components
Keratan Sulfate	Galactose and N-acetylglucosamine
Heparan Sulfate (Option A)	Glucuronic acid (or iduronic acid) and N-sulfated glucosamine
Dermatan Sulfate (Option D)	Iduronic acid and N-acetylgalactosamine
Chondroitin Sulfate (Option C)	Glucuronic acid and N-acetylgalactosamine
Hyaluronic Acid	Glucuronic acid and N-acetylglucosamine

Solution for Question 19:

Correct answer: A) Keratan sulfate

Explanation:

Keratan sulfate is unique among mucopolysaccharides as it is the only GAG that does not contain any uronic acid.

Keratan sulfate (Option A)	Repeating units are galactose and N-acetyl glucosamine in beta linkage.
Chondroitin sulfate (Option B)	Repeating units of glucuronic acid → beta-1,3-N-acetyl galactosamine sulfate → beta-1,4 and so on.
Dermatan sulfate (Option C)	It Contains L-iduronic acid and N-acetyl galactosamine in beta-1, 3 linkages.
Heparin (Option D)	An anticoagulant widely used Repeating units of sulphated glucosamine → alpha-1, 4-L-iduronic acid → and so on Iduronic acid is the oxidized form of idose.

Solution for Question 20:

Correct answer: C) Soluble fiber helps lower LDL cholesterol

Explanation:

Soluble fiber	Insoluble fiber
Lower LDL cholesterol levels by increasing bile acid excretion and interfering with bile acid reabsorption Fermented in the large intestine to short-chain fatty acids (Option A) Ex: Oats, barley, psyllium, brussel sprouts, etc	Adds bulk and promotes bowel movements. (Option D) Passes through the digestive tract largely unchanged (Option A) Ex: whole wheat, walnuts, chickpeas, etc

Dietary fibre is the non-digestible, non-starch carbohydrate and lignin present intact in plants.

Dietary fibre does not provide much energy but provides other benefits. It adds bulk to the diet, leads to early satiety, reduces blood glucose spikes after a meal, lowers total cholesterol and LDL, and reduces constipation, haemorrhoids and diverticulosis .(Option B)

Solution for Question 21:

Correct answer: A) Homoglycans - repeated units of the same monosaccharide;

Heteroglycans - two or more different types of monosaccharide units.

Explanation:

Polysaccharides are classified into :

Homoglycans/ homopolysaccharides	Heteroglycans/ heteropolysaccharides
Made of a single kind of monosaccharide unit. (Option A)	Made of two or more different monosaccharides. (Option A)
Used for both structural and energy storage. (Option B)	Used for both structural and energy storage. (Option B)
Variable solubility (dextran, cellulose) (Option C)	Variable solubility (agar, hyaluronic acid, keratin sulphate) (Option C)
Found in both plants and animals (starch & glycogen) (Option D)	Found in both plants & animals (agar & keratan sulfate) (Option D)
Ex: starch, glycogen and cellulose.	Ex: hyaluronic acid, chondroitin sulfate.

Solution for Question 22:

Correct answer: B) Hydrogen breath test

Explanation:

- The diagnosis here points to lactose intolerance , a digestion defect caused by the enzyme lactase deficiency .
- It presents with symptoms like osmotic diarrhoea, flatulence, bloating, and frothy stools.
- The investigation of choice for lactose intolerance is the Hydrogen breath test and methane breath test.

Other tests include:

- Stool acidity test : Unabsorbed lactose is fermented by colonic bacteria into lactic acid, lowering stool pH.
- Dietary elimination : Elimination of certain foods reduces the symptoms.
- Milk tolerance test : 500ml milk to be consumed and to note the blood glucose level to see malabsorption
- Lactose tolerance test: Checks lactose absorption after consuming a lactose-containing liquid.
- Small intestine biopsy : Done to find secondary causes only.
- Genotyping

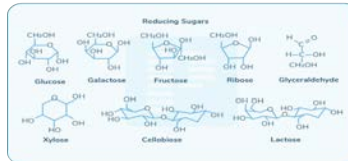
Solution for Question 23:

Correct answer : C) Sucrose

Explanation :

Non-reducing sugars are carbohydrates that lack a free aldehyde or ketone group, preventing them from acting as reducing agents.

Examples include sucrose and trehalose, where the glycosidic bond involves the reducing ends. They do not react with Benedict's or Fehling's solutions. However, after acid hydrolysis, they can be converted to reducing sugars, giving a positive result in these tests. Non-reducing sugars are significant in various biological and nutritional contexts.



Solution for Question 24:

Correct Answer : A) Glucose and Galactose

Explanation:

Disaccharide	Sugars present	Linkage
Maltose (Basic thumb rule)	2 glucose residues - glucose + glucose	alpha(1,4)
Isomaltose (Six atoms)	2 glucose residues - glucose + glucose	alpha(1,6)
Trehalose	2 glucose residues - glucose + glucose	alpha(1,1)
Sucrose	Glucose + fructose	alpha(1,2)
Lactose	Galactose + glucose	beta(1,4)

Fructose and galactose metabolism , HMP pathway

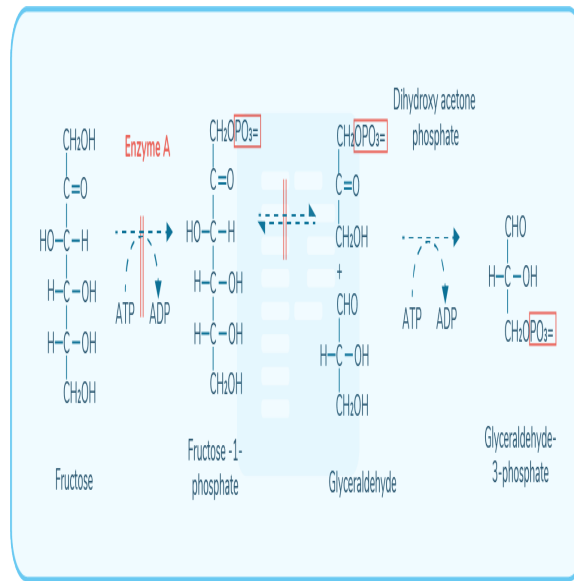
1. Which enzyme is considered the rate-limiting step in galactose metabolism?

- A. Galactokinase
 - B. Galactose-1-phosphate uridylyltransferase
 - C. UDP-hexose-4-epimerase
 - D. Aldolase Reductase
-

2. An infant is diagnosed with classic galactosemia. Which enzyme deficiency is responsible for this condition?

- A. Galactokinase
 - B. Galactose-1-phosphate uridylyltransferase
 - C. UDP-galactose 4-epimerase
 - D. Aldose reductase
-

3. A child is diagnosed with essential fructosuria during a routine health check. Name the enzyme A responsible for this condition



- A. Aldolase B
- B. Hexokinase
- C. Fructokinase
- D. Glucokinase

4. Which of the following statements is true regarding Hereditary Fructose Intolerance (HFI) and Classical Galactosemia?

- A. Both present with hyperglycemia.
- B. Both require lifelong avoidance of lactose.
- C. Both present with hypophosphatemia.
- D. Both have symptoms that appear at birth.

5. A 45-year-old male presents with fatigue and pallor. Blood tests reveal hemolytic anemia. On further investigation, it is found that his erythrocytes are susceptible to oxidative damage. What is the primary role of the pentose phosphate pathway

(PPP) that is deficient in this patient's RBCs?

- A. To produce ATP for energy metabolism
 - B. To generate ribose-5-phosphate for nucleotide synthesis
 - C. To provide NADPH for maintaining reduced glutathione levels
 - D. To synthesize hemoglobin
-

6. Which sequence correctly represents the steps of the oxidative phase of the pentose phosphate pathway (PPP)?

- A. 6-Phosphogluconolactone → Glucose-6-phosphate → 6-Phosphogluconate → Ribulose-5-phosphate
 - B. Glucose-6-phosphate → 6-Phosphogluconate → 6-Phosphogluconolactone → Ribulose-5-phosphate
 - C. Glucose-6-phosphate → 6-Phosphogluconolactone → 6-Phosphogluconate → Ribulose-5-phosphate
 - D. 6-Phosphogluconate → 6-Phosphogluconolactone → Glucose-6-phosphate → Ribulose-5-phosphate
-

7. Which statement is true regarding NADH and NADPH?

- A. Both are produced in the Pentose Phosphate Pathway.
 - B. NADH is used in lipid biosynthesis, while NADPH is not.
 - C. NADPH is used for anabolic reactions, while NADH is primarily for catabolic reactions.
 - D. Both serve the same functions in the cell.
-

8. All the following enzymes catalyze oxidative decarboxylation reactions except?

- A. α -Ketoglutarate dehydrogenase
 - B. Malate dehydrogenase
 - C. 6-Phosphogluconate dehydrogenase
 - D. Isocitrate dehydrogenase
-

9. Which of the following enzymes is not involved in the reversible phase of the pentose phosphate pathway?

- A. Epimerase
 - B. Transketolase
 - C. 6-phosphogluconate dehydrogenase
 - D. Transaldolase
-

10. A 30-year-old woman with glucose-6-phosphate dehydrogenase (G6PD) deficiency, living in a malaria-prone area, presents with fatigue, jaundice, dark urine, and recent consumption of fava beans. Lab tests show hemolytic anemia. How does glucose-6-phosphate dehydrogenase (G6PD) deficiency lead to the observed symptoms?

- A. Reduced NADPH impairs glutathione regeneration, causing hemolytic anemia.
 - B. Increased ribulose-5-phosphate overloads nucleotides, leading to renal failure.
 - C. Impaired glucose-6-phosphate conversion disrupts glycolysis, causing hypoglycemia.
 - D. Decreased ATP production affects muscle function, causing myopathy.
-

11. What is the role of inosine monophosphate in stored blood?

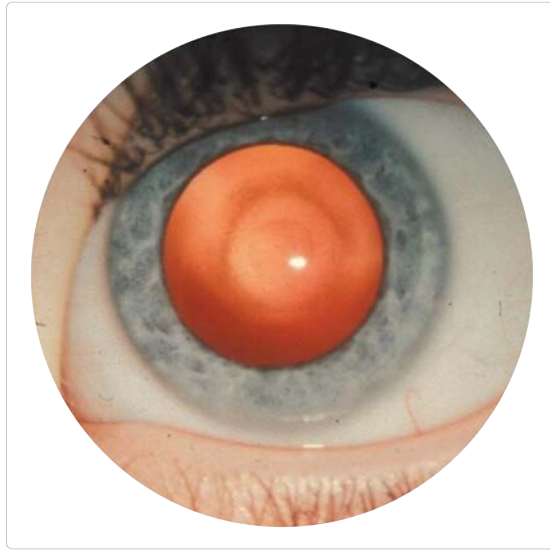
- A. It increases plasma Na and K levels.

- B. It stimulates 2,3-DPG synthesis.
 - C. It prevents lactic acid buildup.
 - D. It stabilizes hemoglobin concentration.
-

12. A patient with diabetes presents with vision impairment. Upon examination, snowflake cataracts are observed. The absence of which enzyme in certain tissues leads to sorbitol accumulation, resulting in conditions such as snowflake cataracts?

- A. Aldose Reductase
 - B. Glucose-6-Phosphate Dehydrogenase
 - C. Sorbitol Dehydrogenase
 - D. Hexokinase
-

13. A newborn presents with bilateral lens opacities and decreased visual acuity. On examination, the characteristic "oil droplet" appearance is noted in the lens on retro illumination. Which metabolic disorder is primarily associated with developing oil-drop cataracts in newborns?



- A. Phenylketonuria
 - B. Galactosemia
 - C. Maple Syrup Urine Disease
 - D. Homocystinuria
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	3
Question 4	3
Question 5	3
Question 6	3
Question 7	3
Question 8	2
Question 9	3
Question 10	1
Question 11	2
Question 12	3
Question 13	2

Solution for Question 1:

Galactose-1-phosphate uridylyltransferase (GALT) is the rate-limiting enzyme in galactose metabolism.

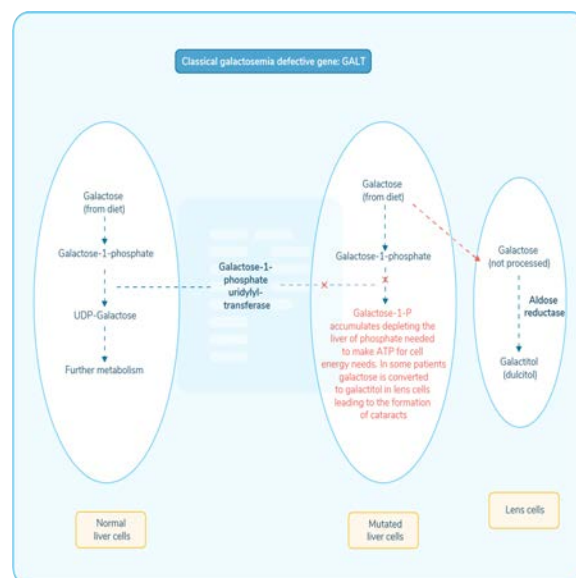
Galactose Metabolism :

Dietary Source	Lactose in milk and milk products.
Structure	Aldohexose and it is the 4th epimer of glucose.
Metabolism Site	Metabolized exclusively by the liver . Therefore, galactose tolerance tests can assess liver function.
Galactose Metabolism :	
1) Phosphorylation Galactose is phosphorylated by galactokinase to form galactose-1-phosphate. (Option A ruled out) ATP is the phosphate donor.	
2) Uridine diphosphate-galactose (UDP-galactose) formation Galactose-1-phosphate converts to UDP-galactose and glucose-1-phosphate via an exchange reaction with UDP-glucose The reaction is catalyzed by Galactose-1-phosphate uridyl transferase (GALT : rate-limiting enzyme in galactose metabolism) (Option B ruled out)	
3) UDP-galactose conversion to UDP-glucose UDP-galactose is isomerized to UDP-glucose by UDP-hexose-4-epimerase. (Option C ruled out) The new UDP-glucose can enter biosynthetic reactions (e.g., glycogenesis) and the GALT reaction.	
4) UDP-galactose in biosynthetic reactions UDP-galactose serves as an active donor of galactose for synthesizing lactose, glycoproteins, glycolipids and glycosaminoglycans. If dietary galactose is unavailable, UDP-galactose can be synthesized from UDP-glucose via UDP-hexose-4-epimerase.	

Aldolase reductase (Option D)- reduces galactose to galactitol and glucose to sorbitol. It is not the rate-limiting of galactose metabolism

Solution for Question 2:

Galactose-1-phosphate uridylyltransferase (GALT) deficiency is seen in classic galactosemia, in which galactose-1-phosphate and, therefore, galactose accumulate.



Galactose-1-phosphate Uridyl transferase (GALT) Deficiency	Severely deficient in individuals with Classic Galactosemia in which galactose-1-phosphate and, therefore, galactose accumulate. (Option B) It is an inborn error of metabolism
Accumulation	Accumulated galactose is shunted into side pathways like galactitol production. Aldose reductase catalyzes it. Galactose-1-phosphate accumulates in the liver, inhibiting galactokinase and glycogen phosphorylase, leading to hypoglycemia.

Bilirubin	Reduced bilirubin uptake and conjugation, increasing unconjugated bilirubin in blood
Symptoms	Hypoglycemia, hypophosphatemia, hyperuricemia, lactic acidosis Hepatomegaly, jaundice, Severe mental retardation Galactosuria (galactose excreted in urine) Cataracts due to dulcitol accumulation in the lens Tubular damage and generalized aminoaciduria
Diagnosis	Clinical manifestations (e.g., congenital cataract), presence of galactose in urine and elevated blood galactose levels Prenatal diagnosis via amniocentesis Detection of heterozygous parents by elevated galactose levels after a galactose load
Complications in untreated cases	E. coli sepsis, Fanconi syndrome Feeding problems, Failure to thrive, Hepatocellular damage, Bleeding
Treatment	Withdrawal of galactose and lactose from the diet.
GALT deficiency is a part of the newborn screening panel	

Galactokinase (Option A) phosphorylate galactose to form galactose 1-phosphate. Its deficiency leads to less severe disorder of galactose metabolism (Benign Galactosemia).

UDP-hexose-4-epimerase (Option C) isomerises UDP-galactose to UDP-glucose. Its deficiency leads to a less severe disorder of galactose metabolism.

Aldolase reductase (Option D) reduces galactose to galactitol and glucose to sorbitol.

Solution for Question 3:

Essential fructosuria is a benign metabolic disorder caused by fructokinase deficiency.

Fructose Metabolism :

Dietary Source	Found in fruits, honey, and high-fructose corn syrup
Structure	Ketohexose
Metabolism Site	Primarily metabolized by the liver
Fructose Metabolism :	

1) Phosphorylation: Fructokinase is the primary enzyme for fructose phosphorylation. It has a high affinity (low Km) and a high Vmax for fructose. It converts fructose to fructose-1-phosphate using ATP. It is found in the liver, kidneys, and small intestine .	
2) Fructose 1-Phosphate Cleavage: Aldolase B cleaves fructose 1-phosphate to dihydroxyacetone phosphate (DHAP) and glyceraldehyde. It is present in the liver, kidneys, and small intestine. DHAP enters glycolysis or gluconeogenesis. Glyceraldehyde is metabolized through various pathways.	
Disorders of Fructose Metabolism	1) Essential Fructosuria: Benign metabolic disorder Caused by fructokinase deficiency Excretion of fructose in urine Low risk of cataracts as aldose reductase does not act on fructose, and sorbitol dehydrogenase has low affinity for it. 2) Hereditary Fructose Intolerance (HFI): Caused by aldolase B deficiency. (Option A ruled out) Symptoms appear when infants consume sucrose or fructose. Effects are due to accumulation of fructose 1-phosphate. Symptoms: Vomiting, appetite loss, failure to thrive, hepatomegaly, jaundice, potential liver damage and death Diagnosis: Detection of fructose in urine, enzyme assay using liver cells or DNA-based testing. Treatment: Remove sucrose, sorbitol and fructose from the diet.

Hexokinase (Option B) – This enzyme is involved in the initial phosphorylation of glucose, not fructose. Hexokinase can phosphorylate fructose to fructose-6-phosphate, but it does not cause the specific symptoms seen in essential fructosuria. Therefore Hexokinase is not the enzyme responsible for essential fructosuria.

Glucokinase (Option D) – This enzyme is involved in glucose metabolism and is primarily found in the liver and pancreas. It is not involved in the metabolism of fructose. Therefore, Glucokinase is not relevant to essential fructosuria.

Solution for Question 4:

- Hereditary Fructose Intolerance (HFI): In HFI, the enzyme aldolase B is deficient. This leads to the accumulation of fructose-1-phosphate in the liver, kidneys, and intestines. The buildup of fructose-1-phosphate interferes with phosphate metabolism, leading to a decrease in available phosphate, resulting in hypophosphatemia.

- In HFI, the enzyme aldolase B is deficient. This leads to the accumulation of fructose-1-phosphate in the liver, kidneys, and intestines.
- The buildup of fructose-1-phosphate interferes with phosphate metabolism, leading to a decrease in available phosphate, resulting in hypophosphatemia.
- Classical Galactosemia: In Classical Galactosemia, the enzyme galactose-1-phosphate uridylyltransferase (GALT) is deficient. This leads to the accumulation of galactose-1-phosphate. Similar to HFI, the accumulated galactose-1-phosphate also interferes with phosphate metabolism, causing hypophosphatemia.
- In Classical Galactosemia, the enzyme galactose-1-phosphate uridylyltransferase (GALT) is deficient. This leads to the accumulation of galactose-1-phosphate.
- Similar to HFI, the accumulated galactose-1-phosphate also interferes with phosphate metabolism, causing hypophosphatemia.
- In HFI, the enzyme aldolase B is deficient. This leads to the accumulation of fructose-1-phosphate in the liver, kidneys, and intestines.
- The buildup of fructose-1-phosphate interferes with phosphate metabolism, leading to a decrease in available phosphate, resulting in hypophosphatemia.
- In Classical Galactosemia, the enzyme galactose-1-phosphate uridylyltransferase (GALT) is deficient. This leads to the accumulation of galactose-1-phosphate.
- Similar to HFI, the accumulated galactose-1-phosphate also interferes with phosphate metabolism, causing hypophosphatemia.

Feature	Hereditary Fructose Intolerance	Classical Galactosemia
Enzyme Deficiency	Aldolase B	Galactose-1-phosphate uridylyltransferase (GALT)
Onset (Option D ruled out)	Symptoms appear when sucrose/fructose is introduced into the diet (around 6 months of age)	Initial signs usually present in the first few weeks of life
Accumulated Metabolite	Fructose-1-phosphate	Galactose-1-phosphate and galactose
Symptoms (Option A ruled out) (Option C)	Hypoglycemia, hypophosphatemia, hyperuricemia, lactic acidosis Hepatomegaly, jaundice, Vomiting, Loss of appetite, Failure to thrive. Aldose reductase cannot act on fructose (very low risk of cataract)	Hypoglycemia, hypophosphatemia, hyperuricemia, lactic acidosis, Hepatomegaly, jaundice, Severe mental retardation, Galactosuria, Cataracts
Diagnosis	Presence of fructose in urine, enzyme assay or by DNA-based testing	Clinical manifestations, elevated blood galactose levels; amniocentesis for prenatal diagnosis
Urinary Findings	Positive Benedict's test for fructose	Positive Benedict's test for galactose

Dietary Management (Option B ruled out)	Sucrose, sorbitol and fructose must be removed from the diet	Galactose and lactose must be removed from the diet
Complication in untreated cases	Renal failure, Hepatic failure, Seizures, Progressive coma	E. coli sepsis, Feeding problems, Failure to thrive, Hepatocellular damage.

Note:

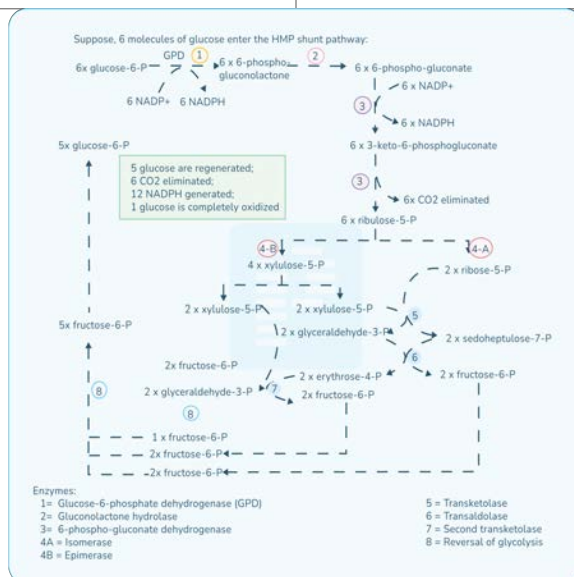
- Benedict's test is positive for reducing sugars.
- Seliwanoff's test/ Rapid furfural test is specific for Fructose.

Solution for Question 5:

The pentose phosphate pathway (PPP) is essential in erythrocytes because it generates NADPH. NADPH is crucial for maintaining glutathione in its reduced form, which is necessary for the activity of glutathione peroxidase. This enzyme removes harmful hydrogen peroxide (H_2O_2), protecting red blood cells from oxidative damage and preventing hemolysis.

Pentose Phosphate Pathway (PPP)	
The Pentose Phosphate Pathway (PPP), also known as the hexose monophosphate shunt (HMP shunt), is an alternative metabolic pathway for glucose that doesn't generate ATP. Location: Cytosol It serves two primary functions:	
NADPH Production	The PPP is a major source of NADPH, vital for biosynthesis (fatty acids and steroids). NADPH Production: RBCs rely on glycolysis for ATP due to the lack of mitochondria and depend solely on the PPP for NADPH, as they lack mitochondrial dehydrogenases. Glutathione Maintenance: NADPH keeps glutathione reduced, enabling glutathione peroxidase to neutralise hydrogen peroxide and prevent oxidative damage. Impact of NADPH Deficiency : Low NADPH impairs glutathione peroxidase, increases oxidative damage , and leads to hemolysis .
Ribose-5-Phosphate (R-5-P) Synthesis	Ribose-5-phosphate is essential for nucleotide synthesis needed for RNA and DNA.

Homeostasis between NADPH and R-5-P production.	When NADPH demand exceeds R-5-P's need, the non-oxidative phase converts R-5-P to glycolytic intermediates (fructose-6-phosphate and glyceraldehyde-3-phosphate). If the demand for R-5-P is higher than for NADPH, the non-oxidative phase generates R-5-P from fructose-6-phosphate and glyceraldehyde-3-phosphate, bypassing the oxidative steps.
Differences from Glycolysis	Uses NADP ⁺ instead of NAD ⁺ and generates CO ₂ , not ATP.



To produce ATP for energy metabolism (Option A): While RBCs do require ATP, their primary energy source is glycolysis. The PPP is not a major ATP-producing pathway.

To generate ribose-5-phosphate for nucleotide synthesis (Option B): Ribose-5-phosphate is a product of the PPP and is essential for nucleotide synthesis. However, mature RBCs lack a nucleus and do not synthesize nucleotides. Therefore this is not the primary function in mature RBCs.

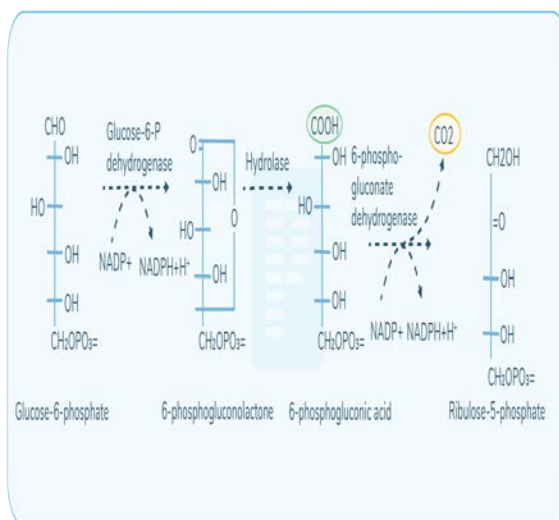
To synthesize haemoglobin (Option D): Haemoglobin synthesis occurs during RBC development in the bone marrow, not through the PPP.

Solution for Question 6:

Oxidative Phase of the Pentose Phosphate Pathway (PPP):

Irreversible process producing NADPH and converting glucose-6-phosphate to ribulose-5-phosphate.

Step	Substrates	Enzyme	Products
1	Glucose-6-phosphate + NADP ⁺	Glucose-6-phosphate dehydrogenase	6-Phosphogluconolactone + NADPH
2	6-Phosphogluconolactone	Gluconolactone hydrolase	6-Phosphogluconate
3	6-Phosphogluconate + NADP ⁺	6-Phosphogluconate dehydrogenase	Ribulose-5-phosphate + NADPH + CO ₂



Overall Reaction:



Fate of products:

- **NADPH:** The two molecules of NADPH produced in this phase are essential for various biosynthetic pathways, such as fatty acid synthesis and steroid synthesis.
- **Ribulose-5-Phosphate:** Used for nucleotide synthesis or converted to other sugars for glycolysis or gluconeogenesis.

Solution for Question 7:

NADH and NADPH Functions:

- NADH: Primarily involved in catabolic reactions such as cellular respiration where it functions as an electron carrier, transferring electrons to the electron transport chain to produce ATP.
- NADPH: Primarily involved in anabolic reactions, such as fatty acid and nucleotide synthesis, and in maintaining the reducing environment of the cell by donating electrons.

NADPH Role and Function:

Feature	NADH	NADPH
Full Name	Nicotinamide Adenine Dinucleotide (reduced).	Nicotinamide Adenine Dinucleotide Phosphate (reduced).
Primary Role	Energy production, electron transport.	Reductive biosynthesis, maintaining reduced states.
Pathways	Primarily generated in catabolic pathways: Citric Acid Cycle, Glycolysis, Fatty Acid oxidation.	Primarily generated in anabolic pathways: Pentose Phosphate Pathway, Malic Enzyme Reaction, Oxidative Decarboxylation - Isoforms of Isocitrate Dehydrogenase.
Key Functions	Donates electrons to the Electron Transport Chain (ETC) for ATP production. The ETC uses NADH for ATP production, yielding 2.5 ATPs.	Donates electrons & provides reducing power for biosynthetic reactions (reductive biosynthesis) like lipid and nucleotide synthesis. Maintains reduced glutathione (G-SH) in RBCs, preventing oxidative damage. Regenerates reduced glutathione (G-SH) from its oxidized form. It is not involved in ATP generation via the ETC.
Production Site	Mitochondria (mainly), Cytosol	Cytosol (mainly), some in mitochondria and plastids

Both are produced in the Pentose Phosphate Pathway (Option A): This is only partially true. NADPH is a key product of the pentose phosphate pathway. NADH is produced in other metabolic pathways, like glycolysis and the citric acid cycle.

NADH is used in lipid biosynthesis, while NADPH is not. (Option B): This is incorrect. NADPH is essential for lipid biosynthesis, providing the necessary reducing power. NADH has a minimal role in that process.

Both serve the same functions in the cell. (Option D): This is incorrect. While both are electron carriers, their main purposes and locations of work within the cell, differ greatly.

Solution for Question 8:

Malate Dehydrogenase: Catalyzes the oxidation of malate to oxaloacetate in the citric acid cycle, producing NADH but not releasing CO₂. This is an oxidation reaction, not an oxidative decarboxylation.



Enzymes that catalyse oxidative decarboxylation reaction:

Enzyme	Reaction (Oxidative decarboxylation of)	Cofactors	Details
Pyruvate Dehydrogenase (PDH)	Pyruvate to acetyl-CoA	Thiamin diphosphate, lipoate, NAD ⁺ , FAD, CoA	Converts pyruvate to acetyl-CoA in mitochondria Similar to the α -ketoglutarate dehydrogenase complex Regulated by end-product inhibition and covalent modification.
α -Ketoglutarate Dehydrogenase (α KGDH) (Option A ruled out)	α -ketoglutarate to succinyl-CoA	Similar to the PDH complex; Equilibrium favors succinyl-CoA formation; Inhibited by arsenite.	
Isocitrate Dehydrogenase (IDH) (Option D ruled out)	Isocitrate to α -ketoglutarate	Mg ²⁺ or Mn ²⁺ ions	Forms oxalosuccinate, which then decarboxylates; Part of the citric acid cycle.
6-Phosphogluconate Dehydrogenase (6-PGDH) (Option C ruled out)	6-phosphogluconate to ribulose-5-phosphate	NADP ⁺	Catalyzes a reaction in the pentose phosphate pathway.
Branched-Chain Ketoacid Dehydrogenase (BCKADH)	Decarboxylation of branched-chain keto acids	Thiamin pyrophosphate (TPP), lipoate, NAD ⁺ , FAD, CoA	Similar to the PDH complex; Defects cause maple syrup urine disease (MSUD).

Solution for Question 9:

- The oxidative phase of the pentose phosphate pathway (PPP) is referred to as the irreversible phase, while the nonoxidative phase is known as the reversible phase.
- 6-phosphogluconate dehydrogenase is part of the oxidative phase and catalyzes the conversion of 6-phosphogluconate to Ribulose-5-phosphate.

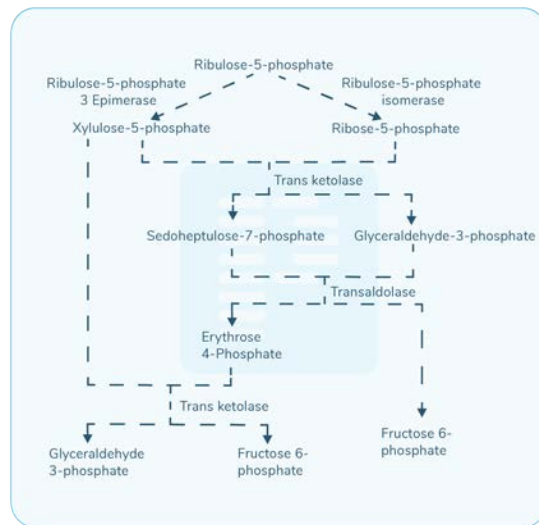
Substrate	Enzyme	Products
Ribulose-5-phosphate	Ribulose-5-phosphate 3- Epimerase (Option A ruled out)	Xylulose-5-phosphate
Ribulose-5-phosphate	Ribose-5-phosphate ketoisomerase	Ribose-5-phosphate
Xylulose-5-phosphate + Ribose-5-phosphate	Transketolase (Option B ruled out) Cofactor: Mg^{2+} , Thiamin diphosphate (Vitamin B1)	Sedoheptulose-7-phosphate + Glyceraldehyde-3-phosphate
Sedoheptulose-7-phosphate + Glyceraldehyde-3-phosphate	Transaldolase (Option A ruled out)	Fructose-6-phosphate + Erythrose-4-phosphate
Xylulose-5-phosphate + Erythrose-4-phosphate	Transketolase (second reaction)	Fructose-6-phosphate + Glyceraldehyde-3-phosphate

Regenerating Glucose-6-Phosphate:

- Two glyceraldehyde-3-phosphate molecules combine to form one fructose-6-phosphate (reversing step 4 of glycolysis).
- Fructose-6-phosphate is then converted back to glucose-6-phosphate (reversing step 2 of glycolysis).

Thiamine deficiency:

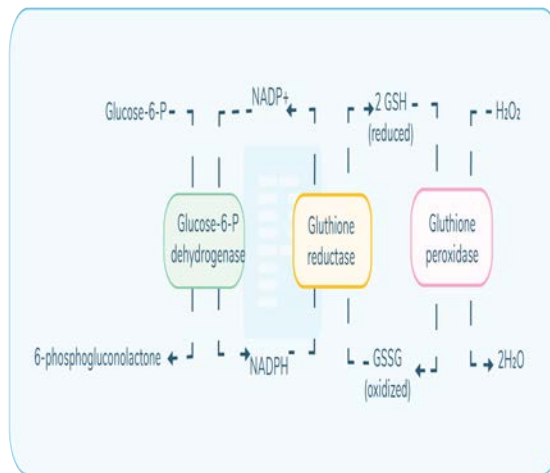
- Transketolase requires thiamine pyrophosphate (TPP) as a coenzyme.
- Thiamine deficiency impairs transketolase activity.
- RBCs rely on the pentose phosphate pathway for NADPH.
- Low RBC transketolase activity can indicate thiamine deficiency.



Solution for Question 10:

- In this patient with G6PD deficiency, the consumption of fava beans triggered oxidative stress. Glucose-6-phosphate dehydrogenase (G6PD) is an enzyme involved in the pentose phosphate pathway, which is crucial for producing NADPH. NADPH is necessary to regenerate glutathione, a potent antioxidant that helps protect red blood cells (RBCs) from oxidative damage.
- In individuals with G6PD deficiency, there is insufficient NADPH production. This makes the RBCs more vulnerable to oxidative stress, such as that caused by fava beans, which contain compounds that can trigger oxidative damage. Without enough NADPH to counteract this stress, RBCs are damaged, leading to hemolysis (destruction of red blood cells).
- As a result of hemolysis, the patient may experience fatigue, jaundice (due to the release of bilirubin from the breakdown of red blood cells), and dark urine (from the presence of hemoglobin or bilirubin in the urine).

Increased ribulose-5-phosphate overloads nucleotides, leading to renal failure (Option B): While ribulose-5-phosphate is a product of the PPP, its accumulation does not directly lead to renal failure. The primary issue in G6PD deficiency is the lack of NADPH and the resulting oxidative damage.



Impaired glucose-6-phosphate conversion disrupts glycolysis, causing hypoglycemia (Option C): While G6PD deficiency affects glucose-6-phosphate, it does not significantly disrupt glycolysis to cause hypoglycemia. The primary issue is oxidative stress, not energy deficiency.

Decreased ATP production affects muscle function, causing myopathy (Option D): While ATP is important, the PPP is not a major ATP-producing pathway in RBCs. The primary problem is the lack of NADPH and the resulting oxidative damage to the red blood cells.

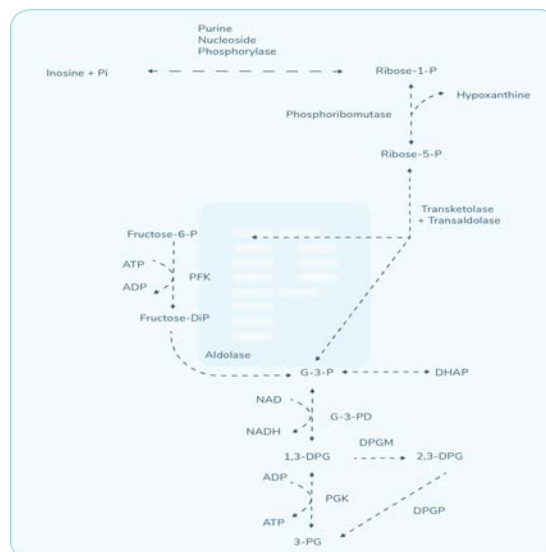
Solution for Question 11:

- **2,3-DPG and Oxygen Delivery:** 2,3-diphosphoglycerate (2,3-DPG) is a molecule found in red blood cells that affects hemoglobin's affinity for oxygen. It binds to hemoglobin, decreasing its affinity for oxygen, which means oxygen is more readily released to tissues. In stored blood, 2,3-DPG levels decline over time. This results in increased hemoglobin affinity for oxygen and impaired oxygen delivery to tissues.
- 2,3-diphosphoglycerate (2,3-DPG) is a molecule found in red blood cells that affects hemoglobin's affinity for oxygen.
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- Inosine Monophosphate (IMP): IMP is a purine nucleotide. In stored blood, IMP can be metabolized by red blood cells to regenerate 2,3-DPG. This helps to maintain adequate 2,3-DPG levels, ensuring that transfused red blood cells can effectively deliver oxygen to tissues.
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Inosine Monophosphate (IMP) is Hypoxanthine + Ribose

Inosine in Stored Blood :



- Inosine stimulates 2,3-DPG synthesis, by getting converted to ribose-1-phosphate and hypoxanthine. (Option B)
- Ribose-1-phosphate is isomerized to ribose-5-phosphate and enters the pentose phosphate pathway, and is recycled back to fructose-6-phosphate or glyceraldehyde-3-phosphate (GA3P).
- GA3P is converted to 1,3-diphosphoglycerate (1,3-DPG) in the presence of GA3P dehydrogenase and NAD.
- 1,3-DPG is converted to either 2,3-diphosphoglycerate (2,3-DPG) or 3-phosphoglycerate
- 2,3 DPG enhances oxygen delivery to tissues.
- Blood low in 2,3-DPG shifts the oxygen-hemoglobin dissociation curve left, reducing oxygen delivery. Reconstitution of 2,3-DPG is rapid, but normalization takes days.
- Inosine does not retard changes in plasma Na, K, Hb, lactic acid, dextrose, or pH during storage.

It increases plasma Na and K levels (Option A): IMP's primary function is related to red blood cell metabolism, specifically 2,3-DPG. It does not directly influence plasma sodium or potassium levels. Electrolyte balance is maintained by other mechanisms, and IMP is not involved in that.

It prevents lactic acid buildup (Option C): Lactic acid buildup in stored blood is a consequence of anaerobic glycolysis due to glucose metabolism. While IMP plays a role in red blood cell metabolism, it does not directly prevent lactic acid accumulation. Other additives in stored blood, like citrate, help to mitigate this by preventing coagulation and slightly altering pH.

It stabilizes hemoglobin concentration (Option D): Hemoglobin concentration in stored blood is primarily determined by the initial blood collection and processing. IMP's role is to improve the functional capacity of hemoglobin (oxygen delivery), not to stabilize its concentration.

Solution for Question 12:

Sorbitol Dehydrogenase is the enzyme responsible for converting sorbitol to fructose. Its absence in tissues like the retina and lens leads to sorbitol accumulation, causing osmotic stress and cellular swelling, resulting in snowflake cataracts in diabetic patients.

Hyperglycemia and Sorbitol Accumulation:

- Occurs in diabetes due to chronic high blood sugar.
- Excess glucose enters insulin-independent cells (retina, lens, kidneys, peripheral nerves).

- Aldose reductase activity increases, leading to more sorbitol production.
- Sorbitol dehydrogenase enzyme is absent in the eye hence sorbitol cannot be broken down further.

Consequences of Sorbitol Accumulation:

Osmotic Stress:

Sorbitol gets trapped inside cells, causing water influx and cell swelling and cataract Formation.

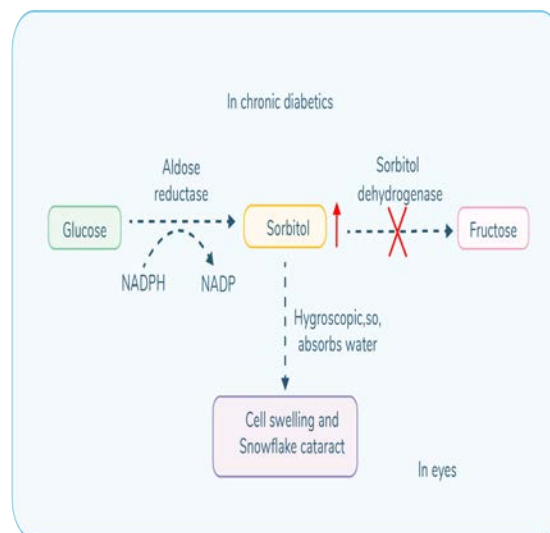
Tissue-Specific Enzyme Presence:

Present:

- Liver: Sorbitol to fructose conversion, enters glycolysis.
- Ovaries: Sorbitol dehydrogenase present.
- Seminal Vesicles: Sorbitol dehydrogenase provides energy (fructose) for sperm cells.

Absent:

- Retina, Lens, Kidneys, Peripheral Nerves: Have aldose reductase but lack sorbitol dehydrogenase, leading to sorbitol accumulation.



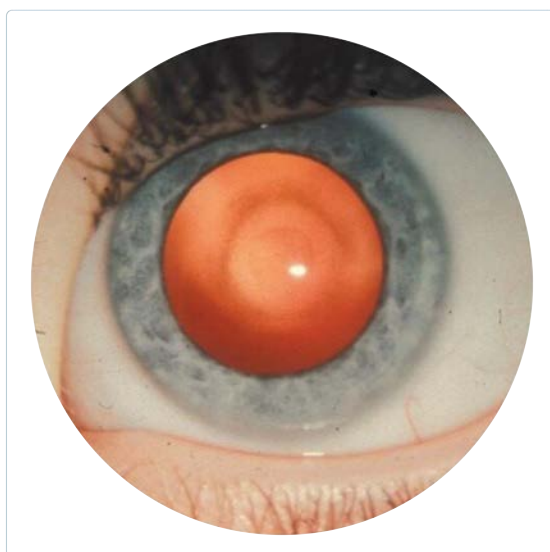
Aldose Reductase (Option A): converts glucose to sorbitol.

Glucose-6-phosphate dehydrogenase (Option B): is a part of the pentose phosphate pathway and does not play a direct role in the sorbitol pathway.

Hexokinase (Option D): Catalyses the first step of glycolysis and is not involved in the sorbitol pathway.

Solution for Question 13:

Galactosemia is a genetic disorder that impairs the body's ability to metabolise galactose due to a defect in the enzyme galactose-1-phosphate uridylyltransferase. This leads to the accumulation of galactitol, which causes osmotic stress and the characteristic "oil drop" cataracts.



Oil Drop Cataracts and Galactosemia:

- Galactosemia is an autosomal recessive genetic disorder caused by a deficiency in the enzyme galactose-1-phosphate uridylyltransferase, impairing galactose metabolism.

Characteristics

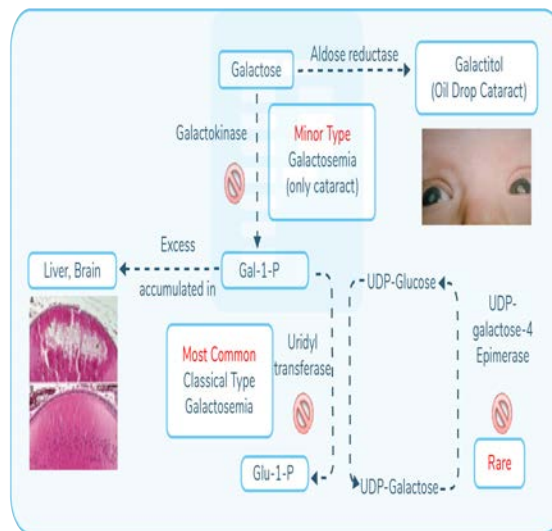
- Appearance: Distinct "oil droplet" look on retro illumination due to dulcitol accumulation.
- Symptoms: Decreased visual acuity, lenticular myopia, potential total lens opacification.

Pathophysiology

- The inability to convert galactose to glucose leads to the buildup of galactitol.
- Affects lens transparency, causing cataracts shortly after birth.

Management and Treatment

- Dietary Restrictions: Eliminate all sources of galactose (milk, dairy) to prevent progression.
- Surgical Intervention: Cataract removal may be necessary if vision is significantly impaired.



Phenylketonuria (Option A) is a metabolic disorder affecting phenylalanine metabolism, leading to intellectual disability if untreated.

Maple Syrup Urine Disease (Option C) is a metabolic disorder affecting branched-chain amino acids, characterized by sweet-smelling urine.

Homocystinuria (Option D) is a disorder affecting methionine metabolism, leading to symptoms such as skeletal abnormalities and lens dislocation.

Fates of pyruvate, PDH complex, Electron Transport Chain

1. In red blood cells under low energy states, which of the following best describes the fate of pyruvate?

- A. Converted to acetyl-CoA to enter the citric acid cycle.
 - B. Converted to oxaloacetate to replenish the citric acid cycle.
 - C. Converted to lactate by lactate dehydrogenase to regenerate NAD + .
 - D. Converted to alanine via transamination.
-

2. Which of the following metabolic disturbances is most likely to be seen in a patient with pyruvate dehydrogenase deficiency?

- A. Hypoglycemia due to impaired gluconeogenesis.
 - B. Hyperglycemia due to increased gluconeogenesis.
 - C. Ketoacidosis due to excessive fatty acid oxidation.
 - D. Lactic acidosis due to impaired conversion of pyruvate to acetyl-CoA.
-

3. During fasting, after how many hours does gluconeogenesis become the primary source of blood glucose?

- A. 8 to 12 hours
 - B. 12 to 24 hours
 - C. 16 to 48 hours
 - D. 48 to 72 hours
-

4. Which of the following is the primary site for peripheral lipolysis?

- A. Liver
 - B. Skeletal muscle
 - C. Adipose tissue
 - D. Pancreas
-

5. A 5-year-old boy presents with severe lethargy, vomiting and ataxia for 24 hours. Lab tests show metabolic acidosis, elevated lactate, and pyruvate. Genetic testing confirms pyruvate dehydrogenase complex deficiency. What is the underlying mechanism for his metabolic derangements?

- A. Impaired conversion of pyruvate to acetyl-CoA.
 - B. Increased conversion of pyruvate to ribose-5-phosphate.
 - C. Reduced activity of the electron transport chain.
 - D. Inhibition of gluconeogenesis in the liver.
-

6. Which of the following regulatory effects best explains the anti-neoplastic effect of the ketogenic diet?

- A. Acetyl CoA stimulates pyruvate carboxylase.
 - B. Malonyl CoA inhibits carnitine acyltransferase-1.
 - C. Citrate inhibits PFK1.
 - D. Acyl CoA stimulates carnitine acyltransferase-1.
-

7. A 25-year-old athlete presents with exercise intolerance and muscle fatigue, particularly during high-intensity training. Genetic testing reveals a deficiency in the enzyme that facilitates the transfer of electrons from cytosolic NADH to mitochondrial FADH₂ in skeletal muscle. Which shuttle system is likely impaired in this athlete?

- A. Malate shuttle
 - B. Glycerol phosphate shuttle
 - C. Citrate shuttle
 - D. Aspartate shuttle
-

8. Which of the following statements regarding the electron transport chain (ETC) is incorrect?

- A. Electrons from NADH enter the ETC through Complex I
 - B. Complex II is the only complex that does not pump protons.
 - C. Coenzyme Q and cytochrome c transfer electrons between complexes
 - D. Complex IV is the only complex that contains iron-sulfur proteins.
-

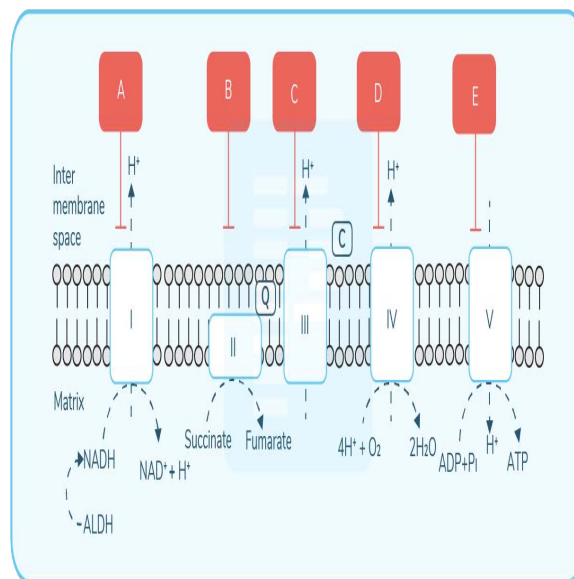
9. A patient, with a history of attempting weight loss, presents with fatigue, muscle weakness, and vision problems. Genetic testing confirms MELAS (mitochondrial encephalopathy, lactic acidosis, and stroke). Which electron transport chain complex is most likely affected in this patient

- A. Complex I
 - B. Complex II
 - C. Complex III
 - D. Complex V
-

10. In the electron transport chain, how many ATP molecules are produced per NADH and FADH₂, respectively? How many protons (H⁺) are translocated across the inner mitochondrial membrane for each?

- A. 2.5 ATP per NADH and 1.5 ATP per FADH₂ ; with 10 H⁺ and 6 H⁺ translocated, respectively.
- B. 3 ATP per NADH and 2 ATP per FADH₂ ; with 12 H⁺ and 8 H⁺ translocated, respectively.
- C. 2 ATP per NADH and 1 ATP per FADH₂ ; with 8 H⁺ and 4 H⁺ translocated, respectively.
- D. 1 ATP per NADH and 0.5 ATP per FADH₂ ; with 6 H⁺ and 3 H⁺ translocated, respectively.

11. The image below shows the sites of action for different ETC inhibitors. Based on the image, which of the following is the site of action of Cyanide?



- A. A
- B. E

C. C

D. D

12. Which of the following is a physiological uncoupler?

A. 2,4-dinitrophenol

B. Uncoupling protein 1 [UCP1]

C. Valinomycin

D. Nigericin

13. A 28-year-old woman presents with recurrent episodes of neurologic disturbances and lactic acidosis, particularly after carbohydrate-rich meals. Genetic testing reveals a mutation affecting one of the components of the pyruvate dehydrogenase complex. Which of the following is most likely to be directly affected by this mutation?

A. Flavin adenine dinucleotide (FAD)

B. Dihydrolipoyl transacetylase

C. Thiamine diphosphate

D. Coenzyme A

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	3
Question 4	3
Question 5	1
Question 6	3
Question 7	2
Question 8	4
Question 9	1
Question 10	1
Question 11	4
Question 12	2
Question 13	2

Solution for Question 1:

Pyruvate can become lactate, acetyl CoA, alanine and oxaloacetate based on the energy state of the cell.

In low energy state:

- In a low-energy state, the fate of pyruvate depends on the oxygenation of the cell:

In aerobic conditions:	In anaerobic conditions:
In the presence of oxygen, aerobic respiration will continue.	In the absence of oxygen, anaerobic glycolysis takes place.
Pyruvate, which is produced at the end of glycolysis, is transported to mitochondria.	It takes place in mitochondria deficit cells such as RBCs and neurons and also in muscle cells during exercise (Option C)
In mitochondria, pyruvate combines with coenzyme A to produce Acetyl CoA, which enters the citric acid pathway (Option A ruled out).	In these cells, pyruvate is converted to lactate by lactate dehydrogenase, using NADH to regenerate NAD ⁺ , which is necessary for ATP production.
The byproducts of this reaction include CO ₂ and NADH	The byproduct is NAD ⁺ .

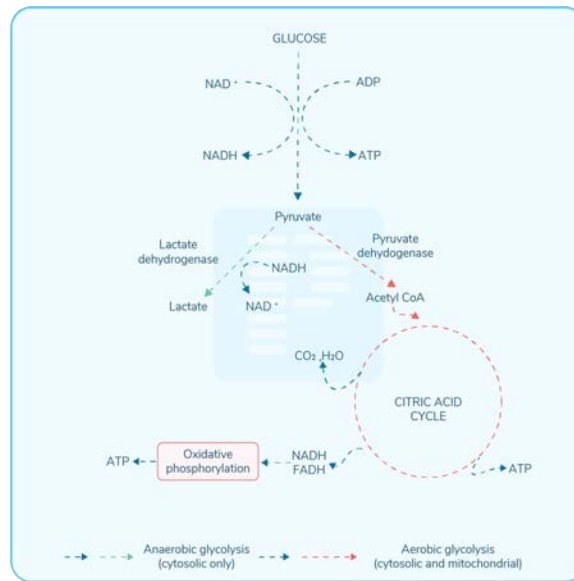
Converted to oxaloacetate to replenish the citric acid cycle (Option B): Oxaloacetate conversion by pyruvate carboxylase mainly occurs in liver and kidney cells, replenishing the citric acid cycle under in high energy state (not in anaerobic conditions). However, RBCs lack mitochondria and do not undergo this conversion.

Converted to alanine via transamination (Option D): Pyruvate can be converted to alanine via transamination, which is seen in the high-energy state of cells.

Solution for Question 2:

Pyruvate dehydrogenase deficiency:

- The most common cause is a mutation in the PDHA1 gene, which is inherited in an X-linked pattern.
- PDH deficiency leads to impaired conversion of pyruvate to acetyl CoA.
- As a result, pyruvate accumulates in the tissues and is shunted towards lactate production, leading to lactic acidosis, particularly after glucose load, a hallmark of PDH deficiency (Option D).
- Glucose levels are normal (Options A&B; ruled out)



- Clinical features include nausea, vomiting, neurologic problems such as ataxia, intellectual disability, seizures, hypotonia, and poor coordination.
- Treatment is based on stimulating PDH with cofactor supplementation such as thiamine, and lipoic acid.

Ketoacidosis due to excessive fatty acid oxidation (Option C): Ketoacidosis is more commonly associated with conditions like diabetic ketoacidosis or inborn errors of metabolism involving fatty acid oxidation, not PDH deficiency. PDH deficiency also presents with ketoacidosis, but the most likely presentation is lactic acidosis

Solution for Question 3:

- Gluconeogenesis becomes the primary source of blood glucose after liver glycogen stores are depleted.
- Glycogenolysis maintains blood glucose for the first 12 to 24 hours .
- After 16 hours , gluconeogenesis gradually increases and becomes the dominant source by 24 to 48 hours .
- Major substrates include lactate, alanine, and glycerol.

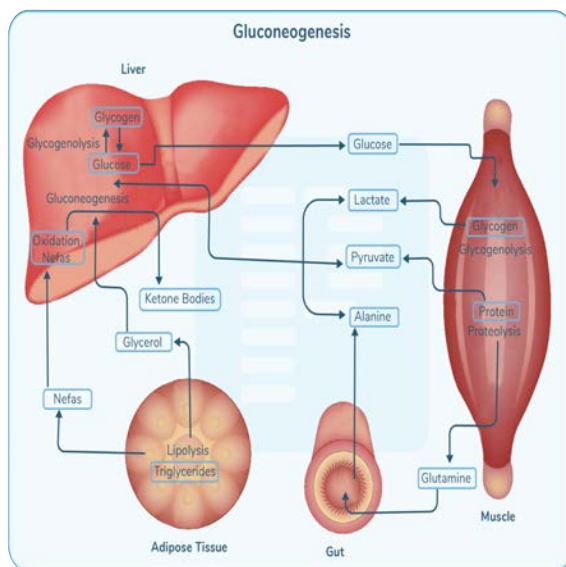
During fasting:

- After 4-16 hrs, the primary source of glucose is glycogenolysis

- After 16-48 hrs, the primary source of glucose is gluconeogenesis.

Gluconeogenesis:

- It is a process in which glucose is synthesized by non-carbohydrate sources such as amino acids, lactate, glycerol, and propionyl CoA.
- The major site of gluconeogenesis the liver > kidney.



Regulation of gluconeogenesis:

- Glucagon activates fructose 1,6-bisphosphatase and inhibits PFK-1, favoring gluconeogenesis and inhibiting glycolysis
- The availability of substrates such as alanine, lactate, and glycerol favors gluconeogenesis
- Acetyl CoA is an allosteric activator of 1st regulatory step of gluconeogenesis (pyruvate carboxylase)

8 to 12 hours (Option A): Incorrect because glycogenolysis is still the primary source of glucose at this stage.

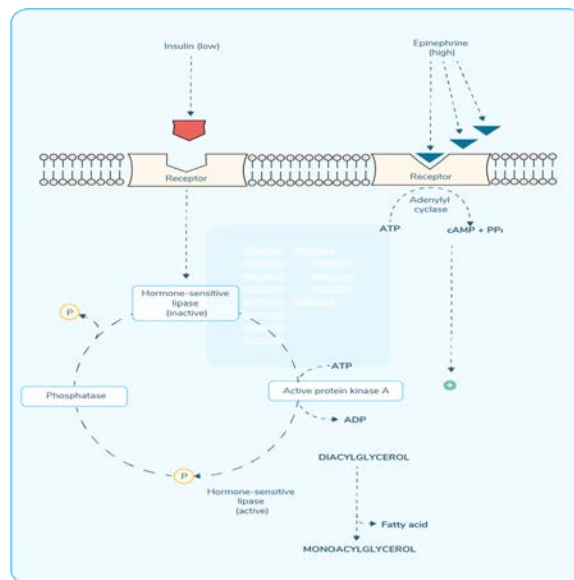
12 to 24 hours (Option B): Incorrect because gluconeogenesis is only starting to rise, but glycogenolysis is still significant.

48 to 72 hours (Option D): Incorrect because gluconeogenesis has already been the dominant glucose source for a long time by this point.

Solution for Question 4:

Peripheral lipolysis:

- It is a process in which triacylglycerols are hydrolyzed into free fatty acids (FFA) and glycerol.
- This process is mediated by hormone-sensitive lipase (HSL).
- This process occurs in adipose tissue (Option C)
- The lipase acts on triacylglycerols and converts them into diacylglycerol, the preferred substrate for HSL.



- The glycerol released from the adipose tissue is transported to the liver and participates in glycolysis or gluconeogenesis.
- Free fatty acids bind to albumin and are transported to muscles, where they are activated into their CoA derivatives and oxidized in mitochondria for ATP.

Regulation:

- HSL is activated when phosphorylated with protein kinase A, which in turn is dependent on cAMP produced in the adipocyte when catecholamines activate adenyl cyclase.
- Insulin dephosphorylates HSL and inactivates it.
- The counter-regulatory hormones of insulin, such as glucagon, epinephrine, and cortisol, stimulate the activity of HSL.

Liver (Option A): Incorrect because the liver processes fatty acids but is not the primary site for peripheral lipolysis.

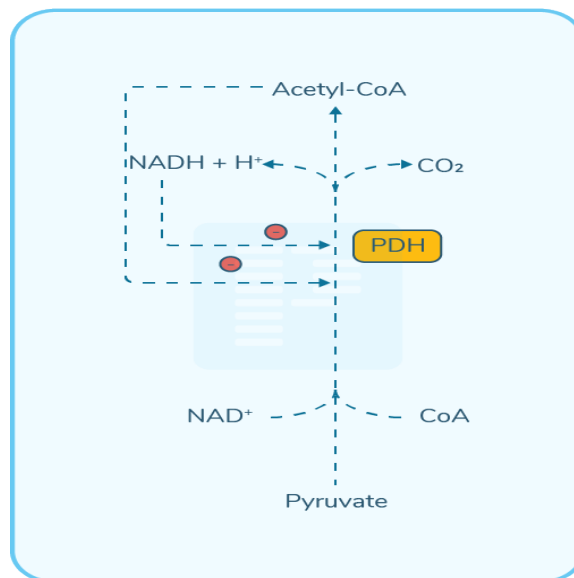
Skeletal muscle (Option B): Incorrect because muscle utilizes fatty acids but does not primarily break down triglycerides.

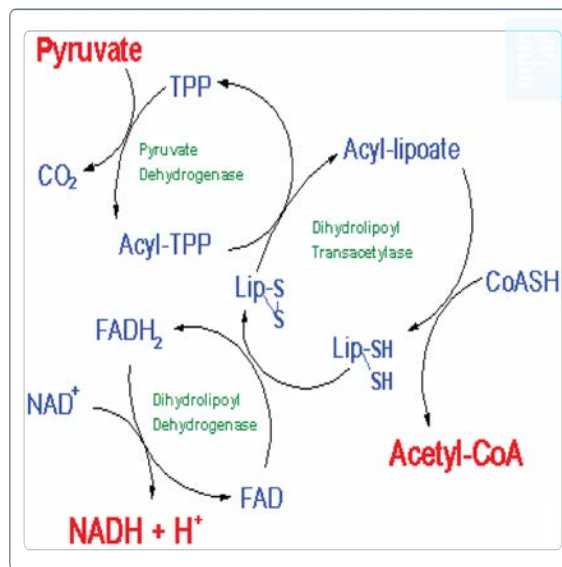
Pancreas (Option D): Incorrect because the pancreas regulates metabolism via insulin and glucagon but does not carry out lipolysis.

Solution for Question 5:

- The pyruvate dehydrogenase complex is responsible for the irreversible conversion of pyruvate to acetyl-CoA, a crucial step in aerobic glucose metabolism.
- A deficiency in this complex leads to the accumulation of pyruvate, which is then shunted towards lactate production, resulting in lactic acidosis.

Pyruvate dehydrogenase (PDH) complex:





- Catalyzes the irreversible conversion of pyruvate to acetyl-CoA in mitochondria.
- A critical step in aerobic glucose metabolism, as it links glycolysis to the citric acid cycle.
- Structure : 5 Coenzymes - thiamine pyrophosphate (TPP), lipoic acid, CoA, FAD, and NAD +. 3 products: NADH, CO₂, Acetyl-CoA. 3 substrates: Pyruvate, NAD, CoA. 3 subunits: E1 (PDH / PDC), E2 (dihydrolipoyl transacetylase), and E3 (dihydrolipoyl dehydrogenase).
- 5 Coenzymes - thiamine pyrophosphate (TPP), lipoic acid, CoA, FAD, and NAD +.
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- 3 substrates: Pyruvate, NAD, CoA.
- 3 subunits: E1 (PDH / PDC), E2 (dihydrolipoyl transacetylase), and E3 (dihydrolipoyl dehydrogenase).
- Irreversible reaction : Pyruvate to acetyl-CoA is highly exergonic and irreversible, preventing direct synthesis of glucose from acetyl-CoA.
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- Regulation : Inhibition: Acetyl-CoA, NADH, and ATP inhibit PDH to conserve glucose. Phosphorylation: Kinase inactivates, and phosphatase activates PDH based on energy status. Hormonal Control: Insulin activates PDH for fat synthesis Calcium: Activates PDH during muscle contraction.
- Inhibition: Acetyl-CoA, NADH, and ATP inhibit PDH to conserve glucose.
- Phosphorylation: Kinase inactivates, and phosphatase activates PDH based on energy status.
- Hormonal Control: Insulin activates PDH for fat synthesis
- Calcium: Activates PDH during muscle contraction.

- Clinical significance : PDH complex deficiencies are rare but serious genetic disorders causing lactic acidosis, neurological issues, and developmental delays. Thiamine deficiency, common in alcoholics, impairs PDH complex activity, leading to pyruvate accumulation and lactic acidosis.
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- Phosphorylation: Kinase inactivates, and phosphatase activates PDH based on energy status.
- Hormonal Control: Insulin activates PDH for fat synthesis
- Calcium: Activates PDH during muscle contraction.
- PDH complex deficiencies are rare but serious genetic disorders causing lactic acidosis, neurological issues, and developmental delays.
- Thiamine deficiency, common in alcoholics, impairs PDH complex activity, leading to pyruvate accumulation and lactic acidosis.

Increased conversion of pyruvate to ribose-5-phosphate (Option B): Pyruvate does not convert to ribose-5-phosphate (R-5-P), which is produced through the pentose phosphate pathway, and it does not involve the PDH complex. The issue is the impaired conversion of pyruvate to acetyl-CoA, not R-5-P production.

Reduced activity of the electron transport chain (Option C) is a potential cause of metabolic acidosis.

Inhibition of gluconeogenesis in the liver (Option D) would result in hypoglycemia but would not explain the accumulation of lactate and pyruvate.

Solution for Question 6:

The ketogenic diet's antineoplastic effect is explained by the Pasteur effect. The ketogenic diet is high fat low carbohydrate diet. This helps in restricting neoplastic growth, as neoplastic cells are dependent on glucose through aerobic glycolysis (Warburg effect).

Effect	Definition	Clinical Significance
Glucose-Sparing Effect	The inhibition of glucose oxidation by fatty acid oxidation.	Neurons and red blood cells exclusively use glucose for fuel. Red and cardiac muscle fibers can utilize glucose or fatty acids. The glucose-sparing effect ensures cardiac muscle prioritizes fatty acids, preserving glucose for neurons and red blood cells.
Pasteur Effect	The inhibition of anaerobic glycolysis by aerobic oxidation products.	The ketogenic diet limits carbohydrate intake, thereby reducing glucose availability and glycolysis. Lower pyruvate reduces citric acid cycle activity. Increased fatty acid oxidation raises acetyl-CoA, forming citrate. Citrate inhibits PFK-1, suppressing glycolysis and limiting glucose use by neoplasia.
Warburg Effect	The phenomenon in cancer cells where, even in the presence of oxygen, glucose is primarily converted to lactate through aerobic glycolysis.	This allows faster ATP production and provides intermediates for neoplastic cell growth and multiplication. It increases NADPH production, protecting neoplastic cells from oxidative stress.

Acetyl-CoA stimulates pyruvate carboxylase (Option A): While acetyl-CoA does stimulate pyruvate carboxylase, this process increases glucose levels, which is counterproductive for someone with neoplasia.

Malonyl-CoA inhibits carnitine acyltransferase-1 (Option B): This is related to fatty acid oxidation but is not directly involved in the antineoplastic effect of a ketogenic diet.

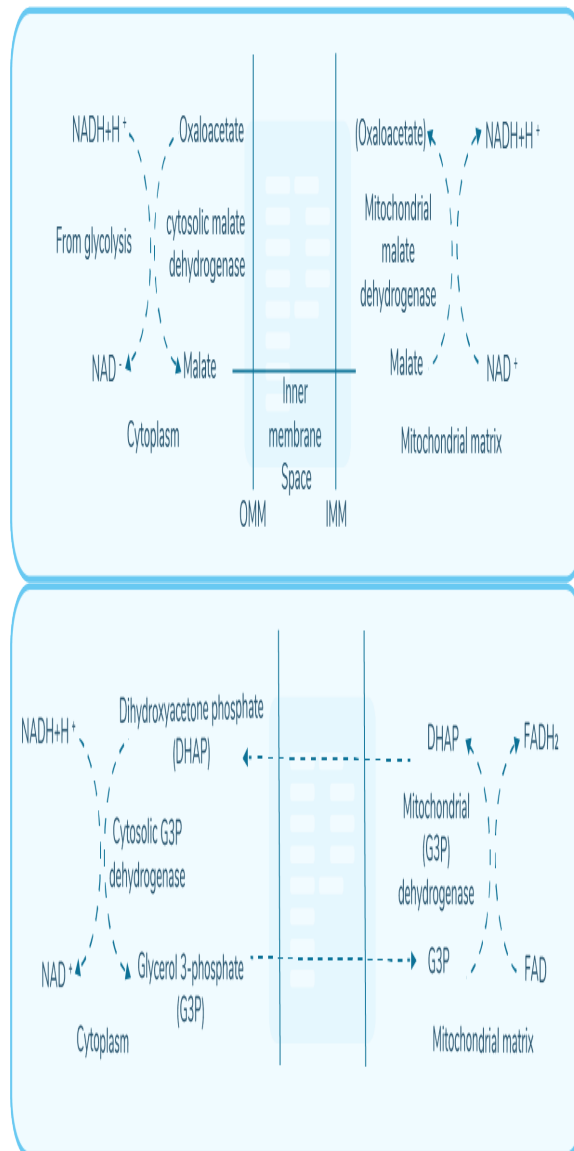
Acyl CoA stimulates carnitine acyltransferase-1 (Option D): This is related to fatty acid oxidation but is not directly involved in the antineoplastic effect of a ketogenic diet.

Solution for Question 7:

The glycerol phosphate shuttle, present in white muscle fibers and neurons, specifically transfers electrons from cytosolic NADH to mitochondrial FADH₂

Electron Transport Chain (ETC):

- Location : Inner mitochondrial membrane. The ADP-ATP translocase in the inner mitochondrial membrane exports ATP to the cytosol and imports ADP into the mitochondria. Atractyloside inhibits this transport process.
- Function : It plays a crucial role in oxidative phosphorylation. Oxidation, which involves the loss of electrons, is central to this process and drives ATP synthesis
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- Oxidation, which involves the loss of electrons, is central to this process and drives ATP synthesis
- Shuttles : NADH Shuttles: Glycolysis, which occurs in the cytoplasm, generates 2 NADH molecules per glucose molecule. NADH cannot directly cross the mitochondrial membrane; it requires shuttles for transport. There are two main shuttle systems: the malate-aspartate shuttle (Image 1) and the glycerol phosphate shuttle (Image 2), which facilitate NADH transport across the mitochondrial membrane
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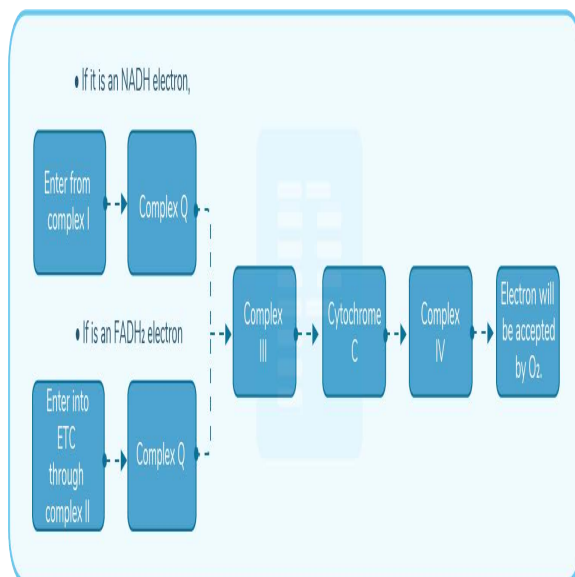
Property	Malate Shuttle (Option A ruled out)	Glycerol Phosphate Shuttle
Location	Present in most cells except white muscle fibers and neurons.	White muscle fibers and neurons
Energy	"No net ATP loss" → Consumes 2 ATP per glucose molecule.	Loss of 2 ATPs per glucose molecule
Electron Carrier	Electron Carrier: Transfers electrons to mitochondrial NADH, maintaining cytosolic NAD^+/NADH balance.	Transfers electrons to mitochondrial FADH_2
Enzymes	Cytosolic and mitochondrial glycerol-3-phosphate dehydrogenase.	Cytosolic & mitochondrial glycerol-3-phosphate dehydrogenase

Citrate shuttle (Option C) transports acetyl-CoA from the mitochondria to the cytosol for fatty acid synthesis, not for transferring electrons from NADH to FADH₂.

The aspartate shuttle (Option D) , also known as the malate-aspartate shuttle, is involved in transferring electrons from cytosolic NADH to mitochondrial NADH.

Solution for Question 8:

All electron transport chain complexes contain iron-sulfur proteins, except Complex IV (cytochrome c oxidase), which contains copper instead.



Complex	Name	Electron Donor	Electron Acceptor	Protons Pumped
Complex I	NADH dehydrogenase	NADH	Coenzyme Q (or Ubiquinone)	4
Complex II	Succinate dehydrogenase	Succinate (FADH ₂)	Coenzyme Q	0
Complex III (Cytochrome b & c ₁)	Q-cytochrome c oxidoreductase	Coenzyme Q (QH ₂)	Cytochrome C	4
Complex IV (Cytochrome a & a ₃)	Cytochrome c oxidase	Cytochrome C	Oxygen	2

Complex V	ATP Synthase			
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Complex V consists of two main parts: The F_0 portion, which is embedded in the membrane and transports protons, and the F_1 portion, which catalyzes ATP formation from ADP and P_i .

Electrons from NADH enter the ETC through Complex I (Option A): Correct because NADH transfers electrons to Complex I, initiating the electron transport chain.

Complex II is the only complex that does not pump protons (Option B): Correct because Complex II transfers electrons to Coenzyme Q but does not contribute to proton pumping.

Coenzyme Q and cytochrome c transfer electrons between complexes (Option C): Correct because these mobile carriers shuttle electrons between different complexes.

Solution for Question 9:

- MELAS (Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like episodes) is caused by mutations in mitochondrial DNA, primarily affecting Complex I of the electron transport chain.
- Complex I (NADH dehydrogenase) is responsible for transferring electrons from NADH to Coenzyme Q, initiating the ETC and contributing to ATP production.
- Dysfunction in Complex I leads to impaired oxidative phosphorylation, decreased ATP generation, and increased lactic acid levels, contributing to the metabolic abnormalities seen in MELAS.
- Symptoms such as fatigue, muscle weakness, and neurological issues (including vision problems and stroke-like episodes) result from energy deficiency in high-demand tissues like muscles and the brain.

Electron Transport Chain Defects in Mitochondrial Disorders

Complex	Role in ETC	Association with Disease
Complex I (NADH dehydrogenase)	Transfers electrons from NADH to Coenzyme Q	MELAS, Leber's Hereditary Optic Neuropathy (LHON)
Complex II (Succinate dehydrogenase)	Transfers electrons from $FADH_2$ to Coenzyme Q	Rarely affected in mitochondrial diseases

Complex III (Cytochrome bc ₁ complex)	Transfers electrons from Coenzyme Q to cytochrome c	Rare disorders (e.g., Mitochondrial Complex III deficiency)
Complex IV (Cytochrome c oxidase)	Transfers electrons from cytochrome c to oxygen	Mitochondrial disorders like Leigh syndrome
Complex V (ATP synthase)	Synthesizes ATP from proton gradient	Neuromuscular disorders, Leigh syndrome

Complex II (Option B): Incorrect because Complex II mutations are not a major cause of MELAS. It primarily functions in the Krebs cycle and does not contribute significantly to mitochondrial DNA-related disorders.

Complex III (Option C): Incorrect because MELAS is not typically caused by defects in Complex III. Mutations here are rare and usually cause other mitochondrial myopathies.

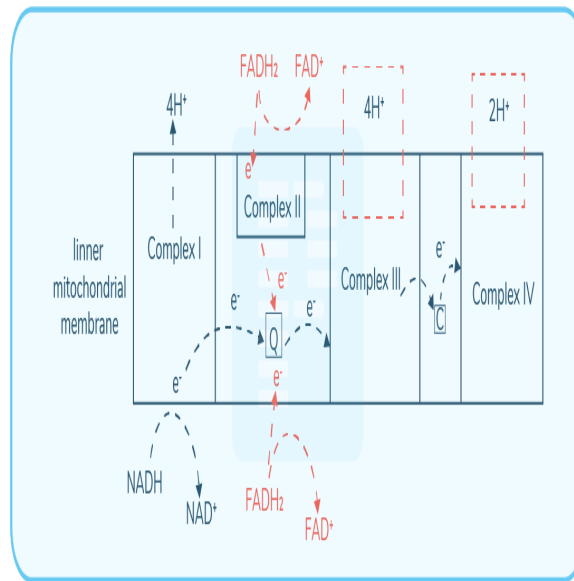
Complex V (Option D): Incorrect because Complex V (ATP synthase) is responsible for ATP production but is not directly linked to MELAS, which primarily affects electron transport.

Solution for Question 10:

- H⁺ ions are pumped into the intermembrane space, increasing the positive potential. This proton pumping drives ATP production via the ATP synthase complex.
- ATP synthase produces 1 ATP per 4 H⁺ ions passing through the F₁ component.
- Since 4 H⁺ generate 1 ATP, 10 H⁺ generate 2.5 ATP. Therefore, NADH yields 2.5 ATP.
- NADH, which pumps 4 protons from Complex I, 4 from Complex III, and 2 from Complex IV, generates approximately 2.5 ATP.

FADH₂, which pumps 4 protons from Complex III and 2 from Complex IV (but none from Complex II), generates approximately 1.5 ATP.

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Solution for Question 11:

Cyanide is a strong inhibitor of Complex IV (cytochrome c oxidase). It blocks electron transfer to oxygen, disrupting the electron transport chain and reducing ATP production, potentially causing cellular damage.

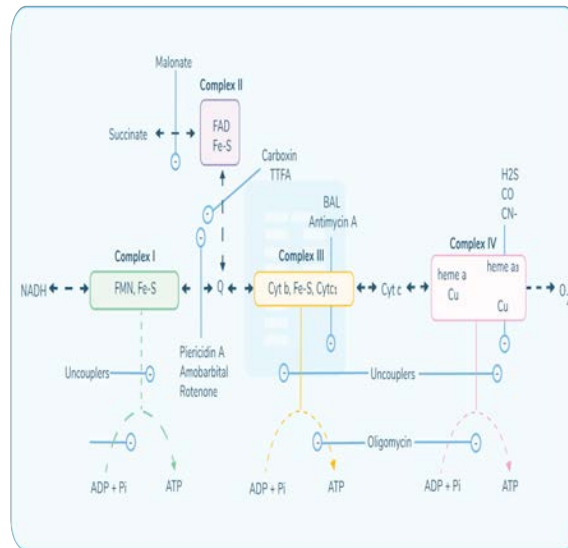
Inhibitors of the Electron Transport Chain (ETC):

Inhibitor	Target Site	Mechanism
Rotenone Amobarbital Piericidin A	Complex I (Option A ruled out) (NADH dehydrogenase)	Blocks electron transfer from NADH in Complex I to ubiquinone.
TTFA (Thenoyltrifluoroacetone) Carboxin	Complex II (Succinate dehydrogenase)	Blocks electron transfer from Complex II to ubiquinone
Malonate	Complex II	Blocks Complex II activity via competitive inhibition.
Antimycin A BAL	Complex III (Option C ruled out) Cytochrome bc1 complex	Blocks electron transfer from Complex III to cytochrome C.
Cyanide (CN ⁻) Carbon monoxide (CO) Hydrogen Sulfide (H ₂ S)	Complex IV (Option D) (Cytochrome c Oxidase)	Inhibits electron transfer from Complex IV to oxygen, halting respiration.

Oligomycin	ATP Synthase (Complex V) (Option B ruled out)	Inhibits the proton channel of ATP synthase, preventing ATP formation.
Atractyloside	Adenine Nucleotide Translocase	Blocks ADP/ATP exchange across the mitochondrial membrane

Note:

- ADP to ATP conversion is inhibited by: Oligomycin
- ADP to ATP transfer is inhibited by: Atractyloside



Solution for Question 12:

UCP1, also known as thermogenin, is the physiological uncoupler found in brown adipose tissue.

Uncouplers:

- In the electron transport chain (ETC), oxidation (electron flow) and phosphorylation (ATP formation) are normally coupled.
- Uncouplers disrupt this coupling, allowing electron transport to occur without ATP synthesis.

Mechanism of Uncoupling:

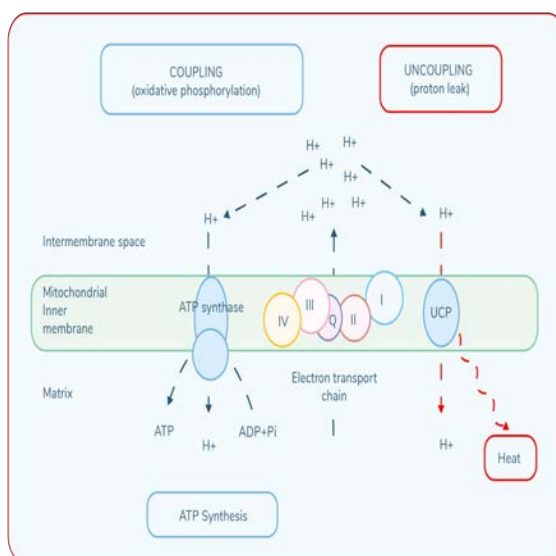
- Uncouplers act as ion channels in the inner mitochondrial membrane (IMM), allowing protons (H⁺) to bypass ATP synthase.

- Energy from proton movement is released as heat rather than being used for ATP synthesis.
- Effects of Uncouplers: No ATP production, increased heat production, and increased basal metabolic rate (BMR).

Physiological uncouplers	Artificial uncouplers
Present in brown adipose tissue Aid in thermogenesis in newborns Stimulated by hormones such as thyroxine Ex: Thermogenin (UCP1) (Option B) Thyroxine Free fatty acids (long chain)	These were once used as anti-obesity drugs. Ex: 2,4 Dinitrophenol (Option A ruled out) Valinomycin (Option C ruled out) Nigericin (Option D ruled out) High dose aspirin

Non-Shivering Thermogenesis:

Brown adipose tissue contains abundant mitochondria with uncoupling proteins (thermogenin/UCP1), facilitating non-shivering thermogenesis, which is crucial for heat production in neonates and hibernating animals.



Solution for Question 13:

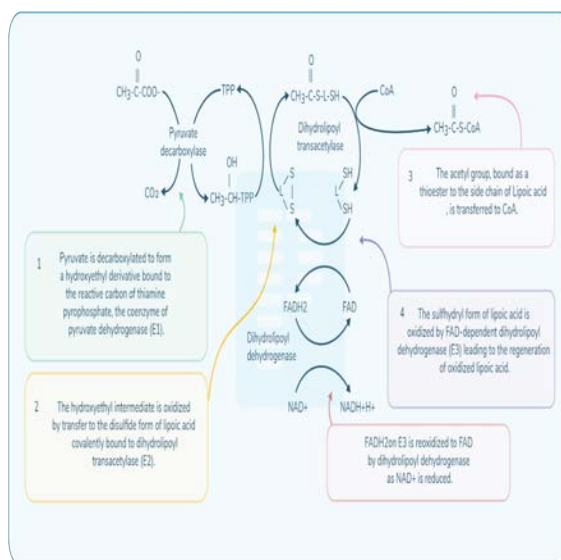
- Dihydrolipoyl transacetylase is one of the key enzymatic components of the pyruvate dehydrogenase complex.

- The pyruvate dehydrogenase complex consists of three enzyme components : pyruvate dehydrogenase (E1), dihydrolipoyl transacetylase (E2), and dihydrolipoyl dehydrogenase (E3) .
- A genetic mutation in any of these enzymes can cause pyruvate dehydrogenase deficiency, leading to lactic acidosis.
- Patients with inherited pyruvate dehydrogenase deficiency, caused by defects in one or more components of the enzyme complex, develop lactic acidosis, especially after a glucose load.
- Because the brain relies on glucose as its primary energy source, these metabolic defects often lead to neurological impairments

Flavin adenine dinucleotide (FAD) (Option A): s a coenzyme involved in the pyruvate dehydrogenase complex reaction, but it is not a protein component.

Thiamin diphosphate (Option C) is a cofactor, not a protein component of the complex. Pyruvate dehydrogenase (E1) catalyzes pyruvate decarboxylation, forming a hydroxyethyl-thiamin diphosphate intermediate.

Coenzyme A (Option D) is a cofactor, not a protein component of the complex. It accepts the acetyl group from acetyl-lipoamide, forming acetyl-CoA and reduced lipoamide.



GLUT transporters and Glycolysis

1. Match the following GLUT transporters with their primary locations: 1. GLUT1 a) Liver 2. GLUT2 b) Neurons 3. GLUT3 c) Erythrocytes 4. GLUT4 d) Skeletal Muscle

1. GLUT1	a) Liver
2. GLUT2	b) Neurons
3. GLUT3	c) Erythrocytes
4. GLUT4	d) Skeletal Muscle

- A. 1-c, 2-a, 3-b, 4-d
- B. 1-d, 2-b, 3-c, 4-a
- C. 1-b, 2-d, 3-a, 4-c
- D. 1-a, 2-c, 3-d, 4-b

2. Which of the following mechanisms is primarily responsible for glucose uptake into enterocytes?

- A. Facilitated diffusion via GLUT5 transporter
- B. Secondary active transport via SGLT1 transporter
- C. Primary active transport via Na⁺/K⁺ ATPase
- D. Simple diffusion through the lipid bilayer

3. Which of the following tissues is unable to transport glucose independently of insulin?

- A. Hepatocytes
 - B. Cardiac muscle
 - C. RBC
 - D. Neurons
-

4. All of the following are true about glycolysis except:

- A. Aerobic glycolysis produces 2 pyruvates and 7 ATPs while anaerobic glycolysis produces 2 lactates and 2 ATPs.
 - B. Retinal and corneal cells can produce energy using glycolysis in hypoxic conditions.
 - C. Tumor cells rely mainly on fatty acid oxidation
 - D. Fast twitch muscle fibers primarily rely on anaerobic glycolysis for ATP sources.
-

5. Which of the following statements is false?

- A. RBCs solely depend on glycolysis for ATP
 - B. White muscle fibers depend on anaerobic glycolysis during muscular contraction
 - C. Myoglobin is the storage form of oxygen in white muscle fibers
 - D. Renal medulla is most affected by ischemic damage to the kidney
-

6. Which of the following statements regarding aerobic glycolysis is true?

- A. Each acetyl CoA molecule yields 5 ATPs by converting to pyruvate
- B. Each molecule of NADPH yields 2.5 ATP
- C. Aerobic glycolysis of one molecule of glucose yields a net total of 7 ATP molecules.
- D. Each turn of the citric acid cycle produces 20 ATP molecules.

7. Which of the following enzymes catalyzes the conversion of glucose to glucose-6-phosphate in the glycolysis pathway?

- A. Hexokinase
 - B. Phosphofructokinase
 - C. Aldolase
 - D. Enolase
-

8. Which of the following steps in glycolysis is irreversible?

- A. Conversion of glucose-6-phosphate to fructose-6-phosphate
 - B. Conversion of 3-phosphoglycerate to 2-phosphoglycerate
 - C. Conversion of phosphoenolpyruvate to pyruvate
 - D. Conversion of fructose-1,6-bisphosphate to glyceraldehyde-3-phosphate and dihydroxyacetone phosphate
-

9. Which of the following statements accurately describes the difference between hexokinase and glucokinase?

- A. Hexokinase has a higher K_m and V_{max} compared to glucokinase.
 - B. Glucokinase is inhibited by glucose-6-phosphate, whereas hexokinase is not.
 - C. Hexokinase is primarily found in the liver and pancreatic β -cells, while glucokinase is found in most tissues.
 - D. Glucokinase has a higher K_m and V_{max} compared to hexokinase.
-

10. Which of the following chemical compounds inhibits glycolysis by inhibiting the enzyme enolase?

- A. Iodoacetate
 - B. Fluoride
 - C. Arsenate
 - D. 2-Deoxyglucose
-

11. What is the primary function of the Rapoport-Luebering Shunt in red blood cells?

- A. To generate ATP from 2,3-bisphosphoglycerate
 - B. To produce 2,3-bisphosphoglycerate, which regulates oxygen binding to hemoglobin
 - C. To convert pyruvate to lactate
 - D. To facilitate the conversion of glucose to glucose-6-phosphate
-

12. Which of the following is a key positive regulator of the glycolytic enzyme phosphofructokinase-1 (PFK-1)?

- A. ATP
 - B. Citrate
 - C. Fructose-2,6-bisphosphate
 - D. Acetyl-CoA
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	3
Question 5	3
Question 6	3
Question 7	1
Question 8	3
Question 9	4
Question 10	2
Question 11	2
Question 12	3

Solution for Question 1:

Correct Answer: A) 1-c, 2-a, 3-b, 4-d

Explanation:

GLUT proteins are distributed in various tissues according to their specific functions.

Glucose Transporters:

Transporter	Tissue Location	Functions
GLUT 1	Brain, kidney, colon, placenta, erythrocytes	Facilitates glucose uptake
GLUT 2	Liver, pancreatic β cells, small intestine, kidney	Enables rapid glucose uptake or release
GLUT 3	Brain, kidney, placenta	Assists in glucose uptake
GLUT 4	Heart and skeletal muscle, adipose tissue	Mediates insulin-stimulated glucose uptake
GLUT 5	Small intestine	Facilitates fructose absorption
SGLT 1	Small intestine and kidney	Actively transports glucose against a concentration gradient

GLUT 1-5 are facilitative bidirectional transporters, whereas SGLT 1 is a sodium-dependent unidirectional transporter.

Solution for Question 2:

Correct Answer: B) Secondary active transport via SGLT1 transporter

Explanation:

- Enterocytes primarily uptake glucose through the secondary active transport mechanism mediated by the Sodium-dependent glucose cotransporter 1 (SGLT1).
- This transporter utilizes the sodium gradient created by the Na^+/K^+ ATPase pump on the basolateral membrane of the enterocyte.
- Sodium ions move down their concentration gradient into the cell, and glucose is co-transported with them into the enterocyte against its concentration gradient.

GLUT5 (Option A) is a transporter for fructose, not glucose. Facilitated diffusion via GLUT transporters does occur for glucose, but in enterocytes via GLUT2.

The Na^+/K^+ ATPase pump (Option C) does not directly transport glucose.

Glucose is a polar molecule and cannot diffuse (Option D) through the lipid bilayer of the cell membrane without assistance.

Solution for Question 3:

Correct Answer: B) Cardiac muscle

Explanation: Cardiac muscle primarily relies on Insulin-dependent GLUT4 transporter for glucose uptake. Rest other tissues use GLUT transporters which are insulin-independent for glucose uptake, as given in the table below

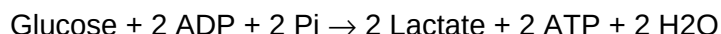
Glut Transporters	Locations
GLUT1	● Neurons, RBCs, and Placenta
GLUT2	● Enterocytes, Hepatocytes, β cells of pancreas, Proximal convoluted tubule
GLUT3	● Neurons, RBCs, and Placenta
GLUT4	● Insulin Dependent Glucose Uptake in Skeletal Muscles, Cardiac Muscles and Adipose Tissues
GLUT 5	● Fructose Absorption along the intestine.

Solution for Question 4:

Tumor cells primarily use anaerobic glycolysis for energy production even in the presence of oxygen, a phenomenon known as the Warburg effect , where glucose is converted to lactate despite sufficient oxygen.

Aerobic glycolysis produces 2 pyruvates and a net gain of 7 ATPs (5 ATP indirectly from 2 NADH and 2 ATP directly from glycolysis itself), while anaerobic glycolysis produces 2 lactates

and only 2 ATPs . (Option A) Anaerobic glycolysis:



Retinal & corneal cells can produce energy via anaerobic glycolysis under hypoxic conditions (Option B) . Also, RBC exclusively uses anaerobic glycolysis as the primary source of energy due to the lack of mitochondria, thereby reducing pyruvate to lactate. Fast twitch muscle (white muscle fibers or Type II fibers) rely on anaerobic glycolysis for a rapid source of ATP during intense, short bursts of activity (Option D) . Also, Slow twitch muscles (red muscle fibers or Type I fibers) have a high mitochondrial number and primarily rely on aerobic pathways for the generation of ATP.

Solution for Question 5:

- Myoglobin is a protein that stores oxygen in red muscle fibers (slow-twitch fibers) rather than in white muscle fibers (fast-twitch fibers). White muscle fibers are less reliant on myoglobin and tend to have fewer of these oxygen-storing molecules. Instead, white fibers rely more on anaerobic glycolysis for energy during muscle contractions.
 - White muscle fibers lack myoglobin and contain few mitochondria. Therefore, they rely on anaerobic glycolysis and creatine phosphate for energy production, especially during muscle contraction when the blood flow to the muscle fiber is reduced. (Option B)
 - Mature erythrocytes do not contain mitochondria. Without mitochondria, erythrocytes cannot utilize fatty acids or ketone bodies as metabolic fuel. Therefore, glycolysis is the only pathway available for erythrocytes to generate ATP . (Option A)
 - The medullary region of the kidney is the most prone to ischemic damage due to high oxygen demand and reduced perfusion due to supply by end arteries. Hence, in case of ischemia, they rely on anaerobic glycolysis for ATP. (Option D)
-

Solution for Question 6:

The complete oxidation of 1 mole of glucose to carbon dioxide and water yields a net of 32 moles of ATP . In aerobic glycolysis:

- Total ATP generated: 5 (from 2NADH oxidation) + 2 (from 1,3-BPG) + 2 (from PEP) = 9 ATP.
- Net ATP yield: 9 (total ATP generated) - 2 (consumed in initial steps) = 7 ATP
- In total, aerobic glycolysis of one molecule of glucose yields a net total of 7 ATP molecules.

Glycolysis in Anaerobic Cells (RBCs)	Glycolysis in Aerobic Cells Neurons
Anaerobic glycolysis.	Aerobic glycolysis.
2 Lactates are formed.	2 Pyruvates are formed.
2 ATPs are generated.	32 ATPs are generated on the complete oxidation of glucose in neurons. ● 7 ATPs from Glycolysis. ● 2 NADH $\times$ 2.5 ATP = 5 ATPs on conversion of 2 Pyruvates $\rightarrow$ 2 Acetyl CoA. With the help of PDH complex. ● Acetyl CoA enters the TCA cycle to produce 10 ATPs per molecule. ● 2 Acetyl CoA $\times$ 10 $\times$ ATPs=20ATPs.

- Each turn of the citric acid cycle starting from Acetyl CoA produces ten ATP molecules, one from substrate-level phosphorylation and nine from oxidative phosphorylation (Option A and D) .
- Each molecule of NADH yields 2.5 ATP, not NADPH (Option B) .

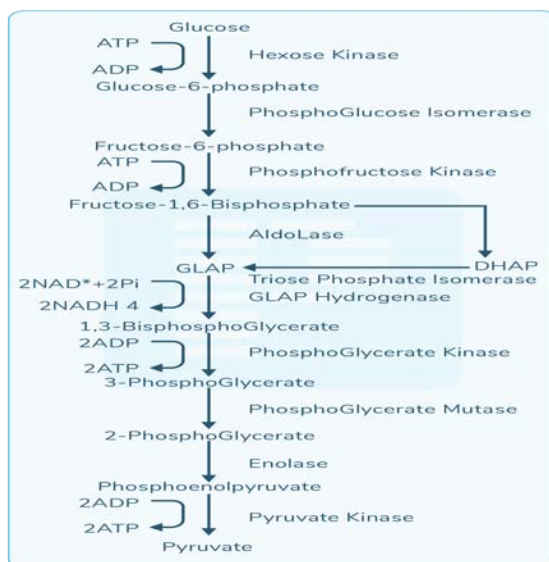
Solution for Question 7:

- Hexokinase: This enzyme catalyzes the first step of glycolysis, converting glucose to glucose-6-phosphate by transferring a phosphate group from ATP.
- Phosphofructokinase (Option B): This enzyme catalyzes the third step of glycolysis, converting fructose-6-phosphate to fructose-1,6-bisphosphate. It is a key regulatory enzyme in glycolysis.
- Aldolase (Option C): This enzyme catalyzes the fourth step of glycolysis, splitting fructose-1,6-bisphosphate into two three-carbon sugars: glyceraldehyde-3-phosphate and

dihydroxyacetone phosphate.

- Enolase (Option D): This enzyme catalyzes the ninth step of glycolysis, converting 2-phosphoglycerate to phosphoenolpyruvate.

Steps of Glycolysis:



Solution for Question 8:

In glycolysis, there are three major irreversible steps:

- Hexokinase/Glucokinase Reaction: This step converts glucose to glucose-6-phosphate and is catalyzed by hexokinase (or glucokinase in the liver). This reaction is irreversible and uses ATP.
- Phosphofructokinase-1 (PFK-1) Reaction: This step converts fructose-6-phosphate to fructose-1,6-bisphosphate and is catalyzed by PFK-1. This reaction is irreversible and is a key regulatory step in glycolysis, also using ATP.
- Pyruvate Kinase Reaction: This step converts phosphoenolpyruvate to pyruvate and is catalyzed by pyruvate kinase. This reaction is irreversible and produces ATP.

Conversion of glucose-6-phosphate to fructose-6-phosphate (Option A), Conversion of 3-phosphoglycerate to 2-phosphoglycerate (Option B), and Conversion of fructose-1,6-bisphosphate to glyceraldehyde-3-phosphate and dihydroxyacetone phosphate

(Option D) are all reversible reactions in glycolysis.

Solution for Question 9:

- Glucokinase has a higher K_m and V_{max} compared to hexokinase: Glucokinase has a lower affinity (higher K_m) for glucose but a higher capacity (higher V_{max}) compared to hexokinase.

Feature	Hexokinase	Glucokinase
Tissue Distribution (Option C)	Most tissues	Liver and pancreatic β -cells
K_m (Affinity for Glucose) (Option A)	Low K_m (high affinity for glucose)	High K_m (low affinity for glucose)
V_{max} (Maximum Rate)	Low V_{max}	High V_{max}
Regulation (Option B)	Inhibited by glucose-6-phosphate	Not inhibited by glucose-6-phosphate
Role	Maintains basal glucose utilization	Regulates blood glucose levels
Function at Low Glucose	Active even at low glucose concentrations	Only active when glucose levels are high
Gene Regulation	Constitutive expression	Inducible by insulin
Substrate Specificity	Broad (can phosphorylate other hexoses)	Specific to glucose

Solution for Question 10:

- Fluoride inhibits the enzyme enolase in the glycolytic pathway, preventing the conversion of 2-phosphoglycerate to phosphoenolpyruvate. By blocking this step, fluoride effectively halts glycolysis, leading to an accumulation of glycolytic intermediates upstream of the enolase reaction.

Chemical Inhibitors of Glycolysis:

Chemical Inhibitor	Target Enzyme	Mechanism of Inhibition
Fluoride	Enolase	Inhibits the conversion of 2-phosphoglycerate to phosphoenolpyruvate.
Iodoacetate (Option A)	Glyceraldehyde-3-phosphate dehydrogenase	Inhibits the conversion of glyceraldehyde-3-phosphate to 1,3-bisphosphoglycerate by covalently modifying the enzyme.
Arsenate (Option C)	Glyceraldehyde-3-phosphate dehydrogenase	Replaces phosphate, forming unstable 1-arseno-3-phosphoglycerate, bypassing ATP generation.
2-Deoxyglucose (Option D)	Hexokinase	Phosphorylated to 2-deoxyglucose-6-phosphate, which accumulates and inhibits hexokinase, halting glycolysis.

Solution for Question 11:

The Rapoport-Luebering Shunt forms 2,3-bisphosphoglycerate (2,3-BPG) through a series of enzymatic reactions involving 1,3-bisphosphoglycerate (1,3-BPG).

- Formation of 1,3-BPG: In glycolysis, glyceraldehyde-3-phosphate is converted to 1,3-bisphosphoglycerate by the enzyme glyceraldehyde-3-phosphate dehydrogenase. This reaction involves the addition of an inorganic phosphate and the reduction of NAD⁺ to NADH.
- In glycolysis, glyceraldehyde-3-phosphate is converted to 1,3-bisphosphoglycerate by the enzyme glyceraldehyde-3-phosphate dehydrogenase.
- This reaction involves the addition of an inorganic phosphate and the reduction of NAD⁺ to NADH.
- Conversion to 2,3-BPG: In the Rapoport-Luebering Shunt, 1,3-BPG is converted to 2,3-BPG by the enzyme bisphosphoglycerate mutase. This reaction shifts the phosphate group from the first carbon to the second carbon of the glycerate molecule, forming 2,3-BPG.
- In the Rapoport-Luebering Shunt, 1,3-BPG is converted to 2,3-BPG by the enzyme bisphosphoglycerate mutase.
- This reaction shifts the phosphate group from the first carbon to the second carbon of the glycerate molecule, forming 2,3-BPG.

- Conversion of 2,3-BPG to 3-phosphoglycerate: 2,3-BPG can be hydrolyzed to 3-phosphoglycerate by the enzyme 2,3-bisphosphoglycerate phosphatase. This reaction removes a phosphate group, resulting in the formation of 3-phosphoglycerate, which then re-enters the glycolytic pathway.
- 2,3-BPG can be hydrolyzed to 3-phosphoglycerate by the enzyme 2,3-bisphosphoglycerate phosphatase.
- This reaction removes a phosphate group, resulting in the formation of 3-phosphoglycerate, which then re-enters the glycolytic pathway.
- In glycolysis, glyceraldehyde-3-phosphate is converted to 1,3-bisphosphoglycerate by the enzyme glyceraldehyde-3-phosphate dehydrogenase.
- This reaction involves the addition of an inorganic phosphate and the reduction of NAD⁺ to NADH.
- In the Rapoport-Luebering Shunt, 1,3-BPG is converted to 2,3-BPG by the enzyme bisphosphoglycerate mutase.
- This reaction shifts the phosphate group from the first carbon to the second carbon of the glycerate molecule, forming 2,3-BPG.
- 2,3-BPG can be hydrolyzed to 3-phosphoglycerate by the enzyme 2,3-bisphosphoglycerate phosphatase.
- This reaction removes a phosphate group, resulting in the formation of 3-phosphoglycerate, which then re-enters the glycolytic pathway.

The Rapoport-Luebering Shunt does not generate ATP; it bypasses an ATP-generating step in glycolysis (Option A) .

The conversion of pyruvate to lactate is a function of lactate dehydrogenase and is not related to the Rapoport-Luebering Shunt (Option C) .

The conversion of glucose to glucose-6-phosphate is catalyzed by hexokinase or glucokinase and occurs at the very beginning of glycolysis, unrelated to the Rapoport-Luebering Shunt (Option D) .

Solution for Question 12:

- Fructose-2,6-bisphosphate is a potent activator of PFK-1. It enhances the enzyme's affinity for its substrate (fructose-6-phosphate) and diminishes the inhibitory effects of ATP, thus

promoting glycolysis.

Hormonal Regulation

- Insulin: Promoter of glycolysis. Increases the expression and activity of glycolytic enzymes like glucokinase, PFK-1, and pyruvate kinase. Enhances glycolysis by promoting glucose uptake and utilization.
- Promoter of glycolysis.
- Increases the expression and activity of glycolytic enzymes like glucokinase, PFK-1, and pyruvate kinase.
- Enhances glycolysis by promoting glucose uptake and utilization.
- Glucagon: Inhibitor of glycolysis. Activates pathways that result in the phosphorylation and inhibition of PFK-2, leading to reduced levels of fructose-2,6-bisphosphate. Decreases glycolysis by reducing the activity of PFK-1 through lower levels of fructose-2,6-bisphosphate.
- Inhibitor of glycolysis.
- Activates pathways that result in the phosphorylation and inhibition of PFK-2, leading to reduced levels of fructose-2,6-bisphosphate.
- Decreases glycolysis by reducing the activity of PFK-1 through lower levels of fructose-2,6-bisphosphate.
- Promoter of glycolysis.
- Increases the expression and activity of glycolytic enzymes like glucokinase, PFK-1, and pyruvate kinase.
- Enhances glycolysis by promoting glucose uptake and utilization.
- Inhibitor of glycolysis.
- Activates pathways that result in the phosphorylation and inhibition of PFK-2, leading to reduced levels of fructose-2,6-bisphosphate.
- Decreases glycolysis by reducing the activity of PFK-1 through lower levels of fructose-2,6-bisphosphate.

Allosteric Regulation

- Fructose-2,6-bisphosphate: Activator of PFK-1. Increases the affinity of PFK-1 for fructose-6-phosphate and reduces the inhibitory effects of ATP. Promotes glycolysis by enhancing the activity of PFK-1.
- Activator of PFK-1.
- Increases the affinity of PFK-1 for fructose-6-phosphate and reduces the inhibitory effects of ATP.
- Promotes glycolysis by enhancing the activity of PFK-1.

- ATP: (Option A) Inhibitor of PFK-1. Binds allosterically to PFK-1, indicating sufficient cellular energy, and decreases its activity. Inhibits glycolysis by reducing the activity of PFK-1.
- Inhibitor of PFK-1.
- Binds allosterically to PFK-1, indicating sufficient cellular energy, and decreases its activity.
- Inhibits glycolysis by reducing the activity of PFK-1.
- Citrate: (Option B) Inhibitor of PFK-1. Binds allosterically to PFK-1, indicating ample biosynthetic precursors and energy. Reduces glycolytic flux by decreasing the activity of PFK-1.
- Inhibitor of PFK-1.
- Binds allosterically to PFK-1, indicating ample biosynthetic precursors and energy.
- Reduces glycolytic flux by decreasing the activity of PFK-1.
- AMP: Role: Activator of PFK-1. Binds to PFK-1, increasing its activity when cellular energy is low. Enhances glycolysis by promoting the activity of PFK-1.
- Role: Activator of PFK-1.
- Binds to PFK-1, increasing its activity when cellular energy is low.
- Enhances glycolysis by promoting the activity of PFK-1.
- H⁺ (Protons): Inhibitor of PFK-1. High concentrations of H⁺ (low pH) inhibit PFK-1 to prevent excessive acidification of the cell. Decreases glycolytic flux to avoid further acidification.
- Inhibitor of PFK-1.
- High concentrations of H⁺ (low pH) inhibit PFK-1 to prevent excessive acidification of the cell.
- Decreases glycolytic flux to avoid further acidification.
- Activator of PFK-1.
- Increases the affinity of PFK-1 for fructose-6-phosphate and reduces the inhibitory effects of ATP.
- Promotes glycolysis by enhancing the activity of PFK-1.
- Inhibitor of PFK-1.
- Binds allosterically to PFK-1, indicating sufficient cellular energy, and decreases its activity.
- Inhibits glycolysis by reducing the activity of PFK-1.
- Inhibitor of PFK-1.
- Binds allosterically to PFK-1, indicating ample biosynthetic precursors and energy.
- Reduces glycolytic flux by decreasing the activity of PFK-1.
- Role: Activator of PFK-1.

- Binds to PFK-1, increasing its activity when cellular energy is low.
 - Enhances glycolysis by promoting the activity of PFK-1.
 - Inhibitor of PFK-1.
 - High concentrations of H^+ (low pH) inhibit PFK-1 to prevent excessive acidification of the cell.
 - Decreases glycolytic flux to avoid further acidification.
 - Acetyl-CoA (Option D): While Acetyl-CoA is not a direct regulator of PFK-1, it generally indicates a high-energy state and leads to a decrease in glycolysis through the activation of pyruvate carboxylase and the inhibition of pyruvate dehydrogenase.
-

Glycogen Metabolism and Glycogen Storage Disorders

1. Which of the following enzymes is responsible for converting the glycogen molecule from a branched structure to a linear structure during glycogenolysis?

- A. Glycogen Phosphorylase
 - B. Branching Enzyme
 - C. Debranching Enzyme
 - D. Glycogen Synthase
-

2. Which glycogen storage disease does not involve hypoglycemia or exercise intolerance?

- A. Von Gierke Disease
 - B. McArdle Disease
 - C. Andersen's Disease
 - D. Tarui Disease
-

3. A 6-month-old infant presents with recurrent episodes of hypoglycemia and elevated ketone levels. Genetic testing was done and diagnosed as Type 0 Glycogen storage disorder. Which of the following statements about Glycogen Storage Disease Type 0 is true?

- A. The enzyme deficiency is glycogen synthase.
- B. The affected genes are GYS2 and GYS1.
- C. It is an autosomal recessive disorder
- D. All of the above

4. A 1 year old child presents with failure to thrive, weak bones, and renal dysfunction. The physician suspected glycogen storage disorder and further identified a mutation in the SLC2A2 gene while genetic testing. Which among the following defects is associated with it?

- A. GLUT 2
 - B. Liver phosphorylase
 - C. Muscle and erythrocyte phosphofructokinase 1
 - D. Muscle Phosphorylase
-

5. A 6-month-old infant presents with muscle hypotonia and heart failure. Laboratory tests reveal an accumulation of glycogen in lysosomes. Which of the following is the most likely diagnosis?

- A. McArdle Disease
 - B. Pompe Disease
 - C. Von Gierke Disease
 - D. Cori Disease
-

6. A 5-year-old child presents with muscle weakness and mild hepatomegaly. On physical examination, the child has mild cardiomegaly. Blood tests reveal mild hypoglycemia and elevated serum creatine kinase levels. Which enzyme deficiency is associated with this glycogen storage disease?

- A. Glycogen synthase
- B. Glucose-6-phosphatase
- C. Debranching enzyme

D. Muscle phosphorylase

7. A 6-month-old infant presents with severe hypoglycemia, hepatomegaly, growth retardation, irritability, and lethargy between feedings. Laboratory tests reveal elevated lactate, triglycerides, and uric acid. The child frequently feeds to manage hypoglycemia. Based on these findings, which condition is most likely responsible for the symptoms?

- A. Von Gierke's disease
 - B. McArdle's disease
 - C. Tarui's Disease
 - D. Hers' Disease
-

8. Match the glycogen storage disorders with their respective enzyme defects. Glycogen Storage Disorders: Enzyme Defects: 1. Pompe Disease a. Debranching enzyme 2. McArdle Disease b. Acid maltase 3. Von Gierke Disease c. Glucose-6-phosphatase 4. Cori Disease d. Myophosphorylase

Glycogen Storage Disorders:	Enzyme Defects:
1. Pompe Disease	a. Debranching enzyme
2. McArdle Disease	b. Acid maltase
3. Von Gierke Disease	c. Glucose-6-phosphatase
4. Cori Disease	d. Myophosphorylase

- A. 1-b, 2-d, 3-c, 4-a
- B. 1-a, 2-c, 3-b, 4-d
- C. 1-d, 2-b, 3-a, 4-c
- D. 1-c, 2-a, 3-d, 4-b

9. A 32-year-old woman is admitted to the hospital for a routine check-up. During the visit, her diet is discussed, and it is noted that she has been consuming a high carbohydrate diet. You are asked to explain the impact of her dietary habits on glycogen metabolism. Which of the following statements best describes glycogen metabolism?

- A. Increased Gluconeogenesis
 - B. Reduced peripheral lipolysis
 - C. Low rate of glycogen synthesis
 - D. High rate of glycogenolysis
-

**10. Which of the following statements is true on the structure of glycogen?
Arranged in 12 concentric layers Glucose residues are connected by α -1,4 linkage
Branching points formed by α -1,6 linkage**

- A. 1,2
 - B. 2,3
 - C. 1,2,3
 - D. 1,3
-

11. A 25-year-old male athlete presents to the emergency department after completing a high-intensity interval training session. He reports feeling extremely fatigued and experiencing muscle cramps. Blood tests indicate elevated levels of lactic acid. What is the net ATP yield when one molecule of muscle glycogen is converted to lactic acid during strenuous exercise?

- A. 1 ATP

- B. 2 ATP
 - C. 3 ATP
 - D. 4 ATP
-

12. Which of the following statements correctly differentiates liver glycogen phosphorylase (LGP) and muscle glycogen phosphorylase (MGP)?

- A. LGP is activated by glucagon, while MGP is activated by epinephrine.
 - B. LGP releases glucose-6-phosphate into the bloodstream, while MGP releases free glucose into the muscle.
 - C. LGP activity is regulated by AMP, while MGP is regulated by glucagon and epinephrine.
 - D. MGP is highly active during fasting compared to LGP.
-

13. Which of the following enzymes catalyzes the rate-limiting step of glycogenolysis

- A. Phosphoglucomutase
 - B. Glucose-6-phosphatase
 - C. Glucan transferase
 - D. Glycogen phosphorylase
-

14. Which of the following statements is false regarding glycogenolysis?

- A. Glucose-6-phosphatase acts in liver
- B. Glycogen phosphorylase cleaves α -1,4 linkages
- C. Phosphoglucomutase converts glucose-1-phosphate into glucose-6-phosphate
- D. Glycogenolysis is the reverse process of glycogenesis.

15. Which of the following enzymes is activated by phosphorylation?

- A. Acetyl-CoA carboxylase
 - B. Glycogen synthase
 - C. Pyruvate dehydrogenase
 - D. Glycogen phosphorylase
-

16. What is the correct order of enzymes that act in glycogenesis?

- A. Hexokinase → Phosphoglucomutase → UDP-glucose pyrophosphorylase
 - B. Phosphoglucomutase → UDP-glucose pyrophosphorylase → Hexokinase
 - C. Hexokinase → UDP-glucose pyrophosphorylase → Phosphoglucomutase
 - D. UDP-glucose pyrophosphorylase → Phosphoglucomutase → Hexokinase
-

17. Assertion (A): Glycogen and amylopectin are both branched polysaccharides composed of glucose units. Reason (R): The branch points in glycogen occur every 8-12 glucose residues, while in amylopectin, they occur every 20-30 glucose residues.

- A. Both the assertion and the reason are correct, and the reason is the correct explanation for the assertion.
 - B. Both the assertion and the reason are correct, but the reason is not the correct explanation for the assertion.
 - C. The assertion is correct, but the reason is incorrect.
 - D. The assertion is incorrect, but the reason is correct.
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	4
Question 4	1
Question 5	2
Question 6	3
Question 7	1
Question 8	1
Question 9	2
Question 10	3
Question 11	3
Question 12	1
Question 13	4
Question 14	4
Question 15	4
Question 16	1
Question 17	1

Solution for Question 1:

Correct Answer: C) Debranching Enzyme

Explanation:

Debranching Enzymes play a crucial role in glycogenolysis by removing the branches from the glycogen molecule.

It has two activities : it transfers a short segment of glucose from a branch to a nearby chain and hydrolyzes the remaining glucose unit at the branch point, thus converting branched glycogen into a more linear form that can be further degraded.

- Transferase Activity : Moves a segment of 3 glucose units from the branch to the main chain
- α -1,6-Glucosidase Activity : Hydrolyzes the remaining α -1,6-glycosidic bond at the branch point ↓

Free Glucose (from the α -1,6-glycosidic bond)

Glycogen Phosphorylase (Option A) catalyzes the cleavage of glycogen into glucose-1-phosphate units by adding a phosphate group. It acts on the linear chains of glycogen, not the branches, and does not convert branched structures into linear ones.

Branching Enzyme(Option B) is involved in the synthesis of glycogen, specifically in the creation of branch points by transferring a segment of the glycogen chain to a new location. It is not involved in converting branched structures into linear structures during glycogenolysis.

Glycogen Synthase(Option D) is responsible for the synthesis of glycogen from glucose-1-phosphate by forming α -1,4-glycosidic bonds. It is involved in building glycogen rather than breaking it down and does not affect the branched structure of glycogen.

Solution for Question 2:

Andersen's Disease (GSD Type IV) does not involve hypoglycemia or exercise intolerance . It is caused by a branching enzyme deficiency , leading to abnormal glycogen accumulation in the liver and other organs, eventually causing cirrhosis.

Andersen's Disease (Glycogen Storage Disease Type IV)

Aspect	Details
Enzyme Deficiency	Branching enzyme (glycogen branching enzyme)
Affected Gene	GBE1
Mode of Inheritance	Autosomal recessive
Key Features	Hepatosplenomegaly, cirrhosis, failure to thrive
Hypoglycemia	Absent (Unlike Von Gierke or Cori disease)
Exercise Intolerance	Absent (Unlike McArdle or Tarui disease)
Prognosis	Progressive liver disease, leading to liver failure

Glycogen Storage Diseases (GSD) Classification

GSD T ype	Name	Enzyme Deficiency	Affected Gene	Key Features	Hypoglycemia	Exercise Intolerance
GSD I	Von Gierke Disease (Option A ruled out)	Glucose-6-phosphatase	G6PC	Severe hypoglycemia , hepatomegaly, lactic acidosis, hyperuricemia, hyperlipidemia	Yes	No
GSD II	Pompe Disease	Lysosomal acid maltase (α 1,4-glucosidase)	GAA	Cardiomegaly, muscle hypotonia, early death (infantile form)	No	Yes
GSD III	Cori Disease (Forbes Disease)	Debranching enzyme	AGL	Mild hypoglycemia , hepatomegaly, muscle weakness, limit dextrin accumulation	Yes	Yes
GSD IV	Andersen Disease	Branching enzyme	GBE1	Hepatosplenomegaly, cirrhosis, no hypoglycemia or exercise intolerance	No	No
GSD V	McArdle Disease (Option B ruled out)	Muscle glycogen phosphorylase	PYGM	Exercise intolerance , muscle cramps, myoglobinuria	No	Yes
GSD VI	Hers Disease	Liver glycogen phosphorylase	PYGL	Mild hypoglycemia , hepatomegaly	Yes	No

GSD VI I	Tarui Disease (Option D ruled out)	Muscle phosphofructokinase-1	PFKM	Exercise intolerance , hemolysis	No	Yes
GSD IX	Phosphorylase kinase deficiency	Phosphorylase kinase	PHKA1, PHKB, PHKG2	Hepatomegaly, mild hypoglycemia , muscle weakness (some subtypes)	Yes	Yes
GSD XI	Fanconi-Bickel Syndrome	GLUT2 transporter defect	SLC2A2	Hepatorenal dysfunction, growth retardation, glucose/galactose intolerance	Yes	No
GSD 0	Glycogen Synthase Deficiency	Glycogen synthase	GYS1, GYS2	Ketotic hypoglycemia , postprandial hyperglycemia, no glycogen storage	Yes	No

Classification Based on Hypoglycemia and Exercise Intolerance

- GSD with Hypoglycemia: Von Gierke (I), Cori (III), Hers (VI), GSD IX, Fanconi-Bickel (XI), GSD 0
- GSD with Exercise Intolerance: Pompe (II), McArdle (V), Tarui (VII), Cori (III), GSD IX
- GSD with Both: Cori (III), GSD IX
- GSD with Neither: Andersen (IV)

Solution for Question 3:

Glycogen Storage Disease Type 0 (GSD 0) is caused by a deficiency of glycogen synthase , which leads to low glycogen storage in the liver and muscles . It is an autosomal recessive disorder caused by mutations in the GYS1 (muscle glycogen synthase) and GYS2 (liver glycogen synthase) genes .

- Enzyme Deficiency: Glycogen synthase deficiency leads to impaired glycogen synthesis, causing fasting hypoglycemia and ketosis.
- Affected Genes: GYS2 affects the liver, and GYS1 affects muscle function.
- Inheritance: Autosomal recessive pattern.

Glycogen Storage Disease Type 0 (GSD 0)

Feature	Details
Cause	Glycogen synthase deficiency (caused by GYS1 or GYS2 mutation)- an autosomal recessive disorder. (Options A, B & C)
Pathophysiology	Inability to synthesize glycogen → Reduced glycogen stores → Fasting hypoglycemia and ketosis
Clinical Features	Fasting hypoglycemia, ketosis, postprandial hyperglycemia, muscle weakness (GYS1 mutation)
Diagnosis	Genetic testing (GYS1, GYS2 mutation), blood glucose and ketone levels
Management	Frequent meals, cornstarch supplementation, avoiding fasting

Solution for Question 4:

The child presents with failure to thrive, weak bones (osteopenia), and renal dysfunction , which are characteristic features of Fanconi-Bickel Syndrome (Glycogen Storage Disease Type XI) .

- Gene involved: SLC2A2 mutation
- Defective protein: GLUT2 transporter (responsible for glucose and galactose transport in the liver, kidney, pancreas, and intestines)
- Key features: Glycogen accumulation in the liver and kidney Glucose and galactose intolerance Renal Fanconi syndrome (proximal renal tubular dysfunction leading to loss of phosphate, amino acids, and glucose in urine) Rickets/osteopenia due to phosphate loss.
- Glycogen accumulation in the liver and kidney
- Glucose and galactose intolerance
- Renal Fanconi syndrome (proximal renal tubular dysfunction leading to loss of phosphate, amino acids, and glucose in urine)
- Rickets/osteopenia due to phosphate loss.
- Glycogen accumulation in the liver and kidney
- Glucose and galactose intolerance
- Renal Fanconi syndrome (proximal renal tubular dysfunction leading to loss of phosphate, amino acids, and glucose in urine)

- Rickets/osteopenia due to phosphate loss.

Fanconi-Bickel Syndrome (Glycogen storage disease Type XI)

Feature	Details
Cause	Mutation in SLC2A2 gene leading to defective GLUT2 transporter
Affected Organs	Liver, kidney, intestines, pancreas
Pathophysiology	Impaired glucose transport → Hepatic glycogen accumulation → Renal Fanconi syndrome
Clinical Features	Failure to thrive , hepatomegaly, renal dysfunction , osteopenia , hypophosphatemic rickets, fasting hypoglycemia
Diagnosis	Genetic testing (SLC2A2 mutation), urine analysis (glycosuria, phosphaturia, aminoaciduria), fasting blood glucose
Management	Supportive care, frequent small meals , vitamin D and phosphate supplementation, uncooked cornstarch to prevent fasting hypoglycemia

Liver phosphorylase (Option B): Incorrect because Hers' disease (GSD VI) is caused by liver phosphorylase deficiency, leading to mild hepatomegaly and hypoglycemia, not renal dysfunction.

Muscle and erythrocyte phosphofructokinase 1 (Option C): Incorrect because Tarui disease (GSD VII) is due to PFKM gene mutation , causing exercise intolerance and hemolysis, not liver and renal dysfunction.

Muscle Phosphorylase (Option D): Incorrect because McArdle disease (GSD V) is due to PYGM gene mutation , leading to exercise-induced muscle cramps and myoglobinuria, with no hepatic or renal involvement.

Solution for Question 5:

- Pompe Disease (Glycogen Storage Disease Type II) is caused by a deficiency of lysosomal acid maltase (α -1,4 and α -1-6 glucosidase) , leading to glycogen accumulation in lysosomes.

- Clinical presentation: Infantile-onset form presents with muscle hypotonia (floppy baby syndrome), cardiomegaly, and heart failure . Death occurs early due to cardiorespiratory failure if untreated.
- Infantile-onset form presents with muscle hypotonia (floppy baby syndrome), cardiomegaly, and heart failure .
- Death occurs early due to cardiorespiratory failure if untreated.
- Laboratory findings: Accumulation of glycogen within lysosomes , confirmed on muscle biopsy. Elevated creatine kinase (CK), lactate dehydrogenase (LDH), and transaminases and lactate levels.
- Accumulation of glycogen within lysosomes , confirmed on muscle biopsy.
- Elevated creatine kinase (CK), lactate dehydrogenase (LDH), and transaminases and lactate levels.
- Infantile-onset form presents with muscle hypotonia (floppy baby syndrome), cardiomegaly, and heart failure .
- Death occurs early due to cardiorespiratory failure if untreated.
- Accumulation of glycogen within lysosomes , confirmed on muscle biopsy.
- Elevated creatine kinase (CK), lactate dehydrogenase (LDH), and transaminases and lactate levels.

Pompe's Disease (Glycogen storage disease Type II)

Enzyme Deficiency	Lysosomal α -1-4 and α -1-6 glucosidase (acid maltase)
Affected Gene	GAA
Clinical Features	Muscle hypotonia, death due to heart failure within 2 years of age (seen in juvenile) Muscle dystrophy (seen in adults)
Mode of Inheritance	Autosomal recessive
Management	Supportive care Enzyme replacement therapy Chest x-ray at regular intervals Obtain an initial echocardiogram to evaluate the extent of cardiomyopathy Medical management may be useful based on the stage of cardiomyopathy. Avoid drastic changes in fluid status, either through dehydration or fluid overload.

Glycogen Storage Disorders

Disorder	Enzyme Deficiency	Key Clinical Features
Pompe Disease (GSD II)	Lysosomal acid maltase	Hypotonia, cardiomegaly, glycogen in lysosomes

McArdle Disease (GSD V) (Option A ruled out)	Muscle phosphorylase	Exercise intolerance, muscle cramps, myoglobinuria
Von Gierke Disease (GSD I) (Option C ruled out)	Glucose-6-phosphatase	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis
Cori Disease (GSD III) (Option D ruled out)	Debranching enzyme	Hepatomegaly, muscle weakness, limit dextrin accumulation

Solution for Question 6:

- The child presents with muscle weakness, mild hepatomegaly, mild cardiomegaly, and elevated creatine kinase (CK) , which suggests McArdle's disease (Glycogen Storage Disease Type V) .
- McArdle's disease is caused by a deficiency of muscle phosphorylase (myophosphorylase) , leading to impaired glycogen breakdown in muscles.
- Key clinical features include: Muscle weakness and exercise intolerance Myoglobinuria (red/brown urine after exercise) "Second-wind" phenomenon (improved performance after initial fatigue) Elevated serum creatine kinase (CK) due to muscle damage Mild hepatomegaly and cardiomegaly in some cases
- Muscle weakness and exercise intolerance
- Myoglobinuria (red/brown urine after exercise)
- "Second-wind" phenomenon (improved performance after initial fatigue)
- Elevated serum creatine kinase (CK) due to muscle damage
- Mild hepatomegaly and cardiomegaly in some cases
- Muscle weakness and exercise intolerance
- Myoglobinuria (red/brown urine after exercise)
- "Second-wind" phenomenon (improved performance after initial fatigue)
- Elevated serum creatine kinase (CK) due to muscle damage
- Mild hepatomegaly and cardiomegaly in some cases

Cori's Disease (Glycogen storage disease Type III)

Enzyme Deficiency	Liver and muscle debranching enzyme
Affected Gene	AGL

Mode of Inheritance	Autosomal Recessive Inheritance
Clinical Features	Hypoglycemia Hepatomegaly seen at infancy Limit dextrin accumulation Muscle weakness
Management	Dietary Management: A diet high in protein and low in carbohydrates is often recommended to manage the disease. Enzyme Replacement Therapy: In some cases, enzyme replacement therapy may be used to supplement the deficient enzyme.

Glycogen synthase (Option A): Incorrect because glycogen synthase deficiency (GSD 0) leads to fasting hypoglycemia without hepatomegaly or muscle weakness.

Glucose-6-phosphatase (Option B): Incorrect because Von Gierke's disease (GSD I) presents with severe hypoglycemia, hepatomegaly, and metabolic abnormalities , but not muscle weakness or elevated CK.

Debranching enzyme (Option C): Incorrect because Cori's disease (GSD III) causes mild hypoglycemia and hepatomegaly , but muscle involvement is less prominent , and cardiomegaly is uncommon.

Solution for Question 7:

- Von Gierke's Disease (Glycogen Storage Disease Type I) is caused by a deficiency of glucose-6-phosphatase , impairing glycogen breakdown and gluconeogenesis.
- The key clinical features in this infant, such as severe hypoglycemia, hepatomegaly, growth retardation, irritability, and lethargy between feedings , strongly suggest impaired glucose release from the liver.
- Laboratory findings of elevated lactate, triglycerides, and uric acid further support Von Gierke's disease due to defective glucose metabolism.
- Frequent feedings are characteristic, as affected infants need continuous glucose intake to prevent hypoglycemia.

Von Gierke's Disease (Glycogen Storage Disease Type I)

Cause	Deficiency of glucose-6-phosphatase enzyme
Pathophysiology	Inability to convert glucose-6-phosphate to glucose; leads to accumulation of glycogen, lactate, and triglycerides

Symptoms	Severe hypoglycemia Hepatomegaly and Renomegaly Growth retardation Lactic acidosis Hyperuricemia and Hyperlipidemia
Diagnosis	Biochemical tests (hypoglycemia, lactic acidosis, hyperuricemia, hyperlipidemia) Genetic testing
Management	Frequent feedings with slow-release carbohydrates Blood glucose monitoring Medications for associated conditions Liver transplantation (in severe cases)

Glycogen Storage Disorders (GSDs)

GSD Type	Name	Enzyme Deficiency	Affected Organs	Key Features
Type I	Von Gierke's Disease	Glucose-6-phosphatase	Liver, Kidney	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis, hyperuricemia, hyperlipidemia
Type V	McArdle's Disease (Option B ruled out)	Muscle phosphorylase	Skeletal Muscle	Exercise intolerance, muscle cramps, myoglobinuria after exertion
Type VII	Tarui's Disease (Option C ruled out)	Muscle phosphofructokinase	Muscle, RBCs	Exercise intolerance, muscle cramps, hemolytic anemia
Type VI	Hers' Disease (Option D ruled out)	Liver phosphorylase	Liver	Hepatomegaly, mild hypoglycemia, milder symptoms than Von Gierke's

McArdle's Disease (Option B): Incorrect because it primarily affects muscle , leading to exercise intolerance and myoglobinuria rather than hypoglycemia.

Tarui's Disease (Option C): Incorrect because it affects muscle and red blood cells , leading to exercise intolerance and hemolytic anemia , not hypoglycemia.

Hers' Disease (Option D): Incorrect because it causes mild hypoglycemia without lactic acidosis , whereas this case presents with severe hypoglycemia and metabolic abnormalities .

Solution for Question 8:

Glycogen Storage Disorder	Enzyme Defect	Pathophysiology
Pompe Disease (Type II)	Acid maltase (Lysosomal α -1,4-glucosidase)	Accumulation of glycogen in lysosomes due to defective degradation
McArdle Disease (Type V)	Myophosphorylase (Muscle glycogen phosphorylase)	Impaired glycogen breakdown in muscle, causing exercise intolerance
Von Gierke Disease (Type I)	Glucose-6-phosphatase	Defective glycogenolysis and gluconeogenesis, leading to severe hypoglycemia
Cori Disease (Type III)	Debranching enzyme (α -1,6-glucosidase)	Abnormal glycogen with short outer chains, leading to mild hypoglycemia and hepatomegaly

Glycogen Storage Disorders (GSDs)

Glycogen storage disorders are a group of inherited metabolic diseases characterized by enzyme deficiencies affecting glycogen metabolism. These disorders cause abnormal glycogen accumulation or impaired glycogen utilization in various tissues.

GSD Type	Name	Enzyme Defect	Affected Organs	Key Features
Type I	Von Gierke Disease	Glucose-6-phosphatase	Liver, Kidney	Severe fasting hypoglycemia, hepatomegaly, lactic acidosis
Type II	Pompe Disease	Acid maltase (Lysosomal α -1,4-glucosidase)	Muscle, Heart	Cardiomegaly, hypotonia, early death in infantile form
Type III	Cori Disease	Debranching enzyme (α -1,6-glucosidase)	Liver, Muscle	Mild hypoglycemia, hepatomegaly, muscle weakness
Type V	McArdle Disease	Myophosphorylase (Muscle glycogen phosphorylase)	Skeletal Muscle	Exercise intolerance, muscle cramps, myoglobinuria

Solution for Question 9:

A high-carbohydrate diet increases blood glucose levels, leading to insulin secretion , which promotes glycogen synthesis and suppresses glycogen breakdown. The body enters the fed state , characterized by active glycogenesis and glucose utilization.

- High glucose levels → Insulin release
- Insulin stimulates glycogen synthase → Increased glycogen synthesis
- Inhibits glycogen phosphorylase → Decreased glycogen breakdown.

Glycogen Metabolism in Different Metabolic States

Glycogen metabolism is regulated by the body's metabolic state, primarily influenced by insulin, glucagon, and epinephrine . The balance between glycogenesis (glycogen synthesis) and glycogenolysis (glycogen breakdown) ensures glucose homeostasis.

Metabolic State	Primary Hormone	Glycogen Metabolism	Key Enzymes Involved
Fed State (Option B ruled out)	Insulin	↑ Glycogenesis , ↓ Glycogenolysis	Glycogen synthase (activated) , Glycogen phosphorylase (inhibited)
Fasted State (Option A ruled out)	Glucagon	↑ Glycogenolysis , ↓ Glycogenesis	Glycogen phosphorylase (activated) , Glycogen synthase (inhibited)
Starvation State (Option C ruled out)	Glucagon, Cortisol	Glycogen nearly depleted , ↑ Gluconeogenesis	Gluconeogenic enzymes (e.g., PEPCK, F-1,6-BPase)
Post-Exercise State (Option D ruled out)	Epinephrine, Insulin (during recovery)	↑ Glycogenolysis during exercise , ↑ Glycogenesis during recovery	Glycogen phosphorylase (exercise), Glycogen synthase (recovery)

Fed State and Glycogenesis (Option B ruled out)

- After a high-carbohydrate meal , blood glucose levels increase , stimulating insulin secretion from the pancreas.
- Insulin activates glycogen synthase , promoting the conversion of glucose into glycogen in the liver and muscle .
- Glycogenolysis is inhibited to store excess glucose.

Fasted State and Glycogenolysis (Option A ruled out)

- Low blood glucose stimulates glucagon release , which activates glycogen phosphorylase , breaking down glycogen into glucose-1-phosphate.

- This provides energy for vital functions, primarily in the liver to maintain blood glucose levels.

Starvation State and Gluconeogenesis (Option C ruled out)

- Glycogen stores are depleted after prolonged fasting.
- The body shifts to gluconeogenesis , converting amino acids and glycerol into glucose.
- Cortisol and glucagon drive this process.

Post-Exercise State and Glycogen Replenishment (Option D ruled out)

- After intense exercise, muscle glycogen is depleted , and the body prioritizes glycogen resynthesis .
- Insulin sensitivity is high , allowing rapid glucose uptake for glycogenesis in muscle cells.

Solution for Question 10:

- A spherical molecule, approximately 21 nm in diameter, observable under electron micrographs
- Arranged in 12 concentric layers (Statement 1)
- The glycogenin protein is present at the center
- Composed of polysaccharide chains, each containing approximately 13 glucose residues
- Branched chains in the inner layers and unbranched chains in the outermost layer
- Short chains of glucose residues are linked by α -1,4 glycosidic bonds (Statement 2)
- Branch points are formed by α -1,6 glycosidic bonds (Statement 3)
- Heavily branched and compact, it can store a large amount of energy in a limited space due to its compact structure

Solution for Question 11:

When one molecule of muscle glycogen undergoes anaerobic glycolysis (glycogen → lactate), the net ATP yield differs from glucose metabolism because glycogen breakdown bypasses the hexokinase step, which normally consumes ATP. Here's a stepwise explanation:

Glycogenolysis (Breakdown of Glycogen)

- Glycogen phosphorylase removes glucose units from glycogen as glucose-1-phosphate (G1P).
- Phosphoglucomutase converts G1P to glucose-6-phosphate (G6P).
- Since G6P is directly obtained, it skips the hexokinase step (which normally consumes 1 ATP in glucose metabolism).

Glycolysis (Conversion of Glucose-6-Phosphate to Pyruvate)

- G6P enters glycolysis and follows the usual pathway.
- Key ATP-related steps: Energy investment: 1 ATP is used at the phosphofructokinase-1 (PFK-1) step. Energy generation: 4 ATP are produced from substrate-level phosphorylation. Net ATP gain from glycolysis: 4 ATP (produced) - 1 ATP (used) = 3 ATP.
- Energy investment: 1 ATP is used at the phosphofructokinase-1 (PFK-1) step.
- Energy generation: 4 ATP are produced from substrate-level phosphorylation.
- Net ATP gain from glycolysis: 4 ATP (produced) - 1 ATP (used) = 3 ATP.
- Energy investment: 1 ATP is used at the phosphofructokinase-1 (PFK-1) step.
- Energy generation: 4 ATP are produced from substrate-level phosphorylation.
- Net ATP gain from glycolysis: 4 ATP (produced) - 1 ATP (used) = 3 ATP.

Anaerobic Conditions (Conversion of Pyruvate to Lactate)

- In anaerobic exercise (e.g., intense training), pyruvate is converted to lactate by lactate dehydrogenase (LDH).
- This step does not produce or consume ATP; its main role is regenerating NAD⁺ for continued glycolysis.

ATP Yield Calculation from Muscle Glycogen Breakdown

Step	Process	ATP Change
Glycogenolysis	Glycogen → Glucose-1-phosphate → Glucose-6-phosphate (no ATP used)	+0 ATP
Glycolysis	Glucose-6-phosphate → 2 Pyruvate	+4 ATP
Energy Consumption	ATP used in glycolysis (Phosphofructokinase-1 step)	-1 ATP

Anaerobic Conversion	2 Pyruvate → 2 Lactate (No ATP change)	0 ATP
Net ATP Yield		3 ATP

Final Net ATP Yield:

- Glycolysis from muscle glycogen produces 3 ATP (instead of 2 ATP when starting from free glucose) because the hexokinase step is bypassed.
- No additional ATP is lost or gained during lactate formation.
- Total ATP yield per glycogen molecule → 3 ATP.

1 ATP (Option A):

- Incorrect because the actual ATP yield is higher . Even free glucose metabolism yields 2 ATP , and muscle glycogen yields 3 ATP due to bypassing hexokinase.

2 ATP (Option B):

- Incorrect because this is the ATP yield when glucose (not glycogen) is used . Muscle glycogen metabolism produces one extra ATP by skipping the ATP-consuming hexokinase step.

4 ATP (Option D):

- Incorrect because glycolysis does produce 4 ATP, but 1 ATP is consumed at the PFK-1 step , leading to a net gain of 3 ATP, not 4 .

Solution for Question 12:

Glycogen phosphorylase is an enzyme crucial for glycogen metabolism. It catalyzes the breakdown of glycogen into glucose-1-phosphate. There are two main isoforms of glycogen phosphorylase:

- Liver Glycogen Phosphorylase (LGP) is primarily regulated by glucagon and epinephrine , which promote glycogen breakdown to maintain blood glucose levels during fasting.
- Muscle Glycogen Phosphorylase (MGP) is activated by epinephrine and allosterically regulated by AMP , which signals low energy and promotes glycogenolysis for ATP production in muscle cells.

Aspect	Liver Glycogen Phosphorylase (Liver-specific isoform)	Muscle Glycogen Phosphorylase (Muscle-specific isoform)
Primary Role	Maintains blood glucose levels by breaking down glycogen during fasting	Provides glucose for ATP production during muscle contraction.
Function	Converts glycogen to glucose-1-phosphate, then to glucose-6-phosphate, which is converted to free glucose by glucose-6-phosphatase and released into the bloodstream	Converts glycogen to glucose-1-phosphate, then to glucose-6-phosphate for energy production within muscle cells (muscle lacks glucose-6-phosphatase)
Hormonal Activation	Activated by glucagon (during fasting) and epinephrine (during stress)	Activated by epinephrine and AMP
Allosteric Regulation	Inhibited by glucose	Activated by AMP, inhibited by ATP and glucose-6-phosphate
Response to Fasting	Increased activity	Not significantly affected
Response to Feeding	Decreased activity	Not significantly affected
Response to Exercise	No significant change	Increased activity

LGP releases glucose-6-phosphate into the bloodstream, while MGP releases free glucose into the muscle (Option B):

- Incorrect because LGP converts glucose-1-phosphate to glucose-6-phosphate, which is then converted to free glucose by glucose-6-phosphatase and released into the bloodstream .
- MGP does not release free glucose since muscle lacks glucose-6-phosphatase . Instead, glucose-6-phosphate enters glycolysis for ATP production within muscle cells.

LGP activity is regulated by AMP, while MGP is regulated by glucagon and epinephrine (Option C):

- Incorrect because AMP primarily regulates MGP, not LGP . AMP increases MGP activity when energy levels are low.
- LGP is regulated by glucagon and epinephrine, not AMP .

MGP is highly active during fasting compared to LGP (Option D):

- Incorrect because LGP is highly active during fasting to maintain blood glucose levels .
- MGP is more active during exercise , when ATP demand is high, not necessarily during fasting.

Solution for Question 13:

- Glycogen phosphorylase is the rate-limiting enzyme of glycogenolysis. It catalyzes the phosphorolytic cleavage of α -1,4 glycosidic bonds in glycogen, releasing glucose-1-phosphate.
- It is regulated by phosphorylation: Activated by phosphorylase kinase and inhibited by phosphoprotein phosphatase.
- Different isoforms exist in the liver, muscle, and brain.
- Requires pyridoxal phosphate (PLP) as a coenzyme.

Glycogenolysis

Feature	Details
Definition	Breakdown of glycogen to release glucose
Location	Primarily in liver and muscle
Purpose	Provides glucose for energy during fasting (liver) and muscle contraction (muscle)
Rate-Limiting Enzyme	Glycogen phosphorylase
End Products	Glucose-1-phosphate (converted to glucose-6-phosphate, then to glucose in liver)

Steps of Glycogenolysis

Glycogen phosphorylase:

- Cleaves α -1,4 glycosidic bonds and releases glucose-1-phosphate .
- Rate-limiting enzyme , activated by phosphorylation.

Debranching enzyme (Glucan transferase & α -1,6-glucosidase):

- Transfers trisaccharide units to the main chain (glucan transferase).
- Hydrolyzes α -1,6 glycosidic bonds to release free glucose.

Phosphoglucomutase:

- Converts glucose-1-phosphate to glucose-6-phosphate .

Glucose-6-phosphatase (Liver only):

- Converts glucose-6-phosphate to free glucose , which is released into the blood.

Regulation of Glycogenolysis

Factor	Effect on Glycogenolysis
Glucagon (Liver)	Stimulates
Epinephrine (Muscle & Liver)	Stimulates
Insulin	Inhibits
AMP (Muscle)	Stimulates
ATP, Glucose-6-P	Inhibits

- Glycogenolysis in muscle provides glucose-6-phosphate for glycolysis , not free glucose.
- Glycogenolysis in liver maintains blood glucose levels during fasting.

Rate-Limiting Enzymes in Carbohydrate Metabolism

Pathway	Rate-Limiting Enzyme	Key Regulation
Glycolysis	Phosphofructokinase-1 (PFK-1)	Activated by AMP, fructose-2,6-bisphosphate; Inhibited by ATP, citrate
Gluconeogenesis	Fructose-1,6-bisphosphatase	Activated by ATP, inhibited by AMP, fructose-2,6-bisphosphate
Glycogenesis	Glycogen synthase	Activated by insulin, glucose-6-phosphate; Inhibited by glucagon, epinephrine
Glycogenolysis	Glycogen phosphorylase	Activated by glucagon, epinephrine, AMP; Inhibited by insulin, ATP, glucose-6-phosphate
Pentose Phosphate Pathway (HMP Shunt)	Glucose-6-phosphate dehydrogenase (G6PD)	Activated by NADP ⁺ ; Inhibited by NADPH
TCA Cycle	Isocitrate dehydrogenase	Activated by ADP, Ca ²⁺ ; Inhibited by ATP, NADH

These rate-limiting enzymes regulate carbohydrate metabolism by controlling the flux of substrates through their respective pathways based on energy demands and hormonal signals.

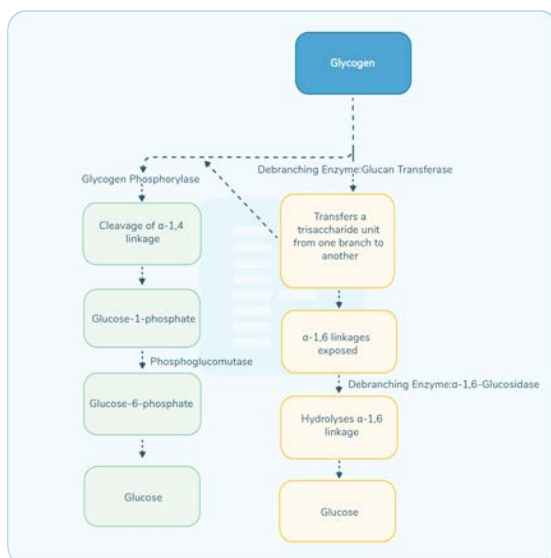
Phosphoglucomutase (Option A): Incorrect because it converts glucose-1-phosphate to glucose-6-phosphate, which is not the rate-limiting step.

Glucose-6-phosphatase (Option B): Incorrect because it converts glucose-6-phosphate to free glucose in the liver but is not involved in the rate-limiting step of glycogenolysis.

Glucan transferase (Option C): Incorrect because it helps in debranching glycogen by transferring trisaccharide units but does not control the rate of glycogen breakdown.

Solution for Question 14:

- Glycogenolysis and glycogenesis are distinct pathways with different enzymes and mechanisms.
- Glycogenolysis involves breaking down glycogen into glucose-1-phosphate, whereas glycogenesis involves synthesizing glycogen from glucose-1-phosphate.



Glucose-6-phosphatase acts in the liver (Option A). This enzyme functions in glycogenolysis by converting glucose-6-phosphate into free glucose, which is then released into the bloodstream.

Glycogen phosphorylase cleaves α -1,4 linkages (Option B):

Glycogen phosphorylase cleaves α -1,4 glycosidic bonds in glycogen during glycogenolysis, releasing glucose-1-phosphate from glycogen.

Phosphoglucomutase converts glucose-1-phosphate into glucose-6-phosphate (Option C) :

Phosphoglucomutase catalyzes the reversible conversion of glucose-1-phosphate into glucose-6-phosphate. Glucose-6-phosphate can enter glycolysis for energy production or be converted into free glucose in the liver by glucose-6-phosphatase.

Solution for Question 15:

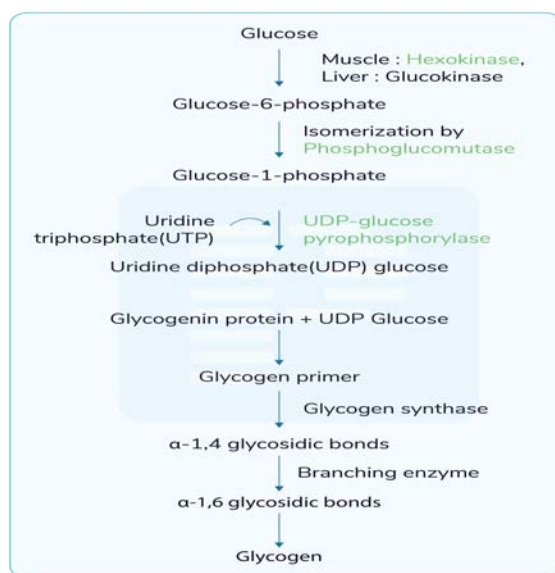
Enzymes that increase blood glucose are stimulated by phosphorylation, whereas enzymes that decrease blood glucose are stimulated by dephosphorylation.

Activated by phosphorylation	Deactivated by phosphorylation
Glycogen phosphorylase Debranching enzyme Pyruvate carboxylase PEPCK (Phosphoenolpyruvate carboxykinase).	Acetyl-CoA carboxylase (Option A ruled out) Glycogen synthase (Option B ruled out) Pyruvate dehydrogenase (Option C ruled out) HMG-CoA reductase

Solution for Question 16:

Glycogenesis

- Process of glycogen synthesis.
- Occurs in liver and muscle, with liver playing a major role in maintaining blood glucose levels



Solution for Question 17:

Glycogen and amylopectin are indeed both branched polysaccharides made up of glucose units.

Branch points occur every 8-12 glucose residues in glycogen, whereas in amylopectin, they occur every 20-30 glucose residues.

This difference in branching frequency affects to the structural and functional properties of the two polysaccharides.

Feature	Amylopectin	Glycogen
Source	Plants	Animals
Structure	Moderately branched, larger molecules	Highly branched, compact molecules.
Branching	Less frequent (every 20-30 glucose units)	More frequent (every 8-12 glucose units)
Glucose Linkages	α-1,4 and α-1,6 glycosidic bonds	α-1,4 and α-1,6 glycosidic bonds
Solubility	Less soluble	More soluble
Granule Shape	Larger, spherical	Smaller, irregular

Hydrolysis Enzyme	Amylase and limit dextrinase	Glycogen phosphorylase and debranching enzyme
Function	Energy storage in plants	Rapid energy mobilization in animals

Citric Acid cycle and Gluconeogenesis

1. Which enzyme is the rate-limiting enzyme of gluconeogenesis?

- A. PEP Carboxykinase
 - B. Pyruvate Carboxylase
 - C. Glucose-6-phosphatase
 - D. Fructose 1,6-bisphosphatase
-

2. In patients with uncontrolled diabetes, which of the following intermediates is depleted, suppressing the TCA cycle?

- A. Acetyl CoA
 - B. Oxaloacetate
 - C. Succinyl-CoA
 - D. Malate
-

3. Which of the following factors regulates the expression of PEPCK in liver cells?

- A. Glucagon
 - B. Insulin
 - C. Citrate
 - D. Acetyl-CoA
-

4. A 45-year-old woman presents with fatigue and muscle weakness. Laboratory tests reveal hypoglycemia and elevated lactate levels. The physician suspects a deficiency in a coenzyme essential for gluconeogenesis. Which coenzyme is most likely deficient in this patient?

- A. Magnesium
 - B. Thiamine Pyrophosphate
 - C. Biotin
 - D. Fluoride
-

5. Which of the following enzymes in gluconeogenesis is considered a "tandem enzyme" with fructose-2,6-bisphosphatase, functioning in the opposite direction?

- A. Pyruvate carboxylase
 - B. Phosphoenolpyruvate carboxykinase
 - C. Phosphofructokinase I
 - D. Phosphofructokinase-2
-

6. A patient presents with symptoms of hypoglycemia shortly after consuming a high-carbohydrate meal. Laboratory tests reveal elevated insulin levels. Which of the following enzymes in the gluconeogenesis pathway is most likely to be strongly inhibited in this scenario?

- A. Pyruvate carboxylase
 - B. Phosphoenolpyruvate carboxykinase (PEPCK)
 - C. Fructose-1,6-bisphosphatase
 - D. Glucose-6-phosphatase
-

7. Which of the following statements about the malate shuttle in gluconeogenesis is not true?

- A. Helps transfer reducing equivalents from the cytoplasm to mitochondria
 - B. Involves the interconversion of oxaloacetate and malate
 - C. Requires the enzyme malate dehydrogenase in both mitochondria and cytosol
 - D. Directly transports NADH across the mitochondrial membrane
-

8. Which of the following enzymes involved in gluconeogenesis is located in the endoplasmic reticulum?

- A. Pyruvate carboxylase
 - B. Phosphoenolpyruvate carboxykinase (PEPCK)
 - C. Fructose-1,6-bisphosphatase
 - D. Glucose-6-phosphatase
-

9. Which of the following is NOT a substrate for gluconeogenesis?

- A. Lactate
 - B. Glycerol
 - C. Alanine
 - D. Acetyl-CoA
-

10. All of the following metabolic pathways occur in the cytoplasm except?

- A. Glycolysis
- B. Pentose Phosphate Pathway

C. Fatty Acid Synthesis

D. Citric Acid Cycle

11. Which of the following best describes the amphibolic nature of the citric acid cycle?

A. It generates ATP and produces citrate for fatty acid synthesis.

B. It exclusively produces NADH for oxidative phosphorylation.

C. It breaks down fatty acids and synthesises only cholesterol.

D. It provides intermediates solely for glucose synthesis.

12. Which of the following is correct regarding anaplerotic reactions?

A. Produce ATP

B. Replenish TCA intermediates

C. Reduction of NAD⁺ to NADH

D. Breakdown of fatty acids

13. ATP inhibits all of the following enzymes in the citric acid cycle except:

A. Citrate synthase

B. Isocitrate dehydrogenase

C. α -Ketoglutarate dehydrogenase

D. Succinate dehydrogenase

14. Which of the following toxins directly inhibits the activity of the enzyme aconitase in the Citric Acid Cycle?

- A. Cyanide
 - B. Fluoroacetate
 - C. Rotenone
 - D. Antimycin A
-

15. Calculate the total amount of ATP produced from the Citric Acid Cycle alone for one glucose molecule.

- A. 30 ATP
 - B. 9 ATP
 - C. 18 ATP
 - D. 20 ATP
-

16. Which of the following by-products from the Krebs Cycle is involved in heme synthesis?

- A. Citrate
 - B. α -Ketoglutarate
 - C. Succinyl-CoA
 - D. Malate
-

17. Which of the following steps in the Citric acid cycle involves oxidative phosphorylation and decarboxylation? Isocitrate $\rightarrow$ α -Ketoglutarate Succinate $\rightarrow$ Fumarate Malate $\rightarrow$ Oxaloacetate α -Ketoglutarate $\rightarrow$ Succinyl-CoA

- A. 1 and 4 are correct
 - B. 2 and 3 are correct
 - C. 1, 3, and 4 are correct
 - D. All are correct
-

18. Which of the following is not a cofactor for the enzyme catalysing α -Ketoglutarate $\rightarrow$ Succinyl-CoA?

- A. Fe^{2+}
 - B. Thiamine
 - C. Lipoic Acid
 - D. Pantothenic acid
-

Correct Answers

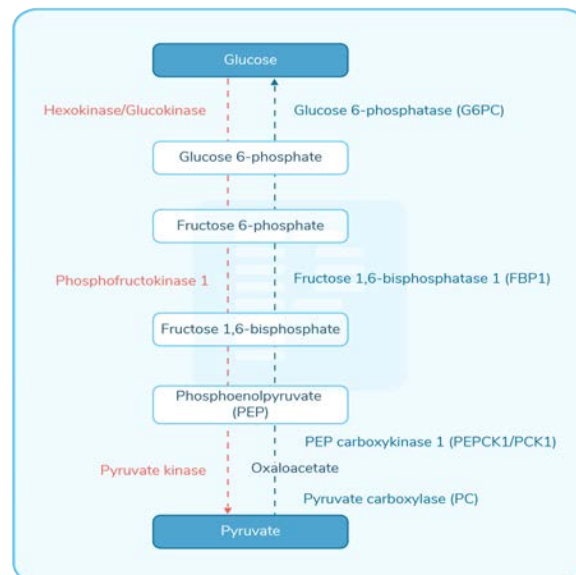
Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	3
Question 5	4
Question 6	3
Question 7	4
Question 8	4
Question 9	4
Question 10	4
Question 11	1
Question 12	2
Question 13	4
Question 14	2
Question 15	4
Question 16	3
Question 17	1
Question 18	1

Solution for Question 1:

Correct Answer: 4) Fructose 1,6-bisphosphatase

Explanation:

The rate-limiting enzyme of gluconeogenesis is Fructose 1,6-bisphosphatase, which plays a crucial role in the regulation of gluconeogenesis, catalysing the conversion of fructose 1,6-bisphosphate to fructose 6-phosphate.



Solution for Question 2:

- In uncontrolled diabetes, gluconeogenesis is upregulated due to low insulin and high glucagon levels, leading to excessive oxaloacetate (OAA) utilization for glucose production.
- This depletes oxaloacetate, a key intermediate in the TCA cycle, causing suppression of the cycle despite an abundance of acetyl-CoA.
- Excess acetyl-CoA is then diverted to ketogenesis, leading to ketoacidosis.

Role of Oxaloacetate in the TCA Cycle

Function	Effect in Uncontrolled Diabetes
Combines with acetyl-CoA to form citrate	Depleted due to increased gluconeogenesis
Essential for TCA cycle continuation	Suppression of the TCA cycle
Deficiency leads to ketone body formation	Ketogenesis → Ketoacidosis

Acetyl-CoA (Option A): Incorrect because acetyl-CoA accumulates due to β -oxidation but cannot enter the TCA cycle without oxaloacetate .

Succinyl-CoA (Option C): Incorrect because its levels are not significantly affected in diabetes; it plays a role in the TCA cycle but is not a primary regulator.

Malate (Option D): Incorrect because malate can be converted to oxaloacetate , but oxaloacetate is preferentially diverted to gluconeogenesis , leading to depletion.

Solution for Question 3:

- Phosphoenolpyruvate carboxykinase (PEPCK) is a key enzyme in gluconeogenesis that converts oxaloacetate to phosphoenolpyruvate (PEP) .
- Glucagon stimulates gluconeogenesis by increasing cAMP levels , activating protein kinase A (PKA) , and enhancing PEPCK gene transcription .
- This ensures glucose production during fasting states to maintain blood glucose levels.

Regulation of PEPCK Expression

Regulator	Effect on PEPCK	Mechanism
Glucagon	Increases	$\uparrow$ cAMP $\rightarrow$ $\uparrow$ PKA $\rightarrow$ $\uparrow$ PEPCK transcription
Insulin	Decreases	$\downarrow$ cAMP $\rightarrow$ $\downarrow$ PKA $\rightarrow$ $\downarrow$ PEPCK transcription
Cortisol	Increases	Activates PEPCK gene transcription
Glucose	Decreases	Inhibits gluconeogenic enzymes
Acetyl-CoA	Activates pyruvate carboxylase	Does not directly regulate PEPCK

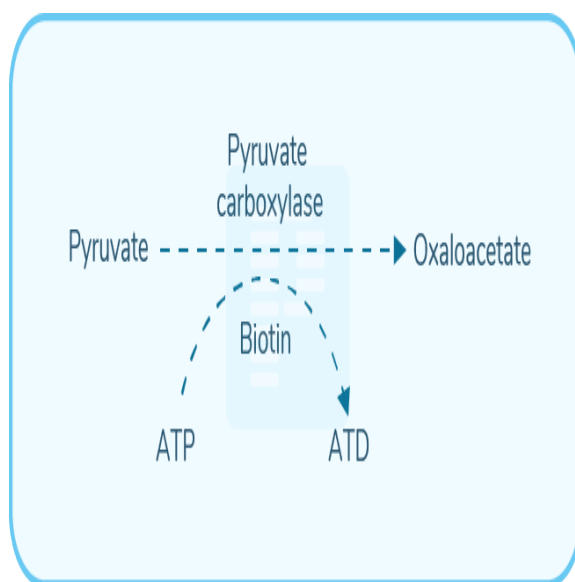
Insulin (Option B): Incorrect because insulin inhibits PEPCK expression by reducing cAMP and promoting glycolysis .

Citrate (Option C): Incorrect because citrate mainly activates gluconeogenesis by inhibiting PFK-1 but does not regulate PEPCK expression.

Acetyl-CoA (Option D): Incorrect because it activates pyruvate carboxylase , not PEPCK.

Solution for Question 4:

- Biotin is a crucial coenzyme for pyruvate carboxylase , which catalyzes the conversion of pyruvate to oxaloacetate, the first regulatory step of gluconeogenesis.
- This reaction occurs in mitochondria and is essential for maintaining blood glucose levels during fasting.



- Biotin also functions as a coenzyme for propionyl-CoA carboxylase , which is involved in odd-chain fatty acid metabolism.
- Biotin deficiency leads to decreased gluconeogenesis , causing hypoglycemia and elevated lactate levels due to impaired conversion of pyruvate to glucose.
- Causes of biotin deficiency include: Severe malnutrition Consumption of raw eggs (contain avidin , which binds biotin and prevents its absorption) Inherited metabolic disorders (e.g., autosomal recessive biotinidase deficiency or holo-carboxylase synthetase deficiency)

- Severe malnutrition
- Consumption of raw eggs (contain avidin , which binds biotin and prevents its absorption)
- Inherited metabolic disorders (e.g., autosomal recessive biotinidase deficiency or holo-carboxylase synthetase deficiency)
- Severe malnutrition
- Consumption of raw eggs (contain avidin , which binds biotin and prevents its absorption)
- Inherited metabolic disorders (e.g., autosomal recessive biotinidase deficiency or holo-carboxylase synthetase deficiency)

Gluconeogenesis & Coenzymes

Step	Enzyme	Required Coenzyme	Location
Pyruvate → Oxaloacetate	Pyruvate carboxylase	Biotin	Mitochondria
Oxaloacetate → PEP	PEPCK	GTP	Cytoplasm/Mitochondria
F-1,6-BP → F-6-P	Fructose-1,6-bisphosphatase	Mg ²⁺ ■	Cytoplasm
G-6-P → Glucose	Glucose-6-phosphatase	Mg ²⁺ ■	Endoplasmic Reticulum

Magnesium (Option A): Required for ATP-dependent reactions but does not function as a coenzyme for pyruvate carboxylase.

Thiamine Pyrophosphate (Option B): Cofactor for pyruvate dehydrogenase , not for gluconeogenesis.

Fluoride (Option D): Inhibits enolase in glycolysis but has no role in gluconeogenesis.

Solution for Question 5:

Tandem enzymes refer specifically to a single bifunctional enzyme with both kinase and phosphatase activity, such as PFK-2/Fructose-2,6-bisphosphatase , rather than two separate enzymes working in opposing pathways.

- Phosphofructokinase-2 (PFK-2) and Fructose-2,6-bisphosphatase (FBPase-2) are two opposing activities within a single bifunctional enzyme , making it a tandem enzyme .
- PFK-2 activity: Converts fructose-6-phosphate to fructose-2,6-bisphosphate , which stimulates glycolysis by activating PFK-1 .

- FBPase-2 activity: Converts fructose-2,6-bisphosphate back to fructose-6-phosphate , reducing glycolysis and promoting gluconeogenesis.
- This regulation allows for reciprocal control of glycolysis and gluconeogenesis in response to insulin and glucagon.

Tandem Enzymes in Glucose Metabolism

Tandem enzymes are pairs of enzymes that catalyze opposite reactions in glycolysis and gluconeogenesis, ensuring precise regulation of glucose metabolism. These enzyme pairs prevent futile cycling and allow metabolic control in response to hormonal signals.

Tandem Enzyme Pairs

Tandem Enzyme Pair	Glycolysis Enzyme	Gluconeogenesis Enzyme	Function & Regulation
PFK-2 / FBPase-2	Phosphofructokinase-2 (PFK-2)	Fructose-2,6-bisphosphatase (FBPase-2)	Regulates fructose-2,6-bisphosphate levels , controlling glycolysis & gluconeogenesis.
Hexokinase / Glucose-6-phosphatase	Hexokinase/Glucokinase	Glucose-6-phosphatase	Controls phosphorylation and dephosphorylation of glucose.
Pyruvate kinase / PEPCK	Pyruvate kinase	Phosphoenolpyruvate carboxykinase (PEPCK)	Controls conversion between PEP and pyruvate, linking glycolysis and gluconeogenesis.

PFK-2 / FBPase-2: The Classic Tandem Enzyme

PFK-2 and FBPase-2 are two opposing activities within a single bifunctional enzyme , which regulates glycolysis and gluconeogenesis.

Condition	PFK-2 Activity	FBPase-2 Activity	Metabolic Effect
Fed state (Insulin ↑)	Active	Inhibited	↑ Fructose-2,6-bisphosphate → ↑ Glycolysis
Fasting state (Glucagon ↑, cAMP ↑, PKA ↑)	Inhibited	Active	↓ Fructose-2,6-bisphosphate → ↑ Gluconeogenesis

- PFK-2 Activity: Converts fructose-6-phosphate into fructose-2,6-bisphosphate, which activates PFK-1 , promoting glycolysis.
- FBPase-2 Activity: Converts fructose-2,6-bisphosphate back to fructose-6-phosphate, inhibiting glycolysis and promoting gluconeogenesis.

Regulation of Tandem Enzymes

Regulatory Factor	Effect on PFK-2/FBPase-2	Metabolic Outcome
Insulin (Fed State)	Activates PFK-2	↑ Glycolysis, ↓ Gluconeogenesis
Glucagon (Fasting State)	Activates FBPase-2	↓ Glycolysis, ↑ Gluconeogenesis
Fructose-2,6-bisphosphate	Stimulates PFK-1 , inhibits FBPase-1	↑ Glycolysis, ↓ Gluconeogenesis
cAMP / PKA Activation	Phosphorylates enzyme → Activates FBPase-2	Promotes gluconeogenesis

Solution for Question 6:

- Fructose-1,6-bisphosphatase is a key enzyme in gluconeogenesis that converts fructose-1,6-bisphosphate to fructose-6-phosphate .
- Elevated insulin levels after a high-carbohydrate meal lead to increased fructose-2,6-bisphosphate , which strongly inhibits fructose-1,6-bisphosphatase , suppressing gluconeogenesis and contributing to hypoglycemia .

Regulation of Gluconeogenesis

Regulatory Mechanism	Effect on Gluconeogenesis	Key Enzymes Affected
Hormonal Regulation		
Insulin (Fed State)	Inhibits gluconeogenesis by increasing fructose-2,6-bisphosphate , which inhibits fructose-1,6-bisphosphatase.	Fructose-1,6-bisphosphatase , PEPCK , Glucose-6-phosphatase
Glucagon (Fasting State)	Stimulates gluconeogenesis by decreasing fructose-2,6-bisphosphate .	Fructose-1,6-bisphosphatase , PEPCK , Glucose-6-phosphatase
Allosteric Regulation		
Fructose-2,6-bisphosphate	Strongly inhibits fructose-1,6-bisphosphatase , suppressing gluconeogenesis.	Fructose-1,6-bisphosphatase
AMP (Low Energy State)	Inhibits gluconeogenesis, signaling low ATP levels.	Fructose-1,6-bisphosphatase

Acetyl-CoA (High Energy State)	Activates pyruvate carboxylase , promoting gluconeogenesis.	Pyruvate carboxylase
Substrate Availability		
Lactate	Enters gluconeogenesis via pyruvate.	Pyruvate carboxylase , PEPCK
Glycerol	Converted to DHAP , feeding into gluconeogenesis.	Glycerol kinase , G3P dehydrogenase
Alanine	Converted to pyruvate for gluconeogenesis.	Alanine aminotransferase , PEPCK
Energy Status		
High ATP/Low AMP	Promotes gluconeogenesis.	Fructose-1,6-bisphosphatase , PEPCK
Low ATP/High AMP	Inhibits gluconeogenesis.	Fructose-1,6-bisphosphatase

Pyruvate carboxylase (Option A): Incorrect because insulin does not directly inhibit this enzyme; it is activated by acetyl-CoA during fasting.

Phosphoenolpyruvate carboxykinase (PEPCK) (Option B): Incorrect because insulin suppresses its gene transcription, but it is not the most strongly inhibited enzyme in this acute setting.

Glucose-6-phosphatase (Option D): Incorrect because it is regulated at the transcriptional level by insulin but is not directly inhibited by fructose-2,6-bisphosphate.

Solution for Question 7:

- The malate shuttle does not directly transport NADH across the mitochondrial membrane.
- Instead, it indirectly transfers reducing equivalents by converting oxaloacetate to malate in the cytoplasm, allowing malate to cross the membrane, and then converting malate back to oxaloacetate in the mitochondria.
- This is necessary because the inner mitochondrial membrane is impermeable to NADH , preventing its direct transport.

Malate Shuttle:

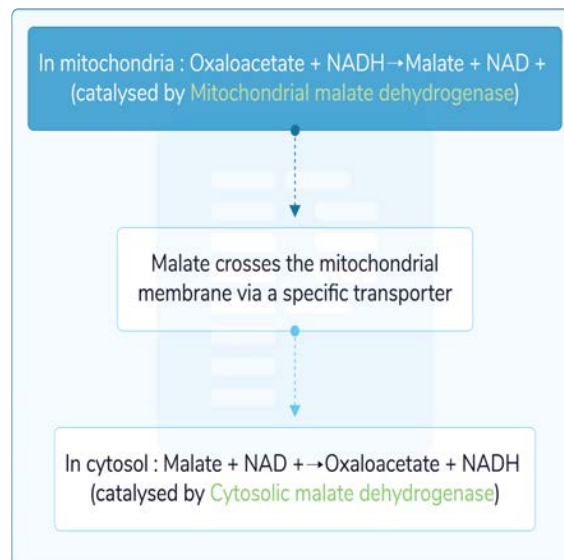
- The malate shuttle, or malate-aspartate shuttle , is a biochemical system that facilitates the transport of electrons from the cytosol to the mitochondria.

- It is crucial for oxidative phosphorylation, as the mitochondrial inner membrane is impermeable to NADH, the primary reducing equivalent of the electron transport chain.
- Malate Shuttle in Gluconeogenesis Step Process Cytosol Oxaloacetate → Malate (via cytosolic malate dehydrogenase, using NADH) Transport Malate crosses mitochondrial membrane via malate- α -ketoglutarate antiporter Mitochondria Malate → Oxaloacetate (via mitochondrial malate dehydrogenase, regenerating NADH) Return Pathway Oxaloacetate → Aspartate (via transamination), which exits mitochondria via the glutamate-aspartate antiporter

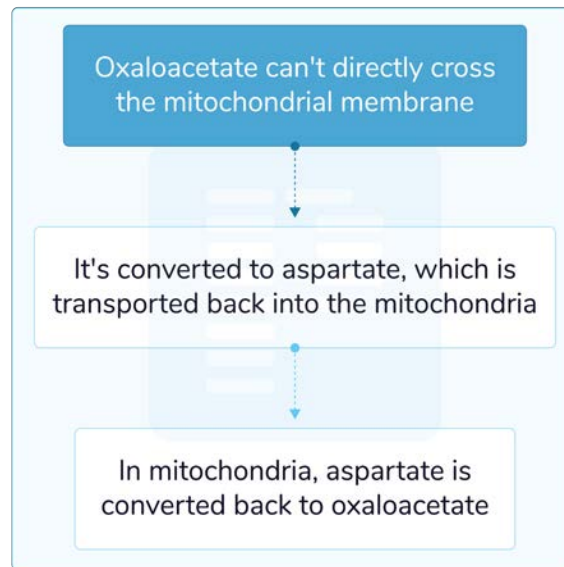
Malate Shuttle in Gluconeogenesis

Step	Process
Cytosol	Oxaloacetate → Malate (via cytosolic malate dehydrogenase, using NADH)
Transport	Malate crosses mitochondrial membrane via malate- α -ketoglutarate antiporter
Mitochondria	Malate → Oxaloacetate (via mitochondrial malate dehydrogenase, regenerating NADH)
Return Pathway	Oxaloacetate → Aspartate (via transamination), which exits mitochondria via the glutamate-aspartate antiporter

In mitochondria:



Oxaloacetate return:



- The malate shuttle is important in both gluconeogenesis and glycolysis

Solution for Question 8:

- Glucose-6-phosphatase is the only gluconeogenic enzyme located in the endoplasmic reticulum (ER) .
- It catalyzes the final step of gluconeogenesis:

Glucose-6-phosphate → Glucose + P_i

- This reaction occurs in the ER lumen , and glucose is then transported into the cytoplasm and released into the bloodstream.

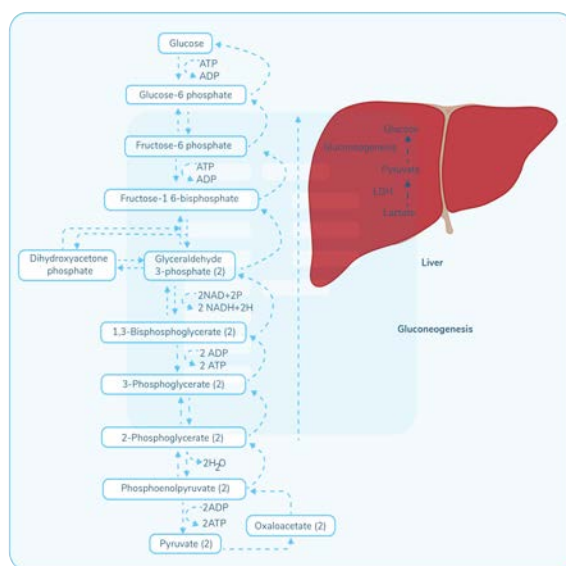
Enzymes of Gluconeogenesis, Their Locations, and Cofactors

Enzyme	Location	Function	Cofactors
Pyruvate Carboxylase (Option A ruled out)	Mitochondria	Converts pyruvate to oxaloacetate	Biotin, ATP, CO ₂
Phosphoenolpyruvate Carboxykinase (PEPCK) (Option B ruled out)	Cytoplasm & Mitochondria	Converts oxaloacetate to PEP	GTP (as a phosphate donor)

Fructose-1,6-bisphosphatase (Option C ruled out)	Cytoplasm	Converts F-1,6-BP to F-6-P	Mg ²⁺
Glucose-6-phosphatase	Endoplasmic Reticulum	Converts G-6-P to free glucose	Mg ²⁺

Solution for Question 9:

Acetyl-CoA is not a substrate for gluconeogenesis. Although acetyl-CoA is an important metabolic intermediate, it cannot be converted to glucose because the reaction catalysed by pyruvate dehydrogenase is irreversible.



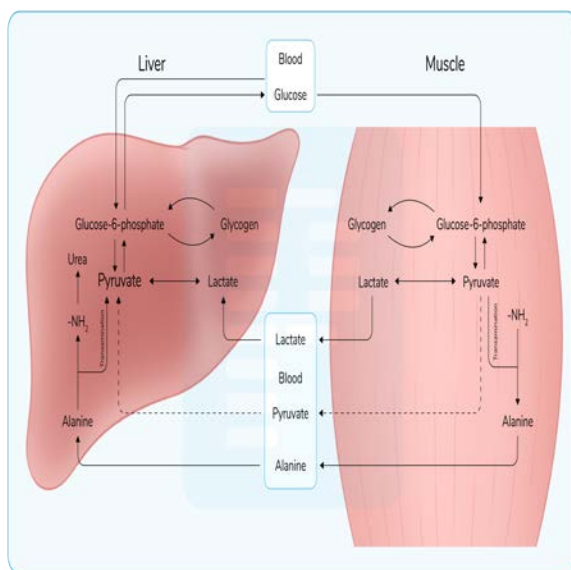
Substrates for Gluconeogenesis

Substrate	Source	Entry Point in Gluconeogenesis
Lactate	Anaerobic glycolysis (muscles, RBCs)	Converted to pyruvate via the Cori cycle
Glycerol	Lipolysis (adipose tissue)	Converted to DHAP
Alanine	Muscle protein breakdown	Converted to pyruvate via transamination

Propionate	Odd-chain fatty acid oxidation	Converted to succinyl-CoA (TCA cycle)
------------	--------------------------------	---------------------------------------

Alanine is converted to pyruvate, which enters the gluconeogenesis pathway as part of the glucose-alanine (Cahill) cycle.

Propionate : Obtained from the metabolism of odd-chain fatty acids.



Cori's cycle/ Glucose Lactate cycle and Cahill cycle/Glucose Alanine cycle

Lactate (Option A): Incorrect because it is a major gluconeogenic substrate via the Cori cycle.

Glycerol (Option B): Incorrect because it enters gluconeogenesis via DHAP formation.

Alanine (Option C): Incorrect because it contributes to gluconeogenesis via pyruvate formation.

Solution for Question 10:

The Krebs Cycle, also known as the Citric Acid Cycle, primarily occurs in the mitochondria, not the cytoplasm.

The other pathways listed (Glycolysis, Pentose Phosphate Pathway, and Fatty Acid Synthesis) primarily occur in the cytoplasm.

Krebs Cycle:

- It's a mitochondrial pathway and a common pathway for carbohydrates, lipids, and protein.
- Strictly aerobic and produces 10 ATP per cycle
- It is an Amphibolic pathway and consists of Anaplerotic reactions that replenish intermediates of the Krebs cycle.

Cellular Metabolism Pathways based on their locations:

Cytoplasm	Mitochondria
Glycolysis (Option A ruled out) Pentose Phosphate Pathway (Option B ruled out) Fatty Acid Synthesis (Option C ruled out) Urea Cycle (partial- Cytoplasm & Mitochondria) HMP Shunt	Krebs Cycle (Citric Acid Cycle) Electron Transport Chain Fatty Acid Oxidation Ketogenesis
Gluconeogenesis occurs in the cytoplasm, mitochondria, and endoplasmic reticulum.	

Solution for Question 11:

The citric acid cycle is amphibolic because it both produces energy (catabolic function) and provides intermediates for biosynthesis, such as amino acids, glucose, and fatty acids (anabolic function).

Amphibolic nature of TCA cycle:

1. Catabolic Role (Energy Production)

- Oxidation of Acetyl-CoA: Acetyl-CoA, derived from the breakdown of carbohydrates, fats, and proteins, enters the TCA cycle and is oxidized to produce NADH, FADH₂, and GTP (converted to ATP).
- Generation of Reducing Equivalents: The cycle generates NADH and FADH₂, which are essential for the electron transport chain and subsequent ATP production.

Oxidation of Acetyl-CoA: Acetyl-CoA, derived from the breakdown of carbohydrates, fats, and proteins, enters the TCA cycle and is oxidized to produce NADH, FADH₂, and GTP (converted to ATP).

Generation of Reducing Equivalents: The cycle generates NADH and FADH₂, which are essential for the electron transport chain and subsequent ATP production.

2. Anabolic Role (Biosynthesis)

- Citrate for Fatty Acid Synthesis : Citrate from the TCA cycle is exported to the cytoplasm and cleaved to provide acetyl-CoA for fatty acid synthesis.
- α -Ketoglutarate for Amino Acid Synthesis : α -Ketoglutarate is transaminated to form glutamate, which can be converted to glutamine, proline, and arginine.
- Succinyl-CoA for Heme Synthesis: Succinyl-CoA and glycine participate in the synthesis of δ -aminolevulinic acid (ALA) , the first step in heme biosynthesis.

Citrate for Fatty Acid Synthesis : Citrate from the TCA cycle is exported to the cytoplasm and cleaved to provide acetyl-CoA for fatty acid synthesis.

α -Ketoglutarate for Amino Acid Synthesis : α -Ketoglutarate is transaminated to form glutamate, which can be converted to glutamine, proline, and arginine.

- Oxaloacetate for Gluconeogenesis: Oxaloacetate can be converted to phosphoenolpyruvate (PEP), which is a key intermediate in gluconeogenesis, the pathway that produces glucose from non-carbohydrate sources.

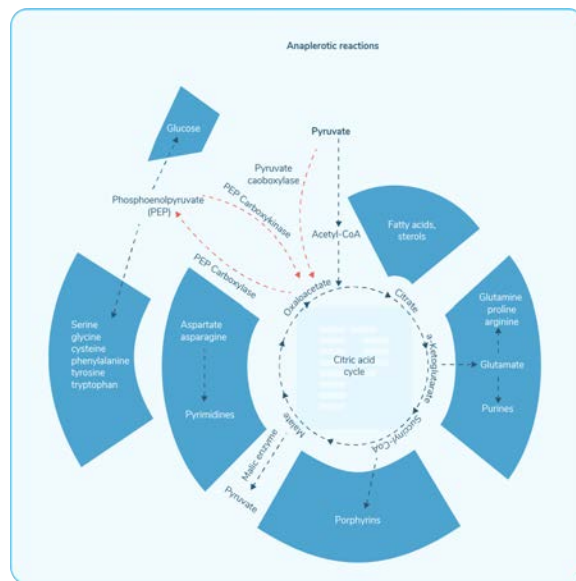
It exclusively produces NADH for oxidative phosphorylation (Option B) : While the citric acid cycle produces NADH, its amphibolic nature also involves generating intermediates for biosynthesis, not just ATP production.

It directly synthesises fatty acids within the cycle (Option C) : The citric acid cycle does not directly synthesise fatty acids. Instead, it produces citrate, which can be transported out of the mitochondria and used in the cytoplasm for fatty acid synthesis.

It provides intermediates solely for glucose synthesis (Option D) : The citric acid cycle provides intermediates for various biosynthetic pathways, not just glucose synthesis.

Solution for Question 12:

- Anaplerotic reactions are biochemical pathways that replenish intermediates of the TCA cycle , ensuring its continuous function.
- These reactions compensate for the loss of intermediates used in biosynthetic pathways (e.g., amino acid synthesis, gluconeogenesis).



Anaplerotic Reactions

Reaction	Enzyme	Intermediate Replenished	Source
Pyruvate → Oxaloacetate	Pyruvate carboxylase	Oxaloacetate	Pyruvate (from glycolysis)
Glutamate → α-Ketoglutarate	Glutamate dehydrogenase	α-Ketoglutarate	Glutamate (from amino acid metabolism)
Propionyl-CoA → Succinyl-CoA	Propionyl-CoA carboxylase	Succinyl-CoA	Odd-chain fatty acids, Valine, Isoleucine, Methionine
Aspartate → Oxaloacetate	Aspartate transaminase	Oxaloacetate	Aspartate (from protein metabolism)

Produce ATP (Option A) : Incorrect because anaplerotic reactions do not generate ATP directly. ATP is primarily produced through substrate-level phosphorylation and oxidative phosphorylation in the TCA cycle.

Reduction of NAD⁺ to NADH (Option C) : Incorrect because anaplerotic reactions do not involve NADH production. NADH is generated during oxidative pathways like glycolysis and the TCA cycle.

Break down fatty acids (Option D) : Incorrect because fatty acid β-oxidation generates acetyl-CoA, which enters the TCA cycle but does not serve as an anaplerotic reaction.

Solution for Question 13:

Succinate dehydrogenase is not regulated by ATP. Its regulation is primarily by the availability of its substrate (succinate) and competitive inhibition by oxaloacetate.

Regulation of Enzymes in the TCA Cycle

Enzyme	Activators	Inhibitors
Citrate Synthase (Option A ruled out)	ADP	ATP, Citrate, NADH, Succinyl-CoA
Isocitrate Dehydrogenase (Option B ruled out)	ADP, NAD ⁺ , Ca ²⁺ ■	ATP, NADH
α-Ketoglutarate Dehydrogenase (Option C ruled out)	ADP, Ca ²⁺ ■	ATP, NADH, Succinyl-CoA
Succinate Dehydrogenase	Succinate availability	Oxaloacetate

Solution for Question 14:

Fluoroacetate itself does not directly inhibit aconitase. Instead, fluoroacetate is converted to fluoroacetyl-CoA, which then forms fluorocitrate. Fluorocitrate inhibits aconitase, leading to citrate accumulation.

Fluoroacetyl-CoA condenses with oxaloacetate to form fluorocitrate, which inhibits the aconitase enzyme, causing citrate accumulation.

Inhibitors of enzymes involved in the citric acid cycle:

Inhibitor	Enzyme	Result
Fluoroacetate	Aconitase	Accumulation of citrate
Arsenite	Alpha-ketoglutarate dehydrogenase complex	Accumulation of alpha-ketoglutarate
Malonate	Succinate dehydrogenase	Accumulation of succinate

Cyanide (Option A) is a very toxic chemical that inhibits cytochrome c oxidase (Complex IV) in the electron transport chain, leading to the failure of aerobic respiration and histotoxic hypoxia.

Rotenone (Option C) is an inhibitor of Complex I (NADH dehydrogenase) of the electron transport chain, and it is not directly associated with the inhibition of the TCA cycle

Antimycin A (Option D) is a direct inhibitor of Complex III (cytochrome bc1 complex) of the electron transport chain, and it is not directly associated with the inhibition of the TCA cycle

Solution for Question 15:

Each glucose molecule produces 2 Acetyl-CoA , and each Acetyl-CoA completes one turn of the TCA cycle .

Molecule Produced	ATP Yield per Molecule	Total ATP for 2 Cycles
NADH (3 per cycle)	2.5 ATP	$3 \times 2.5 \times 2 = 15 \text{ ATP}$
FADH $\blacksquare$ (1 per cycle)	1.5 ATP	$1 \times 1.5 \times 2 = 3 \text{ ATP}$
GTP (Direct ATP) (1 per cycle)	1 ATP	$1 \times 2 = 2 \text{ ATP}$
Total ATP		20 ATP

- Each Acetyl-CoA generates 3 NADH, 1 FADH $\blacksquare$, and 1 ATP .
- Since one glucose produces 2 Acetyl-CoA, we double the ATP yield .
- Total ATP from Citric Acid Cycle per glucose = 20 ATP .

Each Acetyl-CoA generates 3 NADH, 1 FADH $\blacksquare$, and 1 ATP .

Citric Acid Cycle (Krebs Cycle)

The Citric Acid Cycle (TCA cycle, Krebs cycle) is a central metabolic pathway that occurs in the mitochondria and plays a crucial role in energy production, biosynthesis, and metabolic regulation.

Feature	Details
Location	Mitochondrial matrix
Purpose	Oxidation of Acetyl-CoA to generate ATP, NADH, and FADH $\blacksquare$
Entry Molecule	Acetyl-CoA (from glycolysis, fatty acid oxidation, or amino acid metabolism)

End Products per Cycle	3 NADH (Produces 7.5 ATP) 1 FADH ₂ (Produces 1.5 ATP) 1 ATP (GTP) 2 CO ₂ (Byproduct)
ATP Yield per Glucose	20 ATP (from 2 Acetyl-CoA)
Regulatory Enzymes	Citrate Synthase (Inhibited by ATP, NADH, and Succinyl-CoA) Isocitrate Dehydrogenase (Rate-limiting step, inhibited by ATP and NADH) α -Ketoglutarate Dehydrogenase (Inhibited by ATP, NADH, and Succinyl-CoA)
Significance	Provides intermediates for biosynthesis (e.g., amino acids, heme, fatty acids) Generates reducing equivalents (NADH, FADH ₂) for the electron transport chain Integrates carbohydrate, lipid, and protein metabolism

Steps in the Citric Acid Cycle:

Step	Reaction	Enzyme	ATP/NADH/FADH ₂ Produced
1	Acetyl-CoA + Oxaloacetate → Citrate	Citrate Synthase	-
2	Citrate → Isocitrate	Aconitase	-
3	Isocitrate → α -Ketoglutarate + CO ₂	Isocitrate Dehydrogenase	1 NADH (2.5 ATP)
4	α -Ketoglutarate → Succinyl-CoA + CO ₂	α -Ketoglutarate Dehydrogenase	1 NADH (2.5 ATP)
5	Succinyl-CoA → Succinate	Succinyl-CoA Synthetase	1 ATP (GTP)
6	Succinate → Fumarate	Succinate Dehydrogenase	1 FADH ₂ (1.5 ATP)
7	Fumarate → Malate	Fumarase	-
8	Malate → Oxaloacetate	Malate Dehydrogenase	1 NADH (2.5 ATP)

Regulation of the Citric Acid Cycle:

Regulator	Effect on TCA Cycle	Mechanism
ATP	Inhibits	High energy status
NADH	Inhibits	High reducing equivalents
ADP	Activates	Low energy status

Ca ²⁺ ■	Activates	Stimulates muscle contraction and energy demand
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Integration with Other Pathways:

- Gluconeogenesis : Oxaloacetate is a precursor.
- Fatty Acid Synthesis : Citrate provides acetyl-CoA for lipogenesis.
- Amino Acid Metabolism : Intermediates serve as precursors for non-essential amino acids.
- Urea Cycle : Fumarate and Aspartate link TCA with nitrogen metabolism.

The TCA cycle is a central hub for metabolism, connecting multiple biochemical pathways and ensuring efficient ATP production.

Solution for Question 16:

- Succinyl-CoA and glycine are precursors for δ -aminolevulinic acid (ALA), the first step in heme synthesis.
- In the heme synthesis, succinyl-CoA combines with glycine to form δ -aminolevulinic acid (ALA), the first step.
- Succinyl-CoA is replenished via anaplerotic reactions involving amino acids like Valine, Isoleucine, and Methionine, which are converted to Propionyl-CoA before forming Succinyl-CoA.
- Propionyl CoA from odd fatty acid breakdown enters the pathway by converting to Succinyl CoA.

Citrate (Option A) is a 6-carbon molecule formed from acetyl-CoA and oxaloacetate. Citrate serves as an intermediate in the TCA cycle and plays a key role in fatty acid and cholesterol synthesis by transporting acetyl-CoA from the mitochondria to the cytoplasm.

α -Ketoglutarate (Option B) is a 5-carbon molecule involved in amino acid metabolism and serves as an intermediate in the Krebs Cycle.

- It undergoes a Transamination Reaction to form Glutamate, which is a Reversible reaction. (α -Ketoglutarate $\leftrightarrow$ Glutamate)
- Glutamate is derived from α -ketoglutarate and can be synthesized from Histidine, Proline, Arginine, and Glutamine via transamination and deamination reactions.

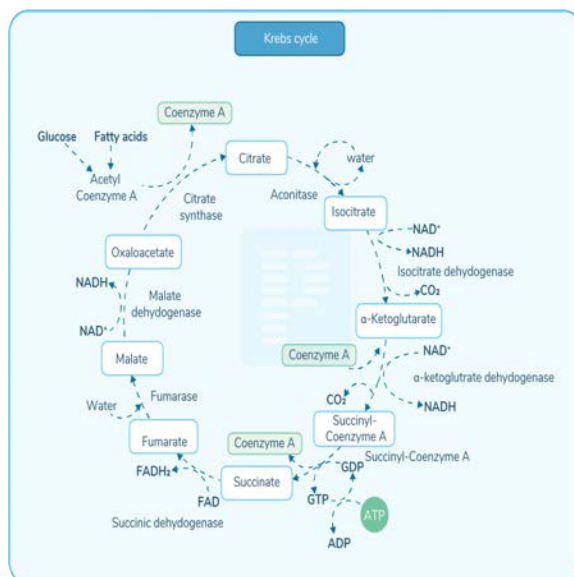
Malate (Option D): Malate is an intermediate in the Krebs Cycle, formed from fumarate and converted into oxaloacetate.

- Malate can be converted to pyruvate by Malic Enzyme, generating NADPH , which is crucial for fatty acid synthesis.
- It is crucial as pyruvate can yield both Acetyl CoA (Pyruvate dehydrogenase) and Oxaloacetate (Pyruvate carboxylase), enabling their entry into the Citric cycle.

Solution for Question 17:

- The enzymes involved in both reactions are Isocitrate Dehydrogenase (Step 1) and α -Ketoglutarate Dehydrogenase (Step 4).
- Both involve oxidation and removal of CO_2 and produce NADH, which is later used in oxidative phosphorylation.
- Steps 3 and 4 are catalysed by Succinate dehydrogenase, and Malate Dehydrogenase catalyzes the oxidation of malate to oxaloacetate, producing NADH, which enters oxidative phosphorylation.

Krebs cycle:



Reactions in Krebs cycle:

Reaction Type	Enzyme	Reaction	Product Formed
Oxidative Decarboxylation	Isocitrate Dehydrogenase	Isocitrate → α -Ketoglutarate	$\text{CO}_2 + \text{NADH}$
α -Ketoglutarate Dehydrogenase	α -Ketoglutarate → Succinyl-CoA	$\text{CO}_2 + \text{NADH}$	
Oxidation (Production of $\text{NADH}/\text{FADH}_2$)	Isocitrate Dehydrogenase	Isocitrate → α -Ketoglutarate	NADH
α -Ketoglutarate Dehydrogenase	α -Ketoglutarate → Succinyl-CoA	NADH	
Succinate Dehydrogenase	Succinate → Fumarate	FADH_2	
Malate Dehydrogenase	Malate → Oxaloacetate	NADH	
Substrate-Level Phosphorylation	Succinyl-CoA Synthetase	Succinyl-CoA → Succinate	GTP (or ATP)
Hydration	Fumarase	Fumarate → Malate	H_2O Addition

- Oxidative Decarboxylation removes CO_2 and generates NADH .
- Oxidation reactions contribute to the Electron Transport Chain by generating $\text{NADH}/\text{FADH}_2$.
- Substrate-level phosphorylation occurs at the Succinyl-CoA to Succinate step.
- Hydration reaction occurs when Fumarase converts Fumarate to Malate.

Solution for Question 18:

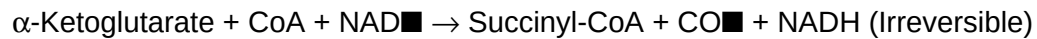
The enzyme responsible for the reaction α -Ketoglutarate → Succinyl-CoA is α -Ketoglutarate Dehydrogenase, in which Fe^{2+} is not a cofactor.

Iron (Fe^{2+}) is a cofactor for Aconitase, which catalyzes the isomerization of citrate to isocitrate.

α -Ketoglutarate Dehydrogenase:

- It is a Multienzyme complex involving 3 subunits and requires 5 Cofactors: Thiamine pyrophosphate (Vit B1) (Option B ruled out) Riboflavin (Vit B2 / FAD) Niacin (Vit B3 / NAD) Lipoic Acid (Option C ruled out) Pantothenic Acid / CoA / Vit B5 (Option D ruled out)
- Thiamine pyrophosphate (Vit B1) (Option B ruled out)

- Riboflavin (Vit B2 / FAD)
- Niacin (Vit B3 / NAD)
- Lipoic Acid (Option C ruled out)
- Pantothenic Acid / CoA / Vit B5 (Option D ruled out)
- Thiamine pyrophosphate (Vit B1) (Option B ruled out)
- Riboflavin (Vit B2 / FAD)
- Niacin (Vit B3 / NAD)
- Lipoic Acid (Option C ruled out)
- Pantothenic Acid / CoA / Vit B5 (Option D ruled out)
- It is involved in oxidative decarboxylation in the Krebs cycle but does not participate in oxidative phosphorylation.
- It catalyses the oxidative decarboxylation of α -ketoglutarate to form succinyl-CoA.



- It is Inhibited by Arsenite (Non-Competitive Inhibition).

Previous Year Questions

1. Which of the following is deficient in the mucopolysaccharidosis associated with Natowicz disease?

- A. Dermatan sulfate
 - B. Heparan sulfate
 - C. Keratan sulfate
 - D. Hyaluronic acid
-

2. What is the composition of lactulose?

- A. Glucose + Galactose
 - B. Glucose + Fructose
 - C. Galactose + Fructose
 - D. Fructose + Ribulose
-

3. A 4-year-old child presents with easy fatigability. The mother also complains that the child has increased hunger between meals, which is relieved after food. Liver biopsy revealed no glycogen. The most likely deficient enzyme is?

- A. Branching enzyme
 - B. Glycogen synthase
 - C. Debranching enzyme
 - D. Glucose-6-phosphatase
-

4. If an individual is following a diet solely comprised of carbohydrates, what would be the respiratory quotient?

- A. 0.7
 - B. 0.8
 - C. 1
 - D. 1.2
-

5. Which metabolic pathway is hindered by Fluoroacetate, a constituent found in pesticides?

- A. TCA cycle
 - B. Glycolytic pathway
 - C. Oxidative phosphorylation
 - D. ETC
-

6. Which of the following is the best method to measure HbA1C?

- A. Isoelectric focusing
 - B. Affinity chromatography
 - C. Ion exchange chromatography
 - D. Electrophoresis
-

7. Which of the following enzymes are involved in gluconeogenesis? A. Pyruvate carboxylase B. Phosphoenol pyruvate kinase C. Hexokinase D. Pyruvate kinase E.

Glucose 6 phosphatase

- A. A, B and E only
 - B. B and C only
 - C. A and E only
 - D. A, B and C only
-

8. The infant is suffering from which of the following abnormalities?



- A. Snowflake cataract
 - B. Oil drop cataract
 - C. Her's disease
 - D. Von gierke's disease
-

9. What is the enzyme that is shared by both glycolysis and gluconeogenesis?

- A. Pyruvate kinase

- B. Phosphofructokinase
 - C. Phosphoenolpyruvate carboxy kinase
 - D. Phosphoglycerate kinase
-

**10. Ketosis is common in type 1 DM due to: Decreased beta-oxidation of fatty acids
Increased mobilization of fat from the adipose cell Decreased acetyl CoA in TCA
cycle Increase beta-hydroxy butyryl synthase**

- A. 1, 2, and 3
 - B. 1,2,3 and 4
 - C. 2 only
 - D. 1 and 2 only
-

11. A patient came to the hospital with a complaint of abdominal pain. Routine investigations were unremarkable. On observation, the urine turned black on standing. Which of the following enzymes is likely to be defective in this patient?

- A. Homogentisate oxidase
 - B. Xanthine oxidase
 - C. Phenylalanine hydroxylase
 - D. Dihydrate orotate dehydrogenase
-

12. Which among the following absorbs the least water?

- A. Mucilage
- B. Cellulose
- C. Gums

D. Pectin

13. In a patient who has experienced prolonged starvation, which of the following tests would yield positive results when conducted on their urine?



- A. Benedict's test
 - B. Rothera's test
 - C. Biuret test
 - D. Emulsion test
-

14. The meal that exhibits the highest thermic effect is characterized by

- A. Protein
- B. Fat
- C. Carbohydrates
- D. Mixed meal having all 3 in equal amount

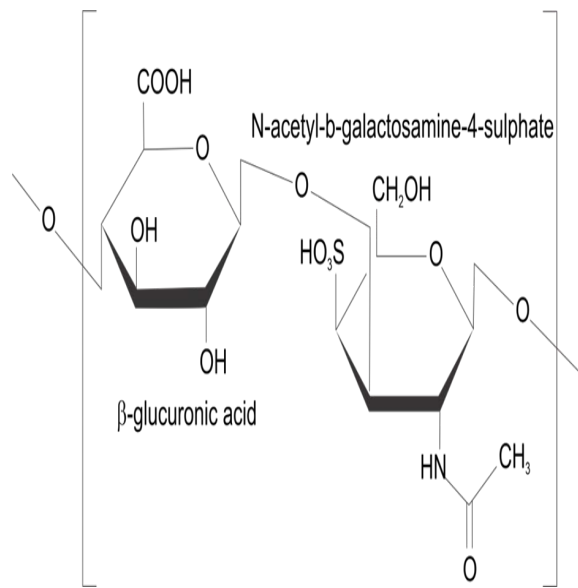
15. How many calories are provided by a TPN bag containing 100g of glucose, 30g of amino acids, and 40g of lipids?

- A. 840 KCal
 - B. 880 KCal
 - C. 640 Kcal
 - D. 680 Kcal
-

16. Glucose is stored in the form of glycogen because

- A. It is compact
 - B. It has multiple reducing ends
 - C. It can be stored at multiple organs
 - D. It can provide glucose for about 1 week when needed
-

17. The given chemical structure is of

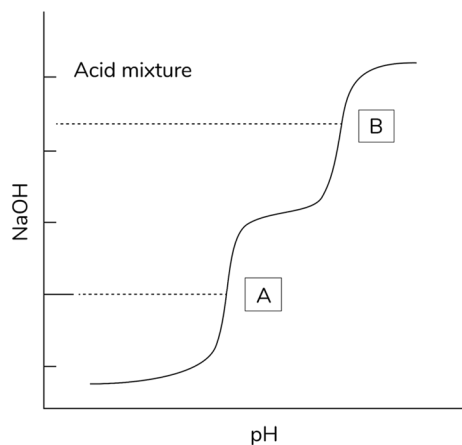


- A. Keratan sulfate
- B. Chondroitin sulfate
- C. Heparin
- D. Dermatan sulfate

18. Which among the following has low glycemic index?

- A. Easily digestible
- B. Increases plasma glucose
- C. Has slower absorption
- D. Increase glycogen deposits

19. The graph shown below is the titration curve of a biochemical compound. Which of the following statement is true?



- A. The maximum buffering capacity of the compound is represented by points A and B
- B. The points A and B represent the range of maximum ionization of the amine and carboxyl group
- C. The compound has three ionizable side chains
- D. The compound has one ionizable group

20. Which of the following statements is false regarding proteoglycans?

- A. Carries negative charge
- B. Chondroitin sulfate is a glycosaminoglycans
- C. Made of amino acids and sugars
- D. Holds less water

21. Which of the options listed below possesses both protective and lubricating qualities?

- A. Collagen
 - B. Mucin
 - C. Albumin
 - D. Immunoglobulin
-

22. Which of the following accurately describes the composition of lactulose

- A. Galactose and glucose
 - B. Glucose and fructose
 - C. Galactose and fructose
 - D. Fructose and mannose
-

23. Which enzymes utilize molecular oxygen as a recipient of hydrogen in order to generate water?

- A. Catalase
 - B. Superoxide dismutase
 - C. Cytochrome c oxidase
 - D. Pyruvate dehydrogenase
-

24. A patient presented to the OPD with liver damage. The picture depicts the patient having their eyes examined. Which of the following subsequent factors is responsible for this condition?



- A. Copper
- B. Glucose
- C. Galactose
- D. Mannose

25. Which of the following substances inhibits cytochrome complex IV, a component of the electron transport chain responsible for ATP synthesis?

- A. Cyanide
- B. Carbon dioxide
- C. Oligomycin
- D. Ouabain

26. What is the cause of elevated H^+ ions in the intermembrane space of mitochondria?

- A. Decreased oxidative phosphorylation
 - B. Increased ATP production
 - C. Reduced chemiosmosis
 - D. Water Retention in inner mitochondrial space
-

27. You are conducting an experiment on mitochondrial respiration. You add malate/pyruvate and respiration is normal. You add succinate and respiration is normal. When you add another substance in the presence of pyruvate/succinate and malate, respiration is blocked. Which of the following substances is most likely added?

- A. Rotenone
 - B. Antimycin A
 - C. Oligomycin
 - D. 2,4-dinitrophenol
-

28. Except for which of the following, all are cofactors for Dehydrogenase?

- A. NADP
 - B. FAD
 - C. NAD
 - D. SAM
-

29. Which of the following enzymes is irreversible step in gluconeogenesis?

- A. Pyruvate carboxylase
- B. Phosphoenolpyruvate carboxylase

- C. Pyruvate kinase
 - D. Glucose-6-phosphatase
-

30. Which specific glucose transporter is present in the skeletal muscle?

- A. GLUT 1
 - B. GLUT 2
 - C. GLUT 3
 - D. GLUT 4
-

31. Which enzyme deficiencies could potentially be the cause of exercise intolerance and cramps in an adolescent male patient?

- A. Myophosphorylase
 - B. Hexokinase
 - C. Glucose-6-phosphatase
 - D. Hepatic glycogen phosphorylase
-

32. What is the enzyme defect if a child with anemia, thrombocytopenia, bony pain, and hepatosplenomegaly shows a "crumpled tissue paper" appearance on light microscopy of bone marrow aspirate?

- A. Beta- glucosidase
 - B. Sphingomyelinase
 - C. Hexosaminidase
 - D. Glucose6-phosphatase
-

33. In a fasting state, glucagon promotes the production of glucose by which of the following pathways?

- A. Glycolysis and gluconeogenesis
 - B. Gluconeogenesis and glycogenolysis
 - C. Glycogenesis and gluconeogenesis
 - D. Only gluconeogenesis
-

34. In Von Gierke's disease, which enzyme is commonly observed to be deficient?

- A. Glucose-6-phosphatase
 - B. Branching enzyme
 - C. Debranching enzyme
 - D. Muscle glycogen phosphorylase
-

35. Which of the subsequent cellular pathways is found solely within the cytoplasm of a cell?

- A. Beta oxidation
 - B. TCA cycle
 - C. Glycolysis
 - D. Urea cycle
-

36. A patient who was in excruciating pain all over his body was taken to the hospital. In recent years, he has experienced these incidents frequently. When

exercising vigorously, the ache begins. Anemia was detected on blood examination along with elongated RBC as opposed to concave ones. It was determined that he had sickle cell anemia. What substitution takes place in sickle cell anemia?

- A. Substitution of valine by glutamic acid at the 6th position
 - B. Substitution of glutamic acid by valine at the 6th position
 - C. Substitution of valine by glutamic acid at the 5th position
 - D. Substitution of glutamic acid by valine at the 5th position
-

37. Glycogen phosphorylase is activated by:

- A. Calcium
 - B. Glucose-6-phosphate
 - C. Insulin
 - D. Protein phosphatase
-

38. What is the source of fructosamine formation through glycation?

- A. Fructose
 - B. Albumin
 - C. Urea
 - D. Glycosaminoglycan
-

39. Which of the following enzyme defect leads to Pompe's disease?

- A. Lysosomal acid alpha glucosidase
- B. β -hexosaminidase-A

- C. β -galactocerebrosidase
 - D. β -glucocerebrosidase
-

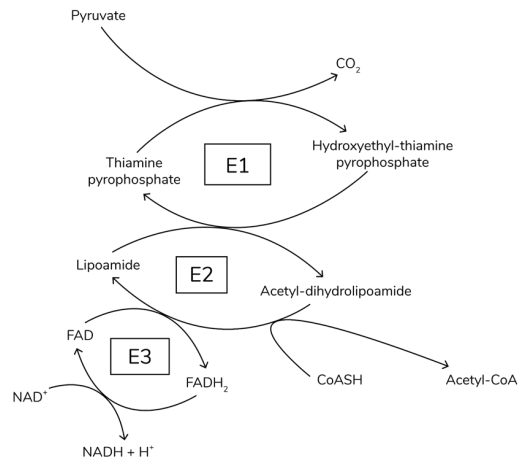
40. Which of the following glycosaminoglycans (GAGs) are responsible for corneal transparency?

- A. Keratan sulfate
 - B. Chondroitin sulfate
 - C. Heparin sulfate
 - D. Hyaluronic acid
-

41. In this scenario, what is the origin of the blood glucose measured at 7 am the following morning for an individual who had dinner at 8 pm the previous night?

- A. Hepatic glycogenolysis
 - B. Hepatic gluconeogenesis
 - C. Muscle glycogenolysis
 - D. Dietary glucose
-

42. In the diagram provided, which enzymes are identified as E1 and E2, responsible for the conversion of pyruvate to acetyl CoA?



- A. Pyruvate dehydrogenase and dihydrolipoyl transacetylase
- B. Pyruvate carboxylase and dihydrolipoyl transacetylase
- C. Pyruvate dehydrogenase and dihydropoyl dehydrogenase
- D. Pyruvate carboxylase and dihydrolipoyl dehydrogenase

43. Gluconeogenesis is inhibited by

- A. Insulin
- B. Glucagon
- C. Lactate
- D. None of above

44. In a patient who hails from a Mediterranean country and travels to Africa, they contract malaria. After receiving treatment with primaquine, the patient subsequently develops hemolytic anemia. Which pathway deficiency in the enzyme could potentially be responsible for this condition?

- A. Glycolysis
 - B. Gluconeogenesis
 - C. Hexose monophosphate (HMP) pathway
 - D. Luebering-Rapoport pathway
-

45. What is the most frequently occurring amino acid in collagen?

- A. Proline
 - B. Lucine
 - C. Arginine
 - D. Glycine
-

46. Which is the enzyme activated by a decrease in insulin: glucagon ratio?

- A. Glucokinase
 - B. Hexokinase
 - C. Phosphofructokinase
 - D. Glucose 6-phosphatase
-

47. Which enzyme is responsible for the deficiency causing classical galactosemia?

- A. Aldose reductase
 - B. Galactokinase
 - C. Galactose-1-phosphate uridyl transferase
 - D. UDP galactose-4-epimerase
-

48. Which one of the following choices does not represent a function of glucagon?

- A. Glycolysis
 - B. Gluconeogenesis
 - C. Glycogenolysis
 - D. Ketogenesis
-

49. A 10-year-old boy rapidly develops hypoglycemia after moderate activity. Blood examination reveals raised levels of ketone bodies, lactic acid, and triglycerides. On examination, the liver and kidneys are found to be enlarged. Which of the following deficiencies would be seen in the liver of this child?

- A. Glucose-6-phosphate
 - B. Glucose-1-phosphate
 - C. Glucose-1,6-diphosphate
 - D. Glycosaminoglycans
-

50. What is the primary source of reducing equivalents for liver fatty acid synthesis?

- A. Hexose monophosphate shunt pathway
 - B. Glycolysis
 - C. Tricarboxylic acid cycle
 - D. Uronic acid pathway
-

51. Which one of the options provided does not serve as a substrate for the synthesis of glucose?

- A. Acetyl coenzyme A
 - B. Glycerol
 - C. Alanine
 - D. Lactate
-

52. What is the main substrate and metabolic fuel used by RBCs during fasting period?

- A. Ketone bodies
 - B. Glucose
 - C. Free fatty acids
 - D. Amino acids
-

53. Which of the following diagnostic markers is/are not correctly matched? 1. Beta glucocerebrosidase Von Gierke disease 2. Lipase Acute pancreatitis 3. Ceruloplasmin Hepatolenticular degeneration 4. High LDH1: LDH2 ratio Myocardial infarction

1. Beta glucocerebrosidase	Von Gierke disease
2. Lipase	Acute pancreatitis
3. Ceruloplasmin	Hepatolenticular degeneration
4. High LDH1: LDH2 ratio	Myocardial infarction

- A. 1 only
- B. 1 and 2
- C. 1, 2 and 4

D. 2, 3 and 4

54. Which enzyme is likely to be deficient in a 4-year-old child who presents with a habit of biting their lips and fingers, and has a missing portion of fingers on their left hand? Further evaluation shows delayed developmental milestones. The parents report that two male cousins on the mother's side have similar symptoms and have been diagnosed with a genetic disorder. Laboratory reports indicate hyperuricemia.

- A. Purine nucleoside phosphorylase
 - B. Hypoxanthine -guanine phosphoribosyl transferase
 - C. Adenine phosphoribosyl transferase
 - D. Adenosine deaminase
-

55. A newborn baby refuses breast milk since the 2nd day of birth but accepts glucose water. He develops vomiting and severe jaundice by the 5th day. Benedict's test was positive for urine, and blood glucose was low. The most likely cause is due to the deficiency of:

- A. Aldose reductase
 - B. Galactokinase
 - C. Galactose-1-phosphate uridyl transferase
 - D. UDP galactose-4- epimerase
-

56. Which of the following is the best screening investigation for metabolic disorders?

- A. Western blot

B. Tandem mass spectrometry

C. PCR

D. Gel electrophoresis

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	2
Question 4	3
Question 5	1
Question 6	3
Question 7	1
Question 8	2
Question 9	4
Question 10	3
Question 11	1
Question 12	2
Question 13	2
Question 14	1
Question 15	2
Question 16	1
Question 17	2
Question 18	3

Question 19	1
Question 20	4
Question 21	2
Question 22	3
Question 23	3
Question 24	1
Question 25	1
Question 26	1
Question 27	3
Question 28	4
Question 29	3
Question 30	4
Question 31	1
Question 32	1
Question 33	2
Question 34	1
Question 35	3
Question 36	2
Question 37	1
Question 38	2
Question 39	1

Question 40	1
Question 41	1
Question 42	1
Question 43	1
Question 44	3
Question 45	4
Question 46	4
Question 47	3
Question 48	1
Question 49	1
Question 50	1
Question 51	1
Question 52	2
Question 53	1
Question 54	2
Question 55	3
Question 56	2

Solution for Question 1:

Answer: C) Keratan sulfate

Explanation:

- Natowicz Disease (Mucopolysaccharidosis Type IX - MPS IX) A rare mucopolysaccharidosis (MPS IX) caused by a deficiency of hyaluronidase . Leads to the accumulation of Keratan sulfate , making it unique among MPS disorders.
- A rare mucopolysaccharidosis (MPS IX) caused by a deficiency of hyaluronidase .
- Leads to the accumulation of Keratan sulfate , making it unique among MPS disorders.
- Clinical Features: Mild phenotype compared to other MPS disorders. Soft tissue masses, joint stiffness, and normal intelligence . No significant skeletal or neurological abnormalities , unlike other MPS types.
- Mild phenotype compared to other MPS disorders.
- Soft tissue masses, joint stiffness, and normal intelligence .
- No significant skeletal or neurological abnormalities , unlike other MPS types.
- Biochemical Deficiency: Hyaluronidase deficiency impairs the degradation of Keratan sulfate , leading to its accumulation.
- Hyaluronidase deficiency impairs the degradation of Keratan sulfate , leading to its accumulation.
- A rare mucopolysaccharidosis (MPS IX) caused by a deficiency of hyaluronidase .
- Leads to the accumulation of Keratan sulfate , making it unique among MPS disorders.
- Mild phenotype compared to other MPS disorders.
- Soft tissue masses, joint stiffness, and normal intelligence .
- No significant skeletal or neurological abnormalities , unlike other MPS types.
- Hyaluronidase deficiency impairs the degradation of Keratan sulfate , leading to its accumulation.

MPS Type	Enzyme Deficiency	Accumulated GAG
MPS I (Hurler/Scheie)	α -L-iduronidase	Dermatan sulfate, Heparan sulfate
MPS II (Hunter)	Iduronate-2-sulfatase	Dermatan sulfate, Heparan sulfate
MPS III (Sanfilippo)	Heparan N-sulfatase, etc.	Heparan sulfate
MPS IV (Morquio)	Galactosamine-6-sulfatase	Keratan sulfate
MPS VI (Maroteaux-Lamy)	Arylsulfatase B	Dermatan sulfate
MPS VII (Sly)	β -Glucuronidase	Dermatan sulfate, Heparan sulfate
MPS IX (Natowicz)	Hyaluronidase	Keratan sulfate

- Heparan sulfate (Option B): Incorrect because it accumulates in MPS I, II, III, and VII , but not in MPS IX (Natowicz syndrome).

- Dermatan sulfate (Option A): Incorrect because it accumulates in MPS I, II, VI, and VII but not MPS IX.
 - Hyaluronic acid (Option D): Incorrect because hyaluronidase deficiency does not lead to hyaluronic acid accumulation in MPS IX ; rather, Keratan sulfate is the primary accumulating GAG.
-

Solution for Question 2:

Correct Option C - Galactose + Fructose:

- Lactulose is a synthetic disaccharide composed of galactose and fructose. It is used primarily as a laxative and in the treatment of hepatic encephalopathy.

Incorrect Options:

Option A - Glucose + Galactose: This combination forms lactose, not lactulose. Lactose is a disaccharide sugar commonly found in milk.

Option B - Glucose + Fructose: This combination forms sucrose, also known as table sugar. Sucrose is a common disaccharide found in many plants.

Option D - Fructose + Ribulose: Ribulose is a five-carbon sugar, and this combination does not form lactulose. Ribulose is involved in the Calvin cycle of photosynthesis.

Solution for Question 3:

Correct Option B - Glycogen synthase:

- The patient exhibits exercise intolerance, hypoglycemia, and a lack of glycogen in the liver. The likely deficient enzyme is glycogen synthase, leading to glycogen storage disorder type 0.
- Glycogen synthase converts UDP glucose into glycogen primer in the liver and muscles, aiding glucose storage. Its deficiency prevents glycogen formation, causing hypoglycemia between meals and exercise intolerance due to absent glycogen in muscles.

Incorrect Options:

Option A - Branching Enzyme: Branching enzyme deficiency leads to glycogen storage disease type IV (Anderson disease), characterized by abnormal glycogen accumulation in the liver, resulting in hepatomegaly and liver cirrhosis.

Option C - Debranching enzyme: Debranching enzyme deficiency results in glycogen storage disease type III (Cori disease), presenting with exercise intolerance, hypoglycemia, and abnormal glycogen accumulation with multiple branches.

Option D - Glucose 6 phosphatase: Deficiency of glucose 6 phosphatase results in glycogen storage disease type I (also known as von Gierke disease), characterized by hypoglycemia, doll-like facies, xanthoma, ketosis, and hepatomegaly.

Solution for Question 4:

Correct Option C - 1

- An exclusive carbohydrate diet means that the individual is only consuming carbohydrates as their source of energy. Since the respiratory quotient is the ratio of CO₂ produced to O₂ consumed, an exclusive carbohydrate diet would result in an RQ of 1. This means that for every molecule of oxygen consumed, an equal number of molecules of carbon dioxide is produced.

Incorrect Options:

Option A - 0.7: An RQ of 0.7 is typically associated with fat metabolism, as fats require more oxygen consumption relative to carbon dioxide production compared to carbohydrates.

Option B - 0.8: An RQ of 0.8 is also commonly associated with fat metabolism, as fats have a slightly lower ratio of CO₂ production to O₂ consumption compared to carbohydrates.

Option D - 1.2: An RQ of 1.2 is not physiologically possible. It would imply more CO₂ production than O₂ consumption, which is not consistent with normal cellular respiration.

Solution for Question 5:

Correct Option A - TCA cycle

- Fluoroacetate is a toxic compound that can be found in certain pesticides. It inhibits the TCA cycle, which is a central metabolic pathway involved in the aerobic metabolism of carbohydrates, fats, and amino acids.

Incorrect Options:

Option B - Glycolytic pathway: The glycolytic pathway, also known as glycolysis, is a separate metabolic pathway that takes place in the cytoplasm of the cell.

Option C - Oxidative phosphorylation: Oxidative phosphorylation is the process that occurs in the mitochondria, where ATP is generated through the transfer of electrons from NADH and FADH₂ to the electron transport chain (ETC).

Option D - ETC (Electron Transport Chain): The electron transport chain is a series of protein complexes located in the inner mitochondrial membrane. It is responsible for the transfer of electrons and the generation of a proton gradient, which drives ATP synthesis.

Solution for Question 6:

Correct Option C - Ion exchange chromatography

- HbA1C, also known as glycated hemoglobin, is a form of hemoglobin that is used as a marker for long-term glucose control in individuals with diabetes.
- Ion Exchange Chromatography is one of the most widely used methods for measuring HbA1C in clinical laboratories.
- It works by separating hemoglobin fractions based on their charge differences, allowing differentiation between glycated and non-glycated hemoglobin.
- High-Performance Liquid Chromatography (HPLC) with ion-exchange resin is considered the gold standard for HbA1C measurement

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Incorrect Options:

Option A - Isoelectric focusing: Isoelectric focusing is a technique used to separate molecules based on their isoelectric points (pI). It is commonly used in protein analysis.

Option B - Affinity chromatography: Affinity chromatography is a technique that utilizes specific interactions between a target molecule and a ligand immobilized on a solid support.

Option D - Electrophoresis : Electrophoresis is a general technique that involves the movement of charged particles in an electric field to separate molecules based on their size, charge, or other properties.

Solution for Question 7:

Correct Option A - A, B, and E only:

- This option includes pyruvate carboxylase, phosphoenolpyruvate kinase, and glucose 6 phosphatase, which are indeed involved in gluconeogenesis.
- The enzymes involved in gluconeogenesis are: Pyruvate carboxylase
Phosphoenolpyruvate carboxykinase Fructose-1,6-bisphosphatase Glucose-6-phosphatase
- Pyruvate carboxylase
- Phosphoenolpyruvate carboxykinase
- Fructose-1,6-bisphosphatase
- Glucose-6-phosphatase
- Pyruvate carboxylase
- Phosphoenolpyruvate carboxykinase
- Fructose-1,6-bisphosphatase
- Glucose-6-phosphatase

Incorrect Options

Option B, C, and D: These options are not involved in gluconeogenesis Pyruvate kinase, and hexokinase are involved in the glycolysis pathway

Solution for Question 8:

Correct Answer B - Oil drop cataract

- The given image is of a Bilateral Oil drop cataract seen in galactosemia patients.

Incorrect Options:

Option A - Snowflake cataract is typically inherited as an autosomal dominant trait, meaning that it can be passed down from one affected parent to their offspring.

Option C - Her's disease: The given image is not of her's disease

Option D - Von Gierke's disease, also known as glycogen storage disease type I (GSD I), is an inherited metabolic disorder that affects the body's ability to break down and use glycogen, a stored form of glucose (sugar).

Solution for Question 9:

Correct Option D - Phosphoglycerate kinase

- The enzyme that is shared by both glycolysis and gluconeogenesis is Phosphoglycerate kinase (PGK) .
- PGK plays a critical role in both glycolysis and gluconeogenesis by catalyzing the conversion of 1,3-bisphosphoglycerate (1,3-BPG) to 3-phosphoglycerate (3-PG) during glycolysis and the reverse reaction during gluconeogenesis.

Incorrect Options:

Option A - Pyruvate kinase: It is incorrect option.

Option B - Phosphofructokinase: Phosphofructokinase is a key regulatory enzyme in glycolysis. It catalyzes the conversion of fructose-6-phosphate to fructose-1,6-bisphosphate.

Option C - Phosphoenolpyruvate carboxykinase: Phosphoenolpyruvate carboxykinase (PEPCK) is an enzyme involved in gluconeogenesis. It catalyzes the conversion of oxaloacetate to

phosphoenolpyruvate (PEP), an important intermediate in gluconeogenesis.

Solution for Question 10:

Correct Option C - 2 only:

Incorrect Options - A, B, and D- These options are incorrect

Solution for Question 11:

Correct Option A - Homogentisate oxidase

- The patient's black urine suggests the presence of a condition called alkaptonuria.
- Alkaptonuria is a rare inherited disorder caused by a deficiency of the enzyme homogentisate oxidase. This enzyme is responsible for the breakdown of homogentisic acid, a product of the metabolism of the amino acids phenylalanine and tyrosine.
- When homogentisate oxidase is deficient, homogentisic acid accumulates in the body and is excreted in the urine, leading to the characteristic dark or black urine on standing.

Incorrect Options:

Option B - Xanthine oxidase: Xanthine oxidase deficiency is associated with a condition called xanthinuria, which results in the accumulation of xanthine and hypoxanthine.

Option C - Phenylalanine hydroxylase: Phenylalanine hydroxylase deficiency is seen in a condition called phenylketonuria (PKU). PKU is characterized by the inability to metabolize phenylalanine properly, leading to its accumulation in the body.

Option D - Dihydrate orotate dehydrogenase: There is no known enzyme called dihydrate orotate dehydrogenase. Therefore, this option is incorrect.

Solution for Question 12:

Correct Option B - Cellulose

Cellulose is a polysaccharide found in the cell walls of plants. It is a complex carbohydrate that forms a rigid structure and is not soluble in water. Due to its structure and composition, cellulose has the ability to absorb the least amount of water compared to the other options.

Incorrect Options:

Option A - Mucilage: Mucilage refers to a sticky substance produced by plants. It has the ability to absorb and hold water, acting as a gel-like substance.

Option C - Gums: Gums, such as gum arabic or guar gum, are complex carbohydrates derived from plants. They have the ability to absorb water and form a gel-like consistency.

Option D - Pectin: Pectin is a polysaccharide found in the cell walls of fruits and vegetables. It is known for its ability to form gels and is commonly used as a thickening agent in food processing.

Solution for Question 13:

Correct Option B - Rothera's test:

- Rothera's test is done for the detection of ketone bodies which are produced as a result of starvation/fasting.
- The purple color ring at the junction of 2 liquids is a positive test. Given positive by acetoacetate and acetone

Incorrect Options:

Option A - Benedict test:

- Used for the detection of reducing sugars in a complex

Option C - Biuret test:

- The Biuret test is used to detect the presence of proteins.

Option D - Emulsion test:

- The emulsion test is a method to determine the presence of lipids using wet chemistry.

Solution for Question 14:

Correct Option A - Protein

Protein has the highest thermic effect among the three macronutrients. Approximately 20-30% of the energy from protein is used during the digestion and metabolism process.

Incorrect Options:

Option B - Fat: This option is incorrect. Fat has a lower thermic effect compared to protein. Only about 0-5% of the energy from fat is used during digestion and metabolism.

Option C - Carbohydrates: This option is incorrect. Carbohydrates have a moderate thermic effect. Approximately 5-10% of the energy from carbohydrates is used during digestion and metabolism.

Option D - Mixed meal having all three in equal amount: This option is incorrect. While a mixed meal containing protein, fat, and carbohydrates will require energy for digestion and metabolism of all three macronutrients, the thermic effect will still be higher for protein compared to the other macronutrients.

Solution for Question 15:

Correct Option B - 880 KCal

To calculate the number of calories delivered in the TPN (Total Parenteral Nutrition) bag, we need to know the caloric value of each macronutrient.

The caloric values per gram for each macronutrient are as follows:

- Glucose: 4 calories per gram
- Amino acids: 4 calories per gram
- Lipids (fats): 9 calories per gram

Now let's calculate the calories for each component of the TPN bag:

Now, let's add up the calories from each component: 400 calories (glucose) + 120 calories (amino acids) + 360 calories (lipids) = 880 calories

Therefore, the correct option is b) 880 Kcal, as it represents the total amount of calories delivered in the TPN bag.

Incorrect Options:

Option A, C, and D- These options don't take into account the calories from lipids, resulting in an incorrect lower value.

Solution for Question 16:

Correct Option A - It is compact

- Glycogen is a polysaccharide composed of many glucose molecules linked together through alpha-1,4-glycosidic bonds and branched with alpha-1,6-glycosidic bonds.
- By storing glucose as glycogen, the body can efficiently pack a large amount of glucose molecules into a compact and readily available form. This compact storage allows for efficient storage of glucose within cells, especially in tissues such as the liver and muscles, where glycogen serves as a reserve energy source.

Incorrect Options:

Option B - It has multiple reducing ends: While it is true that glycogen has multiple reducing ends due to its branched structure, this is not the primary reason why glucose is stored as glycogen.

Option C - It can be stored at multiple organs: Although glycogen can be stored in multiple organs, such as the liver and muscles, this is not the main reason why glucose is stored as glycogen.

Option D - It can provide glucose for about 1 week when needed: The storage capacity of glycogen varies depending on the individual and their glycogen reserves.

Solution for Question 17:

Correct Option B - Chondroitin sulfate:

Chondroitin sulfate is a glycosaminoglycan (GAG) that is a major component of the extracellular matrix and connective tissues in the body. It consists of repeating units of N-acetylgalactosamine (GalNAc) and glucuronic acid (GlcA) or iduronic acid (IdoA). Chondroitin sulfate provides structural support and elasticity to various tissues, including cartilage, tendons, ligaments, and skin.

Incorrect Options:

Options A - Keratan sulfate: Keratan sulfate is another type of glycosaminoglycan. It consists of repeating units of galactose and N-acetylglucosamine. It is commonly found in the cornea of the eye and cartilage, and it helps maintain the structural integrity of these tissues.

Options C - Heparin: Heparin is a different type of glycosaminoglycan that is highly sulfated. It consists of repeating units of glucuronic acid and N-acetylglucosamine. Heparin is an anticoagulant and is primarily produced and stored in mast cells. It plays a role in preventing blood clotting.

Options D - Dermatan sulfate: Dermatan sulfate is another glycosaminoglycan that is composed of repeating units of iduronic acid and N-acetylgalactosamine. It is found in various tissues, including skin, blood vessels, and heart valves. Dermatan sulfate provides structural support and flexibility to these tissues.

Solution for Question 18:

Correct option C - Has slower absorption

- Has slower absorption: This option is correct. Low glycemic index foods are characterized by carbohydrates that are digested and absorbed more slowly by the body. This slow absorption leads to a slower and more controlled release of glucose into the bloodstream, resulting in a more stable and sustained blood glucose response.

Incorrect Options:

Option A - Easily digestible: This option is incorrect. The digestibility of food is not directly related to its glycemic index. The glycemic index measures how quickly carbohydrates in food are broken down and absorbed, which is not solely determined by digestibility.

Option B - Increases plasma glucose: This option is incorrect. Foods with a high glycemic index tend to cause a rapid increase in blood glucose levels, while low glycemic index foods result in a slower and more gradual increase in blood glucose levels.

Option D - Increase glycogen deposits: This option is incorrect. Glycogen deposits refer to the storage form of glucose in the body, primarily found in the liver and muscles. The glycemic index of a food does not directly influence glycogen deposits. Instead, it reflects how quickly carbohydrates from the food are converted into glucose and affect blood glucose levels.

Solution for Question 19:

Correct Option A - The maximum buffering capacity of the compound is represented by points A and B

Option A - The statement "The maximum buffering capacity of the compound is represented by points A and B" is true. Points A and B on the titration curve correspond to the region where the compound acts as a buffer. In this region, the compound can effectively resist changes in pH by accepting or donating protons.

Incorrect Option:

Option B - The statement "The points A and B represent the range of maximum ionization of the amine and carboxyl group" is incorrect. Points A and B do not represent the maximum ionization of the amine and carboxyl groups specifically. They represent the region of maximum buffering capacity, but the overall ionization of the amine and carboxyl groups can occur over a broader pH range.

Option C - The statement "The compound has three ionizable side chains" is incorrect. Glycine, being a simple amino acid, does not have any ionizable side chains. It has a carboxyl group (COOH) and an amine group (NH₂), both of which can undergo ionization.

Option D - The statement "The compound has one ionizable group" is incorrect. Glycine has two ionizable groups: the carboxyl group (COOH) and the amine group (NH₂). These groups can gain or lose protons (H⁺) depending on the pH of the solution.

Solution for Question 20:

Correct Option D - Holds less water

Holds less water: This is the correct option. Proteoglycans have a high capacity to hold water due to the presence of the highly hydrophilic glycosaminoglycan (GAG) chains. The negatively charged GAGs attract water molecules, allowing proteoglycans to maintain hydration within tissues. Therefore, the statement that proteoglycans hold less water is not true.

Incorrect Option:

Option A - Carries negative charge: This is a true statement about proteoglycans. Proteoglycans are heavily glycosylated molecules composed of a core protein and long chains of repeating disaccharides called glycosaminoglycans (GAGs). The GAG chains are negatively charged due to the presence of sulfate and carboxylate groups, giving proteoglycans an overall negative charge.

Option B- Chondroitin sulfate is a good glycosaminoglycan (GAG) that is commonly found as a component of proteoglycans. Chondroitin sulfate chains can attach to a core protein to form a proteoglycan structure.

Option C - Made of amino acids and sugars: This is a true statement about proteoglycans. Proteoglycans consist of a core protein that is composed of amino acids and long chains of sugars called glycosaminoglycans (GAGs). The GAGs are made up of repeating disaccharide units, which include a sugar and an amino sugar.

Solution for Question 21:

Correct Option B - Mucin

- Mucin is a glycoprotein that is a major component of mucus. Mucus is a slimy and viscous substance that lines various surfaces in the body, such as the respiratory tract, gastrointestinal tract, and reproductive tract.
- Mucin plays a crucial role in protecting these surfaces by trapping pathogens, dust, and other foreign particles, preventing them from entering the body.
- Additionally, mucin acts as a lubricant, reducing friction and facilitating the movement of materials across the epithelial surfaces. Therefore, option B is the correct answer.

Incorrect Option:

Option A - Collagen- It provides structural support and strength to tissues and is not primarily known for its protective or lubricating properties.

Option C - Albumin- Albumin is a protein found in blood plasma and is involved in maintaining osmotic pressure and transporting various substances.

Option D - Immunoglobulin- They play a crucial role in the immune response, recognizing and binding to specific antigens to neutralize or eliminate them.

Solution for Question 22:

Correct Option C - Galactose and fructose:

- Lactulose, also known as 1,4- β -galactoside-fructose, is a synthetic disaccharide composed of galactose and fructose .

Incorrect Options

Option A - Galactose and glucose:

- Lactulose is not composed of Galactose and Glucose
- Lactose is a disaccharide composed of d-galactose and d-glucose units.

Option B - Glucose and fructose:

- Lactulose is not composed of Glucose and fructose.
- Sucrose is a disaccharide composed of d-glucose and d-fructose units.

Option D - Fructose and mannose:

- Lactulose is not composed of fructose and mannose.

Solution for Question 23:

Correct Options C - Cytochrome c oxidase:

- During electron transport chain (ETC), cytochrome c oxidase uses molecular oxygen as a hydrogen acceptor to produce water.
- Cytochrome c oxidase is a crucial electron transport chain (ETC) enzyme responsible for producing ATP, the cell's energy currency. This reaction is important for maintaining the proton gradient across the mitochondrial membrane, which is necessary to produce ATP.

Incorrect Options:

- Option A - Catalase: Catalase is a cell enzyme that converts hydrogen peroxide into water and oxygen. This reaction removes toxic hydrogen peroxide from the body and prevents cell oxidative damage.
 - Option B - Superoxide dismutase : Superoxide dismutase is an enzyme that converts superoxide radicals into hydrogen peroxide and molecular oxygen. This reaction is important for preventing cell oxidative damage, which can lead to aging and disease.
 - Option D - Pyruvate dehydrogenase: Pyruvate dehydrogenase is an enzyme that converts pyruvate, a product of glycolysis, into acetyl-CoA, which is important for energy metabolism. This reaction is important for providing the body with energy and is a critical step in the citric acid cycle, the primary pathway for cell energy production.
-

Solution for Question 24:

Correct Options A - Copper

- In Wilson's disease, there is a defect in the transport of copper, leading to its buildup in the liver and subsequent release into the bloodstream .
- Excess copper is then deposited in various organs, including the eyes.
- The copper deposition in the eyes can cause a characteristic finding called Kayser-Fleischer rings, which are golden-brown to greenish-brown rings seen around the cornea.
- The accumulation of copper in the liver can damage the liver, resulting in hepatocellular dysfunction, inflammation, and fibrosis .
- This can manifest as hepatomegaly (enlarged liver), jaundice, fatigue, and other symptoms of liver dysfunction.

Incorrect Options:

- Option B - Glucose is not responsible for the condition described. Glucose is a simple sugar and the body's primary energy source. While liver damage can occur due to various causes, glucose itself is not a direct factor responsible for liver damage in this scenario.
 - Option C - Galactose is not responsible for the condition described. Galactose is a sugar found in milk and dairy products. Galactosemia is a rare genetic disorder characterized by the inability to metabolize galactose, leading to its accumulation in the body.
 - Option D - Mannose is not responsible for the condition described. Mannose is a sugar that plays a role in glycoprotein synthesis and cellular communication.
-

Solution for Question 25:

Correct Option A - Cyanide

- The last enzyme complex in the electron transport chain is cytochrome complex IV, also known as cytochrome c oxidase.
- When oxygen and protons are combined to form water, it receives electrons from cytochrome c.
- The ATP synthase uses the proton gradient created by this mechanism to manufacture ATP.
- A powerful inhibitor of cytochrome complex IV is cyanide. It attaches to the enzyme's active site and stops the enzyme from transferring electrons to oxygen. Due to the blockage of the electron transport chain, reducing agents like NADH and FADH₂ begin to accumulate.

Incorrect Options: Option B - Carbon dioxide:

- Carbon dioxide does not inhibit cytochrome complex IV.
- Bicarbonate ions, which are the predominant type of carbon dioxide carried in the blood, do not directly interact with the electron transport chain.

Option C - Oligomycin:

- Cytochrome complex IV is not inhibited by oligomycin.
- It is an ATP synthase inhibitor that attaches to the F_o subunit of ATP synthase and stops protons from passing through the enzyme, hence preventing the production of ATP.

Option D - Ouabain:

- The cytochrome complex IV inhibitor ouabain does not work.

- The Na⁺/K⁺-ATPase pump, which keeps the electrochemical gradient across the plasma membrane of cells, is inhibited by this cardiac glycoside.

-

Solution for Question 26:

Correct Option A - Decreased oxidative phosphorylation:

- The intermembrane space of the mitochondria is an important site for generating an electrochemical gradient that drives ATP synthesis through oxidative phosphorylation.
- Increased H⁺ ions in the intermembrane space can be due to decreased oxidative phosphorylation, which can be caused by a range of factors, such as mitochondrial dysfunction, genetic mutations affecting mitochondrial function, or exposure to toxins or drugs that affect mitochondrial respiration.

Incorrect Options:

Option B: Increased ATP production: This statement is not true. ATP production through oxidative phosphorylation depends on the electrochemical gradient generated by the transport of H⁺ ions from the intermembrane space to the mitochondrial matrix.

Option C: Reduced chemiosmosis: This statement is not entirely true. Increased H⁺ ions in the intermembrane space can lead to increased chemiosmosis, which is the process by which the H⁺ gradient generated across the inner mitochondrial membrane drives ATP synthesis through oxidative phosphorylation.

Option D: Water retention in the inner mitochondrial space: This statement is untrue. Increased H⁺ ions in the intermembrane space are unrelated to water retention in the inner mitochondrial space. However, increased H⁺ ions in the intermembrane space can lead to decreased water concentration due to changes in pH.

Solution for Question 27:

Correct Option C - Oligomycin

Oligomycin is an antibiotic that acts as an inhibitor of ATP synthase, the enzyme responsible for ATP production in mitochondria. When oligomycin is added, it binds to the F_0 subunit of ATP synthase, blocking the flow of protons back into the mitochondrial matrix.

This prevents ATP synthesis from occurring, leading to a blockage of respiration. Since respiration is blocked in the presence of pyruvate/succinate and malate when another substance is added, oligomycin is the most likely substance introduced.

Incorrect Options:

Option A - Rotenone: If rotenone were added, it would block respiration specifically when malate/pyruvate is present since complex I is involved in the oxidation of these substrates.

Option B - Antimycin A: If antimycin A were added, it would block respiration when succinate is present since complex III is responsible for electron transfer from succinate.

Option D - 2,4-dinitrophenol: 2,4-dinitrophenol (DNP) is an uncoupling agent that disrupts the coupling of electron transport and ATP synthesis in mitochondria. If DNP were added, it would uncouple oxidative phosphorylation from respiration but wouldn't block respiration entirely.

Solution for Question 28:

Correct Option D - SAM

- SAM (S-adenosylmethionine). SAM is not a cofactor for dehydrogenase enzymes. It is involved in a variety of methylation reactions as a methyl group donor, playing a role in epigenetic regulation and other biochemical processes
- Dehydrogenase enzymes are involved in oxidation-reduction reactions, where they catalyze the transfer of hydrogen atoms or electrons from a substrate to an electron acceptor. These enzymes typically require specific cofactors to facilitate their activity.

Incorrect Options:

Option A - NADP (Nicotinamide adenine dinucleotide phosphate): NADP is a cofactor that is commonly involved in oxidation-reduction reactions, including those catalyzed by dehydrogenase enzymes. It serves as an electron carrier during these reactions.

Option B - FAD (Flavin adenine dinucleotide): FAD is another cofactor commonly used by dehydrogenase enzymes. It acts as a carrier of electrons and facilitates the transfer of hydrogen

atoms during oxidation-reduction reactions.

Option C - NAD (Nicotinamide adenine dinucleotide): NAD is yet another cofactor that plays a vital role in the activity of dehydrogenase enzymes. It serves as an electron carrier and participates in oxidation-reduction reactions.

Solution for Question 29:

Correct Option C - Pyruvate kinase:

- Pyruvate kinase is not directly involved in gluconeogenesis. It catalyzes the conversion of phosphoenolpyruvate (PEP) to pyruvate in the final step of glycolysis, which is the breakdown of glucose to produce energy.

Incorrect Option

Option A - Pyruvate carboxylase:

- Pyruvate carboxylase is an enzyme that plays a key role in gluconeogenesis. It catalyzes the conversion of pyruvate to oxaloacetate, which is an important intermediate in the gluconeogenesis pathway.

Option B - Phosphoenolpyruvate carboxylase:

- Phosphoenolpyruvate carboxylase is also involved in gluconeogenesis. It catalyzes the conversion of oxaloacetate to phosphoenolpyruvate (PEP), another crucial intermediate in the gluconeogenesis pathway.

Option D - Glucose-6-phosphatase:

- Glucose-6-phosphatase is an enzyme that is essential for gluconeogenesis. It catalyzes the hydrolysis of glucose-6-phosphate to produce glucose, which can be released into the bloodstream.
-

Solution for Question 30:

Correct Option D - GLUT 4

GLUT 4 is the glucose transporter that is primarily found in skeletal muscle cells, as well as adipose tissue. It plays a crucial role in the uptake of glucose from the bloodstream into these target tissues.

Incorrect Options :

Option A - GLUT 1 is a glucose transporter found in various tissues, including red blood cells, the brain, and endothelial cells. It facilitates the basal uptake of glucose in these cells.

Option B - GLUT 2 is mainly found in the liver, pancreas, and small intestine. It has a role in glucose uptake and release in these organs, such as the uptake of glucose from the intestine and the release of glucose from the liver.

Option C - GLUT 3 is predominantly expressed in neurons and is responsible for glucose transport across the blood-brain barrier. It ensures a constant supply of glucose to meet the high energy demands of the brain.

Solution for Question 31:

Correct Option A - Myophosphorylase:

- McArdle disease, or myophosphorylase deficiency, affects glycogen utilization for energy.
- Myophosphorylase converts glycogen to glucose-1-phosphate in muscle tissue. Insufficient enzyme activity leads to exercise intolerance due to impaired glycogen breakdown.
- Symptoms include muscle cramps, stiffness, and weakness during exertion, alleviated by rest.

Incorrect Options:

Option B - Hexokinase:

- Catalyzes glucose to glucose-6-phosphate conversion in glycolysis.
- Insufficiency affects glucose uptake, leading to exhaustion and weakness, not exercise intolerance or cramps.

Option C - Glucose-6-phosphatase:

- Converts glucose-6-phosphate to glucose in gluconeogenesis. Deficiency causes hypoglycemia, not exercise intolerance or cramps.

Option D - Hepatic glycogen phosphorylase:

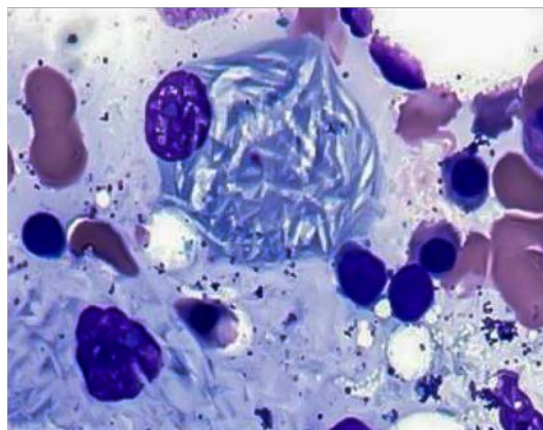
- Breaks down liver glycogen. Deficiency affects liver, not causing exercise intolerance or cramps.

Solution for Question 32:

Correct Option A - Beta- glucosidase:

- Gaucher disease, caused by a defect in β -glucosidase (glucocerebrosidase) , leads to the accumulation of glucosylceramide (glucocerebroside) within macrophages.
- These macrophages, engorged with glucosylceramide, are known as Gaucher cells.
- The accumulation affects the bone marrow and spleen, leading to anemia (due to impaired RBC production and increased destruction) and thrombocytopenia (due to impaired platelet production and increased destruction).
- Gaucher cells have a characteristic appearance, often described as having a "crumpled tissue paper" look. This is due to the accumulation of glucosylceramide in the cytoplasm of macrophages, forming fibril-like structures.

Incorrect Options:



Option B - Sphingomyelinase :

- Niemann-Pick disease, a condition similar to Gaucher disease, results from a deficiency in sphingomyelinase.
- Its bone marrow findings involve foamy histiocytes with lipid-filled cytoplasm, not the "crumpled tissue paper" appearance seen in Gaucher disease.

Option C - Hexosaminidase :

- Tay-Sachs disease: Tay-Sachs disease, characterized by hexosaminidase deficiency, doesn't typically cause anemia, thrombocytopenia, or bone pain.
- Its bone marrow findings may show an "onion skin" appearance, not "crumpled tissue paper" as in Gaucher disease.

Option D - Glucose 6-phosphatase :

- Von Gierke disease, due to glucose 6-phosphatase deficiency, may lead to hepatosplenomegaly and bone pain but not anemia or thrombocytopenia.
- Its bone marrow findings lack the "crumpled tissue paper" appearance seen in Gaucher disease.

Solution for Question 33:

Correct Option B - Gluconeogenesis and glycogenolysis:

- Gluconeogenesis is a metabolic process in the body in which glucose is produced from a non-carbohydrate substrate of carbon.
- Glycogenolysis is biochemical process in which glycogen is broken down into glucose and glucose-1-phosphate.
- During the fasting state, glucose is produced from glycogen in the liver which is called glycogenolysis and also via gluconeogenesis from glycerol and oxaloacetate.

Incorrect Options:

Option A - Glycolysis and gluconeogenesis: Glycolysis is a chain of reactions that produce energy from glucose by breaking into two molecules of three carbon molecule compounds called pyruvates. Hence, this is incorrect.

Option C - Glycogenesis and gluconeogenesis: Glycogenesis is formation of glycogen from primary carbohydrate that is stored in the muscle, liver cells of animals from the glucose molecule. Hence, this is incorrect.

Option D - Only gluconeogenesis: Gluconeogenesis and glycogenolysis are both pathways used in the production of glucose during the fast.

Solution for Question 34:

Correct Option A - Glucose-6-phosphatase:

- Von Gierke's disease, or glycogen storage disease type Ia, results from a deficiency of glucose-6-phosphatase, which is crucial for gluconeogenesis and glycogenolysis.
- Deficiency of glucose-6-phosphatase causes features like hypoglycemia, doll like facies, xanthomas, ketosis, hyperuricemia, lactic acidosis, hepatomegaly, and renomegaly.

Incorrect Options:

Option B - Branching enzyme: This enzyme is vital for glycogen synthesis, forming branching points for compact glucose storage. Mutations lead to glycogen storage disease type IV.

Option C - Debranching enzyme: Important for glycogen breakdown, deficiencies result in glycogen storage disease type III.

Option D - Muscle glycogen phosphorylase: Crucial for glycogen breakdown in muscles, mutations cause glycogen storage disease type V, known as McArdle disease, characterized by muscle weakness and cramping.

Solution for Question 35:

Correct Option C - Glycolysis:

- Glycolysis is the only metabolic pathway among the four that exclusively occurs in the cytoplasm of both eukaryotic and prokaryotic cells.
- Glycolysis is a central pathway of cellular metabolism and is the primary energy source for many cell types. The process involves the breakdown of glucose into pyruvate, which can then be further processed to produce ATP.

Incorrect Options:

Option A - Beta oxidation: Occurs in the mitochondria, breaking down fatty acids into acetyl-CoA for the TCA cycle.

Option B - TCA cycle: Takes place in the mitochondria, generating ATP from carbohydrates, fats, and proteins.

Option D - Urea cycle: Occurs in the liver, converting toxic ammonia into urea for excretion, involving both cytoplasmic and mitochondrial enzymes

Solution for Question 36:

Correct Option B - Substitution of glutamic acid by valine at the 6th position

- In sickle cell anemia, a single-point mutation in the beta-globin gene causes glutamic acid to be substituted with valine at the 6th position of the beta-globin chain.
- This mutation results in abnormal hemoglobin S (HbS) production instead of normal hemoglobin A (HbA). HbS molecules can polymerize under certain conditions, like low oxygen tension, forming rigid fibers inside red blood cells (RBCs).
- This polymerization leads to the characteristic sickle shape of RBCs, reducing flexibility and increasing fragility.

Incorrect Options:

Option A - Substitution of valine by glutamic acid at the 6th position: In sickle cell anemia, there is a substitution of glutamic acid, not valine, at the 6th position of the beta-globin chain of hemoglobin.

Option C - Substitution of valine by glutamic acid at the 5th position: The substitution in sickle cell anemia occurs at the 6th position, not the 5th position.

Option D - Substitution of glutamic acid by valine at the 5th position: The substitution in sickle cell anemia occurs at the 6th position, not the 5th position

Solution for Question 37:

Correct Option A - Calcium:

- Glycogen phosphorylase activation relies on calcium ions, essential for various physiological processes like muscle contraction and enzyme activation.
- Calcium binds to calmodulin, activating phosphorylase kinase. Phosphorylase kinase, in turn, phosphorylates and activates glycogen phosphorylase, facilitating glycogen breakdown into glucose-1-phosphate for cellular energy.

Incorrect Options:

Option B, C and D are incorrect

Solution for Question 38:

Correct Option B - Albumin:

- Fructosamine forms via non-enzymatic glycation of serum albumin, the predominant blood protein.
- Serum albumin's extended half-life makes fructosamine levels a valuable indicator of glycemic control over weeks.
- Fructosamine arises from the reaction between amino groups of lysine and other amino acids in serum albumin and reducing sugars like glucose or fructose.

Incorrect Options:

Options A, C and D are not involved in the formation of fructosamine.

Solution for Question 39:

Correct Option A - Lysosomal acid alpha glucosidase

- Pompe disease, also known as Glycogen Storage Disease Type II, is caused by a deficiency of the enzyme lysosomal acid alpha glucosidase (also called acid maltase or acid alpha-glucosidase). This enzyme is responsible for breaking down glycogen, a complex sugar molecule, into glucose within the lysosomes of cells.
- In Pompe disease, the deficiency or malfunction of lysosomal acid alpha glucosidase leads to the accumulation of glycogen within various tissues and organs, particularly in skeletal and cardiac muscles.

Incorrect Options

The other enzymes mentioned in the options are associated with different disorders:

Option B - β -hexosaminidase-A: Deficiency of this enzyme is seen in Tay-Sachs disease, a rare inherited disorder that affects the central nervous system.

Option C - β -galactocerebrosidase: Deficiency of this enzyme is seen in Krabbe disease, an autosomal recessive disorder characterized by the accumulation of a fatty substance called galactocerebroside.

Option D - β -glucocerebrosidase: Deficiency of this enzyme is seen in Gaucher disease, a lysosomal storage disorder characterized by the accumulation of glucocerebroside in cells.

In summary, Pompe disease is caused by a deficiency of lysosomal acid alpha glucosidase.

Solution for Question 40:

Correct Option A:

- Keratan sulfate is a type of GAG that is specifically responsible for maintaining corneal transparency.
- It is predominantly found within the cornea and helps to regulate the spacing and alignment of collagen fibrils, allowing light to pass through the cornea without significant scattering.
- Keratan sulfate interacts with collagen and other proteoglycans to form a specialized matrix that provides transparency and structural integrity to the cornea.

Incorrect Options:

Option B - Chondroitin sulfate : Chondroitin sulfate is a GAG found in various connective tissues, such as cartilage and tendons. While it plays an important role in these tissues, it is not primarily responsible for corneal transparency.

Option C - Heparin sulfate: Heparin sulfate is a GAG that is found in various tissues and acts as an anticoagulant. It has diverse functions in different organs but is not specifically associated with corneal transparency.

Option D - Hyaluronic acid: Hyaluronic acid is a GAG that is widely distributed in various tissues and is involved in multiple biological processes. While it is present in the cornea and contributes to its hydration and viscoelastic properties, it is not the primary GAG responsible for maintaining corneal transparency.

Solution for Question 41:

Correct Option A - Hepatic glycogenolysis refers to the breakdown of glycogen stored in the liver into glucose. During periods of fasting or between meals, when the body requires a steady supply of glucose to maintain blood glucose levels, the liver plays a crucial role in releasing glucose into the bloodstream through glycogenolysis. The glycogen stored in the liver is broken down into glucose-6-phosphate and subsequently converted into glucose, which is then released into the bloodstream.

Incorrect Options:

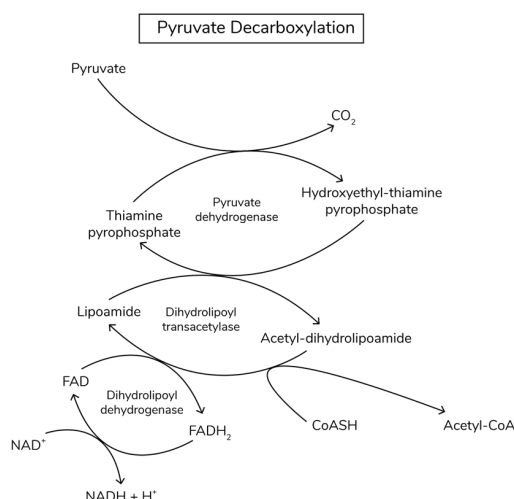
Option B - Hepatic gluconeogenesis: While gluconeogenesis can contribute to maintaining blood glucose levels during prolonged fasting or in certain metabolic states, it is not the primary source of blood glucose immediately after a meal.

Option C - Muscle glycogenolysis: While muscle glycogen can be broken down to provide glucose for energy within the muscle cells, it does not significantly contribute to blood glucose levels.

Option D - Dietary glucose: Dietary glucose refers to the glucose obtained directly from the ingestion of carbohydrates in the diet.

Solution for Question 42:

Correct Option A:



The conversion of pyruvate to acetyl CoA is a key step in cellular respiration and occurs in the mitochondria. It involves a multi-enzyme complex called the pyruvate dehydrogenase complex (PDC).

The enzymes marked as E1 and E2 in the diagram correspond to specific components of the PDC:

Incorrect Options:

Option B - Pyruvate carboxylase and dihydrolipoyl transacetylase: Pyruvate carboxylase is involved in a different metabolic pathway called gluconeogenesis, where it converts pyruvate into oxaloacetate.

Option C - Pyruvate dehydrogenase and dihydrolipoyl dehydrogenase: There is no enzyme called "dihydrolipoyl dehydrogenase" involved in the conversion of pyruvate to acetyl CoA. The correct enzyme involved in transferring electrons in the PDC is dihydrolipoyl dehydrogenase.

Option D - Pyruvate carboxylase and dihydrolipoyl dehydrogenase- This option is also incorrect

Solution for Question 43:

Correct Option - A Insulin

- When insulin levels are high, such as in the fed state, it signals that there is sufficient glucose available for energy metabolism, and there is no need to produce additional glucose.
- Insulin promotes glucose uptake by cells, increases glycogen synthesis, and inhibits the breakdown of glycogen (glycogenolysis) and gluconeogenesis.

Incorrect Options:

Option B - Glucagon, on the other hand, stimulates gluconeogenesis.

Option C - Lactate can be converted into pyruvate through a process called lactate conversion, and pyruvate can be further used in gluconeogenesis.

Option D - None of the above is incorrect

Solution for Question 44:

Correct Option C - Hexose monophosphate (HMP) pathway:

- Primaquine, used to treat malaria caused by *Plasmodium ovale* and *Plasmodium vivax*, disrupts the parasite's Hexose monophosphate (HMP) pathway, leading to the parasite's death.
- However, this pathway is vital for NADPH production, crucial for human red blood cell antioxidant defense. G6PD enzyme deficiency in the HMP pathway can cause hemolytic anemia, a known adverse effect of primaquine in deficient patients.

Incorrect Options:

Option A - Glycolysis :

- Glycolysis converts glucose to pyruvate, generating ATP and NADH. Glycolytic enzyme deficiencies are not commonly associated with hemolytic anemia.

Option B - Gluconeogenesis :

- Deficiencies in gluconeogenic enzymes can lead to hypoglycemia, but hemolytic anemia is rarely observed.

Option D - Luebering-Rapoport pathway :

- This pathway produces 2,3-BPG, aiding oxygen release from hemoglobin. Enzyme deficiencies in this pathway are not typically linked to hemolytic anemia.

Solution for Question 45:

Correct Option D - Glycine

The most common amino acid found in collagen is Glycine. Glycine makes up approximately one-third of the amino acids in the collagen protein. Its small size allows for the close packing of collagen helices, contributing to the structural stability of collagen.

Incorrect Options:

Option A - Proline: Proline plays a crucial role in the formation of collagen helices by introducing kinks in the protein structure.

Option B - Leucine: Leucine is a hydrophobic amino acid and is not typically abundant in collagen.

Option C - Arginine: Arginine residues in collagen contribute to the formation of intermolecular cross-links, which enhance the stability and tensile strength of collagen fibers.

Solution for Question 46:

Correct Option D - Glucose 6-phosphatase

- Glucose 6-phosphatase is responsible for the final step in gluconeogenesis and glycogenolysis, where glucose is released into the bloodstream.
- During times of decreased insulin and increased glucagon, such as fasting or low blood sugar, glucagon stimulates the activation of glucose 6-phosphatase, increasing glucose production.

Incorrect Options:

Option A - Glucokinase: The glucokinase activity decreases with a decrease in the insulin: glucagon ratio.

Option B - Hexokinase: The hexokinase activity decreases with a decrease in the insulin: glucagon ratio.

Option C - Phosphofructokinase: A decrease in the insulin: glucagon ratio decreases its activity. Phosphofructokinase activity is regulated by various factors, including ATP, ADP, and citrate concentration.

Solution for Question 47:

Correct Option C- Galactose-1-phosphate uridyl transferase

- Classical Galactosemia is due to the deficiency of the enzyme Galactose-1-phosphate uridyl transferase (GALPUT).
- It is a rare congenital autosomal recessive disorder. In this condition, galactose levels are increased in circulation and urine. The accumulated galactose-1-phosphate is converted into dulcitol by the enzyme aldose reductase. This accumulation of Dulcitol is associated with the development of cataracts., and impairment of organs such as the liver and kidney.

Incorrect Options:

Option A -Aldose reductase: This enzyme is responsible for the conversion of the already accumulated Galactose-1-phosphate in the cells into dulcitol. Its deficiency does not cause classic Galactosemia.

Option B - Galactokinase: This enzyme is responsible for the conversion of galactose to galactose-1-phosphate. The deficiency of galactokinase causes Galactosemia type II.

Option D - UDP galactose-4-epimerase: This enzyme is responsible for the conversion of UDP-Galactose to UDP-Glucose. The deficiency of this enzyme causes a rare condition called epimerase deficiency galactosemia.

Solution for Question 48:

Correct Option A - Glycolysis

- Glucagon is a hormone secreted by the alpha cells of islets of Langerhans in the pancreas.
- Glucagon does not play a role in Glycolysis which is the process of breakdown of glucose as it reduces blood glucose levels.
- On the contrary, glucagon increases the synthesis of glucose (gluconeogenesis) and the breakdown of glycogen in the liver to release glucose (Glycogenolysis). The levels of Glucagon in blood increase in fasting state.

Incorrect Options:

Option B - Gluconeogenesis : This is the synthesis of glucose from non-carbohydrate sources in the liver. Glucagon acts on the liver and promotes gluconeogenesis. This causes an increase in blood glucose levels.

Option C - Glycogenolysis: This is the process of breakdown of glycogen in the liver to produce glucose. This process is promoted by glucagon.

Option D - Ketogenesis: Glucagon promotes the breakdown of fatty acids to produce ketone bodies. This process is called Ketogenesis.

Solution for Question 49:

Correct Option A - Glucose-6-phosphate

- The given features are consistent with the diagnosis of VON GIERKE DISEASE. In this disease, there is a deficiency of the enzyme glucose 6 phosphatase. It is the most common glycogen storage disorder.
- Features include: Severe Hypoglycemia Doll-like faces Xanthomas Ketosis Lactate acidosis Hyperuricemia
- Severe Hypoglycemia
- Doll-like faces
- Xanthomas
- Ketosis
- Lactate acidosis
- Hyperuricemia
- Severe Hypoglycemia
- Doll-like faces

- Xanthomas
- Ketosis
- Lactate acidosis
- Hyperuricemia

Incorrect Options:

Option B - Glucose-1-phosphate reacts in the presence of phosphorylase with aldoses and ketoses to yield disaccharides and ultimately produce glycogen and phosphoric acid. Also called Cori ester.

Option C - Glucose-1,6-diphosphate is the central regulator in glycogen metabolism, glycolysis, amino sugar formation as well as bacterial cell wall and capsule formation.

Option D - Glycosaminoglycans play an important role in cell signaling and development, angiogenesis, anti-coagulation, tumor progression, axonal growth, and metastasis.

Solution for Question 50:

Correct Option A:

- Hexose monophosphate shunt pathway is the primary source of reducing equivalents for liver fatty acid synthesis
- It does not generate ATP but has two major functions:
 - The formation of NADPH for the synthesis of fatty acids and steroids
 - The synthesis of ribose for nucleotide and nucleic acid formation

Incorrect Options:

Option B - Glycolysis: Glycolysis is the cytosolic pathway of all mammalian cells for the metabolism of glucose (or glycogen) to pyruvate and lactate.

Option C - Tricarboxylic acid cycle: The citric acid cycle is the final common pathway for the aerobic oxidation of carbohydrates, lipids, and proteins because glucose, fatty acids, and most amino acids are metabolized to acetyl-CoA or intermediates of the cycle.

Option D - Uronic acid pathway: In the liver, the uronic acid pathway catalyzes the conversion of glucose to glucuronic acid, ascorbic acid, and pentoses.

Solution for Question 51:

Correct Options - Acetyl coenzyme A

- Substrates of gluconeogenesis are mostly 3C compounds including any TCA cycle intermediate. Acetyl CoA is not a TCA cycle intermediate, it is a part of a link reaction so it cannot act as a substrate for gluconeogenesis.

Incorrect Options:

Option B, C, and D: Glycerol, Alanine, and Lactate are all amino acids that participate in the Krebs cycle and can generate glucose.

Solution for Question 52:

Correct Option B - Glucose

Since ketone bodies and free fatty acids are utilized via beta-oxidation which occurs in mitochondria, RBCs cannot utilize them. Therefore, RBC uses glucose as a substrate for energy production in the fed, fasting, and starvation state.

Incorrect Options:

Option A - Ketone body: They are used as a substrate for energy production during starvation phase by vital organs brain and heart primarily.

Option C - Amino acid: It is used by liver as a fuel in starvation state. It is the major source of fuel for the body in a starvation state.

Option D - Free fatty acid: It is used as a fuel in fasting state by heart, liver, muscle and adipose tissue.

Solution for Question 53:

Correct Option A - 1 only

Von Gierke disease is a glycogen storage disease . The enzyme deficient is glucose 6 phosphatase present in the liver.

Incorrect Choices:

Option B, C, and D

Solution for Question 54:

Correct Option B - Hypoxanthine -guanine phosphoribosyl transferase

- Explanation: Hypoxanthine phosphoribosyltransferase (HPRT) deficiency is an X-linked disorder characterized by uric acid overproduction and variable neurologic impairment. The total deficiency of HPRT is diagnostic of Lesch-Nyhan syndrome manifested by choreoathetosis , spasticity , mental retardation , and self-injurious behaviour .



Incorrect Options :

Option A: Purine nucleoside phosphorylase deficiency is not related to Lesch-Nyhan syndrome. Purine nucleoside phosphorylase deficiency is inherited in an autosomal recessive pattern.

Option C : Adenine phosphoribosyl transferase is inherited in an autosomal recessive manner . Deficiency is characterised by excessive production and renal excretion of 2,8-dihydroxyadenine (DHA), which leads to kidney stone formation and crystal-induced kidney damage (i.e., DHA crystal nephropathy).

Option D: Adenosine deaminase is inherited in an autosomal recessive pattern . Adenosine deaminase (ADA) deficiency damages the immune system and causes severe combined immunodeficiency (SCID) .

Solution for Question 55:

Correct Option C - Galactose-1-phosphate uridyl transferase

- The question presented is a case of galactosemia. This occurs due to the deficiency of galactose-1-phosphate uridyl transferase .
- Due to the deficiency of galactose-1-phosphate uridyl transferase, there is an excess accumulation of Galactose-1-Phosphate in the liver and brain that causes jaundice, vomiting, hepatomegaly, and mental retardation. Galactosemia is seen mainly in the first week of birth with the symptoms of jaundice.

Incorrect Options:

Option A - Aldose reductase: Deficiency of aldose reductase is not associated with the symptoms described in the question, such as refusal of breast milk, vomiting, severe jaundice, positive Benedict's test for urine, and low blood glucose levels.

Option B - Galactokinase: Galactokinase deficiency typically presents with symptoms like the presence of galactose in the blood and urine, cataracts, and sometimes intellectual disability.

Option D - UDP galactose-4-epimerase: UDP galactose-4-epimerase deficiency typically presents with symptoms similar to classic galactosemia, including failure to thrive, hepatomegaly (enlarged liver), jaundice, and developmental delay.

Solution for Question 56:

Correct Option B - Tandem mass spectrometry

- When it comes to screening for metabolic disorders, tandem mass spectrometry (TMS) is considered the best investigation method.
- Tandem mass spectrometry involves ionizing and fragmenting molecules in a controlled manner, followed by measuring the mass-to-charge ratios of the resulting ions. By analyzing the patterns of metabolite fragmentation, TMS can identify and quantify specific metabolites associated with various metabolic disorders.

Incorrect Options:

Option A - Western blot: Western blot is a technique used to detect and analyze specific proteins in a sample.

Option C - PCR (Polymerase Chain Reaction): PCR is a molecular biology technique used to amplify specific DNA sequences. PCR is more commonly used for targeted testing when a specific genetic mutation is suspected.

Option D - Gel electrophoresis: Gel electrophoresis is a technique used to separate molecules based on their size and charge. It is commonly used in protein and nucleic acid analysis.

Lipid Chemistry

1. A 4-month-old infant presents developmental delay. MRI shows demyelination in the central nervous system. Which of the following is the most likely diagnosis, and what enzyme deficiency causes this condition?

- A. Niemann-Pick disease, Sphingomyelinase
 - B. Krabbe disease, β -Galactosidase
 - C. Farber disease, Ceramidase
 - D. Gaucher disease, β -Glucosidase
-

2. Which enzyme in snake venom hydrolyzes only sn-2 acyl bond of phospholipids?

- A. Phospholipase A2
 - B. Phospholipase B
 - C. Phospholipase C
 - D. Phospholipase D
-

3. Which of the following fatty acids is needed for humans from diet as humans cannot create double bonds beyond the 9th carbon from the carboxyl end of a fatty acid chain?

- A. A) Alpha-linolenic acid
 - B. B) Palmitic acid
 - C. C) Oleic acid
 - D. D) Stearic acid
-

4. Which of the following lipids is characterized by its amphipathic nature, allowing it to form micelles in aqueous solutions?

- A. Triacylglycerols
 - B. Cholesteryl esters
 - C. Phospholipids
 - D. Nonpolar lipids
-

5. Which of the following pairs correctly associates a disease with its corresponding enzyme deficiency?

- A. Tay-Sachs disease - α -Galactosidase
 - B. Gaucher disease - β -Glucosidase
 - C. Farber disease - Sphingomyelinase
 - D. Fabry disease - Ceramidase
-

6. A 9-year-old boy presents with episodic pain in his extremities, skin finding as shown in the image, and progressive renal dysfunction. A biopsy of the skin reveals the accumulation of ceramide trihexoside. An α -galactosidase deficiency is suspected. What is the diagnosis?



- A. Tay-Sachs disease
- B. Fabry disease
- C. Krabbe disease
- D. Gaucher disease

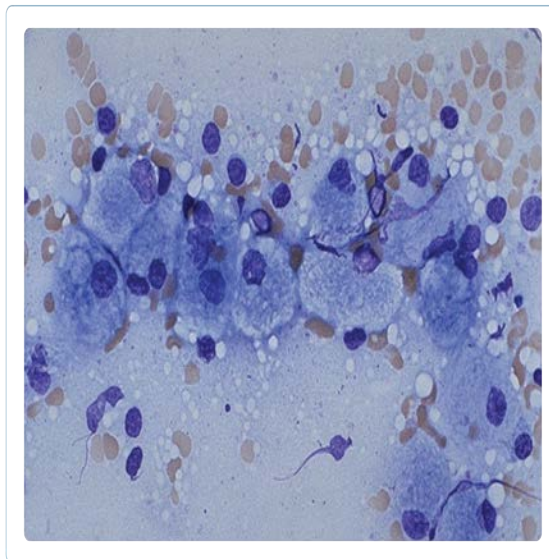
7. A 6-month-old infant presents with a hoarse cry, painful and swollen joints, and skin nodules. The child also shows signs of developmental delay and mental retardation. Histological analysis reveals the accumulation of a certain substance in the tissues, which confirms the diagnosis of Farber disease. Which enzyme deficiency is responsible for this condition?

- A. Hexosaminidase A
 - B. Arylsulfatase A
 - C. Ceramidase
 - D. Sphingomyelinase
-

8. A 6-month-old infant presents with hepatosplenomegaly, progressive neurodegeneration, and a history of frequent respiratory infections. A biopsy shows lipid-laden macrophages and a genetic test confirms the diagnosis. Which of the following enzyme deficiencies is most likely responsible for this condition?

- A. Hexosaminidase A
 - B. Arylsulfatase A
 - C. Sphingomyelinase
 - D. β -Glucosidase
-

9. A patient presents with bone pain, fatigue and hepatosplenomegaly. Blood tests reveal anemia and thrombocytopenia. Bone marrow biopsy shows macrophages with a characteristic "wrinkled paper" appearance. Which substrate accumulates in Gaucher's disease due to the deficient enzyme activity?



- A. Sphingomyelin
- B. Glucosylceramide
- C. Ganglioside
- D. Ceramide

10. Which of the following correctly matches the type of lipid with its description?

- A. Fats - Liquid at room temperature, often containing polyunsaturated fatty acids.
 - B. Phospholipids - Contain glycerol and are the main lipid constituents of cell membranes.
 - C. Glycolipids - Esters of fatty acids that never contain sphingosine.
 - D. Sphingophospholipids - Cerebrosides is a subtype
-

11. A 6-month-old infant has developmental delay, muscle weakness, and on eye examination, a cherry-red spot on the macula is noted. What is the most likely diagnosis, and which enzyme deficiency is responsible?



- A. Niemann-Pick disease, Sphingomyelinase
- B. Gaucher disease, Glucocerebrosidase
- C. Tay-Sachs disease, Hexosaminidase A
- D. Sandhoff disease, Beta-galactosidase

12. Which enzyme deficiency is responsible for GM1 gangliosidosis?

- A. Beta-glucosidase
 - B. Beta-galactosidase
 - C. Alpha-galactosidase
 - D. Hexosaminidase
-

13. Which component is present in gangliosides but not in cerebroside or globosides?

- A. Monosaccharides
 - B. Ceramide
 - C. Fatty acid
 - D. Oligosaccharides
-

14. Which of the following are the correct components of sphingomyelin?

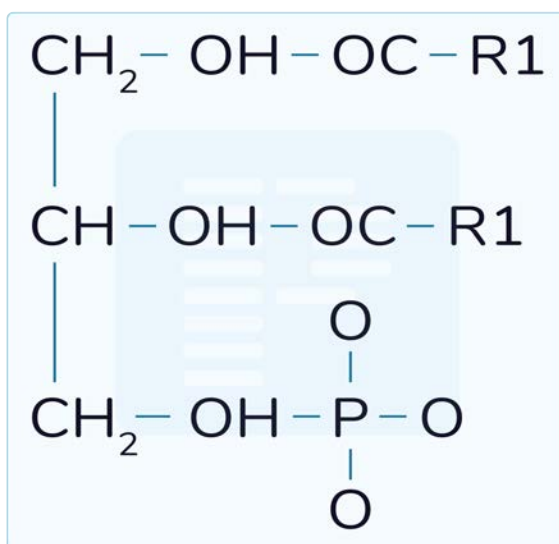
- A. Sphingosine + Fatty acid + Choline
 - B. Ceramide + Phosphate + Choline
 - C. Sphingosine + Phosphate + Choline
 - D. Ceramide + Phosphate + Fatty acids
-

15. A 30-year-old woman presents with a history of four consecutive miscarriages over the past three years. After a thorough evaluation, a diagnosis is made. Which

of the following phospholipids is specifically antigenic in this disease?

- A. Phosphatidylserine
 - B. Phosphatidylinositol
 - C. Cardiolipin
 - D. Phosphatidylethanolamine
-

16. What compound is formed when choline is added to the following structure?



- A. Lecithin
 - B. Cephalin
 - C. Cardiolipin
 - D. Inositol
-

17. A premature baby presents with infant respiratory distress syndrome. The neonatologist says the neonate is yet to produce a type of phospholipid

adequately. All of the following belong to the same class except?

- A. Cephalin
 - B. Cardiolipin
 - C. Lecithin
 - D. Cerebrosides
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	3
Question 8	3
Question 9	2
Question 10	2
Question 11	3
Question 12	2
Question 13	4
Question 14	2
Question 15	3
Question 16	1
Question 17	4

Solution for Question 1:

Correct Answer: B) Krabbe disease, β -Galactosidase

Explanation:

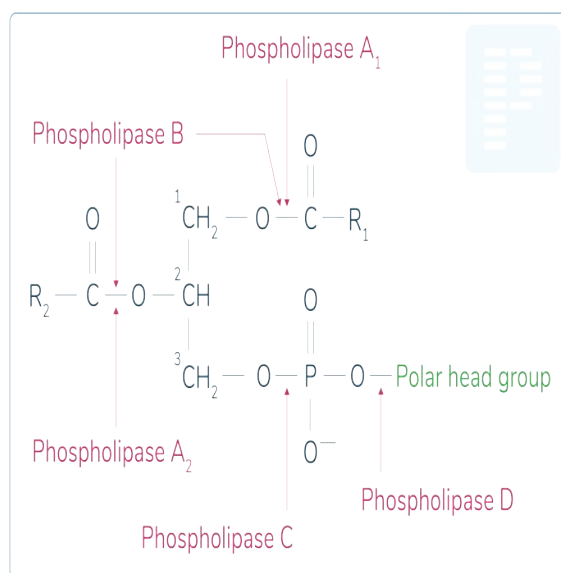
Krabbe disease occurs due to defects in β -Galactosidase, causing demyelination and mental retardation.

Disease	Enzyme deficiency	Features
Tay-Sachs disease	Hexosaminidase A,	Mental retardation Blindness Muscular weakness Cherry red spots in macula
Fabry disease	α -Galactosidase	Skin rash(angiokeratoma) Kidney failure (full symptoms only in males - X-linked recessive)
Krabbe disease (Option B)	β -Galactosidase	Mental retardation Near absence of myelin Predominant neurological manifestation and cranial nerves causing deafness and blindness.
Gaucher disease (Option D ruled out)	β -Glucosidase	Enlarged liver and spleen Erosion of long bones Mental retardation in infants Anemia Thrombocytopenia
Niemann-Pick disease (Option A ruled out)	Sphingomyelinase	Enlarged liver and spleen Mental retardation Fatal in early life Cherry red spot in macula
Farber disease (Option C ruled out)	Ceramidase	Hoarseness Dermatitis Skeletal deformation Mental retardation Fatal in early life

Solution for Question 2:

Correct Answer: A) Phospholipase A2

Explanation: Phospholipase A2 acts on the 2nd carbon atom causing hydrolysis of the ester bond in phospholipids leading to cell membrane disruption and inflammation.



Phospholipase A₂: (Option A)

- Acts at the 2nd carbon atom.
- Hydrolysis of the ester bond in phospholipids
- Release of arachidonic acid.
- Causes cell membrane disruption.

Phospholipase B (Option B) has both PLA₁ and PLA₂ activities. Hence, it can hydrolyze both the sn-1 and sn-2 acyl bonds of phospholipids.

Phospholipase C (Option C) hydrolyzes the phosphodiester bond of phospholipids, particularly phosphatidylinositol.

Phospholipase D (Option D) hydrolyzes the phosphodiester bond at the phosphate group.

Solution for Question 3:

Correct Answer: A) Alpha-linolenic acid

Explanation: Alpha-linolenic acid is an essential fatty acid. It cannot be synthesized by the human body and must be obtained through the diet

Essential fatty acids	Non-essential fatty acids
-----------------------	---------------------------

Cannot be synthesized by the human body and must be obtained through the diet. Humans cannot create double bonds beyond the 9th carbon atom from the carboxyl end of a fatty acid chain. Linoleic Acid (LA): An omega-6 fatty acid. Alpha-linolenic acid (ALA) : An omega-3 fatty acid. (Option A)

Can be synthesized by the human body from other dietary fats Palmitic acid (Option B ruled out) Oleic acid (Option C ruled out) Stearic acid (Option D ruled out)

Solution for Question 4:

Correct Answer: C) Phospholipids

Explanation:

Phospholipids are amphipathic lipids with both hydrophobic and hydrophilic properties. They have a polar head group (hydrophilic) and nonpolar tails (hydrophobic), enabling them to form micelles when in aqueous solutions. This property is crucial for various biological functions, including fat digestion and membrane formation.

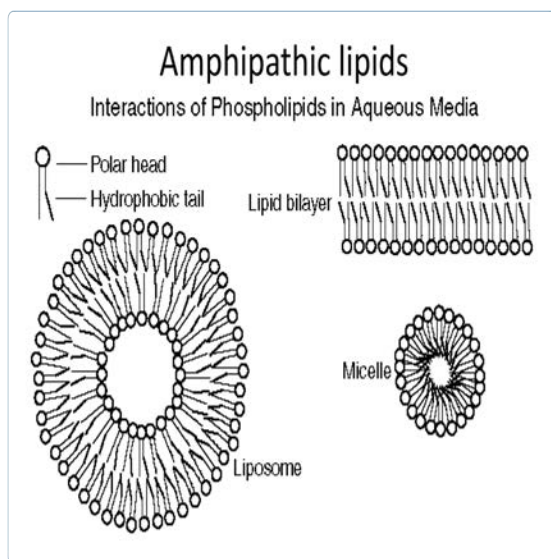
Lipids Categorized by Solubility:

1. Nonpolar Lipids: (Option D ruled out)

- Characteristics: Hydrophobic, insoluble in water, dominated by nonpolar hydrocarbon groups.
- Examples: Triacylglycerols (Triglycerides): Esters of glycerol with three fatty acid chains; primary fat storage in adipose tissue. (Option A ruled out) Cholesteryl Esters: Cholesterol esterified to a fatty acid; used for cholesterol storage and transport. (Option B ruled out)
- Triacylglycerols (Triglycerides): Esters of glycerol with three fatty acid chains; primary fat storage in adipose tissue. (Option A ruled out)
- Cholesteryl Esters: Cholesterol esterified to a fatty acid; used for cholesterol storage and transport. (Option B ruled out)
- Triacylglycerols (Triglycerides): Esters of glycerol with three fatty acid chains; primary fat storage in adipose tissue. (Option A ruled out)
- Cholesteryl Esters: Cholesterol esterified to a fatty acid; used for cholesterol storage and transport. (Option B ruled out)

2. Amphipathic Lipids:

- Characteristics: Both hydrophobic and hydrophilic; contain a polar head group and a nonpolar tail.
- Functions and Structures: Membranes: Bilayer structure with hydrophobic tails inward and hydrophilic heads outward Micelles: Spherical structures with hydrophobic tails in the center and hydrophilic heads on the surface; aid in fat digestion. Liposomes: Spherical vesicles with lipid bilayers enclosing an aqueous compartment Emulsions: Larger particles of nonpolar lipids dispersed in water, stabilized by amphipathic lipids; prevent coalescence.
- Membranes: Bilayer structure with hydrophobic tails inward and hydrophilic heads outward
- Micelles: Spherical structures with hydrophobic tails in the center and hydrophilic heads on the surface; aid in fat digestion.
- Liposomes: Spherical vesicles with lipid bilayers enclosing an aqueous compartment
- Emulsions: Larger particles of nonpolar lipids dispersed in water, stabilized by amphipathic lipids; prevent coalescence.
- Examples: Phospholipids (Option C) Sphingolipids Cholesterol
- Phospholipids (Option C)
- Sphingolipids
- Cholesterol
- Membranes: Bilayer structure with hydrophobic tails inward and hydrophilic heads outward
- Micelles: Spherical structures with hydrophobic tails in the center and hydrophilic heads on the surface; aid in fat digestion.
- Liposomes: Spherical vesicles with lipid bilayers enclosing an aqueous compartment
- Emulsions: Larger particles of nonpolar lipids dispersed in water, stabilized by amphipathic lipids; prevent coalescence.
- Phospholipids (Option C)
- Sphingolipids
- Cholesterol



Solution for Question 5:

Correct Answer: B) Gaucher disease - β -Glucosidase

Explanation:

Gaucher disease is caused by a deficiency of β -Glucosidase, leading to the accumulation of glucosylceramide.

Disease	Enzyme Deficiency
Tay-Sachs disease	Hexosaminidase A, S
Fabry disease	α -Galactosidase
Metachromatic leukodystrophy	Arylsulfatase A
Krabbe disease	β -Galactosidase
Gaucher disease	β -Glucosidase
Niemann-Pick disease types A, B	Sphingomyelinase
Farber disease	Ceramidase

Solution for Question 6:

Correct Answer: B) Fabry disease

Explanation:

Fabry disease is due to a deficiency of α -galactosidase, leading to the accumulation of ceramide trihexoside, resulting in symptoms such as episodic pain, angiokeratomas (as shown in the image), and renal dysfunction.

Disease	Enzyme Deficiency	Lipid Accumulating	Clinical Symptoms
Tay-Sachs disease (Option A ruled out)	Hexosaminidase A, S	Cer—Glc—Gal(NeuAc) GalNAc GM2 Ganglioside	Mental retardation, blindness, muscular weakness Cherry red spot
Fabry disease (Option B)	α -Galactosidase	Cer—Glc—Gal—Gal Globotriaosylceramide	Skin rash (Angiokeratoma), kidney failure (symptoms manifest fully only in males; X-linked recessive)
Metachromatic leukodystrophy	Arylsulfatase A	Cer—Gal—OSO ₃ 3-Sulfogalactosylceramide	Mental retardation and psychological disturbances in adults; demyelination
Krabbe disease (Option C ruled out)	β -Galactosidase	Cer—Gal Galactosylceramide	Mental retardation; myelin almost absent
Gaucher disease (Option D ruled out)	β -Glucosidase	Cer—Glc Glucosylceramide	Enlarged liver and spleen, erosion of long bones, mental retardation in infants
Niemann-Pick disease types A, B	Sphingomyelinase	Cer—P—choline Sphingomyelin	Enlarged liver and spleen, mental retardation; fatal in early life Cherry red spot
Farber disease	Ceramidase	Acyl—Sphingosine Ceramide	Hoarseness, dermatitis, skeletal deformation, mental retardation; fatal in early life

Abbreviations: Cer, ceramide; Gal, galactose; Glc, glucose; NeuAc, N-acetylneuraminic acid; , site of deficient enzyme reaction.

arly life

Solution for Question 7:

Correct Answer: C) Ceramidase

Explanation:

Farber disease is caused by a deficiency of ceramidase, leading to the accumulation of ceramide and presenting with a hoarse cry, painful and swollen joints, skin nodules, and mental retardation.

Disease	Enzyme Deficiency	Lipid Accumulating	Clinical Symptoms
Tay-Sachs disease	Hexosaminidase A, S (Option A ruled out)	Cer—Glc—Gal(NeuAc) GalNAc GM2 Ganglioside	Mental retardation, blindness, muscular weakness
Fabry disease	α -Galactosidase	Cer—Glc—Gal—Gal Globotriaosylceramide	Skin rash, kidney failure (symptoms manifest fully only in males; X-linked recessive)
Metachromatic leukodystrophy	Arylsulfatase A (Option B ruled out)	Cer—Gal—OSO ₃ 3-Sulfogalactosylceramide	Mental retardation and psychological disturbances in adults; demyelination
Krabbe disease	β -Galactosidase	Cer—Gal Galactosylceramide	Mental retardation; myelin almost absent
Gaucher disease	β -Glucosidase	Cer—Glc Glucosylceramide	Enlarged liver and spleen, erosion of long bones, mental retardation in infants
Niemann-Pick disease types A, B	Sphingomyelinase (Option D ruled out)	Cer—P—choline Sphingomyelin	Enlarged liver and spleen, mental retardation; fatal in early life
Farber disease	Ceramidase (Option C)	Acyl—Sphingosine Ceramide	Hoarseness, dermatitis, skeletal deformation, mental retardation; fatal in early life

Abbreviations: Cer, ceramide; Gal, galactose; Glc, glucose; NeuAc, N-acetylneuraminic acid; , site of deficient enzyme reaction.

Solution for Question 8:

Correct Answer: C) Sphingomyelinase

Explanation:

Sphingomyelinase deficiency is responsible for Niemann-Pick disease types A and B, which present with hepatosplenomegaly, and neurodegeneration, biopsy shows lipid-laden macrophages (foam cells) and are often fatal in early life.

Disease	Enzyme Deficiency	Lipid Accumulating	Clinical Symptoms
Tay-Sachs disease	Hexosaminidase A, S (Option A ruled out)	Cer—Glc—Gal(NeuAc) GalNAc GM2 Ganglioside	Mental retardation, blindness, muscular weakness
Fabry disease	α -Galactosidase	Cer—Glc—Gal—Gal Globotriaosylceramide	Skin rash, kidney failure (symptoms manifest fully only in males; X-linked recessive)
Metachromatic leukodystrophy	Arylsulfatase A (Option B ruled out)	Cer—Gal—OSO ₃ 3-Sulfogalactosylceramide	Mental retardation and psychological disturbances in adults; demyelination
Krabbe disease	β -Galactosidase	Cer—Gal Galactosylceramide	Mental retardation; myelin almost absent
Gaucher disease	β -Glucosidase (Option D ruled out)	Cer—Glc Glucosylceramide	Enlarged liver and spleen, erosion of long bones, mental retardation in infants
Niemann-Pick disease types A, B	Sphingomyelinase (Option C)	Cer—P—choline Sphingomyelin	Enlarged liver and spleen, mental retardation; fatal in early life
Farber disease	Ceramidase	Acyl—Sphingosine Ceramide	Hoarseness, dermatitis, skeletal deformation, mental retardation; fatal in early life

Abbreviations: Cer, ceramide; Gal, galactose; Glc, glucose; NeuAc, N-acetylneuraminic acid; , site of deficient enzyme reaction.

Solution for Question 9:

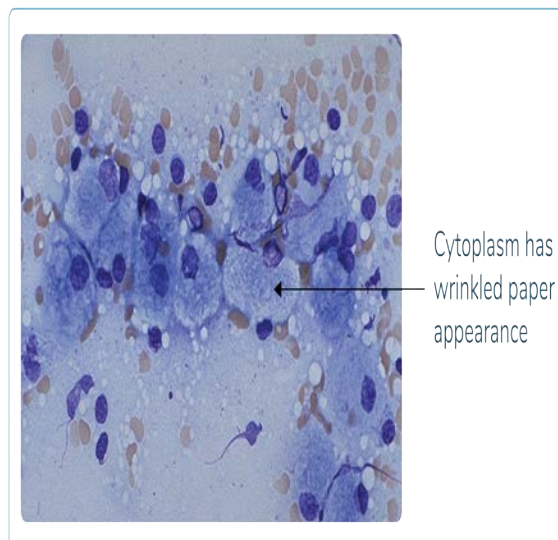
Correct Answer: B) Glucosylceramide

Explanation:

Gaucher's disease is characterized by the accumulation of glucosylceramide in RBCs and platelet membranes due to a deficiency in the enzyme beta-glucosidase, leading to anemia, hepatosplenomegaly, and thrombocytopenia. Bone marrow biopsy shows macrophages with a characteristic "wrinkled paper" appearance.

Gaucher's disease:

- Lysosomal storage disease
- Defective enzyme: β -Glucosidase
- Defect in metabolism: Accumulation of cerebroside (Glucosylceramide) in RBCs and platelet membranes. (Option B)
- Pathophysiology: Abnormal RBCs and platelets are engulfed by macrophages leading to accumulation inside macrophages causing wrinkled paper appearance.



Disease	Enzyme deficiency	Substance accumulation	Features
Tay-Sachs disease	Hexosaminidase A, S	Ganglioside (Option C)	Mental retardation Blindness Muscular weakness Cherry red spots in macula
Niemann-Pick disease	Sphingomyelinase	Sphingomyelin (Option A)	Enlarged liver and spleen Mental retardation Fatal in early life
Farber disease	Ceramidase	Ceramide (Option D)	Hoarseness Dermatitis Skeletal deformation Mental retardation Fatal in early life

Solution for Question 10:

Correct Answer: B) Phospholipids - Contain glycerol or sphingosine

Explanation:

Classification of lipids		
1.Simple Lipids Esters of fatty acids with various alcohols. Further classified into fats, oils & waxes		
Fats	Esters of fatty acids with glycerol; solid at room temperature. (Option A ruled out)	Triacylglycerol with saturated fatty acids.
Oils	Fats in liquid state at room temperature.	Triacylglycerols with polyunsaturated fatty acids.
Waxes	Esters of fatty acids with higher molecular weight monohydric alcohols.	
2.Complex Lipids Esters of fatty acids with alcohol and other groups. Further classified based on additional groups present as Phospholipids, glycolipids and other complex lipids		

Phospholipids: Contain phosphoric acid; Main lipid constituents of membranes. (Option B)	Glycerophospholipids Alcohol is glycerol.	Phosphatidylcholine, phosphatidylethanolamine.
Sphingophospholipids Alcohol is sphingosine	Sphingomyelin, found in myelin sheaths.	
Glycolipids: (glycosphingolipids) Contain fatty acid, sphingosine, and carbohydrate; Involved in cell adhesion and recognition (Option C ruled out)	Cerebrosides (Option D ruled out)	Galactosylceramide, a key component of myelin.
Globosides	Cell adhesion and Pathogen identification	
Gangliosides	Highly concentrated in nerve tissues and function in cell-cell recognition, communication, and as receptors for hormones and bacterial toxins.	
Other Complex Lipids	Includes sulfolipids, amino lipids, and lipoproteins.	
3.Precursor/Derived Lipids	Formed from hydrolysis of simple and complex lipids; Also act as precursors for lipid formation.	Fatty acids, cholesterol, fatty aldehydes, ketone bodies, hydrocarbons, lipid-soluble vitamins, hormones.

Solution for Question 11:

Correct Answer: C) Tay-Sachs disease, Hexosaminidase A

Explanation:

In this scenario, the clinical features and fundoscopic image show cherry red spot in the macula are consistent with the diagnosis of Tay-Sachs disease. It is an inborn error of metabolism due to the failure of degradation of gangliosides, which is caused by a deficiency in the enzyme hexosaminidase A.

Lipid Storage Diseases (Sphingolipidoses):

- They are a group of lysosomal storage diseases.

Disease	Enzyme deficiency	Features
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Tay-Sachs disease (Option C)	Hexosaminidase A	Mental retardation Blindness Muscular weakness Cherry red spots in the macula
Fabry disease	α -Galactosidase	Skin rash Kidney failure (full symptoms only in males X-linked recessive)
Krabbe disease	β -Galactosidase	Mental retardation Near absence of myelin Predominant neurological manifestation and cranial nerves causing deafness and blindness.
Gaucher disease (Option B ruled out)	β -Glucosidase	Enlarged liver and spleen Erosion of long bones Mental retardation in infants Anemia Thrombocytopenia
Niemann-Pick disease (Option A ruled out)	Sphingomyelinase	Enlarged liver and spleen Mental retardation Fatal in early life Cherry red spot in macula
Farber disease	Ceramidase	Hoarseness Dermatitis Skeletal deformation Mental retardation Fatal in early life

Sandhoff disease (Option D):

- Lysosomal storage disease.
- Deficient enzyme: Hexosaminidase B
- Accumulation of GM2 ganglioside and goboside

Solution for Question 12:

Correct Answer: B) Beta-galactosidase

Explanation:

Beta-galactosidase removes galactose in GM 1 ganglioside and converts to GM 2 ganglioside . In the absence of the enzyme, it causes GM 1 gangliosidosis.

Disease	Enzyme deficiency	Features
---------	-------------------	----------

Tay-Sachs disease	Hexosaminidase A (Option D ruled out)	Mental retardation Blindness Muscular weakness Cherry red spots in macula
Fabry disease	α -Galactosidase (Option C ruled out)	Angiokeratomas (reddish purple spots) Kidney failure (full symptoms only in males X-linked recessive)
GM1 gangliosidosis	β -Galactosidase (Option B)	Mental retardation Organomegaly Skeletal deformities
Gaucher disease	β -Glucosidase (Option A ruled out)	Enlarged liver and spleen Erosion of long bones Mental retardation in infants Anemia Thrombocytopenia
Niemann-Pick disease	Sphingomyelinase	Enlarged liver and spleen Mental retardation Fatal in early life

Solution for Question 13:

Correct Answer: D) Oligosaccharides

Explanation:

Gangliosides contain oligosaccharides as part of their structure, whereas cerebrosides contain only a single monosaccharide attached to ceramide.

Ganglioside is a glycolipid, which is made of ceramide, oligosaccharide with NANA (N-acetyl Neuraminic Acid).

Glycolipids:

- Also known as glycosphingolipids as alcohol in glycolipids are sphingosine.
- Glycosphingolipids are a combination of (Option B ruled out) ceramide with one or more sugar residues.
- Examples: cerebroside, globoside, and ganglioside.

Glycolipids	Composition
Cerebroside	Ceramide + monosaccharide (glucose or galactose) (Option A ruled out)

Globoside	Ceramide + oligosaccharide (glucose and galactose) (Option D)
Ganglioside	Ceramide + oligosaccharide + NANA (N-acetyl Neuraminic Acid)

Ceramide:

- Ceramide is synthesized in the endoplasmic reticulum from the amino acid serine.
- Ceramide = Fatty acid (Option C ruled out) + Sphingosine

Importance of glycosphingolipids:

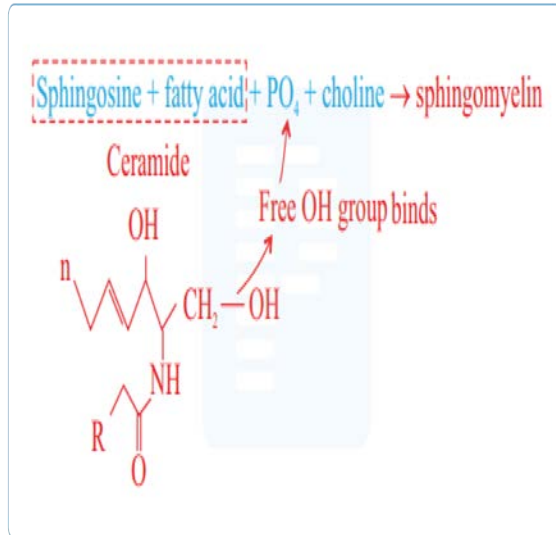
- Constituents of plasma membranes
- Cell adhesion
- Cell recognition
- Signal transduction
- Some are antigens (ABO blood group substances)

Solution for Question 14:

Correct Answer: B) Ceramide + Phosphate + Choline

Explanation:

Sphingosine and fatty acids combine to form Ceramide . The phosphate group attaches to the free OH group in ceramide to form its derivative. Choline when added to this derivative forms Sphingomyelin



- Sphingomyelin is a type of lipid classified as a phospholipid and is an important component of cell membranes. Abundant in the myelin sheath that surrounds nerve fibers. Found in the outer leaflet of the cell membrane lipid bilayer and is a key component of lipid rafts. Plays an important role in cell signaling and apoptosis. Synthesis occurs primarily in the Golgi apparatus,
 - Abundant in the myelin sheath that surrounds nerve fibers.
 - Found in the outer leaflet of the cell membrane lipid bilayer and is a key component of lipid rafts.
 - Plays an important role in cell signaling and apoptosis.
 - Synthesis occurs primarily in the Golgi apparatus,
 - Abundant in the myelin sheath that surrounds nerve fibers.
 - Found in the outer leaflet of the cell membrane lipid bilayer and is a key component of lipid rafts.
 - Plays an important role in cell signaling and apoptosis.
 - Synthesis occurs primarily in the Golgi apparatus,
- (Option A, C, D) are all wrong components of sphingomyelin

Solution for Question 15:

Correct Answer: C) Cardiolipin

Explanation:

- The condition described is Antiphospholipid Syndrome (APS), a disorder characterized by recurrent miscarriages, thrombosis, and the presence of antiphospholipid antibodies.
- Cardiolipin is a phospholipid that becomes antigenic in Antiphospholipid Syndrome, leading to the production of anti-cardiolipin antibodies.
- Other phospholipids (Phosphatidylserine, Phosphatidylinositol, Phosphatidylethanolamine) are not antigenic

Phospholipids	Location	Function	Synthesis	Significance
Phosphatidylserine (Option A ruled out)	Inner leaflet of the plasma membrane	Involved in apoptosis . It flips to the outer leaflet of the plasma membrane, serving as a signal for phagocytes to engulf the dying cell.	Synthesized in the endoplasmic reticulum (ER)	Plays a role in blood clotting by facilitating the binding of clotting factors. Supports neuronal function.
Cardiolipin (Option C)	inner mitochondrial membrane	optimal function of numerous enzymes involved in mitochondrial energy metabolism. Stabilizes the respiratory chain complexes and supercomplexes	Unique dimeric structure consisting of four fatty acid chains Synthesized in the mitochondria from phosphatidylglycerol and cytidine diphosphate-diacylglycerol (CDP-DAG)	Antigenic
Phosphatidylinositol (Option B ruled out)	plasma membrane and intracellular membranes	Involved in cell signaling pathways , particularly through the production of secondary messengers Plays a role in membrane trafficking and cytoskeletal organization.	Produced in the ER and Golgi apparatus from CDP-DAG and inositol.	Crucial for the function of insulin signaling and other receptor-mediated signal transduction pathways. Defects in phosphatidylinositol metabolism are linked to various diseases, including diabetes and cancer.

Phosphatidylethanolamine (Option D ruled out)	All cellular membranes	Important for membrane fusion and the structural integrity of cellular membranes. Serves as a precursor for the synthesis of phosphatidylserine.	Synthesized in the ER via the cytidine diphosphate-ethanolamine (CDP-ethanolamine) pathway or by decarboxylation of phosphatidylserine in the mitochondria.	Plays a role in lipid metabolism and membrane dynamics. Deficiencies can impact cellular functions and are associated with various neurological disorders.
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Phosphatidylserine (Option A) is involved in cell signaling and apoptosis but is not antigenic.

Phosphatidylinositol (Option B) is involved in cell signaling pathways and membrane dynamics but is not antigenic.

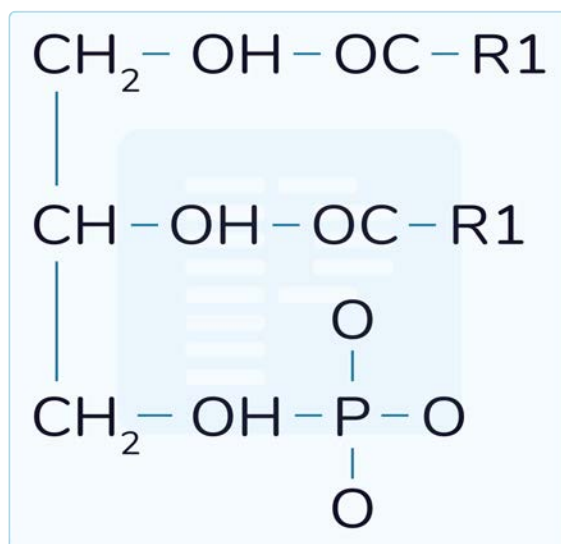
Phosphatidylethanolamine (Option D) is a common phospholipid in cell membranes but is not antigenic.

Solution for Question 16:

Correct Answer: A) Lecithin

Explanation:

This structure is called phosphatidate — phosphatide when combined with choline results in the formation of Phosphatidylcholine also known as Lecithin .



Formation of Phosphatidate:

- When the first two OH groups of a glycerol molecule are esterified with fatty acids.
- And the third OH group bonds with a phosphate group
- This combination forms phosphatidate, which is a precursor to all glycerophospholipids.

Lecithin:

- In phosphatidylcholine, if the first two fatty acids are palmitic acid, the compound is known as Dipalmitoyl phosphatidyl choline.
- This substance is a key component of pulmonary surfactant.
- It is synthesized by type 2 pneumocytes in the lungs.
- In the later stages of the third trimester, it is produced by fully developed fetal lungs.
- Premature neonates often lack this surfactant due to the underdevelopment of their lungs.

Phospholipids:

Phospholipid	Composition
Lecithin (Option A)	Free phosphate + choline
Phosphatidylserine	Free phosphate + Serine
Phosphatidylinositol	Free phosphate + Inositol (Option D ruled out)
Cephalin (Option B ruled out)	Free phosphate + Ethanolamine
Cardiolipin (Option C ruled out)	Combination of Free phosphate group of 2 phosphatidate

Solution for Question 17:

Correct Answer: D) Cerebrosides

Explanation:

The neonate in respiratory distress syndrome fails to produce Lecithin (phosphatidylcholine), a phospholipid crucial for surfactant production. Cephalin, Cardiolipin, and Lecithin are all phospholipids.

Cerebroside is a glycolipid and not a phospholipid.

Classification of Phospholipids(esters containing phosphate groups with fatty acid and alcohol.)		
Glycerophospholipids Alcohol: Glycerol	Phosphatidylcholine (Lecithin) (Option C ruled out)	Most abundant in cell membranes, major source of choline.
Phosphatidylethanolamine (Cephalin) (Option A ruled out)	Found in membranes; differs from phosphatidylcholine by choline substitution.	
Phosphatidylserine	Similar to phosphatidylethanolamine; found in cell membranes.	
Phosphatidylinositol	Contains myoinositol.	
Cardiolipin (Option B ruled out)	Found in mitochondria; synthesized from phosphatidylglycerol.	
Lysophospholipids	Contains one acyl radical; e.g., lysophosphatidylcholine (lysolecithin).	
Plasmalogens	10–30% of phospholipids in brain/heart; ether link instead of ester.	
Sphingophospholipids Alcohol: Sphingosine	Sphingomyelins	Found in myelin sheaths and cell membrane lipid rafts.

Fatty Acid Synthesis and Oxidation

1. Which one of the following is the major regulator of carnitine acyltransferase I (CAT-I) activity?

- A. Insulin
 - B. Glucagon
 - C. Malonyl CoA
 - D. ATP
-

2. Which one of the following does not occur during β -oxidation of fatty acids?

- A. Dehydrogenation of fatty acyl-CoA to trans-enoyl-CoA
 - B. Formation of acetyl-CoA from β -keto fatty acyl-CoA
 - C. Conversion of trans-enoyl-CoA to saturated fatty acyl-CoA
 - D. Hydration of trans-enoyl-CoA to β -hydroxy fatty acyl-CoA
-

3. What is the Total and Net ATP used in the complete oxidation of Arachidonic acid?

- A. 136, 134
 - B. 134, 136
 - C. 120, 122
 - D. 120, 118
-

4. Which of the following is NOT an inhibitor of Carnitine Acyl Transferase-1 (CAT-1)?

- A. Sulfonylureas
 - B. Malonyl-CoA
 - C. AMPK
 - D. Insulin
-

5. A 3-y/o child is brought to the emergency department with symptoms of severe fatigue, vomiting, and episodes of hypoglycemia. Blood tests reveal low glucose levels without elevated ketone bodies, as indicated by a negative Rothera and Benedict tests. Further examination indicates a fatty liver. The child's mother gives h/o acute fatty liver during her pregnancy with the child. Which of the following deficiencies is most likely responsible for these symptoms?

- A. CPT-I Deficiency
 - B. Long-Chain 3-Hydroxyacyl-CoA Dehydrogenase Deficiency
 - C. Jamaican Vomiting Sickness
 - D. Carnitine Deficiency
-

6. Which enzyme requires vitamin B12 for the isomerization of L-methylmalonyl-CoA to succinyl-CoA during odd-chain fatty acid oxidation?

- A. Propionyl-CoA carboxylase
 - B. Methyl malonyl-CoA mutase
 - C. Methyl malonyl-CoA racemase
 - D. Biotin
-

7. Which of the following statements accurately describes the role of glucagon in ketogenesis during starvation?

- A. Glucagon decreases lipolysis and reduces ketogenesis.
 - B. Glucagon inhibits gluconeogenesis and promotes ketogenesis.
 - C. Glucagon increases lipolysis and stimulates ketogenesis.
 - D. Glucagon decreases fatty acid entry into mitochondria and inhibits ketogenesis.
-

8. Which of the following organs can utilize ketone bodies for energy, and which enzyme is involved in their metabolism?

- A. Liver- Thiophorase
 - B. Heart muscle- Acetyl-CoA carboxylase
 - C. Brain- Thiophorase
 - D. Liver- Carboxylase
-

9. Which of the following is an end product of beta-oxidation of odd-chain fatty acids?

- A. Propionyl-CoA
 - B. Malonyl CoA
 - C. Succinyl CoA
 - D. Pyruvate
-

10. Which of the following statements is true about Acetyl CoA metabolism?

- A. It is produced in the cytoplasm from glucose via pyruvate oxidation.
- B. It can be transported directly across the mitochondrial membrane to the cytoplasm.

- C. It is converted to citrate in the mitochondria
 - D. It is involved in the production of NADPH in the cytoplasm.
-

11. Which of the following enzymes is correctly paired with its function in metabolic processes?

- A. Acetyl CoA Carboxylase (ACC) converts acetyl-CoA to citrate, initiating fatty acid synthesis.
 - B. Carnitine Palmitoyltransferase I (CPT-I) regulates entry of all fatty acids into mitochondria for beta-oxidation.
 - C. HMG-CoA Synthase catalyzes ketone body production from acetyl-CoA.
 - D. Acetyl CoA Carboxylase (ACC) converts acetyl-CoA to malonyl-CoA, the rate-limiting step in fatty acid Oxidation
-

12. Which of the following statements about the Fatty acid synthase (FAS) complex is true?

- A. The FAS complex synthesizes fatty acids exclusively in the mitochondria.
 - B. The primary product of the FAS complex is palmitic acid, a 16-carbon saturated fatty acid.
 - C. The FAS complex utilizes the same enzymes and intermediates as those found in β -oxidation.
 - D. Each subunit of the FAS complex consists of a single enzyme responsible for fatty acid synthesis.
-

13. Which of the following statements about the fatty acid synthesis mechanism is NOT TRUE?

- A. Acetyl-CoA is converted to malonyl-CoA by acetyl-CoA carboxylase.

- B. The FAS complex is a homodimer with two identical subunits and six enzymes.
 - C. NADPH is primarily generated by the citric acid cycle.
 - D. Palmitate is the main product, which can be converted into shorter fatty acids.
-

14. Which of the following correctly describes the products and by-products of the palmitic acid synthesis reaction?

- A. 1 Palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADPH
 - B. 1 Palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADP⁺
 - C. 1 Palmitate + 7 CO₂ + 8 H₂O + 8 CoA-SH + 14 NADP⁺
 - D. 1 Palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADP⁺ + 14 H⁺
-

15. Which of the following inhibits Acetyl CoA Carboxylase?

- A. Glucagon
 - B. NADH
 - C. ATP
 - D. Insulin
-

16. Which of the following fatty acid metabolism primarily takes place in the peroxisome?

- A. Very long chain FA
- B. Medium Chain FA
- C. Short chain FA
- D. Long chain FA

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	3
Question 8	3
Question 9	1
Question 10	3
Question 11	3
Question 12	2
Question 13	3
Question 14	2
Question 15	1
Question 16	1

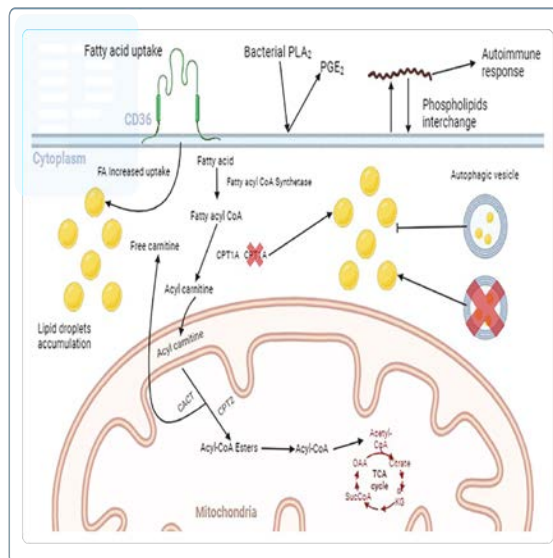
Solution for Question 1:

Correct Answer: C) Malonyl CoA

Explanation:

Malonyl CoA is the major regulator of CAT-I activity. Malonyl CoA inhibits CAT-I, thus preventing the entry of long-chain acyl groups into the mitochondrial matrix

The fate of fatty acid in the cell:



- Fatty acids (FAs) bind to membrane receptors and enter the cytoplasm via transporters.
- Fatty acyl CoA synthetase adds CoA to the fatty acids.
- Carnitine Palmitoyl Transferase 1A (CPT1A) converts fatty acyl CoA to acylcarnitine, which crosses the mitochondrial outer membrane.
- Inside the matrix, acylcarnitine exchanges with free carnitine through Carnitine Acyl Carnitine Transferase (CACT).
- Carnitine Palmitoyl Transferase 2 (CPT2) then converts acylcarnitine back to acyl-CoA esters.

Regulation By CAT-I:

- Mobilized fatty acids enter mitochondria for beta-oxidation.
- Carnitine acyl transferase I (CAT-I) regulates this entry.
- In Low energy states: Conditions like diabetes and starvation, Glucagon levels increase. Increased glucagon decreases malonyl CoA levels. Reduced malonyl CoA levels lead to increased beta-oxidation in the mitochondria.

- Conditions like diabetes and starvation,
- Glucagon levels increase.
- Increased glucagon decreases malonyl CoA levels.
- Reduced malonyl CoA levels lead to increased beta-oxidation in the mitochondria.
- In High energy states: Insulin levels are high. Increased insulin increases malonyl CoA levels. Increased malonyl CoA leads to decreased fatty acid oxidation Hence, Acyl CoA gets stored in the form of TAG in the cells
- Insulin levels are high.
- Increased insulin increases malonyl CoA levels.
- Increased malonyl CoA leads to decreased fatty acid oxidation
- Hence, Acyl CoA gets stored in the form of TAG in the cells
- Conditions like diabetes and starvation,
- Glucagon levels increase.
- Increased glucagon decreases malonyl CoA levels.
- Reduced malonyl CoA levels lead to increased beta-oxidation in the mitochondria.
- Insulin levels are high.
- Increased insulin increases malonyl CoA levels.
- Increased malonyl CoA leads to decreased fatty acid oxidation
- Hence, Acyl CoA gets stored in the form of TAG in the cells

Insulin and glucagon (Option A and B) do not directly influence CAT-1 but their levels indirectly influence it, by increasing or decreasing malonyl COA levels.

ATP (Option D) does not directly regulate CAT-I activity.

Solution for Question 2:

Correct Answer: C) Conversion of trans-enoyl-CoA to saturated fatty acyl-CoA

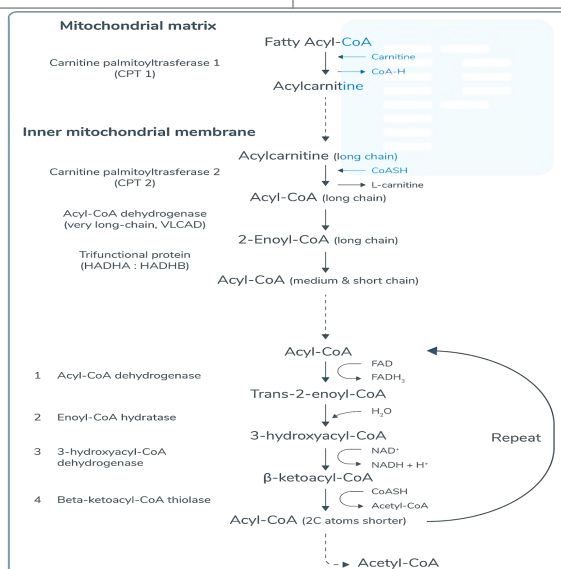
Explanation:

Trans-enoyl-CoA is hydrated to form β -hydroxy fatty acyl-CoA.

No step converts trans-enoyl-CoA directly to a saturated fatty acyl-CoA.

Steps of beta-oxidation:

Step 1: FAD-Linked Dehydrogenation	Catalyzed by Acyl-CoA dehydrogenase Fatty acyl-CoA → trans-enoyl-CoA (FADH ₂ produced). (Option A ruled out) FADH ₂ generates 1.5 ATP in the electron transport chain.
Step 2: Hydration	Catalyzed by enoyl-CoA hydratase. Converts trans-enoyl-CoA to β-hydroxy fatty acyl-CoA (only L isomer formed). (Option D ruled out)
Step 3 :NAD ⁺ -Dependent Dehydrogenation	Catalyzed by β-hydroxy acyl-CoA dehydrogenase β-Hydroxy fatty acyl-CoA → β-keto fatty acyl-CoA (NADH produced). NADH generates 2.5 ATP in the electron transport chain
Cleavage	Catalyzed by Acyl CoA Acetyltransferase (Thiolase) β-Keto fatty acyl-CoA undergoes thiolytic cleavage. Produces one acetyl-CoA and a shorter fatty acyl-CoA for further cycles. (Option B ruled out)



Solution for Question 3:

Correct Answer: A) 136, 134

Explanation:

The oxidation of arachidonic acid occurs in the mitochondria via β -oxidation when C2 units of it degrade once at a time to generate FADH₂, NADH, and Acetyl-CoA.

N = number of carbons in the fatty acid, here, N= 20 (Arachidonic acid is a 20C chain fatty acid)

$N/2$ = of acetyl CoA produced

$N/2 - 1$ = rounds need to fully metabolize the fatty acid, FADH₂, NADH produced.

- Number of Acetyl-CoA produced = $20/2 = 10$
- Number of NADH and FADH₂ produced = $20/2 - 1 = 9$ each.

The 10 Acetyl-CoA then moves in the citrate cycle to produce 3 NADH, 1 FADH₂, and 1 GTP per Acetyl-CoA.

- So, the number of NADH produced = 30,
- The number of FADH₂ produced = 10
- The number of GTP = 10.

The ATP conversions:

- 1 GTP= 1 ATP
- 1 NADH = 2.5 ATP
- 1 FADH₂= 1.5 ATP.

The number of ATPs produced before the Acetyl-CoA moves in the citrate cycle= 9 FADH₂ + 9 NADH = $9(1.5) + 9(2.5) = 36$ ATPs

The number of ATPs produced when the Acetyl-CoA moves in the citrate cycle. 30 NADH + 10 FADH₂ + 1 GTP = $30(2.5) + 10(1.5) + 10(1) = 100$ ATPs

Total ATPs produced= $100 + 36 = 136$

2 equivalents of ATP (ATP $\rightarrow$ AMP + P_i) are used to activate fatty acids to make acyl-CoA; therefore the net yield is 2 fewer ATP molecules.

The true yield is 134 ATP ($136 - 2$ ATP).

Net ATPs produced=134

Solution for Question 4:

AMPK activation leads to increased fatty acid oxidation by promoting mitochondrial biogenesis and improving fatty acid metabolism. It is a metabolic regulator that often enhances Carnitine Acyl Transferase-1 (CAT-1)/ Carnitine Palmitoyl Transferase-1 CPT-1 activity rather than inhibiting it.

CARNITINE ACYL TRANSFERASE-1 (CPT-1)	
STIMULATORS	INHIBITORS
Starvation FFA (Free Fatty Acids) AMPK (Adenosine Monophosphate-Activated Protein Kinase) Glucagon	Well fed state Malonyl-CoA (Option B ruled out) Sulfonylureas (Option A ruled out) Insulin (Option D ruled out) Etomoxir Oxfenicine Perhexiline PPAR α antagonist

Solution for Question 5:

- The scenario describes a child with severe fatigue, vomiting, nonketotic hypoglycemia, and fatty liver, alongside a maternal history of acute fatty liver of pregnancy (AFLP).
- These features are characteristic of Long-Chain 3-Hydroxyacyl-CoA Dehydrogenase Deficiency. This condition affects fatty acid oxidation and can manifest in both the child and mother, particularly with AFLP during pregnancy.
- This condition affects fatty acid oxidation and can manifest in both the child and mother, particularly with AFLP during pregnancy.
- Other deficiencies listed do not specifically present with this combination of symptoms and maternal history.
- This condition affects fatty acid oxidation and can manifest in both the child and mother, particularly with AFLP during pregnancy.

FATTY ACID OXIDATION DEFECTS	
CONGENITAL DEFECTS	ACQUIRED DEFECTS

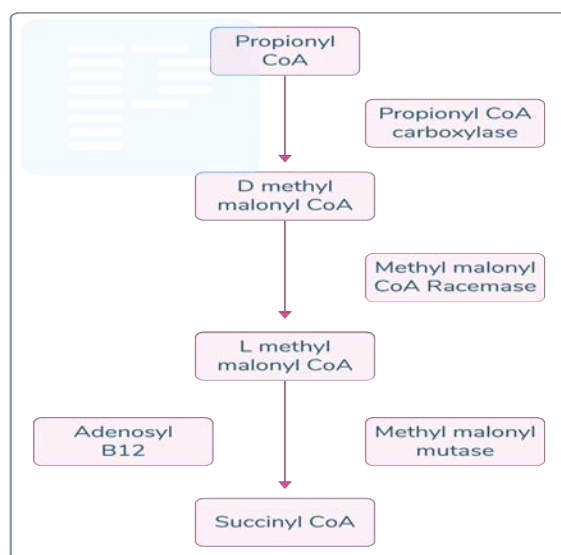
1. CPT-I Deficiency: Site : Liver Pathophysiology : Leading to decreased fatty acid oxidation and ketogenesis, which can result in hypoglycemia (Option A ruled out) 2. CPT-II Deficiency: Site: Primarily affecting skeletal muscle and potentially the liver in severe cases 3. Defects in β-oxidation Enzymes: 2 types: Long and Short-Chain 3-Hydroxyacyl-CoA Dehydrogenase Deficiency: Can cause nonketotic hypoglycemia, coma, and fatty liver. May also contribute to acute fatty liver during pregnancy 3-Ketoacyl-CoA Thiolase and HMG-CoA Lyase Deficiency: These deficiencies not only disrupt fatty acid oxidation but also affect the breakdown of leucine, a ketogenic amino acid.	1. Carnitine Deficiency: Common in preterm babies Cause : Inadequate biosynthesis or renal leakage. Hemolysis Pathophysiology : Impaired fatty acid oxidation→hypoglycemia & Lipid accumulation→muscular weakness (Option D ruled out) 2. Jamaican Vomiting Sickness: Cause : Consuming unripe ackee fruit containing the toxin hypoglycin, Pathophysiology : This condition inhibits medium- and short-chain acyl-CoA dehydrogenase, leading to hypoglycemia (Option C ruled out)
CLINICAL FEATURES	
Hypoglycemia Nonketotic Hypoglycemia Lipid Accumulation Dicarboxylic Aciduria Marked by the excretion of C6—C10 ω-dicarboxylic acids and non-ketotic hypoglycemia It arises from a lack of mitochondrial medium-chain acyl-CoA dehydrogenase. Muscular Weakness due to the buildup of lipids in muscles Coma Fatty Liver Acute Fatty Liver of Pregnancy: Due to Long-chain 3-hydroxyacyl-CoA dehydrogenase deficiency indicating a potential role of hormonal changes influencing enzyme activity. Neurological Symptoms: Defects impacting fatty acid oxidation, especially those also affecting the breakdown of leucine.	

Differential Diagnosis:

Solution for Question 6:

Methylmalonyl-CoA mutase is the enzyme responsible for converting L-methylmalonyl-CoA to succinyl-CoA, a crucial step in the oxidation of odd-chain fatty acids. This reaction requires vitamin B12 as a cofactor.

In vitamin B12 deficiency, the activity of methyl malonyl-CoA mutase is greatly reduced, leading to the accumulation of methyl malonyl-CoA and the subsequent excretion of methylmalonic acid in the urine (methylmalonic aciduria) and neurological side effects leading to mental retardation.



Propionyl-CoA carboxylase (Option A) is another enzyme in the pathway that carboxylates propionyl-CoA to D-methylmalonyl-CoA and requires biotin, not vitamin B12.

Methylmalonyl-CoA racemase (Option C) converts D-methylmalonyl-CoA to L-methylmalonyl-CoA, but this reaction also does not require vitamin B12.

Biotin (Option D) is a cofactor for other carboxylation reactions but is not directly involved in the isomerization of succinyl-CoA.

Solution for Question 7:

Glucagon increases lipolysis, releasing more fatty acids that the liver converts into ketone bodies, thereby stimulating ketogenesis.

Causes of Ketosis:

- Untreated diabetes mellitus causes ketosis due to insulin deficiency, which increases lipolysis and fatty acid oxidation.
- This raises acetyl CoA levels, while reduced oxaloacetate limits TCA cycle activity.
- The body then overproduces ketone bodies as it relies on fat metabolism for energy.
- During starvation, reduced dietary glucose shifts oxaloacetate to gluconeogenesis.
- Increased lipolysis converts excess acetyl CoA to ketone bodies.

- High glucagon levels promote ketogenesis, providing the brain with 75% of its energy from ketones.
- Conditions like hyperemesis in early pregnancy can also cause ketosis.
- Limited glucose boosts free fatty acid mobilization, with the liver converting them to ketones, regulated by carnitine palmitoyltransferase-I (CPT-I).

Explanation for Ketogenesis:

- Glucagon: Increases in glucagon during starvation and diabetes mellitus inhibit glycolysis, activate gluconeogenesis and lipolysis, decrease malonyl CoA, and stimulate ketogenesis. A high glucagon-insulin ratio promotes ketosis.
- Insulin: Opposes glucagon by favoring glycolysis, inhibiting gluconeogenesis, suppressing lipolysis, increasing malonyl CoA, and reducing ketogenesis.
- Lipolysis: Mobilizes fatty acids.
- Fatty Acid Entry to Mitochondria: Regulated by carnitine acyl transferase I (CAT-I) for beta-oxidation.
- Oxidation of Acetyl CoA: Converts acetyl CoA to ketone bodies.

Glucagon decreases lipolysis and reduces ketogenesis. (Option A): Glucagon, in fact, increases lipolysis, which provides more fatty acids for ketogenesis.

Glucagon inhibits gluconeogenesis and promotes ketogenesis. (Option B): Glucagon activates gluconeogenesis and also promotes ketogenesis, but it does not inhibit gluconeogenesis.

Glucagon decreases fatty acid entry into mitochondria and inhibits ketogenesis. (Option D): Glucagon does not decrease fatty acid entry into mitochondria; rather, it promotes ketogenesis by increasing lipolysis and the availability of fatty acids.

Solution for Question 8:

The brain can use ketone bodies as an energy source, and Thiophorase is the enzyme that converts acetoacetate to acetoacetyl CoA , which is utilized by the brain .

KETOSIS	
Ketone Bodies	Produced in the liver Utilized by extrahepatic tissues due to the absence of Thiophorase enzyme in the liver

Tissues that use Ketone Bodies	Almost all tissues except: Liver Red blood cells (RBCs) Tissues that prefer ketone bodies: Heart muscle Renal cortex Placenta Tissues that use ketone bodies only in the absence of glucose: Skeletal muscle Brain Additional Tissues Using Ketone Bodies Intestinal mucosal cells Adipocytes
Enzyme Involved	Thiophorase : Converts acetoacetate to acetoacetyl CoA
Fatty Acids as Primary Energy Source During Starvation	Skeletal muscles Heart Liver
Energy Production Pathway	Acetoacetyl CoA enters the beta-oxidation pathway to produce energy

Liver- Thiophorase (Option A): The liver produces ketone bodies but does not utilize them due to the absence of the enzyme Thiophorase. Thiophorase is involved in ketone body metabolism but is not present in the liver.

Heart muscle- Acetyl-CoA carboxylase (Option B): While the heart muscle can utilize ketone bodies, the enzyme mentioned, acetyl-CoA carboxylase, is not involved in their metabolism. Instead, the correct enzyme is Thiophorase.

Liver- Carboxylase (Option D): The liver does not utilize ketone bodies due to the lack of Thiophorase, and Carboxylase is not involved in ketone body metabolism.

Solution for Question 9:

Propionyl-CoA is the actual end product of beta-oxidation of odd-chain fatty acids. The last cycle of beta-oxidation of odd-chain fatty acids produces propionyl-CoA, which is then converted to succinyl-CoA.

Odd-chain fatty acids undergo beta-oxidation in the same manner as even-chain fatty acids, with successive removal of 2-carbon units (acetyl-CoA). The process continues until the final three-carbon unit, propionyl-CoA, remains.

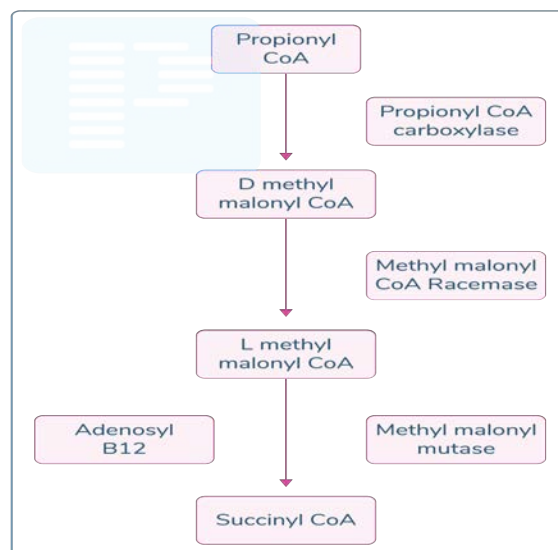
- Carboxylation : Propionyl-CoA is carboxylated to D-methylmalonyl-CoA by a biotin-dependent carboxylase enzyme, using ATP for energy.
- Racemization : D-methylmalonyl-CoA is then converted to L-methylmalonyl-CoA by a racemase enzyme.

- **Rearrangement** : L-methylmalonyl-CoA is rearranged to succinyl-CoA by L-methylmalonyl-CoA mutase, which requires vitamin B12.
- **Entry into the TCA Cycle** : Succinyl-CoA enters the TCA cycle and is converted to oxaloacetate.
- **Gluconeogenesis** : Oxaloacetate from the TCA cycle can be used for gluconeogenesis to synthesize glucose.

Malonyl-CoA (Option B): This molecule is involved in the synthesis of fatty acids, not in their beta-oxidation. Therefore, it is not an end product of beta-oxidation of odd-chain fatty acids.

Succinyl-CoA (Option C): This molecule is produced from propionyl-CoA through a series of reactions, but it is not a direct end product of beta-oxidation. It is an intermediate in the citric acid cycle.

Pyruvate (Option D): This molecule is a key intermediate in glycolysis and gluconeogenesis, but it is not involved in the beta-oxidation of fatty acids.



Solution for Question 10:

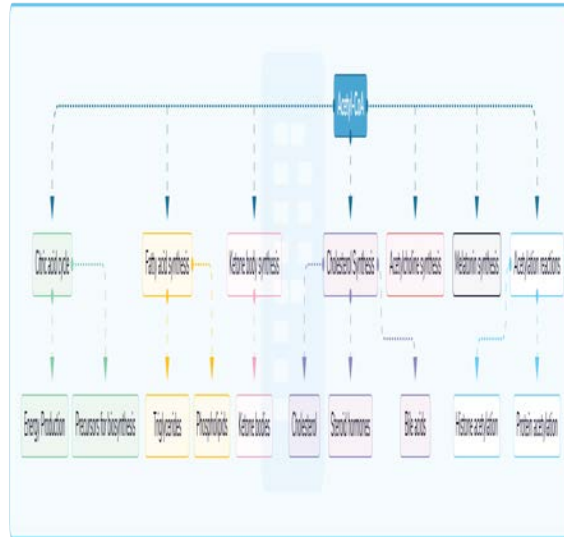
Correct Answer: C) It is converted to citrate in the mitochondria

Explanation

Acetyl CoA combines with oxaloacetate to form citrate in the mitochondria. Citrate can then be transported to the cytoplasm via the tricarboxylate transporter.

Production of Acetyl CoA:

- From Glucose : Acetyl CoA is produced in the mitochondria from glucose via pyruvate oxidation. (Option A ruled out)
- From Fatty Acids : Acetyl CoA is also a product of fatty acid β -oxidation within the mitochondria.
- Citrate Shuttle : Acetyl CoA cannot readily cross the mitochondrial membrane. Instead, it combines with oxaloacetate to form citrate, which is transported to the cytoplasm via the tricarboxylate transporter. (Option B ruled out)
- Conversion in Cytoplasm : In the cytoplasm, citrate is converted back to acetyl CoA and oxaloacetate by ATP-citrate lyase in the presence of CoA and ATP.
- Location : Cytoplasm.
- Process : Acetyl CoA is the primary building block for lipogenesis, regulated by nutritional status and more active in well-fed individuals consuming carbohydrate-rich diets.
- Precursor : Acetyl CoA is a precursor in cholesterol synthesis.
- Activity : This pathway is less active compared to fatty acid synthesis, especially in the liver.
- Citric Acid Cycle : Acetyl CoA can re-enter the mitochondria as citrate and participate in the citric acid cycle for ATP production.
- NADPH Production : Oxaloacetate in the cytoplasm can be converted to malate and then to pyruvate, generating NADPH in the process. Pyruvate returns to the mitochondria and is converted back into acetyl CoA. (Option D ruled out)
- Location : Liver mitochondria.
- A high rate of fatty acid oxidation is common during starvation or in individuals with untreated diabetes mellitus.



Solution for Question 11:

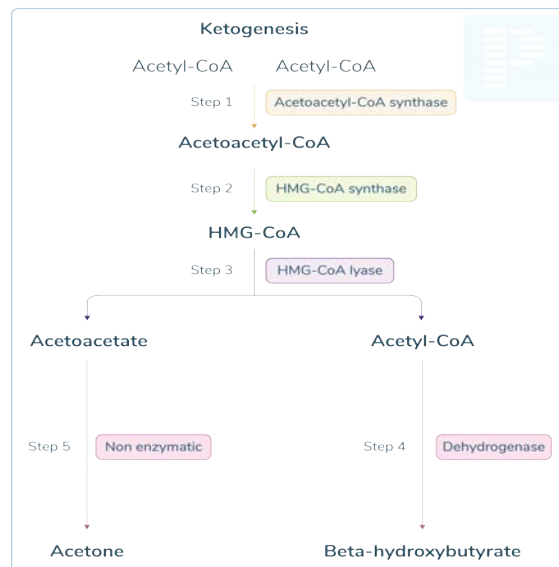
Correct Answer: C) HMG-CoA Synthase catalyzes ketone body production from acetyl-CoA.

Explanation

HMG-CoA Synthase catalyzes a key step in ketogenesis, producing ketone bodies from acetyl-CoA when glucose is limited.

- Acetyl CoA Carboxylase (ACC) : Converts acetyl-CoA to malonyl-CoA, the first committed step in fatty acid synthesis. (Options A & D ruled out) Regulated by citrate (activator) and palmitoyl-CoA (inhibitor). Insulin activates ACC, while glucagon and epinephrine inhibit it.
- Converts acetyl-CoA to malonyl-CoA, the first committed step in fatty acid synthesis. (Options A & D ruled out)
- Regulated by citrate (activator) and palmitoyl-CoA (inhibitor).
- Insulin activates ACC, while glucagon and epinephrine inhibit it.
- Carnitine acyltransferase I (CAT-I) : Regulates the entry of long-chain fatty acids (typically 14-20 carbon atoms) into mitochondria for beta-oxidation. (Option B ruled out) Inhibited by malonyl-CoA to prevent simultaneous synthesis and breakdown of fatty acids. Activated when glucagon levels rise and malonyl-CoA decreases.

- Regulates the entry of long-chain fatty acids (typically 14-20 carbon atoms) into mitochondria for beta-oxidation. (Option B ruled out)
- Inhibited by malonyl-CoA to prevent simultaneous synthesis and breakdown of fatty acids.
- Activated when glucagon levels rise and malonyl-CoA decreases.
- HMG-CoA Synthase : Catalyzes a key step in ketogenesis, producing ketone bodies from acetyl-CoA. Involved in a key regulatory step in ketogenesis.
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- Insulin activates ACC, while glucagon and epinephrine inhibit it.
- Regulates the entry of long-chain fatty acids (typically 14-20 carbon atoms) into mitochondria for beta-oxidation. (Option B ruled out)
- Inhibited by malonyl-CoA to prevent simultaneous synthesis and breakdown of fatty acids.
- Activated when glucagon levels rise and malonyl-CoA decreases.
- Catalyzes a key step in ketogenesis, producing ketone bodies from acetyl-CoA.
- Involved in a key regulatory step in ketogenesis.



Rate limiting enzymes:

S.no	Pathway	Rate limiting enzymes
1	Fatty acid synthesis	Acetyl CoA carboxylase

2	Fatty acid oxidation	Carnitine acyl transferase I
3	Cholesterol synthesis	HMG CoA reductase
4	Ketone body synthesis	HMG Co A lyase > HMG CoA synthase
5	Bile acid synthesis	7- α -hydroxylase

Solution for Question 12:

Correct Answer: B) The primary product of the FAS complex is palmitic acid, a 16-carbon saturated fatty acid.

Explanation:

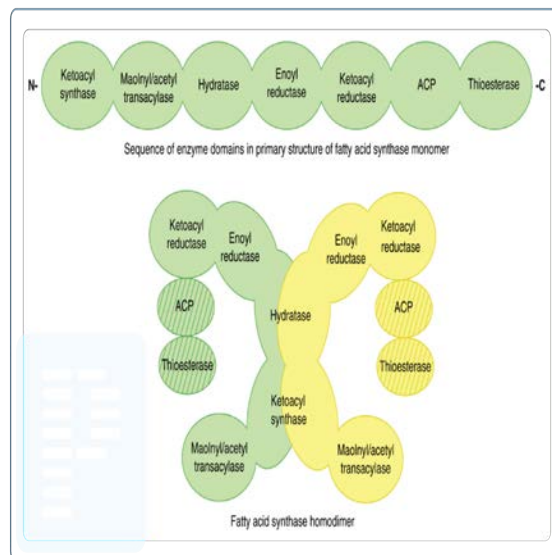
The primary product of the FAS complex is palmitic acid, which is a 16-carbon saturated fatty acid synthesized during fatty acid synthesis.

Fatty Acid Synthase (FAS) Complex Overview:

- Location : Cytoplasm (Option A ruled out) Active in liver, adipose tissue, lactating mammary gland, kidney, and brain.
- Cytoplasm (Option A ruled out)
- Active in liver, adipose tissue, lactating mammary gland, kidney, and brain.
- Actions : De novo synthesis of fatty acids, primarily palmitic acid (16-carbon saturated fatty acid). Not the reverse of β -oxidation; it involves different intermediates, enzymes, coenzymes, and regulatory mechanisms. (Option C ruled out)
- De novo synthesis of fatty acids, primarily palmitic acid (16-carbon saturated fatty acid).
- Not the reverse of β -oxidation; it involves different intermediates, enzymes, coenzymes, and regulatory mechanisms. (Option C ruled out)
- Structure : Multi-enzyme complex organized as a homodimer (two identical subunits). Each subunit contains three domains, six enzymes, and an acyl carrier protein. (Option D ruled out) Enhances efficiency by facilitating interactions between intermediates and active sites.
- Multi-enzyme complex organized as a homodimer (two identical subunits).
- Each subunit contains three domains, six enzymes, and an acyl carrier protein. (Option D ruled out)

- Enhances efficiency by facilitating interactions between intermediates and active sites.
- Subunits : Domain 1 (Condensing Unit) : Acetyl transacylase Malonyl transacylase β -keto acyl synthase (condensing enzyme) Domain 2 (Reduction Unit) : Acyl carrier protein (ACP) β -keto acyl reductase Dehydratase Enoyl reductase Domain 3 (Releasing Unit) : Thioesterase (deacylase)
- Domain 1 (Condensing Unit) : Acetyl transacylase Malonyl transacylase β -keto acyl synthase (condensing enzyme)
 - Acetyl transacylase
 - Malonyl transacylase
 - β -keto acyl synthase (condensing enzyme)
- Domain 2 (Reduction Unit) : Acyl carrier protein (ACP) β -keto acyl reductase Dehydratase Enoyl reductase
 - Acyl carrier protein (ACP)
 - β -keto acyl reductase
 - Dehydratase
 - Enoyl reductase
- Domain 3 (Releasing Unit) : Thioesterase (deacylase)
 - Thioesterase (deacylase)
- Cytoplasm (Option A ruled out)
- Active in liver, adipose tissue, lactating mammary gland, kidney, and brain.
- De novo synthesis of fatty acids, primarily palmitic acid (16-carbon saturated fatty acid).
- Not the reverse of β -oxidation; it involves different intermediates, enzymes, coenzymes, and regulatory mechanisms. (Option C ruled out)
- Multi-enzyme complex organized as a homodimer (two identical subunits).
- Each subunit contains three domains, six enzymes, and an acyl carrier protein. (Option D ruled out)
- Enhances efficiency by facilitating interactions between intermediates and active sites.
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 - Acetyl transacylase
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 - Acyl carrier protein (ACP)

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- Acetyl transacylase
- Malonyl transacylase
- β -keto acyl synthase (condensing enzyme)
- Acyl carrier protein (ACP)
- β -keto acyl reductase
- Dehydratase
- Enoyl reductase
- Thioesterase (deacylase)



Solution for Question 13:

Correct Answer: C) NADPH is primarily generated by the citric acid cycle.

Explanation:

NADPH is primarily generated through the pentose phosphate pathway, not the citric acid cycle.

- Rate-Limiting Step : Formation of malonyl-CoA Enzyme : Acetyl-CoA carboxylase (multienzyme protein) (Option A ruled out) Reaction : Acetyl-CoA + bicarbonate (HCO_3^-) + ATP $\rightarrow$ malonyl-CoA
- Enzyme : Acetyl-CoA carboxylase (multienzyme protein) (Option A ruled out)
- Reaction : Acetyl-CoA + bicarbonate (HCO_3^-) + ATP $\rightarrow$ malonyl-CoA
- Fatty Acid Synthase (FAS) Complex : Structure : Homodimer composed of two identical subunits, each containing six enzymes and an acyl carrier protein (ACP). (Option B ruled out)
- Structure : Homodimer composed of two identical subunits, each containing six enzymes and an acyl carrier protein (ACP). (Option B ruled out)
- Synthesis Process : Elongation Cycles : Each cycle lengthens the fatty acid chain by two carbons. Priming : Attach acetyl-CoA to one ACP and malonyl-CoA to the other ACP of the dimer. Condensation : Acetyl group attacks the methylene group of malonyl residue, catalyzed by 3-ketoacyl synthase, releasing CO_2 and forming a 4-carbon unit. Reduction, Dehydration, Reduction : The 4-carbon unit undergoes a series of reactions using NADPH. Transfer and Binding : The saturated acyl chain is transferred to the other subunit's ACP, and a new malonyl-CoA binds to the free ACP.
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- Elongation Cycle : Repeats six times, producing a 16-carbon palmitate bound to ACP. (Option D ruled out)
- Thioesterase Role : Releases free palmitate from the FAS complex.
- Shorter Fatty Acid Production : Specific thioesterase in mammary glands can release 8, 10, or 12-carbon fatty acids for milk lipid incorporation.
- NADPH's Role : Essential cofactor for reduction steps, mainly sourced from: Pentose phosphate pathway Conversion of malate to pyruvate (malic enzyme) Conversion of isocitrate to alpha-ketoglutarate (extramitochondrial isocitrate dehydrogenase)
- Pentose phosphate pathway
- Conversion of malate to pyruvate (malic enzyme)
- Conversion of isocitrate to alpha-ketoglutarate (extramitochondrial isocitrate dehydrogenase)
- Enzyme : Acetyl-CoA carboxylase (multienzyme protein) (Option A ruled out)
- Reaction : $\text{Acetyl-CoA} + \text{bicarbonate (HCO}_3^-) + \text{ATP} \rightarrow \text{malonyl-CoA}$
- Structure : Homodimer composed of two identical subunits, each containing six enzymes and an acyl carrier protein (ACP). (Option B ruled out)
- Elongation Cycles : Each cycle lengthens the fatty acid chain by two carbons. Priming : Attach acetyl-CoA to one ACP and malonyl-CoA to the other ACP of the dimer. Condensation : Acetyl group attacks the methylene group of malonyl residue, catalyzed by 3-ketoacyl synthase, releasing CO_2 and forming a 4-carbon unit. Reduction, Dehydration, Reduction : The 4-carbon unit undergoes a series of reactions using NADPH. Transfer and Binding : The saturated acyl chain is transferred to the other subunit's ACP, and a new malonyl-CoA binds to the free ACP.
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- The 4-carbon unit undergoes a series of reactions using NADPH.
- The saturated acyl chain is transferred to the other subunit's ACP, and a new malonyl-CoA binds to the free ACP.
- Pentose phosphate pathway
- Conversion of malate to pyruvate (malic enzyme)
- Conversion of isocitrate to alpha-ketoglutarate (extramitochondrial isocitrate dehydrogenase)

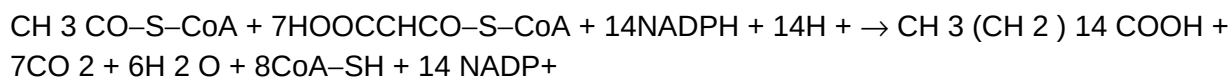
Solution for Question 14:

Correct Answer: B) 1 Palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADP⁺

Explanation:

1 Palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADP⁺ accurately describes the products of palmitic acid synthesis, including 1 palmitate, 7 CO₂, 6 H₂O, 8 CoA-SH, and 14 NADP⁺.

Reaction Equation:



- Synthesis of Palmitate (16-carbon fatty acid): Reactants: 1 molecule of acetyl-CoA (provides carbons 15 and 16) 7 molecules of malonyl-CoA (provides remaining carbons) 14 molecules of NADPH 14 hydrogen protons (H⁺) Products: 1 molecule of palmitate 7 molecules of carbon dioxide (CO₂) 6 molecules of water (H₂O) 8 molecules of coenzyme A (CoA-SH) 14 molecules of NADP⁺
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- 1 molecule of acetyl-CoA (provides carbons 15 and 16)
- 7 molecules of malonyl-CoA (provides remaining carbons)
- 14 molecules of NADPH
- 14 hydrogen protons (H⁺)
- Products: 1 molecule of palmitate 7 molecules of carbon dioxide (CO₂) 6 molecules of water (H₂O) 8 molecules of coenzyme A (CoA-SH) 14 molecules of NADP⁺
- 1 molecule of palmitate
- 7 molecules of carbon dioxide (CO₂)
- 6 molecules of water (H₂O)
- 8 molecules of coenzyme A (CoA-SH)
- 14 molecules of NADP⁺
- Reactants: 1 molecule of acetyl-CoA (provides carbons 15 and 16) 7 molecules of malonyl-CoA (provides remaining carbons) 14 molecules of NADPH 14 hydrogen protons (H⁺)
- 1 molecule of acetyl-CoA (provides carbons 15 and 16)
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- 14 molecules of NADPH
- 14 hydrogen protons (H⁺)
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- 14 molecules of NADP⁺

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- 7 molecules of malonyl-CoA (provides remaining carbons)
- 14 molecules of NADPH
- 14 hydrogen protons (H⁺)
- 1 molecule of palmitate
- 7 molecules of carbon dioxide (CO₂)
- 6 molecules of water (H₂O)
- 8 molecules of coenzyme A (CoA-SH)
- 14 molecules of NADP⁺

1 palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADPH (Option A) NADPH is used during the synthesis, not produced.

1 palmitate + 7 CO₂ + 8 H₂O + 8 CoA-SH + 14 NADP⁺ (Option C) In palmitic acid synthesis, 6 molecules of H₂O are produced, not 8.

1 palmitate + 7 CO₂ + 6 H₂O + 8 CoA-SH + 14 NADP⁺ + 14 H⁺ (Option D) This option shows extra protons (14 H⁺), which are not produced in the reaction, rather they are used up along with 14 NADPH to give 14 NADP⁺.

Solution for Question 15:

Correct Answer: A) Glucagon

Explanation:

- Acetyl CoA Carboxylase is inhibited by Glucagon .
- Glucagon inhibits acetyl CoA carboxylase by keeping the enzyme in the inactive phosphorylated state.
- Glucagon is secreted when the energy charge is low and AMP levels are high.

Acetyl CoA Carboxylase

Stimulated by High energy level states	Inhibited by Low energy level states.
--	---------------------------------------

NADH (Option B ruled out) FADH₂ ATP (Option C ruled out) Citrate Insulin: enhances the uptake of glucose by adipocytes and increases the activity of pyruvate dehydrogenase, acetyl CoA carboxylase, and glycerol phosphate acyl transferase (Option D ruled out)

ADP NAD FAD Glucagon(catabolic hormone) (Option A) Acyl CoA Epinephrine

- The nutritional state of the organism is the main factor regulating the rate of lipogenesis
Diets: Carbohydrate-rich diet: Increases lipogenesis Fat-rich diet: Decreases lipogenesis eg. Keto diet
- Diets: Carbohydrate-rich diet: Increases lipogenesis Fat-rich diet: Decreases lipogenesis eg. Keto diet
- Carbohydrate-rich diet: Increases lipogenesis
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- Carbohydrate-rich diet: Increases lipogenesis
- Fat-rich diet: Decreases lipogenesis eg. Keto diet
- Other situations where lipogenesis is decreased:
 - Restricted caloric intake
 - Diabetes mellitus

Solution for Question 16:

Correct Answer: A) Very long chain FA

Explanation:

Very long-chain fatty acids undergo oxidation in peroxisomes

Fatty acids oxidation

Mitochondria

- Short-chain fatty acids (Option C ruled out)
- Medium-chain fatty acids (Option B ruled out)

- Long-chain fatty acids (Option D ruled out)

Peroxisomes

- Very long-chain fatty acids (>22C)

Functions of peroxisomes

- Breakdown of Very-Long-Chain Fatty Acids (VLCFAs)
 - Energy Production by reducing the length of fatty acid chains, making them suitable for further oxidation in mitochondria.
 - Cholesterol Metabolism : Contribute to the shortening of the side chain of cholesterol in bile acid formation.
 - Involved in the synthesis of ether glycerolipids , which are essential components of cellular membranes.
 - Participate in the synthesis of cholesterol and dolichol , which are important for various cellular functions.
-

Cholesterol, Lipoproteins, and Hypercholesterolemia

1. Which of the following best describes Tangier disease?

- A. Low VLDL levels
 - B. Low HDL levels
 - C. Low LDL levels
 - D. Low LDL receptor activity
-

2. A 12-year-old boy presents with developmental delays, visual impairment, and recurrent episodes of ataxia. Laboratory tests reveal low serum levels of triglycerides, cholesterol, and fat-soluble vitamins. A blood smear shows the presence of acanthocytes. Which of the following findings is most likely present in this patient?

- A. Elevated levels of chylomicrons in the bloodstream.
 - B. Absence of apolipoprotein B-containing lipoproteins.
 - C. Normal absorption of dietary fats.
 - D. High serum levels of low-density lipoprotein (LDL).
-

3. A researcher is studying the mechanisms of lipoprotein lipase (LPL) activity in post-heparinized blood samples. They conduct a mixing study to differentiate between a deficiency of apolipoprotein C-II (Apo C-II) and LPL deficiency. Which of the following outcomes from the mixing study would most accurately indicate an Apo C-II deficiency?

- A. The addition of pooled normal plasma results in no change in LPL activity.
- B. LPL activity normalizes after mixing with normal plasma.

- C. A significant increase in triglyceride levels is observed after mixing.
- D. LPL activity decreases following the addition of normal plasma.

4. Match the following contributing factors with their corresponding effects on fatty liver: A) Choline Deficiency 1) Increases NADH, hindering fatty acid oxidation and promoting triacylglycerol accumulation. B) Orotic Acid Interference 2) Impairs VLDL secretion, resulting in triacylglycerol accumulation and a fatty liver. C) Vitamin E and Selenium 3) Involved in pyrimidine synthesis. Disrupts VLDL glycosylation, hindering release. D) Ethanol Consumption 4) Protect against liver damage from lipid peroxidation

A) Choline Deficiency	1) Increases NADH, hindering fatty acid oxidation and promoting triacylglycerol accumulation.
B) Orotic Acid Interference	2) Impairs VLDL secretion, resulting in triacylglycerol accumulation and a fatty liver.
C) Vitamin E and Selenium	3) Involved in pyrimidine synthesis. Disrupts VLDL glycosylation, hindering release.
D) Ethanol Consumption	4) Protect against liver damage from lipid peroxidation.

- A. A-2, B-3, C-4, D-1
- B. A-3, B-4, C-2, D-1
- C. A-4, B-1, C-3, D-2
- D. A-1, B-2, C-4, D-3

5. Which among the following is a true statement on electrophoretic mobility of lipoproteins seen during serum electrophoresis?

- A. HDL shows the least mobility due to high lipid content.
- B. Lipoproteins with higher protein content move slower towards the anode.
- C. Lipoproteins with higher protein content migrate faster towards the anode

D. Lipoproteins identified by dye color after staining with lipid dyes, not electrophoretic mobility.

6. A 45-year-old male presents with symptoms of peripheral neuropathy, enlarged tonsils, and orange-colored tonsils. His lipid profile shows very low levels of high-density lipoprotein (HDL) cholesterol. Further investigation reveals that the patient has a rare genetic condition leading to severe HDL deficiency. Given these findings, which of the following gene mutations is most likely responsible for this condition?

- A. ABCA1 gene mutation
 - B. LDLR gene mutation
 - C. APOE gene mutation
 - D. PCSK9 gene mutation
-

7. A 45-year-old woman presents with a history of palmar xanthoma. She reports a family history of similar symptoms. Laboratory tests reveal elevated cholesterol and triglyceride levels, with a cloudy appearance of her plasma. What is the most likely diagnosis for this patient?

- A. Type I (Familial chylomicronemia syndrome)
 - B. Type IIa (Familial hypercholesterolemia)
 - C. Type III (Familial dysbetalipoproteinemia)
 - D. Type IV (Familial hypertriglyceridemia)
-

8. Which of the following statements accurately distinguishes Type IIa from Type IIb Hyperlipoproteinemias?

- A. Plasma after 24 hrs- Type IIa: clear; Type IIb: slightly cloudy.

- B. Clinical Features- Type IIa: eruptive xanthoma ; Type IIb: corneal arcus.
- C. Type IIa: VLDL overproduction; Type IIb: LDL and VLDL overproduction.
- D. Management - Type IIa: restrict fat, supplement MCT; Type IIb: weight reduction, drug therapy.
-

9. A 5-year-old boy presents with recurrent episodes of severe abdominal pain and hepatomegaly. Physical examination reveals eruptive xanthomas on his trunk and limbs. Blood tests show a creamy layer over clear plasma after overnight refrigeration. What metabolic defect is most likely present in this patient?

- A. LDL receptor defect
- B. Apo B-100 mutation
- C. Lipoprotein lipase deficiency
- D. Apo E abnormality
-

10. Match the apolipoproteins with their respective functions in lipid metabolism:
Apolipoprotein Function
1) Apolipoprotein A-I a. Enhances lipoprotein lipase activity, facilitating triglyceride hydrolysis.
2) Apolipoprotein B-100 b. Involved in the transport of dietary lipids from the intestine to other tissues.
3) Apolipoprotein C-II c. Helps in reverse cholesterol transport, removing excess cholesterol from tissue back to liver.
4) Apolipoprotein E d. Essential for binding to LDL receptors on various tissues.

Apolipoprotein	Function
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3) Apolipoprotein C-II	c. Helps in reverse cholesterol transport, removing excess cholesterol from tissue back to liver.
4) Apolipoprotein E	d. Essential for binding to LDL receptors on various tissues.

- A. 1-c, 2-d, 3-a, 4-b
 - B. 1-b, 2-c, 3-d, 4-a
 - C. 1-c, 2-d, 3-d, 4-a
 - D. 1-d, 2-a, 3-c, 4-b
-

11. Following are the substrates of cholesterol synthesis. Arrange them in order of appearance in the cholesterol synthesis pathway. Mevalonate Squalene Lanosterol Zymosterol Desmosterol

- A. 1,2,3,4,5
 - B. 5,4,3,2,1
 - C. 4,3,1,2,5
 - D. 1,3,5,2,4
-

12. A 45-year-old male presents with a history of progressive kidney dysfunction and recurrent episodes of hemolytic anemia. On examination, the patient is found to have corneal opacification on slit-lamp examination. His lipid profile shows high levels of cholesterol and lipoprotein X. What is the most likely diagnosis?

- A. Tangier disease
 - B. Familial hypercholesterolemia
 - C. LCAT deficiency
 - D. Niemann-Pick disease
-

13. Which of the following best describes the primary function of Very Low-Density Lipoprotein (VLDL) in the body?

- A. Transport of dietary cholesterol to the liver

- B. Transport of triglycerides from the liver to peripheral tissues
 - C. Removal of cholesterol from peripheral tissues to the liver
 - D. Storage of excess cholesterol in adipose tissue
-

14. The lipid profile of 55 Y/O female is given below: Triglycerides 150 mg/dL Total cholesterol 350 mg/dL HDL 90 mg/dL Calculate LDL level of this patient.

Triglycerides	150 mg/dL
Total cholesterol	350 mg/dL
HDL	90 mg/dL

- A. 250 mg/dL
 - B. 230 mg/dL
 - C. 200 mg/dL
 - D. 270 mg/dL
-

15. Which enzyme facilitates the conversion of discoidal HDL into spherical HDL in HDL metabolism?

- A. Lipoprotein lipase (LPL)
 - B. Lecithin-cholesterol acyltransferase (LCAT)
 - C. Hepatic lipase
 - D. Cholesteryl ester transfer protein (CETP)
-

16. What is Apo B48 primarily associated with?

- A. High-density lipoproteins (HDL)
- B. Chylomicrons
- C. Very low-density lipoproteins (VLDL)
- D. Low-density lipoproteins (LDL)

17. Match the lipoprotein with its function. Lipoprotein Function

1.Chylomicron
a.Transporters triglycerides from the liver to muscle and adipose tissue

2.Very low-density lipoprotein (VLDL)
b.Transporters cholesterol from peripheral tissue to the liver

3. Low-density lipoprotein (LDL)
c.Transporters cholesterol from the liver to peripheral tissues

4. High-density lipoprotein (HDL)
d.Transporters triglycerides from the gut to muscle and adipose tissue

Lipoprotein	Function
1.Chylomicron	a.Transporters triglycerides from the liver to muscle and adipose tissue
2.Very low-density lipoprotein (VLDL)	b.Transporters cholesterol from peripheral tissue to the liver
3. Low-density lipoprotein (LDL)	c.Transporters cholesterol from the liver to peripheral tissues
4. High-density lipoprotein (HDL)	d.Transporters triglycerides from the gut to muscle and adipose tissue

- A. 1-d, 2-a, 3-c, 4-b
- B. 1-a, 2-b, 3-c, 4-d
- C. 1-a, 2-b, 3-d, 4-c
- D. 1-c, 2-d, 3-b, 4-a

18. Which of the following is true regarding bile acid synthesis regulation?

7 α -hydroxylation is the step which is suppressed
Cholic acid suppresses the synthesis
Farnesoid X receptor is activated
Transcription of the CYP7A1 gene is suppressed

- A. 1,2,3
 - B. 1,3,4
 - C. 2,3
 - D. 2,3,4
-

**19. Which of the following statements is true regarding the fate of cholesterol?
Chenodeoxycholic acid and cholic acid are the primary bile acids
Glycine-taurine conjugation happens in liver peroxisomes
Intestinal bacteria produce secondary bile acids**

- A. 1, 2
 - B. 1, 2, 3
 - C. 2, 3
 - D. Only 1
-

20. Which of the following enzymes catalyzes the principal regulatory step of cholesterol synthesis?

- A. HMG CoA synthase
 - B. HMG CoA reductase
 - C. Cytosolic thiolase
 - D. Squalene epoxidase
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	1
Question 5	3
Question 6	1
Question 7	3
Question 8	1
Question 9	3
Question 10	1
Question 11	1
Question 12	3
Question 13	2
Question 14	2
Question 15	2
Question 16	2
Question 17	1
Question 18	2

Question 19	2
Question 20	2

Solution for Question 1:

Tangier disease

- It is marked by extremely low levels of high-density lipoprotein (HDL) cholesterol due to mutations in the ABCA1 gene on chromosome 9q31, which impairs cholesterol transport.
- Defects in ABCA1 are characterized by two clinical hallmarks; enlarged lipid-laden tonsils and low serum HDL.
- Also known as Familial alpha-lipoprotein deficiency/Hypo alpha lipoproteinemia.
- This mutation causes a decreased level of Apo A1, where HDL is not properly formed.
- Leads to the accumulation of cholesterol in various tissues such as the tonsils, liver, spleen, and corneas.
- Clinical features : Include enlarged orange-colored tonsils, peripheral neuropathy, hepatosplenomegaly and early-onset cardiovascular disease.

Low VLDL levels (Option A) are not characteristic of Tangier disease. VLDL primarily carries triglycerides, and its levels are not significantly affected in Tangier disease.

Low LDL levels (Option C) are not specifically affected in Tangier disease. LDL primarily carries cholesterol to tissues, and its levels can vary independently of HDL levels.

Low LDL receptor activity (Option D): This describes familial hypercholesterolemia, not Tangier disease. Familial hypercholesterolemia is characterized by high LDL cholesterol levels due to defective LDL receptors on cell surfaces.

Solution for Question 2:

Abetalipoproteinemia is a rare genetic disorder characterized by the inability to synthesize apolipoprotein B (Apo B), which is essential for the formation of certain lipoproteins.

Pathophysiology :

- Caused by mutations in the microsomal triglyceride transfer protein (MTP) gene.
- Leads to impaired assembly and secretion of lipoproteins, particularly chylomicrons and very low-density lipoprotein (VLDL). (Option A and D ruled out)
- Results in low serum levels of triglycerides, cholesterol, and fat-soluble vitamins (A, D, E, K).

Clinical Features :

- Fat malabsorption leads to steatorrhea (fatty stools). (Option C ruled out)
- Deficiencies in fat-soluble vitamins can cause: Retinitis pigmentosa (leading to blindness). Neurological deficits (e.g., ataxia, developmental delays).
- Retinitis pigmentosa (leading to blindness).
- Neurological deficits (e.g., ataxia, developmental delays).
- Presence of acanthocytes (spiny red blood cells) on blood smear.
- Retinitis pigmentosa (leading to blindness).
- Neurological deficits (e.g., ataxia, developmental delays).

Management :

- Dietary management to provide essential fatty acids and fat-soluble vitamins.
- Supplementation with medium-chain triglycerides (MCTs) may be beneficial.

Solution for Question 3:

- If the LPL activity normalises after mixing, it is an Apo C II defect.
- Post-heparinized blood samples are used because heparin in circulation competes with heparan sulfate on vessel walls, causing enzymes to detach from the vessel wall and attach to circulating heparin.
- If LPL activity is low in post-heparinized individuals, it is either an Apo C II defect or an LPL defect.

Mixing study to differentiate Apo C defect from LPL defect:

- In the patients post heparinized plasma, add an equal amount of pooled normal plasma which acts as a source of apo C II. If LPL activity normalizes it is an apo C II defect and it will respond to FFP administration. If LPL activity does not normalize it is an LPL defect (Option

A).

- If LPL activity normalizes it is an apo C II defect and it will respond to FFP administration.
- If LPL activity does not normalize it is an LPL defect (Option A).
- If LPL activity normalizes it is an apo C II defect and it will respond to FFP administration.
- If LPL activity does not normalize it is an LPL defect (Option A).

An increase in triglyceride levels is not indicative of Apo C-II deficiency and does not relate directly to the expected outcomes of the mixing study. (Option C)

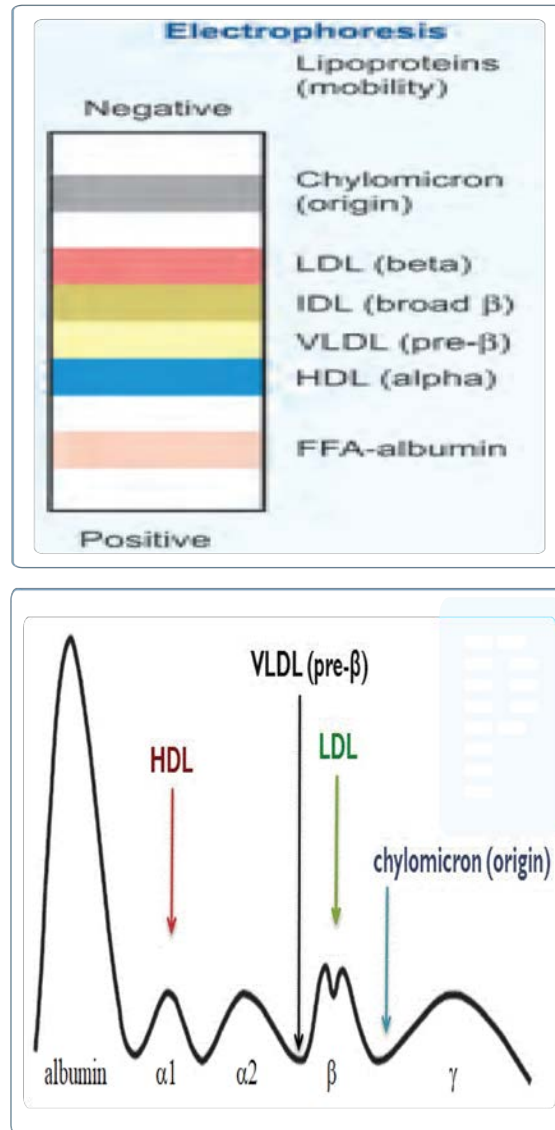
Decrease in LPL activity after mixing would suggest an error in the experiment or another underlying issue, not an Apo C-II deficiency. (Option D)

Solution for Question 4:

Fatty Liver-Contributing Factors	
Choline Deficiency	Choline is crucial for lipid metabolism. Deficiency impairs VLDL secretion , resulting in triacylglycerol accumulation and a fatty liver.
Orotic Acid Interference	Involved in pyrimidine synthesis Disrupts VLDL glycosylation , hindering release.
Nutritional Deficiencies	Deficiencies in essential fatty acids, vitamins (linoleic acid, pyridoxine, pantothenic acid) , and protein contribute to fatty liver.
Ethanol Consumption	Ethanol metabolism increases NADH, hindering fatty acid oxidation and promoting triacylglycerol accumulation.
Vitamin E and Selenium	Protect against liver damage from lipid peroxidation , especially in choline deficiency-induced fatty liver.

Solution for Question 5:

The serum is applied on cellulose acetate, electric current is applied for 2 hours, the strip is dried and stained with lipid dyes, such as Oil Red O. Higher protein content will move faster towards the anode and those with less proteins have minimum mobility.



HDL shows the highest mobility, followed by VLDL, IDL, and LDL. (Option A ruled out)

Lipoproteins with higher protein content move faster towards the anode. (Option B ruled out)

Staining helps visualize lipoproteins, their electrophoretic mobility is crucial for identifying and classifying them. (Option D ruled out)

Solution for Question 6:

The patient presents findings most likely points towards the diagnosis of Tangier disease.

Tangier disease is a rare, autosomal recessive disorder caused by mutations in the ABCA1 (ATP-binding cassette, sub-family A, member 1) gene .

Pathophysiology: The ABCA1 protein is crucial for the efflux of cholesterol and phospholipids to apolipoprotein A-I (ApoA-I), forming HDL particles. Mutations in the ABCA1 gene impair this process, leading to a shortage of HDL and cholesterol accumulation in tissues such as the tonsils, liver, spleen, and peripheral nerves.

Clinical Features:

- Greyish-Orange Tonsils: (Hallmark feature) Tonsils, become enlarged and acquire a greyish-orange hue due to the accumulation of cholesterol esters.



- Hepatosplenomegaly : Due to cholesterol and phospholipid accumulation.
- Peripheral Neuropathy: Multiple nerves can be individually affected due to cholesterol deposition in peripheral nerves, leading to peripheral neuropathy.

LDLR gene mutations (Option B) are associated with familial hypercholesterolemia, which primarily affects low-density lipoprotein (LDL) cholesterol levels, not HDL levels.

APOE gene mutation (Option C) is associated with various lipid disorders and Alzheimer's disease. While APOE influences lipid metabolism, it does not specifically cause the severe HDL

deficiency.

PCSK9 gene mutations (Option D) are linked to familial hypercholesterolemia, affecting LDL cholesterol levels rather than HDL. These mutations result in increased LDL and cholesterol levels and are not associated with HDL deficiency.

Solution for Question 7:

TYPE III (Familial Dysbetalipoproteinemia)	
Lipoprotein Fraction Elevated	Broad β -VLDL, Chylomicrons
Cholesterol Level	Increased
Triglyceride Level	Increased
Appearance of Plasma	Cloudy
Metabolic Defect	Abnormal Apo E Apo CII increased
Features	Palmar xanthomas Increased incidence of vascular disease
Management	Weight reduction Restrict intake of fat and cholesterol Administer polyunsaturated fatty acids (PUFA) and drugs



Type I (Familial chylomicronemia syndrome) (Option A): Characterized by lipoprotein lipase deficiency, causing severe hypertriglyceridemia, eruptive xanthomas, and a creamy layer over clear plasma.

Type IIa (Familial hypercholesterolemia) (Option B): Results from an LDL receptor defect, leading to high LDL levels, atherosclerosis, and tuberos xanthomas.

Type IV (Familial hypertriglyceridemia) (Option D): Associated with elevated VLDL levels, glucose intolerance, and a cloudy or milky appearance of plasma, but not typically with palmar xanthomas.

Solution for Question 8:

Feature	Type IIa (Familial Hypercholesterolemia)	Type IIb (Combined Hyperlipidemia)
Defect	Defective LDL receptors or Apo B-100 ligand region mutation	Increased Apo B, Apo C-II
Lipoprotein Fraction Elevated (Option C)	LDL	LDL and VLDL
Main Lipid Abnormality	Hypercholesterolemia	Combined hyperlipidemia (elevated cholesterol and triglycerides)

Plasma after 24 hours (Option A ruled out)	Clear	Cloudy
Features (Option B ruled out)	Atherosclerosis Coronary artery disease (early onset) Tendon xanthomas	Corneal arcus.
Management (Option D ruled out)	Statins PUFA Lifestyle modifications (diet, exercise)	Statins PUFA Lifestyle modifications (diet, exercise)

Solution for Question 9:

Lipoprotein lipase deficiency : This defect causes familial chylomicronemia syndrome (Type I), leading to hypertriglyceridemia, eruptive xanthomas, hepatomegaly, and a creamy layer over clear plasma.

Hyperlipoproteinemia	Metabolic defect	Lipid profile (Increase in:)	Inheritance
Type I	Lipoprotein lipase deficiency Defect in Apo-C2	Chylomicrons, TG, cholesterol	Autosomal Recessive
Type II A	LDL receptor defect; Apo B-100 mutation (Option A and B)	Cholesterol	Autosomal Dominant
Type II B	Increased Apo B and Apo CII	Cholesterol, LDL, TG	Autosomal Dominant
Type III	Abnormal Apo E; increased Apo CII (Option D)	Chylomicrons, VLDL	Autosomal Recessive

Type IV	Overproduction of VLDL; increased Apo CII	VLDL, TG	Autosomal Dominant
Type V	Secondary to other conditions	VLDL, TG	Autosomal Dominant

Solution for Question 10:

Apolipoprotein	Associated Lipoproteins	Function
Apolipoprotein A-I	HDL (High-Density Lipoprotein)	Major protein component of HDL facilitates reverse cholesterol transport, removing excess cholesterol from tissues and transporting it back to the liver.
Apolipoprotein B-100	LDL (Low-Density Lipoprotein)	Main apolipoprotein of LDL; essential for binding to LDL receptors on various tissues , facilitating cholesterol uptake.
Apolipoprotein B-48	Chylomicrons	Main apolipoprotein of chylomicrons; involved in the transport of dietary lipids from the intestine to other tissues.
Apolipoprotein C-II	Chylomicrons, VLDL (Very-Low-Density Lipoprotein)	Acts as a cofactor for lipoprotein lipase, enhancing triglyceride hydrolysis and aiding in lipoprotein metabolism.
Apolipoprotein C-III	VLDL, HDL	Inhibits lipoprotein lipase activity, affecting triglyceride metabolism and lipoprotein clearance.
Apolipoprotein E	Chylomicrons, VLDL, HDL	Facilitates the uptake of triglyceride-rich lipoproteins by cells through interactions with specific receptors; involved in lipid transport and metabolism.

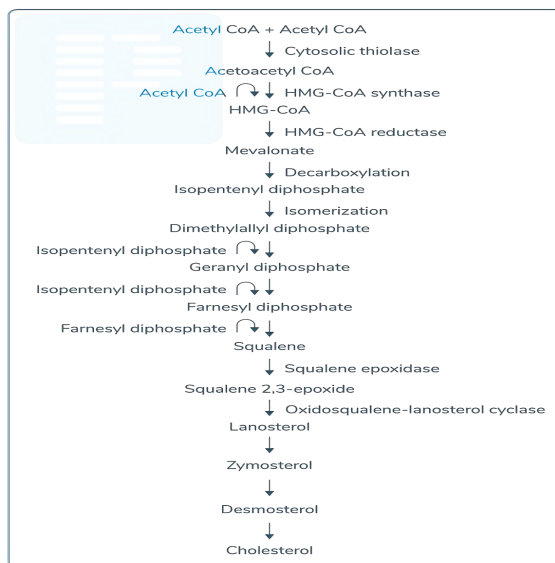
Solution for Question 11:

The substrates of cholesterol synthesis appear in the following order:

So, the correct order of the listed substrates in the cholesterol synthesis pathway is:

Mevalonate → Squalene → Lanosterol → Zymosterol → Desmosterol.

Cholesterol synthesis pathway:



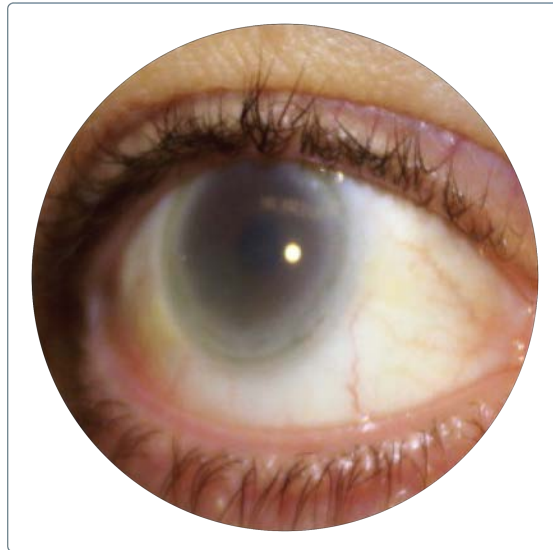
Solution for Question 12:

The most likely diagnosis for the given clinical scenario is LCAT deficiency (Lecithin-Cholesterol Acyltransferase deficiency).

LCAT deficiency exists in two primary forms:

	Partial LCAT Deficiency (Fish eye disease)	Complete LCAT Deficiency (Norum's disease)
LCAT Enzyme Activity	Partial loss, mainly affecting corneal lipid metabolism	Severe impairment or absence, leading to systemic issues

Corneal Appearance	Fish-eye appearance due to cholesterol ester accumulation	Fisheye disease: similar corneal opacification
Systemic Involvement	Less severe, primarily ocular	Severe: hemolytic anemia, CKD, early myocardial infarction, progressive End-stage renal disease



Tangier disease (Option A) is associated with very low HDL levels and characteristic features such as orange tonsils and peripheral neuropathy.

Familial hypercholesterolemia (Option B) typically presents with high LDL levels, xanthomas, and early cardiovascular disease, but not hemolytic anemia or corneal opacification.

Niemann-Pick disease (Option D) is a lysosomal storage disorder involving neurological symptoms and organomegaly.

Solution for Question 13:

VLDL is synthesized in the liver and its main role is to transport triglycerides to peripheral tissues, such as muscle and adipose tissue, where they can be used for energy or stored.

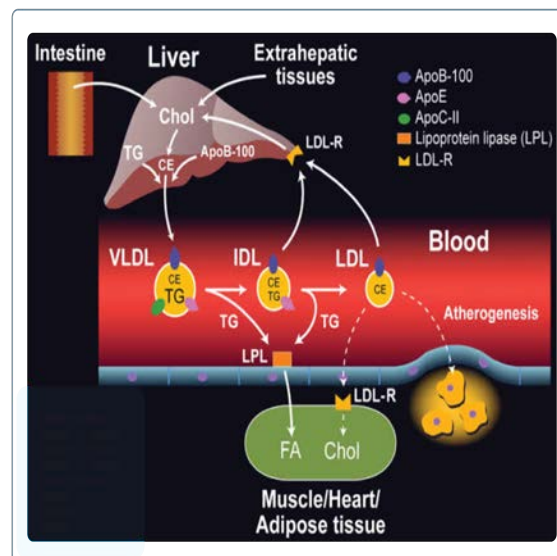
- VLDL is rich in triglycerides and also contains cholesterol and apolipoproteins.
- As VLDL releases triglycerides to cells, it becomes intermediate density lipoprotein (IDL) and eventually low density lipoprotein (LDL).

- This conversion of VLDL to IDL and then to LDL is referred to as lipoprotein cascade pathway

Transport of dietary cholesterol to the liver (Option A) - This function is primarily carried out by chylomicrons, not VLDL. Chylomicrons are responsible for transporting dietary lipids, including cholesterol, from the intestines to the liver.

Removal of cholesterol from peripheral tissues to the liver (Option C) - This function is primarily carried out by high-density lipoprotein (HDL), not VLDL. HDL is involved in reverse cholesterol transport, carrying cholesterol away from the tissues and back to the liver for excretion.

Storage of excess cholesterol in adipose tissue (Option D) - This is not a function of VLDL. Excess cholesterol can be stored in various tissues, but VLDL primarily transports triglycerides, while LDL and other mechanisms are involved in cholesterol storage and transport.



Solution for Question 14:

LDL of the patient can be calculated using

Friedwald's equation:

Total cholesterol = HDL+LDL+TG/5 mg/dL

Substituting the values,

$$350 = 90 + \text{LDL} + 150/5$$

$$\text{LDL} = 350 - 90 - 30$$

$$= 230 \text{ mg/dL}$$

Friedwald's equation is accurate when triglycerides levels are below 400 mg/dL.

Options A, C and D are incorrect.

Solution for Question 15:

Lecithin-cholesterol acyltransferase is the enzyme responsible for the conversion of discoidal HDL (aka Nascent HDL- amphipathic) into spherical HDL (aka Mature HDL- hydrophobic). It catalyzes the transfer of an acyl group from lecithin (phosphatidylcholine) to cholesterol, forming cholesteryl esters.

HDL Metabolism

- **HDL Formation:** Formed in the liver and intestines as nascent, discoidal particles rich in apolipoprotein A-I (apoA-I).
- **Cholesterol Efflux:** ABCA1 Transporter, transfers free cholesterol and phospholipids to nascent HDL. ABCG1 and SR-BI Transporters facilitate cholesterol efflux to HDL.
- **Maturation of HDL:** Lecithin-cholesterol acyltransferase (LCAT) converts discoidal HDL into spherical HDL by forming cholesterol esters that move to the core of HDL particles.
- **HDL Remodeling:** Phospholipid transfer protein (PLTP) and cholesteryl ester transfer protein (CETP) modify HDL size and composition by transferring lipids between lipoproteins.
- **Selective Uptake and HDL Recycling:** Scavenger Receptor Class B Type I (SR-BI) facilitates selective uptake of cholesterol esters from HDL into hepatocytes. CETP transfers cholesteryl esters from HDL to VLDL and LDL for liver uptake.
- **Excretion:** The liver converts cholesterol into bile acids or secretes it into bile for excretion via intestines.

Lipoprotein lipase (LPL) (Option A) is an enzyme that hydrolyzes triglycerides in lipoproteins into free fatty acids and glycerol.

Hepatic lipase (Option C) primarily hydrolyzes triglycerides and phospholipids in intermediate-density lipoproteins (IDL) and low-density lipoproteins (LDL).

Cholesteryl ester transfer protein (Option D) (CETP), facilitates the transfer of cholesteryl esters and triglycerides between lipoproteins.

Solution for Question 16:

- Apolipoprotein B-48 is an essential component of chylomicrons.
- Chylomicrons are lipoproteins that transport dietary lipids from the intestines to other locations in the body.
- Apo B48 is synthesized in the intestines and is integral to the structure and function of chylomicrons, ensuring proper lipid absorption and transport.

Lipoprotein Type	Major Apolipoproteins associated
Chylomicrons	Apo B48, ApoA-I, ApoA-II, ApoC-II, ApoE
VLDL	ApoB-100, ApoC-II, ApoE
LDL	ApoB-100
HDL	ApoA-I, ApoA-II, ApoC-II, ApoE

Solution for Question 17:

Since lipids are insoluble in water, they are transported in blood as lipoproteins

Lipoprotein	Function
Chylomicron	Transports triglycerides from the gut to liver, muscle and adipose tissue
Very low-density lipoprotein (VLDL)	Transports triglycerides from the liver to muscle and adipose tissue
Intermediate-density lipoprotein (IDL)	VLDL remnant

Low-density lipoprotein (LDL)	Transports cholesterol from the liver to peripheral tissues
High-density lipoprotein (HDL)	Transports cholesterol from peripheral tissue to the liver

Solution for Question 18:

Chenodeoxycholic acid is the most potent bile acid which suppresses bile acid production, not cholic acid

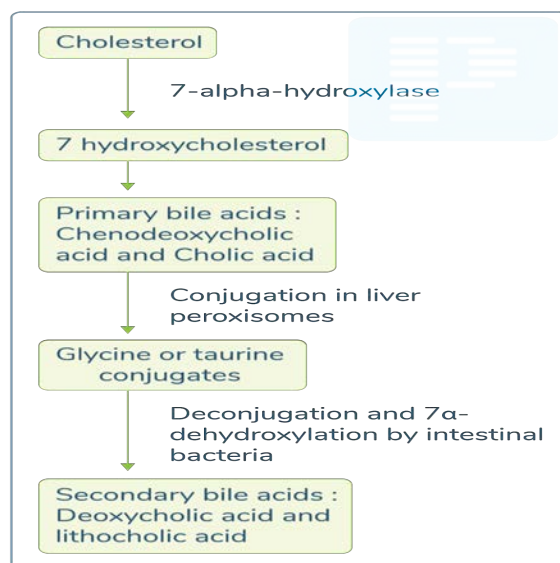
Regulation of bile acid synthesis:

Excess of chenodeoxycholic acid → Farnesoid X receptor (FXR) activation → Suppresses transcription of the CYP7A1 gene → 7- α -hydroxylase enzyme production suppressed → 7 α -hydroxylation of cholesterol (rate-limiting step) is suppressed → Bile acid production is suppressed

Solution for Question 19:

- Chenodeoxycholic acid and cholic acid are the primary bile acids.
- Deoxycholic acid and lithocholic acid are secondary bile acids.
- Primary bile acids are conjugated in liver peroxisomes. They are converted into glycine or taurine conjugates.
- Intestinal bacteria deconjugate and de-hydroxylase the glycine-taurine conjugates to form secondary bile acids.

Fate of cholesterol:



Note: The anions of bile acids and their glycine or taurine conjugates are referred to as bile salts. In the alkaline environment of bile, they are expected to exist in this form.

Solution for Question 20:

- The principal regulatory step in the cholesterol synthesis pathway: HMG-CoA is reduced to mevalonate, catalyzed by HMG-CoA reductase
- Site- cytoplasm and smooth endoplasmic reticulum

Regulation of HMG-CoA reductase:

Category	Regulation Method	Mechanism
Allosteric Modification	Statins (e.g., Lovastatin)	Competitive inhibition of HMG CoA reductase
Covalent Modification	AMP-activated Kinase	Phosphorylates HMG CoA reductase, inactivating it
Protein Phosphatase 1	Dephosphorylates HMG CoA reductase, activating it	
Induction or Repression	Sufficient Cholesterol	Suppresses transcription of the HMG CoA reductase gene

Low Dietary Cholesterol	Increases transcription of the HMG CoA reductase gene	
Insulin and Thyroxine	Increase the activity of HMG CoA reductase (induction)	
Cortisol and Glucagon	Decrease the activity of HMG CoA reductase (repression)	

HMG CoA synthase (Option A) catalyzes the formation of HMG-CoA from acetyl-CoA and acetoacetyl-CoA, which is an upstream step in the cholesterol synthesis pathway but not the rate-limiting step.

- HMG CoA Synthase is also a part of ketone body synthesis pathway (mitochondrial)
- Cholesterol pathway enzyme is cytoplasmic

Cytosolic thiolase (Option C) is involved in an upstream step of cholesterol synthesis by converting two molecules of acetyl-CoA into acetoacetyl-CoA, not directly involved in the regulatory step.

Squalene epoxidase (Option D) catalyzes the conversion of squalene to squalene-2,3-epoxide, a downstream step in the cholesterol synthesis pathway, but it is not a regulatory step.

Previous Year Questions

1. NADPH is used in which of the following process?

- A. Fatty acid synthesis
 - B. Ketone synthesis
 - C. Gluconeogenesis
 - D. Glycolysis
-

2. What is the most frequently occurring end-product of the pathway for fatty acid synthesis among the options provided?

- A. Arachidonic acid
 - B. Oleic acid
 - C. Palmitic acid
 - D. Acetyl CoA
-

3. Which of the following enzyme activity decreases in fasting?

- A. Hormone-sensitive lipase
 - B. Glycogen phosphorylase
 - C. Pyruvate carboxylase
 - D. Phosphofructokinase 1
-

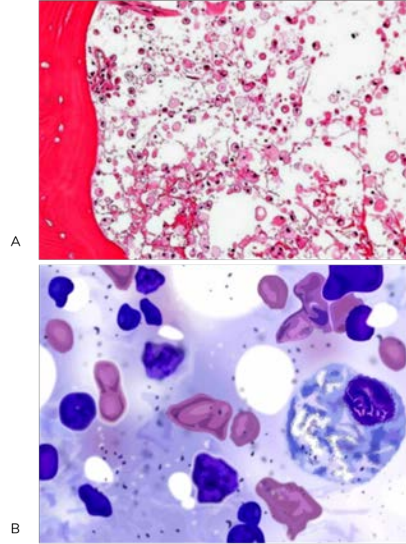
4. Which of the following is the most probable outcome in a 12-year-old girl with poorly managed type 1 diabetes mellitus who is currently experiencing high blood sugar levels?

- A. Glycogenesis in muscle
 - B. Increased protein synthesis
 - C. Increased conversion of fatty acids to acetyl-CoA
 - D. Decreased cholesterol synthesis
-

5. Fat is stored in which form in tissue?

- A. Fatty acid
 - B. Cholesterol
 - C. TGs
 - D. Cholesterol esters
-

6. A 5 year old boy presented with easy fatigability, irritability and inability to concentrate. Labs revealed the following findings a and b (light and oil immersion respectively) on the bone marrow aspiration. Which of the following is the most likely enzyme deficient in this condition?



- A. Hexosaminidase
- B. Glucocerebrosidase
- C. Sphingomyelinase
- D. N-acetylglucosaminidase

7. Which of the following is involved in the synthesis of apolipoprotein B48?

- A. RNA alternate splicing
- B. RNA editing
- C. DNA editing
- D. RNA interference

8. In a patient with obstructive jaundice, which of the following outcomes may be observed in Van der Bergh's reaction?

- A. Direct positive

- B. Indirect positive
 - C. Both positive
 - D. Coproporphyrin
-

9. Which of the following substances is not characterized by amphipathicity?

- A. Sphingolipids
 - B. Phosphoglycerol
 - C. Triglycerides
 - D. Glycolipids
-

10. Which enzyme is not utilized in the synthesis of cholesterol?

- A. Thiolase
 - B. HMG CoA reductase
 - C. HMG CoA synthase
 - D. HMG CoA lyase
-

11. All of the following are lipotropic factors except?

- A. Choline
 - B. Lecithine
 - C. Arginine
 - D. Methionine
-

12. Which coenzyme is required for the de novo synthesis of fatty acids?

- A. NADPH
 - B. TPP
 - C. FAD
 - D. NAD
-

13. Which of the following is the most potent lipid phase antioxidant?

- A. Vitamin A
 - B. Vitamin E
 - C. Vitamin C
 - D. Vitamin K
-

14. Which of the following is cardio protective?

- A. HDL
 - B. LDL
 - C. IDL
 - D. VLDL
-

15. What could be the possible diagnosis for a patient who exhibited tendon xanthomas, with his lipid profile indicating serum cholesterol levels of 283 mg/dL and serum LDL levels of 202 mg/dL, and subsequently initiated on lovastatin treatment?

- A. Familial hypercholesterolemia
- B. Tangier's disease

- C. Lipoprotein lipase deficiency
 - D. Familial hyperchylomicronemia
-

16. Aromatic ring is seen in which of the following amino acids?

- A. Phenylalanine, valine, tryptophan
 - B. Phenylalanine, tryptophan, tyrosine
 - C. Tyrosine, cysteine, tryptophan
 - D. Lysine, methionine, valine
-

17. What is the rate-limiting enzyme of cholesterol synthesis?

- A. HMG CoA reductase
 - B. 7 alpha hydroxylase
 - C. Carnitine acyl transferase 1
 - D. Carnitine acyl transferase 2
-

18. Which of the following possesses cardio-protective properties?

- A. LDL
 - B. VLDL
 - C. HDL
 - D. IDL
-

19. Deficiency of which of the following causes Barth syndrome?

- A. Cardiolipin
 - B. Sphingomyelin
 - C. Glycolipids
 - D. Cerebroside
-

20. In a patient who is tall, obese, known to have hypertension, and has recently been diagnosed with metabolic syndrome during a routine examination, which of the following findings are characteristic findings?

- A. Increase in uric acid and insulin
 - B. Raised blood pressure and normal insulin
 - C. Increase in triglycerides and total cholesterol
 - D. Raised blood sugar and low insulin
-

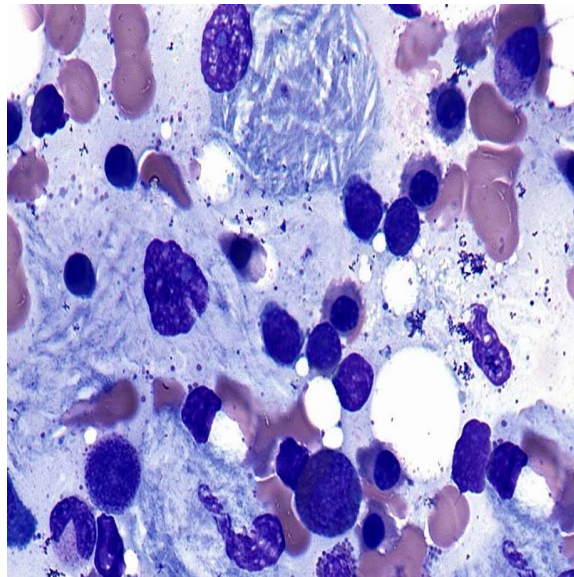
21. A 20-year-old male presented with yellowish discoloration of his skin and sclera. He is otherwise normal. He gives a history of viral infection 10 days ago, which was resolved 2 days back. He also gives a history of similar episodes in the past following any illness. Lab values are given below. What is the most likely diagnosis? Serum bilirubin: 2.4 mg/dL Unconjugated bilirubin: 2.1mg/dL Conjugated bilirubin: 0.3mg/dL Serum AST and ALT: Normal

- A. Dubin-Johnson syndrome
 - B. Crigler-Najar type 1 syndrome
 - C. Gilbert syndrome
 - D. Rotor syndrome
-

22. A child is brought with the inability to walk and difficulty in vision. On funduscopic examination, a cherry red spot is seen. There is no organomegaly. What is the most likely diagnosis?

- A. Tay-Sachs disease
 - B. Niemann-Pick disease
 - C. Hunter syndrome
 - D. Gaucher disease
-

23. What is the probable enzyme deficiency in a child showing symptoms of bone pain and hepatosplenomegaly, as indicated by the findings from a trephine biopsy and aspirate?



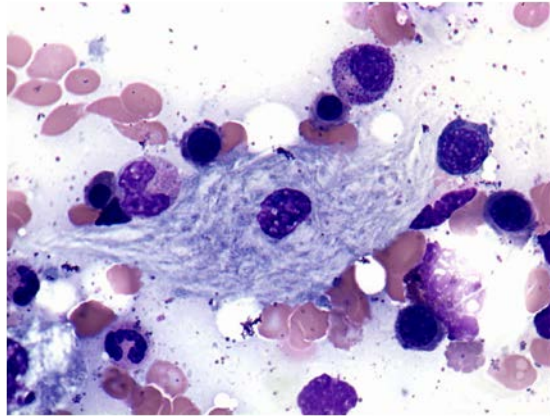
- A. Hexosaminidase
 - B. Glucocerebrosidase
 - C. Sphingomyelinase
 - D. Alpha 1,4-glucosidase
-

24. A patient presented with reduced levels of high-density lipoprotein and an ABCA1 mutation. On examination, tonsils appeared as shown in the image. What is the diagnosis?



- A. Tangier's disease
- B. Abetalipoproteinemia
- C. Hyperlipoproteinemia
- D. Wolman's disease

25. A baby is being evaluated for delayed developmental milestones. On examination, there is hepatosplenomegaly. A microscopic image of the bone marrow evaluation is shown below. What is the diagnosis?



- A. Gaucher disease
 - B. Niemann-pick disease
 - C. Hunter disease
 - D. Hurler disease
-

26. Refsum disease is caused due to the absence of which of the following enzyme?

- A. Absence of alpha oxidation in peroxisomes
 - B. Absence of beta-oxidation in peroxisomes
 - C. Absence of alpha oxidation in lysosomes
 - D. Absence of beta-oxidation in lysosomes
-

27. Zellweger syndrome results from an abnormality in the formation of:

- A. Peroxisomes

- B. Lysosomes
 - C. Mitochondria
 - D. Nucleus
-

28. Which of the options aids in the transportation of fatty acids through the inner mitochondrial membrane?

- A. Acyl carrier protein
 - B. Carnitine
 - C. Lecithin-cholesterol acyltransferase
 - D. Albumin
-

29. Liver produces ketones but cannot use it due to the deficiency of which of the following enzyme?

- A. Thiophorase
 - B. Alkaline phosphatase
 - C. Alanine transaminase
 - D. Thiolase
-

30. Identify the test?



- A. Rothera's test
- B. Benedict's test
- C. Fehling's test
- D. Barfoed's test

31. What is the primary source of cholesterol synthesis?

- A. Malonyl-CoA
- B. Pyruvate
- C. Oxaloacetate
- D. Acetyl-CoA

32. Which of the cycles listed below is absent in muscle tissue?

- A. Glycogenolysis
- B. Ketogenesis

- C. Glycogenesis
 - D. Lipolysis
-

33. A 5-year-old girl was brought to the outpatient department by her mother saying she noted some abnormality in her genital area. On examination, there was the presence of pubic and axillary hair. Which of the following is the least likely diagnosis?

- A. 17-hydroxylase deficiency
 - B. 21-hydroxylase deficiency
 - C. 11 β -hydroxylase deficiency
 - D. 3 beta hydroxy steroid dehydrogenase deficiency
-

34. Which of the following lipase enzyme is inhibited by the insulin?

- A. Lipoprotein lipase
 - B. Pancreatic lipase
 - C. Gastric lipase
 - D. Hormone sensitive lipase
-

35. In cerebrohepatorenal syndrome, an infant exhibits symptoms of hypotonia and seizures. The condition is characterized by the accumulation of which substance in the brain?

- A. Glucose
- B. Long chain fatty acid
- C. Lactic acid

D. Triglycerides

36. After consuming a fatty meal, what is elevated in individuals with lipoprotein lipase deficiency?

- A. Chylomicron
 - B. LDL
 - C. HDL
 - D. Apo A
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	4
Question 4	3
Question 5	3
Question 6	2
Question 7	2
Question 8	1
Question 9	3
Question 10	4
Question 11	3
Question 12	1
Question 13	2
Question 14	1
Question 15	1
Question 16	2
Question 17	1
Question 18	3

Question 19	1
Question 20	3
Question 21	3
Question 22	1
Question 23	2
Question 24	1
Question 25	1
Question 26	1
Question 27	1
Question 28	2
Question 29	1
Question 30	1
Question 31	4
Question 32	2
Question 33	1
Question 34	4
Question 35	2
Question 36	1

Solution for Question 1:

Correct Option A - Fatty acid synthesis:

- NADPH is primarily involved in anabolic processes , including fatty acid synthesis. During fatty acid synthesis, NADPH acts as a reducing agent, providing the necessary reducing equivalents for the synthesis of fatty acids from acetyl-CoA. NADPH is required for the production of malonyl-CoA, the building block of fatty acids, through the action of the enzyme fatty acid synthase.

Incorrect Options:

Option B - Ketone synthesis: Ketone synthesis occurs primarily in the liver mitochondria and doesn't directly rely on NADPH.

Option C - Gluconeogenesis: Gluconeogenesis synthesizes glucose from non-carbohydrate sources. Though NADPH indirectly affects it through other reactions, it's not directly used.

Option D - Glycolysis: Glycolysis breaks down glucose for energy production. NAD⁺ is mainly involved in glycolysis's redox reactions, not NADPH.

Solution for Question 2:

Correct Option C - Palmitic acid

- The most common end-product of the fatty acid synthesis pathway is Palmitic acid.
- Palmitic acid is a 16-carbon saturated fatty acid and serves as a central intermediate in fatty acid metabolism.
- It is synthesized through a series of enzymatic reactions in the fatty acid synthesis pathway, starting from Acetyl CoA.

Incorrect Options :

Option A - Arachidonic acid is a polyunsaturated omega-6 fatty acid stored within lipid bodies in immune cells and is involved in the arachidonic acid synthesis pathway.

Option B - Oleic acid is monounsaturated omega-9 fatty acid acting as an inhibitor in synthesis.

Option D - Acetyl-CoA acts as a precursor for the synthesis of fatty acids.

Solution for Question 3:

Correct Option D - Phosphofructokinase 1:

- During fasting, metabolic adaptations occur to maintain energy balance. A key adaptation is the decrease in phosphofructokinase 1 (PFK-1) activity, a crucial enzyme in glycolysis.
- Glycolysis breaks down glucose to produce ATP, with PFK-1 catalyzing a regulatory step converting fructose-6-phosphate to fructose-1,6-bisphosphate.
- This decrease in PFK-1 activity reduces glycolysis rate, conserving glucose for tissues like the brain while utilizing stored energy sources like glycogen and fatty acids.

Incorrect Options:

Option A - Hormone-sensitive lipase - During fasting, hormone-sensitive lipase breaks down stored fats into free fatty acids and glycerol, providing energy from triglycerides.

Option B - Glycogen phosphorylase - Glycogen phosphorylase increases activity during fasting to break down glycogen into glucose-1-phosphate, maintaining blood glucose levels.

Option C - Pyruvate carboxylase - Pyruvate carboxylase activity increases during fasting to support gluconeogenesis, the synthesis of glucose from non-carbohydrate sources like amino acids and lactate, ensuring a steady glucose supply.

Solution for Question 4:

Correct Option C - Increased conversion of fatty acids to acetyl-CoA

- In inadequately treated type 1 diabetes mellitus, insulin deficiency impairs glucose uptake by cells, leading to increased lipolysis. Fatty acids are metabolized to acetyl-CoA through beta-oxidation, further entering the citric acid cycle for energy production.

Incorrect Options:

Option A - In hyperglycemic conditions like inadequately treated type 1 diabetes, muscle cells undergo glycogenolysis, breaking down glycogen to release glucose into the bloodstream, rather than glycogenesis.

Option B - Type 1 diabetes leads to protein catabolism, not increased protein synthesis. Insulin deficiency affects protein metabolism, causing breakdown for gluconeogenesis and energy.

Option D - Insulin deficiency in type 1 diabetes doesn't directly influence cholesterol synthesis. Cholesterol synthesis is regulated by other factors independent of insulin levels.

Solution for Question 5:

Correct Options C - TGs:

- Fat is primarily stored in the form of triglycerides (TGs) in adipose tissue.
- Triglycerides consist of three fatty acids attached to a glycerol molecule. They serve as the main storage form of fat in the body and are found in specialized fat cells called adipocytes. When energy intake exceeds expenditure, excess dietary fats are converted into triglycerides and stored in adipose tissue for later use.

Incorrect Options:

Option A - Fatty acid: While fatty acids are building blocks of triglycerides and serve as a fuel source during energy demand, they are not the primary form of fat storage in tissues.

Option B - Cholesterol: Cholesterol is essential for various bodily functions but is not primarily stored as fat in tissues. It is synthesized by cells or obtained from dietary sources.

Option D - Cholesterol esters: Cholesterol esters play a role in cholesterol metabolism and transport but are not the primary storage form of fat in tissues. They are found in lipoproteins in the bloodstream.

Solution for Question 6:

Correct Option B - Glucocerebrosidase:

- Gaucher's disease likely involves a deficiency of Glucocerebrosidase, evidenced by clinical presentation and bone marrow aspiration findings. Glucocerebrosidase is responsible for breaking down glucocerebroside, and its deficiency leads to its accumulation in tissues, including bone marrow.

Incorrect Options:

Option A - Hexosaminidase

- Hexosaminidase deficiency causes Tay-Sachs disease, not Gaucher's disease.

Option C - Sphingomyelinase

- Sphingomyelinase deficiency results in Niemann-Pick disease, not Gaucher's disease.

Option D - N-acetylglucosaminidase

- N-acetylglucosaminidase deficiency leads to Sanfilippo syndrome, not Gaucher's disease.
-

Solution for Question 7:

Correct Option B - RNA editing:

- Apolipoprotein B (apoB) is crucial in lipid metabolism, transporting cholesterol and triglycerides in the bloodstream. It has two isoforms: apoB100 synthesized in the liver, part of VLDL, and apoB48 synthesized in the intestine, part of chylomicrons.
- ApoB48 synthesis involves "A-to-I RNA editing," where a specific cytosine is converted to uridine, creating a premature stop codon and producing apoB48.

Incorrect Options:

Option A - RNA alternate splicing: While alternate splicing generates different mRNA isoforms, it's not directly related to apoB48 synthesis.

Option C - DNA editing: ApoB48 synthesis involves RNA editing, not DNA editing.

Option D - RNA interference: RNAi regulates gene expression but doesn't directly impact apoB48 synthesis.

Solution for Question 8:

Correct Option A - Direct positive

- Van der Bergh's reaction: Measures bilirubin levels in blood.
- Two forms of bilirubin: Indirect (unconjugated) and direct (conjugated).
- Obstructive jaundice: Blocks bile flow, causing accumulation of direct bilirubin.
- Van der Bergh's reaction in obstructive jaundice: "Direct positive" result due to increased conjugated bilirubin.

Incorrect Options:

Option B - Indirect positive: Incorrect because obstructive jaundice affects direct, not indirect, bilirubin levels.

Option C - Both positive: Incorrect because obstructive jaundice primarily affects direct bilirubin levels, not both.

Option D - Coproporphyrin: Incorrect because coproporphyrin isn't directly linked to Van der Bergh's reaction or obstructive jaundice.

Solution for Question 9:

Correct Option C - Triglycerides:

- Amphipathic lipid is one which contains a non-polar ester along with polar groups. Triglycerides are neutral lipids /simple lipids as in they contain only FA +alcohol, both being polar compounds only. They are the main storage form of lipids in our bodies.

Incorrect Options:

Option A - Sphingolipids: Diverse group of lipids with a sphingosine backbone, found in cell membranes, important for cell signaling and structure.

Option B - Phospholipid: Contains alcohol, fatty acid, and phosphate; fatty acid is phosphatidic acid, and base can be choline/lecithin or ethanolamine/cephalin. Two types: glycerolipid (parent alcohol: glycerol) and sphingolipid (parent alcohol: sphingosine).

Option D - Glycolipid: Contains alcohol, fatty acid, and carbohydrate; parent alcohol is sphingosine. Phosphate and base are absent.

Solution for Question 10:

Correct Option D: HMG CoA lyase:

- Cholesterol synthesis involves several key enzymes, but HMG CoA lyase is not one of them.
- HMG CoA lyase : This enzyme is involved in ketogenesis , where it converts HMG CoA to acetoacetate (a ketone body), and it is not involved in the cholesterol biosynthesis pathway .

Thiolase (A) : This enzyme catalyzes the first step in cholesterol synthesis, where two acetyl-CoA molecules are combined to form acetoacetyl-CoA.

HMG CoA synthase (C) : This enzyme catalyzes the next step, combining acetoacetyl-CoA with another acetyl-CoA to form HMG CoA (3-hydroxy-3-methylglutaryl-CoA).

HMG CoA reductase (B) : This is the rate-limiting enzyme in cholesterol biosynthesis. It converts HMG CoA to mevalonate , a crucial step in the pathway.

Incorrect Options:

Option A, B and C are utilized in Cholesterol synthesis

Solution for Question 11:

Correct Option C - Arginine.

- Lipotropic factors are compounds that promote the metabolism and transport of fats and lipids in the body. They are involved in various processes, including the breakdown of fats, prevention of fat accumulation in the liver, and promotion of liver health.
- Arginine, while important for protein synthesis and immune function, is not typically considered a lipotropic factor.

Incorrect Options:

Option A - Choline: Essential nutrient crucial for lipid metabolism and liver health, forming phosphatidylcholine, a key component of cell membranes and lipoproteins.

Option B - Lecithin: Phospholipid containing choline, aids in fat breakdown and absorption as an emulsifier in the digestive system.

Option D - Methionine: Essential amino acid involved in protein synthesis and metabolic processes, considered a lipotropic factor for fat metabolism and liver health.

Solution for Question 12:

Correct Option A - NADPH:

- De novo synthesis of fatty acids requires NADPH as a coenzyme. During fatty acid synthesis, NADPH provides reducing equivalents necessary for converting acetyl-CoA into fatty acids. It is utilized by fatty acid synthase (FAS) to reduce intermediate compounds, facilitating fatty acid synthesis.

Incorrect options :

Option B - TPP (Thiamine pyrophosphate) - TPP is involved in decarboxylation reactions but not directly in de novo fatty acid synthesis.

Option C - FAD (Flavin adenine dinucleotide) - FAD functions as a hydrogen and electron carrier but is not specifically required for fatty acid synthesis.

Option D - NAD (Nicotinamide adenine dinucleotide) - NAD participates in redox reactions but is not the coenzyme required for de novo fatty acid synthesis.

Solution for Question 13:

Correct Option B - Vitamin E:

- Among the options given, Vitamin E is the most potent lipid phase antioxidant. It has a crucial role in protecting cell membranes and lipoproteins from oxidative damage by scavenging lipid-based free radicals.

Incorrect Options

Option A - Vitamin A: While important, it's not the most potent lipid phase antioxidant. Vitamin E holds that distinction due to its stronger antioxidant activity in lipids.

Option C - Vitamin C: A potent antioxidant, but its primary action is in the aqueous phase, not the lipid phase, so it's not considered the most potent lipid phase antioxidant.

Option D - Vitamin K: Vital for blood clotting and bone health, but it's not recognized for antioxidant properties in lipids, hence not the most potent lipid phase antioxidant.

Solution for Question 14:

Correct Option A - HDL

- HDL is often referred to as "good" cholesterol because it plays a protective role against cardiovascular disease. HDL helps remove excess cholesterol from the bloodstream and carries it back to the liver for processing and elimination. It has antioxidant and anti-inflammatory properties, and higher levels of HDL are associated with a reduced risk of heart disease. Therefore, option A is the correct answer.

Incorrect Options:

Option B - LDL (Low-Density Lipoprotein): LDL, or "bad" cholesterol, transports cholesterol from the liver to cells and can lead to atherosclerosis if elevated or oxidized. It's a well-known contributor to cardiovascular disease.

Option C - IDL (Intermediate-Density Lipoprotein): IDL is formed from VLDL breakdown and is less studied than LDL or HDL. Though elevated levels can be a risk factor for cardiovascular disease, it's not as recognized as LDL.

Option D - VLDL (Very-Low-Density Lipoprotein): VLDL transports triglycerides from the liver and is a precursor to LDL. Elevated levels can contribute to atherosclerosis and cardiovascular disease.

Solution for Question 15:

Correct Option A - Familial hypercholesterolemia:

- Familial hypercholesterolemia (FH) is a genetic disorder marked by impaired LDL cholesterol clearance, leading to high serum LDL levels and increased cardiovascular risk. Lovastatin, a statin, effectively lowers cholesterol levels, aligning with FH treatment.

Incorrect Options:

Option B - Tangier's disease: Rare disorder with low HDL levels and orange tonsils; unrelated to elevated LDL.

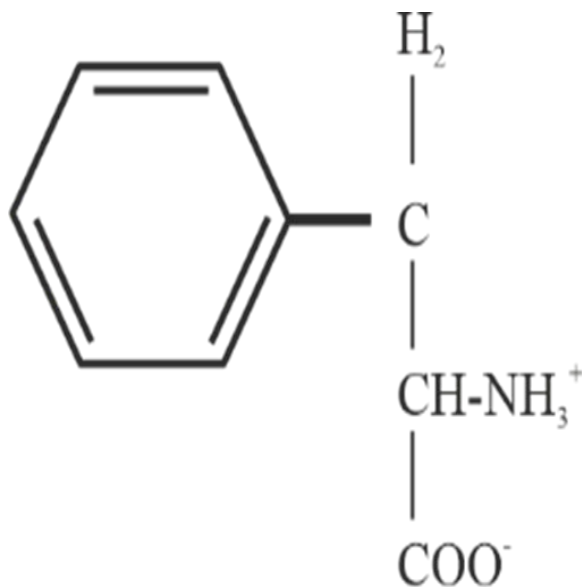
Option C - Lipoprotein lipase deficiency: Causes severe hypertriglyceridemia, not typically elevated LDL.

Option D - Familial hyperchylomicronemia: Results in high triglycerides but not notably elevated LDL, unlike FH.

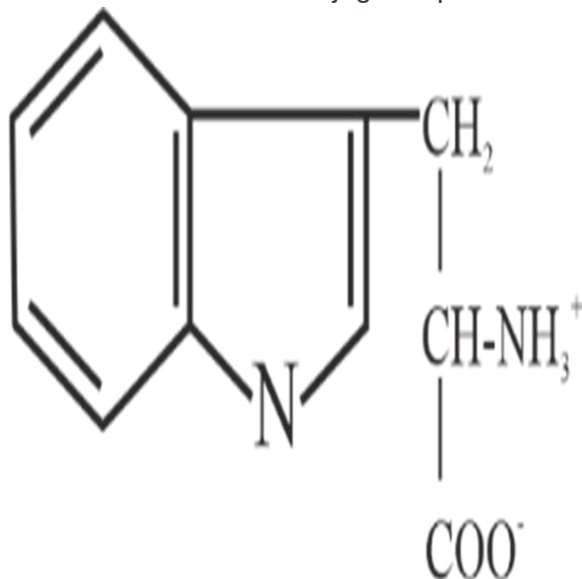
Solution for Question 16:

Correct Option – B Phenylalanine, tryptophan, tyrosine:

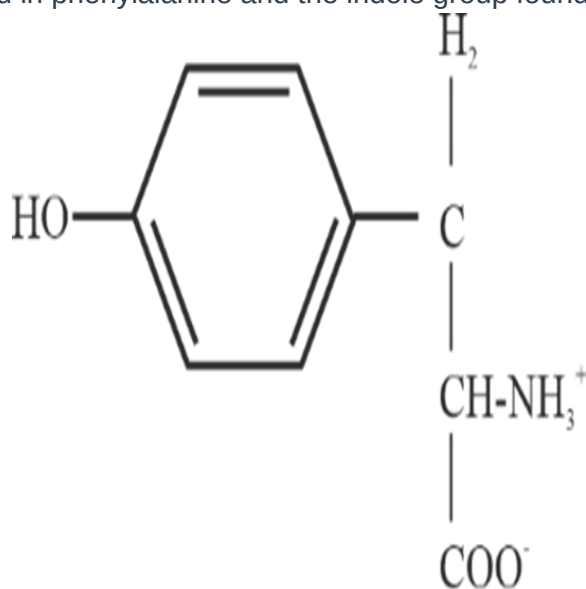
- Phenylalanine: Phenylalanine contains a Benzene group in its side chain, which is a benzene ring. Benzene rings are aromatic due to the delocalization of pi electrons around the ring.



- **Tryptophan:** Tryptophan contains an indole group in its side chain, which is also aromatic. The indole group consists of a benzene ring fused to a five-membered nitrogen-containing ring. This indole ring is aromatic due to its conjugated pi-electron system.



- **Tyrosine:** Tyrosine contains a phenolic group in its side chain, which is aromatic, similar to the benzyl ring found in phenylalanine and the indole group found in tryptophan.

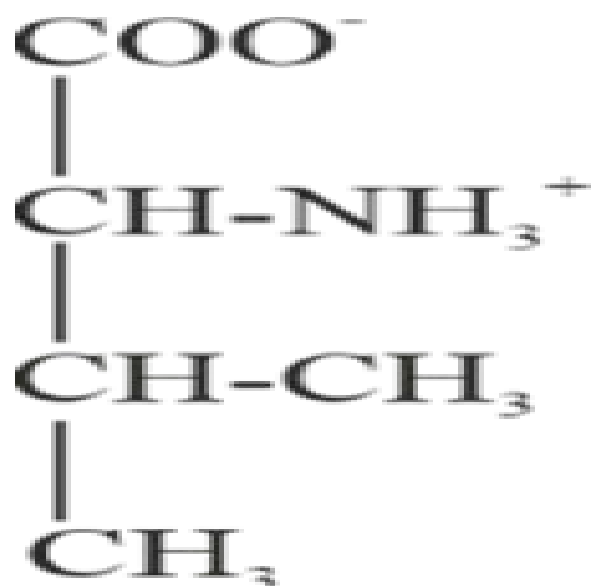


Incorrect Options

Option A - Phenylalanine, valine, tryptophan:

- **Phenylalanine and Tryptophan:** Refer to the explanation of the correct answer.
- **Valine:** Valine does not contain an aromatic ring in its structure. It is a branched chain amino acid.

•



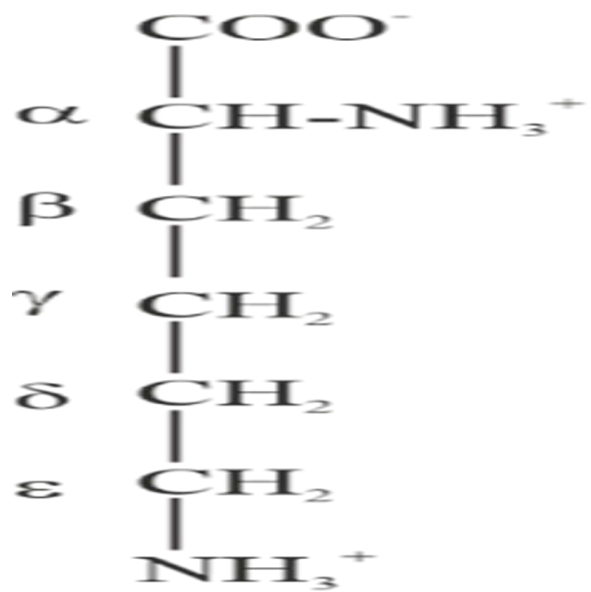
Option C - Tyrosine, cysteine, tryptophan:

- Tyrosine and Tryptophan: Refer to the explanation of the correct answer.
- Cysteine: Cysteine contains a sulfur (-SH) group in its side chain, not an aromatic ring.

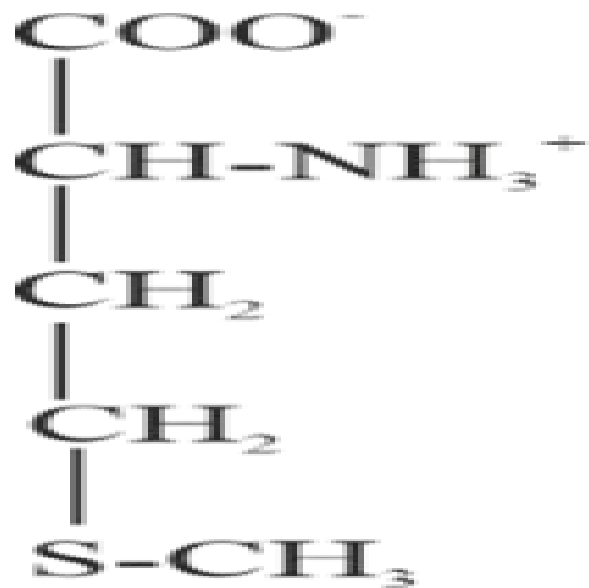


Option D - Lysine, methionine, valine:

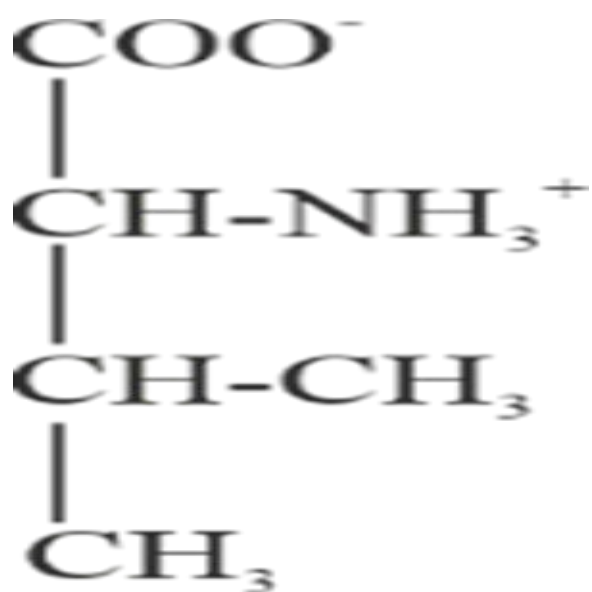
- Lysine: Lysine does not contain an aromatic ring in its structure. It is a dibasic amino acid.
-



- Methionine: Methionine also does not contain an aromatic ring. It is a sulfur containing amino acid.



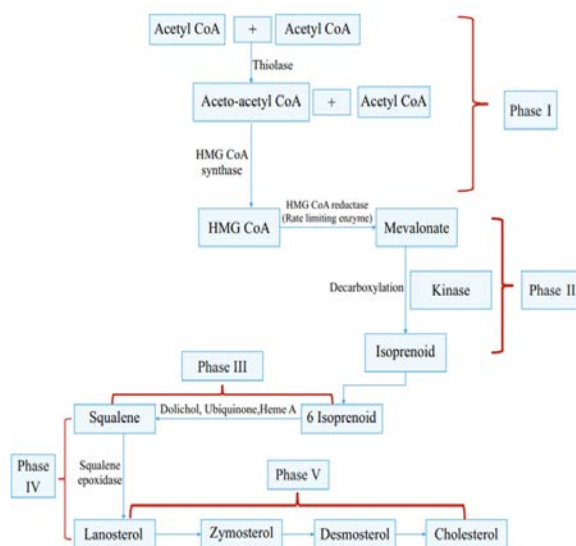
- Valine: Valine does not contain an aromatic ring. It Branched chain amino acid.
-



Solution for Question 17:

Correct Option A - HMG CoA reductase:

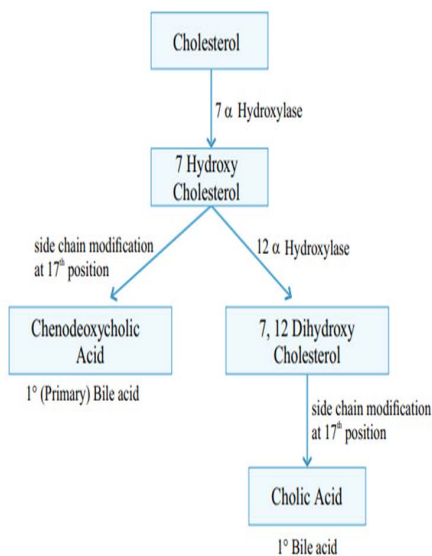
- The reduction of HMG CoA to mevalonate is the rate limiting step of cholesterol synthesis, catalyzed by the enzyme HMG CoA reductase. This enzyme is primarily located in the microsomes, specifically the endoplasmic reticulum.



Incorrect Options

Option B - 7 alpha hydroxylase :

- 7 alpha hydroxylase is an enzyme involved in bile acid synthesis. It catalyzes the conversion of cholesterol to 7-alpha-hydroxycholesterol, which is a precursor in the synthesis of bile acids.



Option C - Carnitine acyl transferase 1:

- Carnitine acyl transferase 1 is an enzyme involved in fatty acid oxidation. It catalyzes the transfer of fatty acids into the mitochondria for beta-oxidation, where they are broken down to produce energy.

Option D - Carnitine acyl transferase 2:

- Carnitine acyl transferase 2 is another enzyme involved in fatty acid oxidation. It is located on the inner mitochondrial membrane and catalyzes the transfer of long-chain fatty acids into the mitochondrial matrix for beta-oxidation.
-

Solution for Question 18:

Correct Option C - HDL:

HDL (High-Density Lipoprotein) is considered cardio-protective due to its beneficial effects on cardiovascular health. Here's why:

Incorrect Options:

Option A - LDL (Low-Density Lipoprotein):

- LDL is often referred to as "bad" cholesterol because elevated levels of LDL can contribute to the development of atherosclerosis and increase the risk of cardiovascular diseases. LDL carries cholesterol from the liver to the tissues, and when LDL levels are high, it can lead to the deposition of cholesterol in the arterial walls.

Option B - VLDL (Very-Low-Density Lipoprotein):

- VLDL is another lipoprotein involved in the transport of triglycerides and cholesterol in the bloodstream. Elevated levels of VLDL are associated with an increased risk of cardiovascular diseases, as VLDL can be converted into LDL particles.

Option D - IDL (Intermediate-Density Lipoprotein):

- IDL is a transitional lipoprotein that is formed during the metabolism of VLDL. It eventually gets converted into LDL. Elevated levels of IDL can be associated with an increased risk of cardiovascular diseases.
-

Solution for Question 19:

Correct Option A - Cardiolipin

- Barth syndrome stems from a cardiolipin deficiency.
- Cardiolipin, found mainly in the inner mitochondrial membrane, is vital for mitochondrial structure and function.
- It participates in crucial mitochondrial processes like the electron transport chain and ATP synthesis.
- A mutation affecting the enzyme tafazzin leads to improper cardiolipin remodeling in Barth syndrome.
- This results in mitochondrial dysfunction, impacting tissues like the heart, skeletal muscles, and immune system.

Incorrect Options:

Option B - Sphingomyelin: Deficiency or abnormal metabolism of sphingomyelin is associated with various sphingolipidoses, such as Niemann-Pick disease.

Option C - Glycolipids: Deficiencies in the enzymes responsible for glycolipid metabolism lead to disorders such as Gaucher disease and Tay-Sachs disease.

Option D - Cerebroside: Cerebrosides are a type of glycosphingolipid. Deficiencies in the enzymes involved in the metabolism of cerebroside result in conditions known as cerebrosideidoses, such as Krabbe disease.

Solution for Question 20:

Correct Option C - Increase in triglycerides and total cholesterol

- Metabolic syndrome is a collection of conditions present in the body that occur combinedly. It increases the risk of stroke, heart disease and type-2 diabetes. The main components of metabolic syndrome include obesity, high blood pressure, high blood triglycerides, low levels of HDL cholesterol and insulin resistance. Increased physical activity and healthy eating are the keys to avoid problems related to metabolic syndrome.

Incorrect Options:

Option A - Incorrect. Insulin resistance is a characteristic feature of metabolic syndrome, but high serum uric acid levels are not associated with it.

Option B - Incorrect. Metabolic syndrome is associated with high blood pressure and insulin resistance.

Option D - Incorrect. While metabolic syndrome increases the risk of type-2 diabetes and insulin resistance, excess weight can contribute to insulin resistance, leading to elevated blood sugar levels.

Solution for Question 21:

Correct Option C - Gilbert syndrome:

- Gilbert Syndrome is a genetic disorder affecting bilirubin processing in the liver due to a mutation in the UGT1A1 gene, which encodes the enzyme UGT.
- Reduced UGT activity leads to bilirubin accumulation, causing jaundice during stress or illness.
- Diagnosis involves blood tests for bilirubin levels, distinguishing it from other syndromes.

Incorrect Option:

Option A - Dubin-Johnson syndrome: Rare disorder impacting bilirubin secretion, resulting in conjugated bilirubin buildup and skin discoloration. Contrastingly, Gilbert's syndrome involves unconjugated bilirubin accumulation.

Option B - Crigler-Najar type 1 syndrome: Rare genetic disorder causing severe unconjugated hyperbilirubinemia, leading to jaundice and complications. Differs from Gilbert's syndrome due to the severity and UDP-glucuronosyltransferase deficiency.

Option D - Rotor syndrome: Rare disorder causing conjugated bilirubin accumulation, akin to Dubin-Johnson syndrome, but distinguished by absence of dark pigments in urine, unlike Dubin-Johnson syndrome.

Solution for Question 22:

Correct Option A - Tay-Sachs disease:

- Tay-Sachs disease is an autosomal recessive disorder caused by the deficiency of beta-hexosaminidase A, leading to GM2 ganglioside accumulation in nerve cells. Symptoms include neurological impairments like vision and hearing loss, muscle weakness, and a cherry red spot on the macula.

Incorrect Options:

Option B - Niemann-Pick disease: Also a lysosomal storage disorder, characterized by acid sphingomyelinase deficiency, resulting in hepatosplenomegaly and neurological deterioration.

Option C - Hunter syndrome: A metabolic disorder caused by iduronate sulphatase deficiency, leading to hepatosplenomegaly. Unlike Tay-Sachs, it lacks visual disturbance and the cherry red spot.

Option D - Gaucher disease: Another lysosomal disorder caused by glucocerebrosidase deficiency, presenting with hepatosplenomegaly, anemia, and bone abnormalities. While it can involve the nervous system, a cherry red spot is not typical

Solution for Question 23:

Correct Option B - Glucocerebrosidase:

- The clinical presentation described, with bone pain and hepatosplenomegaly, is suggestive of Gaucher disease, which is caused by a deficiency of the enzyme glucocerebrosidase. Gaucher disease causes dangerous concentrations of some fats (lipids), primarily the glycolipid glucocerebroside, to build up throughout the body, particularly in the bone marrow, spleen, and liver.

Incorrect Options:

Option A - Hexosaminidase: Associated with Tay-Sachs disease, characterized by neurodegeneration, not hepatosplenomegaly or bone involvement.

Option C - Sphingomyelinase: Deficient in Niemann-Pick disease, presenting with hepatosplenomegaly but not prominent bone pain.

Option D - Alpha 1,4-glucosidase: Associated with Pompe disease, affecting organs like the heart and skeletal muscles, but not causing hepatosplenomegaly.

Solution for Question 24:

Correct Option A - Tangier's disease:

- Tangier's disease is a rare genetic disorder affecting lipid metabolism, resulting in HDL cholesterol deficiency. This deficiency increases the risk of atherosclerosis, leading to cardiovascular issues like heart disease and stroke.
- Symptoms include orange-colored tonsils due to cholesterol accumulation, enlarged liver and spleen, peripheral neuropathy, and cloudy corneas affecting vision.

Incorrect Options:

Options B, C, and D are incorrect.

Solution for Question 25:

Correct Option A - Gaucher disease:

- Gaucher disease results from a deficiency in the enzyme glucocerebrosidase, leading to the accumulation of glucocerebroside in cells, particularly in the spleen, liver, and bone marrow.
- This accumulation causes hepatosplenomegaly, anemia, fatigue, and bone pain.
- Gaucher cells, lipid-laden macrophages with a wrinkled-paper appearance, are characteristic on bone marrow examination.

Incorrect Options - B, C, and D are not associated with the presence of Gaucher cells.

Solution for Question 26:

Correct Option A - Absence of alpha oxidation in peroxisomes:

- Refsum disease results from mutations in the PHYH gene, impacting the production of phytanoyl-CoA hydroxylase enzyme essential for alpha-oxidation of phytanic acid, a fatty acid type.
- Alpha-oxidation occurs in peroxisomes and aids in fatty acid breakdown.
- Insufficient enzyme levels lead to phytanic acid accumulation, notably in the nervous system, triggering symptoms like vision and hearing loss, ataxia, and peripheral neuropathy.

Incorrect Options - B, C and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 27:

Correct Option A - Peroxisomes:

- Zellweger syndrome stems from peroxisomal biogenesis defects. Peroxisomes manage fatty acid breakdown and detoxify harmful substances.
- In Zellweger syndrome, flawed peroxisome development leads to toxic buildup, resulting in developmental delays, vision and hearing issues, and liver dysfunction.

Incorrect Options:

Option B - Lysosomes: Lysosomes handle cellular waste breakdown and component recycling. Biogenesis flaws lead to lysosomal storage disorders, causing substance accumulation and diverse symptoms.

Option C - Mitochondria: Mitochondria generate cellular energy. Biogenesis defects cause mitochondrial disorders, manifesting in symptoms like muscle weakness and neurological issues.

Option D - Nucleus: Nucleus stores genetic material. Biogenesis defects can lead to genetic disorders, including chromosomal abnormalities and single-gene disorders.

Solution for Question 28:

Correct Option B - Carnitine:

- The inner mitochondrial membrane is crossed by the tiny, water-soluble chemical carnitine, which is essential for the oxidation of fatty acids to produce energy.
- Fatty acids cannot pass through the inner mitochondrial membrane, so they must be carried across the membrane using a specialized transport system incorporating carnitine.
- Thus, the movement of fatty acids across the inner mitochondrial membrane, where they are oxidized to produce energy, depends critically on carnitine.

Incorrect Options :

Option A - Acyl carrier protein:

- When fatty acids are synthesized, it is involved in the movement of fatty acids within the cytosol, but not in the movement of fatty acids across the mitochondrial membrane.

Option C - Lecithin-cholesterol acyltransferase:

- The enzyme is not engaged in fatty acid transport across the inner mitochondrial membrane, but rather in the metabolism of phospholipids and cholesterol.

Option D - Albumin:

- Albumin is a protein that functions as a carrier and aids in the transportation of numerous chemicals in the blood, including fatty acids, thus it is not directly engaged in this process.

Solution for Question 29:

Correct Option A - Thiophorase (also known as succinyl-CoA: 3-ketoacid CoA transferase or SCOT):

- This enzyme is responsible for the utilization of ketone bodies in various tissues, including the liver. It plays a crucial role in the breakdown of ketone bodies for energy production. If thiophorase is deficient, the liver may produce ketone bodies but cannot effectively utilize them.

Incorrect Options

Option B - Alkaline phosphatase: Incorrect. This enzyme hydrolyzes phosphate groups but isn't directly related to ketone body utilization.

Option C - Alanine transaminase (ALT): Incorrect. ALT is involved in amino acid metabolism, not ketone body utilization.

Option D - Thiolase: Incorrect. Thiolase catalyzes the final step of ketone body utilization but doesn't affect the liver's ability to produce or use ketones.

Solution for Question 30:

Correct Option A - Rothera's test

- Detects ketone bodies, particularly acetoacetate, in urine.
- Used for diagnosing and monitoring conditions like diabetic ketoacidosis (DKA) and metabolic disorders.
- Involves mixing urine with reagent containing sodium nitroprusside and strong acid.
- Positive result: color change from yellow to purple, intensity correlates with acetoacetate concentration.

Incorrect Options

Option B - Benedict's test:

- Detects reducing sugars like glucose.
- Involves reduction of copper(II) ions to form colored precipitate of copper(I) oxide.
- Result: colored precipitate ranging from green to red, depending on sugar concentration.
- t.

Option C - Fehling's test:

- Also detects reducing sugars.
- Involves reduction of copper(II) ions in Fehling's solution.
- Result: formation of brick-red precipitate of cuprous oxide.

Option D - Barfoed's test:

- Differentiates between monosaccharides and disaccharides.
- Heating test solution with Barfoed's reagent.
- Result: reddish-brown precipitate indicates presence of monosaccharides.

Solution for Question 31:

Correct Option D - Acetyl-CoA

- Acetyl-CoA provides carbon atoms necessary for cholesterol biosynthesis.
- It's required in the first step, forming HMG-CoA, initiating the five-step cholesterol synthesis process.
- Acetyl-CoA is derived from various metabolic sources like fatty acids, carbohydrates, and amino acids.

Incorrect Options:

Option A - Malonyl-CoA - By-product in fatty acid synthesis, not involved in cholesterol synthesis.

Option B - Pyruvate - End product of glycolysis, indirectly converted to Acetyl-CoA, but not directly contributing to cholesterol synthesis.

Option C - Oxaloacetate - Combines with Acetyl-CoA in the citric acid cycle, unrelated to cholesterol synthesis

Solution for Question 32:

Correct Option B - Ketogenesis

- Ketogenesis primarily occurs in liver cell mitochondria and regulated by insulin.
- Fatty acids enter mitochondria via carnitine palmitoyltransferase (CPT-1) and are broken down into acetyl CoA through beta-oxidation.
- It happens when the body lacks sufficient carbohydrates for energy and instead metabolizes fat to produce ketones.
- Ketones serve as an alternative fuel source.

Incorrect Options:

Option A - Glycogenolysis

- Leads to glucose and lactate formation.

- Regulated by glucose-6-phosphatase presence or absence.

Option C - Glycogenesis

- Synthesizes glycogen from glucose.
- Opposite pathway to glycogenolysis.

Option D - Lipolysis

- Releases free fatty acids from adipose tissue.
- These fatty acids are utilized as fuel by most tissues except the brain and erythrocytes.

Solution for Question 33:

Correct Option A - 17-hydroxylase deficiency

- 17 α -hydroxylase deficiency results in low blood levels of estrogens, androgens, and cortisol, leading to compensatory increases in adrenocorticotrophic hormone. This stimulates the production of 11-deoxycorticosterone and corticosterone.
- Symptoms include delayed puberty with primary amenorrhea and hypertension, caused by inefficient sex steroids and excessive mineralocorticoid production. Clinically, females exhibit hyperandrogenism or heterosexual precocious puberty.

Incorrect options:

Option B - 21-hydroxylase deficiency and Option C 11 β -hydroxylase deficiency

- These deficiencies are characterized by hyperandrogenism or heterosexual precocious puberty in females. However, 17 α -hydroxylase deficiency presents with sexual infantilism rather than hyperandrogenism.

Option D - 3 beta hydroxy steroid dehydrogenase deficiency

- This deficiency, part of congenital adrenal hyperplasia, disrupts hormone production and sexual development, but it's distinct from 17 α -hydroxylase deficiency.

Solution for Question 34:

Correct Option D - Hormone sensitive lipase:

- Hormone-sensitive lipase, found in adipose tissue, catalyzes the breakdown of stored fats into fatty acids and glycerol during fasting and starvation, making it a catabolic enzyme.
- It is activated by glucagon, a catabolic hormone, and inhibited by insulin, an anabolic hormone.
- Insulin, released in the fed state to lower blood sugar levels, activates anabolic pathways and enzymes while inhibiting hormone-sensitive lipase.

Incorrect Options:

Option A - Lipoprotein lipase - Found in blood vessel endothelium, it promotes the storage of fats by breaking down circulating triglycerides into fatty acids and glycerol, considered an anabolic process. It is activated by insulin and inhibited by glucagon.

Option B - Pancreatic lipase - Primary enzyme breaking down triglycerides into absorbable forms, activated by insulin.

Option C - Gastric lipase - Present in gastric juice, it breaks down dietary fats into triglycerides for absorption and storage, activated by insulin.

Solution for Question 35:

Correct Option B - Long chain fatty acid:

- Long chain fatty acid : Cerebrohepatorenal syndrome, also known as Zellweger syndrome, is a rare genetic disorder that affects the peroxisomes, which are responsible for the breakdown of very long chain fatty acids (VLCFAs). In this syndrome, there is a deficiency of peroxisomal enzymes, leading to the accumulation of VLCFAs, including in the brain.

Incorrect Options

Option A - Glucose : Glucose is the primary energy source for the brain and is not typically accumulated in the brain in cerebrohepatorenal syndrome.

Option C - Lactic acid : Elevated lactic acid levels can occur in various metabolic disorders, but it is not a characteristic feature of this syndrome.

Option D - Triglycerides : Triglycerides are a form of fat and are not specifically accumulated in the brain in cerebrohepatorenal syndrome.

Solution for Question 36:

Correct Option A - Chylomicron

- Lipoprotein lipase (LPL) deficiency results in impaired triglyceride breakdown in chylomicrons and VLDL, leading to elevated blood levels of these lipoproteins after a fatty meal.

Incorrect Options:

Option B - LDL metabolism is not directly impacted by LPL deficiency, as LDL primarily transports cholesterol from the liver to peripheral tissues.

Option C - HDL metabolism is not directly affected by LPL deficiency; HDL participates in reverse cholesterol transport.

Option D - Apo A levels are not directly influenced by LPL deficiency; Apo A is a component of HDL involved in its structure and function.

Previous Year Questions

1. Which gene defect causes familial hypercholesterolemia?

- A. Apo E
 - B. Apo C II
 - C. LDL receptor
 - D. Apo B48
-

2. Which of the following disorders is characterized by the presence of a cherry red spot on the retina?

- A. Tay-Sachs disease
 - B. Fabry's disease
 - C. Gaucher's disease
 - D. Von Gierke's disease
-

3. A patient presents with xanthomas on the Achilles tendon. Which of the following is the most likely diagnosis?

- A. Familial hypercholesterolemia
 - B. Tangier disease
 - C. Familial hyperchylomicronemia
 - D. Type III hyperlipoproteinemia
-

4. A patient presents to the emergency room with severe abdominal pain, and serum triglyceride levels are 1500 mg/dL. What is the most likely underlying defect?

- A. Apo B-48 defect
 - B. Apo B-100 defect
 - C. Apo C-II defect
 - D. LDL receptor defect
-

5. A patient with multiple tendon xanthomas is found to have a serum cholesterol level of 398 mg/dL and an LDL level of 220 mg/dL. What is the most likely defect?

- A. Lipoprotein lipase deficiency
 - B. LDL receptor defect
 - C. Defective Apo B-100
 - D. PCSK9 gain-of-function mutation
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	3
Question 5	2

Solution for Question 1:

Correct Option C - LDL receptor:

- Familial hypercholesterolemia is most commonly caused by mutations in the gene coding for the LDL receptor.
- This receptor is responsible for the uptake of LDL cholesterol from the bloodstream into the liver.
- Defects in this receptor lead to high levels of LDL cholesterol in the blood.

Incorrect Options:

Option A - Apo E:

- Mutations in the Apo E gene are associated with different lipid disorders, such as familial dysbetalipoproteinemia, but not typically familial hypercholesterolemia.

Option B - Apo C II:

- Apo C II is a cofactor for lipoprotein lipase, and defects in this gene are linked to hypertriglyceridemia, not hypercholesterolemia.

Option D - Apo B48:

- Apo B48 is a form of apolipoprotein B that is involved in chylomicron formation and is not related to familial hypercholesterolemia. The defect in familial hypercholesterolemia involves Apo B100, not Apo B48.
-

Solution for Question 2:

Correct Option A - Tay-Sachs disease:

- This genetic disorder is characterized by a cherry red spot on the retina due to the accumulation of GM2 gangliosides. It presents with neurological symptoms as the disease progresses.

Incorrect Options:

Option B - Fabry's Disease : This is a lysosomal storage disorder caused by a deficiency of the enzyme alpha-galactosidase A. It is not associated with a cherry red spot on the retina. The condition involves skin lesions, pain, and kidney issues.

Option C - Gaucher's Disease: This disorder is caused by a deficiency in the enzyme glucocerebrosidase, leading to the accumulation of glucocerebrosides. It is not associated with a cherry red spot on the retina. Symptoms include hepatosplenomegaly, bone pain, and anemia.

Option D - Von Gierke's Disease: Also known as glycogen storage disease type I, this condition results from a deficiency of the enzyme glucose-6-phosphatase. It does not present with a cherry red spot on the retina but rather with symptoms related to hypoglycemia and liver enlargement.

Solution for Question 3:

Correct Option A - Familial hypercholesterolemia:

- Familial hypercholesterolemia (FH) is an inherited disorder characterized by high levels of low-density lipoprotein (LDL) cholesterol in the blood. Xanthomas, which are deposits of cholesterol in the skin or tendons, are a hallmark of this condition

Incorrect Options:

- Options B, C, and D do not present with tendon xanthomas.
-

Solution for Question 4:

Correct Option C - Apo C-II defect:

- The patient's presentation suggests familial chylomicronaemia syndrome, and it is caused by an Apo-CII defect.

Incorrect Options:

- Options A, B, and D do not cause familial chylomicronaemia.
-

Solution for Question 5:

Correct Option B - LDL receptor defect:

- The most likely defect in a patient with multiple tendon xanthomas, elevated serum cholesterol levels, and LDL levels is an LDL receptor defect. LDL levels in the blood remain high because the liver and other cells cannot efficiently remove LDL from the bloodstream. This leads to elevated cholesterol levels and the formation of xanthomas due to the accumulation of cholesterol in tissues.

Incorrect Options:

- Options A, C, and D do not present with the given findings.
-

Protein Chemistry, Hemoglobin, Myoglobin & EPP Chromatography

1. Which of the following statements about selenocysteine and Pyrrolysine is correct?

- A. Pyrrolysine is incorporated into proteins at UGA codons with the help of a specific tRNA.
- B. The incorporation of selenocysteine requires a special mRNA structure known as the selenocysteine insertion sequence (SECIS).
- C. Selenocysteine is synthesized from cysteine during protein translation.
- D. Pyrrolysine is only found in eukaryotic organisms and not in prokaryotes.

2. Which of the following statements correctly describes the metabolic fate of specific amino acids?

- A. All amino acids can be classified as either glucogenic or ketogenic without exceptions.
- B. Aspartate is a purely ketogenic amino acid.
- C. Phenylalanine can be converted to both glucose and ketone bodies.
- D. Lysine is exclusively glucogenic.

3. An athlete undergoing high-altitude training adapts to decreased oxygen availability by altering haemoglobin to enhance oxygen delivery. Which structural form of haemoglobin is predominantly favoured to enhance oxygen release to tissues?

- A. Taut structure
- B. Relaxed structure
- C. Globular structure

D. Tertiary structure

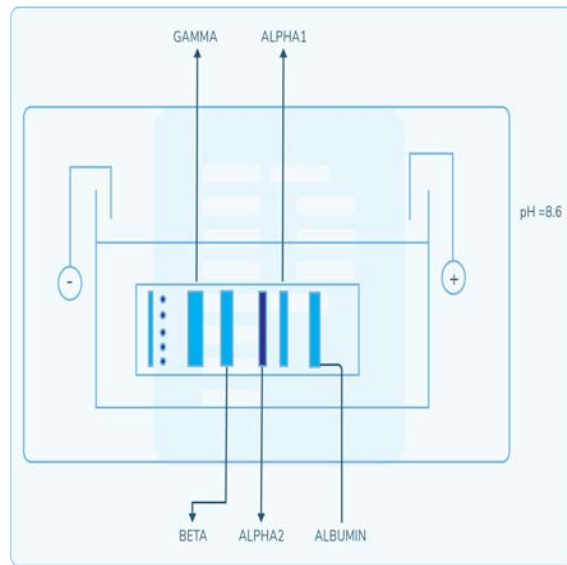
4. Which of the following techniques is most widely used to estimate a patient's HbA1c level for monitoring long-term glucose control?

- A. Cation exchange chromatography
 - B. Anion exchange chromatography
 - C. Gel filtration chromatography
 - D. Paper chromatography
-

5. Based on which characteristic the technique SDS-PAGE is used to separate proteins?

- A. Charge
 - B. Molecular Weight
 - C. Size
 - D. Shape
-

6. The electrophoresis pattern in the image shows a prominent alpha-2 band. Which disease is most likely associated with this finding?



- A. Nephrotic syndrome
- B. Multiple myeloma
- C. Decompensated chronic liver disease
- D. Nephritic syndrome

7. A laboratory technician prepares samples for electrophoresis as the person is suspected to suffer from multiple myeloma. Which sample type is preferred for this test to ensure accurate results?

- A. Serum
- B. Plasma
- C. Whole blood
- D. CSF

8. A 39-year-old male complains of unexplained fatigue and joint pain and visits his physician. The physician suspects chronic inflammatory disease and advises

serum protein electrophoresis, which shows an increase in alpha 2 bands. Which of the following proteins shows an increase in alpha 2 bands?

- A. Albumin
 - B. Macroglobulin
 - C. Gamma globulin
 - D. Transferrin
-

9. Which protein migrates the farthest from the application point towards the anode in serum protein electrophoresis?

- A. Albumin
 - B. Gamma globulin
 - C. Alpha-1 globulin
 - D. Beta globulin
-

10. All the following factors determine the mobility of proteins in electrophoresis except?

- A. Charge
 - B. Isoelectric pH
 - C. Molecular weight
 - D. Shape
-

11. What is the most likely cause of the rightward shift of the oxyhemoglobin dissociation curve?

- A. Decreased 2,3 -BPG

- B. Decreased CO₂
 - C. Increased pH
 - D. Increased temperature
-

12. A newborn is diagnosed with a condition where their haemoglobin has an unusually high affinity for oxygen. What is the physiological basis for the increased oxygen affinity of fetal haemoglobin compared to adult haemoglobin?

- A. Binds more 2,3-BPG, increasing oxygen affinity.
 - B. HbF contains serine that reduces the positive charge of the globin side chain.
 - C. Higher histidine content, reducing oxygen affinity.
 - D. Binds oxygen more loosely than adult Hb
-

13. A patient with anaemia has elevated levels of 2,3-Bisphosphoglycerate (2,3-BPG) in their red blood cells. How does this increased 2,3-BPG level affect the patient's haemoglobin in terms of oxygen delivery?

- A. Forms salt bridges with positively charged residues forming a relaxed structure
 - B. Disrupts salt bridges formed in the taut structure, decreasing oxygen affinity
 - C. Forms salt bridges with positively charged residues stabilising taut structure
 - D. Binds to the heme directly, decreasing the oxygen affinity
-

14. Which of the following statements is TRUE regarding pKa and its implications in biochemistry?

- A. The lower the pKa of the amino group, the weaker the acid.
- B. pKa remains constant and does not change with the medium.

- C. The net charge of a protein does not depend on the pKa value of the functional group.
 - D. When $\text{pH} = \text{pKa} \pm 1$, substances act as buffers.
-

15. Which of the following correctly describes the role of histidine residues in myoglobin in relation to carbon monoxide (CO) binding?

- A. Prevent CO binding by increasing the heme's affinity for oxygen.
 - B. Enhance CO binding, similar to oxygen.
 - C. Inhibit CO binding by sterically hindering the heme iron.
 - D. It does not influence the binding of CO to the heme.
-

16. Which cellular component is primarily responsible for degrading intracellular proteins tagged with ubiquitin?

- A. Lysosomes
 - B. Proteasomes
 - C. Cathepsins
 - D. Endosomes
-

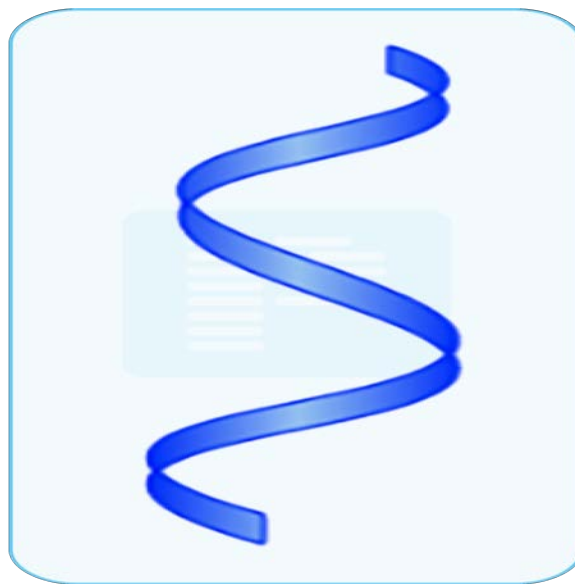
17. Which protein assists in proper folding by binding to hydrophobic regions of other proteins during the folding process?

- A. Proteases
 - B. Chaperonins
 - C. Ubiquitin
 - D. Prions
-

18. Which of the following techniques provides the highest resolution for determining the primary structure of proteins?

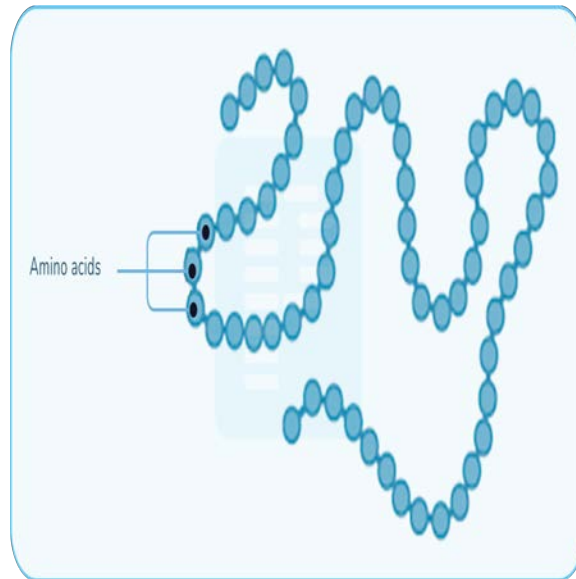
- A. Nuclear Magnetic Resonance (NMR) spectroscopy
 - B. X-ray crystallography
 - C. Sanger's sequencing
 - D. Optical rotatory dispersion
-

19. What type of bond is primarily responsible for stabilising the structure in proteins as shown in the image?



- A. Covalent bonds between side chains
 - B. Ionic bonds between charged amino acids
 - C. Hydrogen bonds between backbone atoms
 - D. Disulfide bridges between cysteine residues
-

20. Which of the following interactions is primarily responsible for maintaining the structure of proteins, as shown in the image?



- A. Peptide bonds
- B. Hydrogen bonds
- C. Disulfide bridges
- D. Van der Waals interactions

21. Match the following protein tests with their primary purpose or detection target:

1.Ninhydrin test a. Detects compounds containing 2 or more peptide bonds

2.Xanthoproteic test b. Detects aromatic amino acids 3.Sakaguchi test c. Detects arginine

4.Biuret test d. Detects alpha-amino acids

1.Ninhydrin test	a. Detects compounds containing 2 or more peptide bonds
2.Xanthoproteic test	b. Detects aromatic amino acids
3.Sakaguchi test	c. Detects arginine

4. Biuret test	d. Detects alpha-amino acids
----------------	------------------------------

- A. 1-b, 2-c, 3-d, 4-a
 - B. 1-d, 2-b, 3-c, 4-a
 - C. 1-a, 2-d, 3-b, 4-c
 - D. 1-c, 2-a, 3-d, 4-b
-

22. Which of the following is true regarding the properties of glycine?

- A. Has asymmetric carbon atoms
 - B. Doesn't exhibit stereoisomerism
 - C. Optically active
 - D. Complex amino acid
-

23. A 22-year-old woman presents with complaints of chronic fatigue and frequent infections. She follows a strict vegetarian diet and consumes less protein-rich food. Laboratory tests reveal a decreased level of a particular amino acid. Which of the following amino acids is likely deficient in this patient?

- A. Threonine and Lysine
 - B. Tyrosine and Tryptophan
 - C. Alanine and Glycine
 - D. Cysteine and Methionine
-

24. Which of the following amino acids contains a hydroxyl group (OH) in its side chain?

- A. Phenylalanine
 - B. Serine
 - C. Cysteine
 - D. Glutamine
-

25. Which of the following is not a characteristic of the isoelectric pH (pI) of a protein?

- A. High solubility
 - B. Low buffering capacity
 - C. No net charge
 - D. Minimal mobility
-

Correct Answers

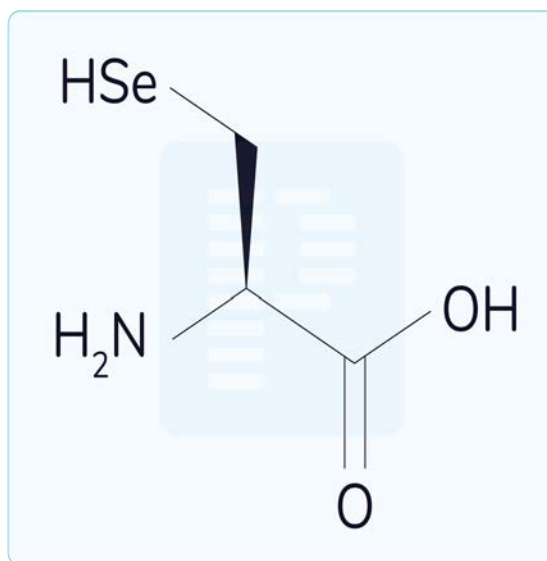
Question	Correct Answer
Question 1	2
Question 2	3
Question 3	1
Question 4	1
Question 5	2
Question 6	1
Question 7	1
Question 8	2
Question 9	1
Question 10	2
Question 11	4
Question 12	2
Question 13	3
Question 14	4
Question 15	3
Question 16	2
Question 17	2
Question 18	3

Question 19	3
Question 20	1
Question 21	2
Question 22	2
Question 23	1
Question 24	2
Question 25	1

Solution for Question 1:

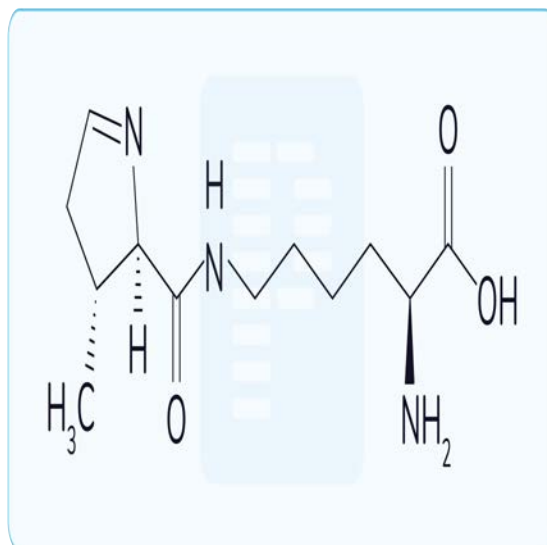
Selenocysteine :

- Classification : Selenocysteine is an L- α -amino acid known as the "21st amino acid."
- Presence : Found in proteins across all domains of life, including approximately 25 selenoproteins in humans (e.g., certain peroxidases, reductases, and selenoprotein P).
- Synthesis: Selenocysteine is synthesized from serine through a series of enzymatic modifications that incorporate selenium and convert it into selenocysteinyl-tRNA , facilitating its incorporation into proteins.
- tRNA : Selenocysteine is incorporated into proteins via a specialized tRNA known as tRNA^{Sec}.
- Codon : It is uniquely coded by the UGA codon, which typically signals a stop in translation. (Option A)
- SECIS Element : Its incorporation requires a specific mRNA structure called the Selenocysteine Insertion Sequence (SECIS) , allowing the ribosome to recognize UGA as coding for selenocysteine rather than as a stop signal.
- Function : Essential for various biological processes, including the conversion of thyroxine (T4) to triiodothyronine (T3) by iodothyronine deiodinases.



Pyrrolysine:

- Classification : Pyrrolysine is often referred to as the 22nd genetically encoded amino acid .
- Presence : It is primarily found in certain archaea and bacteria, specifically in some methanogenic organisms.
- Synthesis : Pyrrolysine is synthesized from lysine through a series of enzymatic modifications that form a pyrroline-carboxylate ring ."
- tRNA : Pyrrolysine is incorporated into proteins using a specific tRNA known as tRNA^{Pyl}.
- Codon : It is uniquely encoded by the UAG codon, which usually signals a stop in protein synthesis. In the context of pyrrolysine, UAG serves as a sense codon when recognized by tRNA^{Pyl}. (Option A)
- In the context of pyrrolysine, UAG serves as a sense codon when recognized by tRNA^{Pyl}. (Option A)
- Role : Pyrrolysine plays a crucial role in specific enzymes involved in methane production, aiding in the metabolism of methylamines.
- In the context of pyrrolysine, UAG serves as a sense codon when recognized by tRNA^{Pyl}. (Option A)



Solution for Question 2:

- Phenylalanine is classified as both glucogenic and ketogenic because its catabolism produces: Fumarate → Enters the TCA cycle , contributing to gluconeogenesis (glucose production). Acetoacetate → A precursor for ketone bodies .
- Fumarate → Enters the TCA cycle , contributing to gluconeogenesis (glucose production).
- Acetoacetate → A precursor for ketone bodies .
- This dual fate allows phenylalanine to contribute to both energy sources.

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Acetoacetate → A precursor for ketone bodies .

This dual fate allows phenylalanine to contribute to both energy sources.

Metabolism of Amino Acids: Glucogenic vs. Ketogenic

Classification of Amino Acids

Amino acids are classified based on their metabolic fate:

Type	Definition	Examples:
Purely Glucogenic	Amino acids that are converted into pyruvate or TCA cycle intermediates (e.g., oxaloacetate, α -ketoglutarate) and contribute to glucose synthesis via gluconeogenesis .	Aspartate, Alanine, Glycine, Serine, Cysteine
Purely Ketogenic	Amino acids that yield only acetyl-CoA or acetoacetate , leading to the formation of ketone bodies .	Leucine, Lysine
Both Glucogenic & Ketogenic	Amino acids that can be converted into both glucose and ketone bodies , contributing to energy metabolism through multiple pathways.	Phenylalanine, Tyrosine, Tryptophan, Isoleucine, Threonine

All amino acids can be classified as either glucogenic or ketogenic without exceptions. (Option A):

- Incorrect because some amino acids (e.g., ornithine and citrulline) are involved in the urea cycle rather than energy metabolism.

Incorrect because some amino acids (e.g., ornithine and citrulline) are involved in the urea cycle rather than energy metabolism.

Aspartate is a purely ketogenic amino acid. (Option B):

- Incorrect because aspartate is purely glucogenic , converting into oxaloacetate , which enters the gluconeogenic pathway .

Incorrect because aspartate is purely glucogenic , converting into oxaloacetate , which enters the gluconeogenic pathway .

Lysine is exclusively glucogenic. (Option D):

- Incorrect because lysine is purely ketogenic , forming acetoacetate , which is used for ketone body synthesis , and does not contribute to glucose production.

Incorrect because lysine is purely ketogenic , forming acetoacetate , which is used for ketone body synthesis , and does not contribute to glucose production.

Solution for Question 3:

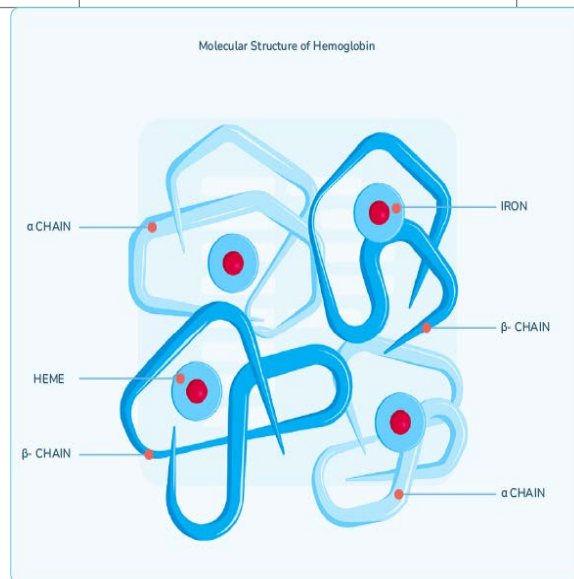
Taut form of hemoglobin has a lower affinity for oxygen, facilitating oxygen release to tissues under conditions of low oxygen and high CO_2/H^+ (Bohr effect).

Structure of Haemoglobin:

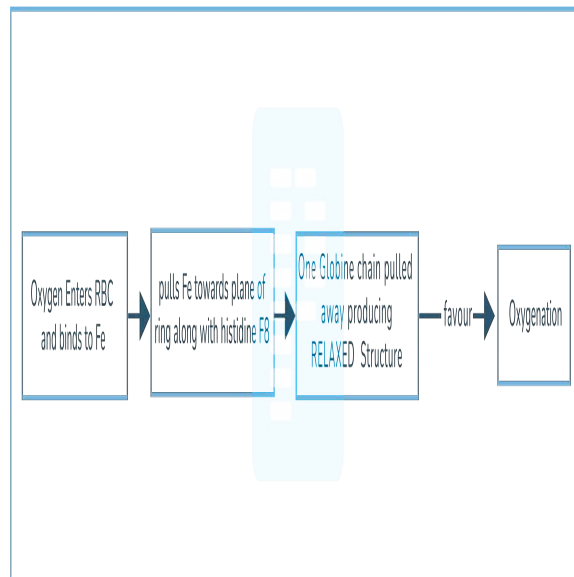
- Hemoglobin consists of four globin chains, each with an associated heme group containing Fe^{2+} .
- Strong polypeptide chain interaction: Taut structure favours deoxygenation.
- Weak polypeptide chain interaction: Relaxed structure favours oxygenation.

Haemoglobin Structure and Oxygen Binding

Structural Form	Description	Effect on Oxygen Binding
Taut (T) form	Low oxygen affinity, stabilized by CO_2 , H^+ , and 2,3-BPG	Favours oxygen release to tissues
Relaxed (R) form	High oxygen affinity, stabilized by oxygen binding	Favours oxygen uptake in lungs

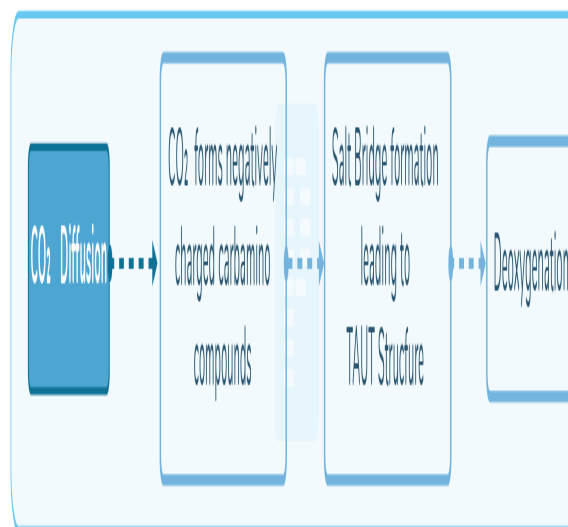


At the Tissue site:



Chloride Shift: Bicarbonate (HCO^-) is exchanged for chloride (Cl^-) via the antiporter, causing RBCs to swell due to increased chloride and water.

In the Lung:



Haldane Effect:

- Reciprocal binding of oxygen and carbon dioxide is Haldane effect.
- It is observed in the lung side

Relaxed structure (Option B): Incorrect because the R form has a high affinity for oxygen and is favored in oxygen-rich environments (lungs), not in tissues needing oxygen.

Globular structure (Option C): Incorrect because hemoglobin's globular nature does not directly impact oxygen release. The T and R states determine its function.

Tertiary structure (Option D): Incorrect because oxygen binding and release are determined by quaternary structural changes ($T \leftrightarrow R$), not just tertiary folding.

Solution for Question 4:

- Chromatography is a separation technique that distinguishes solutes based on their distribution between a stationary and a mobile phase.
- Solutes with a higher affinity for the mobile phase move faster, while those with a higher affinity for the stationary phase move slower or not at all.
- Multiple solutes can be separated using this method.

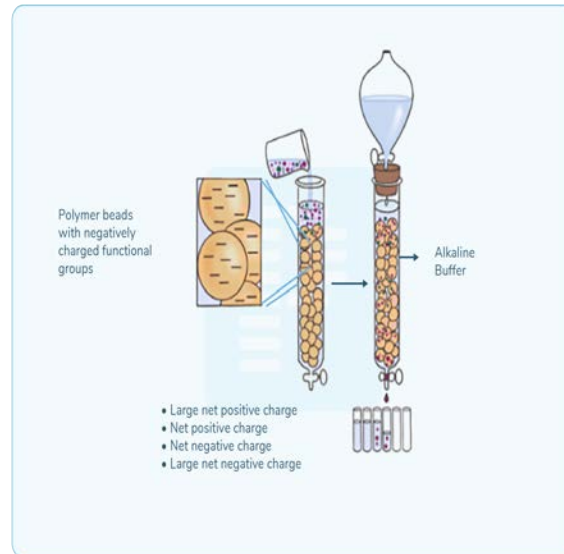
Types of Chromatography:

Types of Chromatography	Description
Cation exchange chromatography (Option A)	Carboxymethyl sepharose is a widely used example of cation exchange chromatography, which is commonly employed for HbA1c estimation. This technique separates positively charged (cationic) molecules using an anionic stationary phase. Initially, the protein mixture is held in the stationary phase to allow impurities to be removed. Once all the impurities are removed, a technique known as elution is used upon the stationary phase to derive the desired result. In this, positively charged proteins bind to the negatively charged stationary phase.
Anion exchange chromatography (Option B)	It is used to separate proteins with a negative charge. The stationary phase consists of cationic groups, such as quaternary ammonium groups. This method retains anions while allowing other components to pass through.
Gel filtration chromatography (Option C)	Also known as size exclusion chromatography, this technique separates proteins based on their size. Smaller proteins get trapped in the pores of the gel matrix, slowing their movement, while larger proteins move quickly through the column.

Paper chromatography (Option D)

It is a type of planar chromatography conducted on a flat surface. This method separates amino acids based on their polarity. Nonpolar amino acids move faster, while polar amino acids stay closer to the application point.

Cation exchange chromatography

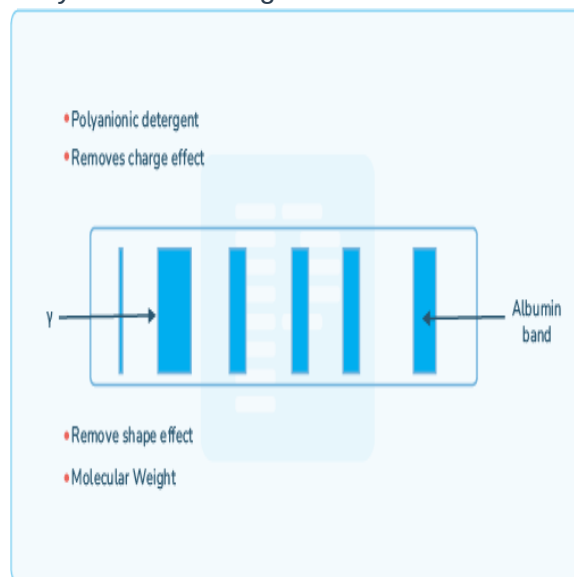


Solution for Question 5:

The SDS-PAGE technique (Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis) separates proteins based on their molecular weight.

- In this technique, proteins are denatured and bound by SDS (Sodium Dodecyl Sulfate), which is an anionic detergent.
- SDS binds to proteins and imparts a uniform negative charge proportional to the protein's mass, effectively masking the protein's original charge and shape. (Option A and D ruled out)
- The denatured proteins are then subjected to an electric field within a polyacrylamide gel matrix.
- Since SDS provides a uniform charge-to-mass ratio, proteins move through the gel based on molecular weight, with smaller proteins migrating faster than larger ones.

- Molecular Weight vs. Size: (Option C ruled out) Molecular weight refers to the mass of a molecule. Size refers to the physical dimensions of a molecule. While larger molecules generally have higher molecular weights, factors like protein shape and conformation can influence their size. SDS coating of the protein removes the influence that shape has on the migration of the protein.
- Molecular weight refers to the mass of a molecule.
- Size refers to the physical dimensions of a molecule.
- While larger molecules generally have higher molecular weights, factors like protein shape and conformation can influence their size.
- SDS coating of the protein removes the influence that shape has on the migration of the protein.
- SDS-PAGE Separation: SDS-PAGE eliminates the influence of protein shape and charge, ensuring that separation is primarily determined by molecular weight.
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- While larger molecules generally have higher molecular weights, factors like protein shape and conformation can influence their size.
- SDS coating of the protein removes the influence that shape has on the migration of the protein.
- SDS-PAGE eliminates the influence of protein shape and charge, ensuring that separation is primarily determined by molecular weight.



Solution for Question 6:

- A prominent alpha-2 band on serum protein electrophoresis (SPEP) is a characteristic finding in nephrotic syndrome .
- In nephrotic syndrome: Albumin is lost in the urine due to increased glomerular permeability. Alpha-2 macroglobulin remains elevated because it is too large to be filtered out, leading to an enhanced alpha-2 band . Other small proteins are lost, leading to a relative increase in alpha-2 globulins (alpha-2 macroglobulin, haptoglobin, and ceruloplasmin) .
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Disorder	EPP Pattern
Multiple Myeloma (Option B)	It is characterised by a narrow and intense band in the gamma region, the M band. This band indicates the presence of a monoclonal protein (M protein), which is a hallmark of multiple myeloma. {{caption_text}}
Nephrotic syndrome (Option A)	It shows a relative prominence of the alpha-2 band. This pattern results from the selective loss of smaller proteins, such as albumin, in the urine while larger proteins, such as alpha-2 macroglobulin, remain in the blood.
Decompensated chronic liver disease (DCLD) (Option C)	It exhibits beta-gamma bridging. This occurs due to polyclonal immunoglobulin production, leading to a continuous smear between the beta and gamma regions.

Nephritic Syndrome (Option D) involves glomerular inflammation leading to hematuria, proteinuria, and decreased kidney function.

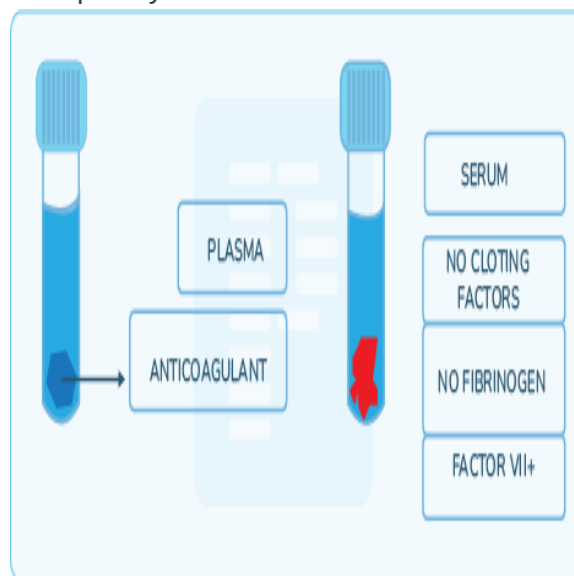
- It may show increased alpha-1 and alpha-2 globulins due to acute phase reactants but not a prominent alpha-2 band.

Solution for Question 7:

- The preferred sample for electrophoresis is Serum due to the absence of fibrinogen.
- Serum (obtained after clotting) lacks clotting factors and fibrinogen.
- When a blood sample is collected in a tube with an anticoagulant, it separates into plasma at the top and blood cells below.
- Conversely, in a tube without an anticoagulant, blood clots due to the coagulation cascade, resulting in serum separation.

Plasma (Option B)

- Fibrinogen present in plasma can migrate to the beta-gamma region, potentially masking monoclonal bands in multiple myeloma.



Whole blood (Option C) contains cellular components (like red and white blood cells) that can interfere with the separation and analysis of proteins.

- Free of these cells, Serum provides a clearer and more accurate representation of the proteins in the blood.

Urine (Option D)

- The protein concentration in urine is much lower compared to blood, which can lead to less reliable results.
- Additionally, urine proteins may vary based on hydration status and can include different proteins not present in serum.

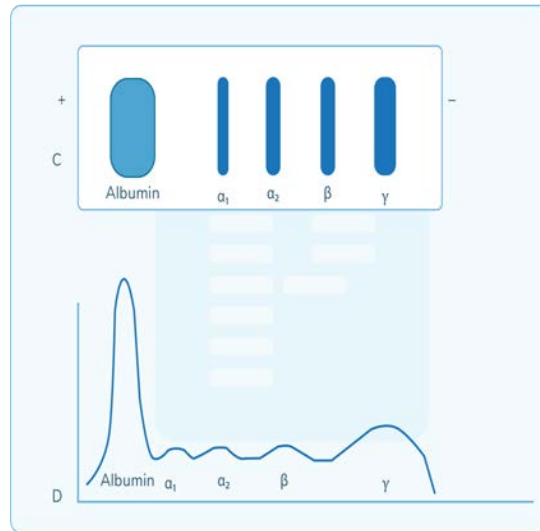
Solution for Question 8:

Macroglobulin is the protein that increases the alpha 2 band during inflammatory diseases.

Serum protein electrophoresis:

- It is a diagnostic technique used to separate and analyse proteins in the blood based on their size, shape, and charge, helping to identify abnormalities in protein levels.
- It is commonly used to diagnose and monitor conditions like chronic inflammatory diseases, multiple myeloma, liver diseases, and nephrotic syndrome by evaluating specific protein bands.

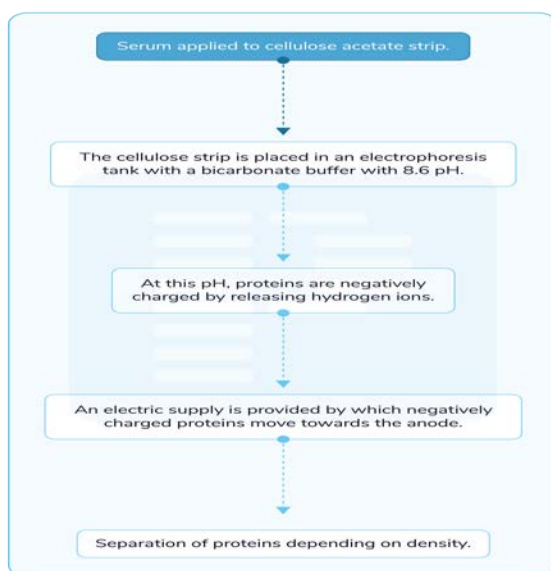
Electrophoresis band	Proteins
Albumin band	Albumin (Option A ruled out)
Alpha 1 band	Alpha 1 antitrypsin, alpha-fetoprotein, transthyretin
Alpha 2 band	Macroglobulin (Option B) , haptoglobin, ceruloplasmin
Beta band	Transferrin (Option D) , low-density lipoproteins
Gamma band	Gamma immunoglobulin (Option C) , fibrinogen



Solution for Question 9:

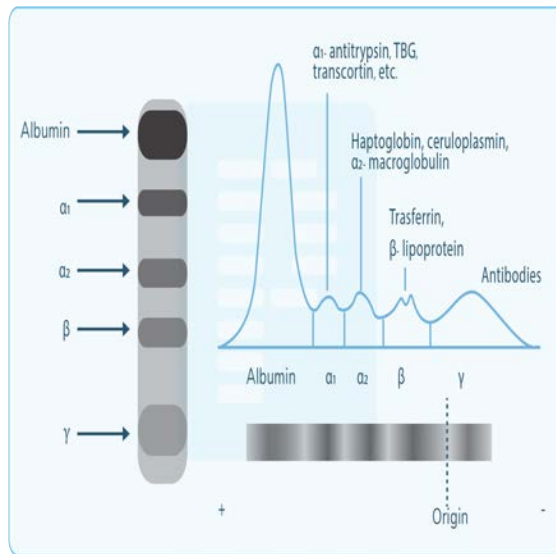
- Albumin has the highest net negative charge at physiological pH (pH ~8.6 in electrophoresis).
- It has a low molecular weight and is the smallest major serum protein, allowing it to migrate the farthest toward the anode (positive electrode) in serum protein electrophoresis (SPEP).
- Other globulin fractions (α , β , and γ) migrate slower due to lower net charge and larger size.

Protein separation on Electrophoresis:



Serum Protein Electrophoresis Migration Pattern

Protein Fraction	Charge	Migration Distance (From Cathode to Anode)
Albumin	Most negative	Farthest migration
Alpha-1 globulin (Option C ruled out)	Less negative	Closer to origin
Alpha-2 globulin	Less negative	Closer to origin
Beta globulin (Option D ruled out)	Intermediate	Closer to albumin
Gamma globulin (Option B ruled out)	Least negative	Closest to the cathode



Solution for Question 10:

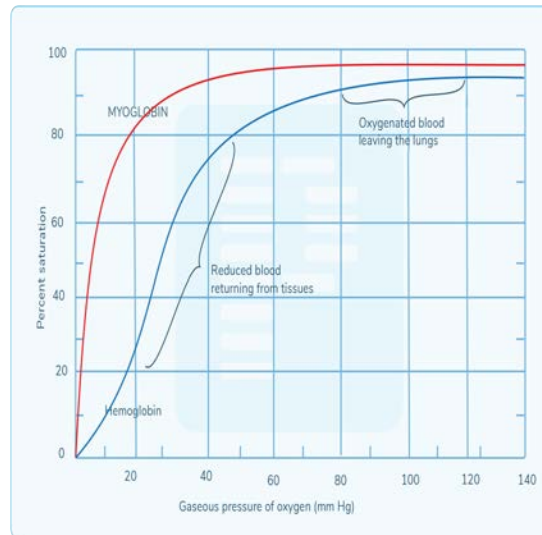
The isoelectric pH is the point at which molecules have no net charge and, therefore, do not migrate during electrophoresis.

The mobility of proteins in electrophoresis depends on

- Charge: Positively charged particles move towards negative electrodes and vice versa. (Option A ruled out)
- Molecular weight: Heavier particles move slowly when compared to lighter particles. (Option C ruled out)
- Shape of the particles: Globular proteins, being more compact, move faster than fibrous proteins, which are elongated and encounter resistance. (Option D ruled out)
- Strength of electric field: A stronger electric field enhances the movement of molecules.
- Properties of the supporting medium: Denser gel decreases the mobility of large molecules.
- Temperature: Higher temperature increases the rate of movement.

Solution for Question 11:

- At higher temperatures, the oxyhemoglobin dissociation curve shifts towards the right, indicating decreased haemoglobin affinity for oxygen and facilitating the release of oxygen to tissues.
- The shape of oxyhemoglobin dissociation is sigmoid due to positive cooperativity.



Factor	Shifts Curve to the Right (↓ O ₂ Affinity, ↑ O ₂ Release)	Shifts Curve to the Left (↑ O ₂ Affinity, ↓ O ₂ Release)
pH (Bohr Effect)	Decreased pH (Acidosis)	Increased pH (Alkalosis) (Option C ruled out)
Carbon Dioxide (CO ₂)	Increased CO ₂ (Hypercapnia)	Decreased CO ₂ (Hypocapnia) (Option B ruled out)
Temperature	Increased temperature (e.g., fever, exercise)	Decreased temperature (e.g., hypothermia)
2,3-Bisphosphoglycerate (2,3-BPG)	Increased 2,3-BPG (e.g., chronic hypoxia, anemia)	Decreased 2,3-BPG (Option A ruled out)
Carbon Monoxide (CO)	-	Presence of CO (Increased oxygen affinity, prevents release)
Fetal Hemoglobin (HbF)	-	Presence of HbF (Higher oxygen affinity)
Exercise	Increased exercise	-

Altitude	Acclimatization to high altitude	-
Growth Hormone	Increases O ₂ release (Shifts right)	-
Thyroxine (T ₄)	Increases O ₂ release (Shifts right)	-

Solution for Question 12:

- Fetal haemoglobin (HbF) contains α and γ globin chains, where γ chains have serine instead of histidine. (Beta chains have more of histidine and lysine).
- This reduces the positive charge in the heme's central cavity, leading to decreased binding of 2,3-Bisphosphoglycerate (2,3-BPG).

Difference between Adult Hemoglobin and Fetal Hemoglobin

Feature	Adult Hemoglobin (HbA)	Fetal Hemoglobin (HbF)
Globin Side Chains	$\alpha\beta$	$\alpha\gamma$
Oxygen Affinity	Lower	Higher
2,3-BPG Binding	Higher	Lower
Physiological Role	Tissue oxygen delivery	Oxygen extraction from maternal blood
Charge in Central Cavity	More positive	Less positive
pH Sensitivity	More sensitive	Less sensitive

Binds more 2,3-BPG, increasing oxygen affinity (Option A): Incorrect- More 2,3-BPG binding decreases oxygen affinity, stabilizing the taut form. HbF binds less 2,3-BPG, increasing its affinity for oxygen.

Higher histidine content, reducing oxygen affinity (Option C): Incorrect- HbF has serine instead of histidine in its γ -chain, which actually increases oxygen affinity.

Binds oxygen more loosely than adult Hb (Option D): Incorrect - HbF binds oxygen more tightly than HbA to facilitate oxygen transfer from maternal blood.

Solution for Question 13:

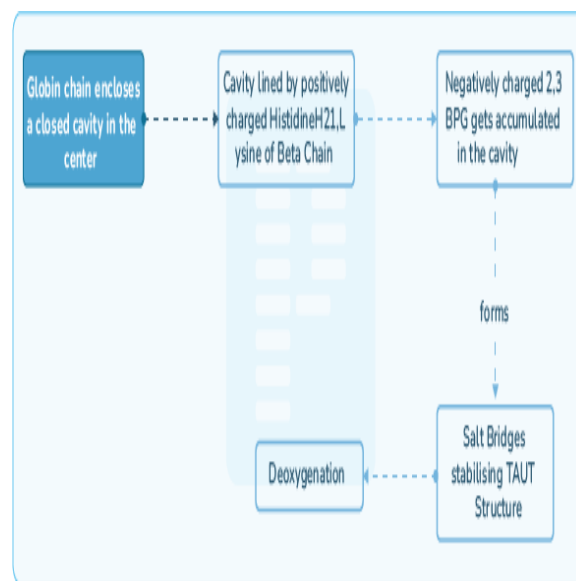
- Increased levels of 2,3-BPG stabilise the taut (T) structure of haemoglobin by forming salt bridges with positively charged residues.
- This stabilisation lowers haemoglobin's affinity for oxygen, promoting oxygen release to tissues, which can be beneficial in conditions like anaemia.

Effect of 2,3 BPG on Hb:

- Reduces the affinity of Hb for oxygen and helps unload oxygen to tissues.
- A by-product of glycolysis. (Rapaport Leubering Cycle)

Mechanism:

2,3-Bisphosphoglycerate (2,3-BPG) binds with a pocket formed by the β chains in the deoxygenated state of hemoglobin.



Forms salt bridges with positively charged residues forming a relaxed structure (Option A): Incorrect because salt bridges stabilize the T form, not the R form.

Disrupts salt bridges formed in the taut structure, decreasing oxygen affinity (Option B): Incorrect because 2,3-BPG forms rather than disrupts salt bridges, stabilizing the T form.

Binds to the heme directly, decreasing the oxygen affinity (Option D): Incorrect because 2,3-BPG does not bind to heme ; it binds between β -chains in deoxygenated Hb.

Solution for Question 14:

- A buffer resists changes in pH when small amounts of acid or base are added.
- This is explained by the Henderson-Hasselbalch equation : $\text{pH} = \text{pK}_a + \log \left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$ pH = Acidity of the solution pKa = The pH at which 50% of the acid is dissociated [A⁻] = Concentration of the conjugate base [HA] = Concentration of the weak acid
- pH = Acidity of the solution
- pKa = The pH at which 50% of the acid is dissociated
- [A⁻] = Concentration of the conjugate base
- [HA] = Concentration of the weak acid
- When pH = pKa , the concentrations of the acid and its conjugate base are equal , maximizing buffering capacity.
- The buffering effect is strongest within ± 1 pH unit of the pKa value .
- pH = Acidity of the solution
- pKa = The pH at which 50% of the acid is dissociated
- [A⁻] = Concentration of the conjugate base
- [HA] = Concentration of the weak acid

pKa and Biochemical Implications

Concept	Explanation
Definition	pKa is the pH at which 50% of a molecule is ionized .
Buffering Capacity	Maximum when $\text{pH} \approx \text{pKa} \pm 1$.
Amino Acids & Proteins	Ionization of functional groups depends on pKa, affecting protein charge and structure .

The lower the pKa of the amino group, the weaker the acid (Option A): Incorrect because lower pKa indicates a stronger acid , not a weaker one.

pKa remains constant and does not change with the medium (Option B): Incorrect because pKa is influenced by the environment , including ionic strength, polarity, and solvent .

The net charge of a protein does not depend on the pKa value of the functional group (Option C): Incorrect because protein charge depends on pKa and the surrounding pH, which affects ionization of amino acid residues.

Solution for Question 15:

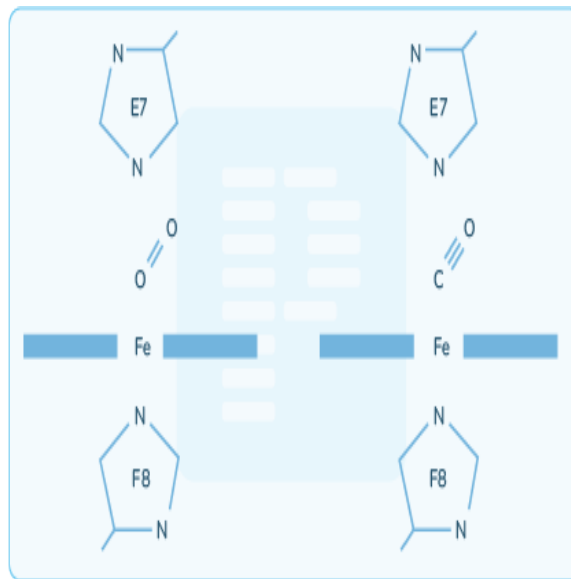
- Histidines E7 and F8 in myoglobin are crucial in regulating CO binding.
- Histidine E7 (distal histidine) particularly creates steric hindrance, which reduces CO's ability to bind effectively to the heme iron.

Structure of Myoglobin:

- Heme Group contains one iron (Fe^{2+}) surrounded by four pyrrole rings. The Fe^{2+} is located slightly below the plane of the heme ring.
- The Fe^{2+} is located slightly below the plane of the heme ring.
- Globin chain comprises a single polypeptide chain of about 150 amino acids (AA), arranged in eight α -helices (labelled A to H).
- Histidine F8: Fe^{2+} is covalently attached to the proximal histidine F8 of the globin chain.
- Histidine E7: Positioned on the opposite side of the heme, it helps create a hindered environment, reducing the risk of carbon monoxide (CO) binding.
- The Fe^{2+} is located slightly below the plane of the heme ring.

Heme and CO:

- Carbon monoxide binds to Fe^{2+} in the heme with 2000 times higher affinity than oxygen (O_2).
- CO displaces O_2 from haemoglobin, reducing oxygen-carrying capacity and leading to anaemic hypoxia.
- The globin chain, with histidine E7, provides a sterically hindered environment around the heme, which reduces CO binding and helps mitigate CO poisoning.



- In Oxygen, the second atom is inclined, so it is easily accommodated, whereas in CO, it is straight, which is hindered by E7.

Prevent CO binding by increasing the heme's affinity for oxygen (Option A): Histidine does not directly increase oxygen affinity but stabilizes O₂ binding via hydrogen bonding .

Enhance CO binding, similar to oxygen (Option B): CO has higher intrinsic affinity for heme than O₂, but histidine hinders CO binding , not enhances it.

It does not influence the binding of CO to the heme (Option D): Incorrect, as distal histidine (His E7) significantly reduces CO binding affinity by enforcing an angled binding geometry.

Solution for Question 16:

- Proteasomes are large protein complexes that degrade ubiquitin-tagged intracellular proteins through a process called ubiquitin-proteasome pathway (UPP) .
- This system is ATP-dependent and ensures regulated protein degradation to maintain cellular homeostasis.
- The 26S proteasome consists of: 20S core particle (catalytic chamber for protein degradation). 19S regulatory particle (recognizes ubiquitinated proteins and unfolds them for degradation).
- 20S core particle (catalytic chamber for protein degradation).

- 19S regulatory particle (recognizes ubiquitinated proteins and unfolds them for degradation).
- 20S core particle (catalytic chamber for protein degradation).
- 19S regulatory particle (recognizes ubiquitinated proteins and unfolds them for degradation).

Ubiquitin-proteasome pathway:

- Major pathway for degrading misfolded, damaged, and regulatory proteins.
- Ubiquitination: Proteins are tagged with ubiquitin molecules by enzymes (E1, E2, E3).
- Proteasome Degradation: Ubiquitin-tagged proteins are transported to the proteasome, a cylindrical structure with proteolytic activity, where they are degraded into small peptides.
- Lysosomes (Option A): Degrade extracellular and some intracellular proteins via autophagy , but not ubiquitin-tagged proteins . Function mainly in bulk degradation rather than targeted proteolysis.
- Degrade extracellular and some intracellular proteins via autophagy , but not ubiquitin-tagged proteins .
- Function mainly in bulk degradation rather than targeted proteolysis.
- Cathepsins (Option C): A group of lysosomal proteases involved in protein degradation inside lysosomes . They do not participate in ubiquitin-proteasome degradation .
- A group of lysosomal proteases involved in protein degradation inside lysosomes .
- They do not participate in ubiquitin-proteasome degradation .
- Endosomes (Option D): Involved in vesicular transport , sorting proteins to lysosomes , plasma membrane recycling , or exocytosis . They do not degrade ubiquitinated proteins directly.
- Involved in vesicular transport , sorting proteins to lysosomes , plasma membrane recycling , or exocytosis .
- They do not degrade ubiquitinated proteins directly.

Lysosomes (Option A):

- Degrade extracellular and some intracellular proteins via autophagy , but not ubiquitin-tagged proteins .
- Function mainly in bulk degradation rather than targeted proteolysis.

Cathepsins (Option C):

- A group of lysosomal proteases involved in protein degradation inside lysosomes .
- They do not participate in ubiquitin-proteasome degradation .

Endosomes (Option D):

- Involved in vesicular transport , sorting proteins to lysosomes , plasma membrane recycling , or exocytosis .

- They do not degrade ubiquitinated proteins directly.

Solution for Question 17:

- Protein folding is the process by which a protein structure assumes its functional three-dimensional shape from a linear chain of amino acids.
- Chaperonins are a class of molecular chaperones that assist in protein folding.
- They form large barrel-shaped complexes that can encapsulate unfolded or partially folded proteins, providing an isolated environment for proper folding.
- Chaperonins bind to hydrophobic regions of unfolded proteins, preventing inappropriate interactions and aggregation.

Role of Chaperonins in Protein Folding

Function	Chaperonin Action
Prevents misfolding	Shields hydrophobic regions to avoid aggregation
Assists refolding	Helps denatured proteins regain functional shape
Provides an isolated chamber	Forms a cavity where proteins fold properly
ATP-dependent process	Uses ATP to facilitate conformational changes

- Proteases (Option A): These are enzymes that degrade proteins by hydrolyzing peptide bonds. They do not assist in proper folding ; instead, they help degrade misfolded proteins.
- These are enzymes that degrade proteins by hydrolyzing peptide bonds.
- They do not assist in proper folding ; instead, they help degrade misfolded proteins.
- Ubiquitin (Option C): A small regulatory protein that tags damaged or misfolded proteins for proteasomal degradation . It does not assist in protein folding .
- A small regulatory protein that tags damaged or misfolded proteins for proteasomal degradation .
- It does not assist in protein folding .
- Prions (Option D): Misfolded proteins that induce abnormal folding in other proteins, leading to neurodegenerative diseases (e.g., Creutzfeldt-Jakob disease). Instead of aiding folding, they promote misfolding .

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Ubiquitin (Option C):

- A small regulatory protein that tags damaged or misfolded proteins for proteasomal degradation .
- It does not assist in protein folding .

Prions (Option D):

- Misfolded proteins that induce abnormal folding in other proteins, leading to neurodegenerative diseases (e.g., Creutzfeldt-Jakob disease).
- Instead of aiding folding, they promote misfolding .

Solution for Question 18:

- Sanger sequencing is typically used for DNA, but it can be adapted for protein sequencing in the form of Edman degradation, which determines the primary structure of proteins.
- This method provides the exact sequence of amino acids, giving the highest resolution of the primary structure.

Protein Structure and Methods of Determination

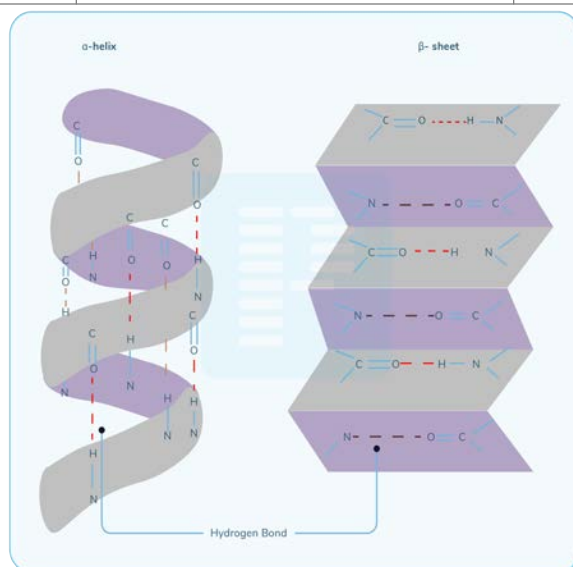
Structure	Methods of Determination
Primary	Edman sequencing Sanger's sequencing (Option C) Reverse sequencing
Secondary	Optical rotatory dispersion (Option D) Ocular dichroism
Tertiary	X-ray crystallography (Option B) UV Spectroscopy NMR spectroscopy (Option A)
Quaternary	SDS PAGE (Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis)

Solution for Question 19:

- The alpha-helix is stabilized primarily by hydrogen bonds between the carbonyl oxygen (C=O) of one amino acid and the amide hydrogen (N-H) of another amino acid, located four residues ahead in the chain.
- This forms a regular helical structure with 3.6 amino acids per turn .

Secondary structure of Protein:

Characteristic	Alpha Helix	Beta-Pleated Sheet
Shape	Spiral or coiled structure	Extended, zigzag structure, pleated
Hydrogen bonding	Within the same strand (intrastrand)	Between different strands (interstrand)
Compactness	More compact	Less compact, more extended
Stability	Stabilised by internal H-bonds	Stabilised by H-bonds between strands
Flexibility	More flexible	Less flexible, more rigid
Examples	Keratin, Myoglobin	Silk fibroin, immunoglobulins



Super secondary Structures:

- These are the segments of polypeptide chains linked to adjacent secondary structures.
- There are 2 types: Turn / Bend: If limited numbers of amino acids are used to link the adjacent structure. Loop: If more numbers of amino acids are used to link the adjacent structure.
- Turn / Bend: If limited numbers of amino acids are used to link the adjacent structure.
- Loop: If more numbers of amino acids are used to link the adjacent structure.
- Turn / Bend: If limited numbers of amino acids are used to link the adjacent structure.
- Loop: If more numbers of amino acids are used to link the adjacent structure.
- They have hydrogen bond linkages.
- Proline and glycine are found in a super secondary structure.

Covalent bonds between side chains (Option A): Incorrect because covalent (peptide) bonds link amino acids but do not stabilize secondary structures.

Ionic bonds between charged amino acids (Option B): Incorrect because ionic bonds contribute to tertiary and quaternary structures , not the stabilization of an alpha-helix.

Disulfide bridges between cysteine residues (Option D): Incorrect because disulfide bonds stabilize tertiary and quaternary structures but are not responsible for alpha-helix formation .

Solution for Question 20:

- The linear sequence of amino acids in a polypeptide chain represents the Primary structure of the protein.
- Peptide bonds are covalent bonds formed between the carboxyl group (-COOH) of one amino acid and the amino group (-NH₂) of the next amino acid.
- These bonds are responsible for linking amino acids together in a specific sequence

Protein Structure	Linkage
Primary Structure: Linear sequence of amino acids encoded by mRNA.	Peptide bonds (Covalent, via dehydration reaction) (Option A)
Secondary structure: Local folding into α -helices & β -sheets due to backbone interactions and is of two types: Alpha-helix Beta-pleated sheet	Hydrogen bonds (Between carbonyl and amide groups) (Option B)

Tertiary structure: Three-dimensional structure of a single polypeptide due to side-chain interactions.	Disulfide bonds (Option C) (Covalent), Hydrophobic interactions, Hydrogen bonds, Ionic bonds, Van der Waals forces
Quaternary structure: Assembly of multiple polypeptide subunits into a functional protein	Non-covalent interactions (Hydrogen bonds, Ionic bonds, Hydrophobic interactions, Van der Waals forces); Rarely Disulfide bonds

Van der Waals interactions (Option D)

- These weak interactions contribute to overall protein stability and folding in tertiary and quaternary structures but do not determine the primary structure.

Solution for Question 21:

Correct Answer: B) 1-d, 2-b, 3-c, 4-a

Explanation:

- Ninhydrin test (d) → Detects α -amino acids by reacting with free amino groups to produce a purple/blue color.
- Xanthoproteic test (b) → Detects aromatic amino acids (tyrosine, tryptophan, phenylalanine) through nitration, producing a yellow color.
- Sakaguchi test (c) → Detects arginine due to the presence of the guanidino group, producing a red color.
- Biuret test (a) → Detects peptide bonds, giving a violet color in the presence of proteins.

Types of Tests for the Detection of Proteins

Test	Principle	Detection Target	Positive Result
Biuret Test	Peptide bonds react with Cu^{2+} in an alkaline medium	Peptide bonds (proteins and polypeptides)	Violet color
Ninhydrin Test	Ninhydrin reacts with free α -amino groups	Free amino acids and proteins with free NH_2	Blue/Purple color
Xanthoproteic Test	Nitration of aromatic rings with HNO_3	Aromatic amino acids (tyrosine, tryptophan, phenylalanine)	Yellow color

Sakaguchi Test	Reaction of α -naphthol with guanidino group in the presence of NaOBr	Arginine	Red color
Lead Acetate Test	Reaction of sulfur-containing amino acids with lead acetate	Sulfur-containing amino acids (cysteine, methionine)	Black precipitate
Hopkins-Cole Test	Reaction of tryptophan with glyoxylic acid in a strong acid medium	Tryptophan	Violet ring
Millon's Test	Reaction of phenol groups with mercuric nitrate in acidic medium	Tyrosine	Red color
Folin-Ciocalteu Test	Phenolic groups react with phosphomolybdic-phosphotungstic acid	Tyrosine & Tryptophan	Blue color
Nitroprusside Test	Reaction of cysteine with sodium nitroprusside	Cysteine	Red color
Heller's Test	Proteins get precipitated by strong acids	Proteins (albumins and globulins)	White precipitate
Test for Casein	Acid precipitation of casein	Casein (milk protein)	White precipitate

Solution for Question 22:

- Glycine is the only amino acid that lacks a chiral center because its side chain is a hydrogen atom (-H), making both groups attached to the α -carbon identical.
- Due to the absence of a chiral center, glycine does not exhibit stereoisomerism and does not exist in D- or L-forms.

Amino Acid Chirality and Stereoisomerism:

Property	Glycine	Other Amino Acids
Chirality	Achiral	Chiral (except glycine)
Stereoisomerism	Absent	Present (D & L forms)
Optical Activity	None	Optically active

Asymmetrical Carbon	No	Yes
---------------------	----	-----

Properties of Glycine:

Property	Description
Simplest Amino Acid	Has two hydrogen atoms attached to the α -carbon.
Polarity	Non-polar; fits into both hydrophilic and hydrophobic environments due to its small size.
Chirality	Lacks a chiral center, making it optically inactive and incapable of stereoisomerism.
Glycine Cleavage System	Produces ammonia (NH_3), carbon dioxide (CO_2), and N,N-methylene tetrahydrofolate.
Biosynthetic Role	Precursor for porphyrins, creatine, glutathione, serine, purines, and heme.

Has symmetrical carbon atoms (Option A): Incorrect because glycine has only one carbon bonded to two identical hydrogen atoms, making it symmetrical.

Optically active (Option C): Incorrect because glycine is achiral and does not rotate plane-polarized light.

Complex amino acid (Option D): Incorrect because glycine is a simple nonpolar amino acid with the smallest side chain ($-\text{H}$).

Solution for Question 23:

- Threonine and Lysine are essential amino acids , meaning they must be obtained from the diet.
- A strict vegetarian diet, especially one low in protein-rich foods like legumes, dairy, and soy, can lead to deficiencies in essential amino acids.
- Lysine deficiency can result in fatigue, impaired immune function, and decreased protein synthesis , which explains the patient's symptoms.
- Threonine deficiency affects immune function and protein metabolism , further contributing to fatigue and increased susceptibility to infections.

Classification of Amino Acids: Essential vs. Non-Essential vs. Conditionally Essential

Amino Acid	Function	Dietary Sources	Deficiency Effects
Threonine	Immune function, collagen formation, protein synthesis	Meat, dairy, eggs, legumes	Impaired immune response, fatigue
Lysine	Protein synthesis, enzyme function, calcium absorption	Meat, fish, dairy, legumes	Poor wound healing, frequent infections, fatigue
Tryptophan (Option B)	Precursor for serotonin and niacin (B3)	Dairy, turkey, nuts, seeds	Mood disturbances, pellagra (niacin deficiency)
Methionine (Option D)	Precursor for S-adenosylmethionine (SAM), involved in methylation reactions	Meat, eggs, dairy	Fatty liver, slow growth
Phenylalanine	Precursor for tyrosine, neurotransmitters (dopamine, norepinephrine)	Meat, dairy, soy, fish	Growth retardation, mental impairment
Leucine	Muscle protein synthesis, energy production	Meat, dairy, eggs, soybeans	Muscle wasting, hypoglycemia
Isoleucine	Muscle metabolism, hemoglobin formation	Meat, fish, eggs, soy	Muscle weakness, dizziness
Valine	Muscle growth, tissue repair	Meat, dairy, legumes	Neurological symptoms, muscle weakness
Histidine	Precursor for histamine, important in immune function	Meat, poultry, fish, dairy	Anemia, impaired immunity
Amino Acid	Function	Synthesized From	Deficiency Effects
Tyrosine (Option B ruled out)	Precursor for neurotransmitters, thyroid hormones	Phenylalanine	Rare deficiency, linked to metabolic disorders
Cysteine (Option D ruled out)	Antioxidant defense (glutathione synthesis)	Methionine	Rare deficiency, oxidative stress
Alanine (Option C ruled out)	Gluconeogenesis, nitrogen transport	Pyruvate	No significant deficiency
Glycine (Option C ruled out)	Collagen synthesis, neurotransmission	Serine	No significant deficiency
Serine	Phospholipid synthesis, neurotransmission	3-Phosphoglycerate	No significant deficiency
Aspartate	Urea cycle, TCA cycle	Oxaloacetate	No significant deficiency

Asparagine	Protein synthesis, nitrogen metabolism	Aspartate	No significant deficiency
Glutamate	Neurotransmitter, precursor for GABA	α -Ketoglutarate	No significant deficiency
Proline	Collagen synthesis, wound healing	Glutamate	No significant deficiency
Amino Acid	Function	Synthesized From	Deficiency Effects
Arginine	Nitric oxide production, wound healing	Glutamate	Poor wound healing, immune dysfunction
Glutamine	Nitrogen transport, immune function	Glutamate	Muscle wasting, immune dysfunction

Solution for Question 24:

Serine contains a hydroxyl (-OH) group in its side chain, making it a polar, uncharged amino acid. This hydroxyl group participates in hydrogen bonding and is a target for phosphorylation in protein regulation.

Structure of serine:

COO⁻

|

CH-NH³⁺

CH₂-OH

Classification of Amino Acids Based on Functional Groups

Amino acids are classified based on the chemical nature of their R (side chain) groups, which determine their biochemical properties. Below is a classification based on their functional groups:

Category	Amino Acids	Functional Group	Properties
Hydroxyl (-OH) group	Serine, Threonine, Tyrosine	Hydroxyl (-OH)	Polar, involved in hydrogen bonding and phosphorylation.

Sulfur-containing	Cysteine (Option B ruled out), Methionine	Thiol (-SH), Thioether (-S-)	Cysteine forms disulfide bonds; Methionine is involved in methyl transfer reactions.
Aromatic	Phenylalanine (Option A ruled out), Tyrosine, Tryptophan	Benzene or Indole rings	Hydrophobic, absorb UV light, involved in protein-protein interactions.
Amide (-CONH $\blacksquare$) group	Asparagine, Glutamine (Option D ruled out)	Amide (-CONH $\blacksquare$)	Polar, involved in hydrogen bonding.
Carboxyl (-COOH) group (Acidic)	Aspartic acid, Glutamic acid	Carboxyl (-COOH)	Negatively charged at physiological pH, important in enzyme active sites.
Amino (-NH $\blacksquare$) group (Basic)	Lysine, Arginine, Histidine	Amine (-NH $\blacksquare$, -C=NH, -NH $\blacksquare$)	Positively charged at physiological pH, involved in salt bridge formation.
Aliphatic (Non-polar hydrocarbon chains)	Glycine, Alanine, Valine, Leucine, Isoleucine, Proline	Hydrocarbon (-CH, -CH $\blacksquare$, -CH $\blacksquare$)	Hydrophobic, contribute to protein stability.

- Serine, Threonine, and Tyrosine contain hydroxyl (-OH) groups and play a role in phosphorylation.
- Cysteine and Methionine contain sulfur and contribute to protein stability (disulfide bonds) and methyl group transfer.
- Aspartic acid and Glutamic acid are acidic due to their carboxyl (-COOH) groups .
- Lysine, Arginine, and Histidine are basic due to their amine (-NH $\blacksquare$) groups .
- Aromatic amino acids (Phenylalanine, Tyrosine, Tryptophan) are hydrophobic and absorb UV light.
- Aliphatic amino acids contribute to protein folding and hydrophobic interactions .

Solution for Question 25:

- Isoelectric pH (pI) is the pH at which a protein has no net charge because its positive and negative charges balance out.
- At pI, proteins tend to aggregate due to reduced electrostatic repulsion, leading to low solubility in aqueous solutions.

Concept of Isoelectric Point (pI):

Characteristic	Explanation
No net charge	At pI, the total positive and negative charges on the protein cancel each other out.
Minimal mobility	Since the protein has no net charge, it does not migrate in an electric field.
Low buffering capacity	Buffers work best when $\text{pH} \neq \text{pI}$; at pI, there is minimal change in charge per unit of pH change.
Low solubility	Electrostatic repulsion is minimal at pI, promoting aggregation and precipitation.

Low buffering capacity (Option B): Correct, because buffering capacity is strongest at pKa values and weakest at pI.

No net charge (Option C): Correct, as proteins do not carry a net charge at pI.

Minimal mobility (Option D): Correct, since proteins do not move in an electric field at pI.

Amino acid Catabolism, Urea cycle and Hyperammonemia

1. A one-year-old boy presents with lethargy, poor muscle tone, and developmental delays. Blood tests reveal elevated levels of glycine. The physician suspects a defect in the glycine cleavage system. Which of the following metabolites, if found LOW, would support the diagnosis of a defective glycine cleavage system?

- A. Serine
- B. N5,N10 methylene tetrahydrofolate
- C. Threonine
- D. Succinyl CoA

2. A patient with a urea cycle disorder is being evaluated. Which urea cycle intermediate serves as the link between the urea cycle and the TCA cycle?

- A. Argininosuccinate
- B. Fumarate
- C. Oxaloacetate
- D. Succinate

3. A 3-week-old infant presents with poor feeding, lethargy, and vomiting. Lab results show elevated ammonia levels, increased glutamine and increased orotic acid in the blood, with no evidence of megaloblastic anaemia. Which enzyme deficiency is likely responsible for this condition?

- A. Ornithine transcarbamylase (OTC)
- B. Ornithine aminotransferase

- C. Uridine monophosphate (UMP) synthase
 - D. Hypoxanthine guanine phosphoribosyltransferase (HGPRT)
-

4. Which enzyme in the histidine catabolism pathway facilitates incorporating the formimino group into the one-carbon pool, forming N5-formimino tetrahydrofolate?

- A. Histidase
 - B. Urocanase
 - C. Glutamate formimino transferase
 - D. Glutamate dehydrogenase
-

5. In the kynurenine pathway of tryptophan catabolism, which of the following compounds directly contributes to the formation of N10-formyl tetrahydrofolate in the one-carbon pool?

- A. Formyl kynurenine
 - B. Kynurenine
 - C. 3- Hydroxy kynurenine
 - D. THFA
-

6. A 62-year-old female patient presents with fatigue, pallor, and glossitis. Laboratory tests indicate macrocytic anaemia, and elevated homocysteine and methylmalonic acid levels. Given these findings, what is the most likely metabolic pathway disrupted by this patient's condition, leading to the observed anaemia?

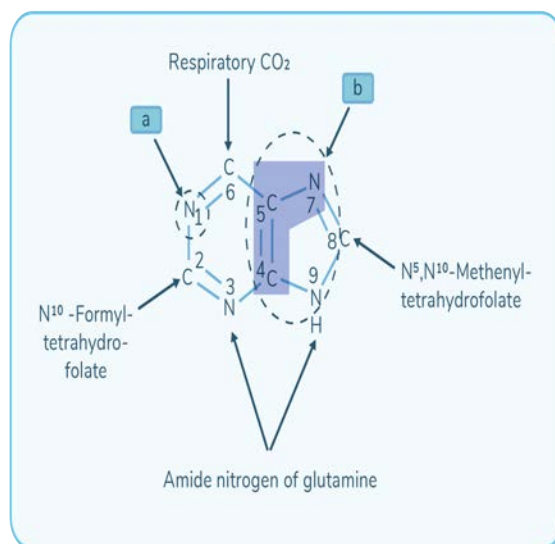
- A. Conversion of homocysteine to methionine
- B. Reduction of folic acid to dihydrofolate
- C. Deamination of histidine to formimino glutamate

D. Conversion of Phenylalanine to Tyrosine

7. Which of the following statements regarding one-carbon derivatives of tetrahydrofolic acid (THFA) and their associated enzymes and reactions is incorrect?

- A. N5-formimino-THFA converts formiminoglutamate (FIGLU) to glutamate.
 - B. N5-methyl-THFA carries the methyl group for converting homocysteine to methionine.
 - C. N5N10-methylene-THFA is required to synthesise thymidylate (dTMP) from deoxyuridylate (dUMP).
 - D. N10-formyl-THFA transfers formyl groups in pyrimidine synthesis.
-

8. In the following figure, identify the sources of the marked components.



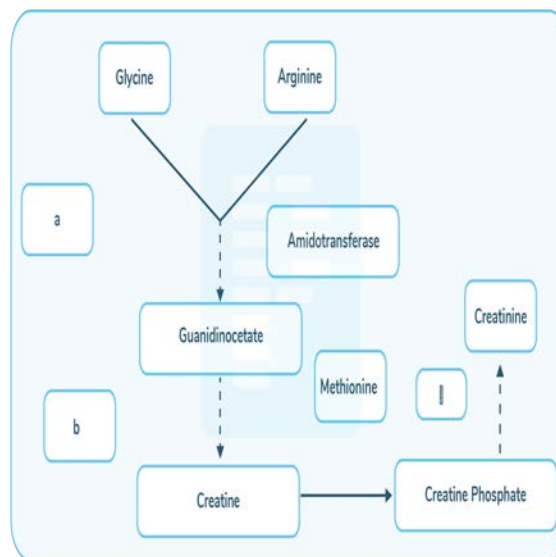
- A. a- aspartate, b- glycine
- B. a- glycine, b-aspartate
- C. a- methionine, b-alanine

D. a- alanine, b- methionine

9. Which of the following diseases is caused by the deficiency of the enzyme ■ ALA synthase?

- A. Porphyria cutanea tarda
- B. X linked sideroblastic anaemia
- C. Acute intermittent porphyria
- D. Congenital erythropoietic porphyria

10. Based on the image given below, what are the sites at which the marked steps of creatinine synthesis occur?



- A. a- brain, b- liver, c- kidney
- B. a- kidney, b- liver, c- muscle
- C. a- liver, b- kidney, c- muscle

D. a- muscle, b- liver, c-kidney

11. Which of the following is not a known use of glycine in humans?

- A. Precursor for metabolites like creatine, glutathione, and heme.
 - B. Enhances the quality of sleep and supports neurological functions.
 - C. Inhibits synthesis of bile acid, reducing lipid absorption.
 - D. Provide structural integrity for collagen structure.
-

12. The glycine cleavage system in the body produces CO₂, ammonia and a one-carbon molecule. Which of the following is a subunit in the system?

- A. Tetrahydrofolate
 - B. Lipoamide
 - C. Tetrahydrofolate-dependent aminomethyltransferase (THFAAMT)
 - D. Pyridoxal phosphate
-

13. A 6-month-old infant presented with lethargy, vomiting, and irritability. Blood tests revealed elevated ammonia levels. The infant was diagnosed with a urea cycle disorder. Which of the following medications, acting as ammonia-scavenging agents, would be most effective in reducing the infant's blood ammonia levels?

- A. Phenylacetate
 - B. N-acetyl glutamate
 - C. Lactulose
 - D. Neomycin
-

14. A 45-year-old man presents to the clinic with complaints of fatigue and abdominal discomfort. Blood tests reveal elevated levels of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). The physician suspects liver disease. Which of the following statements best describes the role of transamination reactions in amino acid metabolism?

- A. Transfer an amino group to one keto acid, producing new amino and keto acids.
 - B. Specific to lysine, threonine, and proline.
 - C. Elevated AST and ALT indicate myocardial infarction.
 - D. Occur exclusively in the liver mitochondria.
-

15. Which of the following is not a source of glycine?

- A. Serine
 - B. Threonine
 - C. choline
 - D. Lysine
-

16. A 45-year-old male with microcytic anaemia is found to have a deficiency of a vitamin essential for the rate-limiting step of heme synthesis. Which vitamin is most likely deficient?

- A. Vitamin B6
 - B. Vitamin B12
 - C. Vitamin C
 - D. Vitamin D
-

17. A 5-month-old boy presents with failure to thrive, developmental delay, and intermittent episodes of vomiting and lethargy. The episodes coincide with the introduction of solid foods. Blood tests reveal elevated ammonia levels and high levels of ornithine in blood. Urine analysis shows increased excretion of homocitrulline. What is the most likely diagnosis for this patient?

- A. Citrullinemia
 - B. Argininosuccinate aciduria
 - C. HHH Syndrome
 - D. Argininemia
-

18. A 5-month-old male infant with a history of vomiting, irritability, and developmental delay. The onset of symptoms coincided with the introduction of formula feed. Physical examination revealed hepatomegaly and signs of encephalopathy. Blood tests revealed elevated ammonia levels and high glutamine, uracil, and orotic acid concentrations. Citrulline levels were normal. What is the most likely diagnosis?

- A. Type 1 Hyperammonemia
 - B. Type 2 Hyperammonemia
 - C. Citrullinemia
 - D. Argininosuccinate aciduria
-

19. Which of the following statements about carbamoyl phosphate synthetases is correct?

- A. CPS-I is activated by PRPP, and CPS-II is inhibited by UTP.
- B. CPS-I is located in cytosol, and CPS-II in mitochondria.

- C. CPS-I is involved in urea cycle, and CPS-II in pyrimidine synthesis.
 - D. Source of nitrogen for CPS-I is glutamine, and CPS-II is ammonia.
-

20. Which of the following is true about the rate-limiting step of the urea cycle?

- A. It involves the formation of citrulline from ornithine.
 - B. It is catalysed by carbamoyl phosphate synthetase I (CPS I).
 - C. It is regulated by argininosuccinate synthetase.
 - D. It is controlled by the enzyme argininosuccinate lyase.
-

21. Which of the following statements about ammonia transport is correct?

- A. Glutamine moves ammonia from liver to brain.
 - B. Alanine transports ammonia from liver to muscles.
 - C. Glutamine and alanine transport ammonia to the liver.
 - D. Alanine does not participate in ammonia transport.
-

22. A 45-year-old man with elevated ammonia levels presents with confusion and disorientation. Which of the following is not a typical symptom of ammonia toxicity?

- A. Slurred speech
 - B. Tremors
 - C. Muscle rigidity
 - D. Blurred vision
-

23. Which of the following pathways is primarily involved in the detoxification of ammonia in the brain?

- A. Conversion of ammonia to urea in neurons
 - B. Direct excretion of ammonia through the blood-brain barrier
 - C. Conversion of ammonia to glutamine via glutamine synthetase in astrocytes
 - D. Storage of ammonia in neurons as glutamate
-

24. Which of the following reactions is not catalysed by glutamate dehydrogenase (GDH)?

- A. Conversion of α -ketoglutarate and ammonia to glutamate
 - B. Oxidative deamination of glutamate
 - C. Formation of aspartate from oxaloacetate and ammonia
 - D. None of the above
-

25. Which of the following pairs of amino acids do not undergo transamination reactions?

- A. Serine and threonine
 - B. Lysine and threonine
 - C. Proline and alanine
 - D. Hydroxyproline and asparagine
-

26. A 50-year-old woman presents with confusion and lethargy. Blood tests show elevated levels of ammonia, suggesting hyperammonemia. Further tests indicate liver dysfunction. Which of the following best describes the process of oxidative deamination in amino acid metabolism?

- A. Removal of an amino group from an amino acid, resulting in the corresponding keto acid and ammonia.
 - B. Involves only transamination reactions in the cytoplasm.
 - C. Generates ammonia, which is directly excreted by the kidneys.
 - D. Predominantly occurs in the liver cytoplasm, producing glutamine.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	3
Question 5	1
Question 6	1
Question 7	4
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	3
Question 13	1
Question 14	1
Question 15	4
Question 16	1
Question 17	3
Question 18	2

Question 19	3
Question 20	2
Question 21	3
Question 22	3
Question 23	3
Question 24	3
Question 25	2
Question 26	1

Solution for Question 1:

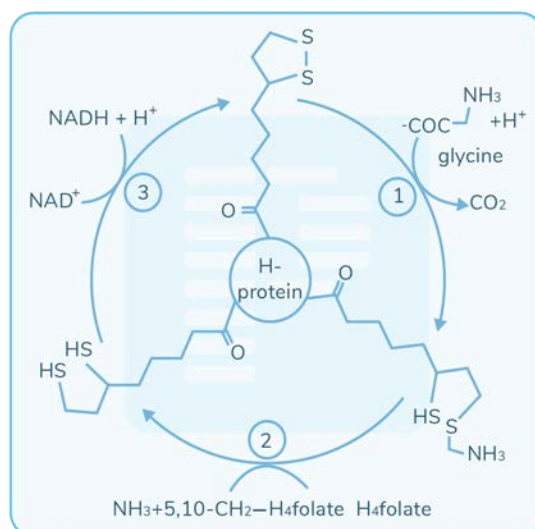
Correct Answer: B) N5,N10 methylene tetrahydrofolate

Explanation:

- The glycine cleavage system directly produces N5,N10-methylene tetrahydrofolate when it breaks down glycine.
- A defect in this system would lead to a decreased production of this metabolite .

Glycine Cleavage System	
Location	Liver mitochondria
Function	Splits glycine into carbon dioxide, and ammonia, and forms N ⁵ , N ¹⁰ -methylenetetrahydrofolate (THFA).
Glycine Cleavage Complex	Composed of three enzymes and an H-protein.
Subunits	T : Tetrahydrofolate-dependent aminotransferase P : Pyridoxal phosphate-dependent glycine decarboxylase L : Lipoamide-dependent dehydrogenase H : Hydrogen or electron transport system
Co-enzymes	Tetrahydrofolate (THF) Pyridoxal phosphate (PLP) Lipoamide

Reaction Mechanism	Glycine is broken down into CO ₂ , NH ₃ , and a methylene group (–CH ₂ –). The methylene group is transferred to THF, forming N ⁵ ,N ¹⁰ -methylenetetrahydrofolate.
Reaction	Glycine + H ₂ -folate + NAD ⁺ → CO ₂ + NH ₃ + 5,10-CH ₂ -H ₂ -folate + NADH + H ⁺
Major Catabolic Pathway	The glycine cleavage system is the major catabolic route for glycine.
Associated Disorders	Nonketotic Hyperglycinemia : Accumulation of glycine in body tissues and CNS. Primary Hyperoxaluria : Failure to break down glyoxylate leads to oxalate formation, kidney stones, and possible renal failure or hypertension. Glycinuria : A defect in kidney reabsorption causes excessive glycine in urine.



Serine levels might even increase as the body tries to compensate for the glycine deficiency. (Option A)

A defective glycine cleavage system is unlikely to directly impact threonine levels. (Option C)

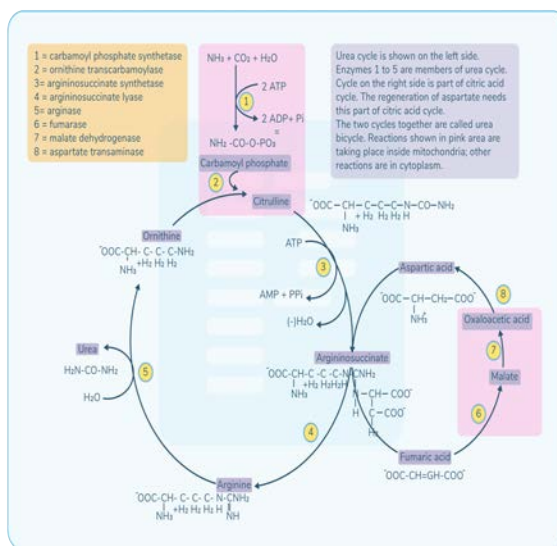
Succinyl CoA is involved in heme synthesis ; defects in glycine breakdown wouldn't impact succinyl CoA levels. (Option D)

Solution for Question 2:

Correct Answer: B) Fumarate

Explanation:

- Argininosuccinate lyase cleaves argininosuccinate to arginine and fumarate.
- Fumarate is converted to malate and then to oxaloacetate in the TCA cycle, which is transaminated to aspartate.
- This links the urea cycle with the TCA cycle.
- The reaction can be summarised as Citrulline + aspartate → Arginine + fumarate.



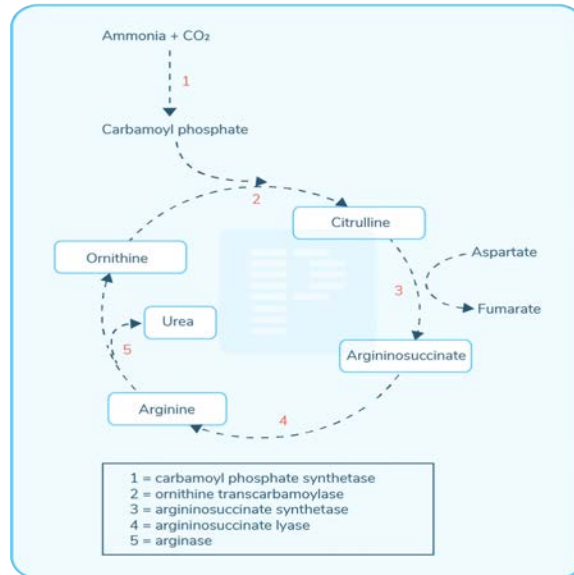
Solution for Question 3:

Correct Answer: A) Ornithine transcarbamylase (OTC)

Explanation:

- Hyperammonemia type 2 is an X-linked deficiency caused by a defect in ornithine transcarbamoylase (this reaction occurs in mitochondria), which forms citrulline in the Krebs cycle.
- Mothers with this condition may have hyperammonemia and avoid high-protein foods.

- Elevated blood, cerebrospinal fluid, and urine glutamine levels are due to increased glutamine synthesis in response to high ammonia levels.
- Orotic aciduria - Due to the channelling of carbamoyl phosphate into pyrimidine synthesis.



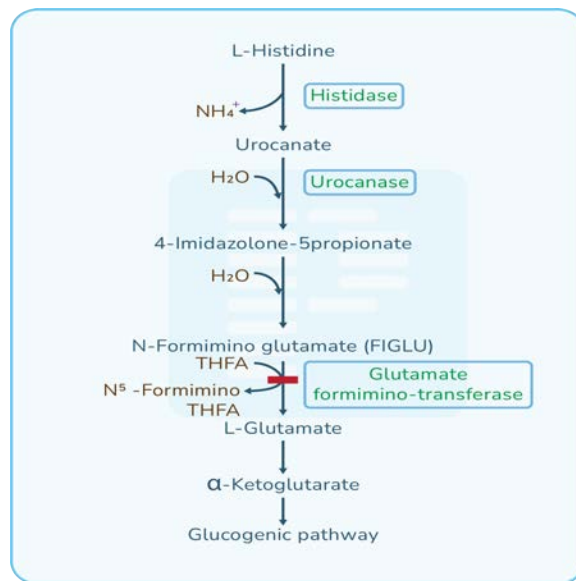
Solution for Question 4:

Correct Answer: C) Glutamate Formimino transferase

Explanation:

Glutamate Formimino transferase catalyzes the transfer of the formimino group from formiminoglutamate (FIGLU) to THFA, forming N⁵-formimino THFA.

Catabolism of Histidine to α -ketoglutarate:



Reaction	Substrate	Enzyme	Product
1.	Histidine	Histidase (Option A)	Urocanate
2.	Urocanate	Urocanase (Option B)	Imidazole propionic acid
3.	Imidazole propionic acid	Hydrolase	Formiminoglutamate (FIGLU)
4.	FIGLU	Glutamate Formimino transferase (Option C)	L-Glutamate
5.	Glutamate	Glutamate dehydrogenase (Option D)	α -Ketoglutarate

Histidine to α -Ketoglutarate Pathway: Role of Tetrahydrofolate (THF)

- Histidine is metabolised to α -ketoglutarate, which subsequently enters the TCA cycle.

Role of Tetrahydrofolate (THF):

- Histidine is broken down into formiminoglutamate (FIGLU).
- THF plays a critical role in converting FIGLU to glutamate.
- THF Function: THF accepts the formimino group (FIG) from FIGLU to become 5-formimino tetrahydrofolate (FITHFA).
- As a result of losing formimino group (FIG), FIGLU becomes glutamate.

Folate Deficiency Consequences:

- Without sufficient THF, FIGLU cannot transfer its formimino group.
- FIGLU builds up and is excreted in urine, especially after a histidine load.

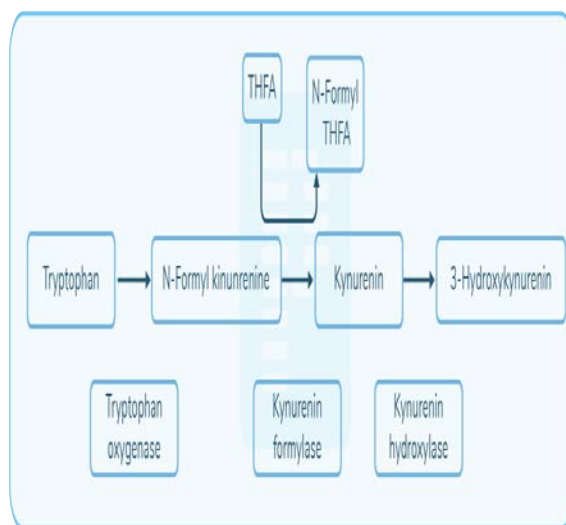
Histidine Load Test: Used to diagnose folate deficiency by detecting elevated levels of FIGLU in urine after a histidine load.

Solution for Question 5:

Correct Answer: A) Formyl kynurenine

Explanation:

- Formyl kynurenine is a key intermediate in the kynurenine pathway.
- It is directly involved in transferring a formyl group, which leads to the production of N10-formyl THFA, an important one-carbon donor in nucleotide biosynthesis■.



Tryptophan Catabolism and One-Carbon Pool Contributions

Compound	Role in Tryptophan Catabolism	Contribution
Formylkynurenine (Option A)	Intermediate; releases formate for N10-formyl THFA	Directly contributes to the formation of N10-formyl THFA
Kynurenine (Option B)	Intermediate; precursor for other metabolites	Participates in the synthesis of other metabolites like quinolinic acid and nicotinic acid■.

3- Hydroxy kynurenine (Option C)	Intermediate; precursor for other metabolites	The product formed from formate derived from formylkynurenine
THFA (Option D)	Carrier of one-carbon units	Not directly derived from tryptophan catabolism

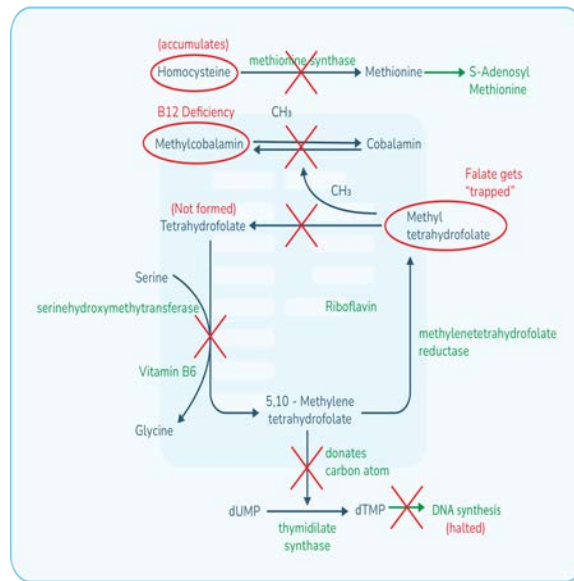
Solution for Question 6:

Correct Answer: A) Conversion of homocysteine to methionine

Explanation:

- Macrocytic anaemia, elevated homocysteine and methylmalonic acid levels are consistent with Vitamin B12 deficiency.
- Conversion of homocysteine to methionine is disrupted in vitamin B12 deficiency.
- Vitamin B12 is a cofactor for methionine synthase, the enzyme responsible for converting homocysteine to methionine.
- This, in turn, forms S-adenosylmethionine, which is necessary for the synthesis of nitrogenous bases in DNA and RNA.
- In the absence of vitamin B12, the N5-methyltetrahydrofolate (N5-methyl THF) is trapped, leading to a functional folate deficiency and, consequently, macrocytic anaemia■.
- This "methyl folate trap" hypothesis explains why folate becomes metabolically inactive.

Folate Trap:



Reduction of folic acid to dihydrofolate (Option B) is catalyzed by dihydrofolate reductase and is unrelated to the vitamin B12-dependent processes.

- It is involved in the conversion of folic acid (the synthetic form) into its active form.

Deamination of histidine to formimino glutamate (Option C) is related to folate metabolism but is more directly associated with diagnosing folate deficiency by the presence of FIGLU in the urine.

Conversion of Phenylalanine to Tyrosine (Option D) is catalyzed by Phenylalanine hydroxylase, which does not require vitamin B12 as a cofactor.

Solution for Question 7:

Correct Answer: D) N10-formyl-THFA transfers formyl groups in pyrimidine synthesis.

Explanation:

N10-formyl-THFA transfers formyl groups in purine synthesis, forming C2 of the purine ring.

One-Carbon Compounds:

One-Carbon Derivatives of THFA	Description	Source	Utilization
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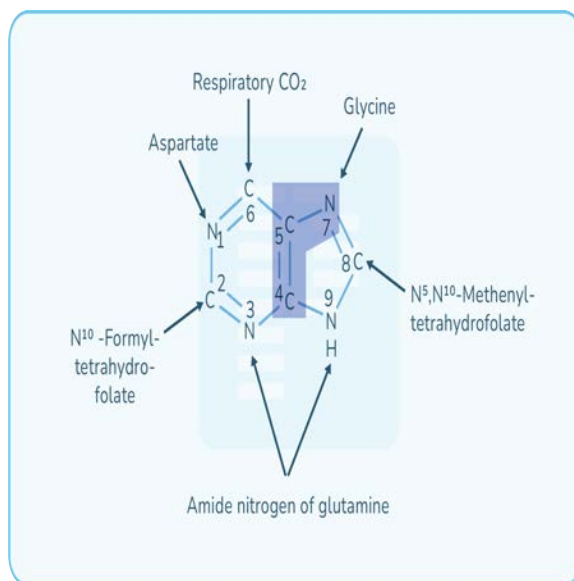
N5-formyl-THFA and N10-formyl-THFA (Option D)	Carry a formyl group (-CHO)	Tryptophan	Synthesis of purine nucleotides, forming C2 of the purine ring.
N5-formimino-THFA (Option A)	Carries a formimino group (-CH=NH)	Histidine via FIGLU	Involved in the conversion of formiminoglutamate (FIGLU) to glutamate, a step in histidine catabolism; Enzyme: Formiminoglutamate hydrolase
N10-hydroxymethyl-THFA	Carries a hydroxymethyl group (-CH ₂ OH)	Choline and betaine	
N5, N10-methylene-THFA (Option C)	Carries a methylene group (-CH ₂ -)	Serine via serine hydroxymethyl transferase, glycine	Glycine synthesis (conversion of serine to glycine), Thymidine synthesis (conversion of dUMP to dTMP); Enzyme: Thymidylate synthase
N5-methyl-THFA (Option B)	Carries a methyl group (-CH ₃)	All one-carbon units ultimately form N5-methyl-THFA.	Methionine synthesis (conversion of homocysteine to methionine); Enzyme: Methionine synthase Precursor for S-adenosylmethionine (SAM) for various transmethylation reactions.

Solution for Question 8:

Correct Answer: A) a-aspartate, b-glycine

Explanation:

- The above-marked nitrogen atom is derived from the amino acid aspartate and the carbon atoms from glycine .
- The major site is the liver .
- Requires three amino acids - Glycine, Aspartate, Glutamine and the cofactor N10-formyl tetrahydrofolate (N10- formyl THF) for purine ring formation.



- The sources of each atom in a purine were found by feeding isotopic precursors of uric acid to pigeons. The C-2 atom is from formyl THF The C-8 atom is from N⁵N¹⁰ methenyl THF The C-6 atom is from respiratory CO₂ . N1 is from aspartate N3 and N9 are from Glutamine C4, C5 and N7 are from Glycine
- The C-2 atom is from formyl THF
- The C-8 atom is from N⁵N¹⁰ methenyl THF
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- The C-8 atom is from N⁵N¹⁰ methenyl THF
- The C-6 atom is from respiratory CO₂ .
- N1 is from aspartate
- N3 and N9 are from Glutamine
- C4, C5 and N7 are from Glycine
- Methionine and alanine have no relation to purine synthesis.

Solution for Question 9:

Correct Answer: B) X linked sideroblastic anaemia

Explanation:

- The deficiency of ALA synthase causes X-linked sideroblastic anaemia.
- Heme synthesis occurs in the bone marrow (85%) and hepatocytes .
- It starts with the condensation of succinyl-CoA and glycine .
- First step: Reaction takes place in mitochondria.
- Substrates used: Glycine.
- Glycine reacts with succinyl CoA (intermediate of the citric acid cycle) in the presence of ALA synthase to form dALA.
- It is a decarboxylation reaction.
- Coenzyme : Pyridoxal phosphate.
- Prerequisite : Glycine should enter into mitochondria.
- Transporter : SLC25A38

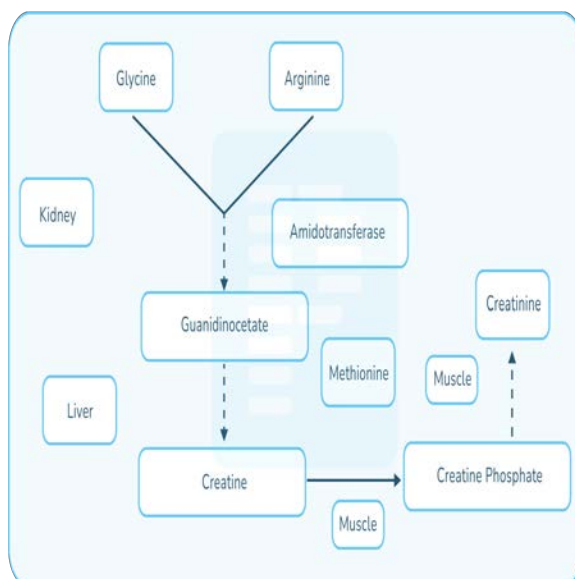
Enzyme Defect	Name of the Disorder
ALA synthase II	X linked sideroblastic anaemia (Option B)
ALA Dehydratase	ALA dehydratase deficient porphyria (ADP)
Porphobilinogen deaminase (Hydroxymethylbilane synthase)	Acute intermittent porphyria (AIP) (Option C)
Uroporphyrinogen III synthase	Congenital erythropoietic porphyria (most severe) (Option D)
Uroporphyrinogen decarboxylase defect	Porphyria cutanea tarda (PCT) (Option A)
Coproporphyrinogen oxidase	Hereditary coproporphyria
Protoporphyrinogen oxidase	Variegate porphyria
Ferrochelatase	Erythropoietic protoporphyria (EPP)

Solution for Question 10:

Correct Answer: B) a- kidney, b- liver, c- muscle

Explanation:

Creatinine synthesis requires the enzyme creatine kinase which is found in the liver, kidney and muscle.



Steps	Enzyme	Site	
1.	The Amidino group of arginine is transferred to glycine to form guanidinoacetic acid.	Amidotransferase	Kidney (a) and pancreas
2.	Guanidoacetic acid is Methylated by S-adenosyl methionine to form creatine.	Methyl transferase	Liver (b)
3.	Creatine is phosphorylated to creatine phosphate.	Creatine kinase	Muscle, brain and liver.
4.	Creatine phosphate may be converted to its anhydride, creatinine	Non- enzymatic	Muscle (c)

Solution for Question 11:

Correct Answer: C) Inhibits bile acid synthesis, reducing lipid absorption.

Explanation:

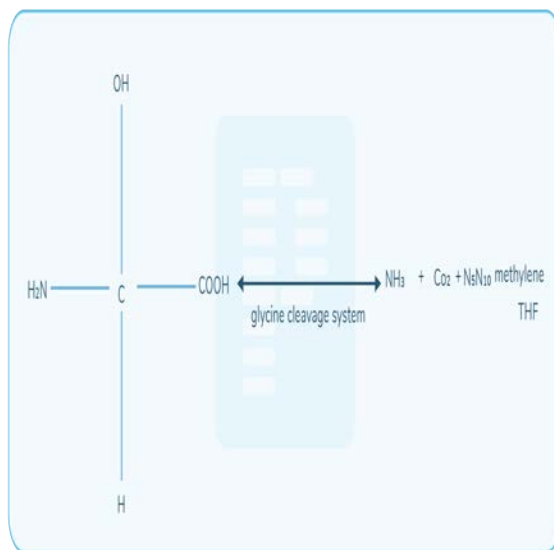
Uses of Glycine	
1) Metabolic Functions :	Precursor for metabolites such as creatine, glutathione, heme , purines, and porphyrins. (Option A) Vital for protein synthesis. Plays a crucial role in collagen structure by occupying every third position in the collagen triple helix. (Option D)
2) Health and Disease Prevention :	Dietary supplementation can prevent or treat metabolic disorders in patients with cardiovascular diseases, inflammatory diseases, obesity, cancers, and diabetes. Supports overall health, growth, and well-being in humans.
3) Neurological and Sleep Benefits :	Acts as a neurotransmitter in the central nervous system, influencing food intake, behaviour, and body homeostasis. Enhances sleep quality and neurological functions. (Option B)
4) Immune Function :	Regulates immune responses by altering intracellular calcium levels. Modulates the production of superoxide and cytokines.
5) Digestive Health :	Facilitates the conjugation of bile acids. Aid in the absorption and digestion of lipid-soluble vitamins and lipids. (Option C)

Solution for Question 12:

Correct Answer: C) Tetrahydrofolate-dependent aminomethyltransferase (THFAAMT)

Explanation:

- The subunit of the glycine cleavage system here is THFAAMT, and the others are the coenzymes that participate in the system.
- The glycine cleavage system takes place in the liver mitochondria.



- In this system, the simple amino acid glycine is split into ammonia and CO₂, and the remaining methylene group (–CH₂–) is transferred by Tetrahydrofolate and converted into N⁵N¹⁰ methylene tetrahydrofolate.

Subunits of the glycine cleavage system	Co-enzymes of glycine cleavage system
Tetrahydrofolate dependant aminotransferase (THFAAMT) (Option C)	THF (Option A)
Pyridoxal phosphate-dependant (PLP) glycine decarboxylase	Pyridoxal phosphate (PLP) (Option D)
Lipoamide dependent dehydrogenase	Lipoamide (Option B)
Hydrogen or electron transport system	

Solution for Question 13:

Correct Answer: A) Phenylacetate

Explanation:

Reducing blood ammonia levels:

- **Ammonia-Scavenging Agent: Sodium benzoate** : It reacts with glycine in the liver to form hippuric acid, which is then excreted in the urine, reducing the availability of glycine. Reducing the availability of glycine shifts the equilibrium to favour glycine formation. Ammonia is used to produce glycine, therefore reducing blood ammonia levels.
- Phenylacetate** : Conjugates with glutamine to form phenylacetylglutamine (PGA), which is then excreted in the urine, depleting glutamine levels. Glutamate production from ammonia and alpha-ketoglutarate increases to compensate for depletion. Each molecule of phenyl acetate utilises two molecules of ammonia: one for the initial glutamine conjugation and one for subsequent glutamate-to-glutamine conversion. Phenylbutyrate is a prodrug metabolised into phenylacetate. This binds with glutamine to form phenylacetylglutamine and is excreted by the kidneys, removing excess nitrogen and ammonia.
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- Phenylbutyrate is a prodrug metabolised into phenylacetate.
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- **Protein degradation** : Adequate caloric intake can help prevent protein breakdown and, subsequently, ammonia production.

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- Urease-producing microorganisms : Certain bacteria in the gut produce urease, an enzyme that breaks down urea into ammonia. To prevent this, Low-dose neomycin : Helps suppress the growth of urease-producing bacteria. (Option D) Lactulose: Converts ammonia into ammonium ions, which cannot be easily absorbed, thus reducing the amount of ammonia. (Option C)
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- Ammonia is used to produce glycine, therefore reducing blood ammonia levels.
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- Phenylbutyrate is a prodrug metabolised into phenylacetate.
- This binds with glutamine to form phenylacetylglutamine and is excreted by the kidneys, removing excess nitrogen and ammonia.
- Adequate caloric intake can help prevent protein breakdown and, subsequently, ammonia production.
- Low-dose neomycin : Helps suppress the growth of urease-producing bacteria. (Option D)
- Lactulose: Converts ammonia into ammonium ions, which cannot be easily absorbed, thus reducing the amount of ammonia. (Option C)

N-acetyl glutamate is used to treat hyperammonemia but does not directly scavenge ammonia . (Option B)

Solution for Question 14:

Correct Answer: A) Transfer an amino group to one keto acid, producing new amino and keto acids.

Explanation:

Transamination reactions transfer an amino group from an amino acid to a keto acid, catalysed by aminotransferases with Pyridoxal Phosphate (PLP) as a coenzyme.

Transamination Process	
Transfer of an α -amino group from an amino acid to one keto acid, producing a new α -keto acid and a new amino acid. (Option A) The first step is the catabolism of most amino acids. All amino acids, except Proline, hydroxyproline, threonine, and lysine, participate in transamination . (Option B)	

Biological Importance	Amino Acid Interconversion: Maintains balance of non-essential amino acids by transferring amino groups to synthesise needed amino acids. Ammonia Disposal: Detoxifies ammonia by funneling amino groups into glutamate, which is deaminated in the liver for excretion.
Glutamate's Role	α -Ketoglutarate: Acts as an acceptor of amino groups from most amino acids, converting to glutamate. Transamination occurs in the cytoplasm; the amino group is transported to the liver as glutamate. (Option D) Oxidative Deamination: Glutamate undergoes rapid oxidative deamination in the mitochondria to form α -ketoglutarate, which can accept another amino group.
Enzymes	Aminotransferases (Transaminases): Catalyse the transfer of amino groups between carbon skeletons. They are not coenzymes. (Option C) Alanine Aminotransferase (ALT): Transfers amino group from alanine to α -ketoglutarate, forming pyruvate and glutamate. Aspartate Aminotransferase (AST): Transfers amino groups from glutamate to oxaloacetate, forming aspartate used in the urea cycle.
Coenzyme	Pyridoxal Phosphate (PLP): Required by all aminotransferases, derived from vitamin B6.
Reversibility	Transamination reactions are reversible, with equilibrium constants near 1.

Solution for Question 15:

Correct Answer: D) Lysine

Explanation:

Serine, Threonine and Alanine are sources of Glycine, whereas Lysine is not.

Glycine

- Simple, non-essential amino acid.
- Acts as a precursor for metabolites like creatine, glutathione, heme, purines, and porphyrins.
- Vital for protein synthesis, collagen structure, neurotransmission, immune response, and lipid digestion.

Sources of Glycine:

- Synthesises glycine from CO₂, NH₄⁺, and a one-carbon unit.
- This multi-enzyme complex requires NAD, lipoamide, tetrahydrofolic acid, and pyridoxal phosphate.
- This is a reversal of the glycine cleavage system.
- Glycine Aminotransferase : Catalyses glycine synthesis from glyoxylate and glutamate or alanine. (Option C)

Solution for Question 16:

Correct Answer: A) Vitamin B6

Explanation:

- Heme biosynthesis begins in mitochondria with the condensation of succinyl Co-A and glycine.
- This reaction produces 5'-aminolevulinic acid (ALA), a key heme intermediate.
- The enzyme aminolevulinic acid synthase (ALAS) requires pyridoxal phosphate (Vitamin B6) to catalyze this step.
- This reaction is the rate-limiting step in heme synthesis.

Solution for Question 17:

Correct Answer: C) HHH Syndrome

Explanation:

This child with hyperammonemia triggered after solid food introduction is most likely to have HHH syndrome.

Hyperornithinemia-Hyperammonemia-Homocitrullinuria Syndrome	
Inheritance	Autosomal recessive
Defective Gene	ORC1
Defective Protein	Mitochondrial Ornithine Transporter: This transporter moves ornithine into the mitochondria and citrulline out of the mitochondria.
Mechanism	Prevents ornithine from entering the mitochondria, which is necessary for the urea cycle. The buildup of ammonia in the blood (hyperammonemia). Ornithine also accumulates in the cytoplasm (hyperornithinemia). Carbamoyl phosphate reacts with lysine to form homocitrulline.
Symptoms	Failure to thrive. Developmental delay. Intermittent episodes of vomiting and lethargy (triggered by protein intake). Seizures. Intellectual disability.
Diagnosis	Elevated blood ammonia Elevated blood ornithine Elevated urinary homocitrulline.
Treatment	Low-protein diet Nitrogen scavenger medications Liver transplant in severe cases.

- Citrullinemia leads to the accumulation of citrulline, which would be elevated in blood tests, unlike in this scenario. (Option A)
- Argininosuccinate aciduria causes the characteristic symptom of trichorrhexis nodosa (brittle, easily broken hair), which is not mentioned in this scenario. (Option B)
- Argininemia involves hyperammonemia. Compared to other urea cycle disorders, its presentation is typically delayed, appearing around 2 years of age, not at 5 months. (Option D)

Solution for Question 18:

Correct Answer: B) Type 2 Hyperammonemia

Explanation:

The infant's symptoms and laboratory findings with the timing of symptom onset after the introduction of the formula are highly suggestive of Type 2 Hyperammonemia .

Feature	Type 1 Hyperammonemia	Type 2 Hyperammonemia
Defective Enzyme	Carbamoyl Phosphate Synthetase I (CPS1)	Ornithine Transcarbamylase (OTC)
Inheritance	Autosomal recessive	X-linked recessive
Mechanism	The inability to synthesize carbamoyl phosphate blocks the urea cycle.	Accumulation of carbamoyl phosphate in the mitochondria, which then spills over into the cytoplasm.
Metabolic Consequences	Elevated ammonia and glutamate Normal orotic acid and uracil Low citrulline. (Option A)	Elevated ammonia, glutamate, and glutamine. High uracil and orotic acid (excess carbamoyl phosphate enters pyrimidine synthesis). Normal citrulline.
Clinical Presentation	Symptoms coincide with increased protein intake. CNS : Lethargy, drowsiness, coma, Convulsions, vomiting, respiratory stimulation, hyperventilation.	Similar to Type 1. Hepatomegaly.

Citrullinemia and Argininosuccinate aciduria (Option C and D) would result in elevated citrulline levels, contradicting the normal citrulline finding in this scenario.

Solution for Question 19:

Correct Answer: C) CPS-I is involved in the urea cycle and CPS-II in pyrimidine synthesis.

Explanation:

CPS-I is involved in the urea cycle , whereas CPS-II is involved in pyrimidine synthesis .

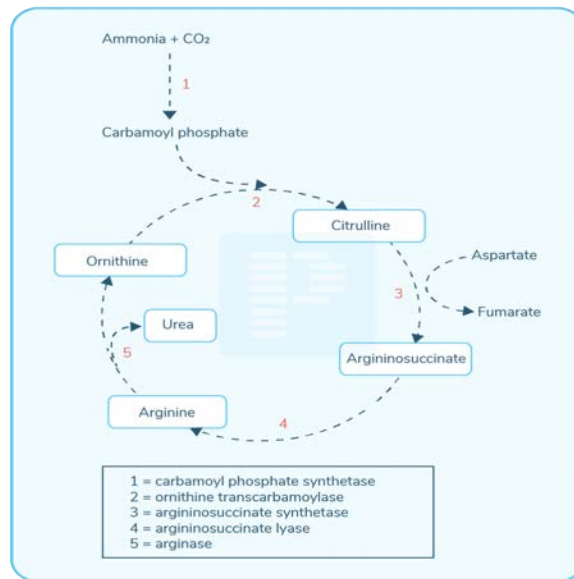
Aspect	Carbamoyl Phosphate Synthetase-I (CPS-I)	Carbamoyl Phosphate Synthetase-II (CPS-II)
Site (Option B)	Mitochondria	Cytosol
Pathway involved (Option C)	Urea cycle	Pyrimidine synthesis
Source of Nitrogen (Option D)	Ammonia	■-Amide group of glutamine
Positive effector (Option A)	N-acetyl glutamate	Phosphoribosyl pyrophosphate (PRPP)
Inhibitor	Nil	Uridine triphosphate(UTP)

Solution for Question 20:

Correct Answer: B) It is catalysed by carbamoyl phosphate synthetase I (CPS I).

Explanation:

- The carbamoyl phosphate synthetase I (CPS I) reaction is the rate-limiting step in the urea cycle.
- The urea cycle is a cycle of biochemical reactions that produce urea from ammonia.
- It is also known as the Krebs-Henseleit urea cycle or the Ornithine cycle.
- It starts in the mitochondria of hepatocytes and ends in the cytoplasm .



Reactions in the Urea Cycle			
Step	Reaction Details	Enzyme Involved and Location	Key Points
Carbamoyl Phosphate Formation	Formation of carbamoyl phosphate from ammonia and CO ₂ using ATP.	Carbamoyl Phosphate Synthetase I (CPS I) Mitochondria	The rate-limiting step of the urea cycle. Irreversible and allosterically regulated by N-acetyl glutamate (NAG).
Citrulline Formation (Option A)	Transfer of carbamoyl phosphate to ornithine to form citrulline.	Ornithine Transcarbamylase (OTC) Mitochondria	Citrulline moves from mitochondria to cytosol for further processing in the urea cycle.
Argininosuccinate Formation (Option C)	Citrulline combines with aspartate to form argininosuccinate.	Argininosuccinate Synthetase Cytoplasm	Requires ATP. Links the urea cycle with the TCA cycle via aspartate.
Argininosuccinate Cleavage (Option D)	Cleavage of argininosuccinate into arginine and fumarate.	Argininosuccinate Lyase Cytoplasm	Fumarate enters the TCA cycle, providing a connection between the urea cycle and other metabolic pathways.
Arginine Cleavage to Ornithine and Urea	Hydrolysis of arginine to produce ornithine and urea.	Arginase Cytoplasm	Ornithine is recycled back into the urea cycle. Urea is excreted by the kidneys, with some urea also entering the intestine.

Fate of Urea	Urea is transported to the kidneys, filtered, and excreted in urine.	Kidneys/ Intestine	Urea excretion is the final step, removing excess nitrogen from the body.
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Solution for Question 21:

Correct Answer: C) Glutamine and alanine transport ammonia to the liver.

Explanation:

Ammonia is transported in a non-toxic form in the circulation as glutamine or alanine.

- Glutamine is formed from glutamate and ammonia by glutamine synthetase, primarily in skeletal muscle, liver, and CNS. Glutamine transports ammonia from the brain and intestine to the liver. (Option A)
- Glutamine is transported in the blood to the liver, where it is cleaved by glutaminase to glutamate and ammonia.
- The glutamate is oxidatively deaminated to ammonia and α -ketoglutarate by glutamate dehydrogenase (GDH), and the ammonia is converted to urea.
- Alanine is the transport form of ammonia from muscle to the liver. (Option B)
- Alanine transports ammonia to the liver via the glucose-alanine cycle. (Option D)
- Alanine is transported in the blood to the liver, where it is transaminated by alanine transaminase (ALT) to pyruvate. The pyruvate is used to synthesise glucose, which can enter the blood and be used by muscles. The glutamate product of ALT can be deaminated by GDH, generating ammonia.
- Thus, both alanine and glutamine carry ammonia to the liver.

Transporter	Formation Process	Source	Destination	Mechanism/Role
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Glutamine	Formed from glutamate and ammonia by glutamine synthetase .	Skeletal muscle, liver, CNS	Liver, kidneys	Transports ammonia from the brain and intestine to the liver. In the liver, glutamine is cleaved by glutaminase to release ammonia and glutamate. Glutamate is then deaminated by glutamate dehydrogenase (GDH) , releasing additional ammonia, which enters the urea cycle for detoxification.
Alanine	Formed from pyruvate and ammonia in muscles during the glucose-alanine cycle .	Muscles	Liver	Transports ammonia from muscle to the liver. In the liver, alanine is transaminated by alanine transaminase (ALT) to form pyruvate and glutamate. The pyruvate is used to synthesize glucose, and the glutamate can be deaminated by GDH to release ammonia for the urea cycle.

Solution for Question 22:

Correct Answer: C) Muscle rigidity

Explanation:

- Neurotoxic effects of hyperammonemia include tremors, slurred speech, drowsiness, vomiting, cerebral oedema, blurred vision, potential coma and death.
- Muscle rigidity is not a symptom of ammonia toxicity.

Hyperammonemia	
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Occurs in advanced liver disease and certain genetic disorders of amino acid metabolism. It can lead to loss of consciousness, coma, convulsions and may be fatal. This is due to: Depletion of α -ketoglutarate from the citric acid cycle to form glutamate and glutamine. Reduced ATP production. Inhibition of α -ketoglutarate and pyruvate dehydrogenases by elevated ammonia levels.	
Neurotoxic effects of hyperammonemia	Tremors (Option B) Slurred speech (Option A) Drowsiness Vomiting Cerebral edema Blurred vision (Option D) Potential coma and death
Causes	Genetic causes : Urea cycle disorders and organic acidurias Hepatic encephalopathy Cor pulmonale Pulmonary emphysema Renal failure
Management	Low-protein/high-calorie diet Antibiotics to reduce bacterial load (■ urease-producing bacteria) Lactulose to decrease intestinal ammonia production

Solution for Question 23:

Correct Answer: C) Conversion of ammonia to glutamine via glutamine synthetase in astrocytes

Explanation:

Glutamine synthetase in astrocytes converts glutamate and ammonia to glutamine, effectively detoxifying ammonia in the brain.

Ammonia Management in Neurons:

- Ammonia can be neurotoxic if accumulated.

Glutamine Synthesis:

- Produced in neurons as a byproduct of amino acid metabolism and neurotransmitter turnover.
- Ammonia reacts with α -ketoglutarate and NADH to form glutamate, catalyzed by glutamate dehydrogenase.



- Glutamate is converted to glutamine by glutamine synthetase, adding another ammonia molecule. Glutamine synthetase is predominantly found in astrocytes.



- This conversion reduces ammonia toxicity by storing it in a non-toxic form.

The fate of glutamine: Release into the Bloodstream:

- Glutamine is released into the extracellular space and enters the bloodstream.
- This transport helps maintain nitrogen balance and prevents ammonia accumulation.

Transport to the Liver:

- Glutamine carries ammonia from the brain to the liver, where it is deaminated to release ammonia.
- Ammonia enters the urea cycle in the liver for detoxification.

Importance of Glutamine in Neuronal Health:

- Acts as a non-toxic ammonia carrier, preventing neurotoxicity.
- Serves as a precursor for neurotransmitters like GABA, maintaining neurotransmitter balance.

Solution for Question 24:

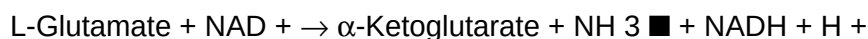
Correct Answer: C) Formation of aspartate from oxaloacetate and ammonia

Explanation:

- The formation of aspartate from oxaloacetate and ammonia is catalyzed by aspartate transaminase, not GDH.
- In the liver, ammonia transforms into the less toxic form, urea, using glutamate and the enzyme glutamate dehydrogenase.
- Location: Found exclusively in liver mitochondria.

Reactions Catalyzed by Glutamate Dehydrogenase:

- Oxidative Deamination (Conversion of Glutamate to α -Ketoglutarate)



- Catalyzed by: GDH (Option A)
- Coenzyme: NAD⁺
- Reductive Amination (Synthesis of Glutamate from α -Ketoglutarate)

α -Ketoglutarate + NH₃ + NADPH + H⁺ → L-Glutamate + NADP⁺

- Catalyzed by: GDH (Option B)
- Coenzyme: NAD⁺

Regulation:

- Activation: ADP activates GDH.
- Inhibition: GTP, ATP, and NADH allosterically inhibit GDH.

Solution for Question 25:

Correct Answer: B) Lysine and threonine

Explanation:

Lysine and Threonine:

- These amino acids do not undergo typical transamination.
- Lysine has an epsilon-amino group that can interfere with the transamination mechanism, and threonine undergoes deamination rather than transamination. Transamination is essential for amino acid catabolism and urea formation. Most amino acids participate in transamination reactions.
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Exceptions:

Amino Acid	Reason
Proline	Has a secondary amino group within its pyrrolidine ring structure. It cannot participate in standard transamination due to its structural characteristics.
Hydroxyproline	Derived from proline through post-translational modification; shares the same structural limitation.

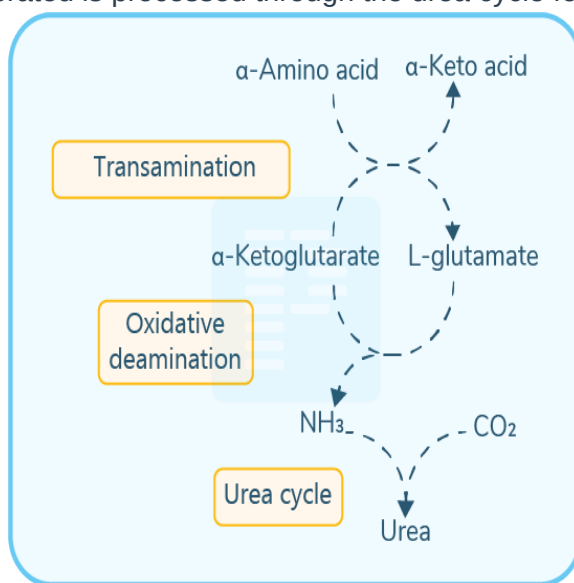
Lysine	Has an epsilon-amino group in its side chain, which can form Schiff bases, potentially interfering with transamination.
Threonine	Loses its α -amino group through deamination rather than transamination. Direct transamination does not occur; instead, it is converted to alpha-ketobutyric acid.

Solution for Question 26:

Correct Answer: A) Removal of an amino group from an amino acid, resulting in the corresponding keto acid and ammonia.

Explanation:

- Oxidative deamination removes the alpha-amino group from an amino acid, producing a keto acid and ammonia.
- This reaction, catalysed by glutamate dehydrogenase in liver mitochondria, is essential for ammonia detoxification and nitrogen balance.
- The ammonia formed is detoxified in the form of glutamate.
- The ammonia generated is processed through the urea cycle for excretion. (Option C)



Oxidative Deamination:

- Process: Removes alpha-amino group from an amino acid, producing a keto acid and ammonia.
- Common Amino Acid: Glutamic acid.
- Enzyme: Glutamate dehydrogenase (GDH).

Transdeamination:

- Step 1: Transamination (all organs) in cytoplasm forms glutamate. (Option D)
- Step 2 (Oxidative decarboxylation): In liver mitochondria, GDH removes the amino group from glutamate, producing alpha-ketoglutarate and ammonia. (Option B)
- Significance: It ensures efficient ammonia disposal, funnelling ammonia into the liver for detoxification via the urea cycle.

Minor deamination Pathways:

- Involve enzymes like L-amino acid oxidase and D-amino acid oxidase.
- Non-oxidative deamination targets specific amino acids, releasing ammonia.

L-Amino Acid Oxidase:

- It deaminates L-amino acids in the kidney.
 - It generates hydrogen peroxide (H_2O_2) and ammonia as by-products.
 - H_2O_2 , a reactive oxygen species (ROS), is broken down into O_2 and H_2O by catalase to mitigate damage from ROS.
-

Phenylalanine & Tyrosine; Amino Acid Metabolism disorders

1. Which of the following statements is true?

- A. Tyrosine is the amino acid forming Serotonin
- B. Niacin deficiency is seen in Carcinoid syndrome
- C. Tryptophan hydroxylase is required for Niacin synthesis
- D. Decarboxylation of 5 hydroxy tryptophan forms niacin.

2. Match the following urinary odours with their corresponding diagnoses: Urinary Odour Diagnosis 1. Mousy odour a. DKA (Diabetic Ketoacidosis) 2. Fruity odour b. Hawkinsinuria 3. Cabbage odour c. Methionine malabsorption 4. Swimming pool odour d. PKU (Phenylketonuria) 5. Oast house odour e. Type 1 Tyrosinemia

Urinary Odour	Diagnosis
1. Mousy odour	a. DKA (Diabetic Ketoacidosis)
2. Fruity odour	b. Hawkinsinuria
3. Cabbage odour	c. Methionine malabsorption
4. Swimming pool odour	d. PKU (Phenylketonuria)
5. Oast house odour	e. Type 1 Tyrosinemia

- A. 1-e, 2-a, 3-d, 4-b, 5-c
- B. 1-d, 2-a, 3-e, 4-c, 5-b
- C. 1-e, 2-a, 3-d, 4-c, 5-b
- D. 1-d, 2-a, 3-e, 4-b, 5-c

3. A 45-year-old patient presents with episodes of palpitations, sweating, and severe headaches. These episodes occur intermittently and are associated with hypertension. Laboratory tests reveal elevated levels of plasma free metanephrines and urinary Vanillyl Mandelic Acid. Which of the following enzymes are directly involved in the synthesis of these metabolites?

- A. Monoamine oxidase, Catechol-O-methyltransferase
 - B. Glutamate decarboxylase, Tyrosine hydroxylase
 - C. Tyrosine hydroxylase, Catechol-O-methyltransferase
 - D. Tyrosine hydroxylase, Monoamine oxidase
-

4. Which of the following statements about Segawa syndrome is incorrect?

- A. Segawa syndrome is autosomal dominant.
 - B. Patients have a defect in the GTP cyclohydrolase enzyme.
 - C. CSF Biopterin levels are elevated.
 - D. Females are more commonly affected than males.
-

5. A newborn screening test reveals elevated phenylalanine levels, prompting further investigation. Genetic testing indicates a defect in dihydrobiopterin reductase. Based on this information, Which type of phenylketonuria does the patient most likely have?

- A. Classic phenylketonuria
 - B. Non-classic phenylketonuria Type II
 - C. Non-classic phenylketonuria Type IV
 - D. Non-classic phenylketonuria Type V
-

6. A 12-year-old boy is brought to the ophthalmologist by his parents due to progressive vision loss, particularly at night. Fundoscopic examination reveals circular areas of chorioretinal atrophy and peripheral retinal degeneration. Laboratory tests indicate markedly elevated plasma ornithine levels. What is the primary biochemical defect in this patient's condition?

- A. Deficiency of ornithine transcarbamylase
 - B. Deficiency of ornithine aminotransferase
 - C. Deficiency of ornithine decarboxylase
 - D. Deficiency of ornithine cyclodeaminase
-

7. A 5-year-old boy is brought to recurrent kidney stones and abdominal pain. Urinalysis reveals hexagonal crystals, and a 24-hour urine test shows elevated levels of cystine. Which condition is the most likely diagnosis?

- A. Cystinuria
 - B. Cystinosis
 - C. Hyperoxaluria
 - D. Nephrocalcinosis
-

8. A 10-year-old girl is evaluated for developmental delay. Physical examination reveals a tall, thin build, scoliosis, and ectopia lentis. Laboratory tests show elevated levels of homocysteine and methionine in the plasma. Treatment is initiated with vitamin B6 (pyridoxine) and folate. Homocysteine levels improve, but methionine levels remain elevated. Based on this scenario, what type of defect does the patient most likely have?

- A. Type 1 Homocystinuria
- B. Type 2 Homocystinuria

- C. Type 3 Homocystinuria
 - D. Marfan Syndrome
-

9. A 7-year-old boy is brought with recurrent episodes of a red, scaly rash that worsens with sun exposure, as well as intermittent episodes of diarrhea and mood changes. A urine sample reveals an abnormal presence of neutral amino acids. The patient's diet is balanced, and there is no history of similar symptoms in the family. Laboratory tests indicate a deficiency in niacin despite adequate dietary intake. What is the underlying cause of the patient's condition?

- A. Deficiency in tryptophan hydroxylase
 - B. Deficiency in the neutral amino acid transporter
 - C. Overactivity of serotonin decarboxylase
 - D. Excessive dietary leucine
-

10. A 45-year-old man presents with symptoms of diarrhea, dermatitis, and dementia. On further investigation, his diet is found to be predominantly corn-based, and he has a history of alcohol use disorder. Laboratory tests reveal low levels of tryptophan and niacin in his blood. Which enzyme deficiency is most likely contributing to the patient's condition?

- A. Tryptophan hydroxylase
 - B. Kynureninase
 - C. Serotonin decarboxylase
 - D. Niacinamide adenine dinucleotide synthetase
-

11. Which of the following statements about the metabolism of phenylalanine is incorrect?

- A. Phenylalanine is converted to tyrosine catalyzed by phenylalanine hydroxylase.
 - B. Phenylalanine on transamination gets converted to phenylpyruvate
 - C. Phenylpyruvate on oxidation yields phenyl lactate
 - D. Phenylpyruvate on oxidative decarboxylation forms phenylacetate.
-

12. Which of the following TCA intermediates is not formed by glucogenic amino acids?

- A. Alpha Ketoglutarate
 - B. Isocitrate
 - C. Fumarate
 - D. Oxaloacetate
-

13. Which of the following amino acids are both glucogenic and ketogenic?

- A. Leucine
 - B. Tryptophan
 - C. Lysine
 - D. Alanine
-

14. A 6-month-old infant has been having developmental delays. Further examination reveals a defect in the enzyme called phenylalanine hydroxylase. Which of the following urine odor is seen in this infant?

- A. Fruity odour
- B. Musty odour
- C. Cabbage odour

D. Burnt sugar

15. A 6-month-old infant presents with a lack of pigmentation in their skin, hair, and eyes. The parents are concerned about the child's sensitivity to sunlight and the involuntary back-and-forth eye movements that the child exhibits. A significant decrease in melanin pigmentation is present. Which of the following enzymes is absent or deficient in this condition?

- A. Alanine aminotransferase (ALT)
- B. Tyrosinase
- C. Phenylalanine hydroxylase
- D. S-adenosylmethionine

16. Tyrosine is involved in the synthesis of multiple neurotransmitters. Match the following neurotransmitters with the corresponding enzymes involved in their synthesis? Neurotransmitter Enzyme 1. Dihydroxy phenylalanine (DOPA) a. Dopamine beta-hydroxylase 2. Dopamine b. N-methyl transferase 3. Norepinephrine c. Tyrosine hydroxylase 4. Epinephrine d. DOPA Decarboxylase

Neurotransmitter	Enzyme
1. Dihydroxy phenylalanine (DOPA)	a. Dopamine beta-hydroxylase
2. Dopamine	b. N-methyl transferase
3. Norepinephrine	c. Tyrosine hydroxylase
4. Epinephrine	d. DOPA Decarboxylase

- A. 1-d, 2-c, 3-b, 4-a
- B. 1-c, 2-b, 3-a, 4-d
- C. 1-d, 2-b, 3-c, 4-a

D. 1-c, 2-d, 3-a, 4-b

17. A 5-month-old infant is brought to the pediatrician due to poor weight gain, vomiting, and yellowing of the skin and eyes. Genetic testing confirms a deficiency in fumarylacetoacetate hydrolase. The physician recommends a low-protein diet and the drug Nitisinone. Which of the following is not an indication of Nitisinone?

- A. Tyrosinemia type I
 - B. Alkaptonuria
 - C. Hawkinsinuria
 - D. Tyrosinemia type II
-

18. A 4-month-old infant presents with jaundice, hepatomegaly, and failure to thrive. The parents also report that the infant has episodes of vomiting and lethargy. Laboratory tests reveal elevated levels of tyrosine and succinylacetone in the blood and urine, along with signs of coagulopathy and renal tubular dysfunction. What is the most likely underlying metabolic disorder?

- A. Phenylketonuria
 - B. Tyrosinemia type II
 - C. Tyrosinemia type I
 - D. Alkaptonuria
-

19. A 2-year-old child is brought to the clinic by his parents due to darkening of urine upon standing and brownish discolouration of the sclera and ear cartilage. The parents also report that their child has been experiencing joint pain. Upon further investigation, the physician suspects a tyrosine metabolic disorder and orders urine tests. Which of the following enzymes is most likely deficient in this

patient?

- A. Parahydroxy phenylpyruvate dioxygenase
 - B. Tyrosine transaminase
 - C. Homogentisate oxidase
 - D. Fumarylacetoacetate hydrolase
-

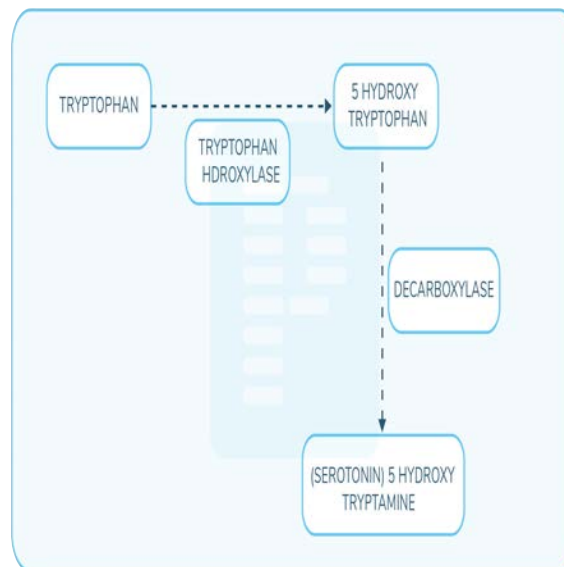
Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	1
Question 8	1
Question 9	2
Question 10	2
Question 11	3
Question 12	2
Question 13	2
Question 14	2
Question 15	2
Question 16	4
Question 17	4
Question 18	3

Solution for Question 1:

- Tryptophan forms both niacin and serotonin. (Option A)
- Carcinoid syndrome causes excess production of serotonin by carcinoid tumors and tryptophan is deficient for the synthesis of niacin causing niacin deficiency.

Synthesis of Serotonin



Other causes of niacin deficiency include Hartnup's disease, in maize eaters, and drugs like Isoniazid

Tryptophan hydroxylase is required for Niacin synthesis (Option C): Tryptophan hydroxylase is required for the synthesis of serotonin

Decarboxylation of 5 hydroxy tryptophan forms niacin (Option D): Decarboxylation of 5 hydroxy tryptophan forms serotonin

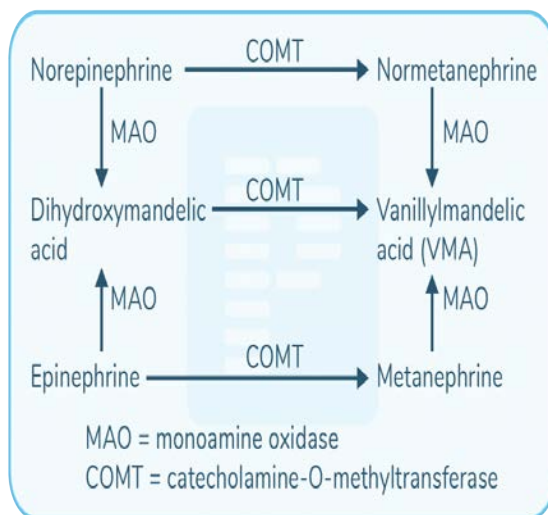
Solution for Question 2:

Urine Odor and Diagnosis

Odor	Disorders
Mousy odour	PKU
Fruity odour	DKA (diabetic ketoacidosis)
Cabbage odour	Type 1 Tyrosinemia
Boiled cabbage odour	Hypermethioninemia
Oast house (urine and sweat smell as oasthouse) or beer baby syndrome	Methionine malabsorption
Swimming pool odour	Hawkinsinuria
Sweaty feet odour	Isovaleric acidemia
Tom cat Urine odour	Multiple carboxylase deficiency
Fish odor odour	Trimethylamine urea

Solution for Question 3:

- Monoamine oxidase & Catechol-O-methyltransferase are directly involved in the metabolism of catecholamines, leading to the formation of metanephrines and vanillylmandelic acid (VMA).
- In Pheochromocytoma, there is elevation of catecholamines leading to increased breakdown of epinephrine and norepinephrine, consequently increased levels of metabolites such as metanephrines and vanillylmandelic acid.



Solution for Question 4:

In Segawa syndrome, blood phenylalanine levels are high . The syndrome is characterized by a defect in the GTP cyclohydrolase enzyme , which leads to reduced levels of tetrahydrobiopterin (BH4) .

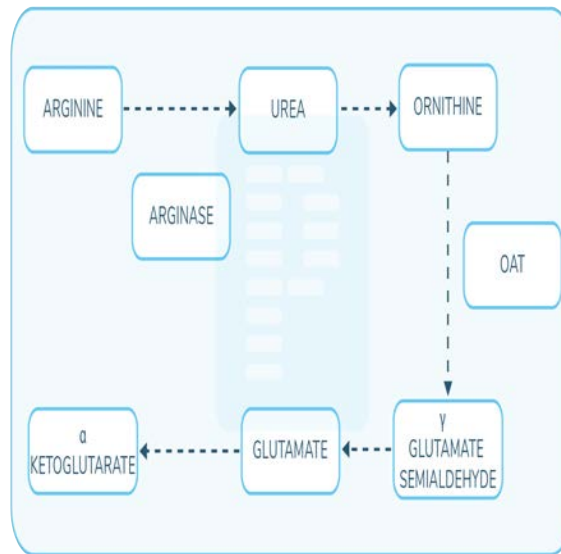
Segawa Syndrome / Dopa-responsive dystonia (DRD)	
Inheritance Pattern	Autosomal dominant (Option A)
Clinical Features	Dystonia with diurnal variation (worsens in the afternoon and evening)
Incidence based on sex	More common in females than males (Option D)
Genetic Mutation	Defect in the GTP cyclohydrolase enzyme (Option B)
Biochemical Findings	Reduced tetrahydrobiopterin (BH4) levels Normal or high blood phenylalanine levels CSF Biopterin levels are low
Diagnosis	Clinical observation of dystonia patterns Genetic testing for GTP cyclohydrolase enzyme mutation

Solution for Question 5:

Type	Enzyme Defect	Description
Classic PKU (Option A)	Phenylalanine hydroxylase (PAH)	Defect in PAH leads to elevated phenylalanine levels and symptoms such as developmental delays and musty odor.
Non-classical PKU Type II & III	Dihydrobiopterin reductase	Defect in dihydrobiopterin reductase, leading to elevated phenylalanine levels.
Non-classical PKU Type IV & V (Option C & D)	Tetrahydrobiopterin (BH4) formation	Defect in the formation of tetrahydrobiopterin, a cofactor needed for phenylalanine hydroxylase, leading to hyperphenylalaninemia.

Solution for Question 6:

- Gyrate atrophy of the retina is caused by a deficiency in the enzyme ornithine aminotransferase (OAT), which leads to the accumulation of ornithine in the plasma.
- Ornithine is toxic to the retinal pigment epithelium and causes gyrate atrophy of the retina.
- This condition manifests with progressive vision loss, especially night blindness, and characteristic fundoscopic findings of chorioretinal atrophy.
- Management includes a Arginine restricted diet and vitamin B6 supplementation, as OAT activity can be B6-dependent in some patients.



Ornithine transcarbamylase (Option A) is involved in the urea cycle, not directly related to gyrate atrophy.

Ornithine decarboxylase (Option C) is involved in polyamine synthesis, not related to ornithine accumulation or retinal degeneration.

Ornithine cyclodeaminase (Option D) is not related to the metabolic pathway implicated in gyrate atrophy.

Solution for Question 7:

•



Cystinuria (Option A) is correct because it is caused by the defect of a transporter that reabsorbs cysteine, ornithine, lysine and arginine in PCT and it causes recurrent kidney stones and hexagonal cystine crystals

Cystinosis (Option B) is incorrect is a lysosomal storage disorder caused by a defect in the transport protein cystinosin, leading to cystine accumulation within cells and formation of crystals in various tissues, including the cornea, causing photophobia.

- This condition also results in kidney damage, causing symptoms like polyuria and polydipsia, and recurrent kidney stones.
- The presence of characteristic crystals in the cornea on slit-lamp examination and hexagonal crystals in the urine supports the diagnosis of cystinosis.

Hyperoxaluria (Option C) is incorrect because it leads to calcium oxalate stones (not cystine stones) and does not cause corneal deposits or Fanconi syndrome.

Nephrocalcinosis (Option D) is incorrect because it refers to calcium deposition in the kidney and does not explain cystine crystals, Fanconi syndrome, or corneal involvement.

Solution for Question 8:

Type 1 vs Type 2 Homocystinuria:

Feature	Type - I Homocystinuria	Type - II Homocystinuria
Enzyme Deficiency	Cystathionine beta-synthase	Methionine synthase
Metabolic Pathway Affected	Homocysteine to cystathionine	Homocysteine to methionine
Elevated Metabolites	Homocysteine and methionine	Homocysteine and elevated levels of methionine
Biochemical Diagnosis	Low levels of cysteine. High levels of methionine.	High levels of cysteine. Low levels of methionine.
Treatment	Vitamin B6 (pyridoxine) supplementation; low-methionine diet; cysteine supplementation	Vitamin B12 and folate supplementation; low-methionine diet

Type 3 Homocystinuria (Option C) is a rare form not commonly linked with elevated methionine and does not match the patient's response to treatment.

Marfan Syndrome (Option D) shares some physical traits (tall stature, scoliosis) but it does not include elevated homocysteine levels or vitamin B6 response.

Solution for Question 9:

Hartnup's disease:

- It is caused by a defect in the neutral amino acid transporter in the kidneys and intestines, leading to poor absorption and increased excretion of neutral amino acids like tryptophan.
- This results in decreased availability of tryptophan for niacin synthesis, leading to symptoms similar to pellagra despite adequate dietary intake.
- The characteristic skin rash, diarrhea, and mood changes are consistent with this diagnosis.

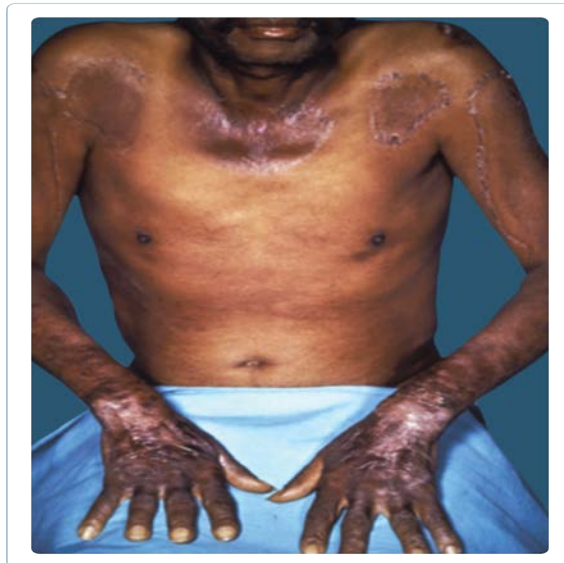
Tryptophan hydroxylase (Option A) is involved in serotonin synthesis, not amino acid transport.

Overactivity of serotonin decarboxylase (Option C) would affect serotonin levels, not amino acid transport.

Excessive dietary leucine (Option D) can interfere with tryptophan metabolism, but it is not the primary cause of Hartnup's disease.

Solution for Question 10:

- Pellagra is a condition that results from a deficiency of niacin (vitamin B3) or tryptophan.
- It is characterized by three main symptoms: dermatitis, diarrhea, and dementia.
- Kynureninase is an enzyme in the kynurenine pathway, responsible for converting tryptophan to niacin.
- A low activity of this enzyme is observed in B6 deficiency, which is caused by alcoholism. This can exacerbate pellagra caused by maize or corn intake.



Causes of Pellagra:

- Diet with Niacin deficiency. Tryptophan (60 mg Tryptophan = 1 mg Niacin) deficiency in the maize-based diet. Tryptophan malabsorption - Hartnup's disease.
- Niacin deficiency.
- Tryptophan (60 mg Tryptophan = 1 mg Niacin) deficiency in the maize-based diet.
- Tryptophan malabsorption - Hartnup's disease.
- Carcinoid Syndrome - Tryptophan is used to form Serotonin leading to niacin deficiency
- Vitamin B6 deficiency - inactivates Kynureninase.
- Leucine pellagra - Quinolinic phosphoribosyltransferase (QPRTASE) enzyme is inhibited in a leucine-rich diet. Example: sorghum diet which is rich in leucine.
- Niacin deficiency.

- Tryptophan (60 mg Tryptophan = 1 mg Niacin) deficiency in the maize-based diet.
- Tryptophan malabsorption - Hartnup's disease.

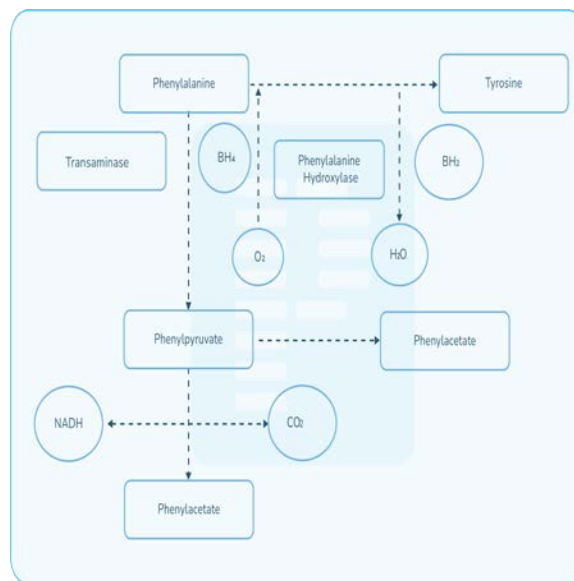
Tryptophan hydroxylase (Option A) converts tryptophan to 5-hydroxytryptophan in the serotonin synthesis pathway, not involved in niacin synthesis.

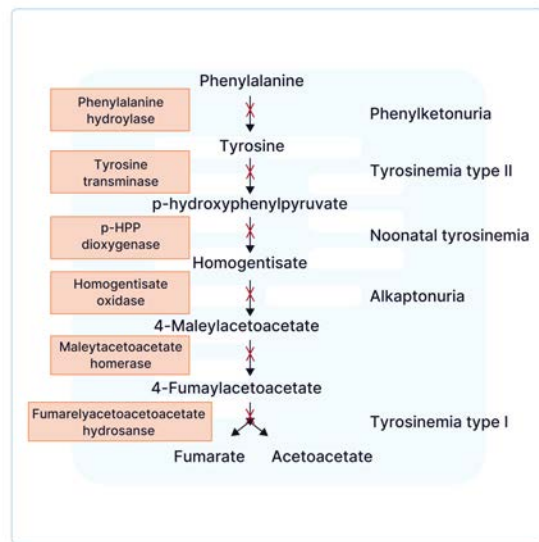
Serotonin decarboxylase (Option C) converts 5-hydroxytryptophan to serotonin, not in niacin synthesis.

Niacinamide adenine dinucleotide synthetase (Option D) converts niacinamide to NAD (nicotinamide adenine dinucleotide), a form of niacin. It is not directly related to the metabolism of tryptophan into niacin.

Solution for Question 11:

Phenylpyruvate on reduction, not oxidation, yields phenyl lactate.

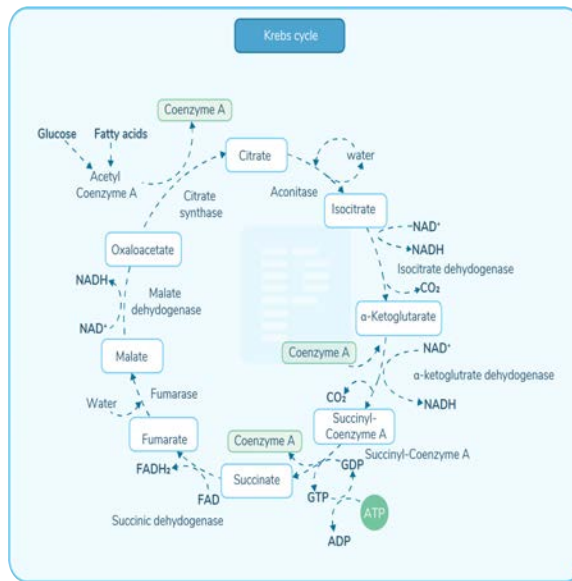




Phenylalanine has 2 metabolic fates in the body:

Major Fate	Phenylalanine is converted into tyrosine by the enzyme phenylalanine hydroxylase. Deficiency of phenylalanine hydroxylase results in Phenylketonuria. It requires tetrahydrobiopterin (BH4) as a cofactor
Minor Fate	Phenylalanine is converted into phenylpyruvate by transamination . Phenylpyruvate can further undergo different pathways: Phenylpyruvate can be reduced to phenyl lactate Phenylpyruvate undergoes oxidative decarboxylation, forming phenylacetate Phenylacetate combines with glutamine to form phenylacetylglutamine , which is excreted in the urine.

Solution for Question 12:



Glucogenic amino acids give rise to 4 intermediates of TCA

- Alpha Ketoglutarate
- Succinyl CoA
- Fumarate
- Oxaloacetate

Amino acids give rise to:

- Alpha-ketoglutarate (on Transamination) Glutamate Glutamine Histidine Proline Arginine
- Glutamate
- Glutamine
- Histidine
- Proline
- Arginine
- Succinyl CoA V- Valine I- Isoleucine M- Methionine (Mnemonic- VIM)
- V- Valine
- I- Isoleucine
- M- Methionine
- (Mnemonic- VIM)
- Fumarate Phenylalanine Tyrosine
- Phenylalanine
- Tyrosine
- Oxaloacetate (on Transamination) Aspartate Asparagine

- Aspartate
- Asparagine
- Glutamate
- Glutamine
- Histidine
- Proline
- Arginine
- V- Valine
- I- Isoleucine
- M- Methionine
- (Mnemonic- VIM)
- Phenylalanine
- Tyrosine
- Aspartate
- Asparagine

Solution for Question 13:

Correct Answer: B) Tryptophan

Explanation:

- Tryptophan is both glucogenic and ketogenic.
- Tryptophan on catabolism gives one of the glycolytic intermediate - Pyruvate or TCA cycle intermediate other than Acetyl CoA.

Classification of Amino Acid Catabolism

Purely Ketogenic	Purely Glucogenic	Both Ketogenic and Glucogenic
On catabolism gives only acetyl CoA	On catabolism gives only pyruvate	Substance on catabolism gives one of the glycolytic intermediate -Pyruvate or TCA cycle intermediate other than Acetyl CoA

Amino acids are: Leucine Lysine	Amino acids are: Alanine: On transamination yields pyruvate. Glycine: Simplest amino acid. 3-hydroxyl group- containing amino acids Serine Threonine Hydroxyproline Sulphur-containing amino acid Cysteine	Amino acids are: Aromatic amino acids Phenylalanine Tyrosine Tryptophan Branched-chain amino acid Isoleucine Threonine
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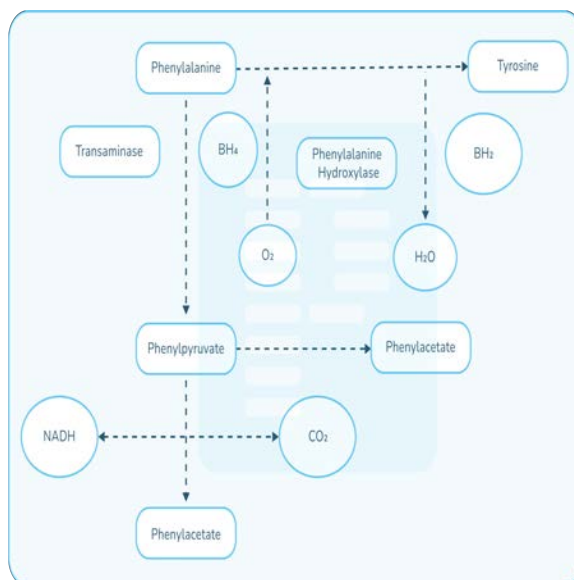
Solution for Question 14:

The clinical scenario is a case of phenylketonuria which causes 3 main clinical features:

- Mental retardation
- Hypopigmentation
- Mousy odour

Phenylketonuria (PKU):

- Phenylketonuria (PKU) is an autosomal recessive inborn error of metabolism caused by a deficiency of phenylalanine hydroxylase or its cofactor tetrahydrobiopterin (BH₄).
- This deficiency leads to an accumulation of phenylalanine and its metabolites, including phenylpyruvic acid.



Mental Retardation:

- Cause: Mental retardation in phenylketonuria (PKU) is due to the accumulation of phenylketones, which compete with other neutral amino acids for transport across the blood-brain barrier.
- Management: To prevent mental retardation, a phenylalanine-restricted diet should be followed, along with tyrosine supplements.

Hypopigmentation:

- Cause: Reduced melanin production in PKU results from a deficiency in tyrosine, a crucial component for melanin synthesis. This leads to lighter skin and hair color compared to individuals without PKU.

Musty/ Mousy urine odor:

- Cause: Due to the accumulation of phenylpyruvic acid (a metabolite of excess phenylalanine) in the urine.

ODOUR	DISORDERS
Musty/ Mousy odour	PKU
Fruity odour	DKA (diabetic ketoacidosis)
Cabbage odour	Type 1 Tyrosinemia
Burnt sugar	Maple syrup urine disease
Oast house or beer baby syndrome	Hypermethioninemia
Swimming pool odour	Hawkinsinuria
Tom-cat urine odour	Multiple carboxylase deficiency
Sweaty feet odour	Isovaleric acidemia
Fish odour	Trimethylaminuria

Solution for Question 15:

- Melanin is a pigment produced by cells called melanocytes, and it plays a critical role in determining the colour of our skin, hair, and eyes.

- Tyrosinase catalyzes the first two steps in the conversion of the amino acid tyrosine to melanin .

Tyrosine → L-Dopa → Dopakinone

- Without enough functional tyrosinase, melanin production is reduced or completely absent, leading to the characteristic features of albinism.

Reaction	Enzyme
Tyrosine to DOPA	Tyrosine hydroxylase/ Tyrosinase
DOPA to Melanin	Non-enzyme catalyzed chemical reaction

Alanine aminotransferase (ALT) (Option A) is a liver enzyme that is released into the bloodstream when the liver is damaged.

Phenylalanine hydroxylase (Option C) is an enzyme that converts phenylalanine to tyrosine. A deficiency in this enzyme leads to phenylketonuria (PKU).

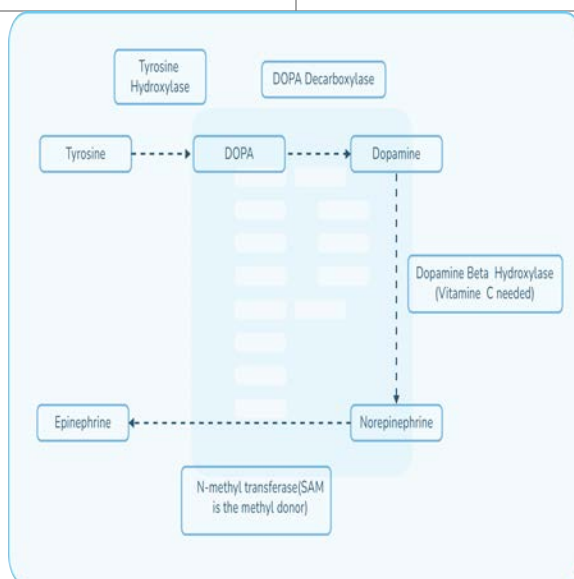
S-adenosylmethionine (Option D) required for:

- Norepinephrine to Epinephrine: Enzyme: Phenylethanolamine N-methyltransferase (PNMT).
- Enzyme: Phenylethanolamine N-methyltransferase (PNMT).
- Conversion of Homocysteine to Methionine: Enzyme: Methionine synthase, using methylcobalamin as a cofactor.
- Enzyme: Methionine synthase, using methylcobalamin as a cofactor.
- Methylation of DNA: SAM donates methyl groups to cytosine residues, affecting gene expression.
- SAM donates methyl groups to cytosine residues, affecting gene expression.
- Synthesis of Creatine: SAM is involved in the methylation of guanidinoacetate to form creatine.
- SAM is involved in the methylation of guanidinoacetate to form creatine.
- Methylation of Neurotransmitters: SAM is also involved in the methylation of other neurotransmitters, including serotonin to melatonin.
- SAM is also involved in the methylation of other neurotransmitters, including serotonin to melatonin.
- Lipid Metabolism: Involved in the synthesis of phosphatidylcholine from phosphatidylethanolamine.
- Involved in the synthesis of phosphatidylcholine from phosphatidylethanolamine.
- Enzyme: Phenylethanolamine N-methyltransferase (PNMT).
- Enzyme: Methionine synthase, using methylcobalamin as a cofactor.

- SAM donates methyl groups to cytosine residues, affecting gene expression.
- SAM is involved in the methylation of guanidinoacetate to form creatine.
- SAM is also involved in the methylation of other neurotransmitters, including serotonin to melatonin.
- Involved in the synthesis of phosphatidylcholine from phosphatidylethanolamine.

Solution for Question 16:

Reaction	Enzyme
Tyrosine to dihydroxy phenylalanine (DOPA)	Tyrosine hydroxylase
DOPA to Dopamine	DOPA Decarboxylase (uses PLP as a coenzyme)
Dopamine to Norepinephrine	Dopamine beta-hydroxylase (depends on Vitamin C)
Norepinephrine to epinephrine	N-methyl transferase (S-adenosylmethionine (SAM) is the methyl donor)



Synthesis of Neurotransmitters from Tyrosine- Dopamine, Norepinephrine and Epinephrine

Solution for Question 17:

Nitisinone is not indicated for Tyrosinemia type II , which is caused by a deficiency in Tyrosine aminotransferase.

- Nitisinone acts by inhibiting the enzyme parahydroxy-phenylpyruvate dioxygenase , thereby inhibiting tyrosine metabolism upstream and prevents the accumulation of toxic metabolites of the tyrosine degradation pathway.
- Nitisinone should be administered with a diet strictly low in phenylalanine and tyrosine.

The main indications of Nitisinone are:

Tyrosinemia type I	Nitisinone inhibits the enzyme Parahydroxy phenylpyruvate dioxygenase, preventing the tyrosine metabolism upstream and reducing the accumulation of toxic maleylacetoacetate and fumarylacetoacetate .
Alkaptonuria	Nitisinone is used in the treatment of Alkaptonuria by inhibiting the formation of Homogentisic acid.
Hawkinsinuria	Nitisinone is used to treat Hawkinsinuria, which is caused by a mutation in 4-hydroxyphenylpyruvate dioxygenase.

Solution for Question 18:

The clinical scenario described, including jaundice, hepatomegaly, failure to thrive, and elevated levels of tyrosine and succinylacetone , is characteristic of Tyrosinemia type I , which is caused by a deficiency in fumarylacetoacetate hydrolase.

Disorders of Tyrosine Metabolism:

Disorder	Deficient Enzyme	Key Clinical Features
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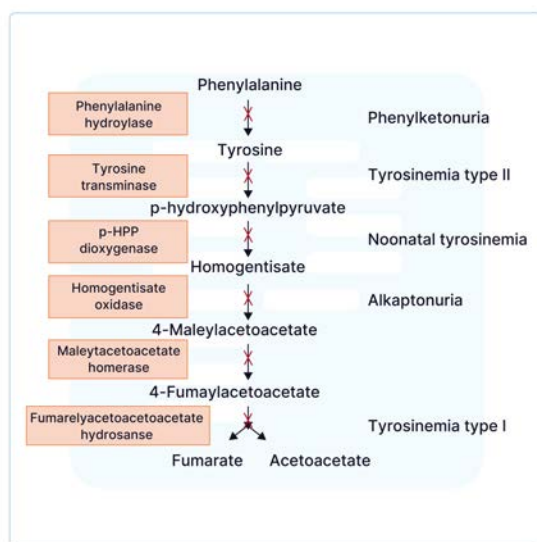
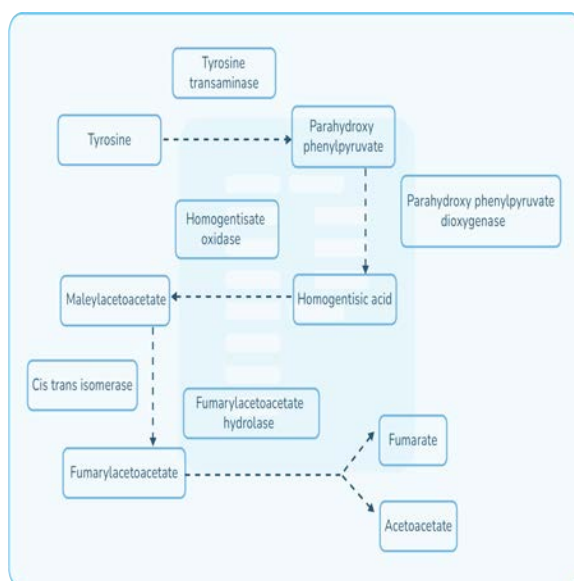
Tyrosinemia type I	Fumarylacetoacetate hydrolase	Jaundice, hepatomegaly, hypoglycemia, HCC, liver cirrhosis Often presents with Fanconi syndrome and renal failure Elevated tyrosine and succinylacetone (pathognomonic)
Tyrosinemia type II	Tyrosine aminotransferase	Painful corneal erosions, palmoplantar hyperkeratosis Also known as oculocutaneous tyrosinemia Elevated tyrosine
Tyrosinemia type III	p-Hydroxyphenylpyruvate dioxygenase	Recurrent seizures, intermittent ataxia Elevated tyrosine and 4-hydroxyphenylpyruvate
Alkaptonuria (Black urine disease)	Homogentisate 1,2-dioxygenase	Dark urine, ochronosis, arthritis Black or brown pigmentation of sclera (Ochronosis) and ear cartilage Elevated homogentisic acid

Solution for Question 19:

The clinical scenario described, including darkening of urine upon standing, brownish discolouration of the sclera and ear cartilage, and joint pain, is characteristic of Alkaptonuria, which is caused by a deficiency of the enzyme homogentisate 1,2-dioxygenase.

Garrod's tetrad in alkaptonuria

1. Homogentisic Aciduria	Presence of homogentisic acid in the urine, which turns dark upon standing or when exposed to air.
2. Ochronosis	Deposition of dark pigment in connective tissues, particularly in the ear cartilage, sclera, and other connective tissues, leading to bluish-black discolouration.
3. Arthropathy	Degenerative joint disease, especially affecting the spine and large joints, leading to chronic pain and stiffness.
4. Cardiac Involvement	Includes heart valve calcification and atherosclerosis due to the deposition of pigment in cardiac tissues.



Reaction	Enzyme	Associated Disorder
Tyrosine → p-Hydroxyphenylpyruvate	Tyrosine transaminase	Tyrosinemia type II
p-Hydroxyphenylpyruvate → Homogentisate	p-Hydroxyphenylpyruvate dioxygenase	Tyrosinemia type III
Homogentisate → Maleylacetoacetate	Homogentisate 1,2-dioxygenase	Alkaptonuria
Maleylacetoacetate → Fumarylacetoacetate	Maleylacetoacetate isomerase	-
Fumarylacetoacetate → Fumarate + Acetoacetate	Fumarylacetoacetate hydrolase	Tyrosinemia type I

- Loss-of-function mutation of p-Hydroxyphenylpyruvate dioxygenase (pHPPD) causes tyrosinemia type III, while a gain-of-function mutation causes H awkinsinuria
-

Previous Year Questions

1. Which of the following amino acids primarily acts as a buffer in blood due to its ability to accept and donate protons at physiological pH?

- A. Histidine
 - B. Arginine
 - C. Tryptophan
 - D. Tyrosine
-

2. A patient presents with arthritis, and arthroscopy reveals blackening of the joints. Which enzyme deficiency is most likely responsible for this condition?

- A. Homogentisate oxidase
 - B. Branched-chain alpha-ketoacid dehydrogenase
 - C. Tyrosinase
 - D. Fumarase
-

3. Zellweger syndrome is associated with which cellular organelle?

- A. Peroxisomes
 - B. Nucleosomes
 - C. Lysosomes
 - D. Ribosomes
-

4. Which of the following neurotransmitter is not derived from amino acid?

- A. GABA
 - B. Creatinine
 - C. Dopamine
 - D. 5 hydroxytryptamine
-

5. What do chaperones assist in?

- A. Ubiquitination of proteins
 - B. Protein modification
 - C. Protein folding
 - D. Protein cleavage
-

6. Nitric oxide is synthesized from:

- A. L-arginine
 - B. L-citrulline
 - C. Glycine
 - D. Lysine
-

7. Ammonia is transported from the brain to the liver in the form of which of the following amino acids?

- A. Glutamate
- B. Alanine
- C. Glycine

D. Glutamine

8. Nitric oxide is derived from:

- A. Arginine
 - B. Histidine
 - C. Glycine
 - D. Aspartate
-

9. Which of the following is the best method to detect HbA1c?

- A. Ion-exchange chromatography
 - B. Electrophoresis
 - C. Immunoassay
 - D. Affinity chromatography
-

10. All of the following are derivatives of tryptophan EXCEPT:

- A. Melatonin
 - B. Serotonin
 - C. Niacin
 - D. Creatinine
-

11. Which of the following amino acids is responsible for buffering action in blood at physiological pH?

- A. Methionine
 - B. Alanine
 - C. Arginine
 - D. Histidine
-

12. The molecule responsible for the transport of ammonia from the brain to the liver is

- A. Glutamine
 - B. Leucine
 - C. Lysine
 - D. Aspartate
-

13. Ammonia in the brain is detoxified to form

- A. Urea
 - B. Glutamine
 - C. Ammonium ions
 - D. Alanine
-

14. Which of the following molecules is universally recognized as a donor of methyl groups in different biochemical reactions?

- A. Glucose-6-phosphate
- B. Acetyl CoA
- C. S-AdenosylMethionine
- D. S-Adenosyl Serine

15. The lab technician observes that a urine sample obtained from a 4-year-old male appeared normal initially, but underwent a change in color as depicted in the image after a few hours. Subsequently, the color intensified over the following hours. Which deficiency can potentially lead to this condition?



- A. Beta-carotene
- B. Tyrosine aminotransferase
- C. L-ascorbic acid
- D. Homogentisic acid oxidase

16. Oxaloacetate is produced from which of the following amino acid?

- A. Alanine
- B. Lysine
- C. Leucine

D. Aspartate

17. A farmer sprayed a chemical in the pool which causes inhibition of complex-III of the electron transport chain. Identify the inhibitor?

- A. Rotenone
 - B. TTFA
 - C. Cyanide
 - D. Antimycin
-

18. Heme is derived from which of the following amino acids?

- A. Glutamine
 - B. Glycine
 - C. Glutamate
 - D. Valine
-

19. Which of the following amino acids are necessary for purine ring formation?

- A. Glycine, aspartate, valine
 - B. Glycine, Aspartate, Glutamine
 - C. Glycine, Aspartate, Glutamate
 - D. Glycine, Asparagine, Glutamine
-

20. Which of the following is caused by the deficiency of the enzyme tyrosine transaminase?

- A. Type I Tyrosinemia
 - B. Type II Tyrosinemia
 - C. Alkaptonuria
 - D. Type III Tyrosinemia
-

21. On centrifugation of a cell, LDH will get deposited in:

- A. Mitochondria
 - B. Lysosome
 - C. Cytoplasm
 - D. Golgi complex
-

22. Activation of trypsinogen is mainly done by:

- A. Chymotrypsinogen
 - B. Enteropeptidase
 - C. Trypsin
 - D. Hexokinase
-

23. Maple syrup urine disease is due to a deficiency of:

- A. α -keto acid dehydrogenase
- B. α -keto acid carboxylase
- C. α -keto acid desulphate

D. α -keto acid oxidase

24. Collagen is rich in which amino acid:

- A. Glycine
 - B. Arginine
 - C. Phenyl - alanine
 - D. Tyrosine
-

25. Which of the following releases histamine?

- A. Mast cells
 - B. Eosinophils
 - C. T — cells
 - D. NK cells
-

26. Vanillyl mandelic acid excreted in the urine in which of the following conditions?

- A. Pheochromocytoma
 - B. Alkaptonuria
 - C. Carcinoid syndrome
 - D. Diabetic ketoacidosis
-

27. Serotonin is also known as?

- A. 5 hydroxy tryptamine
 - B. 5 hydroxytryptophan
 - C. 5 carboxy tryptamine
 - D. 5 carboxy tryptophan
-

28. What is absent in the urine of a patient who presents with recurrent kidney stones? Below in the microscopic examination of the patient's urine sample.



- A. Arginine
 - B. Lysine
 - C. Cystine
 - D. Cysteine
-

29. In which condition is there an excessive excretion of 5-hydroxy indole acetic acid in the urine?

- A. Alkaptonuria
 - B. Albinism
 - C. Carcinoid syndrome
 - D. Phenylketonuria
-

30. Which of the following should be avoided by a patient with fish odor syndrome?

- A. Biotin
 - B. Choline
 - C. Niacin
 - D. Pantothenic acid
-

31. Defective fumarylacetoacetate hydrolase enzyme is associated with_____

- A. Type 1 Tyrosinemia
 - B. Type 2 Tyrosinemia
 - C. Type 3 Tyrosinemia
 - D. Type 4 Tyrosinemia
-

32. A 45-year-old woman with hypertriglyceridemia is considered for niacin therapy. She is a known diabetic managed adequately with sugars well under control. Why should niacin be used cautiously in this patient?

- A. It can cause hyperglycemia.
- B. It can cause scleroderma, which makes injecting insulin difficult
- C. It can increase the metabolism of oral hypoglycemic drugs
- D. It can cause hypoglycemia

33. An infant presented to the OPD with a history of vomiting and poor feeding. Musty odor is present in the baby's sweat and urine. Guthrie test was done and it was found to be positive. All are true regarding this disease except:

- A. Phenylalanine hydroxylase enzyme defect
 - B. White patch of hair due to tryptophan deficiency
 - C. Phenylacetate positive in urine
 - D. Mental retardation is present
-

34. A patient was brought to the OPD with complains of Mental retardation, mousy odor and hypopigmentation. What is the likely diagnosis?

- A. Phenylketonuria
 - B. Alkaptonuria
 - C. Albinism
 - D. MSUD
-

35. Which substrate of the Krebs cycle is diminished by an excess amount of ammonia?

- A. Malate
 - B. Oxaloacetate
 - C. Fumarate
 - D. Alpha ketoglutarate
-

36. From which amino acid is nitric oxide synthesized?

- A. Tryptophan
 - B. Lysine
 - C. Arginine
 - D. Tyrosine
-

37. An infant is brought by his parents with complaints that his urine turns black on standing. Which of the following metabolic disorders is the likely cause?

- A. Phenylketonuria
 - B. Alkaptonuria
 - C. Homocystinuria
 - D. Maple syrup urine disease
-

38. Which immunoglobulin is present in serum/plasma at the highest concentration?

- A. IgG
 - B. IgA
 - C. IgM
 - D. IgE
-

39. In a patient with maple syrup urine disease, all of the following amino acids should be restricted in diet, except?

- A. Methionine
 - B. Valine
 - C. Leucine
 - D. Isoleucine
-

40. Deamination of methylated cytosine forms which of the following?

- A. Uracil
 - B. Cytosine
 - C. Guanine
 - D. Thymine
-

41. A young girl presented to the OPD for examination. She is tall and thin with slender fingers, and dislocated lenses. Workup revealed cystathione beta-synthase deficiency. Which amino acid supplement is required in this case?

- A. Cysteine
 - B. Methionine
 - C. Serine
 - D. Tyrosine
-

42. Replacing alanine by which amino acid, will increase the UV absorbance of protein at 280nm wavelength?

- A. Leucine
- B. Proline
- C. Arginine

D. Tryptophan

43. Which of the following is true about the structural organization of proteins?

- A. Tertiary structure is three-dimensional
 - B. Primary, secondary, and tertiary structures are destroyed by denaturation
 - C. The secondary structure is stabilized by disulfide bonds
 - D. The secondary and tertiary structure depends upon amino acid sequence
-

44. Glycated hemoglobin (HbA1c) is best measured using?

- A. Ion exchange chromatography
 - B. Affinity chromatography
 - C. Isoelectric focusing
 - D. Electrophoresis
-

45. Which of the following options is false in a patient with advanced liver disease?

- A. Hyperammonemia
 - B. Decreased ATP synthesis
 - C. Oxidative carboxylation of alpha-ketoglutarate
 - D. Increased synthesis of glutamine and glutamine from alpha-ketoglutarate
-

46. Which of the following macronutrients requires the most energy for assimilation?

- A. Proteins
 - B. Carbohydrates
 - C. Mono-unsaturated fatty acids
 - D. Polyunsaturated fatty acids
-

47. Which of the following methods cannot be used to precipitate proteins?

- A. Add alcohol and acetone
 - B. Using heavy metal ions
 - C. Moving pH away from isoelectric pH
 - D. Adding trichloroacetic acid
-

48. Which of the following deficiencies leads to an elevation of glutamine levels in the bloodstream?

- A. Arginosuccinate lyase
 - B. Alpha-galactosidase-A
 - C. Ornithine transcarbamoylase
 - D. Arginase
-

49. Which amino acid is essential for the conversion of nor-epinephrine to epinephrine among the following options?

- A. Tyrosine
- B. Tryptophan
- C. Phenylalanine
- D. Methionine

50. Nitrogenous waste is excreted from the body in the form of?

- A. Urea
 - B. Nitric oxide
 - C. Ammonia
 - D. Glutamine
-

51. What is the shared precursor amino acid for urea, creatinine, and nitric oxide?

- A. Glycine
 - B. Arginine
 - C. Aspartate
 - D. Alanine
-

52. What do chaperones assist in?

- A. Ubiquitination of proteins
 - B. Protein modification
 - C. Protein folding
 - D. Protein cleavage
-

53. A female patient presented with pain and redness in the great toe. The serum uric acid level is 9.4mg/dl. The physician prescribed a drug for the treatment. Which of the following enzyme is likely to be inhibited by this drug?

- A. Phosphoribosyl transferase
 - B. Thymidylate synthase
 - C. Xanthine oxidase
 - D. Dihydrofolate reductase
-

54. Which of the following is responsible for the negative charge in fibrinopeptide A?

- A. Glutamate and valine
 - B. Histidine and lysine
 - C. Aspartate and glutamate
 - D. Serine and threonine
-

55. Urea cycle occurs in which part of the cell?

- A. Mitochondria
 - B. Cytoplasm
 - C. Both
 - D. None
-

56. Melanin is synthesized from which of the following?

- A. Tyrosine
- B. Tryptophan
- C. Phenyl alanine
- D. Threonine

57. Which of the following aids in the process of wound healing?

- A. Keratan sulfate
 - B. Dermatan sulfate
 - C. Hyaluronic acid
 - D. Chondroitin sulfate
-

58. What is the reason behind the stability of adenosine receptors, which is attributed to the formation of numerous disulfide bonds between?

- A. Cysteine
 - B. Methionine
 - C. Arginine
 - D. Alanine
-

59. In urea cycle disorders, which of the following can be used to reduce the levels of ammonia?

- A. Glutamate
 - B. Phenylbutyrate
 - C. Isoleucine
 - D. L-carnitine
-

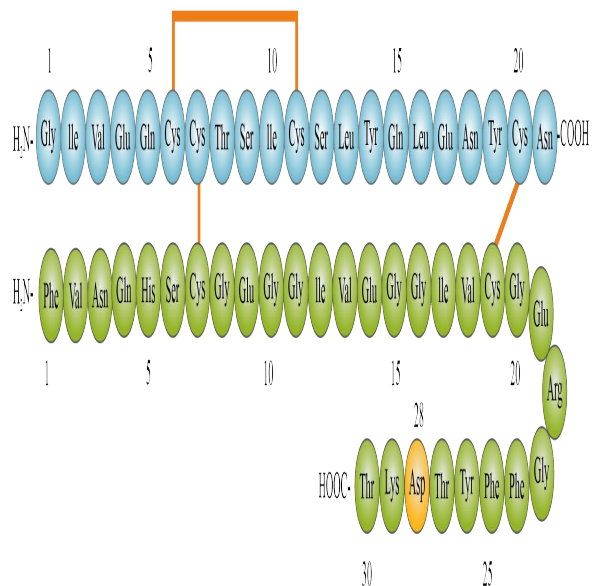
60. The biochemical form of amyloid in hemodialysis-associated amyloidosis is:

- A. AL
- B. AA
- C. Aβ2m
- D. ATTR

61. A body builder starts eating raw eggs for protein. He starts to develop fatigue on moderate exercise. The doctor prescribes a vitamin. Which enzyme is deficient in him?

- A. Glucose 6 phosphate
- B. Pyruvate carboxylase
- C. PEPCK
- D. Glycogen phosphorylase

62. Which modification will lead to a change in the biological activity of the hormone given in the picture?



- A. Hydrolysis of disulphide bonds
 - B. Change in 5-6 in A chain
 - C. Change in 29-30 in B chain
 - D. Change in 14-16 in A chain
-

63. Which of the following would have no effect on the function of the protein product?

- A. Glutamine replaced by asparagine
 - B. Glutamine replaced by alanine
 - C. Glutamine replaced by glutamate
 - D. Glutamine replaced by arginine
-

64. Melanin is derived from which of the following amino acids?

- A. Tyrosine
 - B. Tryptophan
 - C. Phenylalanine
 - D. Alanine
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	1
Question 4	2
Question 5	3
Question 6	1
Question 7	4
Question 8	1
Question 9	1
Question 10	4
Question 11	4
Question 12	1
Question 13	2
Question 14	3
Question 15	4
Question 16	4
Question 17	4
Question 18	2

Question 19	2
Question 20	2
Question 21	3
Question 22	2
Question 23	1
Question 24	1
Question 25	1
Question 26	1
Question 27	1
Question 28	4
Question 29	3
Question 30	2
Question 31	1
Question 32	1
Question 33	2
Question 34	1
Question 35	4
Question 36	3
Question 37	2
Question 38	1
Question 39	1

Question 40	4
Question 41	1
Question 42	4
Question 43	1
Question 44	1
Question 45	3
Question 46	1
Question 47	3
Question 48	3
Question 49	4
Question 50	1
Question 51	2
Question 52	3
Question 53	3
Question 54	3
Question 55	3
Question 56	1
Question 57	3
Question 58	1
Question 59	2
Question 60	3

Question 61	2
Question 62	3
Question 63	1
Question 64	1

Solution for Question 1:

Answer: A) Histidine

Explanation:

- Physiological pH & Buffering Role: Histidine has an imidazole side chain with a pKa of ~6.0 , which is close to physiological pH (7.4). This allows it to accept or donate protons , making it an effective buffer in blood .
- Histidine has an imidazole side chain with a pKa of ~6.0 , which is close to physiological pH (7.4).
- This allows it to accept or donate protons , making it an effective buffer in blood .
- Role in Hemoglobin Buffering: Hemoglobin contains histidine residues that help maintain blood pH by binding or releasing H⁺ ions. This is essential in the Bohr effect , which regulates oxygen delivery.
- Hemoglobin contains histidine residues that help maintain blood pH by binding or releasing H⁺ ions.
- This is essential in the Bohr effect , which regulates oxygen delivery.
- Histidine has an imidazole side chain with a pKa of ~6.0 , which is close to physiological pH (7.4).
- This allows it to accept or donate protons , making it an effective buffer in blood .
- Hemoglobin contains histidine residues that help maintain blood pH by binding or releasing H⁺ ions.
- This is essential in the Bohr effect , which regulates oxygen delivery.

Amino Acid	Buffering Ability	Reason
Histidine	Strong buffer	Imidazole group has pKa ~6.0, close to physiological pH

Arginine	Poor buffer	Guanidinium group has pKa ~12.5, too high to act as a buffer at pH 7.4
Tryptophan	Minimal buffering	Indole group is non-ionizable at physiological pH
Tyrosine	Minimal buffering	Phenol group has pKa ~10.1, too high to be effective

- Arginine (Option B): Incorrect because its guanidinium group (pKa ~12.5) does not donate or accept protons at physiological pH.
- Tryptophan (Option C): Incorrect because its indole ring is not significantly ionizable at pH 7.4, so it lacks buffering ability.
- Tyrosine (Option D): Incorrect because its phenol group (pKa ~10.1) does not participate in blood buffering at physiological pH.

Solution for Question 2:

Answer: A) Homogentisate oxidase

Explanation:

- Arthritis with joint blackening suggests alkaptonuria, a disorder caused by homogentisate oxidase deficiency, leading to the accumulation of homogentisic acid, which deposits in joints and cartilage.
- Ochronosis (dark pigmentation): Homogentisic acid polymerizes and stains connective tissues, cartilage, and joints black, seen during arthroscopy.
- Other features: Black urine (on standing due to oxidation), ochronotic pigmentation in the sclera and ear cartilage.

Feature	Alkaptonuria (Homogentisate Oxidase Deficiency)
Deficient Enzyme	Homogentisate oxidase
Accumulated Metabolite	Homogentisic acid
Key Features	Black urine (on standing), ochronosis (dark connective tissue pigmentation), early-onset arthritis
Affected Areas	Joints, cartilage (ear, sclera, intervertebral discs)

Diagnosis	Urine darkening with alkalization, elevated homogentisic acid in urine
Treatment	Low-protein diet (restricting phenylalanine & tyrosine), vitamin C, symptomatic management

- Branched-chain alpha-ketoacid dehydrogenase (Option B): Incorrect because its deficiency causes Maple Syrup Urine Disease (MSUD), characterized by neurological symptoms, metabolic acidosis, and sweet-smelling urine, not arthritis.
- Tyrosinase (Option C): Incorrect because its deficiency leads to albinism, causing hypopigmentation, not joint issues.
- Fumarase (Option D): Incorrect because its deficiency affects the Krebs cycle, leading to metabolic acidosis, hypotonia, and neurological dysfunction, not ochronosis or arthritis.

Solution for Question 3:

Correct Option A - Peroxisomes:

- Zellweger syndrome is caused by a defect in peroxisomes. Peroxisomes are essential for the oxidation of very long-chain fatty acids, the synthesis of ether lipids (plasmalogens), and the α -oxidation of branched-chain fatty acids. A deficiency or dysfunction in peroxisomes leads to the accumulation of these fatty acids and other toxic substances, causing Zellweger syndrome.

Incorrect Options:

Option B - Nucleosomes: Nucleosomes are units of DNA packaging in the nucleus, consisting of DNA wrapped around histone proteins. They are not involved in Zellweger syndrome.

Option C - Lysosomes : Lysosomes are involved in breaking down waste materials and cellular debris. Disorders of lysosomes, such as Tay-Sachs disease or Gaucher's disease, are not related to Zellweger syndrome.

Option D - Ribosomes : Ribosomes are responsible for protein synthesis. Ribosomal dysfunction can lead to various diseases, but it is not related to Zellweger syndrome.

Solution for Question 4:

Correct Option B – Creatinine:

- Creatinine is a product of glycine and arginine which is an amino acid but creatinine is not a neurotransmitter.

Incorrect Options:

Option A - GABA (Gamma-Aminobutyric Acid):

- GABA is a neurotransmitter that is made from the amino acid glutamate through a decarboxylation reaction catalyzed by the enzyme glutamate decarboxylase.

Option C – Dopamine:

- Dopamine is a neurotransmitter produced from phenylalanine(amino acid) through decarboxylation.

Option D - 5 hydroxytryptamine:

- Serotonin (5-hydroxytryptamine) is a neurotransmitter synthesized from the amino acid tryptophan.
-

Solution for Question 5:

Correct Option C: Protein folding

Chaperones assist other proteins in folding properly during and after synthesis. Folding of over half of mammalian proteins is done by chaperones which can be divided into two types:

- The hsp70 (70-kDa heat shock protein) family of chaperones mainly binds short sequences of hydrophobic amino acids to shield them from solvents in newly synthesized polypeptides. These are produced in response to stress conditions.
- Chaperonins are another type of chaperone that form large protein complexes that provide a protective environment for the folding of the newly synthesized protein.

Chaperones provide an opportunity to form appropriate secondary structural elements and their subsequent coalescence into a molten globule, thus preventing aggregation.

Incorrect Options

Option A : Ubiquitination of proteins happens through ubiquitin-activated enzymes.

Option B : Non-enzymatic pathways result in protein modification.

Option D : Proteases assist in protein cleavage.

Solution for Question 6:

The correct answer is A: L-arginine.

- Nitric oxide (NO) is synthesized from L-arginine, an amino acid. The process of NO synthesis involves the action of an enzyme called nitric oxide synthase (NOS). NOS converts L-arginine into nitric oxide and L-citrulline through a complex biochemical reaction.

Here's a brief explanation of the options provided:

Option A: L-arginine: This is the correct answer. L-arginine serves as the substrate for the synthesis of nitric oxide by nitric oxide synthase.

Option B: L-citrulline: L-citrulline is not directly involved in the synthesis of nitric oxide. However, it is produced as a byproduct of the reaction, along with nitric oxide, during the conversion of L-arginine by nitric oxide synthase.

Option C: Glycine: Glycine is an amino acid, but it is not directly involved in the synthesis of nitric oxide. It has other roles in cellular metabolism.

Option D: Lysine: Lysine is an essential amino acid, but it is not directly involved in the synthesis of nitric oxide. It has various functions in protein synthesis and cellular metabolism.

In summary, nitric oxide is synthesized from L-arginine through the action of nitric oxide synthase. L-citrulline is produced as a byproduct of this reaction.

Solution for Question 7:

Correct Option D - Glutamine:

- Glutamine serves as the major transport form of ammonia from the brain to the liver. The brain is highly sensitive to ammonia toxicity, so it has a specialized mechanism to detoxify ammonia by converting it into glutamine. This conversion is catalyzed by the enzyme glutamine synthetase.
- The synthesis of glutamine involves the following reaction:
- Glutamate + NH₃ + ATP → Glutamine + ADP + Pi

Incorrect Options:

Options A, B, and C are incorrect: Refer to the explanation of the correct option D.

Solution for Question 8:

Correct option : A

- Arginine. Nitric oxide (NO) is derived from the amino acid arginine.
- Nitric oxide is a signaling molecule and a potent vasodilator involved in various physiological processes such as regulation of blood pressure, immune response, and neurotransmission. It is synthesized in the body by the enzyme nitric oxide synthase (NOS), which converts arginine into nitric oxide and citrulline.

Incorrect options :

Option B . Histidine: Histidine is an essential amino acid involved in various biological processes, but it is not directly involved in the synthesis of nitric oxide.

Option C . Glycine: Glycine is an amino acid that plays a crucial role in collagen structure and various other biological functions. However, it is not the precursor for nitric oxide synthesis.

Option D . Aspartate: Aspartate is an amino acid involved in protein synthesis and various metabolic pathways, but it is not directly involved in the synthesis of nitric oxide.

Solution for Question 9:

Correct Choice: A

Glycated hemoglobin (HbA1c) is measured by ion exchange chromatography. This method is based on difference in charge. A mixture of charged proteins is analyzed over carboxymethylcellulose paper which is negatively charged. A positively charged resin bed separates negatively charged HbA1c. It has the advantage of separating hemoglobin variants into peaks. Some hemoglobin variants may interfere with results.

Incorrect Choices:

Option B: Electrophoresis: It is the movement of biomolecules in electric field.

- It depends on charge, size and shape
- It is used to separate the components of a protein mixture
- It denatures the protein structure and coats all the protein with a negative charge.

Option C: Immunoassay: It measures the presence or concentration of a macromolecule in a solution through the use of an antibody (usually) or an antigen, e.g. ELISA

Option D: Affinity chromatography: It is based on affinity between the molecules.

- A mixture of enzymes and other proteins is analysed wherein the other protein elutes first
- It is used to separate enzymes, vitamins, antibodies and hormones.

Solution for Question 10:

Correct Option D - Creatinine:

- Creatinine is not a derivative of tryptophan. It is a waste product of muscle metabolism and is not synthesized from tryptophan.

Incorrect Options:

Option A - Melatonin: Melatonin is a derivative of tryptophan. It is involved in the regulation of the sleep-wake cycle.

Option B - Serotonin: Serotonin, also known as 5-hydroxytryptamine (5-HT), is a neurotransmitter derived from tryptophan. It plays a role in mood regulation and various physiological functions.

Option C - Niacin: Niacin, also known as vitamin B3, is derived from tryptophan. The body can convert tryptophan into niacin, which is essential for various metabolic processes.

Solution for Question 11:

Correct Option D - Histidine:

- Histidine is an amino acid that acts as a buffer in blood at physiological pH. This buffering action is attributed to the presence of the imidazole group in histidine. The imidazole group can accept or donate protons, allowing histidine to act as a weak acid or base, depending on the pH of the surrounding environment.
- Histidine has a pKa value of approximately 6.0, which is close to the physiological pH of the blood. Therefore, histidine is very effective as a buffer in living systems, including blood.

Incorrect Options:

Option A - Methionine:

- Methionine does not have functional groups that can act as weak acids or bases, so it is not involved in buffering reactions.

Option B - Alanine:

- Alanine does not have functional groups that can act as weak acids or bases, so it is not involved in buffering reactions.

Option C - Arginine:

- Arginine buffering capacity is much weaker compared to histidine due to differences in pK values and concentrations.
-

Solution for Question 12:

Correct Option A - Glutamine:

Glutamine is the molecule responsible for transporting ammonia from the brain to the liver in a process called the glutamine cycle. In the brain, ammonia is converted into glutamine by the enzyme glutamine synthetase. Glutamine is then transported through the bloodstream to the

liver, where it is converted back into ammonia by the enzyme glutaminase.

Incorrect Options:

Option B - Leucine: Leucine is an essential amino acid involved in protein synthesis and does not play a direct role in transporting ammonia.

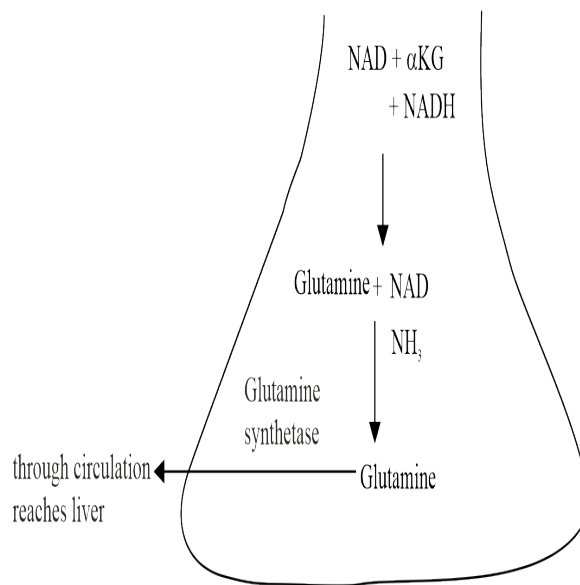
Option C - Lysine: Lysine is also an essential amino acid and does not have a specific role in ammonia transport.

Option D - Aspartate: Aspartate is involved in various metabolic processes but is not primarily responsible for transporting ammonia from the brain to the liver.

Solution for Question 13:

Correct Option B - Glutamine:

- In the brain, excess ammonia (NH_3) is detoxified through a process called the astrocyte-neuron lactate shuttle. Astrocytes take up ammonia from the surrounding brain tissue and combine it with glutamate to form glutamine. This reaction is catalyzed by the enzyme glutamine synthetase.
- Glutamine is then released by astrocytes and taken up by neurons, where it can be utilized for various metabolic processes. Glutamine can also be converted back into glutamate, which is an important neurotransmitter in the brain.



Incorrect Options:

Option A - Urea: Urea is a compound formed in the liver as part of the urea cycle, which occurs primarily in hepatocytes. The urea cycle is responsible for the detoxification of ammonia in the liver, not in the brain.

Option C - Ammonium ions: Ammonium ions (NH_4^+) can be produced from ammonia in the brain, but they are not the final product of detoxification.

Option D - Alanine: Alanine is an amino acid that can participate in the metabolism of ammonia, but it is not the primary detoxification product of ammonia in the brain.

Solution for Question 14:

Correct Option C- S-AdenosylMethionine:

- S-Adenosyl Methionine is a precursor for many reactions that involve methylation. S-AdenosylMethionine is derived from the amino acid Methionine. It is involved in many methylation processes such as DNA methylation.
- S-Adenosylmethionine is highly reactive due to the presence of a positive charge. S-Adenosylmethionine transfers the methyl group to an acceptor and in turn, is converted to S-adenosyl homocysteine.

Incorrect Options:

Option A - Glucose-6-Phosphate: it is an intermediate of the glycolysis cycle. Glucose-6-Phosphate is not a methyl group donor.

Option B - Acetyl CoA: Acetyl CoA is part of many metabolic reactions such as the citric acid cycle and cholesterol synthesis but is not a methyl group donor.

Option D - S-Adenosyl serine: It is a methyl group donor in only certain specific reactions such as the production of serotonin and is not considered a universal methyl group donor.

Solution for Question 15:

Correct Option D - Homogentisic acid oxidase:

- Homogentisic acid (HGA) is a metabolite that is used as the diagnostic substance in the tissues for alkaptonuria. Alkaptonuria is an autosomal recessive disease that occurs rarely.
- When Homogentisic-acid oxidase is found deficient in the tissue, then normal procedure of phenylalanine metabolism gets blocked and Homogentisic acid (HGA) becomes unable to break. Hence, urine appears as black and dark-stained nappies, and can be seen as the earliest signs of it.

Incorrect Options:

Option A - Beta -carotene is a type of carotenoid that is present in carrots, apricots, and sweet potatoes. It is a provitamin and helps the body to make vitamin A. High doses of beta carotene might turn urine bright orange or yellow.

Option B - Tyrosine aminotransferase deficiency in children causes tyrosine to break incompletely. . Ocular manifestations- Painful corneal erosions and Cutaneous manifestation:- Palmar hyperkeratosis

Option C - Excess vitamin C can change urine colour into dark yellow or orange and vice-versa.

Solution for Question 16:

Correct Option D - Aspartate:

- Aspartate is the amino acid that contributes to the production of oxaloacetate through transamination reactions in the citric acid cycle.

Incorrect Options:

Option A - Alanine: Alanine can be involved in the glucose-alanine cycle and can be converted into pyruvate, not oxaloacetate.

Option B - Lysine: Lysine is an essential amino acid, and it is not directly involved in the synthesis of oxaloacetate.

Option C - Leucine: Leucine is also not directly involved in the synthesis of oxaloacetate. It is primarily involved in protein synthesis and can be catabolized to generate acetyl-CoA, which can then enter the citric acid cycle.

Solution for Question 17:

Correct Option D - Antimycin:

- Antimycin is an inhibitor of complex III (cytochrome b-c1 complex) of the electron transport chain. It interrupts the transfer of electrons from ubiquinol to cytochrome c, thereby inhibiting the flow of electrons through this complex.

Incorrect Option:

Option A - Rotenone: Rotenone is an inhibitor of complex I of the electron transport chain. It blocks the transfer of electrons from NADH to ubiquinone in the mitochondrial respiratory chain.

Option B - TTFA: TTFA(Thenoyltrifluoroacetone is an inhibitor of complex II of the electron transport chain.

Option C - Cyanide: Cyanide is an inhibitor of complex IV (cytochrome c oxidase) of the electron transport chain. It prevents the transfer of electrons to oxygen, disrupting the final step of electron transport and therefore blocking the production of ATP.

Solution for Question 18:

Correct Option B - Glycine:

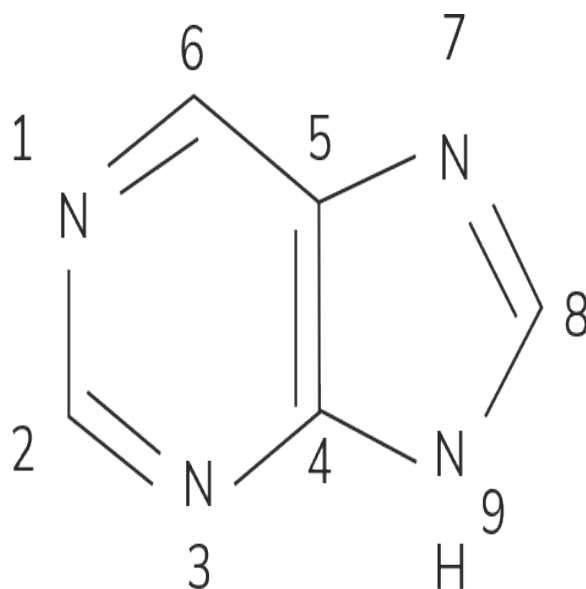
- Heme is a cyclic tetrapyrrole with a hexavalent iron (Fe^{2+}) atom at the center, coordinated with four pyrrole nitrogens. It serves as the final product of the porphyrin synthetic pathway, which initiates from two precursors: glycine and succinyl CoA.

Incorrect Options - A, C, and D are incorrect : Refer to the explanation of option B.

Solution for Question 19:

Correct Option B - Glycine, Aspartate, Glutamine:

- There are many compounds that contribute to the purine ring:
- Aspartate : Provides the nitrogen at position N-1 of the purine ring.
- Tetrahydrofolate: Carries a single carbon moiety that contributes to carbons at positions C-2 and C-8 of the purine ring.
- Glutamine : Contributes the nitrogens at positions N-3 and N-9 of the purine ring.
- CO_2 (Carbon dioxide): Provides the carbon at position C-6 of the purine ring.
- Glycine : Contributes carbons at positions C-4 and C-5, as well as the nitrogen at position N-7 of the purine ring.



- Incorrect Option - A, C, and D are incorrect : Refer to the explanation of option B.

Solution for Question 20:

Correct Option B - Type II Tyrosinemia:

- Tyrosinemia is a group of inherited metabolic disorders characterized by elevated levels of tyrosine in the blood.
- In Type II tyrosinemia, tyrosine transaminase is deficient and presents with symptoms such as painful corneal erosions and plaques, inflammation caused by intracellular crystallization of tyrosine, keratosis of the palmar surface, and mental retardation. Low-tyrosine and low-phenylalanine diets are beneficial in managing this condition.

Incorrect Options:

Option A - Type I Tyrosinemia:

- Fumarylacetoacetate hydrolase is deficient in Type I Tyrosinemia

Option C - Alkaptonuria:

- Alkaptonuria is caused by a deficiency of the enzyme homogentisate oxidase, inherited in an autosomal recessive manner.

Option D - Type III Tyrosinemia:

- Tyrosinemia Type III (neonatal tyrosinemia) results from the absence of the enzyme para-hydroxyphenylpyruvate hydroxylase.
-

Solution for Question 21:

Correct Option C - Cytoplasm:

- LDH, commonly known as lactate dehydrogenase, is a protein that functions as an enzyme. It is a tetrameric enzyme consisting of two monomer types: H (for the heart) and M (for muscle) that combine to yield five LDH isozymes: HHHH (I1), HHMM (I2), b, HMMM (I4), and MMMM (I5). The enzyme of glycolysis, which takes place in the cytoplasm, is lactate dehydrogenase. It serves as a cytoplasmic marker.

Incorrect Options:

Option A - LDH is a cytoplasmic enzyme. Even though LDH is primarily a cytoplasmic enzyme, numerous investigations have proven its mitochondrial presence. A mitochondrial L-lactate dehydrogenase (mL-LDH) was found in yeast, plants, and mammals.

Option B - Lysosomes are membrane-enclosed compartments containing hydrolytic enzymes employed for regulated intracellular macromolecule digestion. LDH is predominantly a cytoplasmic enzyme; various studies do not demonstrate its lysosomal presence.

Option D - The Golgi apparatus serves as a factory where proteins from the ER are further processed and sorted before being transported to their final destinations. Lactate dehydrogenase (LDH) is not one of the enzymes that get deposited in the Golgi apparatus.

Solution for Question 22:

Correct Option B - Enteropeptidase:

- Enteropeptidase, also known as enterokinase, is an enzyme produced by the duodenum. It plays a crucial role in the activation of trypsinogen to trypsin. Once trypsinogen comes into contact with enteropeptidase, it is cleaved to form active trypsin.

Incorrect Options:

Option A - Chymotrypsinogen: Chymotrypsinogen is a precursor of chymotrypsin, another pancreatic enzyme, and is not involved in the direct activation of trypsinogen.

Option C - Trypsin: While trypsin is involved in the activation of other pancreatic zymogens, it is not the primary enzyme responsible for the activation of trypsinogen itself.

Option D - Hexokinase: Hexokinase is an enzyme involved in glucose metabolism and is not related to the activation of trypsinogen.

Solution for Question 23:

Correct Option A - α -keto acid dehydrogenase:

- Maple syrup urine disease (MSUD) is caused by a deficiency of the enzyme α -keto acid dehydrogenase.

Incorrect Options: Options B, C, and D are incorrect.

Solution for Question 24:

Correct Option A - Glycine

- Glycine plays a critical role in collagen structure and function. It is the smallest amino acid and has a unique structural property that allows it to fit tightly into the triple helical structure of collagen.
- Glycine residues are found at every third position in the collagen polypeptide chain, allowing for the close packing of collagen molecules and the formation of stable triple helices.

Incorrect Options :

Option B - Arginine: While arginine is an important amino acid involved in various biological processes, it is not the amino acid that is predominantly present in collagen.

Option C - Phenylalanine: Phenylalanine is an aromatic amino acid, but it is not the primary amino acid found in collagen.

Option D - Tyrosine: Tyrosine is another aromatic amino acid, but it is not the primary amino acid found in collagen.

Solution for Question 25:

Correct Option A - Mast cells

- Mast cells are the primary cells responsible for releasing histamine. They are involved in allergic reactions and immune responses, and the release of histamine from mast cells contributes to the symptoms of allergic conditions.

Incorrect Options:

Option B - Eosinophils: While eosinophils can release certain inflammatory mediators, such as leukotrienes and cytokines, they do not release histamine.

Option C - T-cells: T-cells are a type of white blood cell that plays a crucial role in adaptive immune responses. They help coordinate immune responses and target specific pathogens or infected cells.

Option D - NK cells: NK cells are primarily involved in cell-mediated cytotoxicity and cytokine production. They do not release histamine, which is predominantly released by mast cells.

Solution for Question 26:

Correct Option A - Pheochromocytoma:

- Vanillylmandelic acid is a product of epinephrine and norepinephrine (belongs to the group of similar hormones called catecholamines) metabolism excreted in the urine and produced in the liver.

- Pheochromocytoma , a catecholamine-producing tumor, is a condition in which norepinephrine, epinephrine, and dopamine are released in abnormally high amounts.

Incorrect Options :

Option B - Homogentisic acid is deficient in the urine during alkaptonuria.

Option C - The compound called 5-hydroxy indole acetic acid (5-HIAA) is excreted in the urine in a condition called carcinoid syndrome.

Option D - Ketone is excreted in the urine during a condition known as diabetic ketoacidosis.

Solution for Question 27:

Correct Option A - 5 hydroxytryptamine

- Serotonin is also known as 5 hydroxytryptamine. It is a monoamine neurotransmitter.

Incorrect Options - B, C, and D: These are incorrect

Solution for Question 28:

Correct Option D - Cysteine

- Cystinuria is an inherited AR disease
- It is due to a defective amino acid transporter which is responsible for reabsorbing 4 amino acid along the PCT- Cystine, Ornithine, Lysine and Arginine.
- The image shows cystine in the urine sample in hexagonal shape.

Incorrect Options:

Option A - Urine sample contains arginine.

Option B - Given urine sample contains lysine.

Option C - Given urine sample contains cysteine.

Solution for Question 29:

Correct Option C - Carcinoid Syndrome

- In Carcinoid syndrome, there is an excessive synthesis of serotonin or 5-Hydroxytryptamine. This serotonin, after metabolism, excretes 5-hydroxy indole acetic acid into the urine for excretion.

Incorrect Options:

Option A - In Alkaptonuria, urine turns black on exposure to air by the incomplete breakdown of tyrosine and phenylalanine.

Option C - In Albinism, there is defective production of melanin.

Option D - The buildup of phenylalanine in urine is seen in Phenylketonuria.

Solution for Question 30:

Correct Option B - Choline

- Choline is recommended to be avoided by patients with fish odor syndrome.
- In a normal individual, when choline is ingested, it is reduced to TMA by colonic bacteria. The TMA would diffuse passively across cell membranes and enter the enterohepatic circulation. Here, it is removed by the liver, which FMO3 or flavin monooxygenase 3 enzyme normally oxidizes TMA into odorless TMAO. This process of trimethylamine oxidation is defective or impaired in patients with trimethylaminuria.

Incorrect Options:

Option A - Biotin does not trigger choline or trimethylamine metabolism. So it has no influence on fish odor syndrome.

Option C - Niacin does not trigger choline or trimethylamine metabolism. So the patient will not have aggravation of fish odor syndrome.

Option D - Pantothenic acid does not trigger choline or trimethylamine metabolism. So the patients with fish odor syndrome can include pantothenic acid in diet.

Solution for Question 31:

Correct Option A - Type 1 Tyrosinemia

- Tyrosinemia Type 1 is the most common type of tyrosinemia. It occurs when there is fumarylacetoacetatehydrolase deficiency.
- It results in jaundice, hepatomegaly, hypoglycemia, HCC, and liver cirrhosis.

Incorrect Options:

Option B - Type 2 tyrosinemia occurs when there is tyrosine transaminase deficiency.

Option C - Type 3 tyrosinemia(neonatal tyrosinemia) occurs when there is PHPP hydroxylase deficiency and is the rarest of all the types.

Option D - Type 4 tyrosinemia is not known.

Solution for Question 32:

Correct Option A - It can cause hyperglycemia

- It can cause hyperglycemia: This option is correct. Niacin, also known as vitamin B3, can cause an increase in blood glucose levels.
- It has been shown to impair glucose tolerance and increase insulin resistance, which can lead to elevated blood sugar levels. Therefore, niacin should be used cautiously in patients with diabetes to avoid worsening of glycemic control.

Incorrect Options:

Option B - It can cause scleroderma, which makes injecting insulin difficult: This option is incorrect. Niacin is not known to cause scleroderma, which is a chronic connective tissue disorder characterized by thickening and hardening of the skin.

Option C - It can increase the metabolism of oral hypoglycemic drugs: This option is incorrect. Niacin primarily affects glucose metabolism and insulin resistance.

Option D - It can cause hypoglycemia: This option is incorrect. Niacin is not known to cause hypoglycemia. On the contrary, it is more likely to cause hyperglycemia, as mentioned in the correct option.

Solution for Question 33:

Correct Option B - White patch of hair due to tryptophan deficiency:

- White patch of hair due to tryptophan deficiency: This statement is false. A white patch of hair is not associated with PKU. It is commonly seen in another genetic condition called piebaldism, which is characterized by patches of depigmented skin and hair due to the absence of melanocytes in those areas.

Incorrect Options:

Option A - Phenylalanine hydroxylase enzyme defect: This statement is true. PKU is caused by a defect in the enzyme phenylalanine hydroxylase, which is responsible for converting the amino acid phenylalanine into tyrosine.

Option C - Phenylacetate positive in urine: This statement is true. In PKU, due to the accumulation of phenylalanine, it is metabolized through alternative pathways, leading to the production of phenylacetate. Phenylacetate is then excreted in the urine and can be detected through laboratory testing.

Option D - Mental retardation is present: This statement is true. If left untreated, PKU can lead to severe intellectual disability and developmental delays. The accumulation of phenylalanine in the body, especially during the early stages of brain development, can cause damage to the developing brain, resulting in cognitive impairments.

Solution for Question 34:

Correct Option A - Phenylketonuria:

- Phenylketonuria is an inherited autosomal recessive disorder due to defects in phenylalanine metabolism leading to accumulation of phenyl ketones.
- Clinical features Mental retardation- Due to phenyl ketones crossing BBB. Mousy odor- Due to accumulation of phenylacetate Hypopigmentation- Due to decreased synthesis of melanin
- Mental retardation- Due to phenyl ketones crossing BBB.
- Mousy odor- Due to accumulation of phenylacetate
- Hypopigmentation- Due to decreased synthesis of melanin
- Mental retardation- Due to phenyl ketones crossing BBB.
- Mousy odor- Due to accumulation of phenylacetate
- Hypopigmentation- Due to decreased synthesis of melanin

Incorrect Options:

Options B - Alkaptonuria: Alkaptonuria leads to the accumulation of homogentisic acid, which can cause dark urine, darkening of connective tissues, and joint problems.

Options C - Albinism: Albinism is a genetic condition characterized by a lack of melanin pigment in the skin, hair, and eyes.

Options D - MSUD (Maple Syrup Urine Disease): MSUD is a rare metabolic disorder characterized by the inability to break down certain amino acids, including leucine, isoleucine, and valine. If left untreated, it can lead to severe neurological symptoms.

Solution for Question 35:

Correct Option D - Alpha ketoglutarate

- Alpha ketoglutarate : Excess ammonia can directly reduce alpha ketoglutarate , a key substrate of the Krebs cycle. Ammonia combines with alpha ketoglutarate to form glutamate in a reaction catalyzed by the enzyme glutamate dehydrogenase. This reaction helps in the removal of excess ammonia through the urea cycle.

Incorrect Options

Option A - Malate: Excess ammonia does not directly reduce malate, a substrate of the Krebs cycle. Malate is formed in the Krebs cycle and can be converted back to oxaloacetate.

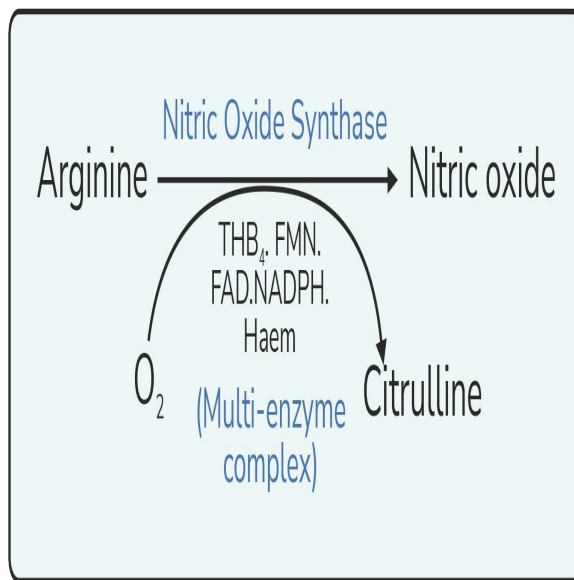
Option B - Oxaloacetate: Excess ammonia does not directly reduce oxaloacetate, another substrate of the Krebs cycle. Oxaloacetate is an intermediate compound in the cycle and can be converted to malate or used for gluconeogenesis.

Option C - Fumarate : Excess ammonia does not directly reduce fumarate, which is also an intermediate compound in the Krebs cycle. Fumarate is produced from the conversion of malate and is further converted to succinate.

Solution for Question 36:

Correct Option C - Arginine:

Nitric oxide (NO) is an important signalling molecule involved in various physiological processes, such as vasodilation and neurotransmission. It is synthesized from the amino acid arginine via the enzyme nitric oxide synthase (NOS).



Incorrect Options:

Option A - Tryptophan: Tryptophan is an essential amino acid converted into various compounds in the body, such as serotonin and melatonin. However, it is not a precursor for the synthesis of nitric oxide.

Option B - Lysine: Lysine is another essential amino acid that plays a role in the synthesis of proteins. However, it is not a precursor for the synthesis of nitric oxide.

Option D - Tyrosine: Tyrosine is a non-essential amino acid that is involved in the synthesis of various compounds, such as dopamine, adrenaline, and thyroid hormones. However, it is not a precursor for the synthesis of nitric oxide.

Solution for Question 37:

Correct Option B - Alkaptonuria:

- Alkaptonuria is a rare autosomal recessive metabolic disorder characterized by a deficiency of the enzyme homogentisate 1,2-dioxygenase (HGD).
- This enzyme is involved in the breakdown of the amino acids phenylalanine and tyrosine. In alkaptonuria, the deficiency of HGD leads to the accumulation of a substance called homogentisic acid (HGA) in the body.
- When exposed to air, HGA undergoes oxidation and polymerization, resulting in the characteristic darkening of urine upon standing.
- This phenomenon is known as "black urine" or "ochronosis".

Incorrect Options:

Option A - Phenylketonuria (PKU): PKU is caused by a deficiency of the enzyme phenylalanine hydroxylase, leading to the accumulation of phenylalanine in the body. It is associated with intellectual disability, developmental delay, and other neurological symptoms if left untreated.

Option C - Homocystinuria: Homocystinuria is a metabolic disorder characterized by the impaired metabolism of the amino acid methionine. It can be caused by deficiencies in enzymes involved in methionine metabolism, such as cystathionine beta-synthase.

Option D - Maple syrup urine disease: It is a rare inherited disorder characterized by the inability to break down the branched-chain amino acids (leucine, isoleucine, and valine). The accumulation of these amino acids can cause a distinct odor in urine resembling maple syrup.

Solution for Question 38:

Correct Option A - IgG

- Immunoglobulin G (IgG) is the immunoglobulin present in serum/plasma at the highest concentration.
- IgG antibodies are the most abundant in the bloodstream, accounting for approximately 75-80% of the total circulating immunoglobulins .

Incorrect Options:

Option B - This option is incorrect. While IgA is an important immunoglobulin, it is not in serum/plasma at the highest concentration. IgA is primarily found in secretions such as tears, saliva, mucus, and breast milk.

Option C - IgM is the first immunoglobulin class produced during the initial immune response to an infection or antigen. It plays a vital role in the primary immune response and is responsible for agglutination (clumping) of antigens.

Option D - IgE is primarily associated with allergic and hypersensitivity reactions. It triggers immediate hypersensitivity reactions by binding to mast cells and basophils, releasing histamine and other inflammatory mediators.

Solution for Question 39:

Correct Option A - Methionine:

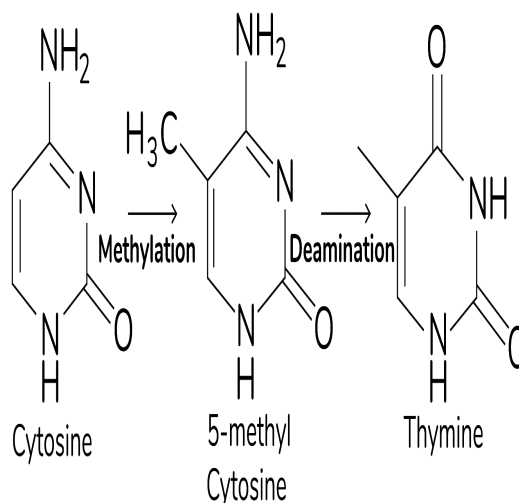
- Maple syrup urine disease is a metabolic defect caused by abnormal activity of the branched-chain alpha-ketoacid dehydrogenase (BCKAD) complex. This complex is responsible for the breakdown of branched-chain amino acids leucine, isoleucine, and valine.
- Methionine is an essential amino acid required for protein synthesis and other metabolic processes.
- It is not restricted in the diet of patients with maple syrup urine disease, and it is not one of the branched-chain amino acids affected by MSUD.

Incorrect Options - B, C, and D must be restricted in the diet of patients with maple syrup urine disease, as they are unable to metabolize it properly.

Solution for Question 40:

Correct Option D - Thymine:

- Deamination of methylated cytosine to thymine is a type of DNA damage that can occur spontaneously or as a result of exposure to environmental factors such as radiation or chemicals. This type of damage can lead to DNA sequence mutations, which can affect gene expression and protein synthesis.



Incorrect Options - A, C, and D are not produced from the deamination of methylated cytosine.

Solution for Question 41:

Correct Option A - Cysteine:

- The clinical presentation of a young girl with tall and thin stature, slender fingers, and dislocated lenses suggests a condition known as Marfan syndrome. However, the mention of

cystathione beta-synthase deficiency indicates that the patient actually has homocystinuria, a rare inherited disorder characterized by defective cysteine metabolism.

- Homocystinuria can result from a deficiency in cystathione beta-synthase, an enzyme involved in the conversion of homocysteine to cystathionine. This leads to the accumulation of homocysteine in the body. The deficiency of cystathione beta-synthase can be caused by genetic mutations.
- By providing exogenous cysteine, the deficiency in cystathione beta-synthase can be bypassed, allowing for the conversion of homocysteine to cysteine.

Incorrect Options:

Option B - Methionine: Methionine is an essential amino acid and serves as the precursor for homocysteine. However, in the case of cystathione beta-synthase deficiency, the conversion of homocysteine to cysteine is impaired, so supplementing with methionine would not address the underlying issue.

Option C - Serine: Serine is a nonessential amino acid and is not directly related to the metabolism of homocysteine or cysteine. It does not play a direct role in addressing the deficiency of cystathione beta-synthase.

Option D - Tyrosine: Tyrosine is a nonessential amino acid and is not involved in the metabolism of homocysteine or cysteine. It is not relevant to the deficiency of cystathione beta-synthase.

Solution for Question 42:

Correct Option D - Tryptophan:

- UV absorbance at 280nm is primarily attributed to the presence of aromatic amino acids, particularly tryptophan and tyrosine, in a protein. Among these, tryptophan has the highest molar absorptivity at 280nm, meaning it can absorb UV light at this wavelength.
- Alanine, on the other hand, is a non-aromatic amino acid with a low absorbance at 280nm. Therefore, substituting alanine with tryptophan would introduce an additional tryptophan residue into the protein sequence, increasing UV absorbance at 280nm.

Incorrect Options:

Option A - Leucine: Leucine is also a non-aromatic amino acid and does not significantly contribute to UV absorbance at 280nm.

Option B - Proline: Proline is an imino acid and absorbs UV light at 260nm. Therefore, it would not increase the UV absorbance at 280nm wavelength.

Option C - Arginine: Arginine is a positively charged amino acid and does not possess an aromatic ring. It does not contribute significantly to UV absorbance at 280nm.

Solution for Question 43:

Correct Option A - Tertiary structure is three-dimensional:

- Tertiary structure refers to the overall three-dimensional arrangement of a protein's polypeptide chain. It is determined by various interactions between amino acid residues, including hydrophobic interactions, hydrogen bonds, electrostatic interactions, and disulfide bonds. These interactions contribute to the folding of the protein into its functional three-dimensional shape.

Incorrect Options:

Option B - Primary structure refers to a protein's specific sequence of amino acids. It is not destroyed by denaturation, as the primary structure remains intact even if the protein loses its native conformation.

Option C - Secondary structure refers to local folding patterns within a protein, such as alpha helices and beta sheets. While disulfide bonds can contribute to the stability of some proteins, they are not the primary stabilizing force in secondary structure formation. Instead, secondary structure is primarily stabilized by hydrogen bonding between amino acid residues.

Option D - Protein's secondary and tertiary structures depend on the amino acid sequence. The primary structure determines the specific sequence of amino acids, and the interactions between these amino acids ultimately drive the formation of the secondary and tertiary structures. Therefore, it is incorrect to say that only the secondary structure depends on the amino acid sequence.

Solution for Question 44:

Correct Option A - Ion exchange chromatography:

- Glycated hemoglobin (HbA1c) is measured by ion exchange chromatography. This method is based on differences in charge. A mixture of charged proteins is analyzed over carboxymethylcellulose paper which is negatively charged.
- A positively charged resin bed separates negatively charged HbA1c. It has the advantage of separating hemoglobin variants into peaks. Some hemoglobin variants may interfere with results.

Incorrect Options:

Option B - Electrophoresis:

- It is used to separate the components of a protein mixture. It denatures the protein structure and coats all the protein with a negative charge.

Option C - Immunoassay: It measures the presence or concentration of a macromolecule in a solution through the use of an antibody (usually) or an antigen, eg. ELISA

Option D - Affinity chromatography: It is used to separate enzymes, vitamins, antibodies and hormones.

Solution for Question 45:

Correct Option C - Oxidative decarboxylation of alpha-ketoglutarate:

- False. In advanced liver disease, there is often a reduction in the activity of the citric acid cycle (also known as the Krebs cycle or TCA cycle), and oxidative decarboxylation of alpha-ketoglutarate is a crucial step in this cycle, not oxidative carboxylation of alpha-ketoglutarate. Therefore, this statement is not true; it is affected in advanced liver disease.

Incorrect Options:

Option A - Hyperammonemia: True. Advanced liver disease, particularly cirrhosis, often leads to a condition called hepatic encephalopathy, which is characterized by elevated ammonia levels in the blood (hyperammonemia). The liver is responsible for detoxifying ammonia to urea, and liver dysfunction impairs this process.

Option B - Decreased ATP synthesis: True. In advanced liver disease, there can be a decrease in ATP synthesis due to impaired liver function. The liver is involved in various metabolic processes, and its dysfunction can result in reduced energy production.

Option D - Increased synthesis of glutamine and glutamine from alpha-ketoglutarate: True. As liver function declines in advanced liver disease, there is an increased synthesis of glutamine. This is because the liver plays a role in the conversion of ammonia (NH_3) into glutamine (NH_4^+) for detoxification. This process is one of the ways the liver attempts to handle the excess ammonia in the blood.

Solution for Question 46:

Correct Option A - Proteins

Proteins consume the highest energy during assimilation as they have the highest thermogenic effect and require maximum energy.

Incorrect Options :

Option B - Carbohydrates : Excess of carbs is converted into fats and stored and it utilizes less energy for assimilation.

Option C - MUFA: They have a single double bond and consume the least energy to be broken down.

Option D - PUFA: They are essential amino acids required by our body and required the least energy to be broken down.

Solution for Question 47:

Correct Option C - Moving pH away from isoelectric pH:

- At $\text{pH} = \text{pI}$, precipitation of proteins occurs (no net charge)
- If $\text{pH} < \text{pI}$, acidic pH = protein will have a positive charge

- If $\text{pH} > \text{pI}$, alkaline $\text{pH} = \text{protein}$ will have a negative charge

Thus, in both these cases if we move pH away from pI , then precipitation will not occur

Incorrect Options:

Option A - Add alcohol and acetone: Alkaloidal reagents are negatively charged. In an acidic medium, proteins are positively charged. So, by neutralization of charges, negative ion precipitates out.

Option B - Using heavy metals and ion: Neutralisation of charges occur which lead to precipitation of positive ion as heavy metals are positively charged.

Option D - Adding trichloroacetic acid: It is negatively charged. In an acidic medium, proteins are positively charged. So, neutralization of charges occurs and the negative ion is separated.

Solution for Question 48:

Correct Option C - Ornithine transcarbamoylase:

- Ornithine transcarbamoylase: Deficiency of ornithine transcarbamoylase (OTC) leads to ornithine transcarbamoylase deficiency, which is an X-linked disorder affecting the urea cycle. It results in the accumulation of ammonia in the blood and increased levels of glutamine.

Incorrect Options

Option A - Argininosuccinate lyase : Deficiency of argininosuccinate lyase leads to argininosuccinic aciduria, which is characterized by the accumulation of argininosuccinic acid in the blood and urine. It does not directly cause increased glutamine levels.

Option B - Alpha-galactosidase-A : Deficiency of alpha-galactosidase-A leads to Fabry disease, which is a lysosomal storage disorder. It results in the buildup of globotriaosylceramide (GL-3) in various organs, but it does not cause increased glutamine levels.

Option D - Arginase : Deficiency of arginase leads to argininemia, which is a rare autosomal recessive disorder. It causes the accumulation of arginine in the blood and does not directly cause increased glutamine levels.

Solution for Question 49:

Correct Option D - Methionine:

- Methionine : Methionine is an essential amino acid that plays a crucial role in the synthesis of epinephrine. It is converted to S-adenosylmethionine (SAM), which serves as a methyl donor in the conversion of norepinephrine to epinephrine.

Incorrect Options

Option A - Tyrosine : Tyrosine is the precursor for the synthesis of norepinephrine and dopamine, but it is not directly involved in the conversion of norepinephrine to epinephrine.

Option B - Tryptophan : Tryptophan is an essential amino acid that is used in the synthesis of serotonin, melatonin, and niacin, but it is not involved in the conversion of norepinephrine to epinephrine.

Option C - Phenylalanine: Phenylalanine is an essential amino acid that is converted to tyrosine in the body. While tyrosine is involved in the synthesis of norepinephrine and epinephrine, phenylalanine itself is not directly involved in this conversion.

Solution for Question 50:

Correct Option A - Urea

- Nitrogenous wastes produce harmful ammonia, which elevates the pH of bodily fluids. Ammonia creation necessitates the use of energy in the form of ATP as well as significant amounts of water to dilute it in a biological system.
- This ammonia gets into the urea cycle, where ammonia is converted to urea which is excreted through urine.

Incorrect Options:

Option B - Excess nitrogen is expelled from the body as nitrogenous waste in the form of urea. There is limited information about the metabolism and elimination of nitric oxide (NO), formerly known as an endothelium-derived relaxing factor (EDRF), from the body.

Option C - Ammonia is a waste substance that the body eliminates through the liver. It can be combined with other compounds to generate glutamine, an amino acid. It can also be used to

make urea , a chemical molecule. The urea is carried by your bloodstream to your kidneys, where it is excreted in your urine. Hence, nitrogenous waste is finally excreted from the body in the form of urea.

Option D - The kidney absorbs glutamine and converts it to ammonia. This process is pH-sensitive and assists in maintaining acid-base balance as well as nitrogen excretion. In this approach, renal glutamine and ammonia metabolism complement hepatic urea production. However, nitrogenous waste is finally excreted from the body in the form of urea.

Solution for Question 51:

Correct Option B - Arginine:

- Urea : Arginine is a precursor for urea synthesis in the urea cycle. It undergoes a series of enzymatic reactions, eventually leading to the production of urea.
- Creatinine: Creatinine is derived from creatine phosphate , which is synthesized from arginine in a two-step process . Creatine is first converted to creatine phosphate in the presence of arginine and glycine, and then creatine phosphate spontaneously converts to creatinine.
- Nitric oxide : Nitric oxide (NO) is synthesized from arginine by the enzyme nitric oxide synthase (NOS). Arginine serves as the substrate for NOS, which catalyzes the production of NO.

Incorrect Options:

Option A - Glycine : Glycine is not a precursor for urea, creatinine, or nitric oxide synthesis.

Option C - Aspartate : Aspartate is involved in the urea cycle as a substrate, but it is not the precursor amino acid for urea, creatinine, or nitric oxide.

Option D - Alanine : Alanine is not directly involved in the synthesis of urea, creatinine, or nitric oxide.

Solution for Question 52:

Correct Option C - Protein folding

- Chaperone proteins participate in the folding of over half of all mammalian proteins.
- The hsp70 family of chaperones binds short sequences of hydrophobic amino acids that emerge while a new polypeptide is being synthesized, shielding them from solvent as polypeptide synthesis continues.

Incorrect Options:

Option A - Ubiquitination of proteins: Ubiquitination is a post-translational modification process in which proteins are tagged with ubiquitin molecules. This process plays a role in protein degradation and regulation, but it is not directly associated with the function of chaperones.

Option B - Protein modification: Protein modification encompasses a broad range of post-translational modifications, including phosphorylation, glycosylation, acetylation, and more.

Option D - Protein cleavage: Protein cleavage refers to the process of breaking peptide bonds within a protein molecule. Chaperones do not directly participate in protein cleavage but rather focus on promoting proper folding and preventing protein misfolding.

Solution for Question 53:

Correct Option: C - Xanthine oxidase:

- Xanthine oxidase is an enzyme involved in the metabolism of purines. It converts hypoxanthine to xanthine and then further converts xanthine to uric acid.
- Inhibition of xanthine oxidase activity leads to a decrease in the production of uric acid.
- Drugs that inhibit xanthine oxidase, such as allopurinol and febuxostat, are commonly used in the treatment of hyperuricemia and gout.

Incorrect Options:

Option A - Phosphoribosyl transferase: Phosphoribosyl transferase is an enzyme involved in the salvage pathway of purine metabolism. It catalyzes the transfer of phosphoribosyl group from phosphoribosyl pyrophosphate (PRPP) to a base, converting it into a nucleotide.

Option B - Thymidylate synthase: Thymidylate synthase is an enzyme involved in DNA synthesis. It catalyzes the conversion of deoxyuridine monophosphate (dUMP) to deoxythymidine monophosphate (dTMP), which is a precursor for thymidine.

Option D - Dihydrofolate reductase: Dihydrofolate reductase is an enzyme involved in the folate metabolism pathway. It catalyzes the conversion of dihydrofolate (DHF) to tetrahydrofolate (THF), which is required for nucleotide synthesis and other cellular processes.

Solution for Question 54:

Correct Option C - Aspartate and glutamate

- Glutamate and aspartate are acidic amino acids that are responsible for negative charge in fibrinopeptide A .

Incorrect Options

Option A - In the case of glutamate and valine, though glutamate is acidic and negatively charged, valine is a nonpolar amino acid and is neutral in charge. So, this pair cannot be responsible for the negative charge in fibrinopeptide A.

Option B - In the case of histidine and lysine, both amino acids are basic and bear a positive charge. Also, they are not part of fibrinopeptide A, proving the b option incorrect.

Option D - In the case of serine and threonine, though serine is part one of the amino acids from which Fibrinopeptide A polypeptide is made, it's neutral. And Fibrinopeptide A molecule does not contain threonine as its part, proving it an incorrect choice.

Solution for Question 55:

Correct Option C - Both

- The urea cycle involves enzymatic reactions that occur in both the cytoplasm and the mitochondria. The synthesis of urea occurs in the cytoplasm, while the ornithine transcarbamylase reaction, which is an important step in the urea cycle, takes place in the mitochondria. Therefore, option C is the correct answer.

Incorrect Options:

Option A - Mitochondria: The urea cycle does involve some enzymatic reactions that occur in the mitochondria, specifically the ornithine transcarbamylase reaction. However, the majority of the urea cycle reactions occur in the cytoplasm. Therefore, option A is not the correct answer.

Option B - Cytoplasm: Many of the enzymatic reactions of the urea cycle take place in the cytoplasm of the cell. These reactions include the synthesis and breakdown of various intermediates involved in the urea cycle.

Option D - None: The urea cycle is a vital biochemical pathway that occurs in the body to remove excess nitrogen in the form of urea. Therefore, option D is not the correct answer.

Solution for Question 56:

Correct Option A - Tyrosine

- Melanin is synthesized from the amino acid tyrosine. Tyrosine is converted into a compound called dihydroxyphenylalanine (DOPA) by the enzyme tyrosinase. Further enzymatic reactions and processes involving tyrosinase lead to the production of melanin. Therefore, option A is the correct answer.

Incorrect Option:

Option B - Tryptophan: Tryptophan is primarily used in the synthesis of other compounds such as serotonin and niacin. Therefore, option B is not the correct answer.

Option C - Phenylalanine: Phenylalanine is primarily used in the synthesis of other important compounds in the body, such as tyrosine and various neurotransmitters. Therefore, option C is not the correct answer.

Option D - Threonine: Threonine is an essential amino acid that plays various roles in protein synthesis and metabolism, but it is not directly involved in the synthesis of melanin. Therefore, option D is not the correct answer.

Solution for Question 57:

Correct Option C - Hyaluronic acid

- Hyaluronic acid is a glycosaminoglycan that is naturally present in the extracellular matrix of various tissues. It plays a crucial role in wound healing due to its ability to retain water, provide hydration to the wound bed, and create a favorable environment for cell migration and tissue repair. It also promotes angiogenesis and the production of growth factors involved in healing processes.

Incorrect Options:

Option A - Keratan sulfate: Keratan sulfate is a sulfated glycosaminoglycan (GAG) found in various tissues, including the cornea and cartilage. While it plays important roles in tissue organization and function, it is not directly involved in wound healing.

Option B - Dermatan sulfate: Dermatan sulfate is another sulfated GAG present in the skin, blood vessels, and connective tissues. Like keratan sulfate, it contributes to tissue structure and function but is not directly involved in wound healing.

Option D - Chondroitin sulfate: Chondroitin sulfate is another sulfated GAG found in cartilage, tendons, and connective tissues. While it has important structural functions in these tissues, it is not directly involved in wound healing.

Solution for Question 58:

Correct Option A - Cysteine:

- Adenosine receptors, like many other proteins, can have disulfide bonds formed between cysteine residues. These disulfide bonds contribute to the stability and structure of the receptor.
- Cysteine contains a thiol group (-SH) that can form a covalent bond with another cysteine residue, resulting in a disulfide bond. These disulfide bonds help maintain the structural integrity of the receptor and are important for its proper function. Therefore, option A is the correct answer.

Incorrect Option:

Option B - Methionine: Methionine is an amino acid that does not participate directly in the formation of disulfide bonds. Therefore, option B is not the correct answer.

Option C - Arginine: Arginine is an amino acid that is positively charged and has a role in protein stability and structure, however, it does not directly participate in the formation of disulfide bonds. Therefore, option C is not the correct answer.

Option D - Alanine: Alanine is a nonpolar, aliphatic amino acid that does not contain a thiol group and is not involved in the formation of disulfide bonds. Therefore, option D is not the correct answer.

Solution for Question 59:

Correct Option B - Phenylbutyrate:

- Phenylbutyrate is a prodrug that is metabolized in the liver to phenylacetate.
- Phenylacetate binds with glutamine (which carries ammonia) to form phenylacetylglutamine.
- Phenylacetylglutamine is then excreted in the urine, effectively removing ammonia from the bloodstream.

Incorrect Option:

Option A - Glutamate:

- Glutamate is an amino acid involved in various metabolic processes, including the detoxification of ammonia.
- It can combine with ammonia to form glutamine, which helps in temporarily storing ammonia.
- However, in urea cycle disorders, this pathway is not sufficient to handle the excess ammonia production.

Option C - Isoleucine:

- Isoleucine is one of the essential branched-chain amino acids involved in protein synthesis and energy production.
- It does not have a direct role in ammonia detoxification or the urea cycle.

Option D - L-carnitine:

- L-carnitine is sometimes used in metabolic disorders to improve mitochondrial function and support energy metabolism.
- However, it is not a primary treatment for reducing ammonia levels in urea cycle disorders.

Solution for Question 60:

Correct Option C - A β 2m

- Amyloidosis is a protein-folding disorder in which normally complete proteins or fragments of normally soluble proteins accumulate as aberrant, insoluble fibrils that damage tissue structure and cause disease.
- A β 2M is a small membrane protein associated with the heavy chains of class I major histocompatibility complex (MHC) on the surface of nucleated cells. Most of the β 2M is typically cleared by glomerular filtration and subsequent proximal tubular reabsorption and catabolism. An increase in the concentration of β 2-macroglobulin (β 2m) in the blood appears to be a requirement for A β 2M amyloidosis, which results in A β 2M amyloid deposition primarily in the osteoarticular tissue.

Incorrect Options:

Option A - AL (primary or light chain (AL) amyloidosis) is linked to renal myeloma . This protein (pink protein) eventually leaks into the urine, causing BJP (Bence Jones Proteinuria).

Option B - AA (amyloid associated) , often known as secondary amyloidosis. AA amyloidosis develops as a result of a persistent inflammatory illness or disease.

Option D - ATTR (Wild-type transthyretin amyloid) , also known as senile systemic amyloidosis, is caused by transthyretin , or TTR, a protein that changes shape and produces fibrous clumps.

Solution for Question 61:

Correct Option B - Pyruvate carboxylase:

- Raw egg whites contain a protein called avidin, which can bind to biotin, a B-vitamin essential for the activity of pyruvate carboxylase. Biotin deficiency can, in turn, affect the function of pyruvate carboxylase, as well as other biotin-dependent enzymes.

- Pyruvate carboxylase: Pyruvate carboxylase is an enzyme involved in the conversion of pyruvate to oxaloacetate, an important step in gluconeogenesis. Deficiency of pyruvate carboxylase can lead to impaired gluconeogenesis and energy production, which could contribute to fatigue during exercise. Therefore, this is the correct answer.

Incorrect Option:

Option A - Glucose 6 phosphate: Deficiency of this enzyme can lead to a condition called glucose-6-phosphate dehydrogenase deficiency (G6PD deficiency), which is not directly related to the body builder's symptoms of fatigue during exercise.

Option C - PEPCK: PEPCK (Phosphoenolpyruvate carboxykinase) deficiency can affect gluconeogenesis and energy production, but it is not directly related to the body builder's symptoms in this scenario.

Option D - Glycogen phosphorylase: Glycogen phosphorylase deficiency can lead to glycogen storage diseases, but it is not directly related to the body builder's symptoms of fatigue during exercise in this context.

Solution for Question 62:

Correct Option C - Change in 29-30 in B chain:

- Insulin undergoes post-translational modification, including the removal of a specific peptide called the C-peptide. A change in the amino acid sequence at positions 29-30 in the B chain could disrupt this processing, leading to altered biological activity.

Incorrect Option:

Option A - Hydrolysis of disulfide bonds: Insulin is a peptide hormone that consists of two polypeptide chains, A and B, connected by disulfide bonds. Hydrolysis of these disulfide bonds would disrupt the structure of insulin, leading to a loss of biological activity.

Option B - Change in 5-6 in A chain: A change in the amino acid sequence at positions 5-6 in the A chain of insulin could potentially affect its structure and binding to insulin receptors. However, the specific impact on biological activity would depend on the nature of the amino acid change.

Option D - Change in 14-16 in A chain: A change in the amino acid sequence at positions 14-16 in the A chain of insulin could potentially affect its structure and binding to insulin receptors. However, the specific impact on biological activity would depend on the nature of the amino acid

change.

Solution for Question 63:

Correct Options A - Glutamine replaced by asparagine

- Glutamine replaced by asparagine: This is the correct option. Glutamine (Gln) and asparagine (Asn) are both polar, uncharged amino acids with similar properties. Therefore, replacing glutamine with asparagine is a conservative substitution that is unlikely to significantly affect the structure or function of the protein.

Incorrect Option:

Option B - Glutamine replaced by alanine: This is an incorrect option. Alanine (Ala) is a nonpolar, aliphatic amino acid, while glutamine (Gln) is a polar, uncharged amino acid. Substituting glutamine with alanine can introduce changes in the protein's structure and interactions, potentially affecting its function.

Option C - Glutamine replaced by glutamate: This is an incorrect option. Glutamate (Glu) is a negatively charged amino acid, while glutamine (Gln) is a polar, uncharged amino acid. Replacing glutamine with glutamate introduces a charge difference, which can impact the protein's structure, stability, and interactions, potentially altering its function.

Option D - Glutamine replaced by arginine: This is an incorrect option. Arginine (Arg) is a positively charged amino acid, while glutamine (Gln) is a polar, uncharged amino acid. Substituting glutamine with arginine introduces a charge difference, which can significantly impact the protein's structure, stability, and interactions, potentially leading to functional changes.

Solution for Question 64:

Correct Option A - Tyrosine:

- Tyrosinase, an enzyme found in melanocytes, catalyzes the oxidation of tyrosine to form dopaquinone, which is then further oxidized and polymerized to form different types of

melanin. Therefore, melanin is derived from the amino acid tyrosine

Incorrect Options:

Option B -Tryptophan: Tryptophan is an amino acid synthesizing proteins and important molecules such as serotonin and melatonin. However, it is not involved in the synthesis of melanin.

Option C - Phenylalanine: Phenylalanine is an amino acid used to synthesize proteins and important molecules such as adrenaline and thyroid hormones. However, it is not involved in the synthesis of melanin.

Option D - Alanine: Alanine is an amino acid involved in the synthesis of proteins and can be used as an energy source by muscle tissue. However, it is not involved in the synthesis of melanin.

DNA and Chromosome Biochemistry: Structure, Replication, and Repair

1. Which of the following is not a requirement for the activity of DNA polymerase during DNA replication?

- A. A primer with a free 3'-OH group
- B. A single-stranded DNA template
- C. Free ribonucleoside triphosphates (rNTPs)
- D. A divalent cation cofactor, typically Mg^{2+}

2. A scientist observes the process of DNA replication in a human stem cell. He takes a sample of double-stranded DNA from the cell and studies it separately. He finds that the total amount of adenine in the sample is 24%. From the given information, what percentage of cytosine is present in the sample?

- A. 52%
- B. 24%
- C. 30%
- D. 26%

3. What is the wavelength at which nucleic acids show maximum absorbance of UV light?

- A. 260 nm
- B. 280 nm
- C. 400 nm

D. 340 nm

4. What sequence is repeated in human telomeres?

- A. 5'-TTAAGG-3'
 - B. 5'-GGTACC-3'
 - C. 5'-TTAGGG-3'
 - D. 5'-AGCTAG-3'
-

5. Which of the following is not caused by a defective NER mechanism?

- A. Xeroderma pigmentosum (XP)
 - B. Cockayne syndrome (CS)
 - C. Trichothiodystrophy (TTD)
 - D. Werner syndrome (WS)
-

6. What is the role of UvrABC exinuclease in nucleotide excision repair (NER)?

- A. It fills in the gap with new nucleotides
 - B. It recognizes and excises damaged DNA segments.
 - C. It seals the DNA strand after repair.
 - D. It unwinds the DNA double helix.
-

7. Mismatch repair defects are mostly associated with which of the following?

- A. Hereditary Nonpolyposis Colon Cancer (HNPCC)
 - B. Familial Adenomatous Polyposis (FAP)
 - C. Li-Fraumeni syndrome
 - D. Retinoblastoma
-

8. Why are methylases important in the mismatch repair mechanism in prokaryotes like *E. coli*?

- A. They add methyl groups to the newly synthesized DNA strand, marking it for repair.
 - B. They methylate the parental DNA strand, allowing the Mut proteins to distinguish it from the daughter strand.
 - C. They remove mismatched bases from the DNA strand.
 - D. They act as exonucleases during the repair process.
-

9. Which of the following enzymes helps remove the defective base during base excision repair (BER)?

- A. DNA polymerase I
 - B. DNA ligase
 - C. DNA glycosylase
 - D. AP endonuclease
-

10. A 6-year-old child presents with recurrent respiratory infections and poor growth despite a normal diet. Genetic testing reveals Cystic fibrosis. What type of mutation causes cystic fibrosis?

- A. Phenotypic heterogeneity

- B. Pleiotropy
 - C. Allelic heterogeneity
 - D. Heteroplasmy
-

11. Which of the following statements accurately describes the role of DNA polymerase δ (Pol δ) in eukaryotic cells?

- A. It synthesizes RNA primers necessary for DNA replication.
 - B. It is responsible for leading-strand DNA synthesis.
 - C. It requires a primer and provides proofreading during lagging-strand synthesis.
 - D. It directly repairs damaged DNA by filling in gaps.
-

12. Which of the following statements about telomeres and telomerase is correct?

- A. Telomeres prevent chromosome ends from degrading but do not shorten during cell division.
 - B. Telomerase adds repetitive DNA sequences to the 5' end of chromosomes
 - C. Telomeres are composed of repetitive sequences that help prevent chromosome fusion.
 - D. The shortening of telomeres occurs due to the continuous synthesis of the lagging strand.
-

13. Which of the following statements about DNA polymerases in *E. coli* is true?

- A. DNA Polymerase I is the primary enzyme for DNA synthesis at the replication fork.
- B. DNA Polymerase II is primarily responsible for gap filling following DNA replication
- C. DNA Polymerase III has the highest rate of chain elongation and is involved in both leading and lagging strand synthesis.

D. All three DNA polymerases perform identical functions in DNA synthesis.

14. Which of the following is not a requirement for DNA replication in eukaryotes?

- A. Origin of replication (ori)
 - B. DNA helicase
 - C. RNA primer
 - D. RNA polymerase
-

15. Which of the following disorders are not caused due to defect in the Non homologous End Joining?

- A. Fanconi's anaemia
 - B. Ataxia telengectasia
 - C. Blooms syndrome
 - D. Severe combined immunodeficiency disorder (SCID)
-

16. A 45 y/o woman presents with visual disturbances, including gradual vision loss and difficulty seeing in low light, along with muscle weakness and fatigue. She has a history of early-onset cardiac conduction issues and worsening symptoms over the years. Her family history shows similar symptoms in her mother. What genetic condition best explains her symptoms?

- A. Leber's Hereditary Optic Neuropathy (LHON)
- B. Myoclonic Epilepsy with Ragged Red Fibers (MERRF)
- C. Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like Episodes (MELAS)
- D. Leigh's Syndrome.

17. Which of the following is a characteristic of mitochondrial DNA (mtDNA)?

- A. It is linear and single-stranded
 - B. It has very few non-coding regions and a higher mutation rate compared to nuclear DNA.
 - C. It uses the standard genetic code, including AGA and AGG as codons for arginine (Arg).
 - D. It is composed of linear strands and encodes for all 67 subunits of the respiratory chain.
-

18. Which histone is least tightly bound to chromatin and can be easily removed with a salt solution?

- A. H2A
 - B. H1
 - C. H3
 - D. H4
-

19. Which of the following factors can cause the denaturation of DNA? a) Increasing temperature b) Increasing salt concentration c) Decreasing temperature d) DNA with a high A-T base composition e) DNA with a high G-C base composition f) Adding Urea g) Adding Formamide h) Decreasing salt concentration

- A. b, c, e, g, h
 - B. c, d, e, h
 - C. a, d, f, g, h
 - D. a, b, e, g
-

20. Which of the following statements about the different types of DNA is TRUE?

- A. B-DNA has a left-handed helix and a full turn with 12 bases.
 - B. A-DNA forms during dehydration and has grooves of the same dimensions.
 - C. Z-DNA forms under normal physiological conditions and has a right-handed helix
 - D. A-DNA is the predominant form found in physiological conditions and has 11 bases per full turn.
-

21. In the analogy of the DNA double helix to a spiral staircase, what do the "handrails" and "steps" represent?

- A. Handrails are nitrogenous bases; steps are phosphodiester bonds
 - B. Handrails are nitrogenous bases; steps are the sugar-phosphate backbone
 - C. Handrails are the sugar-phosphate backbone; steps are nitrogenous bases
 - D. Handrails are hydrogen bonds; steps are deoxyribose sugars
-

22. In the structure of nucleotides, where does the esterification of the phosphate group typically occur?

- A. On the 1' carbon of the pentose sugar
 - B. On the 2' hydroxyl group of the pentose sugar
 - C. On the 5' or 3' hydroxyl group of the pentose sugar
 - D. On the N1 position of the nitrogenous base
-

23. What is the primary role of inosine monophosphate (IMP) in cellular metabolism?

- A. Directly involved in energy transfer like ATP
 - B. Acts as a precursor for both adenosine monophosphate (AMP) and guanosine monophosphate (GMP)
 - C. Inhibits the de novo synthesis of purine nucleotides
 - D. Serves as an end product of the salvage pathway for purine nucleotides
-

24. What type of bond forms between a nitrogenous base and a pentose sugar in a nucleoside?

- A. α -N-glycosidic bond
 - B. β -N-glycosidic bond
 - C. Phosphodiester bond
 - D. Hydrogen bond
-

25. Which of the following is a pyrimidine base found in nucleic acids?

- A. Uracil
 - B. Hypoxanthine
 - C. Xanthine
 - D. Guanosine
-

26. Which of the following is not caused by a defect of ds DNA repair?

- A. Cockayne syndrome (CS)
- B. Fanconi's anemia

- C. Severe Combined Immunodeficiency (SCID)
 - D. Bloom's syndrome
-

27. Which of the following statements about Homologous DNA Repair (HR) is correct?

- A. HR is active during G0/G1 phases of the cell cycle.
 - B. HR is an error-prone process compared to nonhomologous end joining (NHEJ).
 - C. HR uses a homologous DNA template for repair, making it highly accurate.
 - D. Defects in HR-related proteins decrease Ovarian and breast cancers
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	1
Question 4	3
Question 5	4
Question 6	2
Question 7	1
Question 8	2
Question 9	3
Question 10	3
Question 11	3
Question 12	3
Question 13	3
Question 14	4
Question 15	1
Question 16	1
Question 17	2
Question 18	2

Question 19	3
Question 20	2
Question 21	3
Question 22	3
Question 23	2
Question 24	2
Question 25	1
Question 26	1
Question 27	3

Solution for Question 1:

DNA polymerase requires deoxynucleoside triphosphates (dNTPs), not ribonucleoside triphosphates, which are used in RNA synthesis.

DNA Polymerase requirements & function :

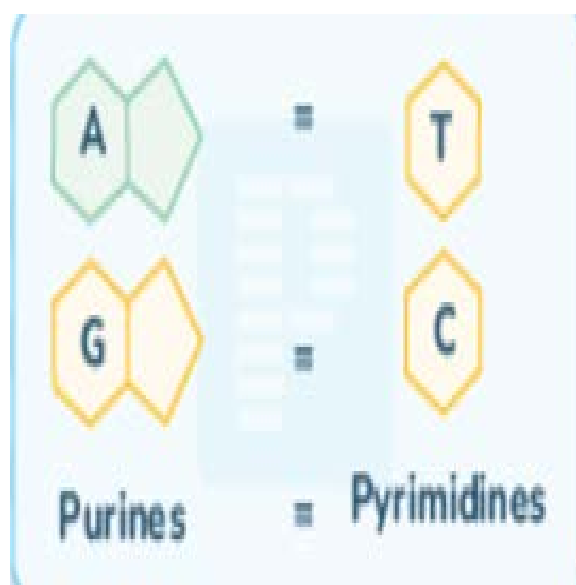
- Synthesis Direction : DNA polymerases synthesize DNA in the 5' to 3' direction, adding nucleotides to the 3' end of a DNA strand.
- Primer Requirement : DNA polymerases require a short RNA primer, synthesized by primase, to initiate DNA synthesis.
- Primer Binding : The 3'-hydroxyl group of the RNA primer initiates the nucleophilic attack on the first deoxynucleoside triphosphate (dNTP).
- Template Requirement : A single-stranded DNA template is necessary to guide the order of dNTP addition. (Option B ruled out)
- dNTPs : The four dNTPs required for DNA synthesis are dATP, dGTP, dCTP, and dTTP.
- Cofactor : DNA polymerases require a divalent cation cofactor, typically Mg^{2+} . (Option D ruled out)

- Properties : Chain elongation, processivity, and proofreading ensure DNA replication and repair accuracy and efficiency.

A primer with a free 3'-OH group (Option A): DNA polymerase cannot initiate DNA synthesis de novo and requires a primer to provide a free 3'-OH group to which it can add nucleotides.

Solution for Question 2:

- According to Chargaff's Rule , DNA follows base-pairing where: Adenine (A) pairs with Thymine (T) Cytosine (C) pairs with Guanine (G)
- Adenine (A) pairs with Thymine (T)
- Cytosine (C) pairs with Guanine (G)
- Given Adenine (A) = 24% , this means Thymine (T) = 24% as well.
- The total percentage of A + T = 24% + 24% = 48% .
- Since DNA is double-stranded , the remaining percentage must be from Cytosine (C) and Guanine (G) . $C + G = 100\% - 48\% = 52\%$. As C and G are equal in amount , we divide: $C = G = 52\% \div 2 = 26\%$.
- $C + G = 100\% - 48\% = 52\%$.
- As C and G are equal in amount , we divide: $C = G = 52\% \div 2 = 26\%$.
- Adenine (A) pairs with Thymine (T)
- Cytosine (C) pairs with Guanine (G)
- $C + G = 100\% - 48\% = 52\%$.
- As C and G are equal in amount , we divide: $C = G = 52\% \div 2 = 26\%$.



Solution for Question 3:

- Nucleic acids (DNA and RNA) absorb UV light maximally at 260 nm due to the presence of aromatic nitrogenous bases (adenine, guanine, cytosine, thymine, and uracil).
- The purines and pyrimidines contain conjugated double bonds that strongly absorb UV light at this wavelength.
- Protein contamination is assessed using the 260/280 nm ratio since proteins absorb maximally at 280 nm due to aromatic amino acids (tryptophan and tyrosine).

UV Absorption of Biomolecules:

Molecule	Maximum Absorbance (nm)	Reason
DNA/RNA	260 nm	Nitrogenous bases (purines & pyrimidines)
Proteins	280 nm (Option B ruled out)	Aromatic amino acids (tryptophan & tyrosine)
NADH/NADPH	340 nm (Option D ruled out)	Reduced nicotinamide ring

Hemoglobin	400 nm (Option C ruled out)	Heme group
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Solution for Question 4:

- Human telomeres consist of repetitive nucleotide sequences that protect chromosome ends from degradation.
- The repeated sequence in human telomeres is 5'-TTAGGG-3' , which is tandemly repeated thousands of times.
- Telomerase, a ribonucleoprotein enzyme, extends the telomeres using an RNA template to prevent progressive shortening.

Telomere Structure and Function:

Feature	Description
Sequence	5'-TTAGGG-3' repeat
Function	Protects chromosome ends from degradation and fusion
Telomerase Activity	Adds repeats using an RNA template
Shortening Effect	Telomeres shorten with each cell division due to incomplete lagging strand synthesis
Associated Diseases	Premature telomere shortening is linked to aging and diseases like dyskeratosis congenita and aplastic anemia

Solution for Question 5:

- Werner syndrome is caused by defective homologous repair and not a defect in the NER mechanism.

Nucleotide Excision Repair (NER):

- Required when chemical damage to a segment occurs.

- Removes bulky DNA lesions, such as pyrimidine dimers and chemical adducts.
- Excises a variable number of nucleotides (typically 24-32 bases), followed by DNA synthesis to fill the gap.

UV-Induced Damage:

- UV radiation causes the covalent joining of adjacent pyrimidines (usually thymine dimers).
- These bulky DNA lesions impede DNA replication and transcription.

Recognition and Excision:

- In bacteria: UvrABC proteins (UvrABC excinuclease) recognize thymine dimers. The damaged strand is cleaved on both the 5' and 3' sides. A short oligonucleotide containing the dimer is excised. DNA polymerase I fills the gap, and DNA ligase seals it.
- UvrABC proteins (UvrABC excinuclease) recognize thymine dimers.
- The damaged strand is cleaved on both the 5' and 3' sides.
- A short oligonucleotide containing the dimer is excised.
- DNA polymerase I fills the gap, and DNA ligase seals it.
- UvrABC proteins (UvrABC excinuclease) recognize thymine dimers.
- The damaged strand is cleaved on both the 5' and 3' sides.
- A short oligonucleotide containing the dimer is excised.
- DNA polymerase I fills the gap, and DNA ligase seals it.

Nucleotide Excision Repair in Humans:

- Repairs UV-induced damage and other bulky lesions throughout the cell cycle.
- Transcription-coupled repair (a type of NER) specifically targets lesions blocking RNA polymerase.
- Defects in NER cause: (Options A, B & C ruled out) Xeroderma pigmentosum (XP) Cockayne syndrome (CS) Trichothiodystrophy (TTD)
- Xeroderma pigmentosum (XP)
- Cockayne syndrome (CS)
- Trichothiodystrophy (TTD)
- Xeroderma pigmentosum (XP)
- Cockayne syndrome (CS)
- Trichothiodystrophy (TTD)

Xeroderma Pigmentosum (XP):

- Pyrimidine dimers form in human skin cells exposed to unfiltered sunlight.

- In XP, the inability to repair UV-induced DNA damage leads to: Accumulation of mutations. Increased risk of early and multiple skin cancers.
- Accumulation of mutations.
- Increased risk of early and multiple skin cancers.
- XP results from defects in genes coding for global genome NER proteins , while CS affects transcription-coupled NER proteins .
- Accumulation of mutations.
- Increased risk of early and multiple skin cancers.



Solution for Question 6:

UvrABC excinuclease is responsible for recognizing bulky DNA lesions and excising the damaged DNA segments during nucleotide excision repair

Nucleotide excision repair:

Nucleotide excision repair is required when chemical damage to a segment occurs. 30 bases are removed and then the correct bases are added

UV-Induced Damage:

- Exposure to UV radiation can cause the covalent joining of adjacent pyrimidines (usually thymine dimers), creating bulky DNA lesions that impede DNA replication.

Recognition and Excision:

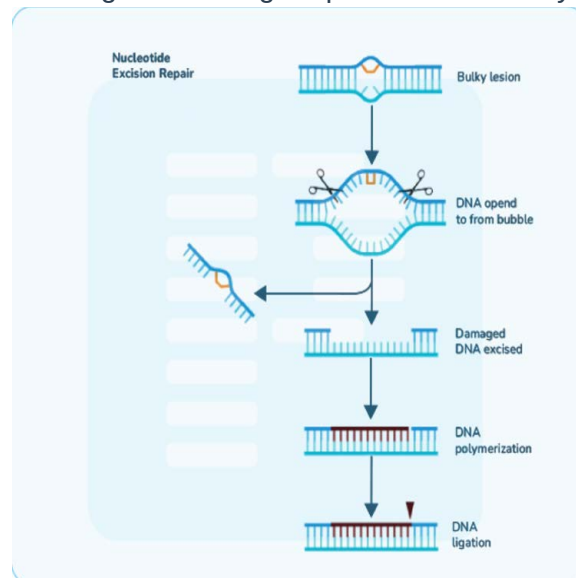
- In bacteria, UvrABC proteins (uvrABC excinuclease) recognize these dimers and cleave the damaged strand on both the 5' and 3' sides of the lesion.
- A short oligonucleotide containing the dimer is excised, leaving a gap in the DNA strand.
- The gap is subsequently filled in by DNA polymerase I and sealed by DNA ligase.

Nucleotide Excision Repair in Humans:

- A related nucleotide excision repair (NER) pathway exists in humans, addressing UV-induced damage encountered during DNA synthesis.
- Transcription-coupled repair, a type of NER, specifically targets lesions that obstruct RNA synthesis.

Clinical application:

- Pyrimidine dimers can form in human skin cells exposed to unfiltered sunlight.
- In xeroderma pigmentosum (XP), a genetic disorder, the inability to repair UV-induced DNA damage leads to the accumulation of mutations and a heightened risk of developing early and multiple skin cancers.
- XP results from defects in genes coding for proteins necessary for NER of UV damage



It fills in the gap with new nucleotides (Option A): While filling in the gap is part of NER, it is the role of DNA polymerase I.

It seals the DNA strand after repair. (Option C): Sealing the DNA strand is the function of DNA ligase.

It unwinds the DNA double helix (Option D): DNA helicase is responsible for unwinding the DNA double helix.

Solution for Question 7:

- Mismatch repair (MMR) defects lead to microsatellite instability (MSI) , which is a hallmark of Hereditary Nonpolyposis Colon Cancer (HNPCC), also known as Lynch syndrome .
- HNPCC is caused by mutations in MMR genes, including MLH1, MSH2, MSH6, and PMS2 .
- Individuals with HNPCC have a higher risk of colorectal cancer, endometrial cancer, and other malignancies .

Syndrome	Genetic Defect	Cancer Risk
HNPCC (Lynch Syndrome)	MMR gene mutations (MLH1, MSH2, MSH6, PMS2)	Colorectal, endometrial, ovarian, gastric cancers
FAP (Familial Adenomatous Polyposis) (Option B ruled out)	APC gene mutation	Multiple colorectal adenomas, colon cancer
Li-Fraumeni Syndrome (Option C ruled out)	TP53 mutation	Breast, brain, sarcomas, leukemias
Retinoblastoma (Option D ruled out)	RB1 gene mutation	Retinoblastoma, osteosarcoma

Solution for Question 8:

DNA adenine methylase (DAM) methylates the parental strand, enabling Mut proteins to identify the parental strand as correct and target the unmethylated daughter strand for repair.

Mismatch Repair:

- The defect occurs as a copying error, 1-5 bases unpaired

- Repair is done by strand cutting and exonuclease digestion

Mismatched Strand Identification:

- MMR corrects errors that escape proofreading during DNA synthesis, significantly reducing the error rate.
- In prokaryotes like *E. coli*, Mut proteins identify and correct mismatches by distinguishing between the newly synthesized (unmethylated) and the parental (methylated) strands.
- GATC sequences in DNA are methylated on adenine by DNA adenine methylase (DAM), with the parental strand being methylated and the newly synthesized strand remaining unmethylated shortly after replication.
- The methylated parental strand is considered correct, guiding the repair of the unmethylated daughter strand.

Repair Procedure:

- Once the mismatched strand is identified, an endonuclease nicks the strand, and an exonuclease removes the mismatched nucleotides along with additional nucleotides at the 5' and 3' ends.
- The gap is then filled using the sister strand as a template by DNA polymerase, typically DNA pol III in prokaryotes.
- DNA ligase seals the nick, completing the repair process.

They add methyl groups to the newly synthesized DNA strand, marking it for repair. (Option A) : Methylation occurs on the parental strand, not the newly synthesized strand, to mark it as the correct template

They remove mismatched bases from the DNA strand. (Option C): Methylases do not remove mismatched bases; they add methyl groups to the DNA.

They act as exonucleases during the repair process. (Option D): Exonucleases remove mismatched nucleotides, but methylases do not have this function.

Solution for Question 9:

DNA glycosylase is the enzyme that recognizes and removes the defective base from the deoxyribose-phosphate backbone, initiating the base excision repair process.

Base excision repair

- A defect occurs as a result of chemical damage to a single base
- The defective base is removed by N-glycosylase and a new base is added

Steps:

Abnormal Base Removal:

- BER targets abnormal bases, such as uracil, which can form from the deamination of cytosine or the incorporation of dUTP instead of dTTP during DNA synthesis.
- Specific DNA glycosylases recognize and cleave these abnormal bases from the deoxyribose-phosphate backbone, creating an apyrimidinic (AP) site if a pyrimidine is removed or an apurinic (AP) site if a purine is removed.

AP Site Recognition and Repair:

- Specific AP endonucleases recognize the missing base and initiate repair by cutting just the 5' side of the AP site. (Option D ruled out)
- A deoxyribose phosphate lyase removes the single, base-free sugar phosphate residue.
- DNA polymerase I fills in the gap with the correct nucleotide. (Option A ruled out)
- DNA ligase seals the nick, completing the repair process. (Option B ruled out)

Solution for Question 10:

Allelic heterogeneity refers to different mutations within the same gene causing the same disease . In cystic fibrosis, many different mutations in the CFTR gene can cause the disease

Phenotypic heterogeneity (Option A ruled out)	Pleiotropy (Option B ruled out)	Allelic heterogeneity (Option C)
Variability in the phenotype that is caused by mutations in the same gene	Disorders where multiple, seemingly unrelated organ systems are affected. A single gene is responsible for several distinct and seemingly unrelated phenotypic effects	Different mutations (changes) in a single gene cause the same disease or condition

Example: Loss-of-function mutations in the RET gene, which encodes a receptor tyrosine kinase, leads to inherited failure of development of colonic ganglia, leading to defective colonic motility and severe chronic constipation (Hirschsprung disease)	Example: The allele causing neurofibromatosis type 1 can produce abnormalities of skin pigmentation, neurofibromas of the peripheral nerves, short stature, macrocephaly, skeletal abnormalities, and fits	Example: Cystic fibrosis is caused by having one of many different mutations in the CFTR gene.
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Heteroplasmy (Option D) refers to the presence of more than one type of mitochondrial DNA within a cell or individual. This is not relevant to cystic fibrosis, which is caused by mutations in the nuclear CFTR gene, not mitochondrial DNA

Solution for Question 11:

- DNA Polymerase δ (Pol δ) is the primary enzyme responsible for lagging-strand synthesis in eukaryotic cells.
- It requires an RNA primer synthesized by DNA polymerase α -primase complex to initiate replication.
- It has 3' $\rightarrow$ 5' exonuclease activity, enabling proofreading and ensuring high fidelity during DNA replication.
- Works in coordination with PCNA (proliferating cell nuclear antigen) for high processivity.

Eukaryotic DNA Polymerases and Their Functions

DNA Polymerase	Function
Pol α	Synthesizes RNA primers and initiates DNA synthesis
Pol δ	Lagging-strand synthesis, proofreading, and high-fidelity replication
Pol ϵ	Leading-strand synthesis and proofreading
Pol β	Involved in base excision repair (BER)
Pol γ	Replicates mitochondrial DNA

It synthesizes RNA primers necessary for DNA replication. (Option A) Incorrect because RNA primers are synthesized by DNA polymerase α -primase, not Pol δ .

It is responsible for leading-strand DNA synthesis. (Option B) Incorrect because leading-strand synthesis is performed by DNA polymerase ϵ .

It directly repairs damaged DNA by filling in gaps. (Option D) Incorrect because DNA repair is primarily performed by DNA polymerase β (base excision repair) and other polymerases in repair pathways.

Solution for Question 12:

- Telomeres are repetitive nucleotide sequences (TTAGGG in humans) at the ends of chromosomes that protect genetic information and prevent chromosome fusion.
- Telomerase is an RNA-dependent DNA polymerase that extends the 3' end of chromosomes, not the 5' end .
- Telomeres shorten with each cell division due to the end-replication problem in the lagging strand, but not solely because of its synthesis.
- While telomeres prevent chromosome degradation , they do shorten with every replication cycle in somatic cells due to incomplete lagging strand synthesis.

Telomeres and Telomerase

Feature	Description
Telomere Composition	Repetitive TTAGGG sequences in humans
Function	Protect chromosome ends from degradation and fusion
Shortening Mechanism	Due to incomplete lagging strand synthesis in DNA replication
Role of Telomerase	Extends the 3' end of DNA using an RNA template to compensate for shortening
Cell Types with Active Telomerase	Germ cells, stem cells, and cancer cells

Telomeres prevent chromosome ends from degrading but do not shorten during cell division. (Option A) Incorrect because telomeres do shorten with each cell cycle in somatic cells.

Telomerase adds repetitive DNA sequences to the 5' end of chromosomes. (Option B) Incorrect because telomerase extends the 3' end , not the 5' end .

The shortening of telomeres occurs due to the continuous synthesis of the lagging strand.
(Option D) Incorrect because telomeres shorten due to incomplete lagging strand synthesis, not its continuous synthesis.

Solution for Question 13:

DNA Polymerase III is the main replicative enzyme in *E. coli*.

- Involved in proofreading and DNA repair.
- Plays a role in gap-filling after DNA replication and recombination.
- Primarily functions in proofreading and DNA repair.
- Main enzyme for DNA synthesis at the replication fork.
- Exhibits the highest chain elongation rate and processivity.
- Participates in both leading and lagging strand synthesis.
- High processivity due to association with β subunits of the sliding clamp, enabling it to add a large number of nucleotides (100 to >50,000) before detaching.

All three DNA polymerases perform identical functions in DNA synthesis (Option D): They have distinct roles, with pol III being the main replicative polymerase while pol I and II focus on proofreading and repair.

Solution for Question 14:

- Origin of replication (ori) : Required to initiate replication. Eukaryotic DNA has multiple origins of replication to ensure efficient duplication.
- DNA helicase : Unwinds the DNA double helix to allow access for replication machinery.
- RNA primer : Required because DNA polymerase cannot initiate synthesis de novo ; RNA primase lays down short RNA primers for DNA polymerase to extend.
- RNA polymerase (Option D): Incorrect because RNA polymerase is involved in transcription, not DNA replication . It synthesizes RNA from a DNA template but does not

play a role in DNA synthesis.

Component	Function
Origin of Replication (ori) (Option A ruled out)	Specific DNA sequences where replication starts
DNA Helicase (Option B ruled out)	Unwinds the DNA double helix
Single-Strand Binding Proteins (SSBPs)	Stabilize unwound DNA strands
Primase (RNA polymerase type)	Synthesizes short RNA primers
DNA Polymerase α , δ , and ϵ	Synthesizes new DNA strands
Sliding Clamp (PCNA)	Enhances DNA polymerase processivity
Topoisomerase	Relieves supercoiling ahead of the replication fork
DNA Ligase	Seals Okazaki fragments on the lagging strand

RNA polymerase (Option D): Incorrect because it is involved in transcription, not DNA replication .

Solution for Question 15:

Fanconi's anemia is caused due to a defect in Homologous recombination and not Non homologous end joining.

- Definition: Double-strand DNA breaks (DSBs) are severe forms of DNA damage that can lead to mutations if not properly repaired. They involve the breakage of both DNA strands.
- Mechanisms : Cells employ specific pathways to repair these breaks, ensuring genetic stability and preventing mutations.
- Sources of DSBs : These can be caused by ionizing radiation, chemotherapeutic agents, X-rays, and natural genetic recombination processes.
- Nonhomologous End Joining (NHEJ)
- Homologous Recombination (HR)
- Definition: NHEJ is a DNA repair mechanism that fixes double-strand breaks (DSBs) by directly joining the broken ends without requiring a homologous template.
- Cell Cycle Phase: Active throughout the entire cell cycle, including G0/G1 phases.

- Process: Involves a group of proteins that recognize, process, and ligate the ends of the broken DNA. Some DNA may be lost during the process, making it error-prone and potentially mutagenic.
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- Some DNA may be lost during the process, making it error-prone and potentially mutagenic.
- Key Proteins: DNA-dependent protein kinase (DNA-PK) and Ku70/80 complex are crucial for recognizing and binding to DNA ends.
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- Clinical Relevance: Defects in NHEJ are associated with an increased risk of cancer and immunodeficiency syndromes due to improper repair and accumulation of mutations Ataxia telangiectasia (Option B) Bloom's syndrome (Option C) Severe Combined Immunodeficiency (SCID) (Option D)
- Defects in NHEJ are associated with an increased risk of cancer and immunodeficiency syndromes due to improper repair and accumulation of mutations
- Ataxia telangiectasia (Option B)
- Bloom's syndrome (Option C)
- Severe Combined Immunodeficiency (SCID) (Option D)
- Involves a group of proteins that recognize, process, and ligate the ends of the broken DNA.
- Some DNA may be lost during the process, making it error-prone and potentially mutagenic.
- DNA-dependent protein kinase (DNA-PK) and Ku70/80 complex are crucial for recognizing and binding to DNA ends.
- Defects in NHEJ are associated with an increased risk of cancer and immunodeficiency syndromes due to improper repair and accumulation of mutations
- Ataxia telangiectasia (Option B)
- Bloom's syndrome (Option C)
- Severe Combined Immunodeficiency (SCID) (Option D)

Solution for Question 16:

- Visual Disturbances : Gradual vision loss and difficulty seeing in low light are characteristic of LHON , a mitochondrial disorder affecting the optic nerve.

- **Muscle Weakness and Fatigue** : Mitochondrial dysfunction commonly leads to neuromuscular symptoms .
- **Cardiac Conduction Issues** : Some cases of LHON involve cardiac arrhythmias , particularly pre-excitation syndromes .
- **Maternal Inheritance** : Mitochondrial diseases, including LHON, follow maternal inheritance , consistent with the family history in her mother.

Mitochondrial Syndromes and Their Features

Syndrome	Genetic Defect	Key Clinical Features
Leber's Hereditary Optic Neuropathy (LHON)	mtDNA mutations in ND1, ND4, ND6 (Complex I subunits)	Painless, progressive vision loss , optic atrophy, cardiac conduction defects , maternal inheritance
Myoclonic Epilepsy with Ragged Red Fibers (MERRF) (Option B ruled out)	mtDNA mutations in MT-TK (tRNA-Lys gene)	Myoclonic epilepsy, muscle weakness , ataxia, hearing loss, ragged red fibers on muscle biopsy
Mitochondrial Encephalopathy, Lactic Acidosis, and Stroke-like Episodes (MELAS) (Option C ruled out)	mtDNA mutations in MT-TL1 (tRNA-Leu gene)	Stroke-like episodes, lactic acidosis , seizures, progressive dementia, diabetes, muscle weakness
Leigh's Syndrome (Option D ruled out)	mtDNA or nuclear DNA mutations (Complex I or IV defects, NDUFS genes)	Neurodegeneration in infancy , hypotonia, brainstem dysfunction, ataxia, respiratory failure
Kearns-Sayre Syndrome (KSS)	Large-scale mtDNA deletions	Progressive external ophthalmoplegia, pigmentary retinopathy, cardiac conduction defects, muscle weakness

Solution for Question 17:

Mitochondrial DNA is characteristic of having very few non-coding regions and a higher mutation rate of 5-10 times compared to nuclear DNA.

Feature	Description
Structure	Circular, double-stranded; composed of heavy (H) and light (L) chains. (Option A ruled out)

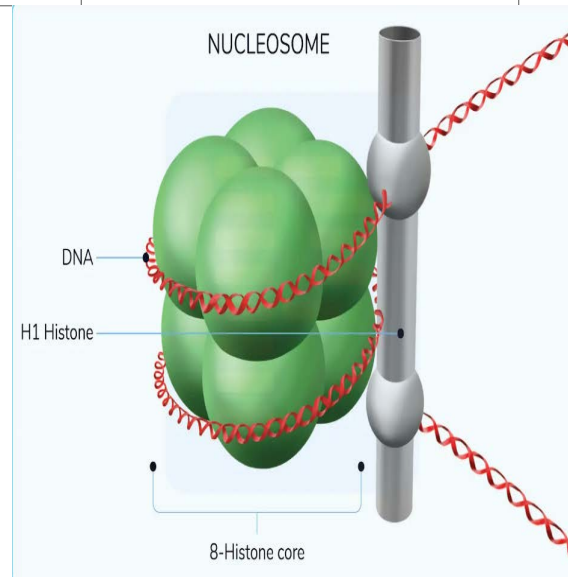
Length	16,569 base pairs (bp).
Protein-Encoding Subunits	Most respiratory chain proteins are encoded by nuclear DNA and imported into mitochondria (Option D ruled out)
Ribosomal RNAs	Encodes 16S (large) and 12S (small) mitochondrial rRNAs.
tRNAs	Encodes 22 mitochondrial tRNA molecules.
Genetic Code Differences	UGA typically a stop codon is read as tryptophan (Trp)
AGA and AGG: normally arginine (Arg) codons are read as stop codons (Option C ruled out)	
Untranslated Sequences	Very few non-coding regions.
Mutation Rate	5-10 times higher than nuclear DNA.
Evolutionary Studies	Comparisons of mtDNA sequences provide insights into the evolutionary history of primates and other species.

Solution for Question 18:

- H1 is a linker histone , not part of the core histone octamer.
- It binds to DNA at nucleosome entry and exit sites , helping in higher-order chromatin organization.
- Least tightly bound to chromatin and can be easily removed with a salt solution , increasing chromatin solubility.

Histone Type	Function	Binding to DNA
H1 (Linker Histone)	Stabilizes nucleosomes, promotes chromatin compaction	Least tightly bound , removable with salt solutions

H2A, H2B (Core Histones) (Option A ruled out)	Form histone octamer, aid in nucleosome structure	Tightly bound
H3, H4 (Core Histones) (Options C & D ruled out)	Critical for nucleosome stability and gene regulation	Tightly bound



Solution for Question 19:

Increasing temperature, Decreasing salt concentration, DNA with a high A-T base composition, and Adding Chaotropic agents like Urea and Formamide cause denaturation of DNA.

Factors Influencing Denaturation of DNA

Denaturation:

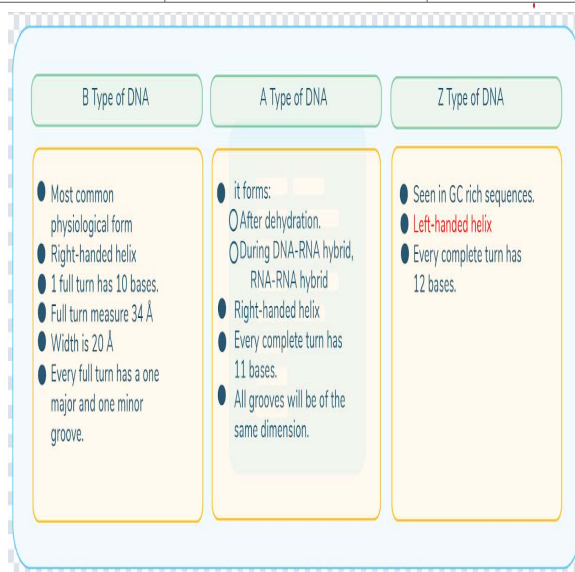
- **In Vivo Denaturation** : In living cells, DNA denaturation and renaturation occur naturally during DNA replication, recombination, repair, and transcription. These processes are mediated by nucleic acid binding proteins, enzymes, and thermal or chemical energy from ATP hydrolysis.
- **Experimental Denaturation** : In the laboratory, a combination of increasing temperature, decreasing salt concentration, and adding chaotropic agents is often used to denature DNA, ensuring the strands separate efficiently without causing significant damage to the nucleotides.

Solution for Question 20:

- A-DNA is a right-handed DNA helix that forms under dehydrated conditions.
- It has 11 base pairs per turn and a more compact structure than B-DNA.
- Unlike B-DNA, A-DNA has grooves of nearly equal dimensions (major and minor grooves are not as distinct).

Explanation of DNA Types:

Type of DNA	Helical Direction	Base Pairs per Turn	Grooves	Conditions
B-DNA	Right-handed	10.5	Major groove is wide and deep, minor groove is narrow	Physiological conditions
A-DNA	Right-handed	11	Grooves are of nearly equal dimensions	Forms during dehydration
Z-DNA	Left-handed	12	Major groove is shallow, minor groove is deep	Forms under high salt or supercoiling



B-DNA has a left-handed helix and a full turn with 12 bases (Option A): Incorrect because B-DNA is right-handed and has 10.5 bases per turn , not 12.

Z-DNA forms under normal physiological conditions and has a right-handed helix (Option C): Incorrect because Z-DNA is left-handed and forms under high salt or supercoiling , not in normal conditions.

A-DNA is the predominant form found in physiological conditions and has 11 bases per full turn (Option D): Incorrect because B-DNA, not A-DNA, is the predominant form under physiological conditions.

Solution for Question 21:

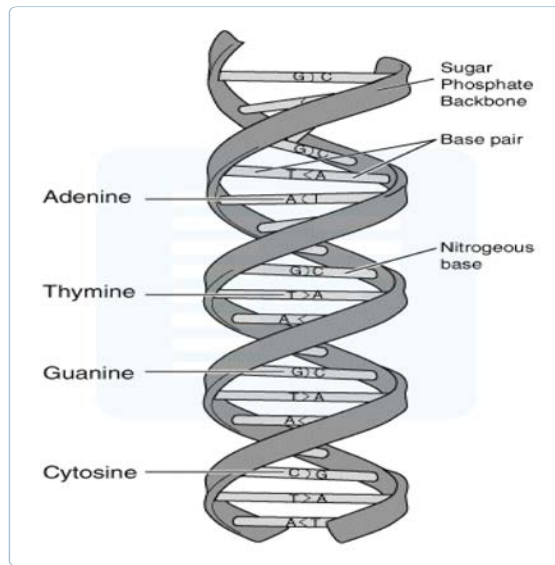
- Handrails (Sugar-Phosphate Backbone): The sugar-phosphate backbone forms the structural framework of the DNA double helix. It consists of deoxyribose sugars connected by phosphodiester bonds.
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- It consists of deoxyribose sugars connected by phosphodiester bonds.
- Steps (Nitrogenous Bases): The nitrogenous bases (adenine, thymine, cytosine, guanine) form the "steps" of the staircase. They pair via hydrogen bonds (A-T with two bonds, G-C with three bonds), connecting the two strands.
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Handrails (Sugar-Phosphate Backbone):

- The sugar-phosphate backbone forms the structural framework of the DNA double helix.
- It consists of deoxyribose sugars connected by phosphodiester bonds.

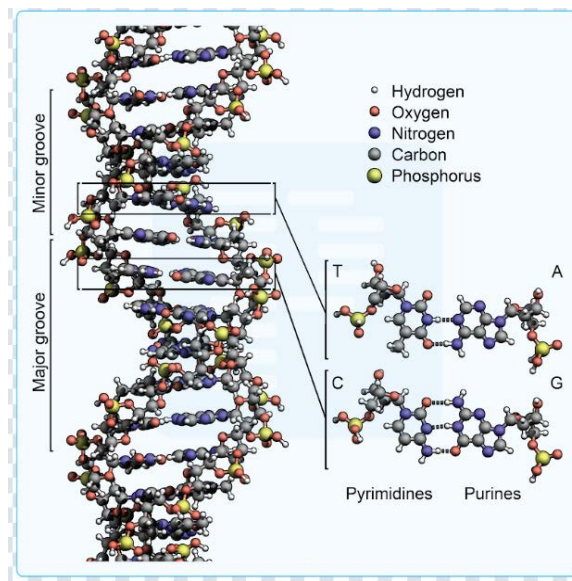
Steps (Nitrogenous Bases):

- The nitrogenous bases (adenine, thymine, cytosine, guanine) form the "steps" of the staircase.
- They pair via hydrogen bonds (A-T with two bonds, G-C with three bonds), connecting the two strands.



Structural Features of the DNA Molecule:

- The DNA molecule consists of two antiparallel strands : one runs 5' to 3' , while the complementary strand runs 3' to 5' .
- It resembles a spiral staircase , where two polynucleotide chains twist together in a right-handed helix .
- The DNA helix has distinct major (2.2 nm) and minor (1.2 nm) grooves , providing protein-binding sites .
- The double helix is stabilized by base stacking , involving hydrophobic interactions and the specific arrangement of base ring systems .
- Each complete turn of the helix spans 3.4 nm and includes ~10.5 base pairs .
- The helical diameter of B-DNA is 2 nm (20 Å) .



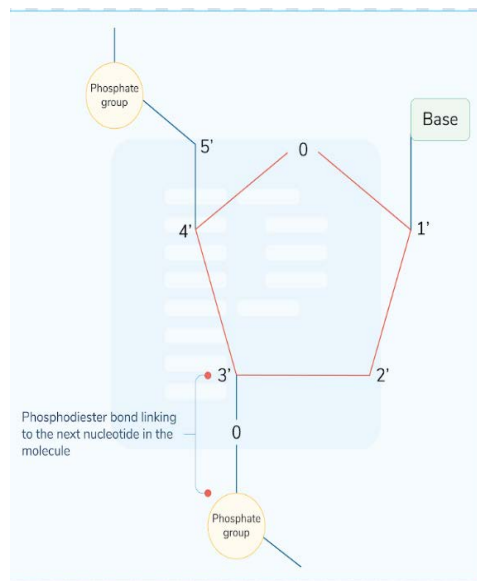
Handrails are nitrogenous bases; steps are phosphodiester bonds (Option A): Incorrect because phosphodiester bonds form the backbone, not the steps.

Handrails are nitrogenous bases; steps are the sugar-phosphate backbone (Option B): Incorrect because nitrogenous bases form the steps, not the handrails.

Handrails are hydrogen bonds; steps are deoxyribose sugars (Option D): Incorrect because hydrogen bonds connect bases, and deoxyribose is part of the backbone, not the steps.

Solution for Question 22:

Phosphate groups in nucleotides are most commonly esterified to the 5' hydroxyl group of the pentose sugar, forming 5'-nucleotides. The 3' hydroxyl group can also participate in forming phosphodiester bonds, linking nucleotides together in nucleic acids.



The 1' carbon of the pentose sugar (Option A): It is involved in forming the β -N-glycosidic bond with the nitrogenous base, not in esterification with the phosphate group.

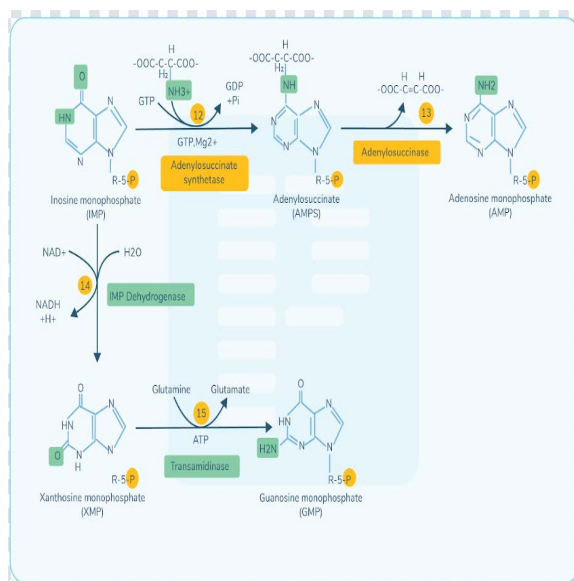
2' hydroxyl group is present in ribose (in RNA) (Option B): it is not typically the site for phosphate esterification. The 2' position plays a role in distinguishing RNA from DNA, as deoxyribose lacks a hydroxyl group at this position.

The N1 position (Option D): It is involved in attaching the base to the pentose sugar in pyrimidines, not in phosphate esterification. In purines, the corresponding attachment occurs at the N9 position.

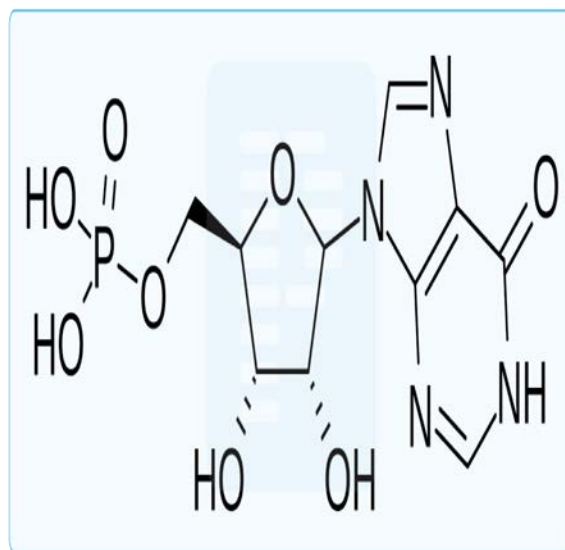
Solution for Question 23:

IMP is a central molecule in purine metabolism, serving as a precursor for AMP and GMP, which are crucial for various cellular processes, including nucleic acid synthesis and energy transfer.

Conversion of IMP to AMP and GMP:



Structure Of IMP:



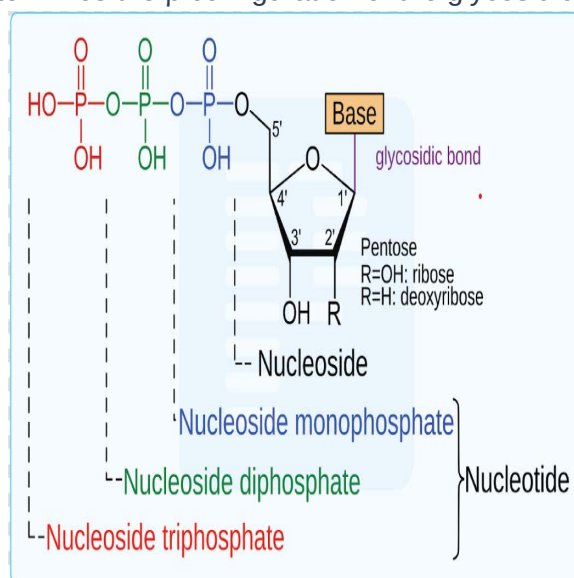
Directly involved in energy transfer like ATP (Option A) : While ATP is essential for energy transfer, IMP does not serve this function directly. IMP's primary role is as an intermediate in purine metabolism.

Inhibits the de novo synthesis of purine nucleotides (Option C) : IMP does not inhibit the de novo synthesis pathway. However, AMP and GMP, derived from IMP, can provide feedback inhibition to regulate their synthesis.

Serves as an end product of the salvage pathway for purine nucleotides (Option D): Incorrect because IMP is an intermediate in the salvage pathway, not the final product.

Solution for Question 24:

- β -N-glycosidic bond is the bond that connects the nitrogenous base to the pentose sugar in nucleosides.
- It links the 1' carbon of the sugar (ribose or deoxyribose) to the nitrogen atom of the base-N9 of purines or N1 of pyrimidines.
- It is called a β linkage because the OH group at the 1' position of ribose or deoxyribose is above the plane of the sugar ring.
- This orientation determines the β configuration of the glycosidic bond.



α -N-glycosidic bond (Option A) : This bond is not found between nitrogenous bases and pentose sugars in nucleosides.

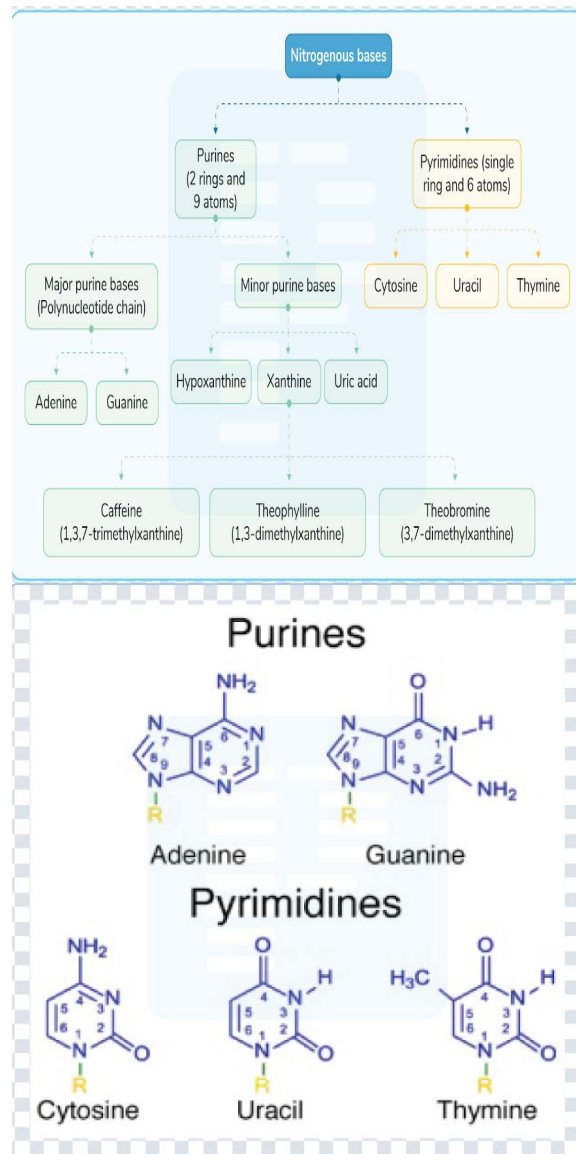
Phosphodiester bond (Option C) : This bond links nucleotides together to form the backbone of nucleic acids (DNA and RNA). It connects the 5' phosphate group of one nucleotide to the 3' hydroxyl group of the next nucleotide.

Hydrogen bond (Option D) : Hydrogen bonds are responsible for the base pairing between complementary nitrogenous bases in DNA or RNA. They do not connect the base to the sugar in a nucleoside.

Solution for Question 25:

- Uracil is a pyrimidine base found in RNA, where it replaces thymine as a complementary base to adenine.
- It is a single-ring nitrogenous base, characteristic of pyrimidines.

Classification of Nitrogenous Bases



Hypoxanthine (Option B): It is a purine derivative , not a primary component of DNA or RNA. It appears in tRNA and purine metabolism as an intermediate.

Xanthine (Option C): It is a purine base involved in purine degradation but not directly incorporated into nucleic acids. It is also a precursor for caffeine, theophylline, and theobromine.

Guanosine (Option D): It is a nucleoside , consisting of guanine (a purine) attached to ribose. While guanine is found in DNA and RNA, guanosine itself is not a base but a nucleoside.

Solution for Question 26:

- Cockayne syndrome (CS) is caused by a defect in Transcription-Coupled Nucleotide Excision Repair (TC-NER) , which is a type of Nucleotide Excision Repair (NER) .
- It does not involve defects in double-strand DNA (dsDNA) repair mechanisms like Homologous Recombination (HR) or Non-Homologous End Joining (NHEJ) .

Disorders Associated with dsDNA Repair Defects

Disorder	Defective Repair Mechanism
Ataxia telangiectasia	Non-Homologous End Joining (NHEJ)
Bloom's syndrome (Option D ruled out)	Homologous DNA Repair (HDR)
Severe Combined Immunodeficiency (SCID) (Option C ruled out)	Non-Homologous End Joining (NHEJ)
Fanconi's anemia (Option B ruled out)	Fanconi Anemia Repair Pathway (Interacts with HDR)
HBOC (Human Breast and Ovarian Cancer Syndrome)	Homologous DNA Repair (HDR) (BRCA1/BRCA2 mutation)

Solution for Question 27:

HR uses enzymes involved in genetic recombination between homologous chromosomes and uses a homologous DNA template for repair, making it highly accurate.

Homologous DNA Repair (HR)

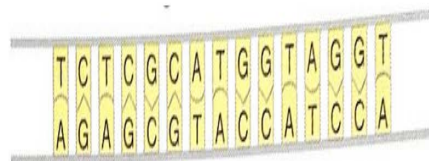
- Definition: HR is a type of Double-strand DNA break repair mechanism used to fix double-strand breaks (DSBs) by utilizing a homologous DNA template for error-free repair.
- Cell Cycle Phase : HR is active during the S and G2 phases of the cell cycle. (Option A ruled out)
- Process: Uses enzymes involved in genetic recombination between homologous chromosomes during meiosis. DNA damage is recognized, and the homologous sequence is used as a template for repair.
- Uses enzymes involved in genetic recombination between homologous chromosomes during meiosis.
- DNA damage is recognized, and the homologous sequence is used as a template for repair.
- Accuracy: HR is less error-prone compared to nonhomologous end joining (NHEJ), making it an "error-free" repair process. (Option B ruled out)
- Clinical Relevance: Mutations in HR-related proteins (e.g., BRCA1/BRCA2) increase the risk for breast and ovarian cancers . (Option D ruled out) Fanconi's anemia HBOC (Human Breast and Ovarian Cancer Syndrome)
- Mutations in HR-related proteins (e.g., BRCA1/BRCA2) increase the risk for breast and ovarian cancers . (Option D ruled out)
- Fanconi's anemia
- HBOC (Human Breast and Ovarian Cancer Syndrome)
- Advantages : Ensures high fidelity in DNA repair by accurately restoring the original DNA sequence using a homologous template.
- Holliday junction is a -cross-shaped structure formed during the process of genetic recombination when two double-stranded DNA molecules become separated into four strands to exchange segments of genetic information.
- Uses enzymes involved in genetic recombination between homologous chromosomes during meiosis.
- DNA damage is recognized, and the homologous sequence is used as a template for repair.
- Mutations in HR-related proteins (e.g., BRCA1/BRCA2) increase the risk for breast and ovarian cancers . (Option D ruled out)
- Fanconi's anemia
- HBOC (Human Breast and Ovarian Cancer Syndrome)

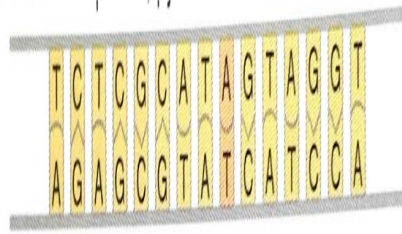
RNA Transcription, Genetic code mutation, Translation inhibitors

1. Original DNA sequence: 5'-ATG GCA TCG GAA UAC-3'. Mutated sequence: 5'-ATG GCA TCG GAA UAA-3'. Identify the mutation.

- A. Frameshift Mutation
- B. Nonsense mutation
- C. Silent mutation
- D. Missense mutation

2. Identify the type of mutation illustrated in the following DNA sequences?
Starting sequence: Mutated sequence:





- A. Point mutation
 - B. Frameshift mutation
 - C. Deletion mutation
 - D. Insertion mutation
-

3. Which technique is commonly used for DNA sequencing to identify mutations by determining the precise order of nucleotides in a DNA sample?

- A. Fluorescence in situ hybridization (FISH)
 - B. Multiplex ligation-dependent probe amplification (MLPA)
 - C. Sanger technique
 - D. Single Strand Conformational Polymorphism (SSCP)
-

4. Which class of antibiotics inhibits bacterial protein synthesis by blocking the attachment of aminoacyl-tRNA to the ribosomal A site?

- A. Tetracyclines
 - B. Chloramphenicol
 - C. Macrolides
 - D. Puromycin
-

5. What is the total number of ATPs required for activating amino acids by aminoacyl tRNA synthetase?

- A. 3
 - B. 2
 - C. 4
 - D. 1
-

6. Which of the following statements is true about aminoacyl tRNA?

- A. Attaches to the A site of 60s ribosomes
 - B. Needs 3 high-energy phosphates for attachment
 - C. It is guided by eIF2C
 - D. None of the above
-

7. Which of the following is NOT TRUE regarding the initiation of translation?

- A. eIF3 and 1A stabilize 60s ribosomal subunit to prevent reassociation with 40s.
- B. Methionine tRNA scans mRNA to identify the initiation AUG codon
- C. ATP dependent helicase eIF4A/4B breaks hydrogen bonds to remove the hairpin loops
- D. Association of mRNA with the 43S preinitiation complex forms the 48S initiation complex

8. Match the following phases of translation (Column A) with the events involved (Column B). Column A Column B

1. Elongation a. RF1 binds with RF3 and GTP

2. Initiation b. Binding of aminoacyl-tRNA to the ribosome's A site

3. Termination c. Methionine tRNA scans mRNA for AUG codon

Column A	Column B
1. Elongation	a. RF1 binds with RF3 and GTP
2. Initiation	b. Binding of aminoacyl-tRNA to the ribosome's A site
3. Termination	c. Methionine tRNA scans mRNA for AUG codon

- A. 1-b, 2-c, 3-a
 - B. 1-c, 2-a, 3-b
 - C. 1-b, 2-a, 3-c
 - D. 1-a, 2-c, 3-b
-

9. Sickle cell anaemia is suspected in a 25-year-old patient who presents with episodes of abdominal pain and fatigue. Genetic analysis reveals GAG in the 6th position is replaced with GTG. Which type of mutation here results in haemoglobin that is still functional but altered?

- A. Unacceptable mutation
 - B. Partially acceptable mutation
 - C. Acceptable mutation
 - D. Nonsense mutation
-

10. A DNA sequence originally reads 5'-ATG GCA TCG TAC-3'. After a mutation, it reads 5'-ATG GCT CGT AC-3'. Identify the mutation that has occurred?

- A. Frameshift mutation
 - B. Point mutation
 - C. Transition mutation
 - D. Insertion mutation
-

11. Which of the following statements accurately reflects a difference between DNA and RNA, considering their roles in genetic information processing and stability?

- A. RNA's 2'-hydroxyl group enhances its stability compared to DNA.
 - B. RNA polymerases need a primer; DNA polymerases do not.
 - C. RNA forms complex structures; DNA remains a double helix
 - D. DNA degrades more rapidly than RNA due to the presence of a 2'-hydroxyl group.
-

12. Which of the following statements are true regarding codons? A, G, T, C are RNA nucleotides A, G, T, C are DNA nucleotides A, G, U, C are RNA nucleotides A, G, U, C are DNA nucleotides

- A. 1 and 3 are correct
 - B. 2 and 3 are correct
 - C. 1 and 4 are correct
 - D. 2 and 4 are correct
-

13. The wobble phenomenon in tRNA decoding is primarily due to:

- A. Inosine forming non-standard base pairs at the third codon position.

- B. The exact matching of all three nucleotides between codon and anticodon.
 - C. Single tRNA binds only to one specific codon.
 - D. The rigid Watson-Crick base pairing at the third position of the codon.
-

14. In the context of genetic code degeneracy, which of the following accurately describes its impact on protein synthesis?

- A. Single codon specifies one amino acid.
 - B. Multiple codons can code for the same amino acid.
 - C. Flexibility occurs at the 2nd codon position.
 - D. Inosine in mRNA contributes to the flexibility in pairing.
-

15. Which of the following best describes the primary mechanisms and applications of RNA interference (RNAi)?

- A. Degradation of mRNA; used for gene manipulation and therapy.
 - B. Enhancement of mRNA synthesis; used for genetic modifications.
 - C. Conversion of mRNA to DNA; used for protein structure studies.
 - D. Stabilisation of mRNA; used for treating infections.
-

16. A 25-year-old student is reviewing the roles of small RNAs in cellular processes. She learns about small nuclear RNAs (snRNAs) and small interfering RNAs (siRNAs). Which of the following statements about snRNAs and/or siRNAs is true?

- A. Both snRNAs and siRNAs are involved in the degradation of mRNA.
- B. snRNAs form complexes with the RNA-induced silencing complex (RISC).

- C. snRNAs are primarily involved in mRNA splicing and are located in the nucleus.
 - D. snRNAs are typically found in the cytoplasm, while siRNAs are found in the nucleus.
-

17. Which of the following statements is true regarding the synthesis of mRNA?

- A. Introns are the coding sequence in a mRNA
 - B. 7-methyl guanosine gets added to the 3' end of mRNA
 - C. Process of splicing occurs in the cytoplasm
 - D. Heteronuclear RNA is not a functional mRNA
-

18. Which of the following steps in the process of tRNA synthesis occurs in the cytoplasm?

- A. Rearrangement of glycosidic linkages giving rise to pseudouridine
 - B. Deamination creating inosine
 - C. Fourth post-transcriptional modification of random methylation
 - D. Aminoacyl-tRNA synthetases attach the amino acid
-

19. Which statement best describes the wobble hypothesis in relation to tRNA's function?

- A. Allows multiple tRNAs to bind to a single codon.
 - B. Explains tRNA's ability to bind to any codon with perfect complementarity.
 - C. Single tRNA recognizes multiple codons due to flexible base pairing.
 - D. Accounts for varying tRNA lengths matching different mRNA sequences.
-

20. Which of the following types of rRNA is associated with the 40S ribosomal subunit in eukaryotic cells?

- A. 5S rRNA
 - B. 18S rRNA
 - C. 28S rRNA
 - D. 16S rRNA
-

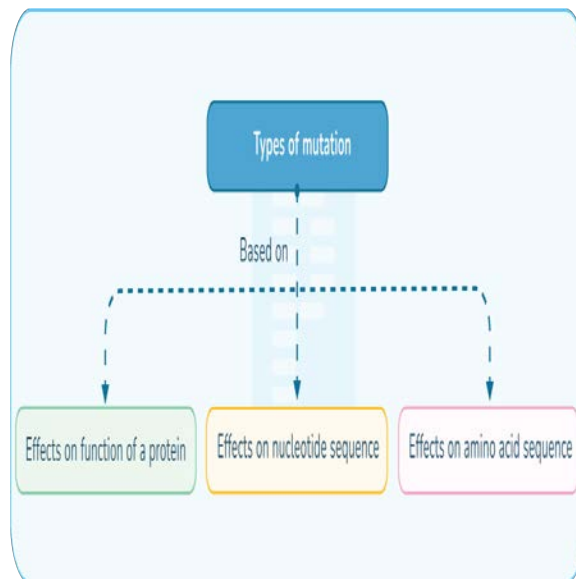
Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	3
Question 4	1
Question 5	2
Question 6	1
Question 7	1
Question 8	1
Question 9	2
Question 10	1
Question 11	3
Question 12	2
Question 13	1
Question 14	2
Question 15	1
Question 16	3
Question 17	4
Question 18	4

Question 19	3
Question 20	2

Solution for Question 1:

UAC mutates to a termination codon (UAA or UAG) leading to premature termination of the protein, and destroying the functional activity. This is called Nonsense mutation.



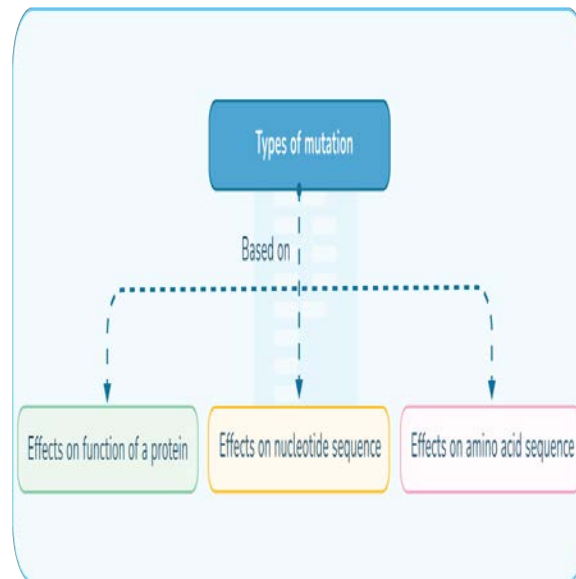
Based on the Effects on Amino acid sequence		
Silent Mutation (Option C ruled out)	Missense Mutation (Option D ruled out)	Nonsense Mutation (Option B)
Mutation may change the codon for one amino acid to a synonym for the same amino acid. It has no effect on the phenotype. Degeneracy of the code.	Different amino acid is incorporated due to change in the nucleotide sequence. It can be acceptable, partially acceptable, and unacceptable .	Codon for an amino acid is mutated to a termination codon (UAA or UAG) causing premature termination of translation.

Frameshift mutation (Option A) is the deletion or insertion of nucleotides leading to altered reading of frame in the mRNA.

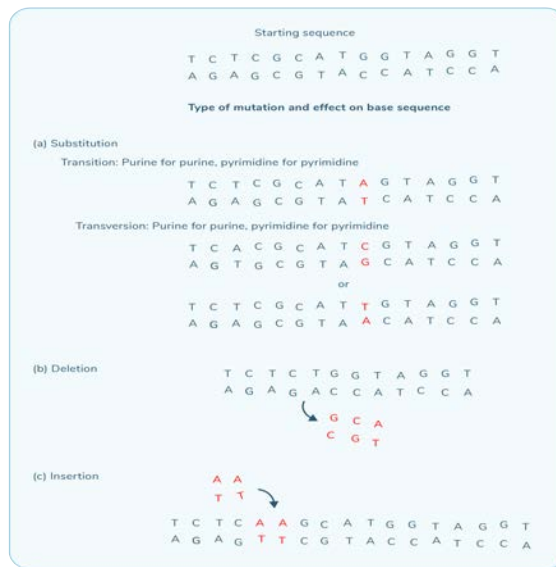
Solution for Question 2:

There is substitution of guanine by adenine and cytosine by thymine from the starting sequence (one nucleotide is substituted by the other), which is a point mutation.

A mutation is defined as a change in nucleotide sequence of DNA.



Based on the Effects on Nucleotide sequence	
Point mutation (Option A)	Frameshift mutation (Option B ruled out)
Substitution of one nucleotide by another. Transition : Purine is replaced by a purine and a pyrimidine is replaced by a pyrimidine. Transversion : Purine is replaced by a pyrimidine and vice versa.	Deletion (Option C ruled out) or insertion (Option D ruled out) of nucleotides leading to altered reading frame in the mRNA.



Solution for Question 3:

The Sanger technique is a DNA sequencing method that uses modified nucleotides to stop DNA synthesis, allowing the sequence of DNA to be read by analyzing fragment lengths.

There are cytogenetics and molecular methods for detection of mutations.

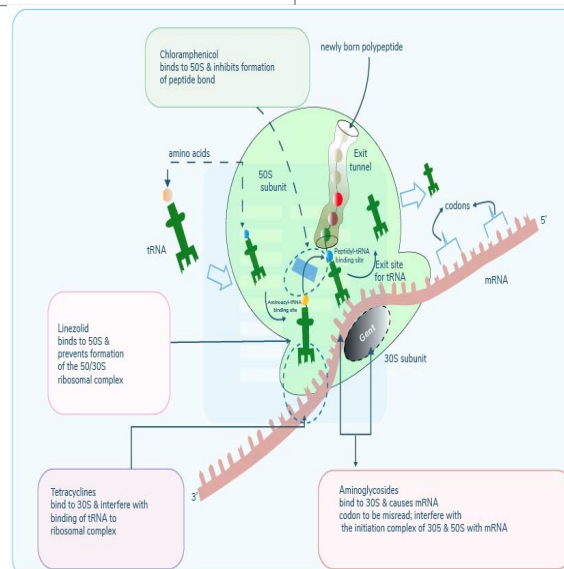
Methods	Uses	Advantages/ disadvantages
Cytogenetics		
Karyotype	Detecting numerical and gross structural aberrations	Low-resolution Time consuming
FISH (Fluorescence in situ hybridization) (Option A ruled out)	Detects copy number variations, structural alterations like translocations and microdeletion	Detects Limitation: Can be used only for known defects
CGH (Comparative genomic hybridization)	Detects copy number variations of genetic material Can be used to detect unknown chromosomal aberrations	Used only for losses and gains
Molecular		

RFLP (Restriction fragment length polymorphism)	Restriction fragments are separated by electrophoresis	Requires availability of enzyme with restriction site at the mutation site
DNA Sequencing	Sanger technique (Option C) is the most widely used. Directly determines the nucleotide sequence of a DNA fragment.	Detects point mutations, small deletions and insertions.
DNA microarray	Microscopic DNA spots attached to a solid surface	Multiple mutations can be detected.
ARMS PCR (Amplification refractory mutation system)	Allele-specific amplification of mutant and normal allele, determination of the genotype of an individual	Highly sensitive May increase time and costs
Multiplex PCR (Polymerase chain reaction)	Amplification of more than one target simultaneously	Reduces time and labour requirements Lower sensitivity and specificity
Real-time PCR	Amplification, detection, and quantification of target	Increased specificity Eliminates post-amplification analyses More expensive
MLPA (Multiplex ligation-dependent probe amplification) (Option B ruled out)	Deletions and duplications	Identifies very small single gene aberrations (50-70 nt)
DGGE (Denaturing Gradient Gel Electrophoresis)	Based on migration within gradient gel electrophoresis	Detects close to 100% of point mutations
SSCP (Single Strand Conformational Polymorphism) (Option D ruled out)	Based on migration of single-stranded conformation on gel electrophoresis	Detects about 80–90% of point mutations
Hetero-duplex analysis	Based on homoduplexes and heteroduplexes motilities in gel electrophoresis	Detects nearly 80% of mutations
OLA (Oligonucleotide ligation assay)	It is based on the ligation of two flanked primers annealed with target sequences	Detects all base exchanges
PTT (Protein truncation test)	It is based on a combination of PCR, transcription, and translation	Detects translation-terminating mutations. Missense mutations are not detected.

Solution for Question 4:

Tetracyclines inhibit the EF1A(prokaryotic elongation factor 1A) and also the attachment of aminoacyl tRNA to the A site.

Inhibitors of Protein Synthesis	Mechanism of action
Reversible Inhibitors in Bacteria (bacteriostatic)	
Tetracyclines (Option A)	Binds to the 30S ribosome of bacteria and inhibit attachment of aminoacyl tRNA to the A site.
Chloramphenicol (Option B ruled out)	Inhibits the peptidyl transferase activity of bacterial ribosomes
Erythromycin (macrolides) (Option C ruled out) and clindamycin	Inhibits the translocation process.
Irreversible Inhibitors in Bacteria (bactericidal)	
Aminoglycoside	Bind to 30S subunit of bacterial ribosomes. Misreading of mRNA. Inhibit the initiation complex formation and totally inhibit protein synthesis.
Inhibitors of Protein Synthesis in Mammals	
Puromycin (Option D ruled out)	Analogue of tyrosine tRNA and attaches to the “A” site of the ribosome causing premature termination.
Cycloheximide	Inhibits peptidyl transferase in 60S subunit
Diphtheria toxin	Inactivation of EF-2 by attachment of ADP to EF-2



Solution for Question 5:

Energetics of translocation of Amino acids:

Steps	Number of ATPs needed
Activating amino acids by aminoacyl tRNA synthetase (Option B)	2
Activating of eEF1 (A attaches amino acyl tRNA to A site)	1
Activating eEF2 (helps in translocation of ribosomes)	1
Total	4

Solution for Question 6:

Initiation methionine tRNA (imet-tRNA)	Aminoacyl tRNA (aa-tRNA)
Involved in the initiation of translation.	Involved in the elongation step of translation.
Guided by eIF2C (Option C ruled out)	Guided by eEF1A
Attaches to the P site of 60s ribosome	Attaches to the A site of 60s ribosome (Option A)
Utilises 3 high energy phosphates for each peptide bond formation (Option B ruled out)	Utilises 4 high energy phosphates for each peptide bond formation

Solution for Question 7:

eIF3 and 1A stabilize 40s (not the 60s subunit) ribosomal subunit to prevent reassociation with 60s.

Steps of initiation:

It involves tRNA, rRNA, mRNA, eukaryotic initiation factors (eIFs), GTP, ATP, and amino acids.

1. Ribosomal Dissociation:

- 80S ribosomes dissociate into 40S and 60S subunits.
- eIF3 and 1A stabilize 40s ribosomal subunit to prevent reassociation with 60s, allowing other translation initiation factors to associate with the 40S subunit. (Option A)

2. Formation of the 43S Preinitiation Complex:

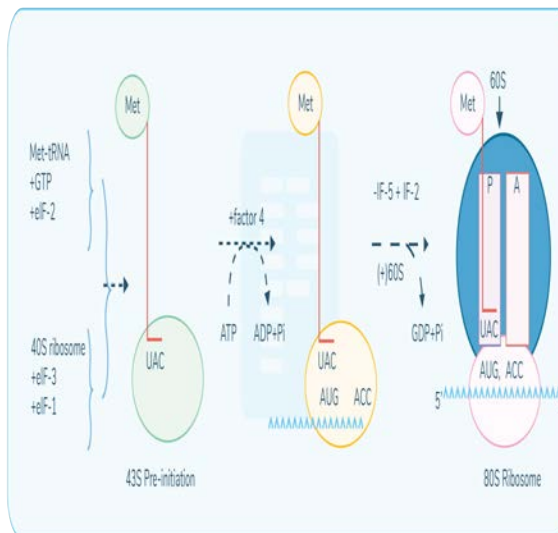
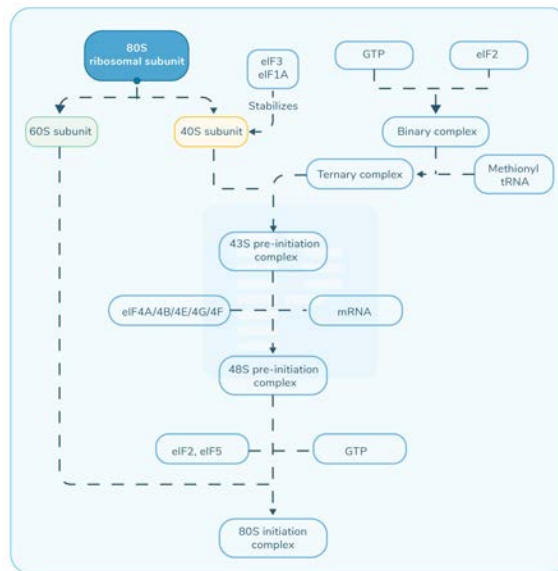
- Binary complex is formed by binding of GTP and eIF-2.
- This binary complex then binds to methionyl tRNA.
- The GTP-eIF-2-tRNA_i ternary complex binds to the 40S ribosomal subunit to form the 43S preinitiation complex.

3. Formation of the 48S Initiation Complex:

- The 5' termini of mRNA molecules in eukaryotic cells are "capped." The binding of mRNA to the 43S preinitiation complex is facilitated by 7meG-cap.
- 4G/4A attaches to the cap through 4E, which forms 4F.
- ATP-dependent helicase eIF4A/4B breaks hydrogen bonds to remove the hairpin loops and make mRNA single-stranded. (Option C ruled out)
- Methionine tRNA scans the mRNA to identify the initiation AUG codon, and attach the tRNA to AUG at the P site. (Option B ruled out)
- The association of mRNA with the 43S preinitiation complex forms the 48S initiation complex. (Option D ruled out)

4. Formation of the 80S Initiation Complex:

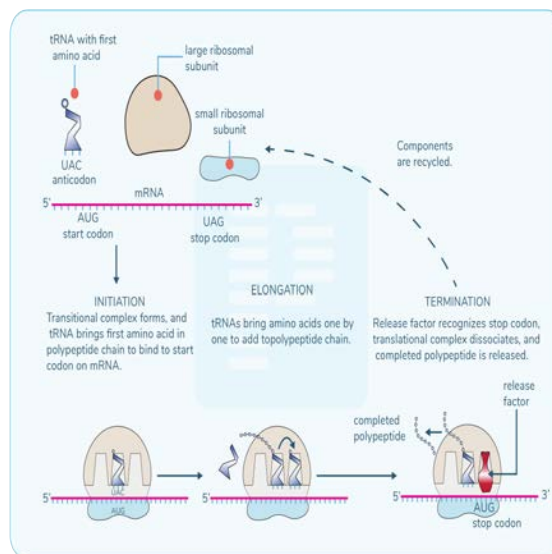
- The 60S ribosomal subunit binds to the 48S initiation complex by hydrolysis of the GTP bound to eIF-2 by eIF-5.
- This results in the release of the initiation factors bound to the 48S initiation complex.
- The rapid association of the 40S and 60S subunits forms the 80S ribosome.
- At this point, the met-tRNA_i is on the P site of the ribosome, ready for the elongation cycle to commence



Solution for Question 8:

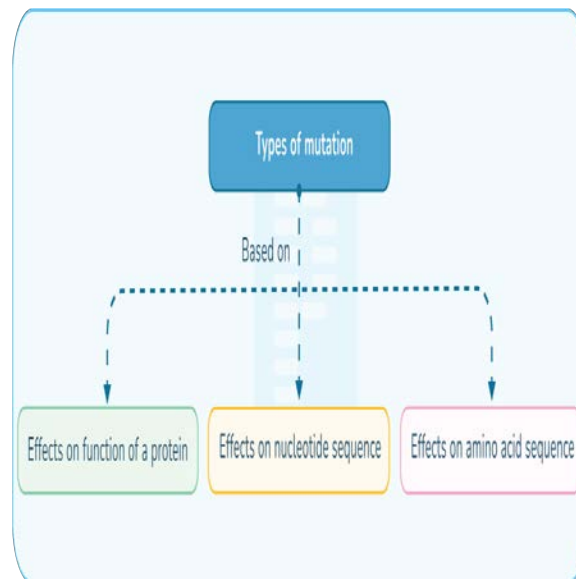
Phases of translation	Key events
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Initiation	Dissociation of 80s ribosome into 60s and 40s, and stabilized by various initiation factors. Formation of the 43S Preinitiation Complex: The 40S ribosomal subunit, in association with initiation factors, binds to the initiator tRNA (carrying methionine in eukaryotes or formyl-methionine in prokaryotes) forming the 43S preinitiation complex. Formation of 48S initiation complex: The 43S complex binds to the 5' end of the mRNA, and scans for start codon which upon recognition, the 48S initiation complex is formed. Formation of the 80S Initiation Complex: Binding of the 60S ribosomal subunit to the 48S initiation. The met-tRNA on the P site of the ribosome is ready for the elongation cycle.
Elongation	Binding of Aminoacyl-tRNA to the A Site Peptide Bond Formation Elongation factor 2 (EF2) binds to and displaces the peptidyl tRNA from the A site to the P site
Termination	Ribosome reaches the terminator codon sequence (UAA, UAG or UGA) Releasing factor(RF1), (RF3) and GTP complex promotes hydrolysis releasing the protein and the tRNA from the P site.



Solution for Question 9:

- The mutation described is a missense mutation , where the GAG codon (which codes for glutamic acid) is replaced by the GTG codon (which codes for valine) at the 6th position of the hemoglobin gene resulting in the production of hemoglobin S (HbS) instead of the normal hemoglobin A (HbA).
- This partially acceptable mutation still allows the hemoglobin to carry oxygen, but with altered efficiency (accumulation of globin chains and causes sickling crisis on exposure to deoxygenation.), leading to symptoms of sickle cell anemia.
- Sickle cell anemia is an example of point mutation, transversion, missense and partially acceptable mutation.



Effects on the functioning of the protein:

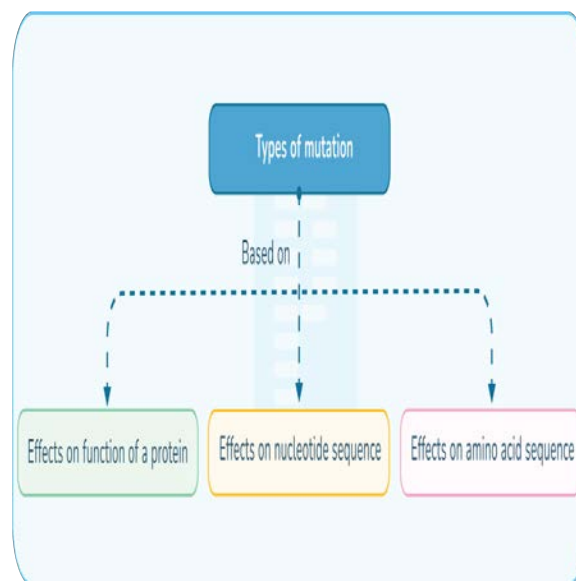
Based on the Effects on the function of protein		
Acceptable mutation: (Option C ruled out)	Partially acceptable mutation: (Option B)	Unacceptable mutation: (Option A ruled out)
Change in the nucleotide sequence causes alteration in the amino sequence that does not affect the functioning of the protein. Example: Hb Sydney	Change in the nucleotide sequence causing alteration in the amino sequence that affects the functioning of the protein but it is not immediately life-threatening. Example: HbS - GAG is replaced by GTG which replaces the glutamic acid to valine.	Change in the nucleotide sequence causes alteration in the amino sequence that affects the functioning of the protein so extensively that it is immediately life-threatening. Example: Congenital methemoglobinemia (HbM) - Histidine (CAU) is replaced with tyrosine (UAU) in alpha-chain.

Nonsense mutation (Option D) : Codon for an amino acid is mutated to a termination codon (UAA or UAG) causing premature termination of translation.

Solution for Question 10:

There is deletion of adenine at the 6th position from the DNA sequence, leading to a shift in the reading frame. This is a frameshift mutation.

A mutation is defined as a change in the nucleotide sequence of DNA.



Based on the Effects on Nucleotide sequence	
Point mutation (Option B ruled out)	Frameshift mutation (Option A)
Substitution of one nucleotide by another. Transition (Option C ruled out) : A purine is replaced by a purine, and a pyrimidine is replaced by a pyrimidine. Transversion : Purine is replaced by a pyrimidine and vice versa.	Deletion (Option A ruled out) or insertion (Option D ruled out) of nucleotides leads to an altered reading frame in the mRNA.

Solution for Question 11:

RNA Structure: RNA is single-stranded but can fold into complex three-dimensional structures due to intrachain base pairing (A-U, G-C). DNA is typically double-stranded with a stable double-helix structure.

Aspect	DNA	RNA
Sugar	Deoxyribose (lacks 2' hydroxyl group) (Option D ruled out)	Ribose (has 2' hydroxyl group)
Bases	Adenine (A), Guanine (G), Cytosine (C), Thymine (T)	Adenine (A), Guanine (G), Cytosine (C), Uracil (U)
Structure	Double-stranded helix, antiparallel strands, complementary base pairing (A-T, G-C) DNA follows both Chargaff rule of base pairing and Watson and Crick base pairing rule.	Single-stranded, can fold into complex structures with intrachain base pairing (A-U, G-C) (Option C) Do not follow Chargaff's Rule of Base Pairing. Watson and Crick Base pairing rule is followed here.
Function	Stores genetic information	Various roles in gene expression: Messenger RNA (mRNA) - Carries genetic information from DNA to ribosomes for protein synthesis Ribosomal RNA (rRNA) - Major component of ribosomes, involved in protein synthesis Transfer RNA (tRNA) - Transports amino acids to ribosomes during protein synthesis
Stability	More stable due to lack of 2'-hydroxyl group B (Option A ruled out)	Less stable, more susceptible to hydrolysis
Degradation Susceptibility	Less prone to degradation, stable over generations	More prone to degradation, shorter lifespan, important for regulating gene expression
Synthesis Primer Requirement	Requires RNA primer for initiation (Option B ruled out)	Can initiate synthesis de novo, no primer needed
Proofreading and Fidelity	High fidelity with 3' to 5' exonuclease activity for proofreading	Higher error rate but can backtrack and cleave incorrect nucleotides.

Solution for Question 12:

A, G, T, and C are nucleotides forming DNA codons, while A, G, U, and C are those forming RNA codons.

Aspect	Details
Codon	Three-nucleotide sequence in mRNA
Codon Functions	Specifies an amino acid sequence in proteins. Start codon (AUG): Signals the initiation of protein synthesis. Stop codons(UAA, UAG, and UGA) : Signals termination of protein synthesis
Total Codons	64 (4 nucleotides, each position can be A, U, G, or C)
DNA Nucleotides	A, G, T, C (Options A, C, D ruled out)
RNA Nucleotides	A, G, U (replaces T), C (Options A, C, D ruled out)
Amino Acid Codons	61 codons encode 20 amino acids
Reading Frame	mRNA sequence read from start codon to stop codon (Open Reading Frame)

Solution for Question 13:

The wobble phenomenon is due to inosine's ability to form non-standard base pairs (with U, C, and A) at the third position of the codon, allowing a single tRNA to recognize multiple codons.

- Definition : Flexible base-pairing between the third nucleotide of a codon (mRNA) and the corresponding nucleotide of the tRNA anticodon. (Option B ruled out)
- Function : Allows one tRNA molecule to bind to multiple codons coding for the same amino acid. (Option C ruled out)
- Inosine (I): A modified nucleotide in some tRNA anticodons, pairs with U, C, and A.
- Non-standard pairing is more flexible than Watson-Crick base pairing, which is strict for the first two codon nucleotides.

Codon Position	Standard Base Pairing	Wobble Pairing	Example
----------------	-----------------------	----------------	---------

1st and 2nd	Watson-Crick rules (Option D ruled out)	Strict	AG A and AG G code for Arginine.
3rd	Non-standard pairing	Flexible	AGA/AGG with UCU, GGU/GGC/GGA with CCI

Solution for Question 14:

Degeneracy allows multiple codons to specify the same amino acid, based on flexible base-pairing rules.

Degeneracy in the Genetic Code :

- Multiple codons can encode the same amino acid. (Option A ruled out) (Option B)
- There are 64 possible codons but only 20 amino acids.
- Example: Six codons (UCU, UCC, UCA, UCG, AGU, AGC) all code for serine.

Causes of Degeneracy :

- Wobble Hypothesis : Flexibility in base-pairing rules between the third nucleotide of mRNA codon and the tRNA anticodon . (Option C ruled out)
- Inosine (I) Role : Inosine in t RNA anticodons can pair with U, C, and A, enhancing code flexibility. (Option D ruled out)

Solution for Question 15:

RNAi primarily functions through mRNA degradation and inhibiting protein synthesis, with gene manipulation and therapy applications.

- Definition : Biological process where double-stranded RNA (dsRNA) silences specific gene expressions.
- Mechanisms : Degradation of messenger RNA (mRNA) molecules corresponding to the target gene. Reduction of expression levels of the target gene.

- Degradation of messenger RNA (mRNA) molecules corresponding to the target gene.
- Reduction of expression levels of the target gene.
- Applications : Tool in genetic research and biotechnology for gene knockdown. Understanding gene function. Developing therapeutic strategies.
- Tool in genetic research and biotechnology for gene knockdown.
- Understanding gene function.
- Developing therapeutic strategies.
- Degradation of messenger RNA (mRNA) molecules corresponding to the target gene.
- Reduction of expression levels of the target gene.
- Tool in genetic research and biotechnology for gene knockdown.
- Understanding gene function.
- Developing therapeutic strategies.

Enhancement of mRNA synthesis; used for genetic modifications (Option B) does not enhance mRNA synthesis; it silences gene expression.

Conversion of mRNA to DNA; used for protein structure studies (Option C) does not involve converting mRNA to DNA; it targets mRNA for degradation

Stabilisation of mRNA; used for treating infections (Option D) causes degradation of mRNA, not stabilisation.

Solution for Question 16:

Small nuclear RNAs (snRNAs) are essential components of the spliceosome, a complex responsible for removing introns from pre-mRNA (hnRNA) during splicing, which occurs in the nucleus.

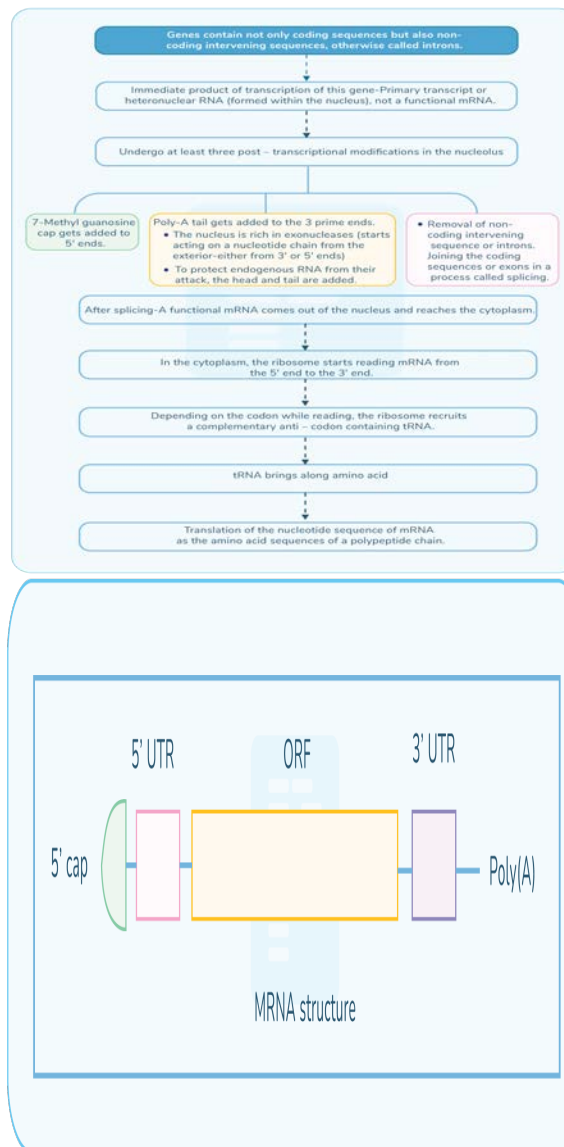
Feature	snRNAs	siRNAs
Main Function	RNA processing	Gene regulation
Location (Option C) (Option D ruled out)	Nucleus	Cytoplasm
Involvement (Option A ruled out)	mRNA splicing, transcription elongation	mRNA degradation

Specific Types	U1, U2, U4, U5, U6 (splicing); U7 (histone mRNAs); 7SK (transcription)	Typically 21-25 nucleotides long
Role in mRNA	Removes introns, generates correct 3' ends of histone mRNAs	Silences gene expression
Source	Endogenous snRNAs	Endogenous/exogenous dsRNAs processed into siRNAs
Processing Mechanism	Forms complexes with proteins (snRNPs) for splicing	Processed by Dicer enzyme into siRNA duplexes
Associated Complex (Option B ruled out)	Small nuclear ribonucleoproteins (snRNPs)	RNA-induced silencing complex (RISC)
Length	Varies; generally 100-300 nucleotides	21-25 nucleotides
Stability	Generally stable, involved in essential RNA processes	Short-lived, rapidly degrades target mRNA
Biogenesis	Transcribed by RNA polymerase II or III	Derived from longer dsRNA precursors
Function in Disease	Defects can lead to splicing disorders (e.g., spinal muscular atrophy)	Utilized in RNAi therapies for gene silencing (e.g., in cancer)
Example Applications	Studying splicing mechanisms, understanding transcription regulation	Therapeutic gene silencing, functional genomics studies

Solution for Question 17:

Heteronuclear RNA (hnRNA) (Primary transcript) is the immediate product of transcription and is the precursor to mRNA, containing introns and exons. It undergoes processing like splicing, capping, and polyadenylation to become functional mRNA.

Synthesis of mRNA:



Introns are the coding sequence in a mRNA (Option A): Exons are the coding sequences, while introns are the non-coding sequences in a mRNA.

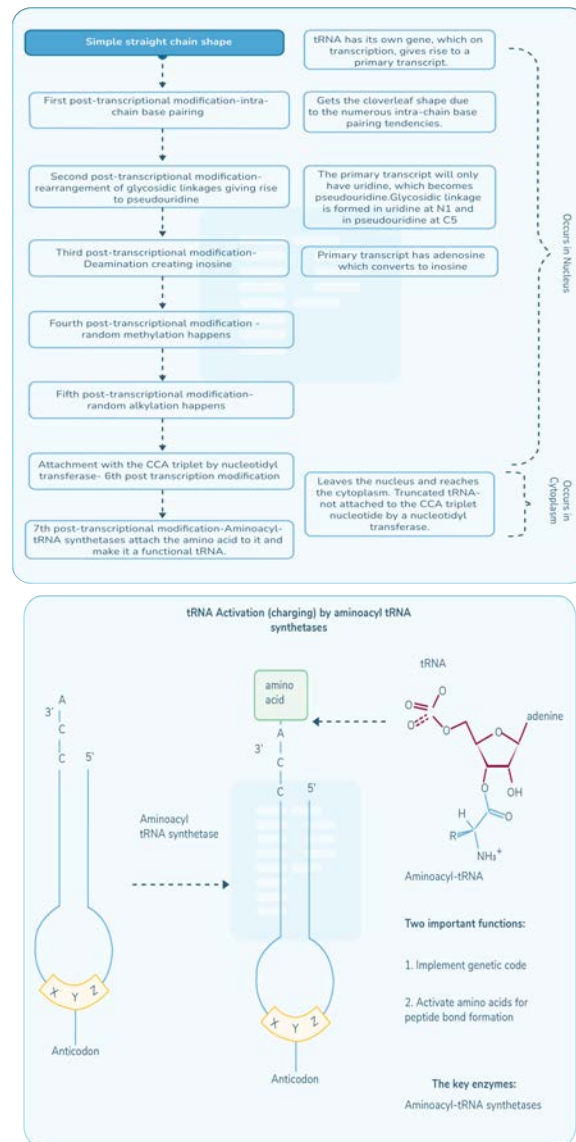
7-methyl guanosine gets added to the 3' end of mRNA (Option B): 7-methyl guanosine gets added to the 5' end of mRNA, while Poly-A tail gets added to the 3' end of mRNA.

Process of splicing occurs in the cytoplasm (Option C): Process of splicing occurs in the nucleus.

Solution for Question 18:

The 7th post-transcriptional modification involves aminoacyl-tRNA synthetases attaching the amino acid to the tRNA, making it functional in the cytoplasm.

Synthesis of tRNA:



Rearrangement of glycosidic linkages giving rise to pseudouridine (Option A) is incorrect because pseudouridine formation occurs in the nucleus, as part of post-transcriptional modifications of tRNA.

Deamination creating inosine (Option B) is incorrect because inosine is formed in the nucleus by the deamination of adenosine during tRNA maturation. Inosine is important for wobble base pairing in codon recognition.

Fourth post-transcriptional modification of random methylation (Option C) is incorrect because methylation of tRNA (as part of modifications like methylation of ribose or bases) occurs in the nucleus, not the cytoplasm.

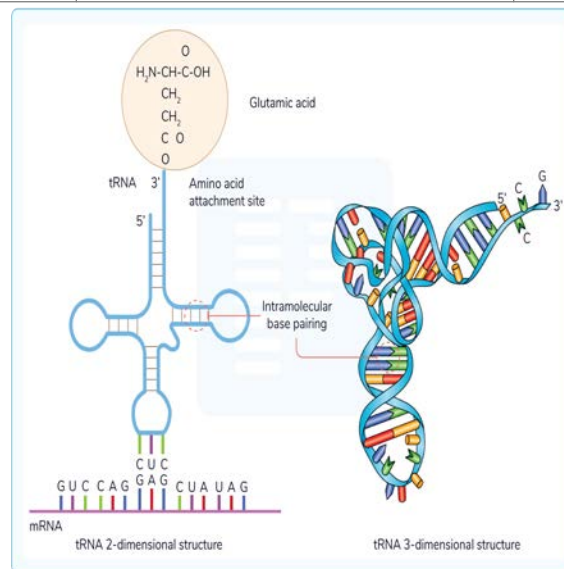
Solution for Question 19:

The Wobble Hypothesis explains that a single tRNA can bind to multiple codons because of flexible base pairing at the third position of the codon.

tRNA / transfer RNA:

Component	Description	Function
Amino Acid Acceptor Arm	Located at the 3' end, ending in CCA, which is added post-transcriptionally. The amino acid is attached to the 3'-OH group of the A moiety.	Charging : Amino acids are attached to the tRNA by aminoacyl-tRNA synthetases for protein synthesis.
D Arm	Contains dihydrouridine, a modified base. Contributes to the overall structure and recognition of tRNA by its corresponding aminoacyl-tRNA synthetase.	Recognition : Ensures correct aminoacylation by interacting with aminoacyl-tRNA synthetase.
T ψ C Arm	Named for its ribothymidine (T), pseudouridine (ψ), and cytidine (C) content. Involved in binding tRNA to the ribosome.	Binding : Positions tRNA correctly at the ribosome for protein synthesis.

Anticodon Loop	Opposite the amino acid acceptor arm, contains a sequence of three nucleotides (anticodon) complementary to the codon on mRNA. Wobble phenomenon Third nucleotide pair formation is non-specific (more than one codon coding for a single amino acid) (Or) Reduced stringency when the third nucleotide pair is formed between codon and anti-codon. (Or) When tRNA is written in 5', 3' direction, the first nucleotide of the anticodon of tRNA is not specific Degeneracy More than one codon coding for a single amino acid because of the wobble phenomenon.	Codon Recognition : Ensures the correct amino acid is added to the polypeptide chain during translation.
Variable Arm	Located between the T ψ C arm and the anticodon loop. Its length varies among different tRNA species and contributes to the tertiary structure of tRNA.	Structural Flexibility : Aids in interactions with the ribosome and other translation factors.



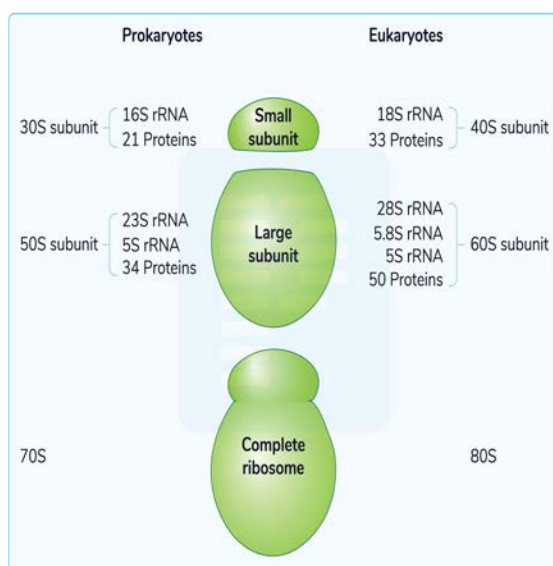
Allows multiple tRNAs to bind to a single codon (Option A): The wobble hypothesis focuses on how a single tRNA can recognize and bind to several codons that code for the same amino acid, not how multiple tRNAs interact with one codon.

Explains tRNA's ability to bind to any codon with perfect complementarity (Option B): The wobble hypothesis involves flexible or non-standard base pairing. It does not imply that tRNAs bind with perfect complementarity to all codons.

Accounts for varying tRNA lengths matching different mRNA sequences (Option D): The hypothesis does not address differences in tRNA lengths or their relationship with mRNA sequences.

Solution for Question 20:

- In eukaryotic cells, the ribosome is composed of two subunits: the small 40S subunit and the large 60S subunit.
- The 40S subunit contains the 18S rRNA, which is essential for the structural integrity and function of the ribosome during protein synthesis.
- The 60S subunit, on the other hand, includes the 5S, 5.8S, and 28S rRNAs (Options A and C ruled out)
- The 16S rRNA is found in the small subunit of prokaryotic ribosomes, not in eukaryotic ribosomes (Option D ruled out)



Type of RNA	Characteristics	Function
mRNA (messenger RNA)	Single-stranded Varies in length depending on the encoded protein Contains exons and introns (in eukaryotes)	Carries genetic information from DNA to the ribosome Encodes the amino acid sequence of proteins Undergoes processing such as capping, polyadenylation, and splicing in eukaryotes

tRNA (transfer RNA)	Small, typically 73-93 nucleotides Cloverleaf structure Contains an anticodon and an amino acid attachment site	Transfers specific amino acids to the ribosome during protein synthesis Recognizes codons on mRNA through its anticodon region Plays a critical role in translating the genetic code into protein
rRNA (ribosomal RNA)	Forms the core of ribosome's structure Eukaryotic ribosomes: 18S (40S subunit), 5.8S, 28S, 5S (components of the 60S subunit) Prokaryotic ribosomes: 16S, 23S, 5S	Major structural and functional components of ribosomes Catalyzes peptide bond formation during protein synthesis
snRNA (small nuclear RNA)	Small, usually less than 200 nucleotides Found in the nucleus Component of the spliceosome	Involved in the processing of pre-mRNA in eukaryotes Essential for splicing introns from pre-mRNA
miRNA (microRNA)	Small, approximately 22 nucleotides Single-stranded Forms hairpin structures	Regulates gene expression post-transcriptionally Binds to complementary sequences on target mRNAs, resulting in translational repression or mRNA degradation Plays a role in various biological processes including development, differentiation, and apoptosis
siRNA (small interfering RNA)	Short, double-stranded RNA, about 20-25 nucleotides Originates from longer double-stranded RNA	Involved in the RNA interference (RNAi) pathway Mediates the degradation of specific mRNAs Used experimentally to silence gene expression

PCR

1. In PCR cloning, the use of proofreading DNA polymerases is often preferred because:

- A. They amplify DNA at a much faster rate than regular Taq polymerase
 - B. They have a lower error rate during DNA synthesis
 - C. They can amplify longer DNA fragments more efficiently
 - D. They are more tolerant to variations in PCR cycling conditions
-

2. Which of the following steps is NOT part of the standard PCR cloning process?

- A. Denaturation
 - B. Annealing
 - C. Ligation
 - D. Elongation
-

3. Which of the following applications is a typical use of Polymerase Chain Reaction (PCR)?

- A. Identifying specific genetic mutations
 - B. Determining protein structure
 - C. Measuring cell growth rates
 - D. Analyzing metabolite levels
-

4. What is the primary purpose of Reverse Transcription Polymerase Chain Reaction (RT-PCR) in molecular biology?

- A. To amplify DNA sequences directly from the genome.
 - B. To convert RNA into complementary DNA (cDNA) and then amplify specific DNA sequences.
 - C. To sequence the entire genome of an organism.
 - D. To separate DNA fragments by size.
-

5. What is the primary advantage of using real-time PCR (qPCR) over traditional PCR techniques?

- A. It allows for the detection of multiple targets in a single reaction.
 - B. It enables the quantification of DNA as the amplification progresses.
 - C. It eliminates the need for a thermal cycler.
 - D. It provides a higher specificity in amplifying DNA.
-

6. Which of the following best describes the main principle behind Sanger sequencing?

- A. Amplification of DNA segments to facilitate sequencing.
 - B. Incorporation of dideoxynucleotides to terminate DNA synthesis at specific bases.
 - C. High-throughput sequencing of millions of DNA fragments simultaneously.
 - D. Insertion of a DNA fragment into a plasmid vector for replication and sequencing.
-

7. A 45-year-old patient diagnosed with COVID-19 has a real-time PCR test result showing a cycle threshold (Ct) value of 35. How should this Ct value be interpreted in the context of their infectivity and clinical management?

- A. The patient has a high viral load and is highly infectious.
 - B. The patient has a low viral load and is less likely to be infectious.
 - C. The patient's Ct value indicates they should be immediately admitted to the ICU.
 - D. The Ct value alone is sufficient to determine the patient's disease severity and infectivity.
-

8. A forensic scientist is analyzing a DNA sample from a crime scene. The DNA is digested with restriction enzymes, separated by gel electrophoresis, and transferred to a nitrocellulose membrane. Labeled DNA probes are then used to detect specific fragments. Which technique is the scientist most likely using?

- A. Restriction Fragment Length Polymorphism (RFLP)
 - B. Polymerase Chain Reaction (PCR)
 - C. Single-Strand Conformation Polymorphism (SSCP)
 - D. DNA Sequencing
-

9. Which of the following statements about Denaturing Gradient Gel Electrophoresis (DGGE) is false?

- A. DGGE can separate DNA fragments based on their melting characteristics.
 - B. DGGE is effective for analyzing DNA segments of 200-300 base pairs.
 - C. DGGE requires the use of a denaturant gradient within a polyacrylamide gel.
 - D. DGGE is commonly used for detecting translocations.
-

10. Which of the following methods is specifically designed to detect differences in DNA sequences based on variations in the electrophoretic mobility of single-stranded DNA fragments?

- A. RFLP (Restriction Fragment Length Polymorphism)
 - B. SSCP (Single-Strand Conformational Polymorphism)
 - C. Sequencing
 - D. All of the above methods are equally effective for detecting DNA sequence variations.
-

11. A researcher is experiencing issues with the accuracy and efficiency of DNA amplification during a PCR experiment. They suspect the problem might be related to magnesium ion concentration. What role does magnesium play in PCR, and how might its concentration affect the experiment?

- A. Magnesium ions serve only to stabilize the DNA template and do not affect enzyme activity.
 - B. Excessive magnesium concentration enhances DNA polymerase activity, reducing the likelihood of errors.
 - C. Insufficient magnesium concentration can lead to decreased enzyme activity and lower DNA amplification efficiency.
 - D. Magnesium ions are not essential for the nucleophilic attack of the primer's 3' hydroxyl group during DNA synthesis.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	1
Question 4	2
Question 5	2
Question 6	2
Question 7	2
Question 8	1
Question 9	4
Question 10	2
Question 11	3

Solution for Question 1:

Correct Answer: B) They have a lower error rate during DNA synthesis

Explanation:

Proofreading DNA polymerases have a lower error rate during DNA synthesis.

Proofreading DNA polymerases, such as Pfu, Phusion, or Q5, are often preferred in PCR cloning for the following reasons:

- Lower error rate: These enzymes possess 3'-5' exonuclease activity, which allows them to detect and remove misincorporated nucleotides during DNA synthesis. This proofreading ability significantly reduces the error rate compared to non-proofreading polymerases like regular Taq.
- Blunt-end products: Many proofreading polymerases generate blunt-end PCR products, which can be advantageous for certain cloning strategies.

Incorrect:

- They amplify DNA at a much faster rate than regular Taq polymerase (Option A) Proofreading polymerases are often slower than Taq polymerase due to their proofreading activity.
- They can amplify longer DNA fragments more efficiently (Option C) While some proofreading polymerases can amplify longer fragments, this is not their primary advantage in PCR cloning.
- They are more tolerant to variations in PCR cycling conditions (Option D) Proofreading polymerases are generally not more tolerant to variations in PCR conditions; in fact, they often require more precise optimization.

Solution for Question 2:

Correct Answer: C) Ligation

Explanation:

Ligation is not a step involved in PCR. It is a step that occurs after PCR when inserting the amplified DNA fragment into a cloning vector. It involves the joining of the DNA fragment with the vector using DNA ligase, and it is not part of the actual PCR amplification process itself.

PCR steps include:

Step 1- Denaturation

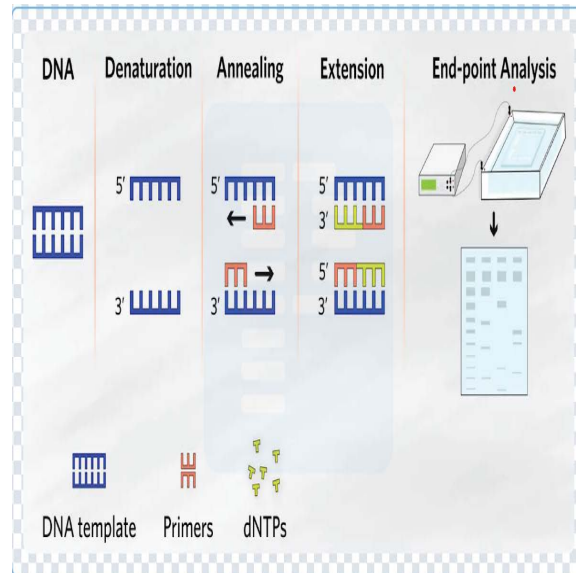
The double-stranded DNA is heated to around 95°C to separate into 2 single strands.

Step 2- Annealing

- Addition of 2 primers, which will anneal to the flanking sequence of the desired fragments.
- An optimal temperature (5 degrees less than the average melting temperature of the primers) should be maintained to avoid non-specific amplification.

Step 3- Elongation

- The enzyme used is thermostable Taq(Thermus Aquaticus)DNA polymerase.
- DNA polymerase elongates new strand in 5'3' direction and optimal temperature required for elongation is 72°.
- If PCR goes through 'N'number of cycle,then '2n' number of products formed



Solution for Question 3:

Correct Answer: A) Identifying specific genetic mutations

Explanation :

PCR applications:

- Medical Diagnostics: PCR can identify pathogens like bacteria and viruses by amplifying their DNA or RNA, making it possible to diagnose diseases such as COVID-19, tuberculosis, and HIV.
- Forensic Science/DNA Fingerprinting: PCR helps amplify DNA samples from crime scenes, enabling identification and comparison of genetic profiles to match suspects or victims.
- Detect Allelic Polymorphisms: PCR can identify allelic polymorphisms, which include single nucleotide polymorphisms (SNPs), insertions and deletions (indels), and gene amplifications.

- Biodiversity Studies: PCR can detect and identify species from environmental samples, helping in the study of biodiversity and monitoring endangered species.

Determining protein structure (Option B): This process typically involves techniques like X-ray crystallography or nuclear magnetic resonance (NMR) spectroscopy, not PCR.

Measuring cell growth rates (Option C): Cell growth is usually measured using methods such as cell counting, growth curves, or assays that assess cell proliferation.

Analyzing metabolite levels (Option D): This involves techniques like mass spectrometry or chromatography to measure the concentrations of metabolites.

Solution for Question 4:

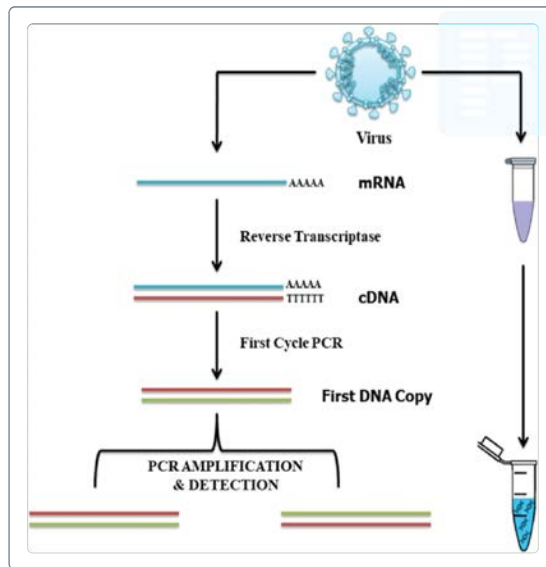
Correct Answer: B) To convert RNA into complementary DNA (cDNA) and then amplify specific DNA sequences

Explanation:

RT-PCR is a technique used to detect and quantify RNA.

It involves two main steps:

- Step 1- Conversion of RNA into complementary DNA (cDNA) using the enzyme reverse transcriptase.
- Step 2- Amplification of specific DNA sequences using PCR.



To amplify DNA sequences directly from the genome (Option A):

This option describes a process related to PCR but not specifically RT-PCR. Standard PCR amplifies DNA sequences directly from a sample, without the intermediate step of converting RNA to cDNA.

To sequence the entire genome of an organism (Option C):

Sequencing the entire genome of an organism is done through techniques such as whole-genome sequencing.

To separate DNA fragments by size (Option D)

This describes gel electrophoresis, not RT-PCR. Gel electrophoresis is used to separate DNA fragments based on size, whereas RT-PCR is used for amplifying and quantifying specific DNA sequences derived from RNA.

Solution for Question 5:

Correct Answer: B) It enables the quantification of DNA as the amplification progresses.

Explanation:

- Real-time PCR, also known as quantitative PCR (qPCR), provides the advantage of quantifying DNA or RNA as the amplification process occurs in real time.
- This is achieved by measuring the fluorescence emitted by DNA-binding dyes or probes that are incorporated during the PCR amplification.
- Traditional PCR only allows for the detection of the PCR product at the end of the process, without the capability to quantify the initial amount of target nucleic acid.

Option A) It allows for the detection of multiple targets in a single reaction.

While real-time PCR can indeed be used to detect multiple targets in a single reaction through the use of different fluorescent probes or dyes, this is not its primary advantage. The main benefit of real-time PCR is its ability to quantify nucleic acids as the amplification occurs.

Option C) It eliminates the need for a thermal cycler.

Real-time PCR still requires a thermal cycler to facilitate the cycling of temperatures necessary for the PCR process.

Option D) It provides a higher specificity in amplifying DNA.

Real-time PCR does not inherently provide higher specificity compared to traditional PCR. Specificity is largely dependent on the design of primers and probes, not the real-time nature of the assay.

Solution for Question 6:

Correct Answer: B) Incorporation of di deoxynucleotides to terminate DNA synthesis at specific bases.

Explanation: Sanger sequencing is a method based on the principle of chain termination, where dideoxynucleotides (ddNTPs) are used to terminate DNA synthesis at specific points, allowing for the determination of the DNA sequence.

Sanger sequencing was developed by Frederick Sanger in 1977. The technique is based on the principle of chain termination and involves several key steps:

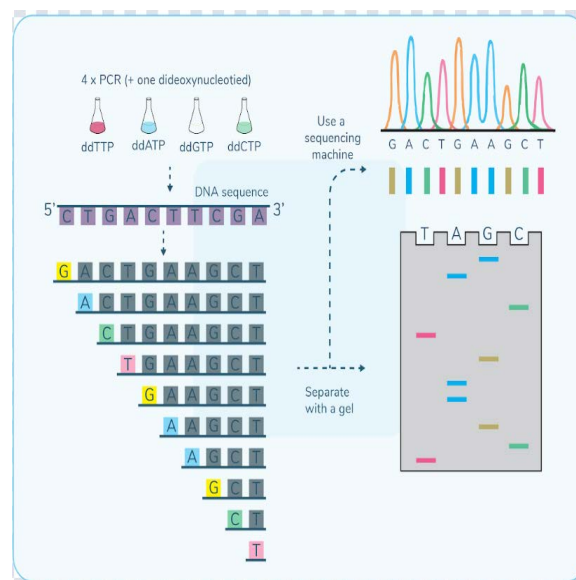
1. Preparation: The DNA of interest is first amplified to produce a large enough quantity for sequencing.
2. Reaction Mix: The DNA is then mixed with DNA polymerase, a primer, and a mixture of deoxynucleotides (dNTPs) and dideoxynucleotides (ddNTPs). The ddNTPs are fluorescently

labeled and act as chain terminators. Each ddNTP corresponds to one of the four DNA bases (A, T, C, G).

3. Chain Termination: As the DNA polymerase extends the DNA strand, it occasionally incorporates a ddNTP instead of a dNTP. This incorporation causes the chain to terminate because ddNTPs lack the 3' hydroxyl group needed for further DNA elongation.

4. Separation: The resulting mixture of terminated DNA fragments of varying lengths is then separated by size using capillary electrophoresis.

5. Detection and Analysis: The fragments are detected as they pass through a laser, and the fluorescent labels indicate the identity of the terminating ddNTP. This information is used to reconstruct the DNA sequence.



Sanger sequencing is known for its accuracy and was historically used for the Human Genome Project. However, next-generation sequencing techniques for high-throughput sequencing have largely supplanted it.

- Amplification of DNA segments to facilitate sequencing (Option A) is done by Polymerase Chain Reaction (PCR).
- High-throughput sequencing of millions of DNA fragments simultaneously (Option C) is done by Next-Generation Sequencing.
- Insertion of a DNA fragment into a plasmid vector for replication and sequencing (Option D) is done by Gene Cloning.

Solution for Question 7:

Higher Ct values (>30) suggest a lower viral load, indicating reduced infectivity.

Cycle Threshold (Ct) Value	
Definition	The number of cycles required for the fluorescent signal to exceed a threshold, indicating detectable target nucleic acid (e.g., viral RNA).
Significance of Ct Values	Viral Load Estimation Lower Ct values (≤ 30) indicate higher viral loads Higher Ct values (>30) suggest lower viral loads (Option B,A) Helps infer potential transmission risk and infection stage. Infectivity Correlation Ct values >24 are linked to reduced infectivity Ct values >34 typically do not yield infectious virus. Clinical Management : Ct values aid in patient management decisions, such as isolation duration. Systematic reporting improves result interpretation and impacts treatment and quarantine measures.
Limitations and Considerations	Variability Across Tests Different PCR platforms may produce varying Ct values due to differences in reagents, protocols, and target genes. Laboratories must establish their own reference ranges and interpretive criteria. Sample Quality Influence Factors like sample collection technique, timing, and storage conditions affect Ct values. Discrepancies can occur between different sample types (e.g., nasal vs. oropharyngeal). Interpretation Challenges (Option C,D) Ct values should not be used alone to determine infectivity or disease severity. Other clinical factors, such as symptoms and exposure context, must also be considered.

Option A: Incorrect. A Ct value of 35 indicates a lower viral load, not a high one.

Option C: Incorrect. A Ct value alone does not necessitate ICU admission without considering other clinical factors.

Option D: Incorrect. Ct values should be interpreted alongside other clinical factors, not in isolation.

Solution for Question 8:

RFLP involves DNA digestion with restriction enzymes, gel electrophoresis, transfer to a nitrocellulose membrane, hybridization with labeled probes, and visualization of fragments.

Restriction Fragment Length Polymorphism (RFLP)

- Variation in DNA fragment lengths due to differences in DNA sequences when digested with restriction enzymes are used as genetic markers.
- Applications: Parenthood Disputes, Population Genetics, Genetic Disease Diagnosis, Genetic Mapping, Cloning, Forensic Medicine.

	Sequential Steps involved using RFLP technique	
	1. DNA Digestion	DNA is cut by restriction enzymes at specific sites.
	2. Electrophoresis	DNA fragments are separated by size using gel electrophoresis.
	3. Denaturation	To enable probe binding, the separated fragments are denatured
	4. Southern Blotting	Fragments are transferred from the gel to a nitrocellulose membrane.
	5. Hybridization	DNA fragments hybridize with labeled DNA probes.
	6. Visualization	Hybridized fragments are visualized to reveal the presence or absence of restriction sites.

Option B: PCR is used to amplify DNA, not to analyze fragment length variations.

Option C: SSCP is used to detect variations in single-stranded DNA based on their conformational differences, not the length of restriction fragments.

Option D: DNA sequencing determines the nucleotide order, not the length variation of fragments after restriction enzyme digestion.

Solution for Question 9:

Correct Answer: D) DGGE is commonly used for detecting translocations.

Explanation:

DGGE is not used for detecting translocations.

Denaturing Gradient Gel Electrophoresis (DGGE)

- Purpose: Separates DNA fragments based on their melting characteristics. (Option A)
- Effective Range: Analyzes DNA segments of 200-300 base pairs, such as bacterial 16S rRNA genes. (Option B)
- Mechanism and Process: Uses a polyacrylamide gel with a gradient of denaturants (urea or formamide). Migration: DNA fragments migrate through the gel and encounter increasing denaturant concentrations. Denaturation: Sequence and structure influence melting behavior, causing fragments to denature at different points. Separation: Differential denaturation results in fragment separation, visible after electrophoresis.
- Uses a polyacrylamide gel with a gradient of denaturants (urea or formamide).
- Migration: DNA fragments migrate through the gel and encounter increasing denaturant concentrations.
- Denaturation: Sequence and structure influence melting behavior, causing fragments to denature at different points.
- Separation: Differential denaturation results in fragment separation, visible after electrophoresis.
- Applications SNP Detection: Identifies single-nucleotide polymorphisms cost-effectively without sequencing. Molecular Fingerprinting: Analyzes microbial communities in environmental microbiology. Clinical Diagnostics: Identifies pathogenic organisms and studies genetic variations related to diseases.
- SNP Detection: Identifies single-nucleotide polymorphisms cost-effectively without sequencing.
- Molecular Fingerprinting: Analyzes microbial communities in environmental microbiology.
- Clinical Diagnostics: Identifies pathogenic organisms and studies genetic variations related to diseases.
- Uses a polyacrylamide gel with a gradient of denaturants (urea or formamide).

- Migration: DNA fragments migrate through the gel and encounter increasing denaturant concentrations.
- Denaturation: Sequence and structure influence melting behavior, causing fragments to denature at different points.
- Separation: Differential denaturation results in fragment separation, visible after electrophoresis.
- SNP Detection: Identifies single-nucleotide polymorphisms cost-effectively without sequencing.
- Molecular Fingerprinting: Analyzes microbial communities in environmental microbiology.
- Clinical Diagnostics: Identifies pathogenic organisms and studies genetic variations related to diseases.

Sequential Steps in Denaturing Gradient Gel Electrophoresis (DGGE)	
1. DNA Extraction	Isolate DNA from the sample.
2. PCR Amplification	Amplify the target DNA region with specific primers.
3. Gel Preparation	Create a polyacrylamide gel with a denaturant gradient. (Option C)
4. Electrophoresis	Load DNA samples into the gel and apply an electric field for separation.
5. Visualization	Stain the gel and visualize bands under UV light.

Solution for Question 10:

Correct Answer: B) SSCP (Single-Strand Conformational Polymorphism)

Explanation:

- SSCP (Single-Strand Conformational Polymorphism) is the method that directly addresses variations in electrophoretic mobility of single-stranded DNA fragments.
- This technique is specifically designed to detect sequence variations by analyzing the conformational changes of single-stranded DNA.
- When DNA is denatured into single strands and then subjected to electrophoresis, differences in the DNA sequence can cause different folding patterns or conformations.

- These conformational changes affect the migration of the DNA through a non-denaturing gel, making SSCP highly effective for detecting small mutations or variations in DNA sequences.

RFLP (Restriction Fragment Length Polymorphism) (Option A):

- RFLP detects variations in DNA sequences by analyzing the pattern of DNA fragments produced after digestion with restriction enzymes.
- This method identifies differences in DNA sequence by comparing the lengths of the resulting fragments on a gel.
- However, it does not specifically address the electrophoretic mobility of single-stranded DNA.

Sequencing (Option C):

- Directly determines DNA sequence but is not based on electrophoretic mobility changes of single-stranded DNA.

Solution for Question 11:

Magnesium ions (Mg^{2+}) are critical in PCR, acting as essential cofactors for DNA polymerases and insufficient concentration can decrease enzyme activity and amplification efficiency.

Role of Magnesium in PCR	
1. Enzymatic Function	Cofactor for DNA Polymerases: Binding of Nucleotides: Magnesium ions facilitate the correct positioning of nucleotides and stabilize the negative charges that develop during the polymerization process. Stabilization of Charges: This stabilization is crucial for maintaining the proper geometry for the nucleophilic attack of the primer's 3' hydroxyl group on the alpha phosphate of the incoming nucleotide. Formation of Catalytic Complexes: Coordination of Substrates: Mg^{2+} ions are involved in the coordination of substrates within the active site of the enzyme, essential for effective catalysis. Active Site Geometry: Magnesium maintains the correct geometry of the DNA polymerase active site, positioning the 3' hydroxyl group for efficient catalysis. Stabilization of Transition State: Magnesium stabilizes the penta-coordinated transition state during the nucleotidyl transfer reaction, reducing errors during replication.

2. Stability and Fidelity:	DNA Structure Stability: Magnesium helps maintain the integrity of DNA strands during PCR. Error Minimization: Proper magnesium concentration minimizes errors in DNA replication. Optimization: Both insufficient and excessive magnesium can lead to decreased efficiency and accuracy in DNA amplification.
----------------------------	---

Option A: Incorrect. Magnesium ions are essential for enzyme activity, not just DNA template stabilization.

Option B: Incorrect. Excessive magnesium concentration can lead to non-specific amplification and errors.

Option D: Incorrect. Magnesium ions are necessary for the nucleophilic attack of the primer's 3' hydroxyl group on the incoming nucleotide during DNA synthesis.

Advanced Molecular Techniques: Recombinant DNA technology, Microarray, CRISPR, Lac operon

1. Which of the following best describes a mechanism involved in epigenetic changes that lead to heterochromatin formation?

- A. Histone acetylation and DNA methylation of adenine residues
 - B. Histone deacetylation and methylation in CG islands
 - C. DNA replication
 - D. Histone phosphorylation
-

2. Which of the following statements about the lac operon is/are correct? Lac I is constitutively expressed. Lac Y is responsible for lactose transport into the cell. The structural genes are expressed when the CAP site is free. The ZYA genes are expressed when the operator site is free.

- A. 1 and 2
 - B. 1, 2, and 3
 - C. 1, 2, and 4
 - D. 1, 2, 3, and 4
-

3. Which of the following statements about cloning vectors is correct? Plasmids can carry DNA inserts up to 10 kilobases in size. Phage vectors can carry DNA inserts up to approximately 50 kilobases. Cosmids are hybrid vectors that combine features of plasmids and phages. Artificial chromosomes are used to clone very large DNA fragments.

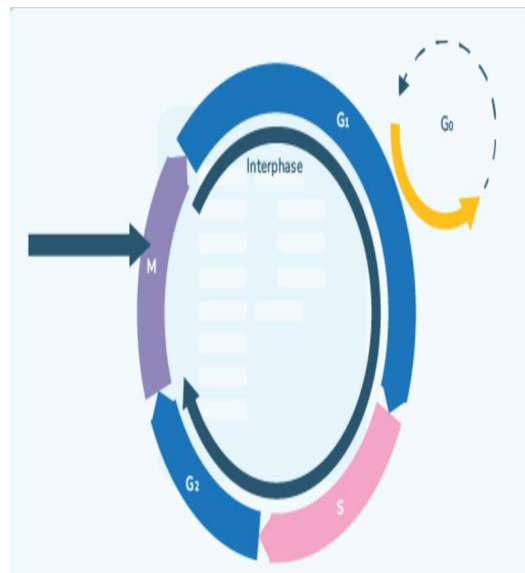
- A. 1, 2, and 3

- B. 1 and 4
 - C. All of the above
 - D. 1, 3, and 4
-

4. In their CRISPR-Cas9 experiment with a mammalian cell line, the research team observes the repair of DNA double-strand breaks. Which repair mechanisms are they most likely observing? Non-Homologous End Joining (NHEJ) Homologous End Joining (HEJ) Nucleotide Excision Repair (NER) Base Excision Repair (BER)

- A. 1 and 2
 - B. 1,2,3
 - C. 1 and 4
 - D. All of the above
-

5. Which of the following is the feature of the 'M' phase in the image of the Cell cycle?



- A. DNA replication occurs to duplicate the genome.
 - B. Chromosomes are highly condensed to facilitate segregation.
 - C. Cell growth and preparation for DNA synthesis take place.
 - D. Proofreading and repair of DNA replication errors are conducted.
-

6. A 32-year-old patient presents with a family history of progressive muscle weakness, cognitive decline, and involuntary movements. Genetic testing reveals a specific type of microsatellite instability disorder. Which of the following trinucleotide repeat sequences is most likely associated?

- A. CAG
 - B. CTG
 - C. CGG
 - D. GAA
-

7. A 5-year-old boy presents with frequent falls, difficulty climbing stairs, and an unusual way of getting up from the floor. Genetic testing reveals a mutation in a gene associated with a severe muscle disorder. Which of the following statements about the involved gene is true?

- A. Located on chromosome 7
 - B. Largest in the human genome.
 - C. Associated with a condition that primarily affects females
 - D. The Gene's product is a component of the sarcomere.
-

8. Satellite DNA has various applications due to its unique repetitive sequences. Which of the following is not an application of satellite DNA?

- A. Forensic identification and criminal investigations
 - B. Determining genetic susceptibility to cardiovascular diseases
 - C. Resolving paternal dispute cases
 - D. Studying evolutionary relationships and genetic diversity
-

9. Which of the following statements is not accurate regarding post-transcriptional modifications of eukaryotic RNA?

- A. The addition of a poly-A tail at the 3' end.
 - B. The 5' capping involves a 5' to 5' triphosphate bridge.
 - C. Methylation of the N6 position of adenine.
 - D. Introns are removed, and exons are spliced within the cytoplasm.
-

10. A researcher aims to clone a large eukaryotic gene fragment (~600 kb) for use in functional genomics studies. Which of the following cloning vectors is the most appropriate for this purpose?

- A. Plasmid
 - B. Bacteriophage λ
 - C. Yeast Artificial Chromosome (YAC)
 - D. Cosmid
-

11. In human somatic cells, which of the following statements accurately describes the chromatin and its characteristics?

- A. Constitutive heterochromatin is always active
- B. Facultative heterochromatin includes the Barr body

- C. Euchromatin is found in centromeric regions
 - D. Heterochromatin is active and less stained
-

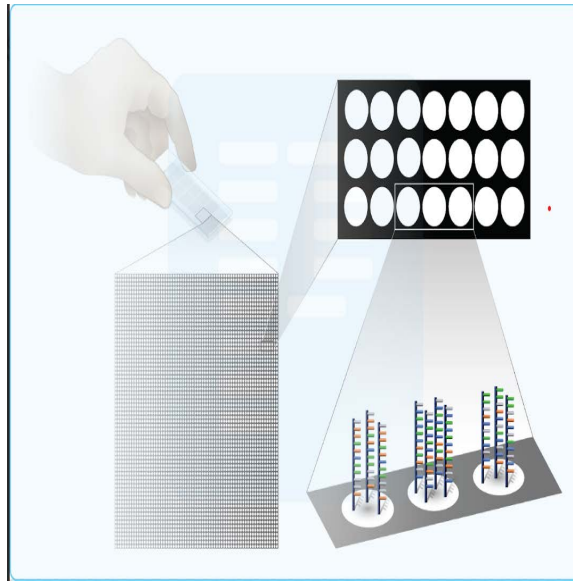
12. In a laboratory setting, E. coli bacteria are cultured in a medium where glucose and lactose is available. To increase the expression of the lac operon, which levels of lactose and cAMP would you expect to be present?

- A. High lactose and low cAMP
 - B. Low lactose and high cAMP
 - C. High lactose and high cAMP
 - D. Low lactose and low cAMP
-

13. Which of the following techniques is specifically designed to introduce specific genetic modifications, such as gene knock-ins or knock-outs, into the genome of an organism?

- A. FISH (Fluorescence In Situ Hybridization)
 - B. CRISPR-Cas
 - C. Recombinant DNA Technology
 - D. Comparative Genomic Hybridization
-

14. Which of the following is true about the significance of technology in the following image? {{caption_text}}



- A. Used to study multiple gene expressions simultaneously
 - B. Not used to deciding the course of treatment
 - C. They have been completely replaced by next-generation sequencing.
 - D. Not used in proteomics and genotyping.
-

15. Which of the following vectors is most suitable for cloning large DNA fragments up to 300 kb in size?

- A. Plasmids
 - B. Bacteriophages
 - C. Bacterial Artificial Chromosomes (BACs)
 - D. Yeast Artificial Chromosomes (YACs)
-

16. Which type of blotting technique is used to detect specific RNA sequences within a sample?

- A. Southern Blotting
 - B. Northern Blotting
 - C. Western Blotting
 - D. Eastern blotting
-

17. Which type of restriction endonuclease creates staggered cuts in the DNA, producing "sticky ends"?

- A. Type I restriction enzymes
 - B. Type II restriction enzymes
 - C. Type III restriction enzymes
 - D. Type IV restriction enzymes
-

18. Which of the following represents the correct sequence of steps in recombinant DNA technology?

- A. Isolation of DNA → Insertion of recombinant DNA into host → Ligation → Cutting of DNA with restriction enzymes → Selection and screening.
 - B. Cutting of DNA with restriction enzymes → Ligation → Isolation of DNA → Insertion of recombinant DNA into host → Selection and screening.
 - C. Isolation of DNA → Cutting of DNA with restriction enzymes → Ligation → Insertion of recombinant DNA into host → Selection and screening.
 - D. Ligation → Isolation of DNA → Cutting of DNA with restriction enzymes → Insertion of recombinant DNA into host → Selection and screening.
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	4
Question 4	1
Question 5	2
Question 6	1
Question 7	2
Question 8	2
Question 9	4
Question 10	3
Question 11	2
Question 12	3
Question 13	2
Question 14	1
Question 15	3
Question 16	2
Question 17	2
Question 18	3

Solution for Question 1:

Correct Answer: B) Histone deacetylation and methylation in CG islands

Explanation: Epigenetic changes that lead to heterochromatin formation are primarily driven by histone deacetylation and methylation of cytidine residues in CG islands.

Epigenetic changes involve stable alterations in gene expression that do not involve changes in the nucleotide sequence.

Epigenetic Modifications:

- DNA Methylation: Represses gene expression by methylation of CpG islands. Involved in ageing, cancer, genomic imprinting, X chromosome inactivation, and dysregulation linked to diseases like Fragile X syndrome.
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- Involved in ageing, cancer, genomic imprinting, X chromosome inactivation, and dysregulation linked to diseases like Fragile X syndrome.
- Histone Methylation: Suppress or activate transcription depending on the location of methyl groups. Methylation of lysine and arginine residues on histones.
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- Methylation of lysine and arginine residues on histones.
- Histone Acetylation: Activates transcription (Option A) The addition of acetyl groups neutralises histone charges, relaxing DNA coiling. Generally increases gene expression; affects processes like thyroid hormone synthesis.
- Activates transcription (Option A)
- The addition of acetyl groups neutralises histone charges, relaxing DNA coiling.
- Generally increases gene expression; affects processes like thyroid hormone synthesis.
- Histone Deacetylation: Represses transcription. Removal of acetyl groups tightens DNA coiling. Typically, it decreases gene expression and is linked to diseases such as Huntington's disease.
- Represses transcription.
- Removal of acetyl groups tightens DNA coiling.
- Typically, it decreases gene expression and is linked to diseases such as Huntington's disease.
- Represses gene expression by methylation of CpG islands.

- Involved in ageing, cancer, genomic imprinting, X chromosome inactivation, and dysregulation linked to diseases like Fragile X syndrome.
- Suppress or activate transcription depending on the location of methyl groups.
- Methylation of lysine and arginine residues on histones.
- Activates transcription (Option A)
- The addition of acetyl groups neutralises histone charges, relaxing DNA coiling.
- Generally increases gene expression; affects processes like thyroid hormone synthesis.
- Represses transcription.
- Removal of acetyl groups tightens DNA coiling.
- Typically, it decreases gene expression and is linked to diseases such as Huntington's disease.

DNA replication and Histone phosphorylation(Option C & D): Not associated with Epigenetic modification.

Solution for Question 2:

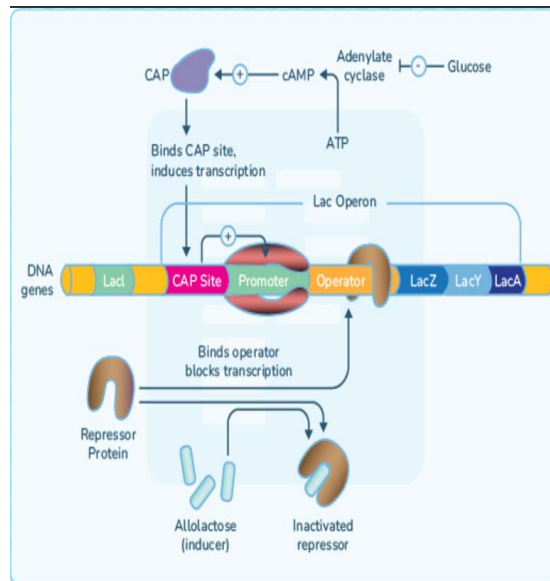
Correct Answer: C) 1, 2, and 4

Explanation:

The structural genes are expressed when the CAP site is Occupied.

Lac Operon:

- The lac operon enables E. coli to metabolise lactose by producing enzymes necessary for lactose uptake and breakdown in the absence of glucose.



Components:

Structural Genes:

- lacZ: Encodes β -galactosidase (Breaks down lactose into glucose and galactose)
- lacY: Encodes lactose permease (Helps lactose enter the cell) (Statement 2)
- lacA: Encodes thiogalactoside transacetylase

Regulatory Elements :

- lacI Gene: Encodes the LacI repressor, which binds to the operator to block transcription when lactose is absent. (Lactose unbinds repressor from Operator site) Constitutively Expressed (Statement 1)
- Constitutively Expressed (Statement 1)
- Promoter (P) : The binding site for RNA polymerase, which initiates transcription of the lac genes.
- Operator (O) : A DNA sequence that the lac repressor protein binds to, blocking transcription when lactose is absent.
- CAP (Catabolite Activator Protein): Bind to the CAP binding site near the promoter and enhance transcription (Statement 3). This requires requires cAMP
- Constitutively Expressed (Statement 1)

(Glucose inhibits Adenylate cyclase which converts ATP to cAMP)

Mechanism: Lactose unbinds repressor from Operator site → Operator site becomes free → Increases Transcription (Statement 4)

Solution for Question 3:

Correct Answer: D) 1, 3, and 4

Explanation:

Phage vectors can carry DNA inserts up to approximately 20 kilobases, not 50 kilobases.

Cloning Vectors are DNA molecules that carry and replicate foreign DNA segments within host cells. They are useful in molecular biology for gene cloning and manipulation.

Vector	Size (Kb)	Features
Plasmids	<10	Small, circular DNA molecules found in Bacteria Commonly used for cloning and expressing small DNA fragments Replicate independently of the bacterial chromosome.
Bacteriophages	10-20	Bacterial viruses with linear DNA genomes Useful for cloning slightly larger DNA fragments Infect bacterial cells to propagate the cloned DNA
Bacterial Artificial Chromosomes (BACs)	50-250	BACs are vectors based on the F-plasmid of E. coli. They can accommodate large DNA inserts Used for constructing genomic libraries
Yeast Artificial Chromosomes (YACs)	>500	YACs are vectors that can accommodate very large DNA inserts, even larger than BACs More prone to chimerism and instability compared to BACs
Cosmids	35 - 50	Hybrid vectors combining plasmid and phage features
Viral Vectors		Derived from viruses that infect mammalian cells Include adenoviral, adenovirus-associated viral (AAV), and retroviral vectors.

Solution for Question 4:

Correct Answer: A) 1 and 2

Explanation:

NHEJ and HEJ are the two mechanisms employed by the CRISPR-Cas9 System. It is a powerful tool for inducing precise double-stranded breaks in DNA.

CRISPR-Cas9 System

- Components: Guide RNA (gRNA): Directs the Cas9 enzyme to a specific DNA sequence. Cas9 Enzyme: Induces double-stranded breaks (DSBs) at the target site.
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- Guide RNA (gRNA): Directs the Cas9 enzyme to a specific DNA sequence.
- Cas9 Enzyme: Induces double-stranded breaks (DSBs) at the target site.
- Repair Mechanisms: Non-homologous end Joining (NHEJ): This technique directly ligates broken DNA ends without a template, often leading to small insertions or deletions. It can also cause gene knockout by disrupting the coding sequence. Homologous End Joining (HEJ): Uses a homologous template for precise repair of Double strand breaks and enables accurate gene knock-in or correction of mutations.
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- Homologous End Joining (HEJ): Uses a homologous template for precise repair of Double strand breaks and enables accurate gene knock-in or correction of mutations.

Other DNA repair mechanism are:

Base Excision Repair: BER is used to repair small, non-helix-distorting base lesions, not double-stranded breaks. It is not related to CRISPR-Cas9 break repair.

Nucleotide Excision Repair: NER repairs bulky DNA lesions (like those caused by UV light), not double-stranded breaks. It is not involved in the repair of CRISPR-Cas9-induced breaks.

Solution for Question 5:

Correct Answer: B) Chromosomes are highly condensed to facilitate segregation.

Explanation:

This is a key feature of the M phase, where chromosomes become highly condensed to ensure proper segregation during mitosis. They are compressed to about 8000 fold during metaphase in M-Phase.

Cell Cycle:

The cell cycle is a series of stages that a cell goes through to divide and produce two daughter cells.

Phases of the Cell Cycle:

Phases	Chromosome State	Features
Gap 1 (G1) Phase Pre-synthetic phase	Uncondensed	Cell growth and synthesis of proteins and organelles. (Option C) DNA proofreading and repair.
Synthesis (S) Phase DNA replication occurs (Option A)	Uncondensed	Replication of the entire genome and formation of sister chromatids. Increase in the levels of DNA polymerases and other replication enzymes.
Gap 2 (G2) Phase Post-synthetic or pre-mitotic phase	Uncondensed	Final proofreading and repair. (Option D)
Mitosis (M) Phase Division of the nucleus and cytoplasm to produce two daughter cells.	Condensed (ensure proper segregation).	Chromosome condensation, alignment, and segregation. Cytokinesis, where the cytoplasm divides, resulting in two separate cells.

Interphase		The combined phases of G1, S, and G2 are where the cell grows, duplicates its DNA and prepares for division.
Gap 0 (G0) Phase (Resting phase)	Condensed	Permanent or Temporary phase based on Cell type.

Solution for Question 6:

Correct Answer: A) CAG

Explanation:

The clinical presentation is consistent with Huntington's Disease, which is characterised by progressive muscle weakness, cognitive decline, and involuntary movements (chorea). The specific trinucleotide repeat expansion associated with this condition is CAG.

Microsatellite instability (MSI) Disorders:

- Genetic conditions are characterised by the accumulation of mutations in microsatellite regions of DNA due to failure in the DNA mismatch repair system.
- Anticipation is seen (Disease severity increases in successive generations)

Trinucleotide Repeat Sequences and Associated Disorders:

Trinucleotide repeat sequences are short DNA sequences in which a specific set of three nucleotides is repeated multiple times in tandem beyond a normal range, resulting in instability and dysfunction.

Disease	Trinucleotide Repeats	Mode of Inheritance
Huntington's Disease	CAG	AD
Myotonic Dystrophy	CTG (Option B)	AD
Fragile X Syndrome	CGG (Option C)	XLD
Friedreich's Ataxia	GAA (Option D)	AR

Note: AD: Autosomal Dominant; AR: Autosomal Recessive; XLD: X-Linked Dominant

Solution for Question 7:

Correct Answer: B) Largest in the human genome

Explanation:

- The clinical presentation with frequent falls, difficulty climbing stairs and unusual way of getting up (Gower sign) points towards Duchenne muscular Dystrophy.
- Inherited in X-linked Recessive pattern - Primarily affects males (Option C).
- The gene affected is Dystrophin, which is the largest gene in the human genome, located on the X chromosome (Option A).
- Dystrophin is a crucial protein found in muscle cells that helps maintain the muscle fibre's structural integrity. It acts as a bridge between sarcolemma and the internal cytoskeleton, which helps in stabilization.

Option D: Dystrophin is not a component of the sarcomere but is crucial for anchoring the cytoskeleton to the muscle cell membrane. Sarcomere components involved in contraction include actin and myosin.

Solution for Question 8:

Correct Answer: B) Determining genetic susceptibility to cardiovascular diseases

Explanation:

Satellite DNA is not used to identify susceptibility to cardiovascular diseases, as it mainly provides information about repetitive DNA sequences rather than disease susceptibility.

Satellite DNA refers to repetitive DNA sequences that are found in specific regions of the genome, notably in centromeric and telomeric regions.

- Types: Highly Repetitive Sequences (HRS) : These sequences repeat >1,000,000 times.
- Simple Sequence Repeats (SSR) : Also known as microsatellites, these are short, repetitive sequences.
- Minisatellite Repeats : Known as Variable Number Tandem Repeats (VNTR),

these are longer repetitive sequences compared to microsatellites.

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- Minisatellite Repeats : Known as Variable Number Tandem Repeats (VNTR), these are longer repetitive sequences compared to microsatellites.
- Applications: Forensic Identification : Satellite DNA is used in forensic science to identify individuals due to its high variability between people uniquely. Paternal Dispute Resolution : It helps resolve paternal disputes by matching genetic profiles through repetitive DNA sequences. Genetic Studies : To study evolutionary relationships and genetic diversity among different species.
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Solution for Question 9:

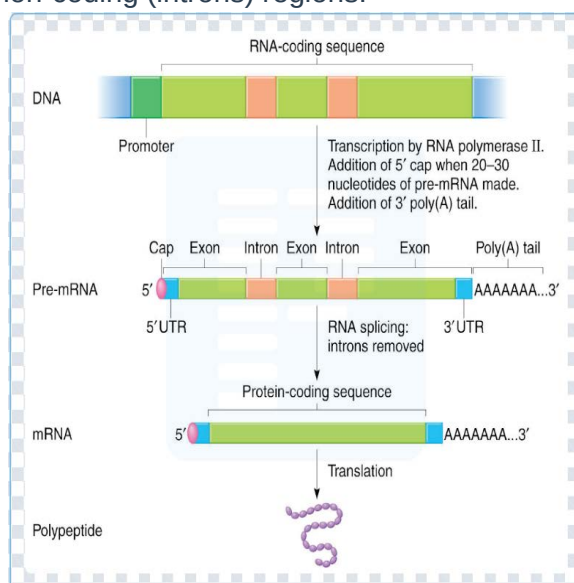
Correct Answer: D) Introns are removed, and exons are spliced within the cytoplasm.

Explanation:

Introns are removed, and exons are spliced together in the nucleus, not the cytoplasm.

Post-Transcriptional Modification of Eukaryotic RNA:

- Primary Transcript (hnRNA): Newly synthesised mRNA directly from DNA; includes both coding (exons) and non-coding (introns) regions.



- Key Modifications:

Poly-A Tailing (3' End)	Addition of a poly-A tail (20-250 nucleotides) in the nucleoplasm. Protects mRNA from degradation.
5' Capping	Addition of a 7-methyl guanosine cap at the 5' end. Involves a 5' to 5' triphosphate bridge. Enhances mRNA stability and translation initiation.
Splicing	Introns (non-coding regions) are removed, and exons (coding regions) are joined together. This splicing occurs in the nucleus and requires energy.
Methylation	Methylation of adenine residues and ribose 2'-hydroxyl groups. Primarily occurs in the cytoplasm for tRNA and rRNA.

Bacterial mRNA:

- Not subject to extensive post-transcriptional modifications.
- Translation can begin before transcription is complete.

Solution for Question 10:

Correct Answer: C) Yeast Artificial Chromosome (YAC)

Explanation:

Yeast Artificial Chromosomes (YACs) are ideal for cloning large DNA fragments (100–1000 kb), making them suitable for large eukaryotic gene fragments (~600 kb)

Cloning Type	Key Features	Insert Size	Applications
Plasmid-Based Cloning (Option A)	Small, circular DNA in bacteria, independent replication, antibiotic resistance genes	~10 kb	Gene cloning, bacterial expression
Phage-Based Cloning (Option B)	Uses bacteriophage (e.g., λ phage), foreign DNA inserted into phage genome	~20 kb	High-efficiency cloning, phage display
Cosmid-Based Cloning (Option D)	Hybrid of plasmid & phage λ , contains cos sites for efficient packaging	~100-300 kb	Genome sequencing, gene mapping
Yeast Artificial Chromosome (YAC)	Functions as a eukaryotic chromosome, contains centromere & telomere	~100-1000 kb	Cloning large eukaryotic DNA fragments
Mammalian Viral Vectors	Used in gene therapy, adenoviral (DNA-based), retroviral (RNA-based)	Varies	Gene expression in mammalian cells, stable integration

Solution for Question 11:

Correct Answer: B) Facultative heterochromatin includes the Barr body

Explanation:

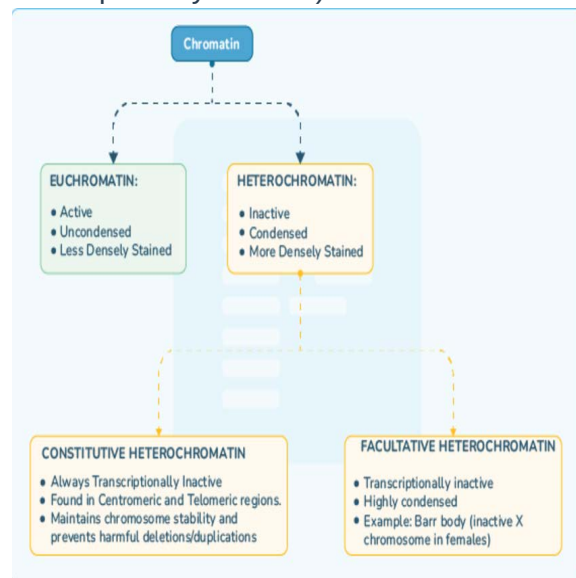
Facultative heterochromatin includes the Barr body.

In somatic cells of females, one of the two X-chromosomes is transcriptionally active while the other is transcriptionally inactive and highly condensed, known as a Barr body.

Chromatin is a complex of DNA and proteins found in the nucleus of eukaryotic cells. It plays a crucial role in packaging DNA into a more compact, dense structure, allowing it to fit within the cell nucleus and regulating gene expression.

Types of Chromatin:

- Euchromatin (Transcriptionally Active)
- Heterochromatin (Transcriptionally Inactive)



Solution for Question 12:

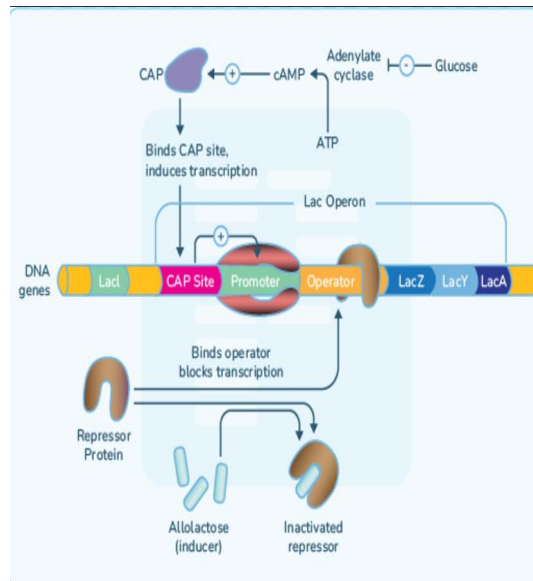
Correct Answer: C) High lactose and high cAMP

Explanation:

In a laboratory setting aiming to maximise the expression of the lac operon, optimal transcription would be expected under conditions with high lactose and high cAMP.

Lac Operon:

- The lac operon enables *E. coli* to metabolise lactose by producing enzymes necessary for lactose uptake and breakdown in the absence of glucose.

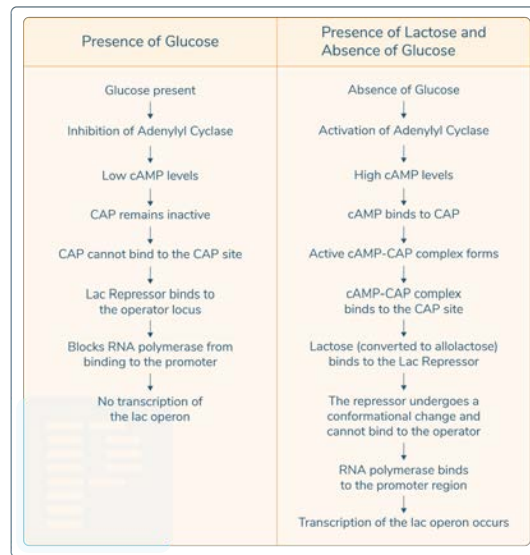


Basic components:

- lacZ: Encodes β -galactosidase, which breaks down lactose into glucose and galactose.
- lacY: Encodes lactose permease, which helps lactose enter the cell.
- lacA: Encodes thiogalactoside transacetylase.
- lacI: Encodes the LacI repressor, which binds to the operator to block transcription when lactose is not present. (Lactose unbinds repressor from Operator site)
- CAP (Catabolite Activator Protein): Requires cAMP to bind to the CAP binding site (CRE) near the promoter and enhance transcription.

(Glucose inhibits Adenylate cyclase, which converts ATP - - - > cAMP)

Regulation:



Solution for Question 13:

Correct Answer: B) CRISPR-Cas

Explanation:

CRISPR-Cas is a genome-editing tool that can be used to introduce specific genetic modifications, including gene knock-ins (adding new genetic material) and gene knock-outs (disrupting or deleting existing genes).

CRISPR-Cas Mechanism:

- Guide RNA : Directs the Cas enzyme to the specific DNA sequence.
- Cas Enzyme : Cuts the DNA at the target location.
- DNA Repair : NHEJ (Non-Homologous End Joining) : Repairs the break by directly ligating the DNA ends, often leading to gene knockout. HDR (Homologous Directed Repair) : Uses a template to repair the break accurately, allowing for gene knock-in.
- NHEJ (Non-Homologous End Joining) : Repairs the break by directly ligating the DNA ends, often leading to gene knockout.
- HDR (Homologous Directed Repair) : Uses a template to repair the break accurately, allowing for gene knock-in.

- NHEJ (Non-Homologous End Joining) : Repairs the break by directly ligating the DNA ends, often leading to gene knockout.
- HDR (Homologous Directed Repair) : Uses a template to repair the break accurately, allowing for gene knock-in.

FISH (Fluorescence In Situ Hybridization) is used for detecting and visualizing specific nucleic acid sequences in cells but does not modify genes (Option A).

Recombinant DNA Technology involves creating new DNA combinations but is not as precise as CRISPR-Cas for making specific gene knock-ins or knock-outs (Option C).

Comparative Genomic Hybridization (CGH) is used to detect chromosomal abnormalities and does not involve gene editing (Option D).

Solution for Question 14:

Correct Answer: A) Used to study multiple gene expressions simultaneously

Explanation;

The given image represents Microarray Technique which can analyze thousands of genes at once facilitating multiple gene expression studies.

- Microarray consist of thousands of microscopic DNA spots (probes) on a slide, used for hybridizing cDNA or cRNA samples.
- Each spot on the microarray chip represents a specific gene, and by measuring the intensity of the signal at each spot, researchers can determine how much of each gene is expressed in the sample.
- Microarrays are crucial for gene expression profiling and can assist in making treatment decisions by helping us to understand the pathogenesis of a disorder (Option B)
- Types: DNA Microarray cDNA Microarray Protein Microarray Array comparative genomic hybridisation (CGH)
- DNA Microarray
- cDNA Microarray
- Protein Microarray
- Array comparative genomic hybridisation (CGH)
- DNA Microarray

- cDNA Microarray
- Protein Microarray
- Array comparative genomic hybridisation (CGH)

Applications of Microarray Technique:

- Genotyping
- Gene sequencing
- Gene Expression Profiling: Used for cancer treatment decisions (e.g., identifying need for chemotherapy).
- Protein Detection by monoclonal antibody or aptamer microarrays, useful in personalised medicine and Proteomics (Option D)
- Clinical Uses: Diagnosing cancer, infectious diseases, and allergies.

Microarrays are still being used alongside next-generation sequencing (Option C).

Solution for Question 15:

Correct Answer: C) Bacterial Artificial Chromosomes (BACs)

Explanation:

Vector	Size	Description
Plasmids (Option A)	Up to 10 kb	Plasmids are small, circular DNA molecules commonly used in recombinant DNA technology. They are easy to manipulate and introduce into bacterial cells. However, they typically can only accommodate small to medium-sized DNA inserts. They are not suitable for very large DNA fragments.
Bacteriophages (Option B)	10-20 kb	Bacteriophages, particularly lambda phage vectors, are used for cloning DNA fragments. While larger than plasmids, they are still not suitable for very large DNA fragments of 300 kb.

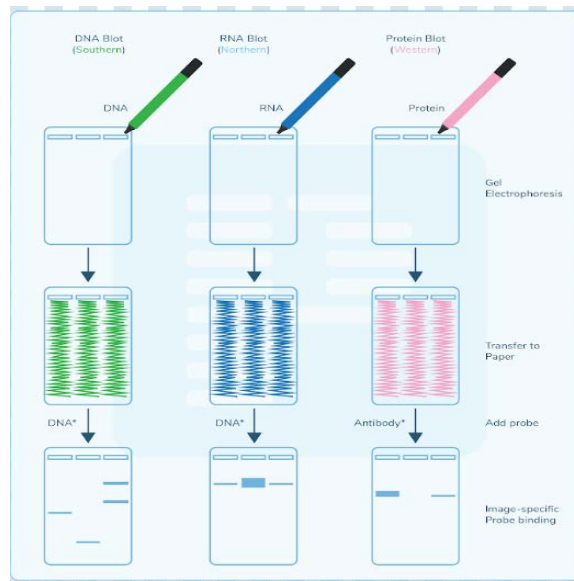
Bacterial Artificial Chromosomes (BACs) (Option C)	50-250 kb	BACs are vectors based on the F-plasmid of E. coli. They can accommodate large DNA inserts and are widely used for constructing genomic libraries.
Yeast Artificial Chromosomes (YACs) (Option D)	>500	YACs are vectors that can accommodate very large DNA inserts, even larger than BACs. While they can certainly handle 300 kb inserts, they are more prone to chimerism and instability compared to BACs.

Solution for Question 16:

Correct Answer: B) Northern Blotting

Explanation:

Types	Target	Features
Southern blotting (Option A)	DNA sequences	It involves separating DNA fragments by gel electrophoresis, transferring them to a membrane, and then using labelled probes to identify specific sequences
Northern blotting (Option B)	RNA sequences	The process is similar to DNA blotting but focuses on RNA.
Western blotting (Option C)	Proteins	This technique specifically detects and analyses proteins in a complex mixture. It involves separating proteins by gel electrophoresis, transferring them to a membrane, and then using specific antibodies to detect the proteins.
Eastern blotting (Option D)	Post-translational modifications	It refers to specialised Western blotting techniques that detect specific post-translational modifications of proteins, such as phosphorylation or glycosylation.



Solution for Question 17:

Correct Answer: B) Type II restriction enzymes

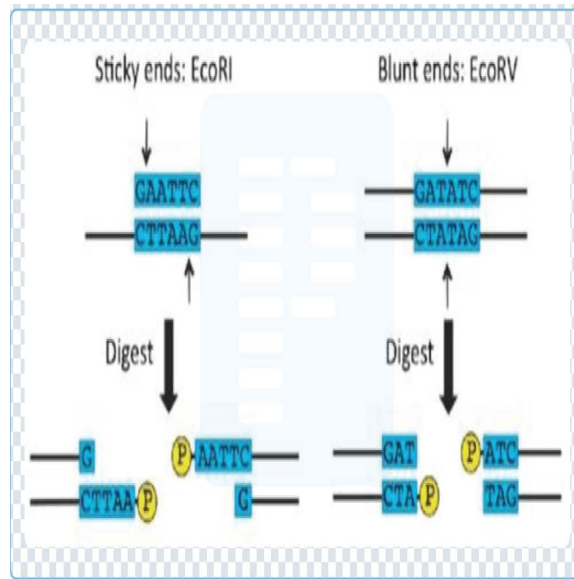
Explanation:

Restriction endonucleases are enzymes that cut DNA at specific recognition sequences.

Characteristic	Type I	Type II
Subunit structure	Complex (3 subunits)	Simple (1-2 subunits)
ATP requirement	Yes	No
Cleavage site	Random, far from recognition site	At or near the recognition site
Recognition sequence	Asymmetric	Usually symmetric

Type II Restriction Enzyme Subtypes :

Sticky End	Blunt End
Cuts both the DNA strands at two different sites.	Cuts both the strands at the same site.
Forms two complexes with overhanging ends.	Forms two complexes with blunt ends.



Type III restriction enzymes(Option C):

Recognise specific DNA sequences but cut at a short distance from the recognition site. They do not produce sticky ends and also require ATP.

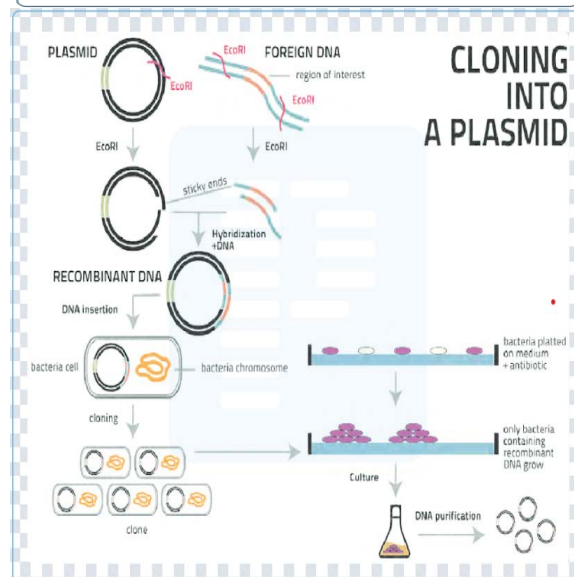
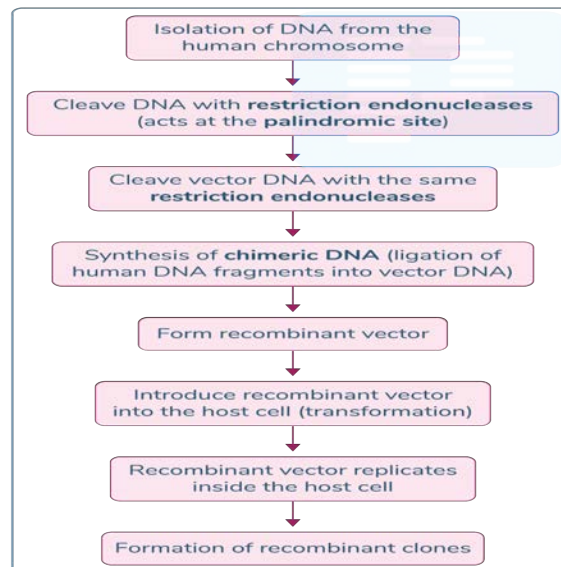
Type IV restriction enzymes(Option D):

Target modified DNA, such as methylated DNA, rather than specific sequences. They do not produce sticky ends and are involved in the recognition of modified bases rather than sequence-specific cleavage.

Solution for Question 18:

Correct Answer: C) Isolation of DNA → Cutting of DNA with restriction enzymes → Ligation → Insertion of recombinant DNA into host → Selection and screening

Steps of recombinant DNA Technology



Previous Year Questions

1. A child presents with micrognathia and low-set ears. These clinical features are commonly associated with which type of genetic abnormality?

- A. Deletion
 - B. Translocation
 - C. Amplification
 - D. Inversion
-

2. What is true about the Hayflick limit?

- A. Human cells can divide a finite number of times only
 - B. Free radical oxidation level is unrestricted
 - C. Telomerase activity is unrestricted
 - D. RNA polymerase can function for a finite number of time only
-

3. Mental retardation is a clinical feature of:

- A. Alkaptonuria
 - B. Albinism
 - C. Hawkinsuria
 - D. Phenylketonuria
-

4. Which of the following is not associated with post-transcriptional modification?

- A. Methylation
 - B. Endonuclease cleavage
 - C. 5' capping
 - D. Glycosylation
-

5. Which type of mutation results in the reversal to the wild type of phenotype when the mutant gene is suppressed?

- A. Frameshift of the mutated gene
 - B. Addition of another normal gene
 - C. Deletion of mutated gene
 - D. Mutation in tRNA
-

6. FISH technique helps to identify all except??

- A. Point mutation
 - B. Translocation
 - C. Deletion
 - D. Gene amplification
-

7. Which of the following uses CRISPR-Cas9 technology?

- A. DNA Editing
- B. Post-transcriptional Modification
- C. DNA Induction
- D. DNA Editing Solution

8. Which of the following post-translational modifications is common to collagen, hemoglobin, and immunoglobulins?

- A. Subunit aggregation
 - B. Splicing
 - C. Poly A tail addition
 - D. Hydroxylation
-

9. A 7-year-old child of short stature presented to the outpatient department with skin patches and learning difficulties. Diagnostic workup indicated a defect in DNA repair specifically related to double-strand breaks. What is the most likely diagnosis?

- A. Bloom Syndrome
 - B. Xeroderma Pigmentosum
 - C. Cockayne Syndrome
 - D. Lynch Syndrome
-

10. A couple brings their 4-year-old child to the outpatient department (OPD) due to self-mutilating behavior and poor growth. Which of the following conditions and associated enzyme deficiencies could explain these symptoms?

- A. HGPRT deficiency
- B. Homogentisic Aciduria
- C. Phenylalanine Hydroxylase deficiency
- D. Benzoyl Acidase deficiency

11. A mother brings her 14-year-old mentally retarded son to the clinic, complaining of joint pain and an irresistible urge for the boy to bite his fingers and lips. Serum uric acid is elevated. What is the most likely diagnosis?

- A. Lesch-Nyhan syndrome
 - B. Gout
 - C. Hyperuricemia
 - D. Juvenile arthritis
-

12. A 13-year-old with acute flaccid paralysis, the pediatric resident orders an RT-PCR on a CSF sample to identify the causative virus. Which of the following described the enzyme correctly?

- A. DNA-dependent RNA polymerase
 - B. RNA-dependent DNA polymerase
 - C. DNA-dependent DNA polymerase
 - D. RNA-dependent RNA polymerase
-

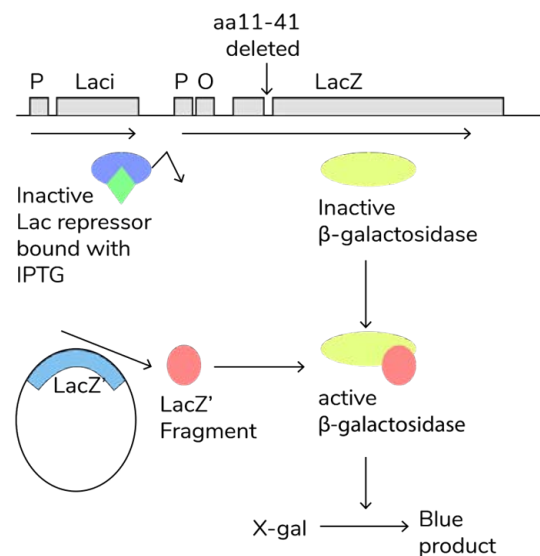
13. Which type of DNA repair mechanism involves the removal and replacement of a damaged DNA strand with the help of an undamaged complementary strand as a template?

- A. Base Excision Repair
- B. Nucleotide Excision Repair
- C. Mismatch Repair
- D. Homologous Recombination

14. The attachment of Eukaryotic mRNA to the ribosome is mediated through

- A. Poly A tail
 - B. tRNA
 - C. Guanyl Cap
 - D. Shine-Dalgarno Sequence
-

15. Identify the phenomenon shown below.



- A. Translation
 - B. Transformation
 - C. Complementation
 - D. Conjugation
-

16. What is the mechanism through which gene repair occurs when CRISPR-Cas9 is utilized?

- A. Base excision
 - B. Non-Homologous End joining
 - C. Homologous End joining
 - D. Mismatch repair
-

17. What is chimeric DNA used for?

- A. Paternity test
 - B. Maternity test
 - C. Personal identification
 - D. Organ transplantation
-

18. Steps of PCR in sequence are?

- A. Extend DNA, anneal primers, Denature DNA
 - B. Anneal primers, extend DNA, Denature DNA
 - C. Denature DNA, anneal primers, Extends DNA
 - D. Denature DNA, Extend DNA, Anneal primers
-

19. Which of the following is a start codon?

- A. UAA

- B. UAG
 - C. UGA
 - D. AUG
-

20. Which of the following methods is frequently utilized for the identification of RNA using molecular analysis?

- A. G banding
 - B. Sanger's method
 - C. RT-PCR
 - D. Western blot
-

21. What is the most accurate test for evaluating the effectiveness of treatment in chronic myeloid leukemia (CML)?

- A. CBC
 - B. Cytogenetic test
 - C. RT PCR
 - D. FISH
-

22. Klenow fragment lacks the activity of

- A. 3'→5' Exonuclease
- B. 5'→3' Exonuclease
- C. 5'→3' DNA polymerase
- D. 3'→5' DNA polymerase

23. Which of the subsequent methods relies on RNA?

- A. RT PCR
 - B. Sanger's technique
 - C. Next generation sequencing
 - D. Western blot
-

24. Which of the following does not favor euchromatin formation due to changes occurring at cytosine residues at CpG islands in DNA?

- A. Methylation
 - B. Phosphorylation
 - C. Alkylation
 - D. Sumoylation
-

25. Which of the following processes does not occur in the direction from 5' to 3'?

- A. Transcription
 - B. DNA replication
 - C. DNA repair
 - D. RNA editing
-

26. What is the specific type of genetic mutation observed in sickle cell anemia?

- A. Insertion
 - B. Deletion
 - C. Point mutation
 - D. Frameshift mutations
-

27. On which of the following will restriction endonucleases act?

- A. AAGAAG
 - B. GAGAGG
 - C. AAGCTT
 - D. TACGAG
-

28. What is the repair mechanism associated with CRISPR-cas9?

- A. Non-homologous end joining
 - B. Base excision repair
 - C. Nucleotide excision repair
 - D. Mismatch repair
-

29. Assertion: According to central dogma, information flows from DNA to RNA, which is then encoded into protein Reason: In retrovirus, reversal of central dogma happens

- A. Both assertion and reason are true and the reason is the correct explanation for the assertion
- B. Both assertion and reason are true, but the reason is correct not the correct explanation for the assertion

- C. Assertion is true, but the reason is false
 - D. Both assertion and reason are false
 - E. The assertion is false but reason is true
-

30. Assertion: According to central dogma, information flows from DNA to RNA, which is then encoded into protein Reason: In retrovirus, reversal of central dogma happens Which is true regarding this scenario?

- A. Both assertion and reason are true and the reason is the correct explanation for the assertion
 - B. Both assertion and reason are true, but the reason is correct not the correct explanation for the assertion
 - C. Assertion is true, but the reason is false
 - D. Both assertion and reason are false
-

31. If the RNA sequence is CAAGGU. What is the sequence of the template strand?

- A. CAAGGT
 - B. GTTCCA
 - C. ACCTTG
 - D. TGGAAC
-

32. Hybridoma technology uses which of the following?

- A. HMP shunt
- B. TCA cycle
- C. Purine Salvage pathway

D. Glycolysis

33. In which of the following, genetic material adenine is not equal to thymine?

- A. Mitochondrial DNA of humans
 - B. E. coli
 - C. HIV
 - D. Herpes
-

34. Which of the following does not contribute to the repair of DNA double-strand breaks?

- A. ATP
 - B. RAD 51
 - C. ATM
 - D. Ku 70
-

35. After 3 days of dengue infection, diagnosis can be made by all except?

- A. ELISA for antigen
 - B. Viral load
 - C. ELISA for antibody
 - D. RT PCR
-

36. What happens to chloride levels in sweat and pancreatic secretions in Cystic fibrosis?

- A. Chloride is raised in sweat
 - B. Chloride is decreased in sweat
 - C. Chloride concentration increases in pancreatic secretion
 - D. Chloride concentration normal in pancreatic secretion
-

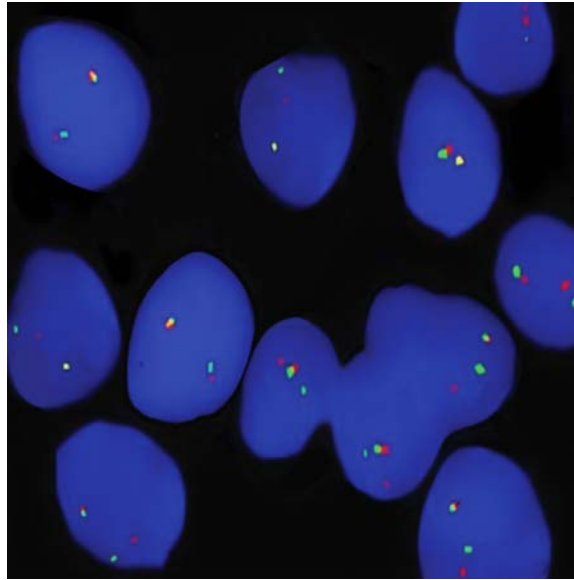
37. Reverse transcriptase PCR analyzes:

- A. DNA
 - B. RNA
 - C. Amino acid
 - D. Protein particle
-

38. What determines the DNA sequence?

- A. Flow cytometry
 - B. FISH
 - C. RFLP
 - D. PCR
-

39. A patient was diagnosed with Ewing sarcoma with ESR1 gene mutation. The technique shown in the image was performed to confirm the diagnosis. Identify the technique shown in the image.



- A. Fluorescent in situ hybridization
 - B. Karyotyping
 - C. Immunohistochemistry
 - D. Sanger's technique
-

40. What is the repair mechanism associated with CRISPR-cas9?

- A. Non-homologous end joining
 - B. Base excision repair
 - C. Nucleotide excision repair
 - D. Mismatch repair
-

41. Wobble hypothesis explains which of the following concepts of the genetic code?

- A. Universal

- B. Unambiguous
 - C. Degeneracy
 - D. Non-punctuated
-

42. Which of the following molecular analysis methods is used to detect RNA?

- A. Sanger's method
 - B. G banding
 - C. Western blot
 - D. RT-PCR
-

43. Which of the following enzymes is deficient in the autosomal recessive type of severe combined immunodeficiency?

- A. Ornithine transcarbamylase
 - B. Phosphomannose isomerase
 - C. Adenosine deaminase
 - D. Hypoxanthine-guanine phosphoribosyl transferase
-

44. If the percentage of thymine residues in DNA is 28%. What is the % of cytosine?

- A. 28%
 - B. 44%
 - C. 36%
 - D. 22%
-

45. All of the following are found in DNA structures except

- A. Guanine
 - B. Adenine
 - C. Cytosine
 - D. Uracil
-

46. A patient who has been diagnosed with multiple gastrointestinal polyps and carcinoma, with a positive family history/history of hereditary non-polyposis colon carcinoma has a defect in which of the following?

- A. Mismatch repair
 - B. Nucleotide excision repair
 - C. Base excision repair
 - D. Point mutation
-

47. Misincorporation of uracil into DNA is seen in deficiency of:

- A. Vitamin B12
 - B. Folate
 - C. Vitamin B6
 - D. Vitamin E
-

48. What is the method employed for banding dicentric chromosomes?

- A. G banding

- B. C banding
 - C. NOR
 - D. R banding
-

49. How does RNAi impact gene expression?

- A. Knock out
 - B. Knockdown
 - C. Knock up
 - D. Knock in
-

50. Which of the following techniques is used to detect DNA-protein interaction?

- A. DNA footprinting
 - B. DNA fingerprinting
 - C. ELISA
 - D. Northern blotting
-

51. Which of the following options best describes the mutation seen in patients with cystic fibrosis?

- A. Tryptophan at 708 position, chloride channel
 - B. Phenylalanine at 708 position, calcium channel
 - C. Phenylalanine at 508 position, chloride channel
 - D. Tryptophan at 508 position, calcium channel
-

52. The best method for the detection of mutations with low allele frequency is:

- A. Droplet digital PCR
 - B. FISH
 - C. Nested PCR
 - D. Sanger sequencing
-

53. Affinity maturation of antibodies is because of ____.

- A. Somatic hypermutation
 - B. Differential mRNA processing
 - C. CD40
 - D. Gene rearrangements
-

54. What is the accurate order of steps involved in extracting a desired protein using recombinant DNA technology? Expression of protein and lysis of the bacterial cell Incorporation of genes into bacteria SDS PAGE Protein elution Hybridization

- A. 2, 1, 3, 5, 4
 - B. 1, 2, 4, 3, 5
 - C. 1, 5, 2, 4, 3
 - D. 2, 4, 5, 3, 1
-

55. Which of the following is true about p53?

- A. It increases telomere length in the liver.
 - B. It synthesizes RNA
 - C. It stabilizes RNA
 - D. It retains the telomerase enzyme in the nucleus
-

56. Which of the following mechanisms can explain the complete translation of the Apo-B gene to ApoB-100 in the liver, and its translation to ApoB-48 in the intestine?

- A. RNA editing
 - B. Gene splicing
 - C. Alternative polyadenylation
 - D. Gene rearrangements
-

57. What is the term for a single mutation in a nucleotide base pair that leads to the formation of a termination codon?

- A. Missense mutation
 - B. Nonsense mutation
 - C. Stop mutation
 - D. Silent mutation
-

58. DNA methylation is not related to?

- A. Mismatch repair
- B. Chromatin remodeling
- C. DNA replication
- D. Gene silencing

59. Which of the subsequent disorders exhibits an autosomal recessive inheritance pattern?

- A. Huntington's disease
 - B. Treacher collins syndrome
 - C. Cystic fibrosis
 - D. Achondroplasia
-

60. In the context of wound healing, which of the subsequent types of collagens are implicated?

- A. Type II
 - B. Type III
 - C. Type IV
 - D. Type V
-

61. Which of the following mechanisms can modify gene expression through methylation and acetylation without altering the gene sequence?

- A. Mutation
 - B. Epigenetics
 - C. Translocation
 - D. Inversion
-

62. Which of the following is true about DNA polymerase I?

- A. It is involved in the creation of Okazaki fragments
 - B. It is found in prokaryotes
 - C. It synthesizes RNA primers
 - D. It is the primary enzyme of DNA synthesis
-

63. DNA methylation is not involved in.

- A. Chromatin remodeling
 - B. RNA splicing
 - C. Gene silencing
 - D. DNA replication
-

64. Which of the following statements about telomerase is false?

- A. Has reverse transcriptase activity
 - B. Present only in Eukaryotes
 - C. Maintain the chromosome length
 - D. Involved in DNA repair
-

65. In a 14-year-old male, a medical condition has been identified as a result of a defect in a gene responsible for DNA repair. The individual is experiencing increased pigmentation on the skin and is at a heightened risk of developing skin cancers, according to the physician. Which of the following mechanisms is most likely to be impaired in this particular condition?

- A. Nonhomologous end joining repair defect

- B. Nucleotide excision repair defect
 - C. DNA base excision repair defect
 - D. DNA mismatch repair defect
-

66. The correct order of steps in a polymerase chain reaction (PCR) is:-

- A. Hybridization, annealing, elongation, denaturation
 - B. Elongation, denaturation, Hybridization, annealing,
 - C. Denaturation, annealing, extension.
 - D. Annealing, extension, denaturation
-

67. Which of the following processes does not necessitate 5' capping?

- A. tRNA of alanine
 - B. U6 snRNA
 - C. mRNA for Histone
 - D. siRNA
-

68. Which of the following is reduced in urine in a patient with Wilson's disease?

- A. Tyrosine
 - B. Phosphorous
 - C. Serine
 - D. 3-methylhistidine
-

69. True regarding mitochondrial genome:

- A. Maternal inheritance
 - B. Low rate of mutation
 - C. Single stranded DNA
 - D. Paternal inheritance
-

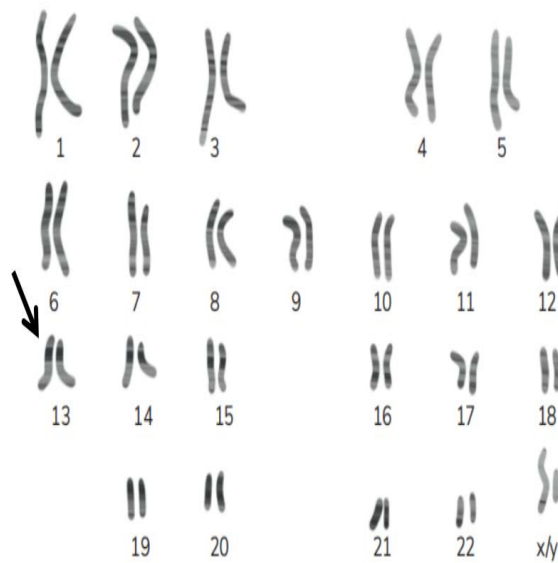
70. Which of the following is autosomal dominant?

- A. Cystic fibrosis
 - B. Duchene muscular dystrophy
 - C. Sickle cell anaemia
 - D. Achondroplasia
-

71. Which of the subsequent diseases demonstrate autosomal recessive inheritance?

- A. Albinism
 - B. G6PD deficiency
 - C. Marfan's syndrome
 - D. Lesch-Nyhan syndrome
-

72. Identify the type of marked chromosome in the given karyotype.



- A. Acrocentric
- B. Metacentric
- C. Submetacentric
- D. Telocentric

73. miRNA binds to which part of the mRNA to inhibit translation?

- A. Gene body
- B. Gene promoter
- C. 3'UTR
- D. 5'UTR

74. Which of the following statements are true regarding mitochondrial DNA?

Higher rate of mutations as compared to the nuclear genome
 Double-stranded circular DNA with about 16,000 base pairs
 A slight change in nucleotide sequence as compared to cellular DNA
 Encodes all electron transport chain proteins

- A. 1 and 3
 - B. 1 and 2
 - C. 2 and 4
 - D. 2 and 3
-

75. Which of the following is the best test for quantitative analysis of an RNA sample? PCR: polymerase chain reaction RT PCR: Reverse transcriptase polymerase chain reaction

- A. Real-time PCR
 - B. Reverse transcriptase PCR
 - C. PCR-restriction fragment length polymorphism
 - D. RTPCR followed by real-time PCR
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	1
Question 3	4
Question 4	4
Question 5	4
Question 6	1
Question 7	1
Question 8	1
Question 9	1
Question 10	1
Question 11	1
Question 12	4
Question 13	4
Question 14	3
Question 15	3
Question 16	2
Question 17	4
Question 18	3

Question 19	4
Question 20	3
Question 21	3
Question 22	2
Question 23	1
Question 24	1
Question 25	4
Question 26	3
Question 27	3
Question 28	1
Question 29	2
Question 30	2
Question 31	3
Question 32	3
Question 33	3
Question 34	1
Question 35	3
Question 36	1
Question 37	2
Question 38	4
Question 39	1

Question 40	1
Question 41	3
Question 42	4
Question 43	3
Question 44	4
Question 45	4
Question 46	1
Question 47	2
Question 48	2
Question 49	2
Question 50	1
Question 51	3
Question 52	1
Question 53	1
Question 54	1
Question 55	4
Question 56	1
Question 57	2
Question 58	3
Question 59	3
Question 60	2

Question 61	2
Question 62	2
Question 63	4
Question 64	4
Question 65	2
Question 66	3
Question 67	1
Question 68	4
Question 69	1
Question 70	4
Question 71	1
Question 72	1
Question 73	3
Question 74	2
Question 75	4

Solution for Question 1:

Answer: A) Deletion

Explanation:

Deletion is the most common genetic abnormality associated with micrognathia (small jaw) and low-set ears . These features are commonly seen in chromosomal deletion syndromes , such as:

- Cri-du-chat syndrome (5p deletion) Micrognathia, low-set ears, high-pitched cry ("cat-like cry") Intellectual disability and growth delay
- Micrognathia, low-set ears, high-pitched cry ("cat-like cry")
- Intellectual disability and growth delay
- DiGeorge syndrome (22q11.2 deletion) Micrognathia, low-set ears, conotruncal heart defects (e.g., TOF) Thymic hypoplasia → immunodeficiency
- Micrognathia, low-set ears, conotruncal heart defects (e.g., TOF)
- Thymic hypoplasia → immunodeficiency
- Treacher Collins syndrome (TCOF1 gene mutation, chromosomal microdeletion) Micrognathia, low-set ears, facial hypoplasia Conductive hearing loss
- Micrognathia, low-set ears, facial hypoplasia
- Conductive hearing loss
- Micrognathia, low-set ears, high-pitched cry ("cat-like cry")
- Intellectual disability and growth delay
- Micrognathia, low-set ears, conotruncal heart defects (e.g., TOF)
- Thymic hypoplasia → immunodeficiency
- Micrognathia, low-set ears, facial hypoplasia
- Conductive hearing loss
- Translocation (Option B) is incorrect because translocations usually cause genetic disorders by gene fusion or misregulation , often leading to cancers (e.g., CML t(9;22)) , rather than micrognathia and low-set ears.
- Amplification (Option C) is incorrect because gene amplification is associated with oncogenesis (e.g., HER2 in breast cancer, N-MYC in neuroblastoma) , not structural anomalies.
- Inversion (Option D) is incorrect because chromosomal inversions rarely lead to major congenital anomalies , unless a critical gene is disrupted.

Solution for Question 2:

Correct Option A - Human cells can divide a finite number of times only:

- The Hayflick limit, discovered by Dr. Leonard Hayflick, states that normal human cells have a finite capacity for replication.

- Telomeres, protective caps at chromosome ends, shorten with each cell division due to incomplete DNA replication.
- Shortened telomeres eventually lead to cellular senescence, preventing uncontrolled cell division and potential cancer.
- Some cells, like stem cells and certain cancers, express telomerase, extending telomeres and bypassing the Hayflick limit.
- In normal human cells, telomerase activity is limited, leading to replicative senescence.

Incorrect Options:

Option B - Free radical oxidation level is unrestricted: Incorrect. The Hayflick limit is not directly related to free radical oxidation levels.

Option C - Telomerase activity is unrestricted: Incorrect. Telomerase activity is typically restricted in normal human cells, unlike stem cells and certain cancers.

Option D - RNA polymerase can function for a finite number of times only: Incorrect. The Hayflick limit primarily involves telomere shortening, not RNA polymerase function.

Solution for Question 3:

Correct Option D - Phenylketonuria (PKU):

- PKU is a metabolic disorder caused by a deficiency of the enzyme phenylalanine hydroxylase, leading to the accumulation of phenylalanine. If untreated, elevated phenylalanine levels can lead to intellectual disability and other neurological problems.

Incorrect Options:

Option A - Alkaptonuria: Alkaptonuria is characterized by the accumulation of homogentisic acid, leading to dark urine and connective tissue pigmentation. It is not typically associated with mental retardation.

Option B - Albinism: Albinism is a genetic condition characterized by a lack of melanin production, resulting in a lack of pigmentation in the skin, hair, and eyes. While it affects vision, it is not primarily associated with mental retardation.

Option C - Hawkinsuria: Hawkinsuria is a rare metabolic disorder characterized by an abnormal excretion of 4-hydroxybutyric acid. It is not typically associated with mental retardation.

Solution for Question 4:

Correct Option D - Glycosylation:

- Glycosylation is a post-translational modification that occurs after translation, not during post-transcriptional modification.

Incorrect Options:

Option A – Methylation:

- Methylation can occur as a post-transcriptional modification, particularly in eukaryotic mRNA. It can affect gene expression and mRNA stability.

Option B - Endonuclease cleavage:

- Endonuclease cleavage typically occurs during RNA processing, where introns are removed from the pre-mRNA to produce mature mRNA. So, it is indeed associated with post-transcriptional modification.

Option C - 5' capping:

- 5' capping is a crucial post-transcriptional modification in eukaryotic mRNA, where a 7-methylguanosine cap is added to the 5' end. This modification protects the mRNA from degradation and aids in translation initiation.

Solution for Question 5:

Correct Option D - Mutation in tRNA:

- This refers to a suppressor mutation, where a second mutation occurs in a tRNA gene.
- The mutant tRNA can recognize the mutant codon and insert the correct amino acid, thus restoring the wild-type phenotype despite the presence of the original mutation in the coding gene.
- This type of mutation effectively suppresses the impact of the original mutation.

Incorrect Options:

Option A - Frameshift of the mutated gene :

- A frameshift mutation involves inserting or deleting a number of nucleotides that are not divisible by three.
- This shifts the reading frame of the codons in the mRNA, often resulting in a completely nonfunctional protein.
- This type of mutation usually cannot be easily reversed by a single mutation and thus is unlikely to lead to a reversion to the wild-type phenotype.

Option B - Addition of another normal gene:

- This would introduce a normal copy of the gene into the genome, which could potentially compensate for the mutant gene. This process is often referred to as gene therapy.
- While this can restore the wild-type phenotype, it is not a mutation but rather an addition of genetic material. Therefore, it does not fit the question's criteria of a mutation leading to reversion.

Option C - Deletion of the mutated gene:

- Deleting a mutant gene would remove its effects but would not necessarily result in a wild-type phenotype unless there is a compensatory mechanism in place.
- Deletion does not count as a mutation that causes reversion to the wild-type phenotype.

Solution for Question 6:

Correct Option A - Point mutation:

- FISH (Fluorescence In Situ Hybridization) is not useful for detecting point mutations.
- FISH is designed to detect larger genetic changes such as deletions, duplications, translocations, and amplifications.
- Point mutations involve a single nucleotide change, which is too small for FISH to detect.

Incorrect Options:

Option B - Translocation:

- FISH is useful for detecting translocations because it can identify specific chromosome rearrangements by using probes that bind to the translocated segments.

Option C - Deletion

- FISH can detect deletions by using probes that bind to the region where the deletion occurs. If the probe does not bind, it indicates a deletion in that area.

Option D - Amplification

- FISH can detect gene amplifications by showing an increased number of signals from the probe, indicating that there are multiple copies of a particular gene or chromosomal region.

Solution for Question 7:

Correct Option A - DNA Editing:

- CRISPR-Cas9 System : CRISPR-Cas9 is a revolutionary technology used for precise gene editing. It involves two key components: Cas9 Enzyme: An endonuclease that introduces a double-strand break at a specific location in the DNA sequence, guided by the guide RNA. Guide RNA (gRNA): This RNA molecule is complementary to the target DNA sequence and directs the Cas9 enzyme to the correct location in the genome.
- Cas9 Enzyme: An endonuclease that introduces a double-strand break at a specific location in the DNA sequence, guided by the guide RNA.
- Guide RNA (gRNA): This RNA molecule is complementary to the target DNA sequence and directs the Cas9 enzyme to the correct location in the genome.
- Cas9 Enzyme: An endonuclease that introduces a double-strand break at a specific location in the DNA sequence, guided by the guide RNA.
- Guide RNA (gRNA): This RNA molecule is complementary to the target DNA sequence and directs the Cas9 enzyme to the correct location in the genome.

Incorrect Options:

Option B - Post-transcriptional Modification: This process involves changes to RNA after it has been transcribed from DNA, such as splicing, capping, and polyadenylation. It does not involve CRISPR-Cas9 technology, which primarily targets DNA rather than RNA.

Option C - DNA Induction: This term is not commonly used in the context of CRISPR-Cas9 technology. CRISPR-Cas9 is focused on directly editing DNA sequences rather than inducing changes.

Option D - DNA Editing Solution: While CRISPR-Cas9 is a powerful tool for DNA editing, the term "DNA Editing Solution" is vague and not specific to the CRISPR-Cas9 system itself.

Solution for Question 8:

Correct Option A - Subunit aggregation:

- This is a post-translational process where multiple protein subunits come together to form a functional protein complex. Collagen, hemoglobin, and immunoglobulins are all examples of multisubunit proteins that undergo subunit aggregation to achieve their functional structures.

Incorrect Options:

Option B - Splicing: This is a post-transcriptional modification where introns are removed from precursor mRNA and exons are joined together. This process occurs before translation and is not specific to the proteins mentioned.

Option C - Poly A tail addition : This is a post-transcriptional modification that involves adding a polyadenine tail to the 3' end of mRNA. It helps in mRNA stability and translation efficiency but does not pertain to the protein structures in question.

Option D - Hydroxylation: This is a post-translational modification involving the addition of hydroxyl groups to amino acids. While hydroxylation is important in collagen synthesis, it is not a common modification for hemoglobin and immunoglobulins.

Solution for Question 9:

Option A - Bloom Syndrome:

- This condition is characterized by short stature, photosensitivity, and a predisposition to cancer. It results from defects in the DNA repair process, particularly in the repair of double-strand breaks. Bloom Syndrome is associated with mutations in the BLM gene, which is involved in the maintenance of genomic stability.

Incorrect Options:

Option B - Xeroderma Pigmentosum: This disorder is caused by defects in nucleotide excision repair, which is a mechanism to repair UV-induced damage. Individuals with Xeroderma Pigmentosum are highly sensitive to sunlight and prone to skin cancers. It does not primarily involve defects in double-strand break repair.

Option C - Cockayne Syndrome : This condition is also related to nucleotide excision repair but differs from Xeroderma Pigmentosum in that it primarily affects the repair of oxidative DNA damage and is characterized by developmental delays and sensitivity to sunlight.

Option D - Lynch Syndrome : This syndrome is associated with defects in mismatch repair rather than double-strand break repair. It increases the risk of various cancers, including colorectal cancer, but does not involve the issues described in the question.

Solution for Question 10:

Correct Option A - HGPRT deficiency:

- This condition refers to a deficiency in the enzyme Hypoxanthine-guanine phosphoribosyltransferase (HGPRT).
- HGPRT is crucial for the purine salvage pathway, which recycles purines from degraded nucleic acids.
- A deficiency in HGPRT leads to Lesch-Nyhan syndrome, characterized by severe behavioral issues such as self-mutilation, along with poor growth and neurological symptoms.

Incorrect Options:

Option B - Homogentisic Aciduria: This condition results from a deficiency in the enzyme homogentisate oxidase. It is associated with alkaptonuria, where homogentisic acid accumulates in the body, causing dark urine and joint problems. It does not typically present with self-mutilation or poor growth.

Option C - Phenylalanine Hydroxylase deficiency: This enzyme deficiency causes Phenylketonuria (PKU). PKU leads to a build-up of phenylalanine, which can result in intellectual disability and other developmental issues if not managed. However, it does not usually present with self-mutilating behavior.

Option D - Benzoyl Acidase deficiency: This is not a well-known metabolic disorder. Benzoyl acidase is not typically associated with significant clinical conditions related to the symptoms described.

Solution for Question 11:

Correct Option A - Lesch-Nyhan syndrome:

- Lesch-Nyhan syndrome is caused by a deficiency of the enzyme hypoxanthine-guanine phosphoribosyltransferase (HGPRT). This enzyme is essential for the salvage pathway of purines, and its deficiency leads to the accumulation of uric acid in the body, causing hyperuricemia. The patient also presents with self-mutilating behaviour.

Incorrect Options:

- Options B, C, and D do not show the given findings.
-

Solution for Question 12:

Correct Option D: RNA-dependent RNA polymerase

- Poliovirus is most commonly recovered from stool specimens. If poliovirus is isolated, the serotype (i.e., 1, 2, or 3) and whether the virus is wild, vaccine (Sabin), or VDPV strain are determined using reverse transcriptase polymerase chain reaction (RT-PCR) and genomic sequencing.
- Because acute flaccid paralysis (AFP) mirrors the clinical look of poliomyelitis, it has been accepted as a critical tool for tracking the global polio eradication effort's success.
- If poliovirus is isolated, the serotype (i.e., 1, 2, or 3) and whether the virus is wild, vaccine (Sabin), or VDPV strain are determined using reverse transcriptase polymerase chain reaction (RT-PCR) and genomic sequencing.
- In ssRNA viruses, RNA genome replication occurs in a membrane-bound replication complex composed of the viral polymerase and other viral proteins, cellular proteins, and

viral RNA. The replicating complex's specific content and nature currently need to be clarified. Plus-sense genomic RNA functions as an mRNA in infected cells to make viral proteins without modifying the viral genome, and hence viral RNA-dependent RNA polymerases are responsible for viral RNA genome replication.

Incorrect Options:

Option A: DNA-dependent RNA polymerase

- RNA-dependent RNA polymerases (RdRp) are enzymes all viruses with RNA genomes require. Poliovirus is a member of the enterovirus subgroup, family Picornaviridae. Picornaviruses are small, ether-insensitive viruses with an RNA genome. DNA-dependent RNA polymerase is not the enzyme for poliovirus.

Option B: RNA-dependent DNA polymerase

- Retroviruses, such as the human immunodeficiency virus (HIV), generate DNA copies of the viral RNA genome via reverse transcriptase, an RNA-dependent DNA polymerase. However, RNA viruses, such as influenza and poliovirus, reproduce their genomes using RNA-dependent RNA polymerases.

Option C: DNA-dependent DNA polymerase

- DNA viruses use DNA-dependent DNA polymerases (also known as DNA polymerases) to replicate their genomes and DNA-dependent RNA polymerases (also known as RNA polymerases) to transcribe mRNA. Poliovirus has an RNA genome and hence cannot use DNA-dependent DNA polymerases.

Solution for Question 13:

Correct Option D - Homologous Recombination:

- Suppose a chromosome has undergone a dsDNA break in the presence of ionising radiation.
- Another copy of the same chromosome will act as a homologous DNA sequence which comes close to the break site.
- Once they find the break site, DNA polymerase will try to create an overhanging end in the homologous DNA repair.
- DNA polymerase uses its exonuclease activity to remove a part of the strand to create overhanging.

- Holliday junction, is a classical sign seen in homologous DNA repair mechanism.

Incorrect Options:

Option A - Base Excision Repair (BER): Base Excision Repair is a DNA repair mechanism that corrects small, non-helix-distorting base lesions, such as damaged or mismatched bases. It involves the removal of the damaged base and its sugar phosphate backbone by specific enzymes, followed by DNA polymerase-mediated repair with the correct base.

Option B - Nucleotide Excision Repair (NER): Nucleotide Excision Repair is a DNA repair mechanism that primarily deals with bulky DNA lesions, such as those caused by UV radiation or certain chemicals. It involves the removal of a segment of damaged DNA, followed by DNA polymerase-mediated repair using the undamaged complementary strand as a template.

Option C - Mismatch Repair (MMR): Mismatch Repair is a DNA repair mechanism that corrects errors that occur during DNA replication, such as mismatched base pairs or small insertion/deletion loops. It identifies and removes the incorrect base and replaces it with the correct one

Solution for Question 14:

Correct Option C - Guanyl Cap:

- The guanyl cap, also known as the 5' cap, is a modified nucleotide structure added to the 5' end of eukaryotic mRNA molecules. It consists of a guanine nucleotide linked to the mRNA through a unique 5' to 5' triphosphate bond.
- The guanyl cap plays a crucial role in the attachment of mRNA to the ribosome during translation initiation. It helps in recognition by the ribosome and initiation factors, facilitating the binding of mRNA to the ribosome for protein synthesis.

Incorrect Option:

Option A - Poly A tail The poly A tail is a stretch of adenine nucleotides added to the 3' end of eukaryotic mRNA molecules. It plays a role in mRNA stability and regulation but is not directly involved in the attachment of mRNA to the ribosome.

Option B - tRNA Transfer RNA (tRNA) molecules are involved in protein synthesis but not in the direct attachment of mRNA to the ribosome.

Option D - Shine-Dalgarno Sequence The Shine-Dalgarno sequence is a specific nucleotide sequence found in prokaryotic mRNA molecules. It plays a role in the initiation of translation in prokaryotes by interacting with the complementary sequence in the small subunit of the ribosome.

Solution for Question 15:

Correct Option C - Complementation:

- In the given image showing the schematic of a blue-white assay used in screening recombinant vectors for inserted DNA fragments, the correct choice would be "Complementation"
- The blue-white assay determines recombinant vector screening based on complementation, where an inserted DNA fragment restores the function of a disrupted lacZ gene.
- This gene encodes β -galactosidase.
- If the DNA fragment restores lacZ function, active β -galactosidase cleaves a colorless substrate (X-gal), resulting in blue colonies.
- Complementation indicates successful restoration of lacZ function, observed as blue coloration in the assay.

Incorrect Options:

Option A - Translation: Incorrect. Translation is the process of protein synthesis and is not directly related to the blue-white assay or screening recombinant vectors.

Option B - Transformation: Incorrect. Transformation is the introduction of foreign DNA into a host organism, but it does not specifically relate to the blue-white assay.

Option D - Conjugation: Incorrect. Conjugation involves genetic material transfer between bacteria through direct contact and is not directly related to the blue-white assay or screening recombinant vectors.

Solution for Question 16:

Correct Option B - Non-Homologous End Joining (NHEJ):

- This is the correct option. NHEJ is a DNA repair pathway that repairs double-strand breaks (DSBs) by directly ligating the broken DNA ends without requiring a homologous template.
- CRISPR-Cas9 can induce targeted DSBs at specific sites in the genome, and NHEJ is often utilized to repair these breaks. However, NHEJ can introduce small insertions or deletions (indels) during the repair process, which may result in gene disruptions or mutations.

Incorrect Option:

Option A - Base excision: Base excision repair (BER) is a DNA repair mechanism that primarily corrects small base lesions, such as single base changes or small chemical modifications.

Option C - Homologous End Joining (HEJ) or Homology-Directed Repair (HDR): HDR is a DNA repair pathway that uses a homologous DNA template to repair DSBs, resulting in precise DNA sequence changes.

Option D - Mismatch repair: Mismatch repair is a DNA repair mechanism that corrects errors in DNA replication, such as mismatches or small insertion/deletion loops.

Solution for Question 17:

Correct Option D - Organ transplantation:

- Chimeric DNA can be used in organ transplantation to analyze compatibility between the donor and recipient. By comparing certain genetic markers, such as human leukocyte antigen (HLA) compatibility, chimeric DNA analysis helps determine the likelihood of a successful organ transplant.

Incorrect Options:

Option A - Paternity test: Paternity tests typically involve analyzing DNA samples from the alleged father, child, and mother (if available) to determine biological parentage. Chimeric DNA is not specifically used for paternity testing.

Option B - Maternity test: Similar to paternity tests, maternity tests involve analyzing DNA samples to determine biological parentage. Chimeric DNA is not specifically used for maternity testing.

Option C- Personal identification: Personal identification can be done using DNA analysis techniques such as DNA fingerprinting or forensic DNA profiling. Chimeric DNA is not specifically used for personal identification.

Solution for Question 18:

Correct option C

- Denature DNA, anneal primers, extend DNA: This is the correct sequence of steps in PCR.
- The first step is denaturation, which involves heating the DNA sample to separate the double-stranded DNA into single strands.
- Next, the temperature is lowered to allow the primers to anneal or bind to the complementary regions of the single-stranded DNA templates.
- Finally, the temperature is raised, and a DNA polymerase extends the primers by adding nucleotides to synthesize new DNA strands.

Incorrect Options:

Option A - Extend DNA, anneal primers, denature DNA: This sequence is incorrect. In PCR, the denaturation step usually comes first to separate the DNA strands. Then, primers are annealed to the single-stranded DNA templates, and finally, DNA extension occurs.

Option B - Anneal primers, extend DNA, denature DNA: This sequence is also incorrect. The denaturation step typically precedes primer annealing.

Option D - Denature DNA, extend DNA, anneal primers: This sequence is incorrect. The annealing of primers typically occurs after the denaturation step but before DNA extension.

Solution for Question 19:

Correct Option D - AUG:

- AUG is the start codon. It codes for the amino acid methionine and also serves as the initiation signal for protein synthesis. It is recognized by the ribosome as the site to begin translation.

Incorrect options:

Option A - UAA: UAA is not a start codon. It is a stop codon that signals the termination of protein synthesis.

Option B - UAG: UAG is also a stop codon and not a start codon. Like UAA, it signals the end of protein synthesis.

Option C - UGA: UGA is yet another stop codon and does not serve as a start codon. It indicates the termination of protein synthesis

Solution for Question 20:

Correct Option C - RT-PCR:

- Reverse transcriptase PCR is used to study RNA as well as DNA. It is a quick, automated, and highly sensitive method to study RNA and also detect mutations.
- RNA with the help of reverse transcriptase enzyme forms complementary DNA (cDNA). This cDNA is amplified using the polymerase enzyme so that multiple copies can be made and studied.

Incorrect Options:

Option A - G banding: While G banding involves staining chromosomes with Giemsa to observe specific banding patterns, it's not specifically designed for detecting monosomy and trisomy. Its primary use is in karyotyping to visualize overall chromosome structure and identify chromosomal abnormalities.

Option B - Sanger's method: It is a DNA sequencing technique also called the dideoxy nucleotide method. It is the process of selective incorporation of chain-terminating dideoxynucleotide by DNA polymerase during in vitro DNA replication.

Option D - Western blot: It is the antigen-antibody reaction (immunoblotting), used for the detection of protein antigen.

Solution for Question 21:

Correct Option C - RT PCR:

- RT-PCR is the most sensitive test for monitoring treatment response in chronic myeloid leukemia (CML).
- It accurately detects and quantifies the BCR-ABL1 fusion gene, allowing for precise assessment of treatment response and minimal residual disease (MRD) status in CML patients.
- Regular monitoring with RT-PCR guides treatment decisions for optimal patient outcomes.

Incorrect Options:

Option A - CBC (Complete Blood Count):

- While CBC indicates changes in blood cell levels, it may not detect MRD or early treatment failure as sensitively as RT-PCR.

Option B - Cytogenetic Test:

- Although cytogenetic tests reveal chromosomal abnormalities like the Philadelphia chromosome, they may not detect low levels of disease or MRD as effectively as molecular techniques like RT-PCR.

Option D - FISH (Fluorescence In Situ Hybridization):

- While FISH detects genetic sequences like the BCR-ABL1 fusion gene, it may not quantify residual disease or assess treatment response as accurately as RT-PCR.
-

Solution for Question 22:

Correct Option B - 5'→3' Exonuclease

- Klenow fragment is a truncated form of DNA polymerase I that lacks the N-terminal region of the enzyme. It retains the DNA polymerase activity and the 3'→5' exonuclease activity, but it lacks the 5'→3' exonuclease activity.

Incorrect Options:

Option A - 3'→5' Exonuclease: Klenow fragment retains the 3'→5' exonuclease activity. This activity allows the enzyme to proofread and remove mismatched nucleotides from the growing DNA chain during DNA synthesis.

Option C - 5'→3' DNA polymerase: Klenow fragment retains the 5'→3' DNA polymerase activity. This activity allows the enzyme to synthesize a new DNA strand using a DNA template.

Option D - 3'→5' DNA polymerase: DNA polymerase I, including its Klenow fragment, does not possess 3'→5' DNA polymerase activity. The DNA synthesis activity of DNA polymerases occurs exclusively in the 5'→3' direction.

Solution for Question 23:

Correct Option A - RT PCR:

- RT-PCR is a technique that involves the reverse transcription of RNA into complementary DNA (cDNA), followed by PCR amplification of the cDNA. It is commonly used to study gene expression by quantifying RNA levels.
- The process begins with the reverse transcription step, where the RNA template is converted into cDNA using the enzyme reverse transcriptase. The resulting cDNA is then amplified through PCR to obtain multiple copies of the target DNA sequence.

Incorrect Options:

Option B - Sanger's technique: Sanger sequencing, or chain termination sequencing, is a method for DNA sequencing that doesn't directly involve RNA.

Option C - Next-generation sequencing: NGS is a high-throughput DNA sequencing method used for various DNA-based applications, although it can also sequence RNA molecules. Although NGS can be used to study RNA by sequencing RNA molecules, the technique itself is not solely based on RNA.

Option D - Western blot: Western blotting detects and quantifies specific proteins in a sample, but it's not directly based on RNA, even though it can be used to analyze proteins encoded by RNA.

Solution for Question 24:

Correct Option A - Methylation:

- Euchromatin refers to the less condensed and transcriptionally active form of chromatin. Changes in cytosine residues at CpG islands, which are regions with a high frequency of cytosine-guanine dinucleotides, can influence chromatin structure and gene expression.

Incorrect Options:

Option B - Phosphorylation: Incorrect. While phosphorylation regulates protein activity and can impact chromatin remodeling, it doesn't directly hinder euchromatin formation.

Option C - Alkylation: Incorrect. Alkylation can alter DNA structure and gene expression, but its effect on euchromatin formation depends on various factors and isn't a consistent rule.

Option D - Sumoylation: Incorrect. Sumoylation affects transcriptional regulation and chromatin organization but isn't specifically linked to hindering euchromatin formation.

Solution for Question 25:

Correct Option D - RNA editing:

- RNA editing is a process that can modify the nucleotide sequence of RNA molecules after transcription. It can involve changes in bases, the addition or removal of nucleotides, or other modifications. While RNA editing can occur in different directions depending on the specific mechanism, it is not universally restricted to the 5' to 3' direction.
- RNA polymerase begins moving down the DNA template strand in the 3' to 5' direction

Incorrect Option:

Option A - Transcription:

- Transcription is the process of synthesizing RNA from a DNA template. It occurs in the 5' to 3' direction, meaning that the RNA molecule is synthesized in the same direction as the template DNA strand.

Option B - DNA replication:

- DNA replication is the process of copying the DNA molecule to produce two identical DNA molecules. It also occurs in the 5' to 3' direction, where the new DNA strand is synthesized in the same direction as the template DNA strand.

Option C - DNA repair:

- DNA repair mechanisms are responsible for fixing damaged DNA. Many DNA repair processes involve the removal and replacement of damaged nucleotides, which generally occur in the 5' to 3' direction. Therefore, DNA repair also occurs in the 5' to 3' direction.

Solution for Question 26:

Correct Option C - Point mutation:

- Point mutation: A point mutation involves the substitution of a single nucleotide with another nucleotide. This can lead to a change in the corresponding amino acid in the protein sequence. In the case of sickle cell anemia, the point mutation is the substitution of Adenine with Thymine, resulting in the replacement of glutamic acid with valine, in the beta globin chain of hemoglobin.

Incorrect Option:

Option A - Insertion: An insertion mutation involves the addition of extra nucleotides into the DNA sequence. This leads to a shift in the reading frame and often results in a frameshift mutation, which can have significant effects on the resulting protein. However, this is not the type of mutation seen in sickle cell anemia.

Option B - Deletion: A deletion mutation involves the removal of nucleotides from the DNA sequence. Similar to an insertion mutation, it can cause a frameshift mutation and alter the resulting protein. However, this is not the type of mutation seen in sickle cell anemia.

Option D - Frameshift mutations: Frameshift mutations occur due to insertions or deletions of nucleotides that cause a shift in the reading frame. This leads to significant changes in the resulting protein sequence. While frameshift mutations can occur in various genetic disorders, it is not the specific type of mutation seen in sickle cell anemia.

Solution for Question 27:

Correct Option C - AAGCTT:

- AAGCTT- This sequence is recognized by the restriction endonuclease EcoRI. EcoRI recognizes the palindromic sequence 5'-GAATTC-3' and cleaves the DNA between the G and the A bases on both strands, resulting in fragments with sticky ends. Therefore, option C is the correct answer.

Incorrect Option:

Option A - AAGAAG This sequence is not a recognized target for any commonly known restriction endonuclease. It does not contain a known recognition site for cleavage. Therefore, option A is incorrect.

Option B - GAGAGG Similar to option A, this sequence does not contain a known recognition site for restriction endonucleases. It does not match the specific sequence required for cleavage. Therefore, option B is incorrect.

Option D - TACGAG This sequence does not contain a known recognition site for restriction endonucleases. It does not match any commonly recognized sequences required for cleavage. Therefore, option D is incorrect.

Solution for Question 28:

Correct Option A - Non-homologous end joining:

- Non-homologous end joining (NHEJ) is one of the main DNA repair pathways utilized by cells to repair double-stranded breaks (DSBs) in DNA. In the context of CRISPR-Cas9, NHEJ is the repair mechanism that is most commonly used to repair the DNA after Cas9 has induced a double-stranded break at the target site.

Incorrect Choices:

- Option B - Base excision repair : Base excision repair is a DNA repair mechanism that primarily deals with the repair of damaged or abnormal bases in the DNA molecule.
- Option C - Nucleotide excision repair: Nucleotide excision repair is a DNA repair pathway that primarily deals with the removal and repair of bulky DNA lesions, such as UV-induced

thymine dimers or certain chemical adducts.

- Option D - Mismatch repair: Mismatch repair is a DNA repair system that corrects errors that occur during DNA replication, such as mismatches or small insertions or deletions that can result in DNA loops.

Solution for Question 29:

Correct Option B - Both assertion and reason are true, but the reason is not the correct explanation for the assertion:

- Both assertion and reason are true, but the reason is not the correct explanation for the assertion: This option is correct. The assertion correctly states that according to the central dogma, information flows from DNA to RNA, which is then encoded into protein. This is the general principle of information flow in most organisms. The reason, on the other hand, states that the reversal of the central dogma happens in retroviruses, which is true. However, this does not provide a correct explanation for the assertion.

Incorrect Options:

Option A - Both assertion and reason are true and the reason is the correct explanation for the assertion: This option is incorrect. While both the assertion and reason are true, the reason provided does not correctly explain the assertion. However, the reason does not explain why the information flow from DNA to RNA is the correct explanation for the central dogma.

Option C - Assertion is true, but the reason is false: This option is incorrect. The assertion is true as per the central dogma, but the reason is also true as retroviruses do reverse the flow of information. Therefore, the reason is not false.

Option D - Both assertion and reason are false: This option is incorrect. Both the assertion and reason are true, as discussed above.

Option E - The assertion is false, but the reason is true: This option is incorrect. The assertion is true, as the central dogma indeed states that information flows from DNA to RNA to protein. The reason is also true, as retroviruses reverse this flow. Therefore, the assertion is not false.

Solution for Question 30:

Correct Option B - Both assertion and reason are true, but the reason is not the correct explanation for the assertion:

- This option is correct. The assertion correctly states that according to the central dogma, information flows from DNA to RNA, which is then encoded into protein. This is the general principle of information flow in most organisms. The reason, on the other hand, states that the reversal of the central dogma happens in retroviruses, which is true. However, this does not provide a correct explanation for the assertion.

Incorrect Option:

Option A - Both assertion and reason are true and the reason is the correct explanation for the assertion:

- This option is incorrect. While both the assertion and reason are true, the reason provided does not correctly explain the assertion. However, the reason does not explain why the information flow from DNA to RNA is the correct explanation for the central dogma.

Option C - Assertion is true, but the reason is false:

- This option is incorrect. The assertion is true as per the central dogma, but the reason is also true as retroviruses do reverse the flow of information. Therefore, the reason is not false.

Option D - Both assertion and reason are false:

- This option is incorrect. Both the assertion and reason are true, as discussed above.
-

Solution for Question 31:

Correct Option C - ACCTTG

- To determine the sequence of the template strand, we need to find the complementary sequence to the given RNA sequence. In RNA, uracil (U) pairs with adenine (A) instead of thymine (T) as in DNA.
- Given RNA sequence: CAAGGU
- Complementary bases: C pairs with G A pairs with U (in RNA) or T (in DNA) U (or T) pairs with A G pairs with C U (or T) pairs with A

- C pairs with G
- A pairs with U (in RNA) or T (in DNA)
- U (or T) pairs with A
- G pairs with C
- U (or T) pairs with A
- The complementary sequence to the given RNA sequence CAAGGU would be: GTTCCA
- But this complementary sequence is 3' → 5' and we always have to write it in 5' → 3'.
- Hence the answer is ACCTTG .
- C pairs with G
- A pairs with U (in RNA) or T (in DNA)
- U (or T) pairs with A
- G pairs with C
- U (or T) pairs with A

Incorrect Option - A, B, and D are incorrect: Refer to the explanation of option B.

Solution for Question 32:

Correct Option C - Purine Salvage pathway:

- Purine synthesis has 2 pathways Denovo pathway - Most active in the liver Salvage pathway - Most active in Brain, RBC and PMNLs Hypoxanthine required for Hybridoma technology comes from the purine salvage pathway. Salvage pathways occur only in b cells making purine in them. Hybridoma technology is commonly used for the production of monoclonal antibodies.
- Denovo pathway - Most active in the liver
- Salvage pathway - Most active in Brain, RBC and PMNLs
- Hypoxanthine required for Hybridoma technology comes from the purine salvage pathway.
- Salvage pathways occur only in b cells making purine in them.
- Hybridoma technology is commonly used for the production of monoclonal antibodies.
- Denovo pathway - Most active in the liver
- Salvage pathway - Most active in Brain, RBC and PMNLs

- Hypoxanthine required for Hybridoma technology comes from the purine salvage pathway.
- Salvage pathways occur only in b cells making purine in them.
- Hybridoma technology is commonly used for the production of monoclonal antibodies.

Incorrect Options:

Option A, B, and D: These options are incorrect because of no involvement in making Hypoxanthine-Aminopterin-Thymidine.

Solution for Question 33:

Correct Option C - HIV:

HIV is a single-stranded RNA virus, In single RNA, Adenine forms a base pair with uracil and not with Thymine.

Incorrect Options:

Option A - Mitochondrial DNA of humans: This option is incorrect as it is a double-stranded circular DNA. In double-stranded DNA, Adenine is usually coupled with Thymine.

Option B - E. coli: This option is incorrect as the genetic material is a double-stranded circular DNA. In double-stranded DNA, Adenine is usually coupled with Thymine.

Option D - Herpes: Herpes virus is a double-stranded DNA virus.

Solution for Question 34:

Correct Option A - ATP:

- Homologous, nonhomologous and joining repair does contribute to the repair of DNA double-strand breaks and ATP gene does come under them.

Correct Options:

Option B - RAD 51: It is involved in the G2 phase of homologous repair, hence it is incorrect.

Option C - ATM: It is involved in the G2 phase of homologous repair, hence it is incorrect.

Option D - Ku 70 -n: It is involved in the G1 phase of nonhomologous repair, hence it is incorrect.

Solution for Question 35:

Correct Option C - ELISA for antibody:

- As the immune system fights the infection, IgM antibodies against dengue virus are detectable starting 4-5 days after onset of symptoms and are reliably detectable for approximately 12 weeks.

Incorrect Options:

Option A - ELISA for antigen: This test is reliable for diagnosis in the first 3 days. The antigens used for this assay are derived from the envelope proteins of the four dengue virus serotypes (DENV-1-4).

Option B - Viral load: Viral load could be an important predictive parameter in dengue related severe symptoms like thrombocytopenia and elevated hematocrit when it goes above a certain threshold ($>10^6$ copies/ mL). It can be done in 1st week.

Option D - RT PCR: Positive results on the CDC DENV-1-4 rRT-PCR multiplex assay indicate current dengue infection. Blood samples from patients suspected of having dengue taken 1-7 days after onset of fever should be tested with this test.

Solution for Question 36:

Correct Option A - Chloride is raised in sweat

- In cystic fibrosis (CF), the dysfunction of the CFTR protein leads to impaired chloride transport across cell membranes.
- The defective CFTR protein results in decreased chloride reabsorption by sweat glands, leading to higher chloride concentrations in sweat.

- Sweat chloride levels above 60 mmol/L are indicative of cystic fibrosis.

Incorrect Options - B, C, and D: These are incorrect

Solution for Question 37:

Correct Option B - RNA:

- RT-PCR is designed to analyze RNA. It is commonly used to study gene expression, detect RNA viruses, and perform other applications where RNA analysis is needed.

Incorrect Options:

Option A - DNA: While traditional PCR (polymerase chain reaction) is used to amplify DNA, reverse transcriptase PCR (RT-PCR) specifically targets RNA. It involves the reverse transcription of RNA into complementary DNA (cDNA) and subsequent PCR amplification of the cDNA.

Option C - Amino acid: RT-PCR does not directly analyze amino acids. It focuses on nucleic acids, specifically RNA.

Option D - Protein particle: RT-PCR is not designed for the analysis of protein particles. It primarily deals with nucleic acids, particularly RNA.

Solution for Question 38:

Correct Option D - PCR:

- The DNA sequence is determined by PCR (Polymerase Chain Reaction).
- Polymerase chain reaction (PCR) is an in vitro DNA amplification procedure in which millions of copies of a particular sequence of DNA can be produced within a few hours.
- It allows for the selective amplification of a target DNA sequence, making it easier to study and analyze.

- PCR can be used in combination with DNA sequencing methods, such as Sanger sequencing or next-generation sequencing, to determine the exact sequence of DNA.
- E.g. In early phases of tuberculosis, the sputum may contain only very few tubercle bacilli, so that usual acid fast staining may be negative. But PCR can detect even one bacillus present in the specimen. Any other bacterial infection can also be detected similarly. The specific nucleotide sequences of the bacilli are amplified by PCR and then detected by Southern blot analysis.

Polymerase chain reaction (PCR) is an in vitro DNA amplification procedure in which millions of copies of a particular sequence of DNA can be produced within a few hours.

It allows for the selective amplification of a target DNA sequence, making it easier to study and analyze.

Incorrect Options:

Option A - Flow cytometry:

- Flow cytometry is a technique used to analyze and sort cells based on their physical and chemical characteristics, such as size, granularity, and fluorescence. It is not directly involved in determining the DNA sequence.

Option B - FISH (Fluorescence in situ Hybridization):

- FISH visualizes and maps specific DNA sequences on chromosomes. It detects chromosomal abnormalities and gene rearrangements but does not determine the complete DNA sequence.

Option C - RFLP (Restriction Fragment Length Polymorphism):

- RFLP is a technique that detects variations in DNA sequences by using specific restriction enzymes to cut the DNA at specific sites. It can be used to identify genetic polymorphisms

Solution for Question 39:

Correct Option A - Fluorescent in situ hybridization:

- Fluorescent in situ hybridization (FISH) is a technique used to visualize and identify specific DNA sequences in cells and tissues. It involves the use of fluorescently labeled probes that are complementary to the target DNA sequence.

- The probes hybridize to the target DNA, allowing it to be visualized and identified using fluorescence microscopy. The technique is based on the principle of hybridization, which refers to the binding of two complementary nucleic acid strands to form a stable double-stranded molecule.

Incorrect Options:

Option B - Karyotyping:

- Karyotyping is a technique used to identify and analyse the chromosomes in a sample of cells.

Option C - Immunohistochemistry (IHC):

- IHC identifies specific proteins in cells and tissues using antibodies labeled with fluorescent or enzyme-linked tags.

Option D - Sanger sequencing:

- Sanger sequencing determines the nucleotide sequence of a DNA molecule by incorporating nucleotide-specific chain-terminating dideoxynucleotides.

Solution for Question 40:

Correct Option A - Non-homologous end joining:

- Non-homologous end joining (NHEJ) is one of the main DNA repair pathways utilized by cells to repair double-stranded breaks (DSBs) in DNA. In the context of CRISPR-Cas9, NHEJ is the repair mechanism that is most commonly used to repair the DNA after Cas9 has induced a double-stranded break at the target site.
- Example of defective NHEJ: Ataxia telangiectasia (AT).

Incorrect Options:

Option B - Base excision repair : Base excision repair (BER): BER repairs damaged or abnormal bases in DNA, not directly associated with CRISPR-Cas9. Example of a BER-related disorder: Xeroderma pigmentosum (XP).

Option C - Nucleotide excision repair : Nucleotide excision repair (NER): NER removes bulky DNA lesions like UV-induced thymine dimers, unrelated to CRISPR-Cas9 repair. Example of an NER-related disorder: Cockayne syndrome (CS).

Option D - Mismatch repair: Mismatch repair corrects replication errors, not directly involved in CRISPR-Cas9 repair. Example of a mismatch repair-related disorder: Lynch syndrome.me

Solution for Question 41:

Correct | Option C - Degeneracy:

- One of the characteristics of the genetic code is that it is degenerate, meaning that multiple codons can code for the same amino acid.
- The Wobble hypothesis suggests that while the first two positions of a codon and anticodon must form specific base pairs (A-U and G-C), the third position allows for flexibility or "wobble" in base pairing.
- This flexibility explains how multiple codons can code for the same amino acid, contributing to the degeneracy of the genetic code.

Incorrect Options:

Option A - Universal: Genetic code is considered universal because humans, animals, plants and bacteria all have the exact same genetic code.

Option B - Unambiguous: The genetic code is unambiguous since a codon always codes for a specific amino acid. It does not code for another amino acid.

Option D - Non-punctuated: No internal punctuation is used in the code. Thus, the code is said to be commaless. Thus once the translation of mRNA begins, the code is read in a non-contiguous fashion i.e one after the other with no breaks between them

Solution for Question 42:

Correct Option D - RT-PCR:

- Reverse transcriptase PCR is used to study RNA as well as DNA. It is a quick, automated, and highly sensitive method to study RNA and also detect mutations.

- RNA with the help of reverse transcriptase enzyme forms complementary DNA (cDNA). This cDNA is amplified using the polymerase enzyme so that multiple copies can be made and studied.

Incorrect Options:

Option A - Sanger's method: Also known as the dideoxy nucleotide method, it's a DNA sequencing technique where chain-terminating dideoxynucleotides are selectively incorporated during DNA replication in vitro.

Option B - G banding: A karyotyping technique ideal for detecting monosomy and trisomy. Cells are arrested in metaphase using colchicine, then stained with Giemsa to visualize G-bands on chromosomes.

Option C - Western blot: It is the antigen-antibody reaction (immunoblotting), used for the detection of protein antigen.

Solution for Question 43:

Correct Option C - Adenosine deaminase:

- Adenosine deaminase (ADA) deficiency causes severe combined immunodeficiency involving the loss of function of T-Cell, B-cells and NK cells. Adenosine deaminase is responsible for the breakdown of adenosine and deoxyadenosine. The deficiency of ADA causes the accumulation of d-ATP. The accumulation is toxic to the lymphocytes as it inhibits ribonucleotide reductase enzyme and prevents DNA replication and DNA synthesis. This deficiency is a genetic disorder of the autosomal recessive type.

Incorrect Options:

Option A - Ornithine transcarbamylase: The deficiency of this enzyme is an X-Linked recessive disorder that causes dysfunction of the urea cycle.

Option B - Phosphomannose isomerase: The deficiency of this enzyme is related to a congenital disorder of Glycosylation.

Option D - Hypoxanthine-guanine phosphoribosyl transferase: The deficiency of this enzyme is seen in a disorder called Lesch-nyhan syndrome which is an X-linked recessive disorder.

Solution for Question 44:

Correct Option D - 22%:

- To understand this, we need to consider Chargaff's rule, which states that in DNA, the amount of adenine (A) is equal to the amount of thymine (T), and the amount of guanine (G) is equal to the amount of cytosine (C).
- Since the total percentage of nucleotides should add up to 100%, and 28% + 28% is already accounted for by thymine and adenine, the remaining percentage for guanine and cytosine is 44%. Since guanine and cytosine are present in equal amounts, the percentage of cytosine would be half of the remaining percentage, which is 22%. Therefore, the correct percentage of cytosine is 22%.

Incorrect Options - A, B, and C: These are incorrect

Solution for Question 45:

Correct Option D - Uracil:

Uracil is not found in DNA structures; it is one of the four nitrogenous bases found in RNA instead. DNA contains the nitrogenous bases Guanine, Adenine, Cytosine, and Thymine. Thymine replaces uracil in DNA and forms base pairs specifically with adenine.

Incorrect Options:

Option A - Guanine: It is one of the four nitrogenous bases found in DNA and forms base pairs with Cytosine.

Option B - Adenine: It is another nitrogenous base found in DNA and forms base pairs with Thymine.

Option C - Cytosine: It is one of the four nitrogenous bases found in DNA and forms base pairs with Guanine.

Solution for Question 46:

Correct Option A - Mismatch repair:

- Hereditary non-polyposis colon carcinoma (HNPCC), or Lynch syndrome, is an autosomal dominant disorder predisposing individuals to colorectal and other cancers.
- It stems from germline mutations in DNA mismatch repair (MMR) genes.
- The mismatch repair (MMR) pathway corrects replication errors and maintains genomic stability.
- HNPCC-associated mutations in MMR genes (e.g., MLH1, MSH2, MSH6, PMS2) lead to error accumulation, increasing cancer risk, particularly colorectal carcinoma.

Incorrect Options:

Option B - Nucleotide excision repair (NER):

- NER is a DNA repair mechanism for bulky DNA lesions from UV radiation or certain chemicals. It's linked to disorders like xeroderma pigmentosum but not hereditary non-polyposis colon carcinoma.

Option C - Base excision repair (BER):

- BER fixes small, non-bulky DNA lesions, like single-base changes or chemical modifications from oxidative stress or alkylating agents. It's associated with genetic disorders like some forms of inherited breast and ovarian cancer, not hereditary non-polyposis colon carcinoma.

Option D - Point mutation:

- Point mutations are changes in individual nucleotides in DNA. While they can contribute to cancer, hereditary non-polyposis colon carcinoma is related to mismatch repair pathway dysfunction, not general point mutations.

Solution for Question 47:

Correct Option B - Folate:

- Folate, also known as vitamin B9 , is important for DNA synthesis and repair . A deficiency of folate can lead to the misincorporation of uracil into DNA and other types of DNA damage. This is because folate is required for the production of thymidine, which is a necessary precursor for DNA synthesis.
- Without sufficient folate, the body may use uracil instead of thymidine, leading to uracil misincorporation into DNA. Folate is also important for DNA repair mechanisms, and deficiency can impair the body's ability to repair DNA damage.

Incorrect Option - A, C and D are incorrect

Solution for Question 48:

Correct Option B - C banding:

- C banding is a chromosome staining technique that uses a specific DNA-binding dye to selectively stain centromeric regions of chromosomes . This technique is particularly useful for identifying dicentric chromosomes, which have two centromeres and can result from chromosomal rearrangements or mutations.

Incorrect Options:

Option A - G banding :

- G banding stains chromosomes with Giemsa's dye, revealing characteristic banding patterns. It's vital in cytogenetics for identifying chromosomal abnormalities and mapping genes and genetic markers on chromosomes.

Option C - NOR:

- NORs are chromosome regions containing genes for rRNA synthesis. Staining techniques targeting NORs visualize these regions, aiding in studying their structure and function.

Option D - R banding:

- R banding is a chromosome staining technique that reveals a reverse banding pattern compared to G banding. It helps identify chromosomal abnormalities like deletions and translocations and maps genes and genetic markers on chromosomes.

Solution for Question 49:

Correct Option B - Knockdown:

- RNA interference technique using miRNA is used to stop the production of any toxic gene, as every cell has the genetic machinery to detoxify harmful toxins. However, this gene is not always active and induced only when required. This operates at the translation level by suppressing or degrading target mRNA, as one RNA is interfering with the function of another RNA. This is called gene knockdown or gene silencing technique, as the gene is present, but its function is silenced or down.

Incorrect Options:

Option A - Knock out :

- Gene knockout involves completely eliminating an organism's gene, resulting in inactive genes termed knockouts (KO). Knockouts are pivotal for studying gene function and the consequences of gene loss.

Option C - Knock-up :

- Gene knock-up is precisely inserting a desired promoter or enhancer to drive the target gene.

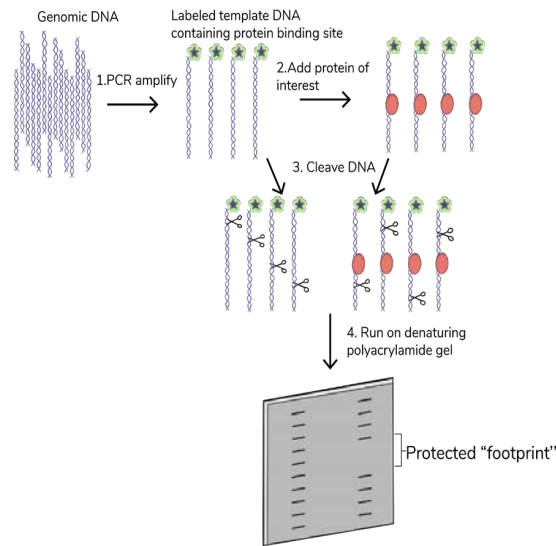
Option D - Knock-in:

- Gene knock-in involves targeted insertion of an exogenous gene at a specific genomic location, modifying the function of an endogenous gene.
-

Solution for Question 50:

Correct Option A - DNA foot printing:

- This technique is used to detect DNA-protein interactions. It involves binding DNA with a protein and then using a nuclease to selectively cleave it at sites not bound by the protein. This creates a pattern of DNA fragments, which can be analyzed to determine the binding sites of the protein.



Incorrect Options:

Option B - DNA finger printing: DNA fingerprinting identifies individuals based on their unique DNA profiles. It utilizes PCR and gel electrophoresis to generate a pattern of DNA fragments for comparison with known samples.

Option C - ELISA : ELISA detects specific proteins or antigens in samples using antibodies linked to enzymes that produce a signal upon interaction with the target.

Option D - Northern blotting: Northern blotting detects RNA molecules by separating them via gel electrophoresis, transferring them to a membrane, and probing with labeled DNA or RNA probes complementary to the target RNA.

Solution for Question 51:

Correct Option C - Phenylalanine at 508 position, chloride channel:

- The most common mutation in cystic fibrosis affects phenylalanine at position 508 of the CFTR protein, a chloride channel found in mucus, sweat, and digestive cells.
- This mutation results in a faulty CFTR protein, causing thick mucus that obstructs airways and ducts.

- Symptoms include chronic lung infections, pancreatic issues, and infertility, varying in severity based on the mutation and other factors.

Incorrect Options - A, B, and D are incorrect.

Solution for Question 52:

Correct Option A - Droplet digital PCR:

- Droplet Digital PCR (ddPCR) is a highly sensitive quantitative technique for detecting mutations with very low allele frequencies.
- It partitions a sample into thousands of droplets, each containing a single DNA molecule.
- PCR is performed on each droplet, and fluorescent probes detect the presence or absence of the mutation.
- Specialized equipment analyzes the droplets to count fluorescent signals, determining mutation frequency.
- ddPCR is highly accurate and valuable for detecting low-frequency mutations in clinical samples.

Incorrect Options - B, C and D are not as sensitive as ddPCR and cannot be used to identify mutations with very low allele frequencies.

Solution for Question 53:

Correct Option A - Somatic hypermutation:

- Somatic hypermutation is a process that occurs in B cells during the immune response.
- In this process, the DNA of the variable regions of the antibody genes is mutated at a very high rate. This process generates a diverse population of B cells, each with a slightly different version of the antibody gene.

- Over time, the B cells with high-affinity antibodies are selected, while those with low-affinity antibodies die. This process is known as affinity maturation and produces high-affinity antibodies that neutralize pathogens.

Incorrect Options:

Options B, C, and D are incorrect.

Solution for Question 54:

Correct Option A - 2, 1, 3, 5, 4:

- The sequence of steps in the isolation of desirable protein using recombinant DNA technology are as follows: 2. Gene incorporation: Desired genes are inserted into bacterial cells using plasmid vectors containing promoter sequences for transcription into mRNA. 1. Protein expression and bacterial cell lysis: Chemical induction activates gene expression, leading to protein synthesis. Bacterial cells are lysed to release proteins, achieved through methods like sonication or enzymatic digestion. 3. SDS-PAGE: Proteins are separated from cellular components using SDS-PAGE, where samples are denatured with SDS, loaded onto a gel, and separated by size via electrophoresis. 5. Hybridization:: Protein identity confirmation is conducted using techniques such as Western blotting, mass spectrometry, or hybridization. 4. Protein elution: Proteins of interest are eluted from the gel, typically through methods like electroelution or diffusion.
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- 1. Protein expression and bacterial cell lysis: Chemical induction activates gene expression, leading to protein synthesis. Bacterial cells are lysed to release proteins, achieved through methods like sonication or enzymatic digestion.
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 - 4. Protein elution: Proteins of interest are eluted from the gel, typically through methods like electroelution or diffusion.
-

Solution for Question 55:

Correct Option D - It retains the telomerase enzyme in the nucleus:

- Telomerase is an enzyme that adds DNA repeats (nucleotides) to the ends of chromosomes, preventing them from shortening during DNA replication.
- p53 has been shown to play a role in regulating telomerase activity.
- It helps retain telomerase in the nucleus, which can act on telomeres, thus maintaining their length and contributing to cellular lifespan.
- By retaining telomerase in the nucleus, p53 helps regulate the balance between cell proliferation and senescence or cell cycle arrest.

Incorrect Options:

Option A - It increases telomere length in the liver:

- Telomere lengthening in the liver is not attributed to p53. Telomeres shorten with cell division, and telomerase, not p53, maintains or lengthens them in specific cells, including stem cells and some cancer cells.

Option B - It synthesizes RNA:

- While p53 regulates gene expression, it is not directly involved in RNA synthesis. RNA polymerases primarily carry out RNA synthesis.

Option C - It stabilizes RNA:

- p53 can indirectly affect RNA levels through gene regulation, p53 does not stabilize RNA molecules.

Solution for Question 56:

Correct Option A - RNA editing:

- RNA editing is a post-transcriptional modification process that alters the nucleotide sequence of RNA molecules and is responsible for the two distinct forms of the ApoB gene. In the liver, the Apo-B gene undergoes RNA editing by the enzyme APOBEC-1, changing a CAA codon to a UAA stop codon, producing ApoB-100.

Incorrect Options: Option B - Gene splicing:

- Gene splicing is involved in generating multiple protein isoforms from a single gene, it's not the mechanism responsible for the distinct forms of ApoB seen in the liver and intestine.

Option C - Alternative polyadenylation:

- Alternative polyadenylation generates multiple mRNA isoforms from a single gene via different polyadenylation signals, impacting mRNA stability, localization, and translational efficiency.

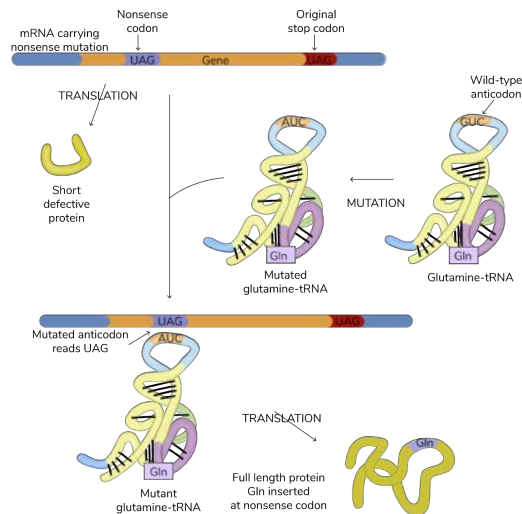
Option D - Gene rearrangements:

- Gene rearrangements primarily occur during the development of lymphocytes and are not relevant to the production of ApoB-100 and ApoB-48.

Solution for Question 57:

Correct Option B - Nonsense mutation:

- Codon coding for an amino acid is replaced by a stop codon.
- There is a premature termination of protein synthesis (truncated proteins are produced).



Incorrect Options:

Option A - Missense mutation:

- Alters protein structure and function by incorporating a new amino acid due to a single nucleotide alteration.

Option C - Stop mutation:

- Rarely used in genetics; if it refers to premature termination of protein synthesis, it's equivalent to a nonsense mutation.

Option D - Silent mutation:

- Doesn't change the protein's amino acid sequence; typically has minimal impact on protein function.

Solution for Question 58:

Correct Option C - DNA replication:

- DNA methylation involves adding a methyl group to DNA, crucial for gene expression regulation. However, it's not directly involved in DNA replication. Instead, it can affect replication regulation by influencing protein binding during initiation or elongation

Incorrect Options:

Option A - Mismatch repair: DNA methylation can be associated with mismatch repair, correcting replication errors like mispaired nucleotides.

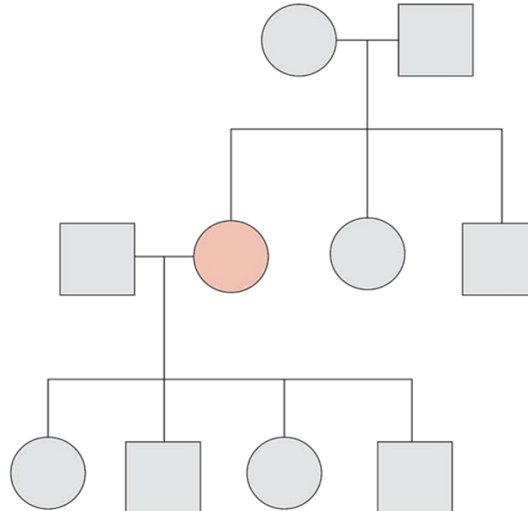
Option B - Chromatin remodeling: DNA methylation influences chromatin structure, impacting gene expression by altering DNA accessibility to cellular machinery.

Option D - Gene silencing: DNA methylation commonly leads to gene silencing by inhibiting transcription factor binding, repressing gene expression.

Solution for Question 59:

Correct Option C - Cystic fibrosis:

- The genetic condition known as cystic fibrosis has an autosomal recessive mode of inheritance.



- In recessive mode of inheritance, a person can have the disorder when they have two copies of the faulty CFTR gene, one from each parent.
- Mutations in the CFTR gene cause the formation of a thick, sticky mucus that can block airways and obstruct the pancreas.

- The CFTR gene encodes a protein that regulates the transfer of salt and water into and out of cells.
- Therefore, Cystic fibrosis follows an autosomal recessive inheritance pattern.

Incorrect Options:

Option A - Huntington's disease

- Huntington's disease is autosomal dominant, requiring only one copy of the mutated gene to manifest the disorder.

Option B - Treacher Collins syndrome

- Treacher Collins syndrome can have autosomal dominant, autosomal recessive, or sporadic inheritance patterns due to mutations in various genes affecting facial bone and tissue development.

Option D - Achondroplasia:

- Achondroplasia is also autosomal dominant, where a single copy of the mutated gene results in the characteristic skeletal abnormalities and short stature.

Solution for Question 60:

Correct Option B - Type III:

- Collagens are vital for tissue structure, including skin and connective tissues.
- Type III collagen, crucial for wound healing, aids in early-stage healing and granulation tissue formation.
- This tissue serves as a scaffold for new blood vessels and fibroblasts.
- Fibroblasts synthesize Type III collagen during the inflammatory phase, providing strength and support for tissue repair.
- Eventually, Type III collagen is replaced by Type I collagen, forming mature scar tissue.

Incorrect Options:

Option A - Type II collagen: Found in cartilage, maintaining its structure; not directly involved in wound healing.

Option C - Type IV collagen: Predominantly in basement membranes, crucial for tissue organization but not directly involved in wound healing.

Option D - Type V collagen: Present in various tissues, contributing to structural integrity but not having a prominent role in wound healing compared to Type I and Type III collagens.

Solution for Question 61:

Correct Option B - Epigenetics:

- Epigenetics means a DNA change but not the DNA code or sequence. It can alter gene expression by methylation and acetylation (chemical DNA modification). The changes are reversible and can be transmitted to the next generation. It can lead to gene activation or gene inhibition.

Incorrect Options:

Option A - Mutation is of many types and the DNA sequence changes in mutation. In point mutation, only one base is changed. Transition, Transversion, silent mutation, missense mutation, and non-sense mutation are all point mutations.

Option B - Translocation occurs when genetic material is exchanged between two different chromosomes.

Option D - Inversion is the alteration in the position of genes within the same chromosome.

Solution for Question 62:

Correct Option B - It is found in prokaryotes:

- DNA polymerase I is found in prokaryotes.
- It removes RNA primers from both leading and lagging DNA strands, leaving single or multiple gaps.
- Its main function is DNA repair, resembling eukaryotic DNA polymerase β .

Incorrect Options:

Option A - It is involved in the creation of Okazaki fragments: DNA polymerase I is not involved in creating Okazaki fragments. It is involved in filling gaps between Okazaki fragments.

Option C - It synthesizes RNA primers: DNA polymerase I do not synthesize RNA primers.

Option D - It is the primary enzyme of DNA synthesis: DNA polymerase I is not the primary enzyme of DNA synthesis. DNA polymerase III is the primary enzyme of DNA synthesis.

Solution for Question 63:

Correct Option D - DNA replication:

- DNA methylation is a biological process that involves the addition of a methyl group to the DNA molecule. DNA methylation is not directly involved in DNA replication itself. Instead, it can influence the regulation of DNA replication by affecting the binding of proteins involved in replication initiation or elongation.

Incorrect Options:

Option A - Mismatch repair: DNA methylation patterns can be recognized by mismatch repair proteins, helping to correct replication errors.

Option B - Chromatin remodeling: Methylation can affect chromatin structure, influencing gene expression by altering DNA accessibility.

Option C - Gene silencing: Methylation of DNA regions, particularly promoters, can repress gene expression by hindering transcription factor binding, leading to gene silencing.

Solution for Question 64:

Correct Option D - Involved in DNA repair:

- Telomerase is an enzyme that plays a crucial role in maintaining the length and stability of telomeres, which are the protective caps at the ends of chromosomes.

- This statement is false. Telomerase is primarily involved in telomere maintenance and lengthening, but it is not directly involved in DNA repair processes.
- DNA repair mechanisms are separate pathways responsible for fixing various types of DNA damage, such as breaks, mismatches, or lesions.

Incorrect Options:

Option A - Has reverse transcriptase activity: True. Telomerase contains a catalytic subunit with reverse transcriptase activity, aiding in synthesizing DNA using an RNA template to maintain telomere length.

Option B - Present only in Eukaryotes: This statement is true. Telomerase are only present in eukaryotes.

Option C - Maintain the chromosome length: True. Telomerase adds telomeric repeats to chromosome ends, preventing shortening during replication, essential for cellular longevity and stability.

Solution for Question 65:

Correct Option B - Nucleotide excision repair defect:

- Xeroderma Pigmentosum (XP) is characterized by extreme sensitivity to UV light and increased skin cancer risk due to a defect in DNA repair.
- The defect most likely involves nucleotide excision repair (NER), crucial for fixing UV-induced DNA damage.
- In XP, impaired NER leads to DNA mutation accumulation and skin cancer risk, along with hyperpigmentation.

Incorrect Options:

Option A - Nonhomologous end joining repair defect: Incorrect. This repair mechanism addresses double-strand breaks, not UV-induced damage like in XP

Option C - DNA base excision repair defect: Incorrect. While this mechanism fixes individual base damage, it's not directly associated with XP symptoms..

Option D - DNA mismatch repair defect: Incorrect. This mechanism corrects replication errors but isn't linked to XP symptoms like UV sensitivity or hyperpigmentation.

Solution for Question 66:

Correct Option C - Denaturation, annealing, extension:

Step 1

- Denaturation- We unwind the two strands of double-stranded DNA to form 2 single strands.
- It is carried out at 94/95 degrees (at this temperature, the hydrogen bonds will be broken).

Step 2

- Annealing-Adding primer which will anneal to the flanking sequences of the desired fragments (the temperature should neither be too high nor too low- it should be optimal)
- It should be 5 degrees < melting temperature of the primers.

Step 3

- Elongation- It should be done by the DNA polymerase, which is thermostable. The enzyme that is used in elongation is Taq. (Thermus aquaticus) DNA polymerase.

Incorrect Option A, B and D: Refer to the explanation given above

Solution for Question 67:

Correct Option A - tRNA of alanine:

- 5' capping is a post-transcriptional modification that involves the addition of a modified guanine nucleotide (7-methylguanosine) to the 5' end of RNA molecules. This modification is important for the stability, processing, and translation of the RNA molecule.

Incorrect Options:

Option B - U6 snRNA: U6 snRNA participates in splicing processes in eukaryotic cells. Its 5' capping ensures stability and proper function during splicing.

Option C - mRNA for Histone: Histone mRNA encodes proteins crucial for DNA packaging. 5' capping of histone mRNA ensures stability and efficient translation into histone proteins.

Option D - siRNA: siRNA molecules are involved in gene silencing. Their 5' capping protects them from degradation and aids in their incorporation into the RNA-induced silencing complex (RISC) for target gene suppression.

Solution for Question 68:

Correct Option D - 3-methylhistidine:

- Wilson's disease, or hepatolenticular degeneration, is a genetic disorder marked by impaired copper metabolism, causing copper accumulation in tissues like the liver, brain, and cornea.
- The defect in copper transport into bile leads to its release into the bloodstream and deposition in organs, causing damage and diverse symptoms.
- 3-methylhistidine levels decrease in urine in Wilson's disease due to impaired copper-dependent lysyl oxidase activity, affecting collagen cross-linking and muscle breakdown.

Incorrect Options:

Option A - Tyrosine is not specifically related to Wilson's disease, and its levels in urine are not typically affected.

Option B - Phosphorus is not directly associated with Wilson's disease, and its levels in urine are not typically altered.

Option C - Serine is not specifically linked to Wilson's disease, and its levels in urine are not typically affected

Solution for Question 69:

Correct Option A - Maternal inheritance:

- Maternal inheritance: Mitochondrial DNA is primarily inherited from the mother . During fertilization, the mitochondria from the sperm are typically not passed on to the offspring, while the mitochondria from the mother's egg are inherited by the child . This results in maternal inheritance of mitochondrial DNA.

Incorrect Options:

Option B - Low rate of mutation : Mitochondrial DNA has a higher mutation rate compared to nuclear DNA. This is because mitochondria lack certain DNA repair mechanisms present in the nucleus. The higher mutation rate contributes to genetic diversity and can lead to mitochondrial diseases.

Option C - Single stranded DNA : Mitochondrial DNA is double-stranded, not single-stranded. It contains both coding and non-coding regions necessary for the production of mitochondrial proteins.

Option D - Paternal inheritance : While the paternal mitochondrial DNA is typically not passed on to the offspring, there have been rare cases where paternal transmission of mitochondrial DNA has been observed.

Solution for Question 70:

Correct Option D - Achondroplasia:

- Autosomal dominant inheritance means that a single copy of the mutated gene is sufficient to cause the disorder.
- Achondroplasia is caused by a mutation in the fibroblast growth factor receptor 3 (FGFR3) gene and follows an autosomal dominant pattern of inheritance . A single copy of the mutated gene is sufficient to cause the characteristic features of achondroplasia, a form of dwarfism.

Incorrect Options:

Option A - Cystic fibrosis is an autosomal recessive disorder.

Option B - Duchene muscular dystrophy is an X-linked recessive disorder.

Option C - Sickle cell anemia is also an autosomal recessive disorder.

Solution for Question 71:

Correct Option A - Albinism:

- Albinism: Albinism is a genetic condition characterized by a lack of pigmentation in the hair, skin, and eyes. It is caused by various mutations in genes involved in the production of melanin. Albinism follows an autosomal recessive pattern of inheritance. Individuals with one copy of the mutated gene are carriers but typically do not show symptoms.

Incorrect Options:

Option B - G6PD deficiency: G6PD deficiency follows an X-linked recessive pattern of inheritance.

Option C - Marfan's syndrome: Marfan's syndrome is inherited in an autosomal dominant manner.

Option D - Lesch-Nyhan syndrome: Lesch-Nyhan syndrome follows an X-linked recessive pattern of inheritance, primarily affecting males. Females can be carriers of the condition.

Solution for Question 72:

Correct Option A - Acrocentric:

- An acrocentric chromosome has the centromere located near one end, resulting in a long arm and a very short (or almost nonexistent) short arm.
- Human chromosomes 13, 14, 15, 21, and 22 are acrocentric.
- Acrocentric chromosomes are characterized by having satellites or small structures called stalks attached to their short arms, which contain multiple copies of ribosomal RNA genes.

Incorrect Options:

Option B. Metacentric:

- A metacentric chromosome has the centromere located approximately in the middle, resulting in two arms of roughly equal length.
- Human chromosomes 1, 3, 16, 19, and 20 are metacentric.
- The centromere's central location allows the chromosome to form a characteristic "V" shape during cell division.

Option C. Submetacentric:

- A submetacentric chromosome has the centromere slightly off-center, resulting in one arm being longer than the other.
- Human chromosomes 2, 4, 5, 6, 7, 8, 9, 10, 11, 12, 17, and 18 are submetacentric.
- These chromosomes have a more "L" shape during cell division due to the unequal arm lengths.

Option D. Telocentric:

- A telocentric chromosome has the centromere at one end, resulting in a chromosome that has only one arm.
- Humans do not have telocentric chromosomes, but they are found in some species such as mice.
- These chromosomes appear to have a straight line shape with no visible short arm.

Solution for Question 73:

Correct Option C - 3'UTR:

- The miRNA along with the proteins forms the RNA-induced silencing complex (RISC complex) and binds to the 3' end of the target mRNA to be degraded. Thus, miRNA is synthesized from DNA and it binds to 3' UTR.

Incorrect Options:

Option A - Gene body: Incorrect. miRNAs do not typically bind to the gene body for translation inhibition.

Option B - Gene promoter: Incorrect. miRNAs do not bind to gene promoters; these regions signal transcription initiation, not translation.

Option D - 5' UTR: Incorrect. While elements in the 5'UTR can affect translation, miRNAs primarily act by binding to the 3'UTR, not the 5'UTR.

Solution for Question 74:

Correct Option B - 1 and 2:

- The mitochondrial DNA is circular ds DNA containing about 16,000 base pairs and 37 genes. Mitochondrial DNA has a high rate of mutations as compared to nuclear DNA due to no introns (introns prevent mutation). There is no DNA repair mechanism in mitochondria. Thus, 1 and 2 are correct.

Incorrect Options:

Option A, C and D:

- 3. Mitochondrial DNA is present only in eukaryotes and resembles prokaryotic DNA
 - 4. The electron transport chain requires 67 proteins, wherein 13 out of the 67 proteins are derived from mitochondrial DNA.
-

Solution for Question 75:

Correct Option D - RTPCR followed by real-time PCR:

- RT-PCR is used to convert RNA into cDNA, and then real-time PCR is performed to quantify the cDNA. Hence, this approach can be suitable for quantitative analysis of RNA.

Incorrect Options:

Option A - Real-time PCR: Real-time PCR can analyze DNA as it is getting synthesized, enabling us to know if there is a problem in the amplification process.

Option B - Reverse transcriptase PCR: RT-PCR converts RNA into complementary DNA (cDNA) using reverse transcriptase enzyme before amplification. While useful for RNA detection and amplification, RT-PCR lacks direct quantitative analysis capability, requiring additional steps like gel electrophoresis or hybridization for quantification.

Option C - PCR-restriction fragment length polymorphism (PCR-RFLP) is a technique used to detect genetic variations by analyzing the length differences in PCR-amplified DNA fragments after digestion with specific restriction enzymes. PCR-RFLP is not suitable for the quantitative analysis of RNA samples.

Enzymes

1. Which of the following examples best illustrates competitive inhibition?

- A. Methotrexate and Dihydrofolate Reductase
 - B. Cyanide and Cytochrome c Oxidase
 - C. Alanine and Pyruvate Kinase
 - D. Citrate and Phosphofructokinase-1(PFK-1)
-

2. Which of the following statements is false on creatine kinase?

- A. Total creatine kinase is estimated in muscular dystrophies
 - B. Starts to rise within 3-6 hours of infarction
 - C. Increased in hemolysis and congestive heart failure
 - D. CK-MB is used as a marker in myocardial infarction
-

3. Which of the following is used to study the kinetics of oligomeric enzymes?

- A. Lineweaver Burke equation
 - B. Michaelis Menten equation
 - C. Hill equation
 - D. Dixon equation
-

4. Which among the following statements regarding the induction and repression of enzymes is NOT true?

- A. Repression acts at the gene level to decrease the amount of enzyme
 - B. Induction relieves repression on the operator site
 - C. Glucokinase is induced by insulin
 - D. ALA(Aminolevulinic acid) synthase is induced by heme
-

5. All of the following enzymes are proteins except?

- A. Aminoacyl tRNA synthetase
 - B. Peptidyl transferase
 - C. RNA polymerase
 - D. Acetyl-CoA synthetase
-

6. In a Lineweaver-Burk plot for an enzyme experiencing non-competitive inhibition, which of the following statements is correct?

- A. The slope of the Lineweaver-Burk plot increases, but the y-intercept remains the same.
 - B. The y-intercept increases while K_m remains constant and the slope increases.
 - C. Both the slope and the y-intercept increase.
 - D. The slope of the Lineweaver-Burk plot decreases while the y-intercept increases.
-

7. Which of the following statements accurately describes the changes in K_m (Michaelis constant) and V_{max} (maximum reaction rate) in the presence of an uncompetitive inhibitor?

- A. K_m increases, V_{max} increases
- B. K_m decreases, V_{max} decreases
- C. K_m stays the same, V_{max} decreases

D. K_m increases, V_{max} stays the same

8. Which of the following statements are true regarding enzymes? a) Enzymes are proteins that can accelerate reactions by factors of 10^6 to 10^9 or more. b) Enzymes act on a wide range of substrates. c) Enzymes stabilize the ground state. d) Enzymes can catalyze both forward and reverse reactions. their e) Enzymes are regulated by pH, temperature, inhibitors, and activators. f) Enzyme are water insoluble

A. a, d, e, f, c

B. a, b, d, e

C. a, d, e

D. a, e, f, c

9. Given the following enzyme kinetics parameters: Maximum velocity (V_{max}): 100 $\mu\text{mol/min}$ Michaelis constant (K_m): 10 μM Initial velocity (V_o): 50 $\mu\text{mol/min}$ What is the substrate concentration $[S]$ of the given enzyme?

A. 5 μM

B. 10 μM

C. 20 μM

D. 25 μM

10. In enzyme kinetics, a double reciprocal plot (Lineweaver-Burk plot) is used to determine kinetic parameters. What do the x-intercept and y-intercept of a double reciprocal plot represent?

A. x-intercept: $-1/K_m$; y-intercept: $1/V_{max}$

- B. x-intercept: $1/K_m$; y-intercept: $1/V_{max}$
 - C. x-intercept: $1/V_{max}$; y-intercept: $-1/K_m$
 - D. x-intercept: $-1/K_m$; y-intercept: $-1/V_{max}$
-

11. Which of the following statements regarding Michaelis Menten equation is true?

- A. Used to calculate the rate of reaction
 - B. Substrate concentration is required to calculate the rate using Michaelis Menten equation
 - C. Michaelis constant is the substrate concentration at which the velocity of the reaction is half the maximal velocity
 - D. All are true
-

12. What is the correct equation to calculate the catalytic efficiency of an enzyme?

- A. K_{cat} / K_m
 - B. K_m / K_{cat}
 - C. V_{max} / S_t
 - D. V_{max} / K_m
-

13. Which of the following statements is false Increase in enzyme concentration increases the rate of reaction Increase in substrate concentration increases the rate of reaction Increase in product concentration increases the rate of reaction

- A. 1, 3
- B. 3
- C. 1, 2, 3

D. None

14. Which of the following statements is/are true regarding the mechanisms of action of enzymes? Enzymes that act by altering the strain of substrates catalyze lytic reactions Enzymes forming covalent bonds are seen commonly in group transfer reactions When the functional groups of aminoacyl side chains of enzymes act as an acid or base, the rate of reaction increases

- A. 1, 2
 - B. 2, 3
 - C. 1, 3
 - D. 1, 2, 3
-

15. A 35-year-old female with chronic diarrhea, weight loss, and fatty stools complains that her symptoms worsen after a high-fat meal. Reduced activity of pancreatic lipase is suspected. Which of the following ions is most likely to be low in this patient?

- A. Calcium
 - B. Copper
 - C. Iron
 - D. Zinc
-

16. A patient presents with symptoms of fatigue and muscle weakness. Laboratory tests indicate a deficiency in riboflavin. Which coenzyme's function is most likely affected?

- A. NAD

- B. FAD
 - C. Coenzyme A
 - D. Folic acid
-

17. Which of the following is not a function of prosthetic groups?

- A. Substrate binding and orientation
 - B. Formation of covalent bonds
 - C. Electron transfer
 - D. Protein degradation
-

18. Match the enzyme class with its corresponding example: Enzyme Class

Example A) Oxidoreductases 1) Hexokinase B) Transferases 2) Lactate dehydrogenase C) Hydrolases 3) DNA ligase D) Lyases 4) Trypsin E) Isomerases 5) Aldolase F) Ligases 6) Triosephosphate isomerase

Enzyme Class	Example
A) Oxidoreductases	1) Hexokinase
B) Transferases	2) Lactate dehydrogenase
C) Hydrolases	3) DNA ligase
D) Lyases	4) Trypsin
E) Isomerases	5) Aldolase
F) Ligases	6) Triosephosphate isomerase

- A. A-2, B-1, C-4, D-5, E-6, F-3
- B. A-1, B-2, C-4, D-5, E-3, F-6
- C. A-2, B-4, C-5, D-1, E-6, F-3

D. A-3, B-2, C-4, D-5, E-6, F-1

19. What is the role of activation energy in a chemical reaction, and how do enzymes influence it?

- A. Activation energy converts molecules from transition to ground state; enzymes increase it.
 - B. Activation energy converts molecules from ground state to transition state; enzymes lower it.
 - C. Activation energy is released during product formation; enzymes do not affect it.
 - D. Activation energy breaks bonds in products; enzymes increase it.
-

20. Which of the following statements about suicide inhibition is incorrect?

- A. The inhibitor covalently binds to the enzyme after being converted to a reactive intermediate.
 - B. Suicide inhibition is a reversible form of enzyme inhibition.
 - C. Suicide inhibitors resemble the enzyme's natural substrate.
 - D. Allopurinol acts as a suicide inhibitor of xanthine oxidase to manage gout.
-

21. Which of the following biochemical reactions is not regulated by allosteric regulation?

- A. Phosphofructokinase-1 in glycolysis
- B. Aspartate transcarbamoylase in pyrimidine biosynthesis
- C. Glycogen phosphorylase in glycogenolysis
- D. Glucokinase in glycolysis

22. Which of the following shapes of the Michaelis-Menten plot is observed for allosteric regulation?

- A. Hyperbolic curve
 - B. Sigmoidal curve
 - C. Linear increase
 - D. Parabolic curve
-

23. Match the following regulation mechanisms with their respective enzymatic reactions:

Regulation mechanisms	Enzymatic reactions
1. Compartmentation	a. Phosphorylation of glycogen phosphorylase
2. Feedback inhibition	b. Negative regulation of phosphofructokinase by ATP
3. Covalent modification	c. Segregation of glycolytic enzymes in the cytosol
4. Proteolytic activation	d. Activation of pepsinogen to pepsin

Regulation mechanisms	Enzymatic reactions
1. Compartmentation	a. Phosphorylation of glycogen phosphorylase
2. Feedback inhibition	b. Negative regulation of phosphofructokinase by ATP
3. Covalent modification	c. Segregation of glycolytic enzymes in the cytosol
4. Proteolytic activation	d. Activation of pepsinogen to pepsin

- A. 1 - c, 2 - d, 3 - a, 4 - b
 - B. 1 - b, 2 - c, 3 - d, 4 - a
 - C. 1 - c, 2 - b, 3 - a, 4 - d
 - D. 1 - d, 2 - a, 3 - b, 4 - c
-

24. Which of the following statements about isoenzymes is incorrect?

- A. Isoenzymes can be distinguished by their different amino acid sequences or structural properties
 - B. Allozymes are produced by different genes and can have distinct K_m values or substrate affinities
 - C. Isoforms result from post-translational modifications, such as changes in sialic acid content in alkaline phosphatase (ALP)
 - D. Electrophoresis is a technique used to separate isoenzymes based on their mobility.
-

25. A 45-year-old male presents with severe pain and swelling in his right calf following an intense workout session two days ago. Laboratory tests, including measurements of lactate dehydrogenase (LDH) isoenzymes, are ordered. Which LDH isoenzyme is most indicative of the patient's condition?

- A. LDH-1
 - B. LDH-2
 - C. LDH-3
 - D. LDH-5
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	4
Question 5	2
Question 6	2
Question 7	2
Question 8	3
Question 9	2
Question 10	1
Question 11	4
Question 12	1
Question 13	2
Question 14	4
Question 15	1
Question 16	2
Question 17	4
Question 18	1

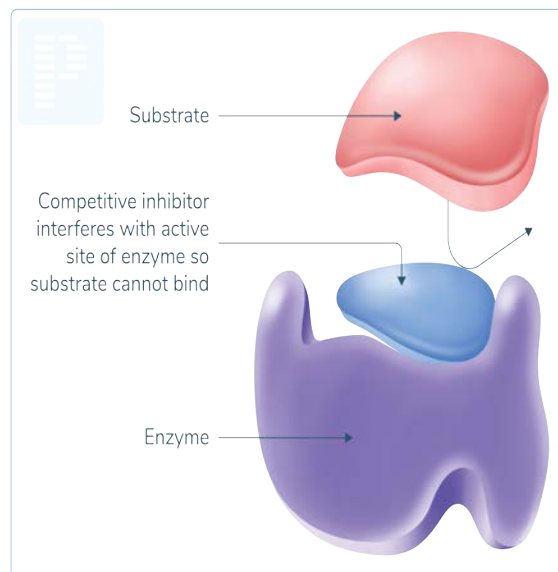
Question 19	2
Question 20	2
Question 21	4
Question 22	2
Question 23	3
Question 24	2
Question 25	4

Solution for Question 1:

Correct Answer: A) Methotrexate and Dihydrofolate reductase

Explanation:

Competitive inhibition occurs when an inhibitor molecule resembles the substrate and competes with the substrate for binding to the active site of the enzyme. It is usually reversible.



Competitive inhibitors

clinical correlation

Methotrexate and Dihydrofolate Reductase	Methotrexate acts as a competitive inhibitor by mimicking dihydrofolate, the natural substrate of DHFR, and binding to its active site, thereby preventing dihydrofolate from binding and inhibiting the enzyme's activity.
Sulfa Drugs and PABA	Sulfa drugs, such as sulfanilamide, competitively inhibit the enzyme dihydropteroate synthase by mimicking para-aminobenzoic acid (PABA), a substrate involved in folic acid synthesis in bacteria.
Ethanol and Alcohol Dehydrogenase	Ethanol can act as a competitive inhibitor of methanol metabolism by competing for the active site of alcohol dehydrogenase. This is used in cases of methanol poisoning.
Captopril and Angiotensin-Converting Enzyme (ACE)	Captopril competes with the natural substrate of ACE, thereby inhibiting the production of angiotensin II, a peptide that increases blood pressure.

Cyanide and Cytochrome c Oxidase (Option B): Cyanide and Cytochrome C Oxidase describe non-competitive inhibitions.

Alanine and Pyruvate Kinase (Option C): Alanine and Pyruvate Kinase describe non-competitive inhibitions.

Citrate and Phosphofructokinase-1 (PFK-1) (Option D): Citrate and Phosphofructokinase-1(PFK-1) describes Allosteric Inhibition.

Solution for Question 2:

Correct Answer: C) Increased in hemolysis and congestive heart failure

Explanation:

Creatine kinase is not increased in hemolysis or congestive heart failure

Creatine kinase:

- Catalyses conversion of creatine phosphate to creatine using ATP
- Normal value is 15-100 U/L for males and 10-80 U/L for females
- Starts to rise within 3-6 hours of infarction (Option B)
- 3 isoenzymes - MM, MB, BB

- Total CK estimated in muscular dystrophy (Option A)
- CK-MB estimated in myocardial infarction (Option D)

Solution for Question 3:

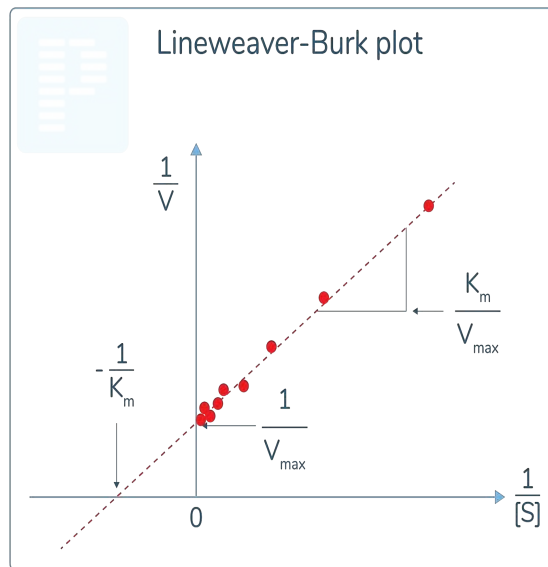
Correct Answer: C) Hill equation

Explanation:

- Hill equation relates reaction velocity to substrate concentration in oligomeric enzymes
- Oligomeric enzymes often exhibit cooperative binding behavior due to their multiple substrate-binding sites
- They show sigmoidal kinetics

Lineweaver-Burk Equation: Michaelis-Menten equation in a linear form (Option A)

$$\frac{1}{V} = \frac{K_m + [S]}{V_{\max} [S]} = \frac{K_m}{V_{\max}} \frac{1}{[S]} + \frac{1}{V_{\max}}$$



Michaelis-Menten Equation (Option B): Relates reaction rate to substrate concentration, might not accurately describe the kinetics of oligomeric enzymes.

Dixon Equation (Option D): A method for determining the inhibition constant of an enzyme inhibitor by plotting the inverse of the reaction rate against the inhibitor concentration at different substrate concentrations

Solution for Question 4:

Correct Answer: D) ALA(Aminolevulinic acid) synthase is induced by heme

Explanation:

- ALA (Aminolevulinic acid) synthase is induced by barbiturates and repressed by heme
- Induction and repression of enzymes increase and decrease enzyme biosynthesis respectively at the gene level

Mechanism	Process	Examples
-----------	---------	----------

Induction	Relieves repression on the operator site, increasing enzyme biosynthesis (Option B)	Lactose-utilizing enzymes in bacteria induced by lactose in the absence of glucose Tryptophan pyrrolase and transaminases induced by glucocorticoids Glucokinase induced by insulin (Option C) ALA (Aminolevulinic acid) synthase induced by barbiturates
Repression	Acts at the gene level, reducing the number of enzyme molecules (Option A)	ALA (Aminolevulinic acid) synthase is repressed by heme

Solution for Question 5:

Correct Answer: B) Peptidyl transferase

Explanation:

Peptidyl transferase is RNA enzymes/Ribozymes while others are proteins

Ribozymes/RNA enzymes:

Point	Description
Definition	Enzymes made up of RNA
Spliceosomes	Ribozymes + protein components (stabilize the structure of ribozymes)
Examples	RNase-P: generates the ends of tRNAs Peptidyl transferase: Present in ribosomes and used for protein biosynthesis

Solution for Question 6:

Correct Answer: B) The y-intercept increases while K_m remains constant and the slope increases.

Explanation:

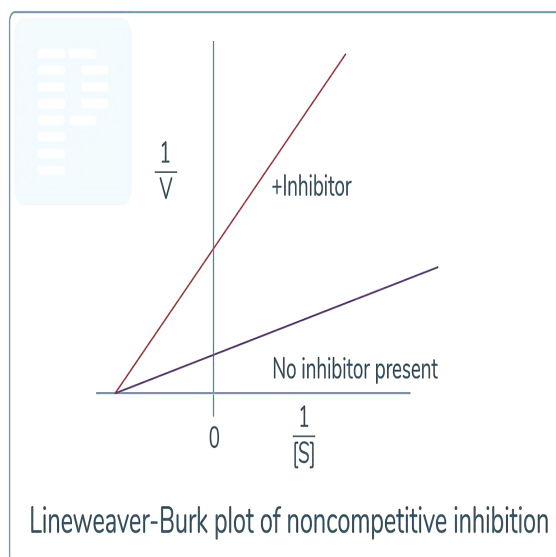
Non-competitive inhibition occurs when an inhibitor binds to an enzyme at a site other than the active site, which is called an allosteric site. This binding changes the enzyme's shape and reduces its activity, regardless of whether the substrate is present.

Effect on the Lineweaver-Burk Plot:

The lineweaver burk plot is graphical representation of enzyme kinetics representing V_{max} and K_m .

- Effect on V_{max} : The y-intercept of the Lineweaver-Burk plot, which represents $1/V_{max}$ increases as V_{max} decreases.
- Effect on K_m : The slope of the plot, which is K_m/V_{max} , remains constant because K_m does not change.

The y-intercept increases while the K_m remains constant for non competitive inhibition.



Examples of non-competitive inhibitors:

- Cyanide acts as a non-competitive inhibitor by binding to a site on cytochrome c oxidase that is not the active site, thereby disrupting its function.
- Fluoride functions as a non-competitive inhibitor by chelating essential metal ions (magnesium or manganese) required for enolase activity, thus inhibiting the enzyme.

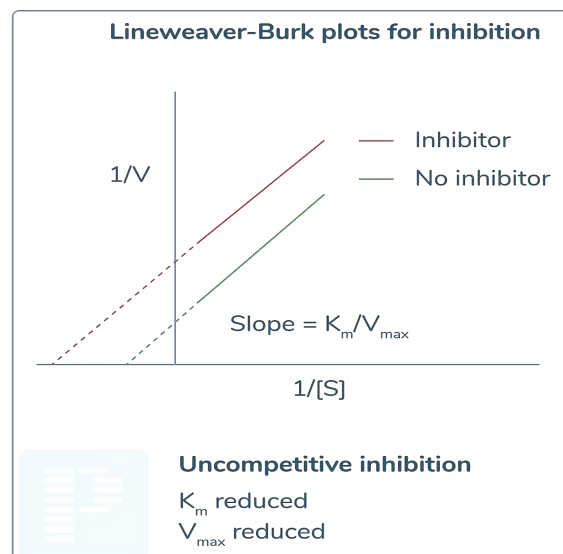
Solution for Question 7:

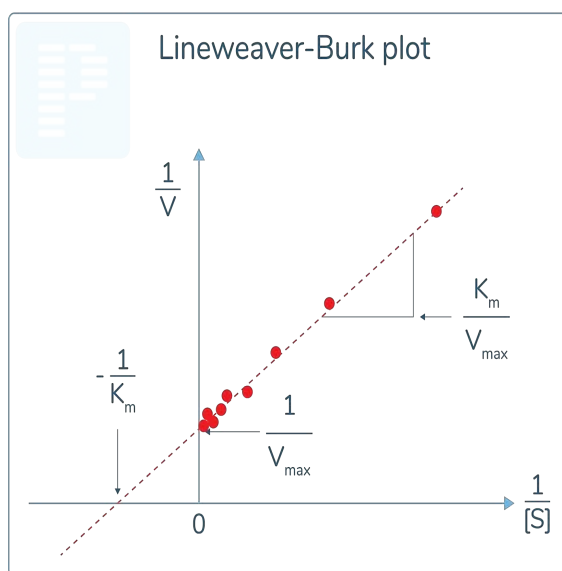
Correct Answer: B) K_m decreases, V_{max} decreases

Explanation:

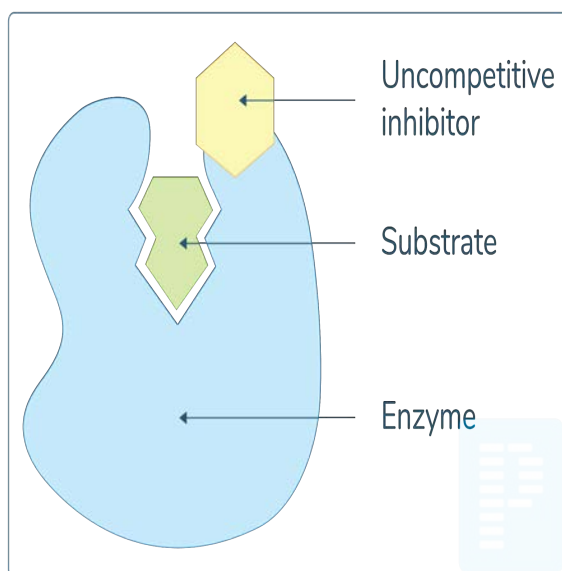
Uncompetitive Inhibition occurs when an inhibitor binds only to the enzyme-substrate complex, preventing it from converting the substrate into product. This type of inhibition results in:

- Decrease in K_m : Since the inhibitor binds to the enzyme-substrate complex, it effectively lowers the concentration of the enzyme available for reaction, making the apparent affinity of the enzyme for the substrate appear higher
- Decrease in V_{max} : The inhibitor reduces the maximum rate of reaction because it prevents the enzyme-substrate complex from proceeding to form the product, thus lowering the overall reaction rate at saturating substrate concentrations.





Therefore, in Uncompetitive Inhibition, both K_m and $V_{\max}$ decrease.



Examples of Uncompetitive inhibition:

Uncompetitive inhibitors	Clinical correlation
Lithium and Inositol Monophosphatase	This inhibition disrupts the phosphatidylinositol signaling pathway, which is important in mood regulation, making lithium effective in treating bipolar disorder.
Phenylalanine and Alkaline Phosphatase	It binds to the enzyme-substrate complex, reducing the enzyme's activity. This mechanism is studied in the context of metabolic disorders involving phenylalanine.

Solution for Question 8:

Correct Answer: C) a, d, e

Explanation:

Properties of Enzyme	
Nature of Enzymes	Enzymes are proteins, heat-sensitive, water-soluble. (Statement f ruled out)
Efficiency	Enzymes can accelerate reactions by factors 10^6 to 10^9 or more. (Statement a)
Specificity	Enzymes act on specific substrates and can distinguish stereoisomers. (Statement b ruled out)
Cofactors/Coenzymes	Some enzymes need non-protein molecules (cofactors or coenzymes) to be active.
Substrate Binding	Enzymes bind substrates at multiple points, creating chiral products from non-chiral substrates.
Transition State	Enzymes stabilize the transition state, lowering activation energy. (Statement c ruled out)
Reversibility	Enzymes can catalyze both forward and reverse reactions. (Statement d)
Regulation	Activity is regulated by pH, temperature, inhibitors, and activators. (Statement e)

Solution for Question 9:

Correct Answer: B) 10 μ M

Explanation:

The substrate concentration [S] represents the amount of substrate available for the enzyme to act upon in a given reaction. It is typically measured in units of concentration, such as micromoles per liter (μM) or millimoles per liter (mM).

To find the substrate concentration [S], you can use the Michaelis-Menten equation:

$$V_o = V_{\text{max}}[S] / K_m + [S]$$

Given:

(V_{max}): $100 \mu\text{mol/min}$

(K_m): $10 \mu\text{M}$

(V_o): $50 \mu\text{mol/min}$

Substitute the given values into the equation:

$$50 = 100[S] / 10 + [S]$$

$$50 (10 + [S]) = 100 [S]$$

Simplify and solve for [S]:

$$500 + 50[S] = 100[S]$$

$$500 = 50[S]$$

$$[S] = 10 \mu\text{M}$$

Therefore, the substrate concentration [S] is $10 \mu\text{M}$.

Solution for Question 10:

Correct Answer: A) x-intercept: $-1/K_m$; y-intercept: $1/V_{\text{max}}$

Explanation :

- Double reciprocal plots typically are used both to distinguish between competitive and non-competitive inhibitors and to simplify evaluation of inhibition constants.
- The Lineweaver-Burk plot is a double reciprocal plot used to linearize the Michaelis-Menten equation. The Michaelis-Menten equation is:

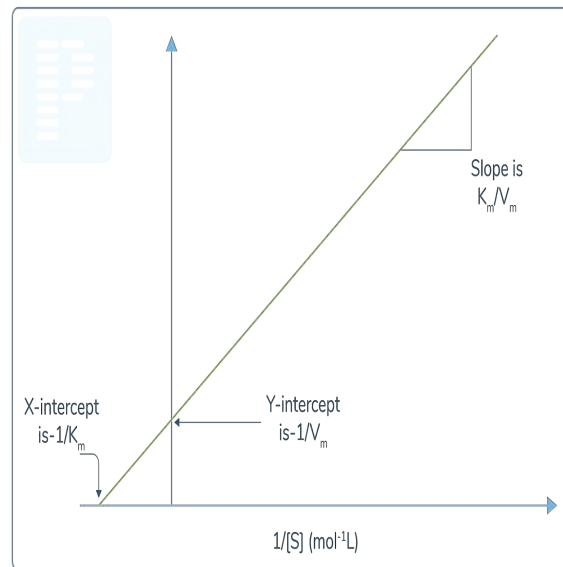
$$V = V_{\text{max}}[S]/K_m + [S]$$

Taking the reciprocal of both sides, we get the Lineweaver-Burk equation:

$$1/V = K_m/V_{max}[S] + 1/V_{max}$$

In this linear form, plotting $1/V$ (y-axis) against $1/[S]$ (x-axis) yields a straight line.

The x-intercept of this plot is $-1/K_m$ and the y-intercept is $1/V_{max}$.



Solution for Question 11:

Correct Answer: D) All are true

Explanation:

Michaelis Menten equation:

$$V = V_{max} \times S / (K_m + S)$$

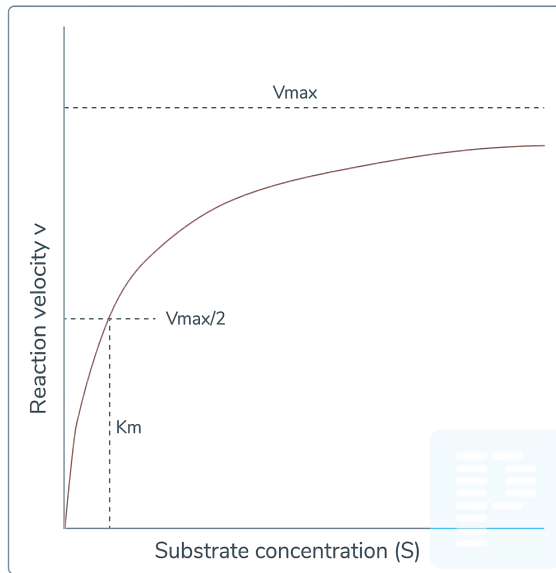
Where,

V = Velocity of reaction

V_{max} = Maximum velocity of the reaction

S = Substrate concentration

K_m = Michaelis constant = Substrate concentration at which the velocity of the reaction is half the maximal velocity ($V_{max}/2$)



Solution for Question 12:

Correct Answer: A) K_{cat} / K_m

Explanation:

The catalytic efficiency of an enzyme is the ratio of K_{cat} to K_m

Catalytic Efficiency = K_{cat} / K_m

Where,

- K_{cat} = Catalytic constant = The catalytic activity of homogeneous enzyme = Ratio of the maximum velocity of the reaction to the number of active sites on the enzymes

$K_{cat} = V_{max} / S_t$

- K_m = Michaelis constant = Substrate concentration at which the velocity of the reaction is half the maximal velocity ($V_{max} / 2$)

Solution for Question 13:

Correct Answer: B) 3

Explanation:

Increased product concentration slows, stops, or reverses the reaction

Factor	Effect on Reaction Rate
Enzyme Concentration	Rate of reaction increases with an increase in enzyme concentration
Substrate Concentration	Initial phases: Velocity increases with increasing substrate concentration V _{max} : Maximum velocity attainable at substrate saturation level Once V _{max} is attained, changes in substrate concentration will not affect the rate of reaction
Product Concentration	Increased product concentration slows, stops, or reverses the reaction
Temperature	Increasing temperature increases reaction velocity Velocity reaches a maximum at an optimum temperature and then falls with further increase
pH	Each enzyme has an optimum pH pH value is nearer to optimum→reaction rate increases

Solution for Question 14:

Correct Answer: D) 1, 2, 3

Explanation:

Different enzymes have different mechanisms to catalyze a reaction

Mechanism	Description
Catalysis by Proximity	Substrates come in bond-forming proximity to enzymes→Enzymes bind substrate molecules at their active site→region of high local substrate concentration created→substrates are ideally positioned to chemically interact→reaction is catalyzed

Acid-Base Catalysis	Ionizable functional groups of aminoacyl side chains and prosthetic groups of enzymes act as acids or bases causing catalysis (Statement 3)
Catalysis by Strain	Enzymes bind substrates in a conformation that weakens the bond targeted for cleavage→ catalyses lytic reactions (Statement 1)
Covalent Catalysis	Formation of a covalent bond between the enzyme and one or more substrates catalyzes the reaction Common among enzymes that catalyze group transfer reactions (Statement 2)

Solution for Question 15:

Correct Answer: A) Calcium

Explanation:

Pancreatic lipase is an ion-activated enzyme. Its activity is enhanced by adding calcium ions, which is its cofactor.

- A cofactor is a collective term that includes coenzymes and metal ions
- Enzymes that require certain metal ions for their activity are called metalloenzymes

Metal-ion (cofactor)	Enzymes
Calcium (Option A)	Lecithinase, lipase
Copper (Option B)	Tyrosinase, cytochrome oxidase, superoxide dismutase
Iron (Option C)	Cytochrome oxidase, catalase, peroxidase, xanthine oxidase
Zinc (Option D)	Carbonic anhydrase, carboxypeptidase, alcohol dehydrogenase
Magnesium	Hexokinase, enolase, glucose-6-phosphatase
Manganese	Phosphoglucosmutase, glycosyl transferases
Molybdenum	Xanthine oxidase

Solution for Question 16:

Correct Answer: B) FAD

Explanation:

FAD is derived from riboflavin and is essential for redox reactions in the electron transport chain.

Coenzyme Type	Source (Vitamin)	Function	Example Reaction
NAD/NADP (Option A)	Nicotinamide (B3)	Redox reactions	Cellular respiration
FAD / FM (Option B)	Riboflavin (B2)	Redox reactions	Electron transport chain
Coenzyme A (Option C)	Pantothenic acid (B5)	Acyl group transfer	Fatty acid metabolism
Folic acid/ Cobamide (Option D)	Folate (B9), B12	One-carbon metabolism	DNA synthesis

Solution for Question 17:

Correct Answer: D) Protein degradation

Prosthetic groups	
Definition	Non-protein molecules tightly bound to proteins via covalent or non-covalent bonds. Essential for enzyme catalytic activity.
Examples	Pyridoxal Phosphate (PLP): Amino acid metabolism. Flavin Mononucleotide (FMN) and Flavin Adenine Dinucleotide (FAD): Redox reactions. Thiamin Pyrophosphate (TPP): Decarboxylation reactions.

Functions	Substrate Binding and Orientation: Proper orientation of substrates for catalytic reaction. (Option A) Formation of Covalent Bonds: Covalent bonds with reaction intermediates. (Option B) Electron Transfer: Facilitates electron transfer in redox reactions. (Option C) Activation of Substrates: Acts as Lewis acids or bases to enhance substrate reactivity.
Characteristics	Stable Incorporation: Tightly bound to enzymes, unlike transient cofactors. Role in Catalysis: Enhance enzyme capabilities beyond amino acid side chains. Versatility: Involved in various reactions like oxidation-reduction, group transfer, covalent bond formation.

Solution for Question 18:

Correct Answer : A) A-2, B-1, C-4, D-5, E-6, F-3

Explanation:

Class	Function	Examples
Oxidoreductases	Catalyze oxidation-reduction reactions	Lactate dehydrogenase, glucose-6-phosphate dehydrogenase
Transferases	Transfer functional groups	Hexokinase, aminotransferase
Hydrolases	Catalyze hydrolytic cleavage of bonds	Trypsin, acetylcholine esterase

Lyases	Cleave bonds by means other than hydrolysis/oxidation	Aldolase, decarboxylases
Isomerases	Catalyze rearrangement of atoms	Triosephosphate isomerase, racemases
Ligases	Join two molecules, using ATP	DNA ligase, acetyl-CoA synthetase

Solution for Question 19:

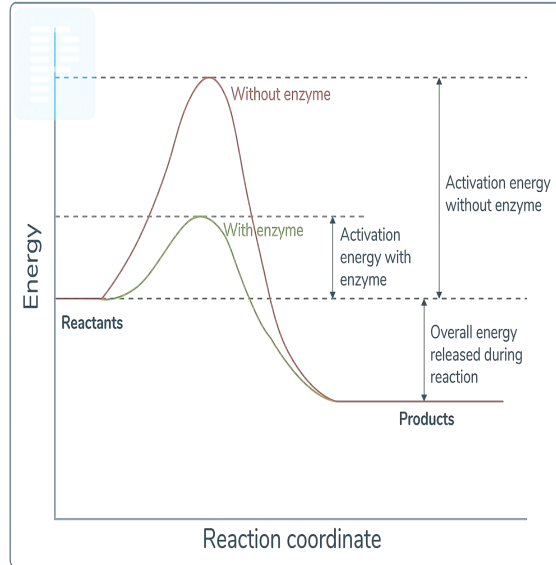
Correct Answer: B) Activation energy converts molecules from ground state to transition state; enzymes lower it.

Explanation

Activation energy	
Definition	Activation energy is the energy required to convert molecules from the ground state to the transition state (An unstable intermediate with partially formed and broken bonds. Activation energy is a barrier that must be overcome for a reaction to proceed; typically positive, indicating an energy requirement.
Role of Enzymes	Enzymes lower activation energy, making it easier for reactions to occur.
Reaction Rate Factors	Reaction rate increases as activation energy decreases; temperature increases kinetic energy, leading to more frequent and energetic collisions.
Visualization	Transition state visualized as neither free substrate nor product, with partial bonds forming and rupturing.
Substrate Binding	Enzymes optimize weak interactions with substrates, providing the driving force for catalysis.
Influencing Factors	Enzyme activity is affected by pH, temperature, and presence of inhibitors or activators.

Example

Acid hydrolysis of sucrose: Activation energy with enzyme (sucrase) is 9,000 cal/mol while without enzyme is around 26,000 cal/mol.



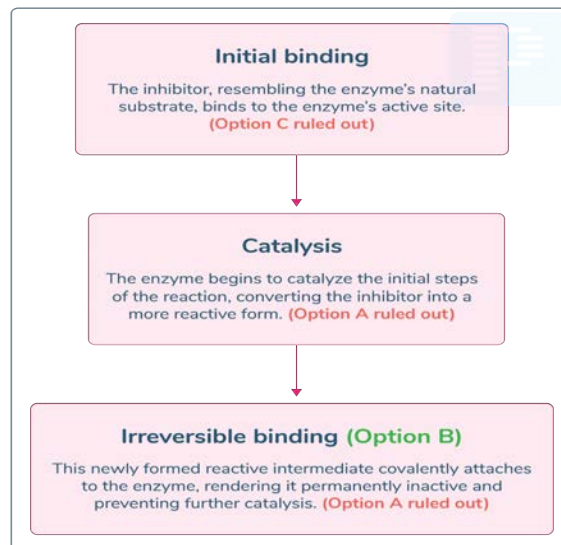
Solution for Question 20:

Correct Answer: B) Suicide inhibition is a reversible form of enzyme inhibition.

Explanation:

Suicide inhibition, is a special type of irreversible enzyme inhibition.

Process of Suicide Inhibition:



Examples of Suicide Inhibition:

Enzyme	Inhibitor	Mechanism	Clinical Application
Ornithine Decarboxylase (ODC)	Difluoromethylornithine (DFMO)	DFMO binds to ODC, is converted to a reactive form, and irreversibly inactivates the enzyme.	Treats trypanosomiasis by stopping polyamine synthesis in Trypanosoma parasites.
Xanthine Oxidase (Option D)	Allopurinol	Allopurinol is oxidized to alloxanthine, which forms an irreversible complex with xanthine oxidase.	Manages gout by reducing uric acid levels.
Cyclo-oxygenase (COX)	Aspirin	Aspirin acetylates a serine residue in COX, irreversibly inhibiting the enzyme.	Reduces inflammation and pain, and prevents heart attacks and strokes.

Solution for Question 21:

Correct Answer: D) Glucokinase in glycolysis

Explanation:

Glucokinase catalyzes the first step of glycolysis, converting glucose to glucose-6-phosphate. Unlike some other enzymes, glucokinase is primarily regulated by product inhibition (glucose-6-phosphate) rather than allosteric regulation.

- Allosteric regulation is a mechanism where an enzyme's activity is modified by the binding of an effector molecule at a site other than the active site, known as the allosteric site.
- Allosteric Site: A specific location on an enzyme distinct from the active site where effector molecules bind, inducing conformational changes that influence enzyme activity.
- Effectors: Molecules that bind to the allosteric site and modify enzyme activity.
- Conformational Changes : Binding of an effector to the allosteric site causes a conformational change in the enzyme, which can alter the enzyme's affinity for its substrate and modify the enzyme's catalytic efficiency.

Examples of Allosteric Regulation

Enzyme	Allosteric Inhibitor	Allosteric Activator	Notes
Phosphofructokinase (Option A)	ATP, citrate	AMP, F-2,6-P	Key regulatory enzyme in glycolysis
Aspartate transcarbamoylase (Option B)	CTP	ATP	Important in pyrimidine synthesis
Glycogen phosphorylase (Option C)	ATP, glucose-6-P	AMP	Regulates glycogen breakdown
Pyruvate kinase	ATP, alanine	F-1,6-BP	Regulates glycolysis

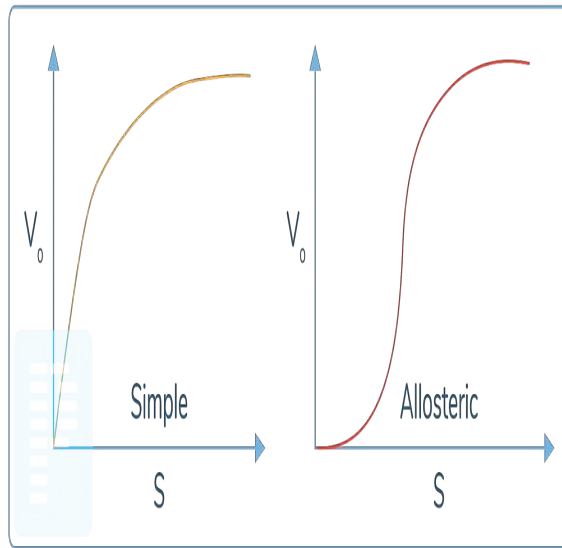
Solution for Question 22:

Correct Answer: B) Sigmoidal curve

Explanation:

A sigmoidal curve, typical of allosteric enzymes, shows cooperative binding. Initially, reaction velocity rises slowly due to low substrate affinity. As more substrates bind, conformational changes enhance enzyme affinity, causing a sharp increase in velocity before saturation.

In enzyme kinetics, a hyperbolic curve in Michaelis-Menten plots represents enzymes with simple kinetics, where reaction velocity increases rapidly at low substrate concentrations and levels off as the enzyme becomes saturated.



Solution for Question 23:

Correct Answer: C) 1 - c, 2 - b, 3 - a, 4 - d

Explanation:

Regulation mechanisms		Enzymatic reactions
1. Compartmentation		c. Segregation of glycolytic enzymes in the cytosol
2. Feedback inhibition		b. Negative regulation of phosphofructokinase by ATP
3. Covalent modification		a. Phosphorylation of glycogen phosphorylase
4. Proteolytic activation		d. Activation of pepsinogen to pepsin
Regulatory Mechanism	Description	Examples
Compartmentation	Separating metabolic processes into different cellular compartments to regulate their activity.	Segregation of glycolytic enzymes in the cytosol. Separation of fatty acid oxidation in the mitochondria from fatty acid synthesis in the cytosol.

Feedback Inhibition	A molecule decreases the activity of an enzyme.	Negative regulation of phosphofructokinase by ATP. Inhibition of acetyl-CoA carboxylase by palmitoyl-CoA in fatty acid synthesis.
Covalent Modification	Addition or removal of a covalent bond of a chemical group to alter an enzyme's activity.	Phosphorylation of glycogen phosphorylase. Acetylation of histones affecting gene expression.
Proteolytic Activation	Converting an inactive enzyme precursor into its active form through proteolytic cleavage.	Activation of pepsinogen to pepsin. Activation of trypsinogen to trypsin.

Solution for Question 24:

Correct Answer: B) Allozymes are produced by different genes and can have distinct K_m values or substrate affinities

Explanation:

Allozymes are variants of the same gene, not produced by different genes. They do exhibit differences in K_m values or substrate affinities.

Isoenzymes

- Isoenzymes or isozymes, are distinct forms of the same enzyme that catalyze the same chemical reaction but differ in their amino acid sequences, structural properties, or regulatory mechanisms (Option A)
- These variations allow isoenzymes to be tissue-specific or associated with different developmental stages, reflecting their adaptation to specific physiological needs or conditions while performing the same fundamental biochemical function.

Variants:

True Isoenzymes: Produced by different genes, e.g., salivary vs. pancreatic amylase.

Allozymes: Variants of the same gene, e.g., glucose-6-phosphate dehydrogenase

Isoforms: Result from post-translational modifications, e.g., sialic acid content in ALP (Option C)

Identification Techniques:

- Electrophoresis: Isoenzymes can be separated based on mobility (Option D)

- Heat Stability: Some isoenzymes are more heat-sensitive.
- Inhibitors: Sensitivity to specific inhibitors can help differentiate isoenzymes.
- Km Value/Substrate Specificity: Different isoenzymes may have distinct Km values or substrate affinities.
- Cofactor Requirements: Isoenzymes might differ in their cofactor needs.
- Tissue Localization: Isoenzymes are often found in specific tissues.

Examples:

- Lactate Dehydrogenase (LDH): 5 isoforms (LDH-1 to LDH-5)
- Creatine Kinase (CK): 3 isoforms (CK-MM, CK-BB, CK-MB)
- Alkaline Phosphatase (ALP): 3 isoforms (Bone, Liver, Placental)

Solution for Question 25:

Correct Answer: D) LDH-5

Explanation:

- Lactate dehydrogenase (LDH) is an enzyme involved in converting lactate to pyruvate during anaerobic glycolysis.
- It consists of four subunits, forming five different isoenzymes that vary in tissue distribution.
- LDH-5, primarily found in skeletal muscle and liver, serves as a biomarker and typically indicates skeletal muscle damage, seen in muscle injury, rhabdomyolysis, or intense physical exertion.

Isoenzyme	Tissue/Organ	Clinical Use
LDH-1 (Option A)	Heart	Elevated levels can indicate myocardial infarction or other tissue damage.
LDH-2 (Option B)	Reticuloendothelial system	Elevated levels can suggest tissue damage or disease.
LDH-3 (Option C)	Lungs	Elevated levels can be associated with lung conditions.
LDH-4	Kidneys and placenta	Elevated levels may indicate kidney or placental issues.

LDH-5	Liver and skeletal muscle	Elevated levels can be related to liver or muscle damage.
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Previous Year Questions

1. Which of the following enzymes is dependent on Vitamin C for its activity?

- A. Lysyl dehydrogenase
 - B. Lysyl oxidase
 - C. Prolyl hydroxylase
 - D. Hydroxyprolyl kinase
-

2. In a given reaction, the concentration of the substrate is 1000 times greater than the K_m value of the enzyme. After 9 minutes, only 1% of the substrate (12 microgram/ml) has been converted into product. If the enzyme concentration is reduced to $1/3$ and the substrate concentration is doubled, what is the new time required to convert the substrate into the same amount of product, i.e., 12 microgram/ml?

- A. 9 minutes
 - B. 4.5 minutes
 - C. 27 minutes
 - D. 13.5 minutes
-

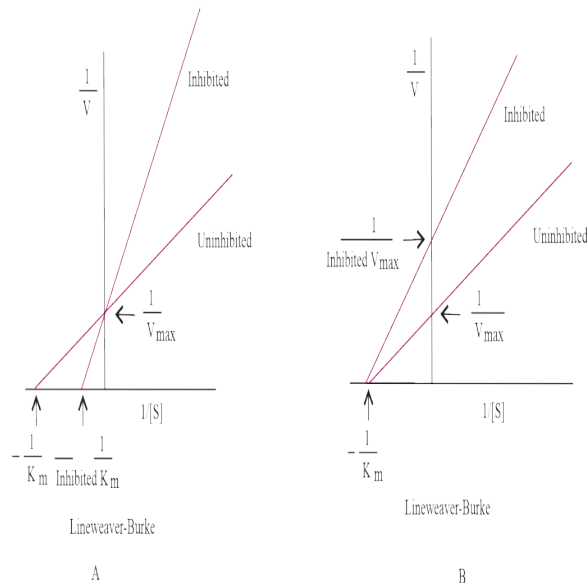
3. Out of the enzymes listed below, which one exhibits the greatest catalytic efficiency?

- A. K_m of an enzyme = 10 micromole and K_{cat} = 20 per sec
- B. K_m of an enzyme = 2000 nanomole and K_{cat} = 50 per sec
- C. K_m of an enzyme = 2 micromole and K_{cat} = 200 per sec
- D. K_m of an enzyme = 4 micromole and K_{cat} = 200 per sec

4. What is the most likely diagnosis of a hypertensive patient presenting with Tanner stage 4 development and hyperpigmentation?

- A. 21 beta hydroxylase deficiency
 - B. 11 Beta hydroxylase deficiency
 - C. 17 beta hydroxylase deficiency
 - D. 17 alpha hydroxylase deficiency
-

5. In the given enzyme kinetics graphs, identify the types of inhibitors labeled as A and B.



- A. Competitive and Non-Competitive
- B. Uncompetitive and Non-competitive
- C. Non-Competitive and Uncompetitive
- D. Non-competitive and competitive

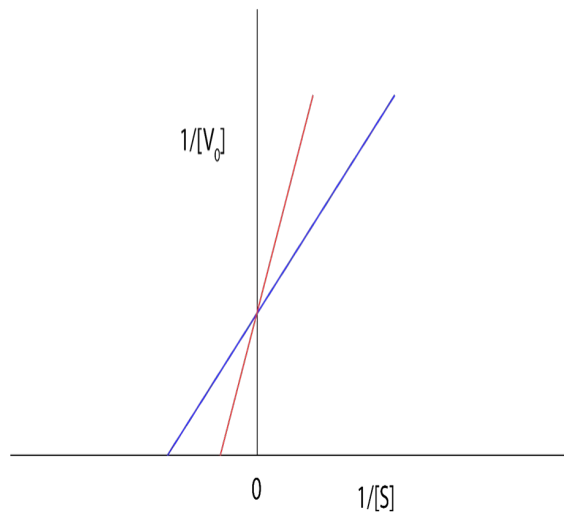
6. A 13-year-old boy presents with jaundice, fatigue, muscle stiffness, tremors, behavioral changes, hepatosplenomegaly, and Kayser-Fleischer rings. Which test is definitive for the diagnosis?

- A. Urinary copper
 - B. Serum ceruloplasmin
 - C. Hepatic parenchymal copper concentration
 - D. Genetic testing for ATP7B mutation
-

7. A patient presents with severe abdominal pain and elevated lipase levels. Imaging reveals a stone in the common bile duct (CBD). Which enzyme is likely to be elevated in this condition?

- A. Gamma-glutamyl transferase (GGT)
 - B. Lactate Dehydrogenase (LDH)
 - C. Alanine Aminotransferase (ALT)
 - D. Aspartate Aminotransferase (AST)
-

8. In the given graph, identify the type of inhibition:



- A. Competitive
- B. Non-competitive
- C. Mixed
- D. None of the above

9. A female was diagnosed with lead poisoning. What is the probable enzyme defect responsible for her condition?

- A. ALA synthase
- B. ALA dehydratase
- C. Heme synthase
- D. Coproporphyrinogen oxidase

10. A 62-year-old male with CAD comes to the physician for a follow-up examination. His blood pressure is 130/80 mm Hg, height of 182 cm, weight of 85 kg, and a BMI of 25 kg/m². Serum studies shows a glucose concentration of 95

mg/dL and homocysteine concentration of $19.3 \mu\text{mol/L}$ (N=5–15). Serum lipid concentrations are within the normal range. which of the following enzyme deficiencies is most likely to be found in this patient?

- A. Homogentisate oxidase
 - B. Cystathionine beta synthase
 - C. Cystathionine gamma synthase
 - D. HGPRTase
-

11. Hormone-sensitive lipase is activated by all of the following EXCEPT?

- A. Glucagon
 - B. Epinephrine
 - C. ACTH
 - D. Insulin
-

12. A 2-month-old infant is brought to the pediatrician due to feeding difficulties, vomiting, and failure to thrive. The child's history reveals that the infant was born full-term and had a normal birth weight. On physical examination, the infant appears jaundiced, and the liver is palpable below the costal margin. Oil drop Cataracts are also noted on eye examination. Laboratory tests show elevated levels of galactose in the blood and urine. Based on these clinical findings, which enzyme deficiency is most likely responsible for this presentation?

- A. G6PD (Glucose-6-phosphate dehydrogenase)
 - B. GALT (Galactose-1-phosphate uridylyltransferase)
 - C. ALDH (Aldehyde dehydrogenase)
 - D. PKU (Phenylalanine hydroxylase)
-

13. Active site of an enzyme is mostly complementary to:

- A. Allosteric activator
 - B. Substrate
 - C. Product
 - D. Transition state
-

14. Which of the following is a rate-limiting enzyme of HMP pathway?

- A. Glucose 6 phosphatase
 - B. Glucose 6 phosphate dehydrogenase
 - C. Phosphofructokinase 1
 - D. Glycogen phosphorylase
-

15. Which of the following statements is false regarding the reduction of activation energy?

- A. GF is lowered
 - B. Stability of transition state is reduced
 - C. It depends upon complementarity of active site of enzyme and substrate
 - D. Conformational change may occur at the active site of enzyme at the time of binding to substrate
-

16. The catalytic efficiency of enzyme is best expressed by which of the following kinetic constant:-

- A. K_{cat}/K_m
 - B. K_m/K_{cat}
 - C. K_m/k_a
 - D. K_a/k_m
-

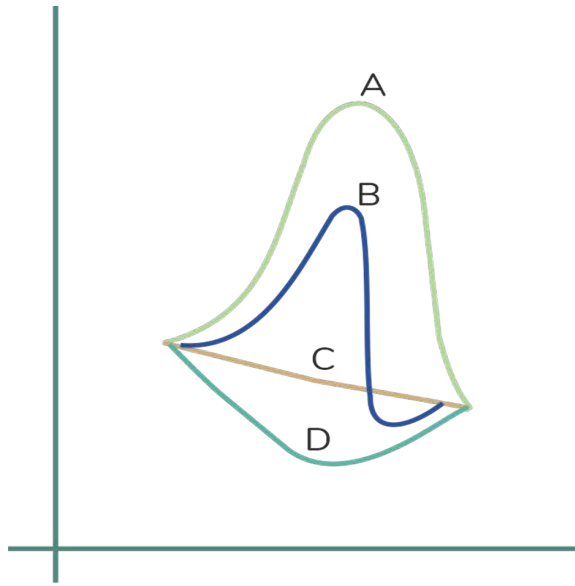
17. Which one of the following is the most specific biomarker for pheochromocytoma, a rare tumor that typically arises in the chromaffin cells (pheochromocytes) of the adrenal gland and is usually benign?

- A. VMA
 - B. Catecholamine
 - C. 5 - HIAA
 - D. Serotonin
-

18. Enolase is inhibited by which of the following substances?

- A. NaF
 - B. Fluoroacetate
 - C. Iodoacetate
 - D. Potassium oxalate
-

19. The Gibbs free energy curve for a reaction catalyzed by an enzyme is demonstrated by

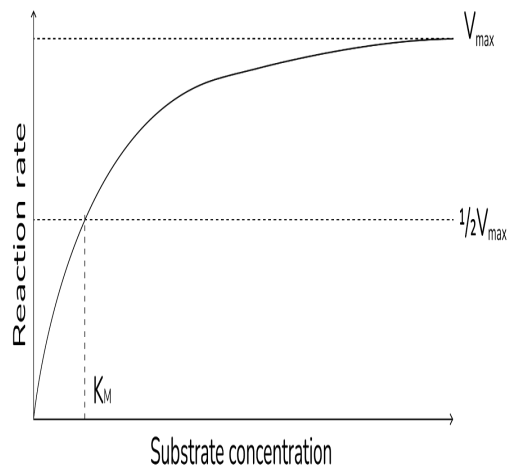


- A. A
- B. B
- C. D
- D. C

20. An adolescent male patient presented with pain in the calf muscles on exercise. On biopsy, an excessive amount of glycogen was found to be present in the muscle. What is the most likely enzyme deficiency?

- A. Muscle-debranching enzyme
- B. Phosphofructokinase
- C. Glucose 6 phosphatase
- D. Muscle glycogen phosphorylase

21. The enzyme kinetics depicted in the graph is suggestive of



- A. Allosteric activation
 - B. Cooperative binding
 - C. Michaelis Menten kinetics
 - D. Sigmoid curve
-

22. By the action of the enzyme aromatase, testosterone is transformed into which of the subsequent hormones?

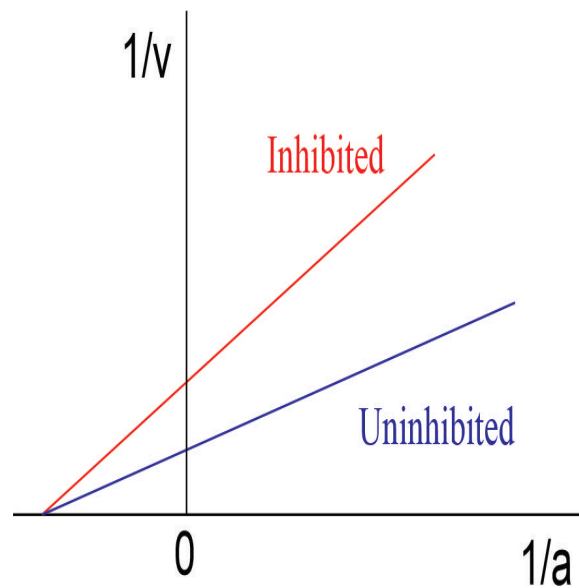
- A. 5-hydroxy testosterone
 - B. Estradiol
 - C. Cortisol
 - D. Cortisone
-

23. What is the substance that is most likely to be secreted by the hypothalamus?

- A. Hexokinase I

- B. Hexokinase II
- C. Hexokinase III
- D. Hexokinase IV

24. Identify the type of inhibition shown in the graph.



- A. Competitive inhibition
- B. Allosteric Inhibition
- C. Uncompetitive inhibition
- D. Noncompetitive inhibition

25. What is the origin and type of cell responsible for testosterone production?

- A. Cholesterol and Sertoli cells
- B. Dihydrotestosterone and Leydig cells
- C. FSH and Leydig cells

D. Cholesterol and Leydig cells

26. A child presents with erythematous scaly and pustular lesions in the perioral region, diarrhea, and impaired epithelial wound healing. What is the most likely mineral deficiency associated with this condition?

- A. Iron deficiency
 - B. Zinc deficiency
 - C. Calcium deficiency
 - D. Copper deficiency
-

27. Which mineral is associated with the functioning of carbonic anhydrase?

- A. Zinc
 - B. Iron
 - C. Calcium
 - D. Potassium
-

28. Which of the following is true regarding competitive inhibition?

- A. The inhibitor binds to an allosteric site on the enzyme.
 - B. Competitive inhibition decreases the apparent V_{max} (maximum reaction rate) of the enzyme.
 - C. Increasing substrate concentration can overcome competitive inhibition.
 - D. Competitive inhibition decreases the K_m for the substrate
-

29. Which mechanism is responsible for the inhibition of the acetylcholinesterase enzyme by organophosphorus compounds?

- A. Non-competitive inhibition
 - B. Reversible competitive inhibition
 - C. Irreversible competitive inhibition
 - D. Suicidal inhibition
-

30. Which of the statements below accurately describes the characteristics of the Michaelis-Menten graph?

- A. K_m is directly proportional to substrate concentration
 - B. K_m doesn't vary with the enzyme concentration
 - C. K_m is indirectly proportional to substrate concentration
 - D. K_m has no relation to substrate concentration
-

31. Which enzymes are affected in Menke's disease?

- A. Lysyl hydroxylase
 - B. Lysyl oxidase
 - C. Prolyl oxidase
 - D. Prolyl hydroxylase
-

32. Which of the following is not a copper-containing enzyme?

- A. Mitochondrial superoxide dismutase

- B. Cytosolic superoxide dismutase
 - C. Tyrosinase
 - D. Lysyl oxidase
-

33. What is the enzyme that is responsible for the defect causing Maple syrup urine disease?

- A. Branched-chain α -ketoacid dehydrogenase
 - B. Isovaleryl CoA dehydrogenase
 - C. Pyruvate carboxylase
 - D. Acetyl CoA carboxylase
-

34. What diagnostic tool is used to determine the cause of chest pain at rest for 6 hours and an electrocardiogram indicating myocardial ischemia in a patient?

- A. LDH 2
 - B. LDH 3
 - C. LDH 1
 - D. LDH 4
-

35. A child with mental retardation, blindness, and muscle weakness was investigated and was found to be deficient for Hexosaminidase-A enzyme. She is most likely suffering from..

- A. Krabbe's disease
- B. Fabry's disease
- C. Tay-sachs disease

D. Gaucher's disease

36. Which statement accurately describes an allosteric modifier?

- A. Desaturates the enzyme
 - B. This causes the enzyme to work faster only
 - C. Binds to the catalytic site
 - D. Participates in feedback regulation
-

37. The flipped LDH ratio implies_____

- A. LDH 1 > LDH 2
 - B. LDH 2 > LDH 1
 - C. LDH 2 > LDH 3
 - D. LDH 3 > LDH 2
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	3
Question 3	3
Question 4	2
Question 5	1
Question 6	2
Question 7	1
Question 8	1
Question 9	2
Question 10	2
Question 11	4
Question 12	2
Question 13	2
Question 14	2
Question 15	2
Question 16	1
Question 17	1
Question 18	1

Question 19	1
Question 20	4
Question 21	3
Question 22	2
Question 23	4
Question 24	4
Question 25	4
Question 26	2
Question 27	1
Question 28	3
Question 29	3
Question 30	1
Question 31	2
Question 32	1
Question 33	1
Question 34	3
Question 35	3
Question 36	4
Question 37	1

Solution for Question 1:

Answer: C) Prolyl hydroxylase

Explanation:

- Role of Prolyl Hydroxylase and Vitamin C: Prolyl hydroxylase catalyzes the hydroxylation of proline residues in collagen. This process is essential for collagen stability via hydrogen bonding. Vitamin C (ascorbic acid) is a cofactor for prolyl hydroxylase, aiding in the regeneration of the enzyme's Fe^{2+} active site .
- Prolyl hydroxylase catalyzes the hydroxylation of proline residues in collagen.
- This process is essential for collagen stability via hydrogen bonding.
- Vitamin C (ascorbic acid) is a cofactor for prolyl hydroxylase, aiding in the regeneration of the enzyme's Fe^{2+} active site .
- Vitamin C Deficiency and Scurvy: Deficiency leads to impaired collagen synthesis , causing fragile blood vessels, poor wound healing, and gingival bleeding (Scurvy).
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- Prolyl hydroxylase catalyzes the hydroxylation of proline residues in collagen.
- This process is essential for collagen stability via hydrogen bonding.
- Vitamin C (ascorbic acid) is a cofactor for prolyl hydroxylase, aiding in the regeneration of the enzyme's Fe^{2+} active site .
- Deficiency leads to impaired collagen synthesis , causing fragile blood vessels, poor wound healing, and gingival bleeding (Scurvy).

Enzyme	Function	Vitamin C Dependency
Prolyl hydroxylase	Hydroxylates proline residues in collagen for stability.	Yes
Lysyl hydroxylase	Hydroxylates lysine residues in collagen for cross-linking.	Yes
Lysyl oxidase	Cross-links collagen via aldehyde formation.	No (Requires copper)
Hydroxyprolyl kinase	Phosphorylates hydroxyproline.	No

- Lysyl dehydrogenase (Option A): Incorrect because this enzyme is not involved in collagen synthesis and does not require Vitamin C.
- Lysyl oxidase (Option B): Incorrect because it requires copper (Cu^{2+}), not Vitamin C , for collagen cross-linking.
- Hydroxyprolyl kinase (Option D): Incorrect because it phosphorylates hydroxyproline and is not dependent on Vitamin C.

Solution for Question 2:

Correct Option C:

- The Michaelis-Menten equation describes the relationship between the reaction rate, substrate concentration, and the kinetic parameters K_m (Michaelis constant) and V_{max} (maximum reaction rate) of an enzyme.
- According to the given information, the substrate concentration is 1000 times the K_m value of the enzyme. This implies that the substrate concentration is at a high level compared to the K_m , indicating that the enzyme is operating under conditions of substrate excess. In this scenario, the initial rate of the reaction can be considered as approximately V_{max} .
- After 9 minutes of reaction, 1% of the substrate is converted to product, which corresponds to 12 microgram/ml. From this information, we can calculate the initial concentration of the substrate (S_0) using the equation:
- $S_0 = (100 / \text{conversion percentage}) * \text{product concentration} = (100 / 1) * 12 \text{ microgram/ml} = 1200 \text{ microgram/ml}$ the changes in enzyme and substrate concentrations in the modified scenario:
- The concentration of the enzyme is changed to $1/3$ of its initial value.
- The concentration of the substrate is doubled.
- Given that the initial substrate concentration (S_0) is 1200 microgram/ml, after doubling, it becomes 2400 microgram/ml. The new enzyme concentration (E_1) is $1/3$ of the initial enzyme concentration.
- To determine the time taken to convert the substrate into the same amount of product (12 microgram/ml), we can use the formula:

$$t = (1 / k) * \ln((S_0 - S) / S_0)$$

Where: t = time taken k = reaction rate constant S_0 = initial substrate concentration S = final substrate concentration

Since the initial rate of the reaction is approximately V_{max} , we can assume that the reaction rate constant (k) remains constant in both scenarios.

Using the given data, the time taken to convert the substrate into the same amount of product can be calculated as follows:

$$t = (1 / k) * \ln((1200 - 12) / 1200) = (1 / k) * \ln(1188 / 1200)$$

- 27 minutes: This is the correct answer for the modified scenario. It represents the time taken to convert the substrate into the same amount of product (12 microgram/ml) with the modified enzyme and substrate concentrations.

Incorrect options:

Option A: 9 minutes: This is the time taken in the initial scenario, but it's not the correct answer for the modified scenario.

Option B: 4.5 minutes: This option is not correct.

Option D: 13.5 minutes: This option is not correct.

Solution for Question 3:

Correct Option C - K_m of an enzyme = 2 micromole and K_{cat} = 200 per sec:

- Catalytic efficiency, also known as specificity constant, is a measure of how effectively an enzyme converts substrate to product. It is calculated as the ratio of the catalytic rate constant (K_{cat}) to the Michaelis constant (K_m).
- K_m of an enzyme = 2 micromole and K_{cat} = 200 per sec: Catalytic efficiency = $K_{cat} / K_m = 200 / 2 = 100$ per micromole per sec

Incorrect Options:

Option A - K_m of an enzyme = 10 micromole and K_{cat} = 20 per sec: Catalytic efficiency = $K_{cat} / K_m = 20 / 10 = 2$ per micromole per sec.

Option B - K_m of an enzyme = 2000 nanomole and K_{cat} = 50 per sec: Catalytic efficiency = $K_{cat} / K_m = 50 / 2000 = 0.025$ per nanomole per sec..

Option D - K_m of an enzyme = 4 micromole and K_{cat} = 200 per sec: Catalytic efficiency = $K_{cat} / K_m = 200 / 4 = 50$ per micromole per sec

Solution for Question 4:

Correct Option B - 11 Beta hydroxylase deficiency:

- This enzyme deficiency leads to an accumulation of 11-deoxycorticosterone (DOC), which has mineralocorticoid activity, causing hypertension.
- Patients with 11-beta hydroxylase deficiency typically present with signs of virilization (such as Tanner stage 4) and hyperpigmentation due to increased ACTH stimulation.

Incorrect Options:

Option A - 21 beta hydroxylase deficiency:

- This deficiency typically causes virilization and hyperpigmentation but is associated with hypotension, not hypertension.

Option C - 17 beta hydroxylase deficiency:

- This enzyme deficiency usually presents with hypertension and ambiguous genitalia in males or lack of secondary sexual characteristics in females, but not virilization (Tanner stage 4).

Option D - 17 alpha hydroxylase deficiency:

- This condition leads to hypertension and sexual infantilism, but not virilization or hyperpigmentation.

Solution for Question 5:

Correct Option A - Competitive and Non-Competitive:

Competitive Inhibition (Inhibitor A):

- In a Lineweaver-Burk plot, competitive inhibition is characterized by lines that intersect on the y-axis. This is because the presence of a competitive inhibitor increases the apparent K_m (Michaelis constant), but the V_{max} (maximum velocity) remains unchanged. The inhibitor competes with the substrate for the active site of the enzyme.

Non-Competitive Inhibition (Inhibitor B):

- In a Lineweaver-Burk plot, non-competitive inhibition is shown by lines that intersect on the x-axis. This occurs because non-competitive inhibitors decrease the V_{max} but do not affect the K_m . The inhibitor binds to an allosteric site (not the active site), affecting the enzyme's function regardless of the substrate concentration.

Incorrect Options:

Option B - Uncompetitive and Non-Competitive:

- Uncompetitive inhibition shows parallel lines in a Lineweaver-Burk plot, as both V_{max} and K_m are reduced proportionally. This does not match the provided data for inhibitor A.

Option C - Non-Competitive and Un-competitive:

- This option incorrectly identifies the inhibitors in reverse order. The data provided for inhibitor A matches competitive inhibition, not uncompetitive.

Option D - Non-competitive and competitive:

- Like option C, this option reverses the identification of inhibitors A and B.
-

Solution for Question 6:

Correct Option B - Serum ceruloplasmin:

- The clinical presentation suggests Wilson's disease, which is diagnosed by serum ceruloplasmin test.

Incorrect Options:

- Options A, C, and D are less preferred to detect Wilson's disease.
-

Solution for Question 7:

Correct Option A - Gamma-glutamyl transferase (GGT):

- Gamma-glutamyl transferase levels are elevated in bile duct obstruction, such as a stone in the common bile duct, along with alkaline phosphatase (ALP).

Incorrect Options:

Options B, C, and D are not elevated in a patient with common bile duct stones.

Solution for Question 8:

Correct Option A – Competitive:

- In competitive inhibition, the inhibitor competes with the substrate for binding to the active site of the enzyme, often resulting in an increased K_m without affecting V_{max} .

Incorrect Options:

Options B, C, and D are incorrect, as explained.

Solution for Question 9:

Correct Option B - ALA dehydratase:

- This is the correct option. ALA dehydratase is inhibited by lead in cases of lead poisoning. This enzyme is responsible for the conversion of delta-aminolevulinic acid (ALA) to porphobilinogen. Inhibition of ALA dehydratase disrupts heme synthesis, leading to the accumulation of ALA and porphobilinogen, which are neurotoxic.

Incorrect Options:

Option A - ALA synthase: ALA synthase is an enzyme involved in the first step of heme synthesis, where delta-aminolevulinic acid (ALA) is formed. Lead poisoning primarily affects the later steps of heme synthesis and does not directly inhibit ALA synthase.

Option C - Heme synthase: Heme synthase, also known as ferrochelatase, is involved in the final step of heme synthesis, where ferrous iron is inserted into protoporphyrin IX to form heme. Lead poisoning primarily affects earlier steps in heme synthesis, such as the inhibition of ALA dehydratase.

Option D - Coproporphyrinogen oxidase: Coproporphyrinogen oxidase is an enzyme involved in the conversion of coproporphyrinogen III to protoporphyrinogen IX in the heme synthesis pathway. While lead poisoning can disrupt heme synthesis, the inhibition is primarily associated with ALA dehydratase rather than coproporphyrinogen oxidase.

Solution for Question 10:

Correct Option B -Cystathionine beta synthase:

- Cystathionine beta synthase deficiency is associated with homocystinuria, a genetic disorder characterized by elevated levels of homocysteine in the blood. The clinical manifestations of homocystinuria can include cardiovascular issues (such as early-onset atherosclerosis), skeletal abnormalities, intellectual disabilities, and other systemic complications. The patient in the question has an elevated homocysteine concentration, which is characteristic of homocystinuria.

Incorrect Options:

Option A - Homogentisate oxidase: Homogentisate oxidase deficiency leads to alkaptonuria, a condition characterized by the accumulation of homogentisic acid, causing darkening of urine and connective tissue. This condition is not consistent with the patient's presentation and lab values.

Option C - Cystathionine gamma synthase: Cystathionine gamma synthase deficiency is not a well-known enzyme deficiency associated with a specific clinical disorder. Cystathionine gamma synthase is involved in the transsulfuration pathway, converting cystathionine to cysteine, but its deficiency doesn't typically lead to a recognized metabolic disorder.

Option D - HGPRTase (Hypoxanthine-guanine phosphoribosyltransferase): HGPRTase deficiency is seen in Lesch-Nyhan syndrome, which is characterized by neurological and behavioral abnormalities, along with uric acid overproduction leading to gout. The patient's presentation and lab values do not match this disorder.

Solution for Question 11:

Correct Option D - Insulin:

- Insulin is a hormone produced by the pancreas that plays a crucial role in glucose metabolism. It promotes glucose uptake into cells and storage as glycogen in the liver and muscle tissues.
- However, insulin inhibits the activation of hormone-sensitive lipase. When insulin levels are high (e.g., after a meal), lipolysis (breakdown of fats) is suppressed, and the body primarily uses glucose as an energy source.

Incorrect Options:

Option A - Glucagon: It also stimulates the activation of hormone-sensitive lipase, which leads to the breakdown of stored triglycerides (lipids/fats) into fatty acids in adipose tissue.

Option B - Epinephrine: It activates hormone-sensitive lipase in adipose tissue, causing the breakdown of triglycerides into fatty acids and glycerol.

Option C - ACTH (Adrenocorticotrophic Hormone): While cortisol plays a role in metabolism, it does not directly activate hormone-sensitive lipase. Instead, cortisol can indirectly influence lipid metabolism by affecting other regulatory pathways.

Solution for Question 12:

Correct Option B - GALT (Galactose-1-phosphate uridylyltransferase) deficiency:

- GALT deficiency, also known as galactosemia, is a genetic disorder that impairs the ability to metabolize galactose properly. Galactose is a sugar found in milk and other dairy products.
- In individuals with GALT deficiency, galactose-1-phosphate accumulates, leading to various symptoms, including feeding difficulties, vomiting, failure to thrive, jaundice, and liver enlargement.
- Cataracts are also a characteristic feature of galactosemia. This condition requires the removal of galactose from the diet to prevent complications.

Incorrect Options:

Option A - G6PD (Glucose-6-phosphate dehydrogenase) deficiency: G6PD deficiency is a genetic disorder that leads to a deficiency of the enzyme glucose-6-phosphate dehydrogenase, which is important for protecting red blood cells from oxidative damage.

Option C - ALDH (Aldehyde dehydrogenase) deficiency: ALDH deficiency is associated with a condition called alcohol-induced flushing, where individuals who carry certain genetic variants of

the aldehyde dehydrogenase enzyme experience facial flushing and other symptoms when they consume alcohol.

Option D - PKU (Phenylalanine hydroxylase) deficiency: PKU is a genetic disorder characterized by the deficiency of the enzyme phenylalanine hydroxylase, which is responsible for converting phenylalanine to tyrosine.

Solution for Question 13:

Correct Option B - Substrate:

- A substrate is a molecule that an enzyme reacts with. The enzyme's active site, or the location where weak bonds between the two molecules can form, is loaded with a substrate.
- An enzyme substrate complex is formed, and the enzyme's pressures on the substrate drive it to react and become the planned reaction's result. It is the most complementary to the active site.

Incorrect Options:

Option A - Allosteric activator: Allosteric activators bind to locations on an enzyme away from the active site, inducing a conformational change that increases the affinity of the enzyme's active site(s) for its substrate(s).

Option C - Product: The product is the molecule that is produced by the chemical reaction catalyzed by the enzyme. In order for a product molecule to be generated, a substrate molecule must first bind to the active site of an enzyme and be transformed into a product molecule.

Option D - Transition state: The transition state is the intermediary state of the reaction, when the molecule is neither a substrate or product. The transition state has the highest free energy, making it a rare and un-stable intermediate.

Solution for Question 14:

Correct Option B - Glucose 6 phosphate dehydrogenase:

- Glucose 6-phosphate dehydrogenase (G6PD) catalyzes the irreversible conversion of glucose 6-phosphate to 6-phosphogluconolactone. This enzymatic reaction requires NADP⁺ (nicotinamide adenine dinucleotide phosphate) as a coenzyme, which acts as an acceptor of the reducing equivalents that are removed from the glucose 6-phosphate molecule. As a result, NADP⁺ is reduced to NADPH (nicotinamide adenine dinucleotide phosphate, reduced form).

Incorrect Options:

Option A - Glucose 6 phosphatase:

Glucose 6 phosphatase is an enzyme involved in gluconeogenesis and glycogenolysis.

Option C - Phosphofructokinase 1 (PFK-1):

Phosphofructokinase 1 is a key regulatory enzyme in glycolysis.

Option D - Glycogen phosphorylase:

- Glycogen phosphorylase is an enzyme involved in glycogenolysis, the breakdown of glycogen into glucose molecules.

Solution for Question 15:

Correct Option B - Stability of the transition state is reduced: This statement is incorrect. In reality, reducing the activation energy involves stabilizing the transition state rather than reducing its stability. This stabilization makes it easier for the reactants to overcome the energy barrier and proceed with the reaction.

Incorrect Option:

Option A - G_f is lowered: This statement is correct. Reducing the activation energy involves lowering the Gibbs free energy of the transition state. By lowering the activation energy, the reaction becomes more favorable, and the transition state is easier to reach.

Option C - It depends upon complementarity of the active site of the enzyme and substrate: This statement is correct. Enzymes catalyze reactions by providing a specific active site that is complementary to the substrate. The complementarity allows for optimal binding and orientation of the substrate, which lowers the activation energy required for the reaction to occur.

Option D - Conformational change may occur at the active site of the enzyme at the time of binding to the substrate: This statement is correct. Enzymes often undergo conformational

changes upon binding to the substrate. These changes can help to stabilize the transition state or induce strain in the substrate, making the reaction more favorable and lowering the activation energy.

Solution for Question 16:

Correct Option A - K_{cat}/K_m

- This option represents the ratio of the turnover number (K_{cat}) to the Michaelis constant (K_m).
- The turnover number (K_{cat}) is a measure of how many substrate molecules an enzyme can convert to product per unit of time when the enzyme is saturated with the substrate.
- The Michaelis constant (K_m) represents the substrate concentration at which the reaction rate is half of its maximum.
- The ratio K_{cat}/K_m provides a measure of how efficient the enzyme is at converting substrate to product, taking into account both the rate of the reaction (K_{cat}) and the affinity of the enzyme for the substrate (K_m). A higher K_{cat}/K_m value indicates a more efficient enzyme.

Incorrect Option:

Option B - K_m/K_{cat} The ratio K_m/K_{cat} does not provide a meaningful measure of catalytic efficiency, as it would yield a value that is the reciprocal of K_{cat}/K_m . It does not accurately represent the efficiency of the enzyme.

Option C - K_m/k_a The rate constant for the association of the enzyme and substrate (k_a) is not directly related to catalytic efficiency. Therefore, this option does not express the catalytic efficiency of the enzyme accurately.

Option D - k_a/K_m Similarly to option C, the rate constant for the association of the enzyme and substrate (k_a) is not directly related to catalytic efficiency. Hence, this option does not accurately express the catalytic efficiency of the enzyme.

Solution for Question 17:

Correct option A - VMA

- Vanillylmandelic acid (VMA) is the most specific biomarker for pheochromocytoma. It is a metabolite of catecholamines, which are secreted by pheochromocytomas. Measurement of VMA levels in urine is a common diagnostic test for pheochromocytoma.

Incorrect options

Options: B - Catecholamine: Catecholamines, such as epinephrine and norepinephrine, are secreted by pheochromocytomas, but they are not the most specific biomarker. Measurement of catecholamines in urine or blood can support the diagnosis but is not considered the most specific test.

Option C - 5-HIAA (5-Hydroxyindoleacetic acid): 5-HIAA is a metabolite of serotonin and is mainly used as a biomarker for carcinoid tumors, not pheochromocytomas. It is not the most specific biomarker for pheochromocytoma.

Option D - Serotonin: Serotonin is not the most specific biomarker for pheochromocytoma. While serotonin-secreting tumors (carcinoid tumors) can occasionally arise from chromaffin cells, the most specific biomarker for pheochromocytoma is VMA (Vanillylmandelic acid)

Solution for Question 18:

Correct Option A - NaF:

- Enolase is an enzyme involved in the glycolytic pathway, specifically in the conversion of 2-phosphoglycerate to phosphoenolpyruvate. It plays a crucial role in energy metabolism.

Incorrect Options

Option B - Fluoroacetate: Fluoroacetate inhibits the enzyme aconitase, which is involved in the citric acid cycle (also known as the Krebs cycle or TCA cycle). It does not directly inhibit enolase.

Option C - Iodoacetate: Iodoacetate is an inhibitor of the enzyme glyceraldehyde 3-phosphate dehydrogenase (GAPDH), which is an enzyme involved in the glycolytic pathway. It does not directly inhibit enolase.

Option D - Potassium oxalate: Potassium oxalate is not known to directly inhibit enolase. It is primarily used as an anticoagulant for blood collection.

Solution for Question 19:

Correct Option A - This option is correct. (A) In enzyme-catalyzed reactions, the Gibbs free energy curve typically shows a lower activation energy compared to the uncatalyzed reaction (represented by a higher activation energy barrier). This is one of the key effects of enzymes, as they lower the activation energy, allowing the reaction to proceed more rapidly.

Incorrect Options:

Option B - B: All the reactions have a transition state, but not all reactions have an intermediate state.

Option C - D: Transition energy intermediate never has lower energy compared to the product

Option D - C: Free energy change cannot be a straight line. It has to pass through a high energy intermediate state

Solution for Question 20:

Correct Option D - Muscle glycogen phosphorylase

- The primary symptom of McArdle disease is exercise intolerance, which typically manifests as muscle cramps, weakness, and fatigue during physical activity. Other common symptoms include myalgia (muscle pain), stiffness, and myoglobinuria (the presence of myoglobin in the urine, giving it a dark color).
- Muscle glycogen phosphorylase is responsible for breaking down glycogen into glucose-1-phosphate during exercise or periods of increased energy demand.
- Deficiency or dysfunction of this enzyme leads to impaired glycogen breakdown, resulting in the accumulation of glycogen in the muscle.

Incorrect Options:

Option A - Muscle-debranching enzyme (also known as amylo-1,6-glucosidase): Deficiency in this enzyme leads to a different type of GSD called Glycogen Storage Disease Type III (GSD III), also known as Cori disease.

Option B - Phosphofructokinase: Deficiency in phosphofructokinase leads to a different type of GSD called Glycogen Storage Disease Type VII (GSD VII), also known as Tarui disease.

Option C - Glucose 6 phosphatase: Deficiency in glucose 6 phosphatase leads to Glycogen Storage Disease Type I (GSD I), also known as von Gierke disease.

Solution for Question 21:

Correct Option C - Michaelis Menten kinetics:

- Michaelis-Menten kinetics, the enzyme-substrate reaction rate initially increases linearly with substrate concentration before reaching a plateau. The curve saturates at high substrate concentrations, indicating that all enzyme active sites are occupied.
- The Michaelis constant (K_m) represents the substrate concentration at which the reaction rate is half of V_{max} , reflecting enzyme-substrate affinity.

Incorrect Options :

Option A. Allosteric activation. In allosteric inhibition, enzyme kinetics graphs typically exhibit a sigmoidal shape due to cooperative binding of substrates.

Option B. Cooperative binding: One substrate binding to an enzyme increases the affinity for other substrates for enzymes.

Option D. Sigmoid curve: In the presence of allosteric modifiers the graph of enzyme-catalyzed reaction obtained is a sigmoid / S-shaped graph.

Solution for Question 22:

Correct Option B - Estradiol:

- Aromatase enzyme is used in the process of androgen (such as testosterone) to estrogen (such as 17 β -estradiol) conversion. Aromatase enzyme is used as a therapeutic target for the treatment process of endocrine-responsive breast cancer. Aromatase also converts androstenedione to estrone.

Incorrect Options:

- Option A - 5-hydroxy testosterone is a metabolite of testosterone, but it is not converted to this hormone by the aromatase enzyme. 5-hydroxy testosterone is produced by the action of the enzyme 5-alpha reductase on testosterone.
 - Option C - Cortisol is a steroid hormone that is produced in the adrenal gland, not in the testis where testosterone is produced. Cortisol is synthesized from cholesterol through a series of enzymatic reactions, but aromatase is not involved in its production.
 - Option D - Cortisone is a steroid hormone that is closely related to cortisol, but it is not produced directly from testosterone. Cortisone is a metabolite of cortisol, and its production involves a series of enzymatic reactions that are different from those involved in the conversion of testosterone to estradiol.
-

Solution for Question 23:

Correct Option D - Hexokinase IV:

- A hexose phosphate is produced from hexose which is a six-carbon sugar with the help of a hexokinase enzyme. Glucose is present as the most important substrate for the hexokinases that further produce glucose-6-phosphate as the most important product.
- Hexokinases I, II, and III are isozymes that have a “low-K_m” value; hence, it has a high-affinity process for glucose, even at the low concentrations of it.
- The hypothalamus releases Hexokinase IV likely.
- HK IV or Glucokinase is also expressed in the pancreas and liver.

Incorrect Choices:

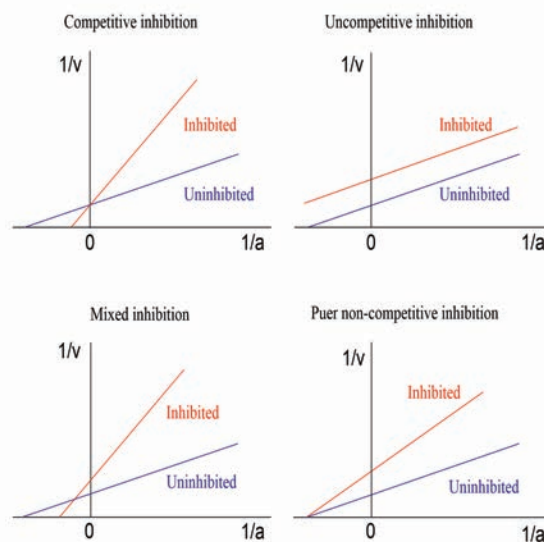
- Option A - Hexokinase I: HK1 is the most expressed isoform of hexokinase among the four hexokinases ubiquitously. It is expressed in many tissues majorly in the kidney, brain, and red blood cells (RBCs).

- Option B - Hexokinase II enzyme is encoded by the HK2 gene present on chromosome 2 in humans. It is the most active isozyme of the hexokinase family. It is expressed in the insulin-sensitive tissues.
- Option C - Hexokinase III is predominantly produced from myeloid cells. HKI and II are able to bind to mitochondria and protect cells against cell death. HKIII is not able to bind mitochondria. Very little is known about its cytoprotective effect.

Solution for Question 24:

Correct Option D - Noncompetitive inhibition:

- In noncompetitive inhibition, the inhibitor binds to the enzyme at a site other than the active site, causing a conformational change that reduces the enzyme activity. Unlike competitive inhibition, the inhibitor does not compete with the substrate for binding to the enzyme. In a graph showing noncompetitive inhibition, the V_{max} decreases, but the K_m remains the same. This means that the enzyme is less active, and it requires the same amount of substrate to reach its maximum velocity.



Incorrect Options:

Option A - Competitive inhibition: In competitive inhibition, the inhibitor molecule competes with the substrate for binding to the enzyme's active site.

Option B - Allosteric inhibition: In allosteric inhibition, the inhibitor molecule binds to the enzyme at a site other than the active site, causing a conformational change in the enzyme that reduces its activity.

Option C - Uncompetitive inhibition: In uncompetitive inhibition, the inhibitor binds to the enzyme-substrate complex, preventing the release of the product.

Solution for Question 25:

Correct Option D - Cholesterol and Leydig cells

- Testosterone is a hormone that is primarily produced in the testes of males and ovaries in females. In males, testosterone is produced in the Leydig cells, which are located in the testes. Leydig cells are stimulated by luteinizing hormone (LH) produced by the pituitary gland, which triggers the Leydig cells to convert cholesterol to testosterone.
- Cholesterol is the precursor molecule for testosterone synthesis, and Leydig cells have all the enzymes necessary to convert cholesterol to testosterone. Testosterone synthesis also requires several enzymes, including 17α -hydroxylase, $17, 20$ -lyase, and 3β -hydroxysteroid dehydrogenase.

Incorrect Options:

Option A - Cholesterol and Sertoli cells: Sertoli cells are supportive cells in the testes that provide nourishment and support for developing sperm cells.

Option B - Dihydrotestosterone and Leydig cells: Dihydrotestosterone (DHT) is a hormone derived from testosterone responsible for developing male secondary sexual characteristics.

Option C - FSH and Leydig cells: Follicle-stimulating hormone (FSH) is a hormone produced by the anterior pituitary gland and plays a crucial role in the development and maturation of sperm cells in males and follicles in females.

Solution for Question 26:

Correct Option B - Zinc deficiency:

- The clinical presentation is suggestive of Acrodermatitis enteropathica characterized by perioral or perianal dermatitis with diarrhea caused by zinc deficiency.
- Zinc is an essential mineral that plays a crucial role in various physiological processes, including wound healing, immune function, and maintaining the integrity of the skin and mucous membranes.

Incorrect Options:

Option A - Iron deficiency: Iron deficiency primarily affects the production of red blood cells and can lead to anemia.

Option C - Calcium deficiency: Calcium deficiency is associated with skeletal abnormalities, such as rickets in children, osteoporosis in adults, and muscle cramps.

Option D - Copper deficiency: Copper deficiency can cause various manifestations, including anemia, neurologic abnormalities, and connective tissue abnormalities.

Solution for Question 27:

Correct Option A - Zinc:

Zinc is an essential mineral that plays a crucial role in the activity of carbonic anhydrase, an enzyme involved in the conversion of carbon dioxide (CO_2) and water (H_2O) to carbonic acid (H_2CO_3) in various tissues and cells of the body.

Incorrect Options:

Option B - Iron: Iron is not directly linked to the activity of carbonic anhydrase. It primarily functions in oxygen transport and storage as a component of hemoglobin in red blood cells.

Option C - Calcium: While calcium plays important roles in various physiological processes such as bone health, muscle contraction, and nerve function, it is not directly associated with carbonic anhydrase activity.

Option D - Potassium: Similar to calcium, potassium is not directly linked to the activity of carbonic anhydrase. Potassium is an essential mineral involved in maintaining proper cell function, nerve transmission, and fluid balance.

Solution for Question 28:

Correct Option C - Increasing substrate concentration can overcome competitive inhibition

- This statement is true. Increasing the concentration of the substrate can overcome competitive inhibition. When the substrate concentration is high, it is more likely to outcompete the inhibitor for binding to the active site, allowing the enzyme to function at its maximum rate.

Incorrect Options

- Option A - The inhibitor binds to an allosteric site on the enzyme: This statement is not true. In competitive inhibition, the inhibitor binds to the same active site on the enzyme where the substrate binds, not to an allosteric site. This binding blocks the substrate from accessing the active site.
 - Option B - Competitive inhibition decreases the apparent V_{max} of the enzyme: This statement is not true. Competitive inhibition does not affect the maximum reaction rate (V_{max}) of the enzyme.
 - Option D - Competitive inhibition decreases the K_m for the substrate: This statement is not true. Competitive inhibition increases the K_m (Michaelis-Menten constant) for the substrate. K_m represents the substrate concentration at which the reaction rate is half of V_{max} .
-

Solution for Question 29:

Correct Option C - Irreversible competitive inhibition

- Organophosphorus compounds, such as certain pesticides and nerve agents, can irreversibly inhibit the activity of acetylcholinesterase (AChE).
- Acetylcholinesterase is an enzyme that breaks down the neurotransmitter acetylcholine in the synaptic cleft, terminating its action.
- By inhibiting AChE, organophosphorus compounds lead to an accumulation of acetylcholine and overstimulation of cholinergic receptors, which can have various toxic effects.

Incorrect Options:

- Option A - Reversible competitive inhibition occurs when an inhibitor molecule competes with the substrate for binding to the active site of the enzyme.
 - Option B - Non-competitive inhibition occurs when an inhibitor binds to a site on the enzyme other than the active site, altering its conformation and reducing its activity.
 - Option D - Suicidal inhibition is another term used to describe the irreversible nature of inhibition caused by organophosphorus compounds.
-

Solution for Question 30:

Option A - K_m is directly proportional to substrate concentration:

- K_m stands for the dissociation constant of enzyme. A high K_m represents weak binding of enzymes with the substrate while a low K_m stands for strong binding with the substrate. Hence, dissociation of substrate does not occur if a low K_m value is present.
- According to Michaelis-Menten equation, $v = V_{max} [S] / K_m + [S]$ where v is the velocity of enzyme, S is substrate, K_m is Michaelis-Menten constant, and V_{max} is maximum velocity achieved by the enzyme. Hence, the rate of enzyme and substrate concentration is directly proportional to each-other. The reaction belongs to first-order kinetics.
- $v = V_{max} [S] / K_m + [S]$
- where v is the velocity of enzyme, S is substrate, K_m is Michaelis-Menten constant, and V_{max} is maximum velocity achieved by the enzyme. Hence, the rate of enzyme and substrate concentration is directly proportional to each-other. The reaction belongs to first-order kinetics.
- $v = V_{max} [S] / K_m + [S]$
- where v is the velocity of enzyme, S is substrate, K_m is Michaelis-Menten constant, and V_{max} is maximum velocity achieved by the enzyme. Hence, the rate of enzyme and substrate concentration is directly proportional to each-other. The reaction belongs to first-order kinetics.

Incorrect Options - B, C, and D: These are incorrect

Solution for Question 31:

Correct Option B - Lysyl oxidase

- Lysyl oxidase is the enzyme affected in Menke's disease. It plays a crucial role in the cross-linking of collagen and elastin fibers, which are essential components of connective tissues.
- In Menke's disease, the defective copper transport system impairs the delivery of copper to lysyl oxidase, leading to decreased activity of the enzyme. This results in abnormal connective tissue formation.

Incorrect Options:

Option A - Lysyl hydroxylase

- Lysyl hydroxylase is an enzyme involved in the hydroxylation of lysine residues in collagen molecules. It plays a role in collagen maturation. However, it is not directly affected by Menke's disease.

Option C - Prolyl oxidase

- Prolyl oxidase is not directly involved in the copper transport system or collagen and elastin cross-linking. Prolyl oxidase catalyzes the oxidation of proline residues in collagen, contributing to collagen stability and structure.

Option D - Prolyl hydroxylase

- Prolyl hydroxylase is an enzyme involved in the hydroxylation of proline residues in collagen molecules. It is responsible for collagen stability and proper folding.
-

Solution for Question 32:

Correct Option A - "Mitochondrial superoxide dismutase."

- Mitochondrial superoxide dismutase (MnSOD) is a manganese-containing enzyme, not a copper-containing enzyme.
- MnSOD plays a vital role in the mitochondria by converting superoxide radicals into hydrogen peroxide, protecting cells from oxidative damage.

Incorrect Options:

Option B - Cytosolic superoxide dismutase (CuZnSOD): It is a copper-containing enzyme responsible for scavenging superoxide radicals.

Option C - Tyrosinase: It is a copper-dependent enzyme involved in melanin production.

Option D - Lysyl oxidase: It is a copper-containing enzyme involved in the cross-linking of collagen and elastin fibers in connective tissues.

Solution for Question 33:

Correct Option A - Branched-chain α -ketoacid dehydrogenase:

- Explanation: Also, known as branched-chain Ketoaciduria Maple syrup urine disease (MSUD) is a rare genetic disorder. It is an aminoacidopathy. It causes a deficiency of branched-chain alpha-keto acid dehydrogenase . The enzyme breaks down the three branched-chain amino acids (BCAAs) leucine, isoleucine and valine , in the body. Very high levels of these amino acids are harmful. One of the characteristic symptoms of MSUD is sweet-smelling urine, which gives the condition its name.

Incorrect Options :

Option B - A deficiency of isovaleryl CoA dehydrogenase is not the cause of maple syrup urine disease. Isovaleryl CoA dehydrogenase deficiency causes a rare condition called isovaleric acidemia.

Option C - Pyruvate carboxylase deficiency is not the cause of maple syrup urine disease.

Option D - Acetyl-CoA carboxylase 1 deficiency causes severe brain damage, persistent myopathy, and poor growth, not maple syrup urine disease (MSUD).

Solution for Question 34:

Correct Option C - LDH1

- LDH (lactate dehydrogenase) is an enzyme that plays a role in energy production within cells. It exists as a tetramer composed of two types of subunits, namely H (heart) and M (muscle). LDH is found in various tissues, including the heart and skeletal muscle
- . During myocardial ischemia (insufficient blood supply to the heart muscle), LDH levels may change due to tissue damage.
- During myocardial ischemia, an increase in LDH1 and LDH2 levels may indicate cardiac tissue damage. LDH1 is mainly localized in the heart, and its elevation can suggest ongoing damage to cardiac cells.

Incorrect Options:

Option A - LDH 2- LDH2 is also present in the heart, but it is more commonly associated with tissue breakdown and damage.

Option B - LDH 3- Elevated levels of LDH3 and LDH4 may indicate involvement of other tissues, such as the lungs or liver, which can occur in conditions associated with myocardial infarction.

Option D - LDH 4- Elevated levels of LDH3 and LDH4 may indicate involvement of other tissues, such as the lungs or liver, which can occur in conditions associated with myocardial infarction.

Solution for Question 35:

Correct Option C - Tay-sachs disease:

- Based on the given information, the child with mental retardation, blindness, and muscle weakness, and a deficiency of Hexosaminidase-A enzyme is most likely suffering from Tay-Sachs disease.
- Tay-Sachs disease is a rare genetic disorder characterized by the accumulation of a fatty substance called GM2 ganglioside in the cells of the brain and nervous system. It is caused by a deficiency of the Hexosaminidase-A enzyme, which is responsible for breaking down this fatty substance. Without sufficient enzyme activity, GM2 ganglioside accumulates and causes progressive damage to the nerve cells.

Incorrect Options:

Option A - Krabbe's disease: Krabbe's disease is a rare genetic disorder characterized by the deficiency of the enzyme galactocerebrosidase.

Option B - Fabry's disease: Fabry's disease is an X-linked genetic disorder characterized by the deficiency of the enzyme alpha-galactosidase A.

Option D - Gaucher's disease: Gaucher's disease is an autosomal recessive genetic disorder caused by a deficiency of the enzyme glucocerebrosidase.

Solution for Question 36:

Correct Option D - Participates in feedback regulation

- Allosteric modifiers are a group of regulatory compounds that bind reversibly and non-covalently to the allosteric site of an enzyme and can alter the affinity and efficacy of other substances acting on enzymes.
- These allosteric modifiers are generally small metabolites or co-factors.
- In addition to active sites, allosteric enzymes generally have one or more regulatory or allosteric sites for binding the modifiers.
- Each modulator is specific to its regulatory site.
- Thus, a feedback regulation occurs when a product of the reaction binds to an allosteric site of an enzyme and affects its catalytic activity.

Incorrect Options :

Option A - Allosteric modifiers never desaturate the enzyme.

Option B - An increase in temperature causes the enzyme to work faster.

Option C - Allosteric modifiers do not bind to the catalytic site.

Solution for Question 37:

Correct Option A - LDH 1 > LDH 2

- In normal people LDH 2 > LDH1 but in those with myocardial infarction , LDH 1 is greater than LDH 2.

- This is known as a flipped ratio because the ratio of LDH 2 and LDH 1 are reversed in these patients compared to the normal patients.

Incorrect Options - B, C, and D: These are incorrect

Heme Synthesis and Porphyrrias

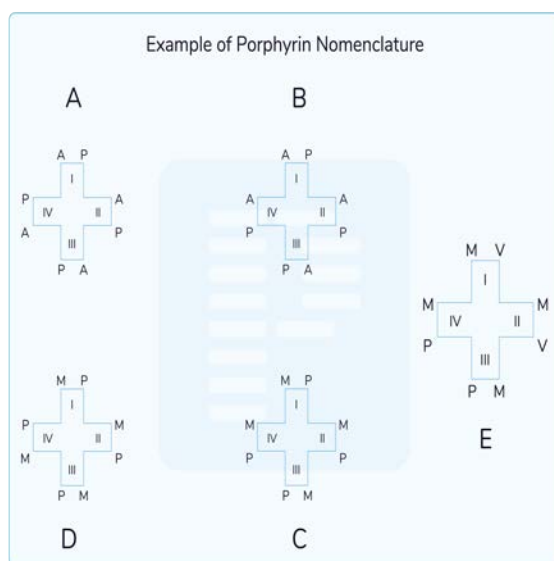
1. Which of the following statements about the protoporphyrin ring and its associated metals is incorrect?

- A. Heme is a derivative of porphyrin, a cyclic compound formed by the fusion of 4 pyrrole rings with a metallic group in the center
- B. Heme-containing proteins such as myoglobin have Fe^{3+} as the metallic group.
- C. Heme-containing enzymes are catalase, peroxidase, and nitric oxide synthase.
- D. The metallic group in Chlorophyll is magnesium, while in cytochrome a/a_3 is copper.

2. A 7-year-old boy is brought to the clinic by his parents due to severe photosensitivity and recurrent skin lesions that worsen upon sun exposure. The parents are paint factory workers living in an industrial complex. The child's urine is noted to be dark red in color. Laboratory tests confirm elevated levels of porphyrins in the blood. Which of the following statements about the diagnosis is incorrect?

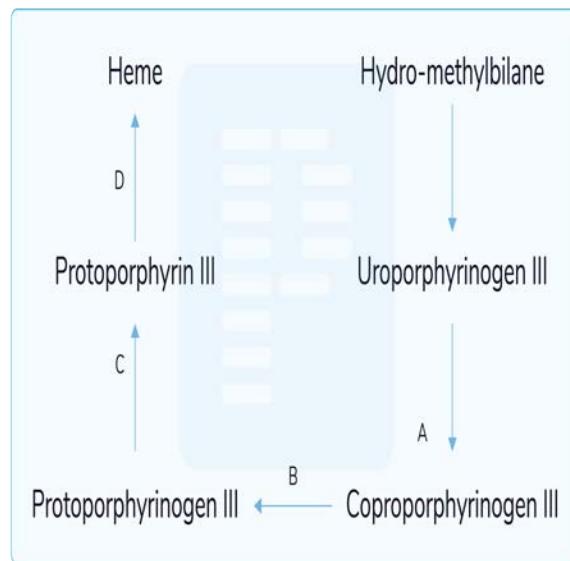
- A. The condition can occur due to exposure to lead, leading to decreased activity of ALA synthase.
- B. The red fluorescence of the teeth under UV light is referred to as erythrodontia, due to emission of 600 nm wavelength by porphyrin.
- C. Porphyrin has a specific absorption band called Soret band corresponding to 400 nm.
- D. Photosensitivity due to the accumulation of porphyrins in the skin which release superoxide radicals causing photooxidative damage.

3. Which of the following structures correctly represents the type of porphyrin? A=acetyl, P=propionyl, M=methyl, V=vinyl



- A. A- Uroporphyrinogen I, B- Uroporphyrinogen III, E- Protoporphyrinogen I
 B. D- Coproporphyrinogen III, C- Coproporphyrinogen I, E- Protoporphyrinogen I
 C. A- Uroporphyrinogen III, B- Uroporphyrinogen I, E- Protoporphyrinogen III
 D. D- Coproporphyrinogen I, C- Coproporphyrinogen III, E- Protoporphyrinogen III

4. According to the given image of heme synthesis, which of the following statements is incorrect?



- A. Steps c and d occur in Mitochondria, while steps a and b occur in the cytosol.
- B. The enzyme involved in step a) is Uroporphyrinogen decarboxylase
- C. 4 molecules of CO₂ are liberated at step a)
- D. Ferrochelatase is the enzyme involved in step d)

5. Which of the following statements about heme synthesis is incorrect?

- A. ALA synthase requires pyridoxal phosphate as a cofactor.
- B. ALA dehydratase contains zinc as a cofactor and is inhibited by lead.
- C. Pyridoxal phosphate deficiency can lead to anemia due to impaired ALA synthase activity.
- D. ALA dehydratase is the rate-limiting step of the heme synthesis pathway.

6. A 35-year-old male presents with microcytic anemia and elevated delta-aminolevulinic acid (ALA). Which statement correctly describes the regulation of ALA synthase I (ALAS I) and ALA synthase II (ALAS II)?

- A. ALAS I is primarily expressed in erythroid tissues and regulated by iron.
- B. ALAS II is ubiquitously expressed and inhibited by heme.
- C. ALAS I is inhibited by heme and primarily expressed in the liver.
- D. ALAS II is regulated by heme and is key in hepatic heme synthesis.

7. A 25-year-old male presents with fatigue, weakness, and pallor. Blood tests reveal microcytic anemia with elevated serum iron and ferritin levels. Bone marrow biopsy shows ringed sideroblasts. Genetic testing confirms a mutation in the ALAS2 gene. What is the most likely diagnosis?

- A. Iron deficiency anemia
- B. Thalassemia
- C. X-linked sideroblastic anemia
- D. Autosomal recessive sideroblastic anemia

8. Match the following disorders with the enzyme defects in context to relation synthesis. Enzyme defect Disorder A. Uroporphyrinogen III synthase i) Porphyria cutanea tarda B. Uroporphyrinogen decarboxylase defect ii) Congenital erythropoietic porphyria C. ALA synthase II iii) Acute intermittent porphyria D. Porphobilinogen deaminase iv) X-linked sideroblastic anemia

Enzyme defect	Disorder
A. Uroporphyrinogen III synthase	i) Porphyria cutanea tarda
B. Uroporphyrinogen decarboxylase defect	ii) Congenital erythropoietic porphyria
C. ALA synthase II	iii) Acute intermittent porphyria
D. Porphobilinogen deaminase	iv) X-linked sideroblastic anemia

- A. A-ii, B-i, C-iv, D-iii

- B. A-iii, B-iv, C-ii, D-i
- C. A-iv, B-iii, C-i, D-ii
- D. A-i, B-ii, C-iv, D-iii

9. Match the following disorder with its clinical classification: Disorder Clinical Manifestations
A.Acute intermittent porphyria i) Neuropsychiatric manifestations
B.Porphyrin Cutanea Tarda ii) Photosensitivity
C.Hereditary coproporphyrin iii) Both Neuropsychiatric manifestation and photosensitivity
D.Congenital erythropoietic porphyria

Disorder	Clinical Manifestations
A.Acute intermittent porphyria	i) Neuropsychiatric manifestations
B.Porphyrin Cutanea Tarda	ii) Photosensitivity
C.Hereditary coproporphyrin	iii) Both Neuropsychiatric manifestation and photosensitivity
D.Congenital erythropoietic porphyria	

- A. A-i, B-ii, C-iii, D-ii
- B. A-i, B-ii, C-ii, D-iii
- C. A-ii, B-i, C-iii, D-iii
- D. A-ii, B-i, C-ii, D-iii

10. Lead poisoning can cause acquired porphyria by inhibiting several enzymes in the heme synthesis pathway. Which of the following enzymes is not affected by lead poisoning?

- A. ALA dehydratase
- B. Coproporphyrinogen oxidase
- C. Ferrochelatase

D. ALA synthase

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	4
Question 4	1
Question 5	4
Question 6	3
Question 7	3
Question 8	1
Question 9	1
Question 10	4

Solution for Question 1:

Correct Answer: B. Heme-containing proteins such as Myoglobin have Fe^{3+} as the metallic group.

Explanation:

Heme is composed of a protoporphyrin ring with a metallic group (Fe^{2+}) in the center. (Option A)

Porphyrin derived proteins	Metallic group in the center
----------------------------	------------------------------

Heme Heme-containing proteins - Hemoglobin, Myoglobin, some cytochromes Heme-containing enzymes- catalase, peroxidase, nitric oxide synthase (Option C)	Iron (Fe ²⁺)
Hematin (Oxidized form of Heme)	Iron (Fe ³⁺)
Cytochrome a/a ₃ (Option D)	Copper
Chlorophyll	Magnesium

Solution for Question 2:

Correct Answer: A. The condition can occur due to exposure to lead, leading to decreased activity of ALA synthase.

Explanation:

The condition described is more likely a type of porphyria, which involves elevated porphyrins and can occur due to lead exposure. However, the enzyme ALA synthase is not affected by lead.

Lead primarily inhibits 3 enzymes of heme synthesis:

- Ferrochelatase
- ALA dehydratase
- Copro-porphyrinogen oxidase

Option B:

- The 4 pyrrole rings in Porphyrins are connected by methenyl bridges (double bonds) capable of absorbing light at 400 nm and emitting light at 600 nm.
- This 600 nm wavelength appears as the red fluorescence of the teeth under UV light, which is referred to as Erythrodontia . This property of porphyrins is being used in cancer phototherapy



- Since tumors often take up more porphyrins than do normal tissues, hematoporphyrin or related compounds are administered to a patient with an appropriate tumor. The tumor is then exposed to an argon laser to excite the porphyrins, producing cytotoxic effects.

Option C: The light absorbed by porphyrins has a specific absorption band called the Soret band , which corresponds to around 400 nm as seen on spectrometry.

Option D:

- Photosensitivity in porphyria is due to the accumulation of porphyrins in the skin.
- Porphyrins absorb light of shorter wavelengths (higher energy) and emit longer wavelengths (lower energy), which in turn releases superoxide radicals causing photooxidative damage.

Solution for Question 3:

Correct Answer: D. D- Coproporphyrinogen I, C- Coproporphyrinogen III, E- Protoporphyrinogen III

Explanation:

Porphyrins can exist in multiple forms depending on the group in the possible 8 positions of 4 pyrrole rings.

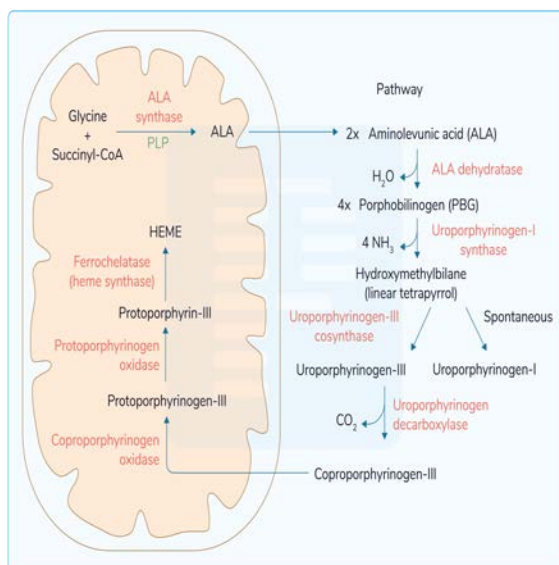
Uroporphyrinogen	Alternating acetyl (A) and propionyl (P) groups at all 8 positions. Uroporphyrinogen I is when there is symmetry in all the rings. Uroporphyrinogen III is when there is an asymmetry in the last ring.
Coproporphyrinogen	Instead of acetyl, methyl (M) groups are present. The conversion from the acetyl to the methyl group is done by decarboxylation. Uroporphyrinogen decarboxylase converts uroporphyrinogen to coproporphyrinogen.
Protoporphyrinogen	In this case, a methyl (M)-vinyl (V) group is present only in rings 1 and 2. Coproporphyrinogen oxidase converts coproporphyrinogen to protoporphyrinogen. Protoporphyrinogen can be converted into protoporphyrin by oxidation (removal of hydrogen) by protoporphyrinogen oxidase.

Solution for Question 4:

Correct Answer: A. Steps c and d occur in Mitochondria, while steps a and b occur in the cytosol.

Explanation:

Step b represents the synthesis of Protoporphyrinogen III from Coproporphyrinogen III, which occurs in mitochondria .



Heme Synthesis:

Step	Reaction	Enzyme	Location	Key Points
1	Succinyl CoA + Glycine → Delta-aminolevulinic acid (ALA)	ALA Synthase	Mitochondria	Rate-limiting step ; requires pyridoxal phosphate as a cofactor
2	2 ALA → Porphobilinogen (PBG)	ALA Dehydratase	Cytoplasm	It involves removal of 2 molecules of water; the enzyme contains zinc and is inhibited by lead
3	4 PBG → Hydroxymethylbilane (HMB)	Uroporphyrinogen I Synthase	Cytoplasm	-
4a	HMB → Uroporphyrinogen I	Spontaneous cyclization	Cytoplasm	-
4b	HMB → Uroporphyrinogen III	Uroporphyrinogen III Synthase	Cytoplasm	-
5	Uroporphyrinogen III → Coproporphyrinogen III	Uroporphyrinogen Decarboxylase (Option B)	Cytoplasm	Decarboxylation, removal of 4 molecules of CO ₂ (Option C)
6	Coproporphyrinogen III → Protoporphyrinogen III	Coproporphyrinogen Oxidase	Mitochondria	
7	Protoporphyrinogen III → Protoporphyrin III	Protoporphyrinogen Oxidase	Mitochondria	
8	Protoporphyrin III → Heme	Ferrochelatase (Option D)	Mitochondria	Insertion of Fe ²⁺

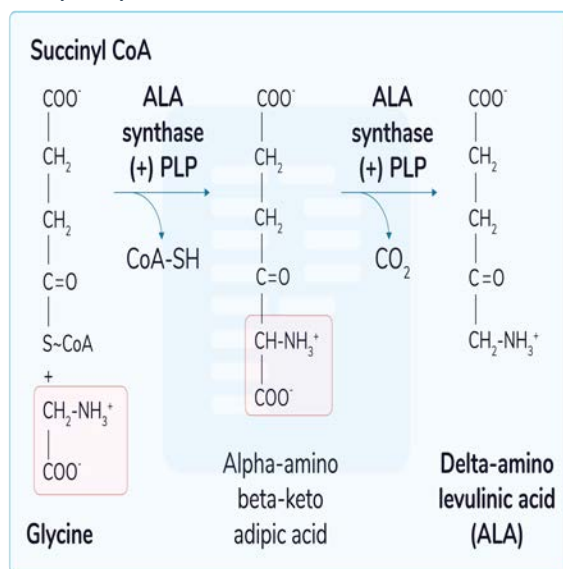
Solution for Question 5:

Correct Answer: D. ALA dehydratase is the rate-limiting step of the heme synthesis pathway.

Explanation:

ALA dehydratase is not the rate-limiting step of heme synthesis; ALA synthase is the rate-limiting enzyme in this pathway.

- Option A: ALA synthase requires pyridoxal phosphate as a cofactor for the synthesis of delta-aminolevulinic acid (ALA).



- Option B: ALA dehydratase contains zinc as a cofactor and its activity is inhibited by lead.
- Option C: Pyridoxal phosphate deficiency can lead to anemia because it impairs ALA synthase activity, reducing heme synthesis.

Solution for Question 6:

Correct Answer: C) ALAS I is inhibited by heme and primarily expressed in the liver.

Explanation:

Regulation of ALA Synthase I and II:

Feature	ALAS I (Non-Erythroid)	ALAS II (Erythroid)
Expression Location	Primarily in the liver	Erythroid tissues
Regulation by Heme	Inhibited by heme	Not regulated by heme
Regulation by Iron	Not regulated by iron	Regulated by iron availability

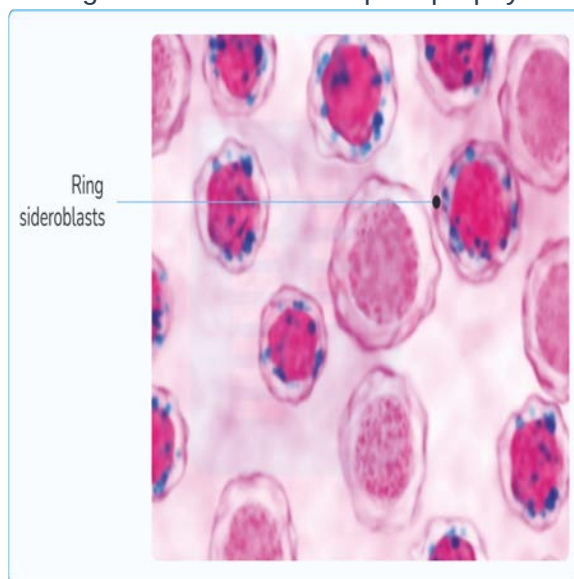
Role in Heme Synthesis	The rate-limiting enzyme in hepatic heme synthesis	Key enzyme in erythroid heme synthesis
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Solution for Question 7:

Correct Answer: C. X-linked sideroblastic anemia

Explanation:

- X-linked sideroblastic anemia (XLSA) is caused by mutations in the ALAS2 gene, which is responsible for the enzyme 5-aminolevulinate synthase 2, essential for the first step in heme synthesis.
- This condition typically presents with microcytic, hypochromic anemia, elevated serum iron and ferritin levels, and ringed sideroblasts in the bone marrow.
- The confirmed mutation in the ALAS2 gene in the patient points to XLSA, distinguishing it from other forms of anemia.
- Overactivation of ALAS2 gene causes x linked protoporphyria



Differences Between X-Linked and Autosomal Recessive Sideroblastic Anemia:

Characteristic	X-Linked Sideroblastic Anemia (XLSA)	Autosomal Recessive Sideroblastic Anemia (ARSA) (Option D)
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Genetic Cause	Mutation in the ALAS2 gene	Mutations in various genes involved in glycine transport into mitochondria, including SLC25A38 and GLRX5
Gender Predominance	Primarily affects males	It affects both males and females equally
Clinical Presentation	Microcytic, hypochromic anemia; ringed sideroblasts; increased serum iron and ferritin	Microcytic anemia; ringed sideroblasts; variable symptoms
Pathway Affected	Heme synthesis (5-aminolevulinate synthase 2)	Mitochondrial function, iron-sulfur cluster biogenesis
Common Symptoms	Fatigue, pallor, weakness	Fatigue, pallor, weakness; potential systemic symptoms
Diagnosis	Genetic testing for ALAS2 mutations	Genetic testing for specific gene mutations
Management Options	Pyridoxine, iron chelation, blood transfusions	Iron chelation, blood transfusions, supportive therapies

- Iron Deficiency Anemia (Option A) is characterized by low serum iron and ferritin levels, which do not match the elevated levels seen in the case.
- Thalassemia (Option B) involves defective globin chain production and normal iron levels, unlike the elevated iron levels and ringed sideroblasts present here.

Solution for Question 8:

Correct Answer: A) A-ii, B-i, C-iv, D-iii

Explanation:

Every defect in the enzyme involved in heme synthesis is associated with a disorder, as shown below:

Enzyme defect	Disorder
ALA synthase II	X-linked sideroblastic anemia
ALA dehydratase	ALA dehydratase deficient porphyria

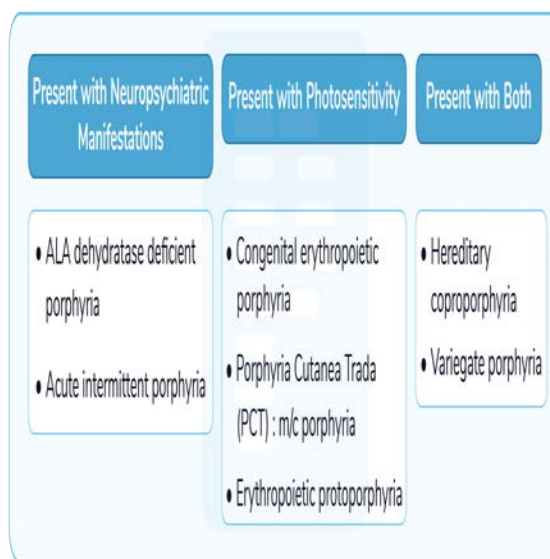
Porphobilinogen deaminase (Hydroxymethylbilane synthase)	Acute intermittent porphyria
Uroporphyrinogen III synthase	Congenital erythropoietic porphyria
Uroporphyrinogen decarboxylase	Porphyria cutanea tarda
Coproporphyrinogen oxidase	Hereditary coproporphyria
Protoporphyrinogen oxidase	Porphyria variegata
Ferro chelataase	Erythropoietic protoporphyria

Solution for Question 9:

Correct Answer: A) A-i, B-ii, C-iii, D-ii

Explanation:

Porphyria can be classified into 3 categories based on their important clinical manifestations:



- Cutaneous photosensitivity is due to accumulation of porphyrins in the skin, which enhances the production of free radicals upon exposure to light.
- Neurological complications often arise from severe episodes of acute attacks. These attacks can also trigger neuropsychiatric symptoms like confusion, hallucinations, anxiety,

and psychosis. However, these symptoms are typically self-limiting.

- CEP and PCT present with blistering photosensitivity whereas EP and X linked protoporphyria present with non blistering protoporphyria\



Werewolf appearance of blistering cutaneous manifestations

Solution for Question 10:

Correct Answer: D. ALA synthase

Explanation: Lead poisoning can lead to acquired porphyria by inhibiting 3 enzymes of the heme synthesis pathway:

Enzyme inhibited	Features
ALA dehydratase (Option A)	ALA elevation
Coproporphyrinogen oxidase (Option B)	Coproporphyrin elevation
Ferrochelatase (Option C)	Elevated Zinc protoporphyrin levels (Most sensitive marker)

Fat-soluble vitamins

1. A 62-year-old male on warfarin therapy for atrial fibrillation presents with uncontrolled bleeding. Laboratory tests show prolonged PT and INR. Which of the following mechanisms best explains the effect of warfarin on coagulation?

- A. Inhibition of Vitamin K epoxide reductase
 - B. Inhibition of thrombin activity directly
 - C. Binding to calcium ions
 - D. Direct inhibition of clotting factors
-

2. A 40 year old woman presents with complaints of persistent nausea, vomiting, and frequent urination over the past few weeks. She has a history of osteomalacia and has been taking supplements for several months without proper medical supervision. Her blood tests reveal elevated serum calcium levels. What is the most likely diagnosis?

- A. Hyperparathyroidism
 - B. Vitamin D deficiency
 - C. Vitamin D toxicity
 - D. Hypercalcemia of malignancy
-

3. Which of the following is not a commonly associated symptom with vitamin E deficiency?

- A. Spinocerebellar ataxia
- B. Megaloblastic anemia
- C. Peripheral neuropathy

D. Hemolytic anemia

4. A 45-year-old man presents with fatigue, frequent urination, constipation, and confusion. His laboratory results show hypercalcemia and low serum parathyroid hormone level. He reports taking high-dose, over-the-counter vitamin D supplements for the past year. Which of the following conditions is most likely associated with his current presentation?

- A. Increased risk of hypertension
 - B. Increased risk of colon cancer
 - C. Increased risk of breast cancer
 - D. Increased risk of insulin resistance
-

5. A 38-year-old female with a history of chronic kidney disease presents with persistent fatigue, bone pain, and muscle weakness. Blood work reveals elevated alkaline phosphatase (ALP), high serum phosphorus, and low serum calcium. What will the result of 1,25 dihydroxy vitamin D likely reveal?

- A. Elevated 1, 25 dihydroxy vitamin D levels.
 - B. Estimating 1, 25 dihydroxy vitamin D is unnecessary.
 - C. Decreased 1, 25 dihydroxy vitamin D.
 - D. Normal 1, 25 dihydroxy vitamin D.
-

6. A 2-year-old child presents with progressive bowing of the legs and tenderness in the lower limbs. The child has a history of limited outdoor play and a diet low in dairy products. Which of the following limb deformities is most likely present in this child?

- A. Rickets

- B. Blount's disease
 - C. Osteomalacia
 - D. Osteogenesis imperfecta
-

7. A 35-year-old woman presents with fatigue, muscle weakness, and bone pain for the past several months. She has a history of hypothyroidism and is currently taking levothyroxine. She lives in North India and avoids dairy products. Physical examination reveals mild tenderness over her shins. Laboratory tests reveal: Serum calcium 8.2 mg/dL (normal range: 8.5-10.5 mg/dL) Serum phosphorus 2.0 mg/dL (normal range: 2.5-4.5 mg/dL) Parathyroid hormone (PTH) Elevated Which of the following is the most likely diagnosis?

Serum calcium	8.2 mg/dL (normal range: 8.5-10.5 mg/dL)
Serum phosphorus	2.0 mg/dL (normal range: 2.5-4.5 mg/dL)
Parathyroid hormone (PTH)	Elevated

- A. Vitamin D deficiency
 - B. Primary hyperparathyroidism
 - C. Hypoparathyroidism
 - D. Chronic kidney disease
-

8. A 52-year-old woman presents with muscle cramps and tingling in her hands and feet. She follows a strict vegetarian diet high in whole grains and has a history of fat malabsorption due to IBS. Lab tests reveal hypocalcemia and low vitamin D levels. What is the most likely cause of her hypocalcemia?

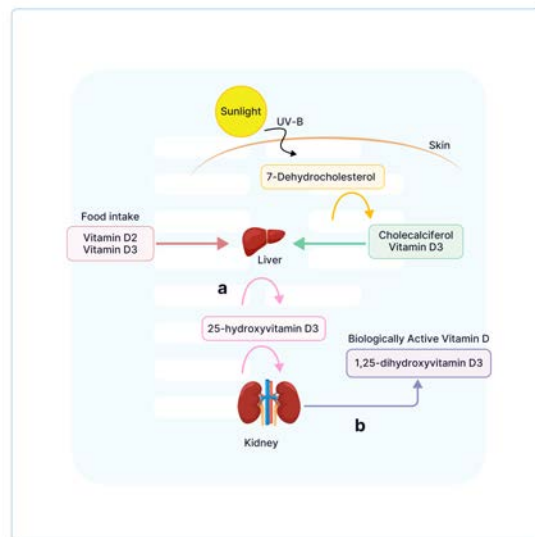
- A. Low intake of oxalate
- B. Low fatty acid in intestine
- C. High phytate intake from cereals

D. Increased calbindin level

9. Which of the following factors does not inhibit 1α -hydroxylase activity?

- A. Hyperphosphatemia
- B. Hypercalcemia
- C. High parathormone
- D. 1,25-dihydroxycholecalciferol

10. Identify the enzymes marked as a and b respectively in the flow chart, as related to vitamin D synthesis:



- A. α -25 alpha hydroxylase, liver; β -1 alpha hydroxylase, kidney
- B. α -1 alpha hydroxylase, kidney ; β -25 alpha hydroxylase, liver
- C. α -1 alpha hydroxylase, liver; β -25 alpha hydroxylase, kidney
- D. α -25 alpha hydroxylase, kidney; β -1 alpha-hydroxylase, liver

11. Which of the following features is NOT associated with a deficiency of Vitamin K?

- A. Hemorrhagic disease of the newborn
 - B. Prolonged prothrombin time
 - C. Hyperbilirubinemia
 - D. Mucous membrane hemorrhage
-

12. Which vitamins are least likely to accumulate in the body's adipose tissue due to its solubility characteristics?

- A. Retinol
 - B. Ergocalciferol
 - C. Tocopherol
 - D. Pantothenic acid
-

13. Which of the following statements about Vitamin K is incorrect?

- A. Vitamin K is crucial for the synthesis of blood clotting factors.
 - B. Vitamin K₂ is a lipid-soluble form produced by intestinal bacteria.
 - C. Menadione is a natural water-soluble form of Vitamin K.
 - D. Vitamin K aids in the carboxylation of bone proteins.
-

14. All of the following are deficiency manifestations of Vitamin E except:

- A. Spinocerebellar ataxia
 - B. Ophthalmoplegia
 - C. Hemorrhagic tendency
 - D. Peripheral neuropathy
-

15. Which of the following is NOT a chain-breaking antioxidant?

- A. α -Tocopherol
 - B. β -Carotene
 - C. Superoxide dismutase
 - D. Glutathione Peroxidase
-

16. A 17-year-old girl presents with severe acne on her face and back. She was previously treated with benzoyl peroxide and oral doxycycline and found no improvement. Physical examination shows comedones, pustules, inflammatory papules, and facial scarring. Which of the following is the next best treatment?

- A. All trans-retinoic acid
 - B. Tocopherol
 - C. Phylloquinone
 - D. 13 cis-retinoic acid
-

17. Which of the following is most likely observed in patients with vitamin A toxicity?

- A. Decreased CSF pressure
- B. Decreased bone mineral density

- C. Thinning of long bones
 - D. Increased appetite
-

18. Which of the following statements about Vitamin A deficiency is not correct?

- A. Causes rough skin and atrophy of epithelial tissues.
 - B. Prolonged deficiency causes keratomalacia.
 - C. Preventable cause of blindness in children under 5.
 - D. Increased retinol-binding protein synthesis is a cause
-

19. A 35-year-old woman reports difficulty seeing in low light. She has a poor diet, lacking in fruits and vegetables. Dark adaptation is delayed on testing. What is the most likely cause of her symptoms?

- A. Impaired retinol storage
 - B. Reduced 11-cis-retinal regeneration
 - C. Defective retinoic acid transport
 - D. Accumulation of all-trans-retinal
-

20. Which enzyme is responsible for the isomerization of all-trans retinyl ester to 11-cis retinol in the retinal pigment epithelium (RPE)?

- A. Lecithin-retinol acyltransferase
 - B. All-trans-retinol dehydrogenase
 - C. Retinoid isomerohydrolase
 - D. 11-cis-retinol dehydrogenase
-

21. Which form of vitamin A is stored in the liver and must bind to retinol-binding protein (RBP) for transport to other tissues?

- A. Retinoic acid
 - B. Retinal
 - C. Retinol
 - D. β -carotene
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	2
Question 4	1
Question 5	3
Question 6	1
Question 7	1
Question 8	3
Question 9	3
Question 10	1
Question 11	3
Question 12	4
Question 13	3
Question 14	3
Question 15	4
Question 16	4
Question 17	2
Question 18	4

Question 19	2
Question 20	3
Question 21	3

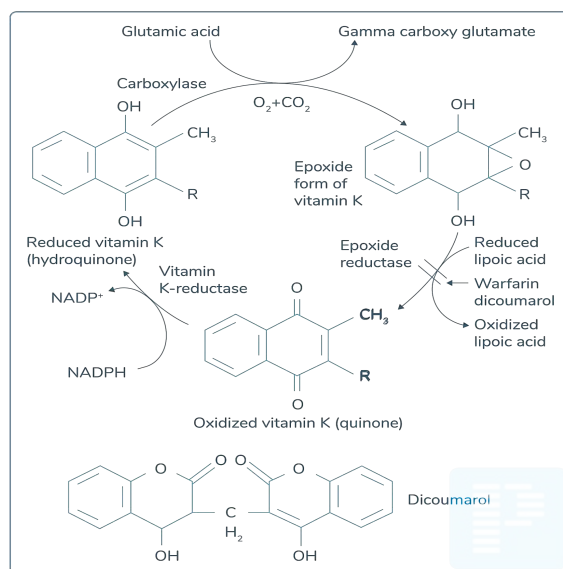
Solution for Question 1:

Warfarin acts as an anticoagulant by inhibiting the enzyme Vitamin K epoxide reductase, which is crucial for the Vitamin K cycle.

It is responsible for the reduction of Vitamin K epoxide back to Vitamin K hydroquinone, the active form of Vitamin K. This inhibition disrupts the carboxylation of Vitamin K-dependent clotting factors II, VII, IX, and X, impairing their function and leading to prolonged PT and INR, which manifests as a bleeding tendency.

Warfarin decreases both Protein C and Protein S, temporarily increasing the risk of thrombosis when treatment is initiated. Careful monitoring is essential during this period.

Vitamin K Cycle:



Inhibition of thrombin activity directly (Option B): Warfarin does not directly inhibit thrombin activity. Instead, it reduces the synthesis of thrombin's precursor, prothrombin (Factor II), necessary for its activation.

Binding to calcium ions (Option C): Vitamin K-dependent clotting factors require calcium ions for activation, but warfarin does not work by binding to calcium.

Direct inhibition of clotting factors (Option D): Warfarin indirectly prevents the synthesis of clotting factors by inhibiting their necessary γ -carboxylation, a process dependent on Vitamin K recycling.

Solution for Question 2:

Vitamin D toxicity aligns with the patient's history of high supplement intake, most probably containing Vitamin D, and her symptoms of nausea, vomiting, frequent urination, and hypercalcemia.

Comparison of Vitamin D Toxicity and Deficiency:

Aspect	Vitamin D Deficiency	Vitamin D Toxicity
Causes	Inadequate dietary intake, insufficient sun exposure, malabsorption syndromes, and certain medical conditions affecting metabolism.	Excessive supplementation. Dietary intake and sun exposure rarely cause toxicity.
Symptoms	Bone pain, muscle weakness, and increased risk of fractures; in children, it leads to rickets; in adults, it leads to osteomalacia or osteoporosis.	Nausea, vomiting, weakness, frequent urination, and hypercalcemia; can lead to kidney stones and kidney damage.
Serum Levels	Defined as 25-hydroxyvitamin D (25(OH)D) < 30 ng/mL, indicating insufficient levels.	Occurs when 25(OH)D > 150 ng/mL, leading to adverse health effects.
Treatment	Managed with dietary supplementation via food or medications and increased sun exposure.	Requires discontinuation of vitamin D intake, intravenous fluids, and medications, such as corticosteroids or bisphosphonates.

Hyperparathyroidism (Option A) typically presents with elevated serum calcium due to increased parathyroid hormone but is less likely given the context of high supplement intake.

Vitamin D deficiency (Option B) would not cause hypercalcemia; rather, it leads to bone pain, muscle weakness, and increased fracture risk.

Hypercalcemia of malignancy (Option D) can cause elevated serum calcium but does not fit the clinical context of excessive Vitamin D supplementation.

Solution for Question 3:

Megaloblastic anemia is caused due to deficiency of vitamin B9 (folic acid) and B12 (cobalamin)

Vitamin E Deficiency:

- Rare but seen in severe fat malabsorption, cystic fibrosis and some forms of chronic liver disease, as these conditions are unable to absorb or transport vitamin E.
 - Premature infants are born with inadequate vitamin E reserves.
 - Symptoms: Hemolytic anemia: Due to lipid peroxidation and fragile erythrocyte membranes. (Option D ruled out) Neurological problems due to poor nerve conduction:
 - Hemolytic anemia: Due to lipid peroxidation and fragile erythrocyte membranes. (Option D ruled out)
 - Neurological problems due to poor nerve conduction:
 - Spinocerebellar ataxia (Option A ruled out)
 - Retinopathy
 - Peripheral neuropathy (Option C ruled out)
 - Myopathies.
 - Hemolytic anemia: Due to lipid peroxidation and fragile erythrocyte membranes. (Option D ruled out)
 - Neurological problems due to poor nerve conduction:
-

Solution for Question 4:

The patient's symptoms and laboratory findings are consistent with vitamin D toxicity. Excessive vitamin D intake can lead to hypercalcemia, which can manifest as fatigue, frequent urination, constipation, and confusion. Hypercalcemia causes high blood pressure. Vitamin D toxicity also

causes low serum parathyroid hormone levels.

Vitamin D Toxicity:

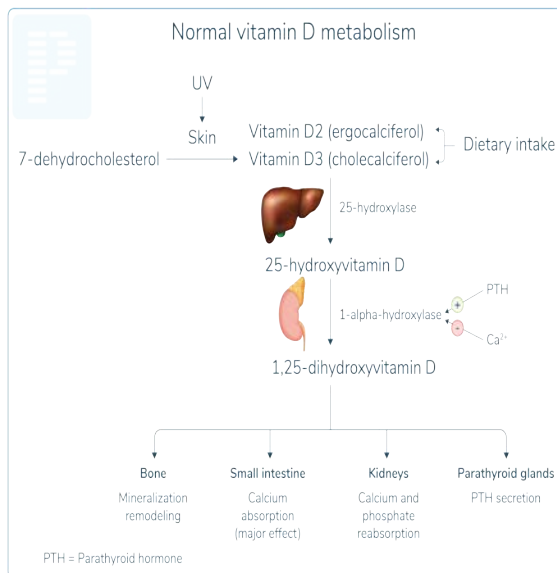
- Vitamin D toxicity is defined as Adults > 4000 IU and children > 2000 IU.
- Calcium homeostasis: Hypercalcemia Thickening of the long bones Calcification of soft tissues (Calcinosis). Calcium deposition in Blood vessels (Monckeberg medial calcific sclerosis, atherosclerosis, etc)
- Hypercalcemia
- Thickening of the long bones
- Calcification of soft tissues (Calcinosis).
- Calcium deposition in Blood vessels (Monckeberg medial calcific sclerosis, atherosclerosis, etc)
- Hypercalcemia: Contraction of blood vessels and High blood pressure. (Option A)
- While excessive dietary vitamin D is toxic, overexposure to sunlight does not cause vitamin D poisoning.
- Hypercalcemia
- Thickening of the long bones
- Calcification of soft tissues (Calcinosis).
- Calcium deposition in Blood vessels (Monckeberg medial calcific sclerosis, atherosclerosis, etc)

Beneficial effects of High Vitamin D status:

- Protects against prediabetes and metabolic syndrome. (Option D ruled out)
- Protect against various cancers: Prostate cancer Colorectal cancer (Option B ruled out) Breast cancer (Option C ruled out)
- Prostate cancer
- Colorectal cancer (Option B ruled out)
- Breast cancer (Option C ruled out)
- Prostate cancer
- Colorectal cancer (Option B ruled out)
- Breast cancer (Option C ruled out)

Solution for Question 5:

Patients with CKD often have reduced 1α -hydroxylase, leading to lower levels of active vitamin D: 1,25 dihydroxy vitamin D.



1,25 dihydroxy Vitamin D in Chronic kidney disease:

- In chronic kidney disease (CKD), estimating 1,25 dihydroxy vitamin D levels is essential due to the kidneys' role in vitamin D metabolism. (Option B ruled out)
- The kidneys are the primary site where 1α -hydroxylase converts inactive vitamin D (25 hydroxy cholecalciferol) into its active form, 1,25 dihydroxy cholecalciferol.
- Patients with CKD often have reduced 1α -hydroxylase, leading to lower levels of active vitamin D: 1,25 dihydroxy vitamin D. (Options A and D ruled out)
- Impact on Calcium Homeostasis : Low levels in CKD causes hypocalcemia.
- Secondary Hyperparathyroidism : To compensate for low calcium, PTH increases and has negative consequences for bone health.
- Renal Osteodystrophy : Abnormal bone turnover, as evidenced by the elevated alkaline phosphatase in the clinical scenario.
- Diagnostic Value : 1,25 dihydroxy cholecalciferol has a longer half-life and remains stable. Therefore, it provides a more reliable test for vitamin D status in CKD and helps guide appropriate treatment.
- Treatment: Dialysis. Calcimimetic drugs - Cinacalcet and Etelcalcetide.
- Dialysis.
- Calcimimetic drugs - Cinacalcet and Etelcalcetide.
- Dialysis.

- Calcimimetic drugs - Cinacalcet and Etelcalcetide.

Solution for Question 6:

The scenario describes a child with classic signs of Rickets associated with vitamin D deficiency. This condition results in leg deformities in children with inadequate sun exposure and dietary intake of vitamin D.

Clinical manifestations of Vitamin D deficiency	
Rickets (Option A)	Occurs in children. Inadequate bone mineralization due to poor calcium absorption. Bones become soft and weak, leading to deformities under pressure. Clinical signs : Muscle weakness Delayed walking Skeletal deformities - "pigeon chest" (pectus carinatum) and genu varus (bowed legs), Genu valgum, windswept deformity. Rickety rosary
Osteomalacia (Option C ruled out)	Occurs in adults. Demineralization of bone, making them weak and prone to fractures. Risk factors: limited sun exposure, women after multiple pregnancies.
Renal osteodystrophy	Long-standing secondary hyperparathyroidism can result in renal osteodystrophy, increasing the risk of pathological fractures.





Blount's disease (Option B) (tibia vara) is an acquired genu varus deformity typically developing later in childhood, while nutritional rickets, caused by vitamin D deficiency from poor diet and limited sun exposure, is more likely in this young child.

Osteogenesis imperfecta (Option D) is unlikely because it is a genetic disorder affecting collagen synthesis, leading to brittle bones and frequent fractures from birth, rather than the progressive bowing of the legs seen in nutritional rickets.

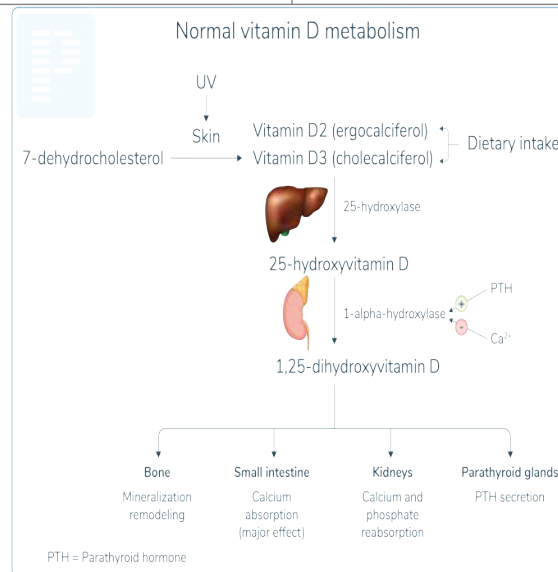
Solution for Question 7:

Based on the above scenario, the patient's symptoms and laboratory findings are most likely to be Vitamin D deficiency.

Blood findings of Vitamin D deficiency

Biochemical Feature	Explanation
Hypocalcemia	Due to decreased calcium absorption in the intestine.
Hypophosphatemia	Due to decreased phosphate absorption in the intestine and decreased reabsorption of phosphate in the kidneys.

Secondary Hyperparathyroidism	In response to low calcium levels, the parathyroid hormone (PTH) increases, which increases calcium reabsorption from bone and the kidneys.
Elevated Alkaline Phosphatase (ALP)	Due to Increased bone turnover as a result of increased PTH.
25-hydroxy cholecalciferol	Decreased.



Prolonged secondary hyperparathyroidism can cause renal osteodystrophy, which may lead to pathological fractures.

Primary hyperparathyroidism (Option B) leads to excess PTH production, causing hypercalcemia.

Hypoparathyroidism (Option C) presents with low PTH and hypocalcemia. Hypothyroidism is commonly associated with vitamin D deficiency, potentially due to immune dysfunction and impacts on calcium metabolism.

Chronic kidney disease (Option D) causes secondary hyperparathyroidism due to impaired phosphate excretion and low 1,25-dihydroxy vitamin D levels, it is typically associated with hyperphosphatemia.

Solution for Question 8:

Phytates, found in high amounts in whole grains and cereals, bind calcium in the intestine, preventing its absorption. Since the patient's diet is rich in these foods, the phytates are likely reducing her calcium absorption.

Vitamin D and Calcium Absorption:

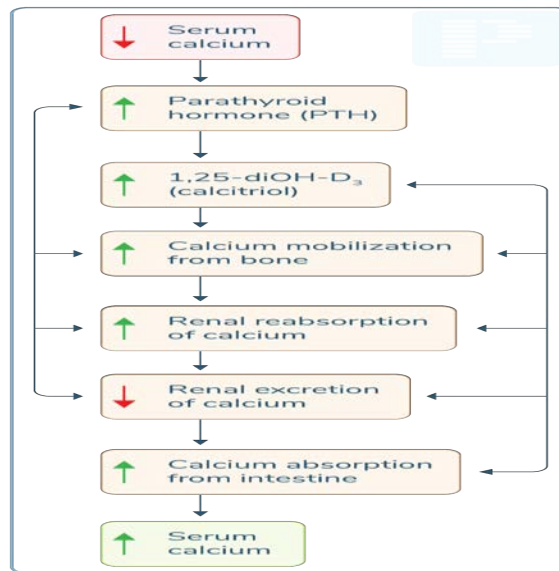
- Vitamin D is essential for intestinal calcium absorption.
- It induces the synthesis of calbindin, a protein crucial for calcium absorption. (Option D ruled out)
- Vitamin D recruits calcium transporters to the cell surface, enhancing absorption rapidly, independent of new protein synthesis.

Factors Inhibiting Calcium Absorption:

- Phytic Acid: Found in cereals, it binds calcium in the intestine, preventing absorption.
- Unleavened Whole-Wheat Products: High in phytate, which binds to calcium, preventing absorption. (Option C)
- Fatty Acids: High levels in the intestine can form insoluble calcium salts, reducing absorption. (Option B ruled out)
- Oxalate: High intake can cause calcium deficiency by forming insoluble calcium oxalate. (Option A ruled out)

Role of vitamin D in calcium metabolism:

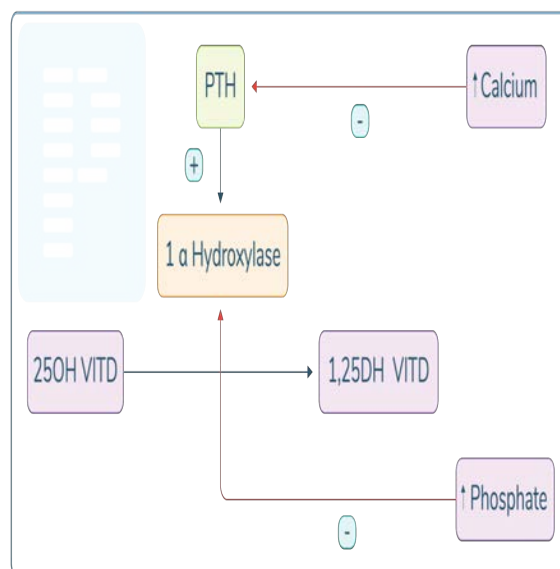
Intestine	Enterocytes directly absorb calcium by active transport (Option B ruled out) Upregulates calbindin (Option C)
Bone	Osteoblasts get activated directly and help in the mineralization of the bone Indirectly activates osteoclasts
Kidney	Increases calcium and phosphate reabsorption



Solution for Question 9:

The parathyroid hormone has a positive effect on 1 alpha-hydroxylase enzyme, which enhances the conversion of 25(OH)2-D3 to 1,25(OH)2-D3.

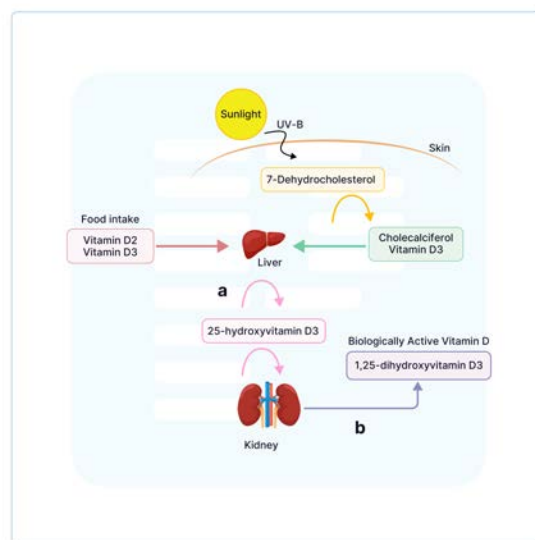
Regulation of Vit-D synthesis:



- The rate-limiting step of vitamin D synthesis is the conversion of 25(OH)₂-D₃ to 1,25(OH)₂-D₃ via 1- α hydroxylase enzyme.
- 1- α hydroxylase gets activated by parathyroid hormone (Option C).
- High calcium and phosphate levels inactivate 1- α hydroxylase activity . (Options A and B ruled out)
- 1,25-DH vit D increases calcium levels, which in turn inactivates 1- α hydroxylase (Option D ruled out).

Solution for Question 10:

Vitamin D synthesis:



(a: α - 25 alpha hydroxylase; b: β -1 alpha hydroxylase)

- Skin: Most of the precursors for 1,25(OH)₂-D₃ synthesis is produced in the malpighian layer of epidermis from 7-dehydrocholesterol via UV light-mediated photolysis reaction. Conversion efficiency is affected by UV exposure and skin pigmentation. Aging reduces 7-dehydrocholesterol levels.
- Most of the precursors for 1,25(OH)₂-D₃ synthesis is produced in the malpighian layer of epidermis from 7-dehydrocholesterol via UV light-mediated photolysis reaction.
- Conversion efficiency is affected by UV exposure and skin pigmentation.

- Aging reduces 7-dehydrocholesterol levels.
- Liver: Vitamin D is transported to the liver by vitamin D-binding protein. In the liver, vitamin D is hydroxylated to 25(OH)D₃ (25 dihydroxy cholecalciferol) by 25 alpha-hydroxylase. This process requires magnesium, NADPH, molecular oxygen and an uncharacterized cytoplasmic factor. Two enzymes involved: NADPH-dependent cytochrome P450 reductase and cytochrome P450. 25(OH)D₃ is the main circulating form of vitamin D and is transported to the kidney.
- Vitamin D is transported to the liver by vitamin D-binding protein.
- In the liver, vitamin D is hydroxylated to 25(OH)D₃ (25 dihydroxy cholecalciferol) by 25 alpha-hydroxylase.
- This process requires magnesium, NADPH, molecular oxygen and an uncharacterized cytoplasmic factor.
- Two enzymes involved: NADPH-dependent cytochrome P450 reductase and cytochrome P450.
- 25(OH)D₃ is the main circulating form of vitamin D and is transported to the kidney.
- Kidney: 25(OH)D₃ is converted to the active form, 1,25(OH)₂D₃ (1,25 dihydroxy cholecalciferol), in the kidneys by the enzyme 1 alpha-hydroxylase. This conversion involves a three-component monooxygenase system requiring NADPH, magnesium, molecular oxygen, and three enzymes: renal ferredoxin reductase, renal ferredoxin, and cytochrome P450. 1,25(OH)₂D₃ is the most potent active metabolite of vitamin D.
- 25(OH)D₃ is converted to the active form, 1,25(OH)₂D₃ (1,25 dihydroxy cholecalciferol), in the kidneys by the enzyme 1 alpha-hydroxylase.
- This conversion involves a three-component monooxygenase system requiring NADPH, magnesium, molecular oxygen, and three enzymes: renal ferredoxin reductase, renal ferredoxin, and cytochrome P450.
- 1,25(OH)₂D₃ is the most potent active metabolite of vitamin D.
- Most of the precursors for 1,25(OH)₂D₃ synthesis is produced in the malpighian layer of epidermis from 7-dehydrocholesterol via UV light-mediated photolysis reaction.
- Conversion efficiency is affected by UV exposure and skin pigmentation.
- Aging reduces 7-dehydrocholesterol levels.
- Vitamin D is transported to the liver by vitamin D-binding protein.
- In the liver, vitamin D is hydroxylated to 25(OH)D₃ (25 dihydroxy cholecalciferol) by 25 alpha-hydroxylase.
- This process requires magnesium, NADPH, molecular oxygen and an uncharacterized cytoplasmic factor.
- Two enzymes involved: NADPH-dependent cytochrome P450 reductase and cytochrome P450.

- 25(OH)2-D3 is the main circulating form of vitamin D and is transported to the kidney.
- 25(OH)D3 is converted to the active form, 1,25(OH)2D3 (1,25 dihydroxy cholecalciferol), in the kidneys by the enzyme 1 alpha-hydroxylase .
- This conversion involves a three-component monooxygenase system requiring NADPH, magnesium, molecular oxygen, and three enzymes: renal ferredoxin reductase, renal ferredoxin, and cytochrome P450.
- 1,25(OH)2D3 is the most potent active metabolite of vitamin D.

Solution for Question 11:

Hyperbilirubinemia is not associated with vitamin K deficiency. Instead, it is related to vitamin K toxicity , particularly in newborns, leading to conditions like hemolysis and kernicterus.

Vitamin K Deficiency	Vitamin K Toxicity
Causes: Fat malabsorption (e.g., cystic fibrosis) Pancreatic insufficiency	Causes: High doses of vitamin K Vitamin E deficiency
Manifestations: Bleeding mucous membrane (Option D ruled out) Increased PT and Bleeding time. (Option B ruled out) Hemorrhagic disease of the newborn (Option A ruled out)	Manifestations: Hemolytic anemia similar to Vitamin E deficiency Hemolytic anemia → Hemolytic jaundice → Kernicterus and brain damage.

Solution for Question 12:

Pantothenic acid is a water-soluble vitamin (B5) and is not stored in significant amounts in adipose tissue. Water-soluble vitamins are generally excreted in the urine and do not accumulate in the body.

Vitamin	Sources
A Retinol Retinal Retinoic acid Beta-carotene	Milk, butter, cream, cheese, egg yolk, Beta carotene, carrots, papaya
D Ergocalciferol (Vitamin D2) Cholecalciferol (Vitamin D3)	Fish liver oil, fish, egg yolk, milk
E Tocopherol (specifically, alpha-tocopherol)	Vegetable oil, Wheat germ oil, sunflower oil, safflower oil, cottonseed oil
K Phylloquinone (Vitamin K1) Menaquinone (Vitamin K2) Menadione (Vitamin K3, a synthetic form)	Phylloquinone: Green vegetables Menaquinones: Synthesized by intestinal bacteria

Solution for Question 13:

Menadione is a water-soluble form of Vitamin K widely used in clinical practice. It can be metabolised to Phylloquinone, but it's a Synthetic form, not a natural form.

Vitamin K:

- Fat-soluble vitamin that acts as a coenzyme or prosthetic group for gamma-carboxylase.
- Calcium-dependent proteins that rely on Vitamin K: Clotting factors II, VII, IX, and X. (Option A ruled out) Anticoagulant factors like protein C and S. Osteocalcin (promotes bone mineralization by depositing calcium).
- Clotting factors II, VII, IX, and X. (Option A ruled out)
- Anticoagulant factors like protein C and S.
- Osteocalcin (promotes bone mineralization by depositing calcium).
- Forms of Vitamin K: Phylloquinone (K1): Found in green vegetables, this form has an isoprenoid side chain with 20 carbons. Menaquinones (K2): Synthesized by intestinal bacteria, these have varying side chain lengths and an isoprenoid side chain containing 30 carbons, which is lipid soluble. (Option B ruled out) Menadione (K3): A synthetic form which is water-soluble. Menadione and menadione diacetate can be metabolized into phylloquinone.

- Phylloquinone (K1): Found in green vegetables, this form has an isoprenoid side chain with 20 carbons.
- Menaquinones (K2): Synthesized by intestinal bacteria, these have varying side chain lengths and an isoprenoid side chain containing 30 carbons, which is lipid soluble. (Option B ruled out)
- Menadione (K3): A synthetic form which is water-soluble. Menadione and menadione diacetate can be metabolized into phylloquinone.
- Vitamin K exists in two forms: Epoxide and Hydroquinone form.
- Functions : Blood Clotting. Bone Health - Carboxylating glutamate to glutamic acid in proteins like osteocalcin is involved in bone mineralization. (Option D ruled out)
- Blood Clotting.
- Bone Health - Carboxylating glutamate to glutamic acid in proteins like osteocalcin is involved in bone mineralization. (Option D ruled out)
- Clotting factors II, VII, IX, and X. (Option A ruled out)
- Anticoagulant factors like protein C and S.
- Osteocalcin (promotes bone mineralization by depositing calcium).
- Phylloquinone (K1): Found in green vegetables, this form has an isoprenoid side chain with 20 carbons.
- Menaquinones (K2): Synthesized by intestinal bacteria, these have varying side chain lengths and an isoprenoid side chain containing 30 carbons, which is lipid soluble. (Option B ruled out)
- Menadione (K3): A synthetic form which is water-soluble. Menadione and menadione diacetate can be metabolized into phylloquinone.
- Blood Clotting.
- Bone Health - Carboxylating glutamate to glutamic acid in proteins like osteocalcin is involved in bone mineralization. (Option D ruled out)

Solution for Question 14:

Vitamin E Toxicity:

Vitamin E Deficiency	
----------------------	--

1. Hemolytic anemia	Due to the destruction of red blood cells by oxidative stress. Vitamin E acts as an antioxidant that protects cell membranes from damage by free radicals■.
2. Spinocerebellar ataxia (Option A ruled out)	Degeneration of the cerebellum and spinal cord leading to imbalance, incoordination and ataxia.
3. Peripheral Neuropathy (Option D ruled out)	Insufficient antioxidant protection damages peripheral nerves, leading to symptoms such as numbness, tingling, and muscle weakness■.
4. Eyes	Pigment retinopathy Ophthalmoplegia (Option B ruled out) Nystagmus

Solution for Question 15:

Glutathione Peroxidase is a preventive antioxidant, not a chain-breaking antioxidant.

Antioxidants are substances that help control and reduce lipid peroxidation by neutralizing free radicals, which can cause cellular damage.

Classification of Antioxidants:

1. Preventive Antioxidants:	Reduce the rate of chain initiation. Examples: Glutathione Peroxidase (Option D) Catalase Selenium
2. Chain-Breaking Antioxidants (Option A, B and C ruled out):	Interfere with chain propagation by stabilizing free radicals. Examples: α -Tocopherol (Vitamin E) β -Carotene (Vitamin A) Vitamin C Superoxide Dismutase (SOD)

Solution for Question 16:

13 cis retinoic acid (isotretinoin) is used to treat nodulocystic acne, which is refractory to treatment with topical medication and oral antibiotics.

13 cis retinoic acid (isotretinoin):

Mechanism	Inhibits sebum production Inhibit keratinocyte proliferation
Use	Nodulocystic acne
Adverse effects	Dry skin Headache Alopecia Idiopathic intracranial hypertension Teratogenic
Contraindications	Pregnancy Liver disease
Precautions	Negative pregnancy test Two contraception methods

All trans-retinoic acid (Option A):

- Used for Acute promyelocytic leukemia (AML M3).
- Binds to retinoic acid receptors, inducing cell differentiation.

Tocopherol (Option B):

- Antioxidant (prevents free radical damage).
- Used in Non-alcoholic steatohepatitis. (NASH)

Phylloquinone (Option C):

- Known as vitamin K1 (Other forms: Menadione, menaquinone (K2))
- Cofactor for gamma-carboxylation of coagulation.
- Deficiency leads to bleeding disorders.

Solution for Question 17:

Vitamin A toxicity

Causes:

- Excessive vitamin A supplementation
- Increased polar bear liver consumption

Mechanism:

- Excessive vitamin A intake leads to accumulation beyond the capacity of binding proteins, causing membrane lysis and tissue damage.
- Higher concentrations of retinol increase lysosomal enzymes, leading to cellular death.

Symptoms:

1. Central Nervous System:

- Headache
- Nausea, vomiting
- Ataxia
- Anorexia (Option D ruled out)
- Increased cerebrospinal fluid pressure, which mimics symptoms of brain tumor (pseudotumor cerebri) (Option A ruled out)

2. Liver:

- Hepatomegaly with histological changes
- Hyperlipidemia
- Cirrhosis

3. Calcium Homeostasis:

- Thickening of long bones (Option C ruled out)
- Hypercalcemia
- Calcification of soft tissues

4. Skin:

- Excessive dryness
- Desquamation
- Alopecia

5. Bone Health:

- Decreased bone mineral density (Option B)
- Increased fracture risk
- Bony exostosis.

6. Pregnancy: Increased risk of teratogenesis: Congenital malformation in developing fetuses.

Solution for Question 18:

Causes for Vitamin A Deficiency :

- Decreased intake.
- Obstructive jaundice: Defective absorption.
- Liver cirrhosis: Reduced synthesis of RBP (retinol binding protein).
- Severe malnutrition: Lack of amino acids for RBP synthesis. (Option D)
- Chronic nephrosis: RBP excretion through urine.

Vitamin A Deficiency Manifestations :

1. Ocular manifestations:

Vitamin A deficiency is the most significant preventable cause of blindness, especially in children below 5 in developing countries. (Option C ruled out)

Early Signs:

- Loss of sensitivity to green light.
- Difficulty adapting to dim light.
- Night blindness.

Prolonged Deficiency:

a) Xerophthalmia:

- Dry, thick, keratinized conjunctiva and cornea. If untreated, it leads to corneal ulceration and ultimately results in blindness due to the formation of opaque scar tissue.
- WHO grading of xerophthalmia:

XN	Night blindness.
X1	X1A: Conjunctival xerosis. X1B: Bitot spots.
X2	Corneal xerosis.
X3	X3A: Corneal ulceration, less than one-third. X3B: Corneal ulceration, more than one-third.
XS	Corneal scar.
XF	Xerophthalmic fundus.

b) Bitot's Spots: Grayish-white triangular plaques firmly adherent to the conjunctiva.

c) Keratomalacia: When xerophthalmia persists for a longer time, it progresses to keratomalacia (corneal softening) (Option B ruled out) .

2) Immune Function:

- Deficiency increases susceptibility to infections.
- Infection reduces retinol-binding protein synthesis, lowering vitamin A levels and impairing immune responses. (Option D)

3) Skin and Mucous Membrane Lesions:

- Follicular Hyperkeratosis (Phrynoderma): Causes rough skin due to hyperkeratinization of the follicular epithelium.
- Keratinizing Metaplasia: Affects respiratory, gastrointestinal, and genitourinary tracts.
- Epithelium is atrophied. (Option A ruled out)

4) Other General Manifestation :

- Growth retardation due to defective synthesis of chondroitin sulfate.
- Decreased protein synthesis.

Solution for Question 19:

- Vitamin A Deficiency - ↓ Recycling of All-trans-Retinal; ↓ Regeneration of 11-cis-Retinal; ↑ Prolonged Dark Adaptation; ↑ Night Blindness
- Vitamin A deficiency impairs the recycling of all-trans-retinal and slows the regeneration of 11-cis-retinal, leading to prolonged dark adaptation and night blindness.

Impact of Vitamin A on the Visual Cycle :

Adequate Vitamin A

- 11-cis-Retinal Availability: With sufficient vitamin A, 11-cis-retinal is readily available to bind with opsin, forming rhodopsin. This is crucial for phototransduction, where light is converted into electrical signals in the retina.
- Efficient Dark Adaptation: The rapid conversion of 11-cis-retinal to rhodopsin supports quick adaptation to low-light conditions, enabling effective dark adaptation.

Adequate Vitamin A: ↑ Recycling of All-trans-Retinal; ↑ 11-cis-Retinal & Rhodopsin Formation; ↓ Rapid Dark Adaptation; ↓ Night blindness

Effects of Vitamin A Deficiency

- Impaired Recycling of All-trans-Retinal: In vitamin A deficiency, the recycling of all-trans-retinal, which detaches from opsin after photon absorption, is impaired. Normally,

all-trans-retinal is converted back to 11-cis-retinal through: Retinol Dehydrogenases (RDH) : Convert all-trans-retinal to all-trans-retinol. Further Conversion: All-trans-retinol is then converted to 11-cis-retinal.

- Retinol Dehydrogenases (RDH) : Convert all-trans-retinal to all-trans-retinol.
- Further Conversion: All-trans-retinol is then converted to 11-cis-retinal.
- Slower Regeneration: The regeneration of 11-cis-retinal from all-trans-retinol is significantly slowed, leading to: Prolonged Dark Adaptation: The delay in regenerating 11-cis-retinal results in slower adaptation to dark conditions. Night Blindness: The visual system's reduced ability to adjust to low light conditions leads to symptoms such as night blindness.
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- Night Blindness: The visual system's reduced ability to adjust to low light conditions leads to symptoms such as night blindness.
- Retinol Dehydrogenases (RDH) : Convert all-trans-retinal to all-trans-retinol.
- Further Conversion: All-trans-retinol is then converted to 11-cis-retinal.
- Prolonged Dark Adaptation: The delay in regenerating 11-cis-retinal results in slower adaptation to dark conditions.
- Night Blindness: The visual system's reduced ability to adjust to low light conditions leads to symptoms such as night blindness.

Impaired retinol storage (Option A) could lead to vitamin A deficiency, but this option is vague and doesn't specifically explain the mechanism affecting night vision. The critical issue in night blindness is the regeneration of 11-cis-retinal, which is directly involved in the visual cycle.

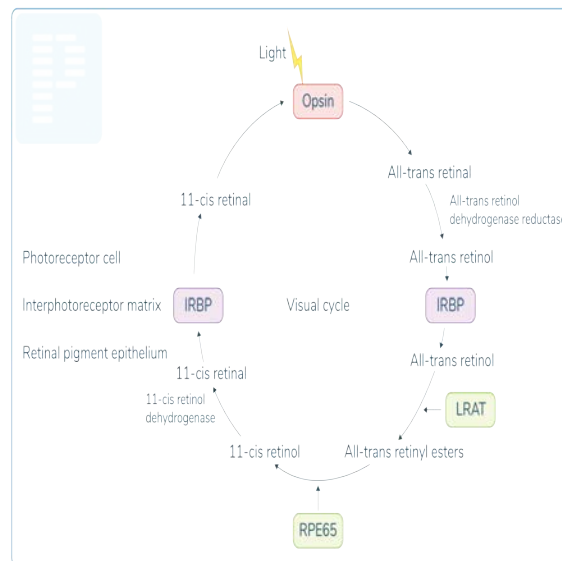
Defective retinoic acid transport (Option C): Retinoic acid is involved in gene expression and cellular differentiation, not in the visual cycle. A defect in its transport would likely lead to other systemic symptoms, not specifically night blindness or delayed dark adaptation.

Accumulation of all-trans-retinal (Option D) could potentially cause oxidative stress, but this is more related to photoreceptor damage rather than the primary issue in night blindness, which is the lack of 11-cis-retinal necessary for vision in low light.

Solution for Question 20:

Retinoid isomerohydrolase (RPE65) is the enzyme responsible for the isomerization of all-trans retinyl ester to 11-cis retinol, a crucial step in the visual cycle.

The visual cycle, also known as Wald's visual cycle, involves the conversion of light into electrical signals in the photoreceptors of the retina and the regeneration of visual pigments to maintain visual sensitivity.



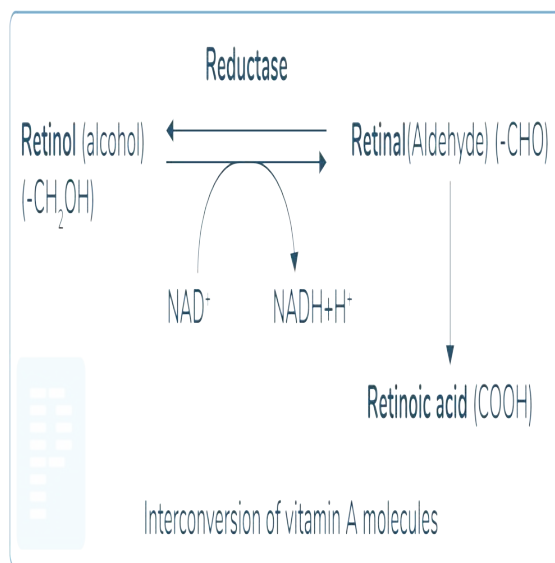
Phototransduction Process/ Photobleaching			
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Light photons are absorbed by visual pigments (rhodopsin in rods, opsins in cones) in the retina. The absorption of a photon causes the 11-cis-retinal (a chromophore bound to opsin) to isomerize to all-trans-retinal. The isomerization induces a conformational change in the opsin protein, activating it (metarhodopsin II). Activated opsin (metarhodopsin II) → Activates the G-protein transducin → Activates PDE, which hydrolyzes cyclic GMP (cGMP) to GMP. Decreased cGMP levels → Leads to the closure of Na ⁺ channels in the photoreceptor cell membrane → Hyperpolarization → Generation of an electrical signal → Visual pathway → Vision occurs			
Visual Cycle Regeneration - Regenerated 11 cis retinal from All-trans retinal			
Step	Site	Reaction	Enzyme/Mechanism
1	Photoreceptor outer segment	All-trans-Retinal + NADPH → All-trans-Retinol + NADP ⁺	All-trans-Retinol Dehydrogenase (Option B)
2	Retinal pigment epithelium (RPE) cells.	Transport to retinal pigment epithelium (RPE) cells.	Cellular Retinoid Binding Proteins
3	Esterification : All-trans-Retinol + Fatty Acyl CoA → All-trans-Retiny Ester + CoA	Lecithin-Retinol Acyltransferase (LRAT) (Option A)	
4	Isomerization and Hydrolysis: All-trans-Retiny Ester → 11-cis-Retinol	RPE65 (retinoid isomerohydrolase) (Option C)	

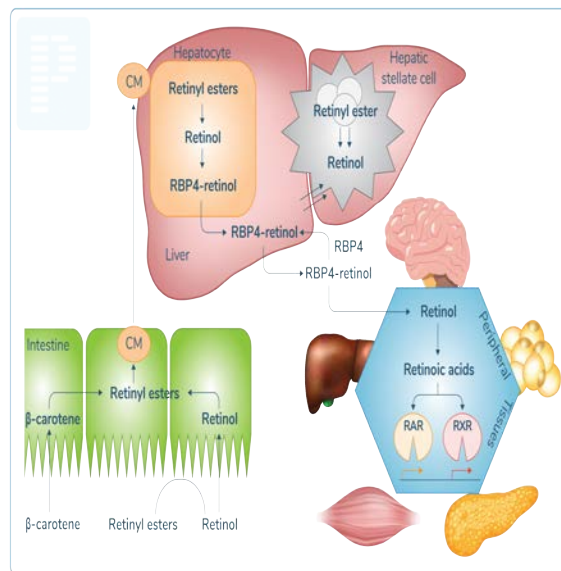
5	Oxidation: 11-cis-Retinol + NAD ⁺ → 11-cis-Retinal + NADH + H ⁺	11-cis-Retinal Dehydrogenase (Option D)	
6	Photoreceptor outer segment	Transport Back to Photoreceptors	Cellular Retinoid Binding Proteins
7	11-cis-Retinal + Opsin → Rhodopsin (or Cone Opsin)	-	

Solution for Question 21:

Retinol is the form of vitamin A that is stored in the liver as retinyl esters. For transport to other tissues, retinol must bind to retinol-binding protein (RBP).



Absorption and Transport of Retinoids:



Retinoids	Structurally related molecules with vitamin A activity.
Functions	Vision, reproduction, growth, epithelial tissue maintenance, immune function.
Types of Retinoids	
Retinol	Primary alcohol containing a β -ionone ring with an unsaturated side chain Storage form of vitamin A as retinyl ester with long-chain fatty acid.
Retinal (Option B)	Aldehyde derived from oxidation of retinol. Retinal and retinol can readily be interconnected.

Retinoic Acid (Option C)	Acid form derived from oxidation of the retinal. Cannot be reduced in body, so can't give rise to retinal or retinol. Binds to retinoic acid receptors (RARs) in target tissue nuclei. Regulates gene expression by modulating RNA synthesis and protein production. 2 Forms: ATRA & Isotretinoin All-trans Retinoic Acid (ATRA) Promotes growth and differentiation of epithelial cells. Mild acne treated with topical tretinoin (all-trans retinoic acid). Treatment in acute promyelocytic leukemia (APL), a subtype of acute myeloid leukemia (AML). 13-cis Retinoic Acid (Isotretinoin) Suppresses keratinization; used to treat resistant acne and Harlequin ichthyosis. Induces apoptosis in sebaceous glands, reducing sebum production.
Functions of Retinoids	
Vision	Essential for visual pigments in rod and cone cells.
Epithelial Cell Maintenance	Supports healthy differentiation and mucus secretion.
Reproduction	Crucial for spermatogenesis and preventing fetal resorption.
Vitamin A Deficiency	
Phrynoderma (Follicular Hyperkeratosis)	Small, papillary lesions with keratin plugs on the limbs; impaired suppression of keratinization.
Conjunctival Xerosis	Dry and keratinized conjunctiva; manifests as Bitot's spots (triangular, foamy white or yellowish spots on the sides of the eyes).

Beta-carotene (Option D): It is found in plant-based foods, β -carotene is a provitamin A. The human body uses an intestinal enzyme to cleave one beta-carotene molecule to produce two retinal molecules, which can then be reduced to retinol.

Carotenemia is a condition characterized by yellow-orange discoloration of the skin due to elevated levels of carotenoids in the blood, primarily from high dietary intake of carotenoid-rich foods like carrots and sweet potatoes.

Water-Soluble Vitamins: Roles, Deficiencies, and Alcohol Impact

1. Which of the following statements about vitamin B12 absorption is true?

- A. Intrinsic factor is secreted by the pancreas and binds to inactive forms of vitamin B12.
 - B. Cobalophilin in the saliva binds vitamin B12 initially and is hydrolyzed in the duodenum.
 - C. Gastric acid and pepsin are not necessary for Vitamin B12 absorption
 - D. Ileal receptors bind directly to vitamin B12 without the need for intrinsic factors.
-

2. A 35-year-old woman comes to the clinic complaining of delayed wound healing, frequent infections, and loss of taste. She also reports fatigue and swollen, bleeding gums. Her diet is low in fruits, vegetables, and protein sources. Laboratory tests confirm a diagnosis of vitamin C deficiency (scurvy). Given that certain nutritional deficiencies can cause similar symptoms, which of the following should also be considered as a possible issue?

- A. Iron Deficiency
 - B. Zinc Deficiency
 - C. Folate Deficiency
 - D. Vitamin B12 Deficiency
-

3. Which of the following statements regarding the avidin-biotin system is not true?

- A. Avidin has a higher affinity for biotin than typical antigen-antibody interactions.
- B. Avidin can bind up to four molecules of biotin, enhancing assay sensitivity.
- C. Consumption of raw egg whites may lead to biotin deficiency due to avidin-binding biotin.

D. Avidin is heat-stable and retains its binding capacity after boiling eggs.

4. Which of the following enzymes is not classified as a biotin-dependent carboxylase?

- A. Acetyl CoA carboxylase
 - B. Pyruvate carboxylase
 - C. Propionyl CoA carboxylase
 - D. Malic enzyme
-

5. A 55-year-old man with a history of hyperlipidemia was started on high-dose niacin (vitamin B3) to improve his lipid profile. After several weeks, he reports experiencing blurred vision and discomfort in his eyes. Upon examination, retinal imaging reveals macular oedema. Which of the following best describes the mechanism by which niacin toxicity can lead to macular oedema?

- A. Increased prostaglandin synthesis
 - B. Inhibition of retinal blood flow
 - C. Direct toxicity to retinal pigment epithelium
 - D. Enhanced inflammation in the retina
-

6. A 28 year old pregnant woman presents with complaints of fatigue and weakness. Her dietary history reveals she follows a strict vegetarian diet. Blood tests reveal megaloblastic anaemia. Which of the following complications is most likely associated with her condition if left untreated?

- A. Increased risk of gestational diabetes
- B. Neurological problems in the mother and cognitive impairments in the foetus

- C. Higher incidence of hypertension during pregnancy
 - D. Increased risk of maternal infections
-

7. After consuming illicit homemade alcohol at a local event, numerous individuals were hospitalized with severe abdominal pain, vomiting, malaise and vision changes. Which of the following cannot be used as an antidote for methanol poisoning?

- A. Ethanol
 - B. Fomepizole
 - C. Folinic acid
 - D. Formic acid
-

8. A 50 year old male with a history of chronic alcohol dependence presents with severe agitation, visual hallucinations, tremors of the tongue and eyelids, and sweating. He reports that he stopped drinking alcohol abruptly three days ago. His vital signs show tachycardia and hypertension. Which of the following management strategies is the most appropriate for his condition?

- A. Initiate antipsychotic medication
 - B. Prescribe selective serotonin reuptake inhibitors (SSRIs)
 - C. Administer benzodiazepines for sedation
 - D. Begin opioid replacement therapy
-

9. A 45-year-old man with a history of heavy alcohol consumption for the past 10 years, presents with fatigue, abdominal discomfort, weight gain, and hepatomegaly. Laboratory tests show elevated liver enzymes and a liver ultrasound reveals fatty liver. What commonly causes fatty liver in chronic alcohol

consumption?

- A. Decreased acetyl CoA levels
 - B. Increased pyruvate oxidation
 - C. Accumulation of acetaldehyde
 - D. Increased fatty acid oxidation
-

10. A 45-year-old male presents to the emergency department with confusion, tremors, and diaphoresis. He has a long history of chronic alcohol use and has been drinking heavily for the past week. His family reports that he has not eaten well during this time. Upon examination, you note signs of malnutrition and his blood glucose level is found to be 50 mg/dL. Which of the following statements regarding the effects of chronic alcoholism on nutritional deficiencies and blood glucose levels is FALSE?

- A. Chronic alcohol use can cause vitamin B12 deficiency despite normal blood levels.
 - B. Folate deficiency in alcoholics leads to megaloblastic anemia.
 - C. Chronic alcohol use causes hypoglycemia by increasing the NADH/NAD⁺ ratio.
 - D. Chronic alcohol use causes hypoglycemia by decreasing the NADH/NAD⁺ ratio.
-

11. A 50-year-old patient presents with symptoms of fatigue and pallor. The physician suspects a vitamin B12 absorption issue and decides to perform the Schilling test. Which of the following statements about the Schilling test is correct?

- A. The Schilling test involves administering one dose of vitamin B12 to assess absorption.
 - B. An abnormal result in stage 1 and a normal result in Stage 2 of the Schilling test typically suggests a lack of intrinsic factor.
 - C. Radioactive vitamin B12 injection is given to saturate body stores.
 - D. Abnormal Stage 2 suggests intrinsic factor deficiency.
-

12. A 35 year old woman presents with severe breathlessness, palpitations, and swelling in her legs and face. She primarily eats white rice and lacks fruits and vegetables in her diet. On examination and investigations she has distended neck veins, peripheral edema, and elevated lactate and pyruvate levels. Thiamine deficiency is suspected. What is the most likely diagnosis, and which enzyme activity should be tested to confirm it?

- A. Dry beriberi; pyruvate dehydrogenase
- B. Maple Syrup Urine Disease (MSUD); branched-chain ketoacid dehydrogenase
- C. Wet beriberi; transketolase
- D. Lactic acidosis; lactate dehydrogenase

13. Match the following vitamins with their respective functional assessment tests:

1. Vitamin B1 (Thiamine) a) Measure urinary methylmalonic acid
2. Vitamin B2 (Riboflavin) b) Measure transketolase activity in red blood cells
3. Vitamin B6 (Pyridoxine) c) Measure glutathione reductase activity in red blood cells
4. Vitamin B12 (Cobalamin) d) Measure activation of red blood cell transaminases with pyridoxal phosphate

1. Vitamin B1 (Thiamine)	a) Measure urinary methylmalonic acid
2. Vitamin B2 (Riboflavin)	b) Measure transketolase activity in red blood cells
3. Vitamin B6 (Pyridoxine)	c) Measure glutathione reductase activity in red blood cells
4. Vitamin B12 (Cobalamin)	d) Measure activation of red blood cell transaminases with pyridoxal phosphate

- A. 1-b, 2-c, 3-d, 4-a
- B. 1-a, 2-b, 3-c, 4-d
- C. 1-b, 2-d, 3-c, 4-a
- D. 1-a, 2-a, 3-b, 4-d

14. A 4-year-old child presents with developmental delay, irritability, and poor feeding. On examination, the child is noted to have a macrocytic anemia, peripheral neuropathy, and a decreased reflex response. Laboratory results reveal elevated homocysteine levels, but normal methylmalonic acid levels. Serum vitamin B12 levels are within the normal range. Which of the following is the most likely diagnosis?

- A. Methylcobalamin deficiency
 - B. Adenosylcobalamin deficiency
 - C. Folate deficiency
 - D. Vitamin B12 deficiency due to dietary insufficiency
-

15. Which of the following consequences of vitamin B6 deficiency directly contributes to an increased risk of seizures in a neonate with refractory seizures?

- A. Decreased GABA synthesis
 - B. Elevated glutamate levels
 - C. Increased serotonin production
 - D. Enhanced myelin formation
-

16. Which of the following metabolic processes is not directly supported by vitamin B6 in its active form, pyridoxal 5'-phosphate (PLP)?

- A. Transamination of amino acids
 - B. Decarboxylation of amino acids
 - C. Breakdown of glycogen to glucose-1-phosphate
 - D. Conversion of glucose to glycogen
-

17. A 15 year old child presents with skin rashes, gastrointestinal issues, aminoaciduria and cognitive disturbances. Which of the following mechanisms explains the symptoms observed?

- A. Hartnup disease causes a direct deficiency in niacin, leading to pellagra symptoms.
 - B. Impaired tryptophan absorption → ↓production of niacin → pellagra-like symptoms.
 - C. Tryptophan excess → disrupts niacin metabolism → pellagra-like symptoms
 - D. Increases the metabolism of tryptophan to serotonin, depleting niacin stores
-

18. A 45 year old woman from a rural area presents with a bright red rash on her feet, ankles, and face that worsens with sun exposure. She also reports frequent diarrhea and recent episodes of irritability and difficulty concentrating. Her diet consists mainly of maize and sorghum. Based on these symptoms and dietary habits, which condition is most likely causing her symptoms?

- A. Vitamin B12 deficiency
 - B. Pellagra
 - C. Rickets
 - D. Iron deficiency anemia
-

19. Which of the following is the earliest sign of riboflavin (Vitamin B2) deficiency?

- A. Cheilosis
 - B. Glossitis
 - C. Seborrheic dermatitis
 - D. Proliferation of the bulbar conjunctival capillaries
-

20. Which of the following enzymes does not utilize FAD as a coenzyme?

- A. Succinate Dehydrogenase
 - B. NADH Dehydrogenase
 - C. Xanthine Oxidase
 - D. Pyruvate Dehydrogenase
-

21. A 52 year old chronic alcoholic presents to the ED with confusion, difficulty walking, and abnormal eye movements. He reports having consumed large amounts of alcohol daily for many years and has had poor dietary intake. On examination, he has nystagmus and an unsteady gait. What is the most likely diagnosis for this patient?

- A. Wernicke encephalopathy
 - B. Korsakoff psychosis
 - C. Lactic acidosis
 - D. Dry beri-beri
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	4
Question 4	4
Question 5	1
Question 6	2
Question 7	4
Question 8	3
Question 9	3
Question 10	4
Question 11	2
Question 12	3
Question 13	1
Question 14	1
Question 15	1
Question 16	4
Question 17	2
Question 18	2

Question 19	4
Question 20	2
Question 21	1

Solution for Question 1:

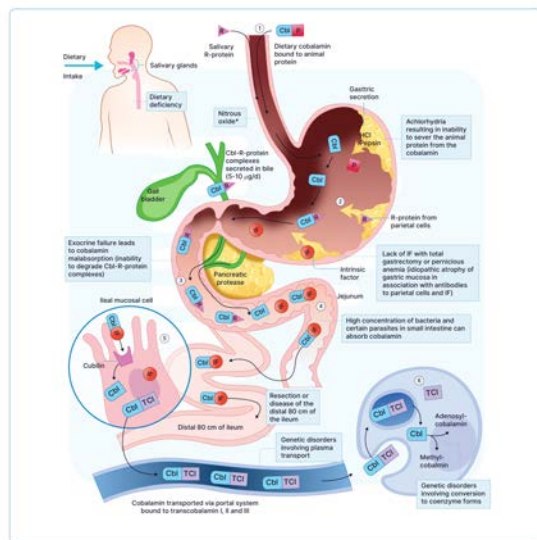
- Cobalophilin (Haptocorrin) is a salivary glycoprotein that binds to vitamin B12 (cobalamin) in the stomach.
- In the duodenum , pancreatic proteases hydrolyze cobalophilin , freeing vitamin B12 to bind intrinsic factor (IF) for further absorption.
- This step is essential for efficient vitamin B12 absorption in the ileum .

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Step	Description
1. Release in Stomach	Gastric acid and pepsin release vitamin B12 from dietary proteins.
2. Binding to Cobalophilin (Haptocorrin)	Salivary cobalophilin (R-protein) binds vitamin B12 for protection from acidic degradation.
3. Release in Duodenum	Pancreatic enzymes hydrolyze cobalophilin, releasing vitamin B12.
4. Binding to Intrinsic Factor (IF)	Intrinsic factor (IF) secreted by gastric parietal cells binds vitamin B12.
5. Ileal Absorption	The IF-B12 complex binds to specific receptors in the terminal ileum for absorption.
6. Transport in Blood	Vitamin B12 binds to transcobalamin II for systemic distribution.



Intrinsic factor is secreted by the pancreas and binds to inactive forms of vitamin B12. (Option A)

- Intrinsic factor (IF) is secreted by parietal cells of the stomach , not the pancreas.
- It binds active vitamin B12, facilitating ileal absorption.

Intrinsic factor (IF) is secreted by parietal cells of the stomach , not the pancreas.

It binds active vitamin B12, facilitating ileal absorption.

Gastric acid and pepsin are not necessary for vitamin B12 absorption. (Option C)

- Gastric acid and pepsin are essential to release vitamin B12 from dietary proteins in the stomach.
- Without them, B12 remains bound to food and cannot be absorbed .

Gastric acid and pepsin are essential to release vitamin B12 from dietary proteins in the stomach.

Without them, B12 remains bound to food and cannot be absorbed .

Ileal receptors bind directly to vitamin B12 without the need for intrinsic factors. (Option D)

- Vitamin B12 must be bound to intrinsic factor (IF) to be absorbed.
- The ileal receptors recognize the IF-B12 complex , not free vitamin B12.

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The ileal receptors recognize the IF-B12 complex , not free vitamin B12.

Solution for Question 2:

- Zinc plays a critical role in wound healing, immune function, and taste perception .
- A diet deficient in protein sources , as mentioned in the question, increases the risk of zinc deficiency.
- Vitamin C and zinc both contribute to collagen synthesis and immune defense , so their deficiencies may present with similar symptoms like poor wound healing and frequent infections.

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Zinc and Vitamin C Deficiency:

Aspect	Vitamin C Deficiency (Scurvy)	Zinc Deficiency
Wound Healing	Impaired collagen synthesis	Impaired epithelial and tissue repair
Immune Function	Increased infections, weak immune response	Frequent infections, impaired immunity
Gum Symptoms	Swollen, bleeding gums (scurvy)	Less common but can cause oral lesions
Taste Perception	Not a primary feature	Hypogeusia (loss of taste)
Skin Changes	Petechiae, ecchymoses (bleeding under skin)	Dermatitis, rough skin
Common Risk Factors	Low fruit/vegetable intake	Low protein/zinc intake (meat, dairy, legumes)

Iron Deficiency (Option A): Incorrect.

- Causes microcytic anemia and fatigue but does not significantly affect wound healing or taste perception .

Causes microcytic anemia and fatigue but does not significantly affect wound healing or taste perception .

Folate Deficiency (Option C): Incorrect.

- Primarily associated with megaloblastic anemia and neural tube defects in pregnancy .

- No direct impact on wound healing or immune function .

Primarily associated with megaloblastic anemia and neural tube defects in pregnancy .

No direct impact on wound healing or immune function .

Vitamin B12 Deficiency (Option D): Incorrect.

- Causes neurological symptoms (paresthesia, ataxia) and megaloblastic anemia , not delayed wound healing or taste loss .

Causes neurological symptoms (paresthesia, ataxia) and megaloblastic anemia , not delayed wound healing or taste loss .

Solution for Question 3:

- Avidin is a glycoprotein found in raw egg whites that binds biotin with high affinity.
- Biotin binding by avidin prevents its absorption in the intestine, potentially leading to deficiency.
- Avidin is heat-labile; cooking denatures it, rendering it incapable of binding biotin.
- Since avidin loses its function when exposed to heat, it does not retain its binding capacity after boiling eggs.

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Since avidin loses its function when exposed to heat, it does not retain its binding capacity after boiling eggs.

Aspect	Details
Avidin	Glycoprotein found in egg whites; binds biotin with high affinity.
Biotin	Water-soluble vitamin (B7) essential for carboxylation reactions in metabolism.
Affinity (Option A ruled out)	Avidin-biotin binding is among the strongest non-covalent interactions ($K_d \sim 10^{-14}$ M).

Binding Sites (Option B ruled out)	Each avidin molecule has four binding sites, making it useful in high-sensitivity biochemical assays.
Biotin Deficiency (Option C ruled out)	Chronic raw egg consumption can lead to deficiency, causing dermatitis, alopecia, and neurological symptoms.
Heat Stability	Avidin is heat-labile and denatured by cooking, preventing it from binding biotin.
Applications	Used in biochemical assays like ELISA, immunohistochemistry, and affinity chromatography.

Solution for Question 4:

- Biotin-dependent carboxylases catalyze carboxylation reactions, where CO₂ is added to a substrate.
- The major biotin-dependent carboxylases include: Acetyl CoA carboxylase → Fatty acid synthesis
Pyruvate carboxylase → Gluconeogenesis
Propionyl CoA carboxylase → Odd-chain fatty acid metabolism
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- Malic enzyme, on the other hand, is not a carboxylase. It converts malate to pyruvate by decarboxylation, generating NADPH in the process.

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Enzyme	Function	Pathway
Acetyl CoA carboxylase (Option A ruled out)	Converts acetyl CoA → malonyl CoA	Fatty acid synthesis
Pyruvate carboxylase (Option B ruled out)	Converts pyruvate → oxaloacetate	Gluconeogenesis
Propionyl CoA carboxylase (Option C ruled out)	Converts propionyl CoA → methylmalonyl CoA	Odd-chain fatty acid metabolism
Malic enzyme (Not biotin-dependent)	Converts malate → pyruvate (Decarboxylation)	NADPH generation

Solution for Question 5:

- Niacin toxicity can cause macular edema due to increased prostaglandin synthesis , leading to vasodilation and increased vascular permeability in the retina.
- Prostaglandins promote fluid leakage from retinal capillaries , contributing to macular edema .
- Niacin-induced flushing and vasodilation are mediated by prostaglandins , which explains both the systemic and ocular effects.
- Discontinuation of niacin or treatment with prostaglandin inhibitors (e.g., aspirin, NSAIDs) can help resolve the condition.

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Discontinuation of niacin or treatment with prostaglandin inhibitors (e.g., aspirin, NSAIDs) can help resolve the condition.

Niacin Toxicity and Ocular Complications

Effect	Mechanism	Clinical Manifestations
Flushing	Increased prostaglandins → Vasodilation	Redness, warmth, tingling
Macular Edema	Increased vascular permeability due to prostaglandins	Blurred vision, central vision loss
Hepatotoxicity	Disrupted lipid metabolism	Elevated liver enzymes, jaundice
Hyperglycemia	Insulin resistance	Increased blood glucose levels
Hyperuricemia	Reduced renal excretion of uric acid	Increased risk of gout

Inhibition of retinal blood flow (Option B): Incorrect because niacin increases rather than decreases retinal blood flow due to vasodilation.

Direct toxicity to retinal pigment epithelium (Option C): Incorrect because niacin does not cause direct retinal pigment epithelial damage .

Enhanced inflammation in the retina (Option D): Incorrect because macular edema in niacin toxicity is not due to inflammatory cell infiltration , but rather increased prostaglandin-induced vascular leakage .

Solution for Question 6:

- The patient's fatigue, weakness, pallor, mild jaundice, and megaloblastic anemia suggest vitamin B12 deficiency , which is common in strict vegetarians as B12 is found primarily in animal-based foods .
- Vitamin B12 is essential for DNA synthesis, myelin formation, and neurological function , making its deficiency particularly dangerous during pregnancy.

Vitamin B12 Deficiency in Pregnancy

Aspect	Details
Importance	Essential for DNA synthesis, red blood cell formation, and neurological function .
Increased Demand	Higher requirement due to fetal development and maternal blood volume expansion .

Complications of Untreated Maternal Vitamin B12 Deficiency:

Category	Effects
Maternal Neurological Problems	Peripheral neuropathy (numbness, tingling, weakness) Memory disturbances Depression and cognitive decline
Fetal Cognitive Impairments	Poor brain development Increased risk of neural tube defects (NTDs) Developmental delay
Hematological Effects	Megaloblastic anemia Increased risk of preterm birth

Increased risk of gestational diabetes (Option A): Incorrect . Vitamin B12 deficiency is not directly linked to gestational diabetes.

Higher incidence of hypertension during pregnancy (Option C): Incorrect. No strong evidence links vitamin B12 deficiency to gestational hypertension or preeclampsia.

Increased risk of maternal infections (Option D): Incorrect. While B12 deficiency weakens the immune system, it does not significantly increase maternal infections.

Solution for Question 7:

Methanol poisoning occurs due to the metabolism of methanol into formic acid and formate , which are highly toxic and lead to:

- Metabolic acidosis (high anion gap)
- Optic nerve toxicity → Vision changes ("snowfield vision")
- Central nervous system depression
- Severe abdominal symptoms

Metabolic acidosis (high anion gap)

Optic nerve toxicity → Vision changes ("snowfield vision")

Central nervous system depression

Severe abdominal symptoms

Antidotes work by preventing methanol metabolism or enhancing its detoxification.

Antidote	Mechanism of Action
----------	---------------------

Ethanol (Option A ruled out)	Competitive inhibition of alcohol dehydrogenase (prevents methanol metabolism).
Fomepizole (Option B ruled out)	Direct inhibitor of alcohol dehydrogenase (preferred treatment).
Folinic acid (Leucovorin) (Option D ruled out)	Enhances formate metabolism into non-toxic byproducts.
Hemodialysis	Used in severe cases to remove methanol and formate.

Solution for Question 8:

- The patient presents with delirium tremens (DTs) , a severe alcohol withdrawal syndrome characterized by: Severe agitation, hallucinations, tremors, autonomic instability (tachycardia, hypertension) Onset: 48–96 hours after alcohol cessation Risk: Seizures, cardiovascular collapse, and death if untreated
- Severe agitation, hallucinations, tremors, autonomic instability (tachycardia, hypertension)
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- Risk: Seizures, cardiovascular collapse, and death if untreated
- Benzodiazepines (e.g., diazepam, lorazepam, or chlordiazepoxide) are the first-line treatment as they: Enhance GABAergic inhibition , reducing excitatory overactivity. Prevent progression to seizures and cardiovascular complications . Provide sedation and symptom control.
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Enhance GABAergic inhibition , reducing excitatory overactivity.

Prevent progression to seizures and cardiovascular complications .

Provide sedation and symptom control.

Severity	Symptoms	Treatment
Mild	Anxiety, insomnia, tremors	Supportive care, thiamine, hydration
Moderate	Tachycardia, hypertension, diaphoresis, hallucinations	Benzodiazepines (e.g., lorazepam, diazepam)
Severe (Delirium Tremens)	Confusion, hallucinations, seizures, autonomic instability	High-dose benzodiazepines, ICU monitoring

Initiate antipsychotic medication (Option A): Incorrect

- Antipsychotics (e.g., haloperidol) can worsen seizures and prolong withdrawal .
- Used only as an adjunct for persistent agitation after benzodiazepine administration.

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Used only as an adjunct for persistent agitation after benzodiazepine administration.

Prescribe SSRIs (Option B): Incorrect

- SSRIs have no role in managing acute alcohol withdrawal or DTs.
- Mood stabilization is considered after withdrawal resolution .

SSRIs have no role in managing acute alcohol withdrawal or DTs.

Mood stabilization is considered after withdrawal resolution .

Begin opioid replacement therapy (Option D): Incorrect

- Opioid therapy is for opioid withdrawal, not alcohol withdrawal .
- Alcohol withdrawal is managed with benzodiazepines .

Opioid therapy is for opioid withdrawal, not alcohol withdrawal .

Alcohol withdrawal is managed with benzodiazepines .

Solution for Question 9:

- Chronic alcohol consumption leads to excessive production of acetaldehyde , a toxic intermediate of alcohol metabolism.
- Acetaldehyde disrupts mitochondrial function and promotes fatty liver (hepatic steatosis) by: Impairing beta-oxidation of fatty acids. Increasing triglyceride synthesis and storage in the liver. Causing mitochondrial damage and oxidative stress.
- Impairing beta-oxidation of fatty acids.
- Increasing triglyceride synthesis and storage in the liver.
- Causing mitochondrial damage and oxidative stress.
- This results in hepatomegaly, abnormal liver function tests, and progression to alcoholic liver disease.

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Causing mitochondrial damage and oxidative stress.

This results in hepatomegaly, abnormal liver function tests, and progression to alcoholic liver disease.

Pathophysiology of Alcohol-Induced Fatty Liver

Effect	Mechanism
Increased Acetaldehyde (Option A ruled out)	Alcohol metabolism (via alcohol dehydrogenase) generates toxic acetaldehyde, which disrupts liver function.

Inhibition of Fatty Acid Oxidation (Option D ruled out)	High NADH/NAD ⁺ ratio inhibits beta-oxidation, leading to fat accumulation in hepatocytes.
Increased Lipogenesis	Excess NADH promotes triglyceride synthesis, worsening hepatic steatosis.
Mitochondrial Dysfunction	Acetaldehyde causes oxidative stress and mitochondrial damage, impairing lipid metabolism.

Increased pyruvate oxidation (Option B): Incorrect. Pyruvate is converted to lactate (not oxidized) due to excess NADH, leading to lactic acidosis, not fatty liver.

Solution for Question 10:

- Chronic alcohol use and metabolism: Alcohol metabolism increases the NADH/NAD⁺ ratio, shifting metabolic pathways. Excess NADH promotes the conversion of pyruvate to lactate and oxaloacetate to malate, depleting key gluconeogenic precursors. This inhibits gluconeogenesis, leading to hypoglycemia, especially in malnourished individuals. The false statement is Option D, which states that alcohol decreases the NADH/NAD⁺ ratio, which is incorrect.
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The false statement is Option D, which states that alcohol decreases the NADH/NAD⁺ ratio, which is incorrect.

Alcohol-Induced Metabolic Effects and Nutritional Deficiencies

Effect	Mechanism
Increased NADH/NAD ⁺ Ratio (Option C ruled out)	Alcohol metabolism (via alcohol dehydrogenase and aldehyde dehydrogenase) generates excess NADH, disrupting metabolic pathways.
Hypoglycemia (Option C ruled out)	Excess NADH inhibits gluconeogenesis, contributing to hypoglycemia. Pyruvate and oxaloacetate are diverted to lactate and malate, depleting gluconeogenic substrates.
Lactic Acidosis	Excess NADH promotes pyruvate to lactate conversion.
Vitamin B12 Deficiency (Option A ruled out)	Alcohol impairs B12 absorption despite normal blood levels.
Folate Deficiency (Option B ruled out)	Alcohol interferes with folate absorption and storage, leading to megaloblastic anemia.

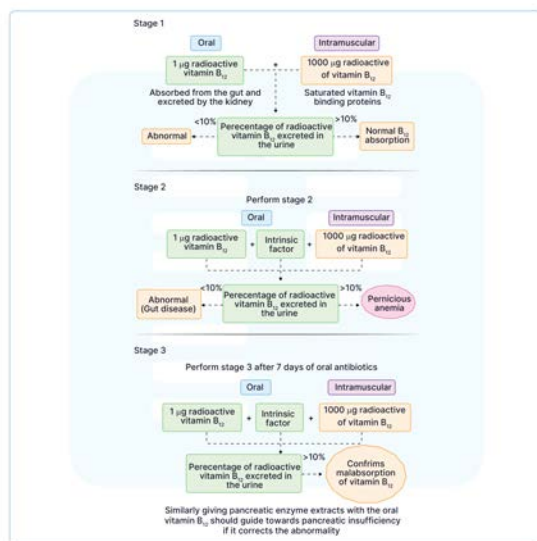
Solution for Question 11:

- The Schilling test assesses vitamin B12 absorption by measuring urinary excretion of radioactive vitamin B12 .
- It consists of four stages , each designed to pinpoint different causes of B12 malabsorption .
- If stage 1 is abnormal but stage 2 is normal , the issue is due to intrinsic factor deficiency , commonly seen in pernicious anemia .

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Schilling Test Stages and Interpretations:

Stage	Procedure	Normal Interpretation	Abnormal Interpretation	Possible Cause of Abnormal Result
Stage 1	Oral radioactive vitamin B12 + 24-hour urine collection	≥8% excretion of B12 in urine (Normal absorption)	<8% excretion of B12 in urine (Malabsorption)	General B12 malabsorption (cause unknown)
Stage 2	Oral radioactive vitamin B12 + intrinsic factor	Excretion normalizes (suggests absorption issue due to IF deficiency)	Still abnormal (no improvement)	Intrinsic factor deficiency (pernicious anemia)
Stage 3	Oral radioactive vitamin B12 after antibiotics	Excretion normalizes	Still abnormal	Bacterial overgrowth syndrome
Stage 4	Oral radioactive vitamin B12 after pancreatic enzymes	Excretion normalizes	Still abnormal	Pancreatic insufficiency

The Schilling test involves administering one dose of vitamin B12 to assess absorption (Option A): The test consists of multiple stages to evaluate different causes of B12 malabsorption.

Radioactive vitamin B12 injection is given to saturate body stores (Option C): The intramuscular injection of non-radioactive B12 is given only to prevent liver uptake , not for saturation.

Abnormal Stage 2 suggests intrinsic factor deficiency (Option D): A normal result in Stage 2 suggests intrinsic factor deficiency (pernicious anemia), whereas an abnormal result in Stage 2 suggests another cause (e.g., bacterial overgrowth or pancreatic insufficiency).

Solution for Question 12:

Thiamine Deficiency

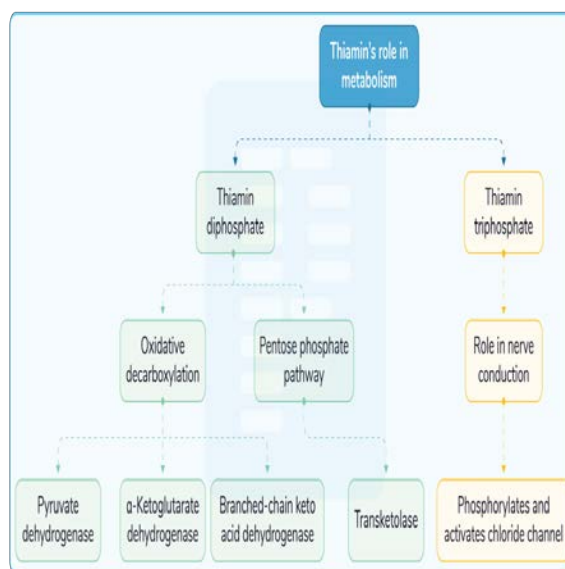
- Aleurone layer of cereals (food grains) is a rich source of thiamine. Whole wheat flour and unpolished hand-pound rice have better nutritive value than completely polished refined foods.
- Polishing grains removes the aleurone layer, reducing thiamine content.

Deficiency Manifestations

- More common CNS manifestations
- Pain, paresthesia, and loss of reflexes
- Peripheral neuritis with sensory disturbances (Option A ruled out)
- Cardiovascular symptoms : breathlessness, palpitations, and peripheral edema
- Distended neck veins observed
- High-output cardiac failure due to vasodilation (Option C)

Biochemical Parameters

- Thiamine deficiency affects transketolase , an enzyme in the pentose phosphate pathway .
- Measuring erythrocyte transketolase activity is a specific diagnostic test for thiamine deficiency (Option C).
- Earliest biochemical change : Reduced erythrocyte transketolase activity, even before clinical symptoms appear.
- Recommended Daily Allowance : 1–1.5 mg/day



Condition	Biochemical Role of Thiamine	Deficiency Manifestations
Dry Beriberi (Option A ruled out)	Essential for carbohydrate metabolism and nerve function	Peripheral neuritis, sensory disturbances, difficulty walking, paralysis due to CNS involvement.
Wet Beriberi	Thiamine is required for pyruvate dehydrogenase (PDH) complex ; deficiency leads to lactic acidosis , causing vasodilation and cardiovascular symptoms	Edema, palpitations, breathlessness, heart failure
Maple Syrup Urine Disease (Option B ruled out)	Cofactor for decarboxylation of branched-chain keto acids	Accumulation of branched-chain keto acids , causing convulsions, mental retardation, vomiting, acidosis, coma, and death
Lactic Acidosis (Option D ruled out)	Cofactor for pyruvate dehydrogenase , which converts pyruvate to acetyl-CoA	Impaired pyruvate metabolism , leading to increased lactate and pyruvate , causing lactic acidosis

Solution for Question 13:

- Vitamin B1 (Thiamin) → Transketolase Activity (b) Thiamin acts as a coenzyme for transketolase in the pentose phosphate pathway. Measuring transketolase activity in red blood cells after adding thiamin pyrophosphate helps assess thiamin status.

- Thiamin acts as a coenzyme for transketolase in the pentose phosphate pathway.
- Measuring transketolase activity in red blood cells after adding thiamin pyrophosphate helps assess thiamin status.
- Vitamin B2 (Riboflavin) → Glutathione Reductase Activity (c) Riboflavin functions as a coenzyme for glutathione reductase (FAD-dependent). Measuring glutathione reductase activity in RBCs helps assess riboflavin status.
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- Measuring glutathione reductase activity in RBCs helps assess riboflavin status.
- Vitamin B6 (Pyridoxine) → Activation of RBC Transaminases (d) Pyridoxal phosphate (PLP) is a coenzyme for transaminases in amino acid metabolism. Measuring activation of RBC transaminases after PLP supplementation assesses B6 status.
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- Measuring activation of RBC transaminases after PLP supplementation assesses B6 status.
- Vitamin B12 (Cobalamin) → Urinary Methylmalonic Acid (a) B12 is required for methylmalonyl-CoA mutase , which converts methylmalonyl-CoA to succinyl-CoA. B12 deficiency leads to increased urinary methylmalonic acid , making it a reliable marker.
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Vitamin B1 (Thiamin) → Transketolase Activity (b)

- Thiamin acts as a coenzyme for transketolase in the pentose phosphate pathway.
- Measuring transketolase activity in red blood cells after adding thiamin pyrophosphate helps assess thiamin status.

Thiamin acts as a coenzyme for transketolase in the pentose phosphate pathway.

Measuring transketolase activity in red blood cells after adding thiamin pyrophosphate helps assess thiamin status.

Vitamin B2 (Riboflavin) → Glutathione Reductase Activity (c)

- Riboflavin functions as a coenzyme for glutathione reductase (FAD-dependent).
- Measuring glutathione reductase activity in RBCs helps assess riboflavin status.

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Measuring glutathione reductase activity in RBCs helps assess riboflavin status.

Vitamin B6 (Pyridoxine) → Activation of RBC Transaminases (d)

- Pyridoxal phosphate (PLP) is a coenzyme for transaminases in amino acid metabolism.

- Measuring activation of RBC transaminases after PLP supplementation assesses B6 status.

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Measuring activation of RBC transaminases after PLP supplementation assesses B6 status.

Vitamin B12 (Cobalamin) → Urinary Methylmalonic Acid (a)

- B12 is required for methylmalonyl-CoA mutase , which converts methylmalonyl-CoA to succinyl-CoA.
- B12 deficiency leads to increased urinary methylmalonic acid , making it a reliable marker.

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Vitamin	Functional Assessment Test	Key Enzyme/Process Assessed	Clinical Relevance
Vitamin B1 (Thiamin)	Transketolase activity in RBCs	Transketolase (Pentose Phosphate Pathway)	Reduced activity indicates thiamin deficiency (e.g., Beriberi, Wernicke-Korsakoff syndrome)
Vitamin B2 (Riboflavin)	Glutathione reductase activity in RBCs	Glutathione Reductase (FAD-dependent)	Low activity suggests riboflavin deficiency, leading to oxidative stress disorders
Vitamin B6 (Pyridoxine)	RBC transaminase activity (after PLP addition)	Aminotransferases (ALT, AST)	Decreased enzyme activity suggests B6 deficiency, leading to neuropathy and anemia
Vitamin B12 (Cobalamin)	Urinary methylmalonic acid measurement	Methylmalonyl-CoA Mutase	Elevated levels indicate B12 deficiency, which can cause megaloblastic anemia and neuropathy

Solution for Question 14:

- The child's symptoms of developmental delay, irritability, and peripheral neuropathy, along with the macrocytic anemia, are indicative of a cobalamin deficiency.

- Elevated homocysteine and normal methylmalonic acid levels suggest folate deficiency, while methylcobalamin deficiency typically leads to elevated homocysteine with increased or normal methylmalonic acid levels.
- Methylmalonic acid levels remain normal in folate deficiency, unlike in vitamin B12 deficiency, where they are elevated.
- The normal methylmalonic acid level helps differentiate it from Adenosylcobalamin deficiency, which typically results in elevated methylmalonic acid.

Vitamin B12-Dependent Enzymes:

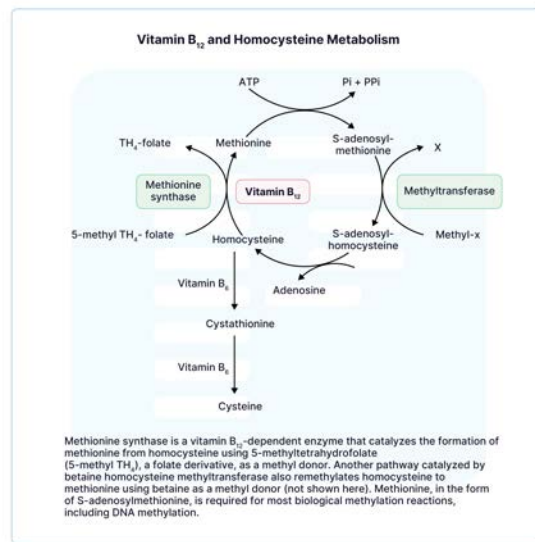
1. Methyl-malonyl-CoA Mutase:

- Function: Converts methylmalonyl-CoA to succinyl-CoA.
- Methyl-malonyl CoA is an intermediate in the breakdown of valine, cholesterol, odd-chain fatty acids, propionate, and carboxylation of propionyl-CoA.
- Deficiency Impact: Leads to accumulation of methylmalonyl-CoA and increased urinary methylmalonic acid, indicating B12 deficiency status.
- Methylmalonic acid will be normal in megaloblastic anaemia caused by folate deficiency.

2. Methionine Synthase:

- Function: Converts homocysteine to methionine.
- Role: Impairment affects folic acid metabolism, leading to functional folate deficiency.
- Deficiency Impact: Causes pernicious anaemia and spinal cord degeneration due to failed methylation of arginine in myelin basic protein. This is a result of methionine deficiency in the CNS.

Folate trap:



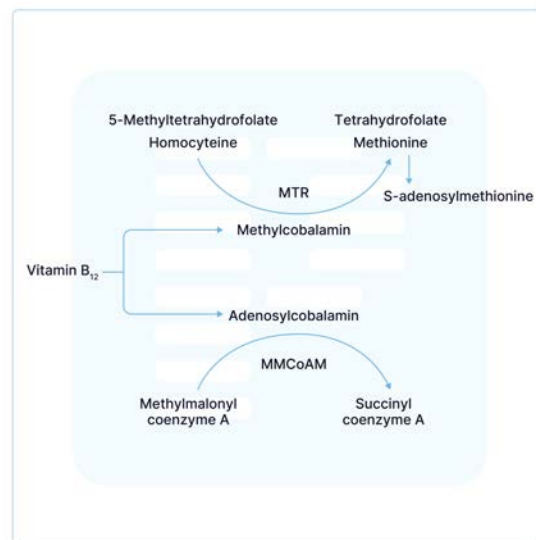
The 2 Co-enzyme forms of vitamin B-12 are:

1. Adenosylcobalamin Deficiency:

- **Function:** Coenzyme form of vitamin B12 essential for energy metabolism in mitochondria.
- **Deficiency Impact:** Impaired Energy Production: Reduces ATP production, causing chronic fatigue, muscle weakness, and weight loss. Neurological Damage: Accumulation of neurotoxic methylmalonic acid impairs fatty acid synthesis for myelin sheath maintenance, resulting in nerve degeneration and neurological disorders. Disrupts amino acid and hormone metabolism, leading to systemic metabolic disorders.
- **Impaired Energy Production:** Reduces ATP production, causing chronic fatigue, muscle weakness, and weight loss.
- **Neurological Damage:** Accumulation of neurotoxic methylmalonic acid impairs fatty acid synthesis for myelin sheath maintenance, resulting in nerve degeneration and neurological disorders.
- **Disrupts amino acid and hormone metabolism,** leading to systemic metabolic disorders.
- **Impaired Energy Production:** Reduces ATP production, causing chronic fatigue, muscle weakness, and weight loss.
- **Neurological Damage:** Accumulation of neurotoxic methylmalonic acid impairs fatty acid synthesis for myelin sheath maintenance, resulting in nerve degeneration and neurological disorders.
- **Disrupts amino acid and hormone metabolism,** leading to systemic metabolic disorders.

2. Methylcobalamin Deficiency:

- **Function:** The bioactive form of vitamin B12 is crucial for the methylation cycle and methionine synthesis from homocysteine.
- **Deficiency Impact:** Elevated Homocysteine Levels: Impairs conversion of homocysteine to methionine, raising homocysteine levels and cardiovascular risk. Neurological Issues: Leads to neuropathy, cognitive decline, and other neurological disorders. Anaemia: Causes megaloblastic anaemia with large, ineffective red blood cells.
- **Elevated Homocysteine Levels:** Impairs conversion of homocysteine to methionine, raising homocysteine levels and cardiovascular risk.
- **Neurological Issues:** Leads to neuropathy, cognitive decline, and other neurological disorders.
- **Anaemia:** Causes megaloblastic anaemia with large, ineffective red blood cells.
- **Elevated Homocysteine Levels:** Impairs conversion of homocysteine to methionine, raising homocysteine levels and cardiovascular risk.
- **Neurological Issues:** Leads to neuropathy, cognitive decline, and other neurological disorders.
- **Anaemia:** Causes megaloblastic anaemia with large, ineffective red blood cells.



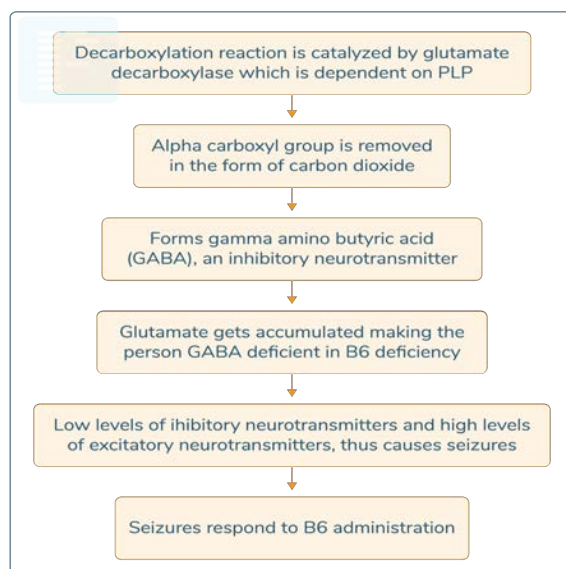
Solution for Question 15:

- Vitamin B6 (Pyridoxal Phosphate, PLP) is a coenzyme for glutamate decarboxylase , which converts glutamate to GABA (an inhibitory neurotransmitter).
- Deficiency of vitamin B6 impairs GABA synthesis , leading to an imbalance between excitatory (glutamate) and inhibitory (GABA) signals, resulting in seizures, particularly in neonates with refractory seizures .
- Pyridoxine-dependent epilepsy (PDE) is a rare condition where B6 supplementation is required to control seizures.

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Vitamin B6 Deficiency and Neurological Consequences

Effect of Vitamin B6 Deficiency	Mechanism	Clinical Manifestation
Decreased GABA synthesis	Impaired glutamate decarboxylase activity	Increased excitability, leading to seizures
Elevated glutamate levels	Reduced conversion of glutamate to GABA	Excitotoxicity, neuronal damage

Impaired serotonin synthesis	PLP is required for serotonin production	Mood disturbances, depression
Disrupted myelin formation	Impaired sphingolipid metabolism	Neuropathy, cognitive deficits

Elevated glutamate levels (Option B): Incorrect because while B6 deficiency can lead to increased glutamate, the primary cause of seizures is GABA deficiency, not glutamate excess .

Increased serotonin production (Option C): Incorrect because vitamin B6 deficiency reduces serotonin synthesis due to impaired decarboxylation of 5-hydroxytryptophan.

Enhanced myelin formation (Option D): Incorrect because vitamin B6 deficiency impairs , rather than enhances, myelin synthesis by disrupting sphingolipid metabolism.

Solution for Question 16:

- Vitamin B6 (Pyridoxal phosphate, PLP) is a coenzyme involved in amino acid metabolism and glycogen breakdown , but not in glycogen synthesis .
- PLP supports: Transamination (e.g., ALT, AST) Decarboxylation (e.g., neurotransmitter synthesis) Glycogenolysis (via glycogen phosphorylase)
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- Decarboxylation (e.g., neurotransmitter synthesis)
- Glycogenolysis (via glycogen phosphorylase)
- Glycogen synthesis (glycogenesis) is catalyzed by glycogen synthase, which does not require PLP.

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Decarboxylation (e.g., neurotransmitter synthesis)

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Glycogen synthesis (glycogenesis) is catalyzed by glycogen synthase, which does not require PLP.

Vitamin B6- Coenzyme Roles

Metabolic Process	Enzyme Involved	Role of Vitamin B6 (PLP)
Transamination (Option A ruled out)	ALT, AST	Transfers amino groups between amino acids and keto acids
Decarboxylation (Option B ruled out)	Glutamate decarboxylase	Produces neurotransmitters (e.g., GABA, serotonin, dopamine)
Glycogenolysis (Option C ruled out)	Glycogen phosphorylase	PLP is a coenzyme for glycogen phosphorylase, aiding glycogen breakdown
Heme synthesis	δ -aminolevulinate synthase	Required for the first step in heme production
Niacin synthesis	Kynureninase	Converts tryptophan to niacin
Sphingolipid synthesis	Serine palmitoyltransferase	Involved in sphingomyelin metabolism
Homocysteine metabolism	Cystathionine β -synthase	Helps convert homocysteine to cystathionine

Solution for Question 17:

- Hartnup's disease is a disorder of amino acid transport, particularly affecting tryptophan absorption in the intestines and kidneys.
- Tryptophan is a precursor for niacin synthesis , and its deficiency results in pellagra-like symptoms , including dermatitis, diarrhoea, and cognitive disturbances .
- Aminoaciduria (loss of amino acids in urine) occurs due to defective renal reabsorption.

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Aminoaciduria (loss of amino acids in urine) occurs due to defective renal reabsorption.

Hartnup's Disease and Tryptophan Metabolism

Aspect	Details
Cause	Mutation in SLC6A19 gene , affecting neutral amino acid transporter in intestines and kidneys.
Deficiency	Tryptophan (precursor for niacin) leading to niacin deficiency .
Clinical Features	Pellagra-like symptoms (Dermatitis, Diarrhoea, Dementia) Aminoaciduria (loss of neutral amino acids in urine) Ataxia, mood changes due to CNS involvement.
Diagnosis	Increased neutral amino acids in urine Decreased tryptophan and niacin levels
Treatment	Niacin supplementation High-protein diet to compensate for amino acid losses

Hartnup's disease causes a direct deficiency in niacin, leading to pellagra symptoms (Option A):
Incorrect because Hartnup's disease primarily affects tryptophan absorption , not niacin itself.
Niacin deficiency is secondary to tryptophan deficiency.

Tryptophan excess → disrupts niacin metabolism → pellagra-like symptoms (Option C):
Incorrect because excess tryptophan does not disrupt niacin metabolism ; instead, it increases niacin synthesis.

Increases the metabolism of tryptophan to serotonin, depleting niacin stores (Option D):
Incorrect because this mechanism occurs in carcinoid syndrome , where tryptophan is preferentially converted to serotonin, leading to niacin deficiency . However, this is not the mechanism in Hartnup's disease .

Solution for Question 18:

Pellagra is characterized by the classic triad of symptoms:

- Dermatitis Bright red erythema- in sun-exposed areas (face, ankles, feet) Casal's necklace
- ↑pigmentation around the neck
- Bright red erythema- in sun-exposed areas (face, ankles, feet)

- Casal's necklace - ↑pigmentation around the neck
- Diarrhoea, and
- Dementia.
- Bright red erythema- in sun-exposed areas (face, ankles, feet)
- Casal's necklace - ↑pigmentation around the neck

The patient's symptoms, combined with a diet high in maize and sorghum, both known to contribute to niacin and tryptophan deficiencies suggest pellagra as the most likely diagnosis.

Sorghum has a high leucine content, which inhibits the QPRTase enzyme required for the synthesis of niacin from tryptophan. Additionally, maize is low in tryptophan.

Causes of pellagra :

- Dietary Deficiency- Maize and Sorghum based diet
- Impaired Tryptophan Metabolism Factors Affecting Tryptophan to Niacin Conversion: Vitamin B6 Deficiency Medications: Isoniazid (INH), OCPs Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism. Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Factors Affecting Tryptophan to Niacin Conversion: Vitamin B6 Deficiency Medications: Isoniazid (INH), OCPs Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism. Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Vitamin B6 Deficiency
- Medications: Isoniazid (INH), OCPs
- Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism.
- Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Carcinoid Syndrome
- Other Contributing Factors: Alcoholism: Interferes with nutrient absorption, including niacin and tryptophan. Malabsorption Syndromes
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- Malabsorption Syndromes
- Factors Affecting Tryptophan to Niacin Conversion: Vitamin B6 Deficiency Medications: Isoniazid (INH), OCPs Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism. Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Vitamin B6 Deficiency

- Medications: Isoniazid (INH), OCPs
- Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism.
- Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Vitamin B6 Deficiency
- Medications: Isoniazid (INH), OCPs
- Hormonal Influences: Estrogen metabolites hinder tryptophan metabolism.
- Genetic Disorders: Inherited conditions like Hartnup disease can impede tryptophan absorption in the intestines and reabsorption in the kidneys
- Alcoholism: Interferes with nutrient absorption, including niacin and tryptophan.
- Malabsorption Syndromes



Vitamin B12 deficiency (Option A) causes fatigue, neurological issues, and anaemia but is not typically associated with the combination of dermatitis, diarrhoea, and cognitive disturbances seen in this case.

Rickets (Option C) resulting from a deficiency in vitamin D, calcium, or phosphate, present with bone deformities and growth retardation. It is not associated with dermatitis, diarrhoea, and neurological symptoms characteristic of pellagra.

Iron deficiency anaemia (Option D) can cause fatigue, pallor, and shortness of breath but does not typically present with skin rash, gastrointestinal symptoms, or cognitive issues associated with pellagra.

Solution for Question 19:

- The proliferation of bulbar conjunctival capillaries, which refers to the visible dilation of blood vessels in the eyes, is often the earliest clinical sign of riboflavin deficiency.
- This sign appears before more obvious symptoms like cheilosis or glossitis develop.

Riboflavin Deficiency		
Symptom	Feature	Image
Cheilosis	Cracks and inflammation at the corners of the mouth (Option A ruled out)	
Glossitis	Inflammation and changes in the appearance of the tongue. It becomes smooth and glossy. It may appear purplish red (Option B ruled out)	
Seborrheic dermatitis	Skin condition with scaly patches, redness and dandruff (Option C ruled out)	
Proliferation of bulbar conjunctival capillaries	Earliest sign. It is the visible dilation of blood vessels in the eyes.	
VITAMINS		ASSAYS TO DETECT DEFICIENCY
Vitamin B1	Transketolase	
Vitamin B2	Glutathione reductase	
Vitamin B3	Xanthurenic acid/ Tryptophan load test	
Vitamin B6	FIGLU/ Histidine load test	
Vitamin B12	Methylmalonate	

Solution for Question 20:

- NADH Dehydrogenase (Complex I of the Electron Transport Chain) does not use FAD as a coenzyme ; instead, it primarily relies on FMN (Flavin Mononucleotide) as its prosthetic group.
- It transfers electrons from NADH to Coenzyme Q (Ubiquinone) through a series of iron-sulfur clusters.
- Unlike FAD-containing enzymes, Complex I utilizes FMN for its redox reactions.

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Unlike FAD-containing enzymes, Complex I utilizes FMN for its redox reactions.

FMN-Dependent vs. FAD-Dependent Enzymes

Category	FMN-Dependent Enzymes	FAD-Dependent Enzymes
Derived from Riboflavin	Yes	Yes
Coenzyme Form	Flavin Mononucleotide (FMN)	Flavin Adenine Dinucleotide (FAD)
Electron Transport	Accepts two electrons and donates them one at a time	Accepts and donates two electrons and two protons
Role in Metabolism	Electron transport in oxidative phosphorylation	Redox reactions in various metabolic pathways
Location	Mitochondrial ETC (Complex I)	Mitochondrial matrix, peroxisomes, cytoplasm
Examples	NADH Dehydrogenase (Complex I) – Transfers electrons from NADH to ubiquinone L-Amino Acid Oxidase – Used in amino acid metabolism	Succinate Dehydrogenase – Converts succinate to fumarate (TCA cycle) Acyl CoA Dehydrogenase – Converts Acyl-CoA to α,β -unsaturated acyl-CoA (Lipid metabolism) Xanthine Oxidase – Converts xanthine to uric acid Pyruvate Dehydrogenase – Converts pyruvate to acetyl-CoA (Carbohydrate metabolism) Alpha-Ketoglutarate Dehydrogenase – Converts alpha-ketoglutarate to succinyl-CoA Mitochondrial Glycerol-3-Phosphate Dehydrogenase – Converts glycerol-3-phosphate to dihydroxyacetone phosphate

Solution for Question 21:

The patient's symptoms of confusion, difficulty walking (cerebellar ataxia), and abnormal eye movements (nystagmus) are classic signs of Wernicke encephalopathy, which is caused by thiamine deficiency often seen in chronic alcoholics.

Wernicke-Korsakoff Syndrome (Cerebral Beriberi)	
Clinical Manifestations	Wernicke Encephalopathy: Symptoms: Ophthalmoplegia, nystagmus, cerebellar ataxia, global confusion.
Korsakoff Psychosis: Symptoms: Anterograde and retrograde amnesia, sensory agnosia, and confabulation. (Option B ruled out)	
Causes of Thiamine Deficiency in Alcoholics	Increased thiamine requirements due to alcohol consumption. Impaired intestinal absorption of thiamine due to alcohol. Increased thiamine utilization and inadequate dietary intake. Inhibition of intestinal thiamine absorption by alcohol.
Metabolic Implications	Impaired conversion of pyruvate to acetyl-CoA. Accumulation of pyruvate and lactate in the blood. Potential for life-threatening lactic acidosis.
Progression and Risk Factors	Progression from Wernicke encephalopathy to Korsakoff psychosis if untreated. Associated with severely compromised nutritional status, often due to chronic alcoholism.
MRI findings	Symmetrical T2/FLAIR hyperintensities in the mammillary bodies, dorsomedial thalami, periaqueductal gray matter, and around the third ventricle.

Lactic acidosis (Option C): While thiamine deficiency can lead to an accumulation of pyruvate and lactate, resulting in lactic acidosis, the clinical presentation of confusion, ataxia, and nystagmus is more indicative of Wernicke encephalopathy. Lactic acidosis might be present but is not the primary diagnosis based on the symptoms described.

Dry beriberi (Option D) primarily affects the peripheral nervous system, causing peripheral neuritis, paresthesia, foot drop, muscle weakness, and distal sensory loss.

Previous Year Questions

1. Which deficiency is most likely in a patient with delayed wound healing and alopecia?

- A. Zinc
 - B. Copper
 - C. Calcium
 - D. Vitamin D
-

2. What vitamin is responsible for the development of macular edema and macular cysts when administered in excessive amounts?

- A. Vitamin A
 - B. Vitamin D
 - C. Vitamin E
 - D. Vitamin B3
-

3. In a patient with alcoholism, who presented with pruritic lesions displayed in the provided image, along with a red, beefy tongue, which of the following additional inquiries would you like to make regarding their medical history?



- A. Ask for diarrhea and a history of memory loss
 - B. Ask for diarrhea, thiamine deficiency features, and detailed dietary history
 - C. History of allergy
 - D. Ask for diarrhea and conjunctivitis
-

4. A farmer has come in with symptoms of loose stools and rashes on sun-exposed areas. Upon further investigation of their diet, it is revealed that they consume a significant amount of maize. Based on these findings, which deficiency would you consider as a possible cause?

- A. Niacin
 - B. Pyridoxine
 - C. Thiamine
 - D. Biotin
-

5. Features of Korsakoff Psychosis are all except Amnesia Confabulation Ophthalmoplegia Polyneuropathy

- A. 1, 2,3
 - B. 2,3,4
 - C. 1,3
 - D. 3,4
-

6. Which of the following statements is not true regarding folate deficiency?

- A. Folate supplementation in pregnancy prevents NTD
 - B. High Homocysteine and Normal Methylmalonic acid
 - C. Hemolytic anemia is a feature
 - D. Treat coexisting B12 deficiency before initiating folate therapy
-

7. The image illustrates which vitamin deficiency?



- A. Vitamin A
 - B. Vitamin E
 - C. Vitamin C
 - D. Niacin (Vitamin B3) Deficiency
-

8. A chronic alcoholic patient presents with agitation, confusion, and gait abnormalities. What vitamin deficiency is most likely responsible for these symptoms?

- A. Nicotinamide (Vitamin B3) deficiency
 - B. Folate (Vitamin B9) deficiency
 - C. Methylcobalamin (Vitamin B12) deficiency
 - D. Tyrosinase
-

9. A 16-year-old patient presents with pellagra-like skin eruptions that occur during the spring season and have been occurring for the past 5 years. Laboratory tests reveal normal levels of 5-Hydroxyindoleacetic Acid (HIAA) and high aminoaciduria, with normal urinary indole levels. What is the most likely diagnosis?

- A. Carcinoid tumor
 - B. Nutritional pellagra
 - C. Hartnup disorder
 - D. Essential fructosuria
-

10. A 50-year-old man with a history of chronic alcoholism presents with skin lesions. His diet primarily consists of polished rice. Based on his symptoms and dietary history, what is the most likely diagnosis?

- A. Phrynoderma
 - B. Beriberi
 - C. Psoriasis
 - D. Eczema
-

11. Which micronutrient deficiency causes anemia, bleeding gums, and Frenkel's white line on X-ray?

- A. Vitamin K
 - B. Vitamin A
 - C. Vitamin C
 - D. Vitamin E
-

12. A patient presents with gum bleeding, petechiae, poor wound healing, and other related symptoms. Which of the following vitamin deficiencies is most likely responsible for this presentation?

- A. Vitamin A
 - B. Vitamin B6
 - C. Vitamin B3
 - D. Vitamin C
-

13. A baby presents with the clinical features shown in the image, including characteristic perioral and perianal rash along with a history of diarrhea. Which of the following is the most likely deficiency responsible for this condition?

- A. Zinc Deficiency

- B. Niacin Deficiency
 - C. Selenium Deficiency
 - D. Magnesium Deficiency
-

14. A 45-year-old patient with a known case of homocystinuria presents with joint pain and weakness. Which vitamin is required for the treatment?

- A. Vitamin B6
 - B. Vitamin B12
 - C. Vitamin B1
 - D. Vitamin B7
-

15. Which of the following are metabolites of methanol poisoning?

- A. Formic acid + Oxalic acid
 - B. Formic acid + Lactic acid
 - C. Glycolic acid + Oxalic acid
 - D. Lactic acid + Glycolic acid
-

16. A child presents with deformities, normal serum calcium, normal PTH, low phosphate, and elevated ALP. Which is the most likely diagnosis?

- A. Hypophosphatemic rickets
- B. Vitamin D Dependent Rickets Type 1 (VDDR type 1)
- C. Vitamin D Dependent Rickets Type 2 (VDDR type 2)
- D. Renal osteodystrophy

17. In a male patient, aged 34, who mainly consumes maize as his primary food, the symptoms observed are diarrhea, dementia, and photosensitive dermatitis in sun-exposed regions. The question inquires about the vitamin that is most likely deficient in this individual.

- A. Niacin
 - B. Vitamin B6
 - C. Ascorbic acid
 - D. Biotin
-

18. A patient presents with anemia, neutropenia, and hypopigmentation of skin and hair. Which micronutrient deficiency is most likely?

- A. Zinc
 - B. Fluoride
 - C. Copper
 - D. Selenium
-

19. A patient presents with follicular hyperkeratosis on the extensor aspect of the forearm and has a dietary history of Vitamin A and C deficiency. What is the diagnosis?

- A. Phrynoderma
- B. Scurvy
- C. Pellagra
- D. Keratosis pilaris

20. What is the recommended treatment for a 45-year-old male patient with a history of chronic alcoholism, who is currently experiencing symptoms of confusion, ataxia, and diplopia?

- A. Vitamin B1 injection followed by glucose infusion
 - B. Glucose infusion followed thiamine injection
 - C. Glucose infusion alone
 - D. Thiamine infusion alone
-

21. Which enzymes listed below depend on vitamin B12? Methyl malonyl CoA isomerase Homocysteine synthase Folate reductase Methionine synthase

- A. 1, 2, 3 and 4
 - B. 1 and 4 only
 - C. 1 and 3 only
 - D. 1, 2 and 3
-

22. Which statement regarding vitamin K is accurate?

- A. A deficiency of vitamin K leads to hypercoagulability
 - B. It is a water-soluble vitamin
 - C. Prolonged use of antibiotics leads to vitamin K deficiency
 - D. Menaquinone is the dietary source of vitamin K found in green vegetables.
-

23. The enzyme responsible for regenerating Vitamin K in its coenzyme form is:

- A. Glutathione reductase
 - B. Quinone reductase
 - C. Vitamin K reductase
 - D. Epoxide reductase
-

24. A middle-aged woman presents with fissured tongue and peripheral neuropathy. Investigations showed reduced glutathione reductase activity. Which vitamin deficiency is the likely cause of this?

- A. Vitamin B1
 - B. Vitamin B2
 - C. Vitamin B6
 - D. Vitamin B12
-

25. Vitamin C cannot be synthesized by the human body due to a deficiency in

- A. L-gulonolactone oxidase
 - B. Xylitol reductase
 - C. Pyruvate dehydrogenase
 - D. UDP glucose dehydrogenase
-

26. Which mineral in excess causes alkali disease in livestock?

- A. Selenium
- B. Manganese

C. Molybdenum

D. Iron

27. Which of the options below promotes the uptake of iron in the body?

A. Vitamin A

B. Vitamin C

C. Thiamine

D. Riboflavin

28. What is the pattern of inheritance for hemophilia A?

A. X linked recessive

B. X linked Dominant

C. autosomal dominant

D. autosomal recessive

29. Wernicke encephalopathy caused by deficiency of which vitamin?

A. Vit B1

B. Vit B2

C. Vit B7

D. Vit B12

30. What is the role of zinc in the treatment of diarrhea?

- A. Reduces the risk, duration and severity of diarrheal episodes
 - B. Enhances immune response
 - C. Regulates intestinal transport & absorption of water and electrolytes
 - D. All
-

31. Which of the following shows functional assessment of B1 deficiency?

- A. RBC Transketolase
 - B. RBC glutathione reductase
 - C. Serum thiamine levels
 - D. RBC glutathione peroxidase
-

32. The sequence of vitamin D synthesis is?

- A. Liver-skin-kidney
 - B. Skin-liver-kidney
 - C. Skin-kidney-liver
 - D. Liver-kidney-skin
-

33. Which one of the following is not classified as a micromineral?

- A. Calcium
- B. Selenium
- C. Iodine
- D. Copper

34. What is the deficiency in a patient who exhibits the following features: -



- A. Phenylalanine
- B. Tryptophan
- C. Glutamine
- D. Arginine

35. What is the probable vitamin deficiency in a patient who displays symptoms of diarrhea, memory impairment, and a distinctive rash as described?



- A. Thiamine
 - B. Niacin
 - C. Cyanocobalamin
 - D. Ascorbic acid
-

36. A 42-year-old female came to the emergency department with a bleeding condition and dental caries. Which vitamin supplementation from the list below would be best for treating this condition?

- A. Vitamin B1
 - B. Vitamin B9
 - C. Vitamin D
 - D. Vitamin K
-

37. Xanthurenic aciduria may be treated with high doses of which vitamin

- A. Vitamin B6
 - B. Vitamin B12
 - C. Vitamin B5
 - D. Vitamin B7
-

38. What is the cause of the following condition due to?



- A. Vitamin C
 - B. Vitamin A
 - C. Vitamin B
 - D. Vitamin E
-

39. Due to a prolonged period of consuming maize as a main food source, a 34-year-old man developed diarrhoea and exhibited lesions on his neck, as depicted in the accompanying image. This condition is caused by a deficiency in:



- A. Niacin
- B. Zinc
- C. Thiamine
- D. Iron

40. Allopurinol and Vitamin C- to reduce uric acid production and prevent gout-like symptoms. Which among the following helps in the maturation of collagen?

- A. Proline
- B. Copper and zinc
- C. Ascorbic acid
- D. Phenylalanine

41. A 37 year old patient who is a chronic alcoholic presented with Muscle weakness tingling or numbness in the hands and feet, loss of muscle coordination, and difficulty in walking. Which of the following is least likely to cause Beriberi?

- A. Chronic use of diuretics
 - B. Vegan diet
 - C. Bariatric surgery
 - D. Chronic alcohol intake
-

42. A male patient presented with symptoms of imbalance, foot numbness, and muscle weakness. During the history-taking process, the patient disclosed that he had been consuming an excessive amount of multivitamin tablets for the past year. Based on this information, what is the most probable cause of the symptoms observed in this patient?

- A. Vitamin D toxicity
 - B. Vitamin B6 toxicity
 - C. Vitamin C toxicity
 - D. Vitamin A toxicity
-

43. Which clotting factor is required for the activity of gamma-glutamyl carboxylase among the options provided?

- A. Factors II, VIII, IX, X
 - B. Factors II, VII, IX, and X
 - C. Factors II, VII, IX, XI
 - D. Factors II, VIII, X, XI
-

44. In a patient, if they exhibit symptoms of diarrhea, dermatitis, and dementia, which vitamin deficiency would you consider as a potential cause?

- A. Vitamin B3 deficiency

- B. Vitamin B1 deficiency
 - C. Vitamin B6 deficiency
 - D. Vitamin B12 deficiency
-

45. A family consumes only polished rice. Which of the following vitamin deficiencies will be seen, and which investigation is used to detect this vitamin deficiency?

- A. Riboflavin-glutathione reductase
 - B. Thiamine-transketolase
 - C. Thiamine-transaminase
 - D. Riboflavin-transketolase
-

46. What vitamin deficiency might be observed in a chronic alcoholic patient who arrives at the emergency department experiencing confusion, ataxia, and painful eye movements?

- A. Vitamin B1 deficiency
 - B. Vitamin A deficiency
 - C. Vitamin D deficiency
 - D. Vitamin C deficiency
-

47. A child was born with pedal edema and cheilosis. Cardiomegaly was present. What is the vitamin deficiency associated with this clinical presentation?

- A. Riboflavin
- B. Thiamine

C. Pyridoxine

D. Niacin

48. A chronic alcoholic patient is diagnosed with gouty arthritis. What biochemical changes will be seen in this patient?

A. Decreased NADH/NAD⁺ ratio

B. Increased urea and urate levels

C. Increased urate and lactate levels

D. Increased urea and lactate levels

49. What is the biochemical basis of scurvy?

A. Low calcium

B. Defective collagen formation

C. Inhibition of clotting factors

D. Increased keratinization of epithelium

50. A child presents with thinning of hair. Her parents report that she consumes raw egg whites in her diet regularly. The given manifestations are due to the deficiency of which of the following vitamins?

A. Niacin

B. Pyridoxine

C. Biotin

D. Thiamine

51. What are the consequences of a deficiency in Vitamin A?

- A. Beriberi
 - B. Pellagra
 - C. Xerophthalmia
 - D. Neuropathy
-

52. An elderly patient who follows a vegetarian diet and exhibits symptoms of easy tiredness and paleness was examined. A peripheral blood smear revealed signs of macrocytic anemia. What could be the likely reason for anemia in this individual? Additional laboratory results are as follows: Hemoglobin - 9 g/dl Mean corpuscular volume (MCV)- 110 fL Serum homocysteine – elevated Serum methymalonic acid – Normal

- A. Vitamin B6 deficiency
 - B. Alcohol-induced macrocytosis
 - C. Vitamin B12 deficiency
 - D. Thiamine deficiency
-

53. A patient has come in with bleeding gums and the examination results are as follows. This particular condition can be observed in the case of a deficiency in



- A. Vitamin A
 - B. Vitamin B12
 - C. Vitamin C
 - D. Vitamin D
-

54. Which of the following contains the maximum amount of vitamin B12?

- A. Animal products
 - B. Green leafy vegetables
 - C. Roots and tubers
 - D. Vegetarian diet
-

55. What is the probable reason for the worsening of neurologic symptoms in a patient who initially showed improvement in blood picture after receiving a dose of 5mg folate for peripheral neuropathy and macrocytic anemia?

- A. Malabsorption of folate
 - B. Treatment with folate unmasking pyridoxine deficiency
 - C. Deficiency of folate reductase in CNS
 - D. Folate therapy using the B12 stores
-

56. Due to a prolonged consumption of maize as a primary food source, a 34-year-old male has developed diarrhea and a visible neck lesion, as depicted in the provided image. This condition arises as a result of a deficiency in:



- A. Niacin
 - B. Zinc
 - C. Thiamine
 - D. Iron
-

57. What is the most likely cause of the worsening neurological symptoms in a 30-year-old woman who presented with complaints of fatigue, difficulty breathing

during exertion, weight loss, and frequent falls, and was found to have bilateral decrease in vibration sense, a hemoglobin level of 8.2 g% with high MCV, and was initially treated with folate which improved her anemia?

- A. Folate is not well absorbed
 - B. Folic acid therapy unmasked pyridoxine deficiency
 - C. Deficiency of folate reductase in CNS
 - D. Folate therapy caused rapid use of vitamin B12 stores aggravating symptoms
-

58. Which of the subsequent stages in the vitamin D metabolism necessitates exposure to sunlight?

- A. 1,25 dihydroxy cholecalciferol to 7-dehydrocholesterol
 - B. 7-dehydrocholesterol to cholecalciferol
 - C. Cholecalciferol to 25-hydroxycholecalciferol
 - D. 25-hydroxycholecalciferol to 1,25 dihydroxycholecalciferol
-

59. What could be the potential diagnosis for a 34-year-old male who has been consuming maize as his main diet for an extended period and is now experiencing symptoms such as abdominal pain, decreased appetite, and diarrhea? Additionally, upon examination, glossitis (inflammation of the tongue) and a pigmented rash in sun-exposed areas were observed, along with a lesion on his neck.

- A. Pellagra
 - B. Scurvy
 - C. Megaloblastic anemia
 - D. Beriberi
-

60. Which of the following vitamins is incorrectly paired with their respective enzymes?

- A. Vitamin B1 (Thiamine): RBC Trans-ketolase
 - B. Vitamin B2 (Riboflavin): RBC glutathione reductase
 - C. Vitamin B9 (Folic acid): FIGLU excretion/Histidine load test
 - D. Vitamin C: Serum methylmalonate level
-

61. To evaluate a deficiency of vitamin B2 (riboflavin), one can assess the activity of:

- A. Transketolase
 - B. Pyruvate dehydrogenase
 - C. Glutathione reductase
 - D. Isocitrate dehydrogenase
-

62. Which antioxidant deficiency can cause the symptoms of iron-deficiency anemia, painful and tender joints, and bleeding manifestations in a 34-year-old woman who primarily consumes packaged and reheated meals?

- A. Glutathione
 - B. β -carotene
 - C. Vitamin C
 - D. Zinc
-

63. Methyl-tetrahydro-folic acid gets accumulated in deficiency of which of the following?

- A. B2
 - B. B12
 - C. B1
 - D. B6
-

64. Which of the following is associated with a defect in collagen maturity?

- A. Scurvy
 - B. Rickets
 - C. Beriberi
 - D. Night blindness
-

65. Which enzyme is affected in a patient with thiamine deficiency who develops lactic acidosis?

- A. Pyruvate carboxylase
 - B. Pyruvate dehydrogenase
 - C. Phosphoenolpyruvate carboxykinase
 - D. Transketolase
-

66. What is the underlying cause of the presenting symptoms in a 21-year-old individual with chronic alcoholism who is admitted to the emergency department with confusion, tremors, ataxia, and ophthalmoplegia?

- A. Vitamin B12
- B. Vitamin B9
- C. Vitamin B1

D. Vitamin B3

67. Which of the following enzyme activities can be estimated in red blood cells to diagnose vitamin B2 deficiency?

- A. Transketolase
 - B. Glutathione reductase
 - C. Kynureninase
 - D. Pyruvate dehydrogenase
-

68. How does Dicoumarol inhibit the function of vitamin K?

- A. Suicide inhibition
 - B. Competitive inhibition
 - C. Non-competitive inhibition
 - D. Uncompetitive inhibition
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	1
Question 4	1
Question 5	4
Question 6	3
Question 7	4
Question 8	3
Question 9	3
Question 10	2
Question 11	3
Question 12	4
Question 13	1
Question 14	1
Question 15	2
Question 16	1
Question 17	1
Question 18	3

Question 19	1
Question 20	1
Question 21	2
Question 22	3
Question 23	4
Question 24	2
Question 25	1
Question 26	1
Question 27	2
Question 28	1
Question 29	1
Question 30	4
Question 31	1
Question 32	2
Question 33	1
Question 34	2
Question 35	2
Question 36	4
Question 37	1
Question 38	2
Question 39	1

Question 40	3
Question 41	2
Question 42	2
Question 43	2
Question 44	1
Question 45	2
Question 46	1
Question 47	2
Question 48	3
Question 49	2
Question 50	3
Question 51	3
Question 52	3
Question 53	3
Question 54	1
Question 55	4
Question 56	1
Question 57	4
Question 58	2
Question 59	1
Question 60	4

Question 61	3
Question 62	3
Question 63	2
Question 64	1
Question 65	2
Question 66	3
Question 67	2
Question 68	2

Solution for Question 1:

Correct Answer: A) Zinc

Explanation:

Zinc plays a crucial role in wound healing, protein synthesis, and maintenance of skin integrity, including hair follicle health. The combination of delayed wound healing and alopecia is particularly characteristic of zinc deficiency, as zinc is essential for both collagen synthesis and cell division necessary for tissue repair and hair growth.

Zinc	
Function	Component of metalloenzymes Involved in protein, DNA, and RNA synthesis Necessary for steroid hormone receptors Essential for spermatogenesis and fetal growth
Clinical Features of Zinc Deficiency	Diarrhea Alopecia Muscle wasting Depression Irritability Rash Delayed wound healing Scaly dermatitis
Diagnostic Level of Zinc Deficiency	Serum zinc level <12 µmol/L (70 µg/dL)

- Copper (Option B): While copper is important for collagen cross-linking and wound healing, it is not typically associated with alopecia as a primary manifestation. The primary features of copper deficiency include anemia and neutropenia. Copper deficiency more commonly

presents with neurological symptoms and hematological abnormalities.

- Calcium (Option C): Calcium deficiency primarily affects bone metabolism and neuromuscular function. It does not typically cause delayed wound healing or alopecia. The classic manifestations of calcium deficiency include tetany, muscle cramps, and bone abnormalities.
- Vitamin D (Option D): Vitamin D deficiency primarily affects calcium homeostasis and bone metabolism. While it can have wide-ranging effects, delayed wound healing and alopecia are not characteristic features. The primary manifestations of vitamin D deficiency include bone pain, muscle weakness, and increased risk of fractures.

Solution for Question 2:

Correct Option D:

Supraphysiological doses of niacin, typically administered for therapeutic purposes, have been associated with ocular complications such as macular edema and macular cysts. Niacin, when used in high doses for the treatment of certain medical conditions, can disrupt retinal fluid balance and lead to fluid accumulation in the macula, resulting in macular edema. This can manifest as blurred vision, distorted vision, or central vision loss.

Incorrect options:

Option A . Vitamin A: High doses of vitamin A, while potentially toxic, are not specifically associated with the development of macular edema or macular cysts.

Option B. Vitamin D: Supraphysiological doses of vitamin D are generally not linked to macular edema or macular cysts. However, excessive vitamin D intake can cause other adverse effects, such as hypercalcemia.

Option C. Vitamin E: There is no evidence to suggest that high doses of vitamin E cause macular edema or macular cysts. Vitamin E is typically associated with other health benefits and is not known to be directly linked to ocular complications.

Solution for Question 3:

Correct Option A:

Ask for diarrhea and a history of memory loss:

- Given the patient's history of alcoholism and the clinical finding of dermatitis, Vitamin B-3 deficiency is a possibility.
- Pellagra/B3 deficiency is characterized by a triad of dermatitis, dementia, and diarrhea.

Incorrect Options:

Option B: Ask for diarrhea, thiamine deficiency features, and detailed dietary history: While thiamine deficiency is a potential concern in alcoholics, this option goes beyond the symptoms described. The focus is primarily on assessing diarrhea, thiamine deficiency features (muscle weakness, fatigue, cardiovascular symptoms), and obtaining a detailed dietary history. However, these aspects do not directly correlate with the specific symptoms mentioned.

Option C: History of allergy : Although allergies can cause skin manifestations, including itching, the specific symptoms mentioned in this case are indicative of a condition associated with alcohol abuse and thiamine deficiency. Therefore, a history of allergy is less likely to provide relevant information regarding the patient's current presentation.

Option D: Ask for diarrhea and conjunctivitis: Conjunctivitis (pink eye) and diarrhea are symptoms that can be associated with certain infections. However, these symptoms are not specifically linked to the presentation of itchy lesions resembling Cassal's necklace and a red-colored beefy tongue in an alcoholic patient.

Solution for Question 4:

Correct Option A:

- In this case, the deficiency that is most likely to be suspected based on the symptoms and diet history is niacin deficiency.
- Niacin (Vitamin B3) deficiency: Niacin deficiency can occur due to a diet rich in maize (corn) because maize is low in niacin and contains niacin-bound compounds that are not readily absorbed by the body. The clinical manifestations of niacin deficiency include diarrhea, dermatitis (rashes), and dementia (known as pellagra). The rashes typically occur in sun-exposed areas due to the increased sensitivity of the skin.

Incorrect Options

Option B: Pyridoxine (Vitamin B6) deficiency: Pyridoxine deficiency is unlikely to be the cause of the symptoms described in this scenario. Pyridoxine deficiency can lead to various neurological symptoms such as peripheral neuropathy, but it is not associated with specific skin rashes or gastrointestinal symptoms like loose stools.

Option C: Thiamine (Vitamin B1) deficiency: Thiamine deficiency typically presents with symptoms related to the nervous system, such as peripheral neuropathy, cardiovascular abnormalities, and Wernicke-Korsakoff syndrome. Loose stools and rashes over sun-exposed areas are not characteristic features of thiamine deficiency.

Option D: Biotin deficiency: Biotin deficiency is rare and usually occurs in individuals with a genetic disorder or those consuming large quantities of raw egg whites, which contain a protein called avidin that binds to biotin and impairs its absorption. Biotin deficiency can lead to symptoms such as hair loss, skin rash, and neurological abnormalities. However, loose stools and rashes over sun-exposed areas are not specific to biotin deficiency.

Solution for Question 5:

Correct Option D - 3, 4:

Incorrect Options:

Option A, B, and C: Refer to explanation of Option D

Solution for Question 6:

Correct Option C - Hemolytic anemia is a feature:

- Hemolytic anemia is not a feature of folate deficiency.
- Folate deficiency typically causes megaloblastic anemia, characterized by the presence of large, immature red blood cells.

- Hemolytic anemia involves the destruction of red blood cells and is not directly related to folate deficiency.

Incorrect Options:

Option B - High Homocysteine and Normal Methylmalonic Acid:

- In folate deficiency, homocysteine levels are elevated because folate is necessary for its conversion to methionine.
- However, methylmalonic acid levels remain normal because MMA accumulation is specifically associated with vitamin B12 deficiency.

Option A - Folate supplementation in pregnancy prevents NTD :

- Folate, also known as folic acid when it's in its synthetic form, is crucial during pregnancy for preventing neural tube defects (NTDs) in the developing fetus.

Option D - Treat coexisting B12 deficiency before initiating folate therapy :

- B12 deficiency indirectly causes folate deficiency.
- It's important to treat coexisting B12 deficiency before initiating folate therapy.
- Folate supplementation can improve the production of red blood cells, which may mask the anemia symptoms of B12 deficiency.
- This can lead to a delay in diagnosing and treating the underlying B12 deficiency, which, if left untreated, can cause irreversible nerve damage.

Solution for Question 7:

Correct Option D - Niacin (Vitamin B3) Deficiency:

- The image depicts symptoms characteristic of Pellagra, which is caused by a deficiency in niacin. Pellagra is often recognized by the "3 D's": Dermatitis, Diarrhea, and Dementia. The term "castles necklace" refers to a specific type of dermatitis that appears as dark, rough patches on the skin, commonly associated with niacin deficiency.

Incorrect Options:

Option A - Vitamin A Deficiency: This typically leads to issues such as night blindness and xerophthalmia (dry eyes), but it does not cause the symptoms shown in the image.

Option B - Vitamin E Deficiency: Vitamin E deficiency can result in neurological problems and muscle weakness, but it does not cause the symptoms depicted in the image.

Option C - Vitamin C Deficiency: This leads to scurvy, which is characterized by bleeding gums, joint pain, and skin issues, but is not associated with the symptoms shown in the image.

Solution for Question 8:

Correct Option C - Methylcobalamin (Vitamin B12) deficiency :

- Symptoms : Vitamin B12 deficiency can lead to neurological symptoms such as gait abnormalities, confusion, and agitation. It often causes a condition known as Subacute Combined Degeneration (SACD) of the spinal cord. This condition includes: Sensory neuropathy: Tingling and numbness in the extremities. Motor neuropathy: Weakness and gait disturbances. Cranial nerve involvement: Potentially affecting cognitive functions and causing confusion.
- Sensory neuropathy: Tingling and numbness in the extremities.
- Motor neuropathy: Weakness and gait disturbances.
- Cranial nerve involvement: Potentially affecting cognitive functions and causing confusion.
- Sensory neuropathy: Tingling and numbness in the extremities.
- Motor neuropathy: Weakness and gait disturbances.
- Cranial nerve involvement: Potentially affecting cognitive functions and causing confusion.

Incorrect Options:

Option A - Nicotinamide (Vitamin B3) Deficiency: Deficiency in Vitamin B3 can lead to pellagra, characterized by the "three D's": dermatitis, diarrhea, and dementia.

Option B - Folate (Vitamin B9) Deficiency: Folate deficiency is often associated with macrocytic anemia and can lead to neuropsychiatric symptoms such as irritability and depression. However, it is less commonly associated with gait abnormalities compared to Vitamin B12 deficiency.

Option D - Tyrosinase: Tyrosinase is an enzyme involved in melanin production, not a vitamin. It is not related to the symptoms described.

Solution for Question 9:

Correct Option C - Hartnup disorder:

- Hartnup disorder is an inborn error of amino acid metabolism characterized by the malabsorption of neutral amino acids in the intestines, which leads to high levels of aminoaciduria. The skin manifestations similar to pellagra and the normal levels of HIAA and indole levels are consistent with this disorder.

Incorrect Options:

Option A - Carcinoid tumor: This option is unlikely because carcinoid tumors are typically associated with elevated levels of 5-Hydroxyindoleacetic Acid (HIAA) in the urine, which is not the case here. Additionally, carcinoid syndrome often does not present with pellagra-like symptoms.

Option B - Nutritional pellagra : Nutritional pellagra results from a deficiency of niacin (vitamin B3) and is commonly associated with a diet lacking in niacin, such as a maize-based diet. The patient's symptoms occurring seasonally and the laboratory findings do not strongly support this diagnosis.

Option D - Essential fructosuria: This condition is caused by a deficiency in the enzyme fructokinase and results in the excretion of fructose in the urine. It does not cause pellagra-like skin symptoms, making it an unlikely diagnosis in this case.

Solution for Question 10:

Correct Option B - Beriberi:

- Beriberi is a condition caused by thiamine (vitamin B1) deficiency. It is commonly seen in individuals with diets low in thiamine, such as those relying heavily on polished rice, which lacks thiamine.
- Chronic alcoholism can also impair thiamine absorption and utilization, exacerbating the deficiency. Beriberi can present with various symptoms, including skin lesions, and is linked to a lack of thiamine in the diet.

Incorrect Options:

Option A - Phrynoderma: Phrynoderma (also known as "toad skin") is a condition associated with vitamin A deficiency, not vitamin B1. It typically presents with rough, bumpy skin, but it is not related to the patient's dietary history of polished rice.

Option C - Psoriasis: Psoriasis is a chronic skin condition characterized by red, scaly patches. It is not directly linked to thiamine deficiency and is unrelated to the patient's dietary habits.

Option D - Eczema: Eczema is a general term for skin inflammation that can result from various factors, including allergies and irritants. It is not typically associated with thiamine deficiency or a diet high in polished rice.

Solution for Question 11:

Correct Option C - Vitamin C:

- Vitamin C influences the absorption of iron, so in case of vitamin C deficiency, iron absorption will be hampered, causing anemia. Other manifestations of scurvy include bleeding and swollen gums. The white line of Frankel is also seen on the X-ray.

Incorrect Options:

- Options A, B, and D do not present with the given clinical condition.

Solution for Question 12:

Answer: D) Vitamin C

Explanation:

- Gum bleeding & petechiae: Classic signs of scurvy, caused by vitamin C deficiency, due to impaired collagen synthesis leading to fragile capillaries.
- Poor wound healing: Vitamin C is crucial for collagen formation and connective tissue repair, and its deficiency results in defective wound healing.

- Other symptoms of scurvy: Fatigue, corkscrew hairs, anemia, and subperiosteal hemorrhages.

Feature	Vitamin C Deficiency (Scurvy)
Cause	Poor dietary intake (malnutrition, alcoholism, elderly, infants on cow's milk)
Key Symptoms	Bleeding gums, petechiae, bruising, corkscrew hairs, poor wound healing
Other Signs	Subperiosteal hemorrhage, joint pain, anemia, weakness
Diagnosis	Clinical features, low plasma vitamin C levels
Treatment	Vitamin C supplementation

Vitamin A Deficiency (Option A): Incorrect because it causes night blindness, dry skin, and keratomalacia, not bleeding gums or petechiae.

Vitamin B6 Deficiency (Option B): Incorrect because it presents with irritability, glossitis, peripheral neuropathy, and seizures, not connective tissue defects.

Vitamin B3 (Niacin) Deficiency (Option C): Incorrect because it causes pellagra (3 Ds- diarrhea, dermatitis, dementia) with photosensitive rash, not gum bleeding.

Solution for Question 13:

Answer: A) Zinc Deficiency

Explanation:

- Characteristic rash: The image likely shows acrodermatitis enteropathica, which presents with perioral, perianal, and acral dermatitis—a hallmark of zinc deficiency.
- History of diarrhea: Zinc plays a crucial role in intestinal function and immunity; its deficiency leads to chronic diarrhea.
- Other symptoms of zinc deficiency: Growth retardation, poor wound healing, alopecia, and immune dysfunction.

Feature	Zinc Deficiency
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Cause	Congenital (SLC39A4 mutation) or acquired (malnutrition, chronic diarrhea)
Key Skin Features	Perioral, perianal, and acral dermatitis with erythematous, scaly plaques
Other Symptoms	Growth retardation, diarrhea, alopecia, immune dysfunction
Diagnosis	Low serum zinc levels
Treatment	Zinc supplementation

Niacin Deficiency (Option B) is incorrect because niacin deficiency leads to pellagra , characterized by the "3 Ds"— dermatitis, diarrhoea, and dementia —but the rash is photosensitive and occurs in sun-exposed areas, not perioral/perianal.

Selenium Deficiency (Option C) is incorrect because selenium deficiency primarily affects the cardiovascular system (e.g., Keshan disease , causing cardiomyopathy) and does not cause a perioral/perianal rash.

Magnesium Deficiency (Option D) is incorrect because magnesium deficiency presents with neuromuscular symptoms (tetany, muscle cramps, seizures) rather than a characteristic rash.

Solution for Question 14:

Correct Option A - Vitamin B6:

- Homocystinuria is often due to a deficiency in the enzyme cystathionine β -synthase, which requires Vitamin B6 as a cofactor. Supplementing with Vitamin B6 can help reduce homocysteine levels and alleviate symptoms like joint pain and weakness.

Incorrect Options:

Options B, C, and D are not preferred in the management of this patient.

Solution for Question 15:

Correct Option B - Formic acid + Lactic acid:

- In methanol poisoning, metabolites like formic acid and lactic acid accumulate and cause toxic effects like histotoxic hypoxia and optic neuritis.

Incorrect Options:

Options A, C, and D do not accumulate in methanol poisoning.

Solution for Question 16:

Correct Option A - Hypophosphatemic rickets:

- Hypophosphatemic rickets is characterized by low phosphate levels, normal calcium and PTH, and elevated ALP, leading to bone deformities.

Incorrect Options:

Option B and C - VDDR type 1 & 2 present with low calcium and elevated PTH.

Option D - Renal osteodystrophy will present with underlying renal disease.

Solution for Question 17:

- A diet primarily of maize may not provide enough Niacin, leading to Niacin deficiency. The symptoms of Niacin deficiency include diarrhea, dementia, and photosensitive dermatitis in sun-exposed areas. Based on the patient's symptoms, the most likely vitamin deficient is Niacin, also known as Vitamin B3.
- Niacin is an atypical vitamin, i.e., it can be synthesized in the body from tryptophan.
- It is also involved in DNA repair and cell communication. Niacin is found in many foods, including meat, fish, and whole grains. However, a diet primarily of maize may not provide enough Niacin, leading to Niacin deficiency.

Incorrect Choices:

- Option b. Vitamin B6 : Vitamin B6 is a water-soluble vitamin involved in many metabolic processes, including amino acid metabolism and the synthesis of neurotransmitters. Vitamin B6 deficiency can lead to various symptoms, including anemia, depression, confusion, and skin inflammation. However, the symptoms in the question stem, such as diarrhea and photosensitive dermatitis, are not typically associated with vitamin B6 deficiency.
 - Option c. Ascorbic acid : Ascorbic acid, also known as vitamin C, is an essential nutrient with antioxidant properties and plays a role in collagen synthesis and the immune system. Vitamin C deficiency can lead to scurvy, characterized by symptoms such as fatigue, muscle weakness, joint pain, and bleeding gums. However, the symptoms in the question stem, such as diarrhea and photosensitive dermatitis, are not typically associated with vitamin C deficiency.
 - Option d. Biotin is important for the metabolism of fats, carbohydrates, and proteins and is involved in energy production. A deficiency of biotin can lead to skin rashes, hair loss, and neurological symptoms. However, the symptoms described in the patient are inconsistent with biotin deficiency. Biotin deficiency is most commonly seen in individuals who consume large amounts of raw egg whites, as egg whites contain avidin. This protein binds to biotin and prevents its absorption.
-

Solution for Question 18:

Correct Option C – Copper:

- Anaemia, neutropenia, and hypopigmented skin and hair are seen in copper deficiency.

Incorrect Options:

Options A, B, and D do not present with the given findings.

Solution for Question 19:

Correct Option A – Phrynoderma:

- Phrynoderma: Also known as follicular hyperkeratosis, this condition is characterized by small, rough papules, typically on the extensor surfaces of limbs, due to Vitamin A deficiency.

Incorrect Options:

Option B - Scurvy is caused by Vitamin C deficiency; it presents with symptoms like bleeding gums, bruising, and joint pain, not follicular hyperkeratosis.

Option C - Pellagra is caused by niacin (Vitamin B3) deficiency, presenting with the classic triad of dermatitis, diarrhea, and dementia.

Option D - Keratosis pilaris is not associated with Vitamin A deficiency.

Solution for Question 20:

Correct Option A - Vitamin B1 injection followed by glucose infusion

- This treatment protocol is commonly known as the "Banana Bag" or "Wernicke-Korsakoff cocktail." The rationale behind this approach is to address both thiamine deficiency and potential hypoglycemia that can occur in individuals with chronic alcoholism.
- The initial administration of thiamine (vitamin B1) through intravenous injection is crucial to rapidly replenish thiamine levels in the body. This helps prevent or reverse Wernicke encephalopathy, a severe neurological condition associated with thiamine deficiency.
- Following thiamine injection, glucose infusion is provided to address the potential risk of hypoglycemia. Alcohol can disrupt glucose metabolism and deplete glycogen stores in the liver, leading to low blood sugar levels. Glucose infusion helps normalize blood sugar levels and provides adequate energy supply to the brain.

Incorrect Options - B, C, and D: These are incorrect.

Solution for Question 21:

Correct Option B - 1 and 4 only

- Methyl malonyl CoA isomerase: This enzyme is dependent on vitamin B12. It catalyzes the isomerization of methyl malonyl CoA to succinyl CoA, which is an important step in the metabolism of certain fatty acids and amino acids.
- Homocysteine synthase: This enzyme, also known as methionine synthase reductase, is not directly dependent on vitamin B12. It utilizes methylcobalamin, a form of vitamin B12, as a cofactor to catalyze the conversion of homocysteine to methionine. However, it is not strictly considered a vitamin B12-dependent enzyme.

Incorrect Options:

Option A, C, and D: These are incorrect options.

Folate reductase: This enzyme is not directly dependent on vitamin B12. It is involved in the metabolism of folate (vitamin B9) and plays a role in the synthesis of DNA and other important cellular processes.

Methionine synthase: This enzyme is dependent on vitamin B12. It uses methylcobalamin as a cofactor to catalyze the conversion of homocysteine to methionine, a critical step in the synthesis of methionine and the recycling of homocysteine.

Solution for Question 22:

Correct Option C: Prolonged use of antibiotics leads to vitamin K deficiency

- Antibiotics used for the long term not just kills harmful bacteria but also kills the bacteria that make vitamin K.
- Vitamin K is a group of vitamins required for blood clotting, mainly encompassing phylloquinone (nutrition K1) and menaquinone (nutrition K2). It is a fat-soluble vitamin. It's rare to have Vitamin K deficiency due to the presence of leafy vegetables and also being synthesized by the bacteria found in small intestines. Thus, sometimes due to consumption of antibiotics may kill the bacteria and lead to mild deficiency in people with low levels, to begin with.

Incorrect Options :

Option A - Deficiency of vitamin K leads to vitamin K deficiency bleeding (VKDB).

Option B - Vitamin K is a fat-soluble vitamin.

Option D - Menaquinone is the dietary source of vitamin K found in meat, dairy foods, and eggs.

Solution for Question 23:

Correct Option D: Epoxide reductase.

- After vitamin K is used in the carboxylation of these clotting factors, it undergoes oxidation and is converted into an inactive form called vitamin K epoxide. In order to be reused and regenerated back into its active coenzyme form, vitamin K epoxide needs to be reduced.
- Epoxide reductase is the enzyme responsible for this reduction process. It catalyzes the reduction of vitamin K epoxide back to its active form, allowing it to participate in further cycles of the clotting cascade.

Incorrect Options:

Option A - Glutathione reductase: Glutathione reductase is an enzyme involved in the regeneration of reduced glutathione, an important antioxidant molecule. It does not directly participate in the regeneration of vitamin K.

Option B - Quinone reductase: Quinone reductase, also known as NAD(P)H:quinone oxidoreductase, is an enzyme involved in the reduction of quinones, which are a class of organic compounds. It is not involved in the regeneration of vitamin K.

Option C - Vitamin K reductase: While the term "vitamin K reductase" might sound relevant, there is no specific enzyme known as vitamin K reductase that is responsible for the regeneration of vitamin K. Instead, it is the epoxide reductase enzyme that performs this role.

Solution for Question 24:

Correct Option B - Vitamin B2:

- Fissured tongue and peripheral neuropathy are clinical manifestations associated with riboflavin deficiency, also known as vitamin B2 deficiency.
- Reduced glutathione reductase activity is a biochemical finding often observed in cases of riboflavin deficiency. Glutathione reductase is an enzyme that depends on flavin adenine

dinucleotide (FAD), a coenzyme derived from riboflavin, for its activity. A deficiency of riboflavin leads to reduced levels of FAD and subsequently impairs the activity of glutathione reductase.

Incorrect Options:

Option A - Vitamin B1 (thiamine): Thiamine deficiency can lead to various neurological manifestations, including peripheral neuropathy, but it is not specifically associated with fissured tongue. Thiamine deficiency is commonly seen in conditions such as Wernicke-Korsakoff syndrome and beriberi.

Option C - Vitamin B6 (pyridoxine): Vitamin B6 deficiency can also cause peripheral neuropathy, but it is not typically associated with fissured tongue. Pyridoxine deficiency is often seen in individuals with malnutrition or certain medical conditions.

Option D - Vitamin B12 (cobalamin): Vitamin B12 deficiency can cause peripheral neuropathy, as well as glossitis (inflammation of the tongue), but it does not specifically result in a fissured tongue. Vitamin B12 deficiency is commonly associated with pernicious anemia and malabsorption disorders.

Solution for Question 25:

Correct Option A - L-gulonolactone oxidase:

- Humans lack the enzyme L-gulonolactone oxidase, which is required for the synthesis of vitamin C (ascorbic acid) from glucose. L-gulonolactone oxidase catalyzes the final step in the biosynthesis pathway of vitamin C in most animals.
- However, due to a genetic mutation, humans have lost the ability to produce this enzyme, resulting in a deficiency of endogenous vitamin C synthesis.

Incorrect Options:

Option B - Xylitol reductase: Xylitol reductase is not directly involved in the synthesis of vitamin C. It is an enzyme involved in the metabolism of xylitol, a sugar alcohol derived from xylose.

Option C - Pyruvate dehydrogenase: Pyruvate dehydrogenase is an enzyme complex involved in the conversion of pyruvate to acetyl-CoA in the process of cellular respiration. It is not directly related to the synthesis of vitamin C.

Option D - UDP glucose dehydrogenase: UDP glucose dehydrogenase is an enzyme involved in the synthesis of glycogen, a storage form of glucose. It is not directly involved in the synthesis of vitamin C.

Solution for Question 26:

Correct Option A - Selenium

- Alkali disease, also known as blind staggers or blindness, is a condition that affects livestock, particularly cattle, and is caused by an excess of certain minerals in their diet. Selenium toxicity is known to be one of the causes of alkali disease in livestock.
- Excessive selenium levels can lead to oxidative stress and damage to cells, particularly in the nervous system.
- The symptoms of alkali disease include neurological abnormalities such as impaired vision, lack of coordination, tremors, and muscle weakness. In severe cases, blindness may occur. These symptoms are associated with the toxic effects of excess selenium on the nervous system.

Incorrect Options:

Option B - Manganese: Manganese toxicity is not associated with alkali disease in livestock. Excess manganese intake can lead to manganese toxicity, which primarily affects the nervous system. However, it does not cause the specific condition known as alkali disease.

Option C - Molybdenum: Molybdenum toxicity is not directly associated with alkali disease in livestock. However, high levels of molybdenum in the diet can lead to a condition called molybdenosis, which primarily affects ruminant animals such as cattle. Molybdenosis can interfere with copper metabolism, leading to copper deficiency and associated health problems.

Option D - Iron: Iron toxicity is not related to alkali disease in livestock. While excessive iron intake can cause iron toxicity in animals, it does not result in the specific condition described as alkali disease.

Solution for Question 27:

Correct Option B - Vitamin C

- Vitamin C enhances iron absorption in the intestines. Iron exists in two forms: non-heme iron (found in plant-based foods) and heme iron (found in animal-based foods). Non-heme iron, which is the predominant form in most diets, is less easily absorbed by the body compared to heme iron.
- Vitamin C acts as a reducing agent, converting ferric iron (Fe^{3+}) to ferrous iron (Fe^{2+}), which is more readily absorbed. Additionally, it forms a soluble complex with iron, protecting it from oxidation and improving its absorption in the intestines.

Incorrect Options:

Option A - Vitamin A: Vitamin A is not directly involved in enhancing iron absorption. However, it is known to play a role in the regulation of iron metabolism and the storage of iron in the liver.

Option C - Thiamine: Thiamine, also known as vitamin B1, is not directly involved in enhancing iron absorption. Thiamine is primarily involved in energy metabolism and the functioning of the nervous system.

Option D - Riboflavin: Riboflavin, also known as vitamin B2, is not directly involved in enhancing iron absorption. Riboflavin is involved in various cellular processes, including energy metabolism and antioxidant defense.

Solution for Question 28:

Correct Answer : A

- The inheritance pattern of hemophilia A is X-linked recessive. Hemophilia A is caused by a mutation in the F8 gene, which is located on the X chromosome. Since males have one X chromosome and females have two, the condition is more commonly observed in males. If a male inherits the affected X chromosome from his mother, he will have hemophilia A. On the other hand, females are usually carriers of the condition, meaning they have one normal X chromosome and one affected X chromosome but do not typically display severe symptoms. However, females can pass the affected X chromosome to their children, potentially leading to the condition in future generations.

Incorrect options:

Option B. X linked Dominant is incorrect

Option C. autosomal dominant is incorrect

OptionD. autosomal recessiveis incorrect

Solution for Question 29:

Correct option A .

- Wernicke encephalopathy is caused by a deficiency of Vitamin B1 (Thiamine) . Thiamine is necessary for normal neurological functioning , and its deficiency can result in various neurological symptoms.

Incorrect options

Options: B . Vit B2 (Riboflavin): Wernicke encephalopathy is caused by a deficiency of Vitamin B1 (Thiamine), not Vitamin B2 (Riboflavin). While both vitamins are essential for normal neurological functioning, a deficiency of Vitamin B1 specifically leads to Wernicke encephalopathy.

Option C . Vit B7 (Biotin): Wernicke encephalopathy is caused by a deficiency of Vitamin B1 (Thiamine), not Vitamin B7 (Biotin). Vitamin B7 is involved in various metabolic processes but is not associated with Wernicke encephalopathy.

Option D . Vit B12 (Cobalamin): Vitamin B12 (Cobalamin) deficiency can lead to neurological symptoms, but it is not the cause of Wernicke encephalopathy. Wernicke encephalopathy is specifically associated with a deficiency of Vitamin B1 (Thiamine).

Solution for Question 30:

Correct Option D - All

- Zinc supplementation addresses multiple aspects related to diarrhea. It reduces the risk, duration, and severity of diarrheal episodes, enhances immune response, and helps regulate

intestinal transport and absorption of water and electrolytes. Therefore, option D is the correct answer.

Incorrect Options:

Option A - Reduces the risk, duration, and severity of diarrheal episodes: Zinc supplementation has been shown to reduce the risk, duration, and severity of diarrheal episodes, particularly in children. It helps in restoring the integrity of the intestinal lining and reducing the loss of fluids and electrolytes during diarrhea.

Option B - Enhances immune response: Zinc plays a crucial role in supporting the immune system. It helps in the production and function of immune cells, such as T cells and natural killer cells, and contributes to the body's defense against pathogens. Zinc deficiency can impair immune function, making individuals more susceptible to infections, including those causing diarrhea.

Option C - Regulates intestinal transport & absorption of water and electrolytes: Zinc is involved in the regulation of intestinal transport and absorption of water and electrolytes. It helps maintain the integrity of the intestinal epithelial cells, which are responsible for the absorption of nutrients and water. Zinc deficiency can disrupt these processes and contribute to diarrhea.

Solution for Question 31:

Correct Option:

Option A. RBC Transketolase: This test assesses the activity of the enzyme transketolase, which requires thiamine (vitamin B1) as a cofactor. A decreased activity indicates a deficiency of thiamine.

Incorrect Option:

Option B. RBC glutathione reductase: This test measures the activity of the enzyme glutathione reductase, which is not specifically related to thiamine deficiency. It is used to evaluate the function of the glutathione system and is not directly indicative of B1 deficiency.

Option C. Serum thiamine levels: This test directly measures the levels of thiamine in the blood. Low serum thiamine levels may suggest a deficiency, but it is not a functional assessment of B1 deficiency.

Option D. RBC glutathione peroxidase: This test measures the activity of the enzyme glutathione peroxidase, which is not specifically related to thiamine deficiency. It is involved in the detoxification of reactive oxygen species and is not directly indicative of B1 deficiency.

Solution for Question 32:

Correct Option B:

The synthesis of active vitamin D, also known as calcitriol or 1,25-dihydroxyvitamin D₃, involves a sequential process that occurs primarily in three organs: the skin, liver, and kidneys.

Incorrect Options:

Option A: Liver-skin-kidney: This sequence is incorrect because the initial step of vitamin D synthesis occurs in the skin upon exposure to UVB radiation, followed by the involvement of the liver and kidneys.

Option C: Skin-kidney-liver: This sequence is also incorrect because the liver is involved in the hydroxylation of previtamin D₃ to calcidiol before the kidneys further hydroxylate it to form the active vitamin D.

Option D: Liver-kidney-skin: This sequence is incorrect as it does not follow the correct order of vitamin D synthesis. The initial step occurs in the skin, followed by the involvement of the liver and kidneys.

Solution for Question 33:

Correct Option A - Calcium

- Calcium is considered a macromineral because it is required by the body in larger amounts compared to microminerals. Macrominerals are needed in quantities greater than 100 milligrams per day.
- Minerals are divided into two types: macrominerals and microminerals. Macrominerals, including calcium, phosphorus, magnesium, are required in amounts larger than 100 mg

daily . Microminerals , which include electrolytes including sodium , potassium , chloride , iron , copper , zinc , selenium , and iodine , are required at levels less than 100 mg per day .

- Macrominerals , including calcium , phosphorus , magnesium , are required in amounts larger than 100 mg daily .

- Microminerals , which include electrolytes including sodium , potassium , chloride , iron , copper , zinc , selenium , and iodine , are required at levels less than 100 mg per day .

- Macrominerals , including calcium , phosphorus , magnesium , are required in amounts larger than 100 mg daily .

- Microminerals , which include electrolytes including sodium , potassium , chloride , iron , copper , zinc , selenium , and iodine , are required at levels less than 100 mg per day .

Incorrect Options:

Option B. Selenium

- Classified as a micromineral . Selenium is needed in very small amounts, typically less than 100 milligrams per day. It is important for antioxidant function and the maintenance of thyroid hormone metabolism.

Option C. Iodine

- Classified as a micromineral . Iodine is essential in trace amounts for the production of thyroid hormones, which regulate metabolic processes in the body. It is required in small amounts, thus fitting the classification of a micromineral.

Option D. Copper

- Classified as a micromineral . Copper is required in trace amounts for various enzymatic functions, including iron metabolism and the formation of connective tissue. It is needed in small quantities, making it a micromineral.

Solution for Question 34:

Correct Ans. B

- Tryptophan Tryptophan deficiency can lead to pellagra, a condition characterized by a deficiency of niacin (vitamin B3). Pellagra can present with various symptoms, including dermatitis, diarrhea, dementia, and the characteristic Casal's necklace, which refers to a dark pigmentation around the neck.

Incorrect Option:

Option A: Phenylalanine Phenylalanine is also an essential amino acid, and its deficiency can lead to a condition called phenylketonuria (PKU). However, PKU does not manifest with the specific symptom of Casal's necklace.

Option C: Glutamine Glutamine is a non-essential amino acid, meaning it can be synthesized by the body. While glutamine deficiency can occur in certain situations, such as severe illness or malnutrition, it is not directly associated with the specific symptom of Casal's necklace.

Option D: Arginine Arginine is a semi-essential amino acid, meaning it can be produced by the body but may become essential in certain conditions, such as during periods of growth or illness. Arginine deficiency is rare and not typically associated with the characteristic symptom of Casal's necklace.

Solution for Question 35:

Correct Option B: Niacin

- Niacin or B3 can be synthesized via metabolic pathways . It is vital for macronutrient metabolism (carbohydrate, protein, and fat) since it is a component of the NAD and NADP coenzymes . Niacin deficiency causes a decrease in NAD and NADP coenzymes .
- The oxidation of glucose generates a considerable amount of chemical energy. NAD/NADH transports electrons through a route that collects energy by forming high-energy phosphate bonds. This adenosine triphosphate (ATP) provides the energy required for subsequent intermediate metabolic steps while also rebuilding NAD from the decreased NADH. A portion of this cofactor is transformed into NADP/NADPH, which serves numerous functions.
- Other pathways also contribute to niacin insufficiency. Carcinoid syndrome has altered tryptophan metabolism, Hartnup disease has impaired tryptophan absorption, and long-term use of certain medications may decrease tryptophan production (isoniazid) or inhibit the conversion of tryptophan to niacin (azathioprine, 6-mercaptopurine, or 5-fluorouracil).
- Pellagra is a disorder characterized by a lack of niacin or its precursor, tryptophan . Pellagra involves the "three Ds," dermatitis, dementia, and diarrhea, and can be fatal. Brown discoloration and skin lesions on sun-exposed skin , such as hands, elbows, knees, and feet, are evident, similar to a sunburn. Initially, neurological changes such as anxiety, poor attention, exhaustion, and sadness may develop, but as pellagra progresses, dementia and delirium may follow. There may also be gastrointestinal issues such as glossitis, cheilosis,

stomatitis, nausea, vomiting, diarrhea, or constipation.

Incorrect Options:

Option A: For thiamine (Vitamin B1) deficiency, common conditions to consider are delirium, depression, folic acid deficiency, hyperthyroidism, cardiomyopathy secondary to other causes such as alcohol or diabetic heart disease, delusional disorder, nerve entrapment syndromes, other psychiatric disorders, or diabetic ketoacidosis. None of these signs or symptoms are present in this case. Therefore, thiamine deficiency can be ruled out.

Option C: specific skin manifestations, such as hyperpigmentation, cutaneous lesions, vitiligo, angular stomatitis, and hair and nail changes characterize Cobalamine (Vitamin B12) deficiency. The patient here shows no signs or symptoms that can suggest cobalamine deficiency.

Option D: Ascorbic acid (Vitamin C) deficiency leads to skin pigmentation, pale skin, poor wound healing, gingival swelling with loss of teeth, mucocutaneous petechiae, ecchymosis, and hyperkeratosis. None of these signs and symptoms were reported by the patient in this case. Hence, vitamin C deficiency can be ruled out.

Solution for Question 36:

Correct Option D - Vitamin K :

- Vitamin K is essential for blood clotting and proper wound healing. It plays a critical role in the production of clotting factors that prevent excessive bleeding. A deficiency in vitamin K can lead to bleeding disorders, where even minor injuries can result in prolonged bleeding. Additionally, vitamin K is involved in maintaining dental health by supporting the proper mineralization of teeth. Therefore, in this case, the best choice for treating the bleeding condition and dental caries would be vitamin K supplementation.

Incorrect Options:

Option A - Vitamin B1 (Thiamine) : Vitamin B1 is important for converting food into energy and maintaining proper nerve function. It is not directly associated with bleeding conditions or dental caries.

Option B - Vitamin B9 (Folate) : Vitamin B9, also known as folate or folic acid, is crucial for cell division and DNA synthesis. It is particularly important during pregnancy and for overall cell growth. While a deficiency in folate can lead to anemia and other health issues, it is not directly related to bleeding conditions or dental caries.

Option C - Vitamin D : Vitamin D is important for maintaining bone health, immune function, and overall well-being. It helps the body absorb calcium and phosphorus from the intestines. While vitamin D deficiency can lead to weakened bones, it is not directly linked to bleeding conditions or dental caries.

Solution for Question 37:

Option A - Vitamin B6

- In the context of xanthurenic aciduria, the deficiency of Vitamin B6 impairs the normal metabolism of tryptophan, an essential amino acid. Tryptophan is normally metabolized to produce various compounds, including serotonin and niacin (Vitamin B3). However, in the absence of sufficient Vitamin B6, tryptophan is diverted to an alternative metabolic pathway, leading to the production of xanthurenic acid. This results in the accumulation of xanthurenic acid in the urine, leading to the condition known as xanthurenic aciduria.

Incorrect Options

The other vitamins mentioned in the options are associated with different functions and deficiencies:

Option B - Vitamin B12: A deficiency of Vitamin B12 can lead to megaloblastic anemia and neurological problems. It is involved in DNA synthesis and nerve cell function.

Option C - Vitamin B5: Also known as pantothenic acid, Vitamin B5 is involved in energy metabolism, the synthesis of various molecules, and the functioning of the nervous system.

Option D - Vitamin B7: Also known as biotin, Vitamin B7 is important for metabolism and the synthesis of fatty acids, glucose, and amino acids.

Solution for Question 38:

Correct Option B - Vitamin A:

- The disorder known as conjunctival xerosis is brought on by a vitamin A deficiency. Conjunctival xerosis can develop when the body does not receive enough vitamin A, causing the conjunctiva to become dry, rough, and irritating.

Incorrect Options : Option A - Vitamin C:

- Scurvy, which is characterized by bleeding gums, joint discomfort, and skin rash, can result from a vitamin C deficit.

Option C - Vitamin B:

- A class of water-soluble vitamins known as the B vitamins are essential to numerous metabolic activities.

Option D - Vitamin E:

- Lack of vitamin E can cause neurological issues like coordination loss and muscle weakness.

Solution for Question 39:

Correct Option A - Niacin

- The above picture presented above suggests Vitamin B3/niacin deficiency according to clinical features. Pellagra disease is caused by the excessive consumption of a diet made up of maize. Maize/corn has low levels of tryptophan and niacin.
- Niacin deficiency causes dermatitis, diarrhoea, and dementia.
- The neck lesion shows characteristic Casal's necklace pattern, as seen in Pellagra. The Casal necklace or Casal collar is a clinical sign of pellagra that shows erythematous pigmented rash of skin near the collarbone region (dermatomes C3 and C4).

Incorrect Options:

Option B - Zinc deficiency is found in people with severe diarrhea, liver cirrhosis, alcoholism causes

Option C - Prolonged deficiency of Thiamine or Vitamin B1 in patients may support sensation loss at extreme level, heart failure symptoms along with hands or feet swelling, demand ischemia related chest pain, double vision, memory loss,

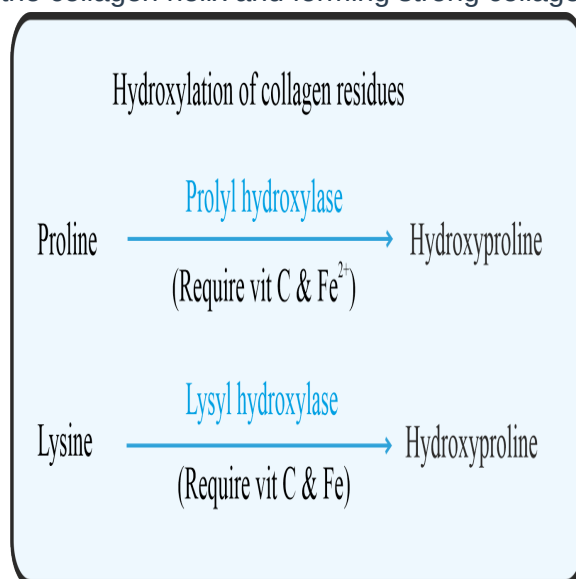
Option D - Iron deficiency in the body causes anemia. Hence, red blood cells are unable to produce enough hemoglobin in the body that allows them to carry oxygen. Shortness of breath

and tiredness are caused as a result of iron deficiency anemia.

Solution for Question 40:

Correct Option C - Ascorbic acid

- Ascorbic acid (Vitamin C) is required for the hydroxylation of proline and lysine residues in collagen, stabilizing the collagen helix and forming strong collagen fibers.



Incorrect Options:

Option A - Proline: Proline stabilizes collagen fibers, forming cross links with other amino acids in collagen. However, proline does not directly help in the maturation of collagen.

Option B - Copper and zinc: They are essential minerals required for many enzymatic reactions in the body, including those involved in collagen synthesis. However, copper and zinc do not directly help collagen maturation.

Option D - Phenylalanine: It is an essential amino acid precursor to tyrosine, another amino acid involved in collagen synthesis. However, phenylalanine does not directly help in the maturation of collagen.

Solution for Question 41:

Correct Option B - Vegan diet

- Beriberi is a condition caused by a thiamine (Vitamin B1) deficiency.
- Thiamine is an essential nutrient required for the metabolism of carbohydrates, the functioning of the nervous system, and the production of energy.
- Thiamine is found in various foods, including meat, fish, and whole grains. However, vegan diets do not cause Beriberi.

Incorrect Options:

Option A - Chronic use of diuretics: Chronic use of diuretics can cause thiamine deficiency by increasing the excretion of thiamine in the urine. This can lead to Beriberi, especially in people with a marginal thiamine status.

Option C - Bariatric surgery: This can lead to malabsorption of essential vitamins and minerals, including thiamine, which is absorbed in the small intestine. Therefore, bariatric surgery can cause Beriberi, especially if the patient does not take thiamine supplements.

Option D - Chronic alcohol intake: Alcohol also interferes with thiamine absorption, storage, and metabolism, leading to thiamine deficiency and Beriberi. Alcoholics who replace food with alcohol are at high risk of developing Beriberi.

Solution for Question 42:

Correct Option B - Vitamin B6 toxicity:

- While multivitamin tablets provide all necessary vitamins, an overdose can lead to toxicity and manifest symptoms. The symptoms, such as numbness, muscle weakness, and difficulty walking, result from nerve damage caused due to high doses of Vitamin B6 supplements. This is because vitamin B6 can accumulate in the body and cause toxicity, particularly in the peripheral nerves.

- Other symptoms of vitamin B6 toxicity may include skin lesions, photosensitivity, and gastrointestinal disturbances, such as nausea and vomiting. In severe cases, it can lead to neuropathy, a type of nerve damage that can cause problems with movement, sensation, and balance problems.

Incorrect Options:

- Option A - Vitamin D toxicity : Symptoms of Vitamin D toxicity may include high blood calcium levels, nausea, vomiting, weakness, polyuria, intense thirst, difficulty in speaking, weight loss.
 - Option C - Vitamin C toxicity : Vitamin C is important for immune function and skin health, but taking too much of it can cause gastrointestinal disturbances, such as diarrhea and nausea.
 - Option D - Vitamin A toxicity : Symptoms of Vitamin A toxicity may include nausea, vomiting, dizziness, and blurred vision. Extremely high doses of Vitamin A can also lead to liver damage and birth defects in pregnant women.
-

Solution for Question 43:

Correct Option B - Factors II, VII, IX, and X

- The gamma-glutamyl carboxylase (GGCX) enzyme contributes significantly to the manufacture of vitamin K-dependent clotting factors II, VII, IX, and X by carboxylating protein-bound glutamate residues.
- This carboxylation process is essential for the functional activation of these factors.
- The addition of the COO group increases the negative charge on these proteins, helping their interaction with more calcium ions and resulting in coagulation.

Incorrect Options:

Option A - Factors II, VIII, IX, X: Factor VIII does not require gamma-glutamyl carboxylase for activation. It is activated by a different mechanism involving thrombin and other factors. Therefore, this option is incorrect.

Option C - Factors II, VII, IX, XI: Factor XI does not require gamma-glutamyl carboxylase for activation. It undergoes limited proteolysis to become an active enzyme. Hence, this option is incorrect.

Option D - Factors II, VIII, X, XI: As mentioned earlier, Factor VIII does not require gamma-glutamyl carboxylase for activation. So, this option is incorrect.

Solution for Question 44:

Correct Option A - Vitamin B3 deficiency:

- The patient's symptoms of diarrhea, dermatitis, and dementia are characteristic of a condition known as pellagra, which is primarily caused by a deficiency in Vitamin B3, also known as niacin.

Incorrect Options:

Option B - Vitamin B1 deficiency: Vitamin B1 deficiency leads to a condition called beriberi, which primarily affects the cardiovascular (wet beriberi) and nervous systems (dry beriberi).

Option C - Vitamin B6 deficiency: Vitamin B6 deficiency can lead to a condition known as pyridoxine deficiency resulting in seizures.

Option D - Vitamin B12 deficiency: Vitamin B12 deficiency can lead to a condition called pernicious anemia. It also presents with paraesthesia, sensory and motor neuropathy.

Solution for Question 45:

Correct Option B - Thiamine-transketolase:

- Thiamine is present in the aleurone layer of the rice; polishing rice results in the removal of the aleurone layer and thiamine deficiency.
- Thiamine acts as a coenzyme for transketolase. So, in thiamine deficiency, the RBC transketolase activity is low.

Incorrect Options:

Option A - Riboflavin-glutathione reductase

- Rice is not a rich source of riboflavin (vitamin B2). Glutathione reductase is an enzyme that uses FAD as a cofactor, and the activity of this enzyme indirectly assesses riboflavin levels.

Option C - Thiamine-transaminase

- Transaminase is an enzyme involved in amino acid metabolism dependent on pyridoxine (B6).

Option D - Riboflavin-transketolase

- Transketolase is indeed a thiamine-dependent enzyme, not riboflavin-dependent.

Solution for Question 46:

Correct Option A - Vitamin B1 deficiency:

- The symptoms described in the patient, including confusion, ataxia (loss of muscle coordination), and painful eye movements, are indicative of Wernicke encephalopathy, which is commonly caused by acute deficiency of thiamine (Vitamin B1).
- Alcoholism is a known risk factor for thiamine deficiency due to several factors, including poor dietary intake, malabsorption, impaired storage, and increased requirements.
- Chronic thiamine deficiency can lead to Korsakoff syndrome, characterized by amnesia, confabulation, and sensory agnosia.

Incorrect Options:

Option B - Vitamin A deficiency: Vitamin A deficiency typically presents with symptoms related to the eyes, such as night blindness, dry eyes, Bitot's spot, and follicular hyperkeratosis.

Option C - Vitamin D deficiency: Vitamin D deficiency is associated with calcium homeostasis and bone metabolism. Its deficiency can lead to hypocalcemia (secondary hypothyroidism) and hypophosphatemia, resulting in osteomalacia and osteoporosis.

Option D - Vitamin C deficiency: Vitamin C deficiency can lead to various symptoms, including fatigue, weakness, gum bleeding, and joint pain (scurvy).

Solution for Question 47:

Correct Option B - Thiamine

- The clinical presentation of pedal edema (gathering of fluid in lower leg and feet), cheilosis (cracked lips), and cardiomegaly (enlarged heart) are suggestive of a vitamin B1 (thiamine) deficiency, also known as Wet beriberi.
- Thiamine is essential for the body to break down and digest food, support metabolism, and help the muscles and nervous system to work efficiently.

Incorrect Options:

Option A - Riboflavin: Riboflavin (Vitamin B2) deficiency does not typically present with pedal edema, cheilosis, or cardiomegaly. Riboflavin deficiency is associated with symptoms such as sore throat, stomatitis (inflammation of the mouth), and skin and eye changes.

Option C - Pyridoxine: Pyridoxine (vitamin B6) deficiency primarily affects the nervous system and can lead to symptoms such as peripheral neuropathy, dermatitis, and impaired cognitive function. It does not typically present with pedal edema, cheilosis, or cardiomegaly.

Option D - Niacin: Niacin (vitamin B3) deficiency can result in a condition called pellagra, which is characterized by the "3 Ds": dermatitis, diarrhea, and dementia. It does not typically present with pedal edema, cheilosis, or cardiomegaly.

Solution for Question 48:

Correct Option C - Increased urate and lactate levels:

- In chronic alcoholic patients, alcohol is metabolized by alcohol dehydrogenase followed by aldehyde dehydrogenase and the production of NADH⁺, resulting in a very high NADH⁺/NAD ratio. Due to increased NADH⁺, all pyruvate will be converted into lactate, hence increased lactate and decreased glucose will be observed.
- Increased intake of alcohol causes increased uric acid synthesis and interferes with uric acid secretion in the renal tubules, leading to hyperuricemia, causing the build-up of urate crystals in the joint, resulting in gout.

Incorrect Options:

Option A - Decreased NADH/NAD⁺ ratio: Chronic alcohol consumption leads to an increased NADH/NAD⁺ ratio, not a decreased ratio. Alcohol metabolism generates excess NADH, which inhibits the normal balance of NADH/NAD⁺ ratio.

Option B - Increased urea and urate levels: Alcohol metabolism does not directly affect urea levels. Urea is primarily produced in the liver during protein metabolism. However, increased urate levels (uric acid) are observed in gouty arthritis due to impaired excretion of uric acid by the kidneys.

Option D - Increased urea and lactate levels: Alcohol metabolism does not directly affect urea levels. However, increased lactate levels are observed.

Solution for Question 49:

Correct Option B - Defective collagen formation:

- Scurvy is a condition caused by a deficiency of vitamin C (ascorbic acid), characterized by anemia, bleeding gum, easy bruising, tooth loss, petechiae, and glossitis.
- This is because vitamin C acts as a coenzyme of proline and lysine hydroxylase required for the cross-linkage of the collagen, resulting in defective collagen formation..

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 50:

Correct Option C - Biotin:

- Raw egg white is rich in avidin, which binds to the biotin and prevents its absorption.
- Biotin is a water-soluble B7 vitamin, which is a coenzyme for carboxylases. So, patients present with alopecia, reduced nail growth, and hypoglycemia due to decreased carboxylase activity.

Incorrect Options:

Option A - Niacin: Niacin deficiency (vitamin B3 deficiency) can cause a condition called pellagra, which is characterized by the "3 Ds" - dermatitis, diarrhea, and dementia. Thinning of hair and scaling of the scalp are not typical features of niacin deficiency.

Option B - Pyridoxine: Pyridoxine deficiency (vitamin B6 deficiency) can lead to various symptoms, including dermatitis, glossitis (inflammation of the tongue), and peripheral neuropathy. While hair thinning and scalp scaling are associated with pyridoxine deficiency, the key feature in this case is the consumption of raw eggs, indicating biotin deficiency.

Option D - Thiamine: Thiamine deficiency (vitamin B1 deficiency) is commonly associated with neurological symptoms, such as peripheral neuropathy and Wernicke-Korsakoff syndrome. Hair and scalp manifestations are not characteristic of thiamine deficiency.

Solution for Question 51:

Correct Option C - Xerophthalmia:

- Xerophthalmia is a condition associated with severe vitamin A deficiency .
- Vitamin A is crucial for maintaining healthy eyes and vision.
- A deficiency of vitamin A can lead to dryness, damage, and inflammation of the cornea and conjunctiva, causing symptoms such as dry, gritty, and red eyes, night blindness, and, in severe cases, corneal ulcers and blindness.

Incorrect Options:

- Option A - Beri-beri is a condition caused by thiamine (vitamin B1) deficiency, not vitamin A. Thiamine is essential for properly functioning the nervous system and energy metabolism. A thiamine deficiency can lead to symptoms such as muscle weakness, peripheral neuropathy, cardiovascular problems, and impaired mental function.
- Option B - Pellagra is a condition caused by niacin (vitamin B3) deficiency, not vitamin A. Niacin is important for energy production, DNA repair, and maintaining healthy skin and nervous systems. Pellagra is characterized by the "4 Ds" - dermatitis (skin inflammation), diarrhea, dementia (mental confusion), and death if left untreated.
- Option D - Neuropathy refers to nerve damage, which can have various causes, such as diabetes, alcohol abuse, certain medications, and vitamin deficiencies. However, vitamin A deficiency is not typically associated with neuropathy. Other B vitamins, such as B12 or B6,

are more commonly linked to neuropathic conditions

Solution for Question 52:

Correct Option C - Vitamin B12 deficiency:

- The probable cause of anemia in this elderly patient with easy fatigability, pallor, macrocytic anemia, and specific laboratory findings is Vitamin B12 deficiency.
- Methyl THFA cannot donate its methyl group and is trapped as such.
- This causes the functional deficiency of methyl THFA and is called methyl folate trap
- Occurs due to the low activity of methionine synthase/homocysteine methyltransferase
- Indirectly B12 deficiency causes folate deficiency, which in turn leads to macrocytic anemia.
- With the decreased activity of methionine synthase, homocysteine accumulates, causing
- homocystinuria

Incorrect Options:

Option A - Vitamin B6 deficiency: Vitamin B6 deficiency can cause various neurological and dermatological symptoms but does not typically result in macrocytic anemia with elevated serum homocysteine levels.

Option B - Alcohol-induced macrocytosis: Excessive alcohol consumption can lead to macrocytosis, however, the specific laboratory findings mentioned (elevated serum homocysteine and normal serum methylmalonic acid) are more indicative of a vitamin B12 deficiency rather than alcohol-induced macrocytosis.

Option D - Thiamine deficiency: Thiamine (Vitamin B1) deficiency can lead to a condition called beriberi, which primarily affects the nervous system and cardiovascular system. It does not typically cause macrocytic anemia with elevated serum homocysteine levels.

Solution for Question 53:

Correct Option C - Vitamin C

The condition of bleeding gums seen in the patient can be associated with a deficiency of Vitamin C.

- Vitamin C, also known as ascorbic acid, is an essential vitamin that plays a crucial role in several bodily functions, including collagen synthesis, wound healing, and maintenance of healthy gums and teeth. Deficiency of Vitamin C can lead to a condition called scurvy, which is characterized by various symptoms, including bleeding gums.

Incorrect Options:

Option A - Vitamin A: Vitamin A deficiency is associated with several ocular manifestations, such as night blindness and dry eyes, as well as skin and immune system issues. It is not directly related to bleeding gums.

Option B - Vitamin B12: Vitamin B12 deficiency can lead to megaloblastic anemia and neurological symptoms, but it is not specifically associated with bleeding gums.

Option D - Vitamin D: Vitamin D deficiency can cause bone and muscle problems, such as rickets in children and osteomalacia in adults. While Vitamin D plays a role in dental health, bleeding gums are not typically associated with its deficiency.

In the given scenario, the presence of bleeding gums suggests a possible deficiency of Vitamin C. Vitamin C is important for maintaining the health of the gums and preventing gum disease. Inadequate intake of Vitamin C can weaken the blood vessels in the gums, leading to bleeding and other symptoms associated with scurvy.

Solution for Question 54:

Correct Option A - Animal products

- Animal products such as meat (beef, chicken, pork), fish, eggs, and dairy products (milk, cheese) are the richest sources of vitamin B12. These foods contain naturally occurring vitamin B12 or are fortified with synthetic vitamin B12.

Incorrect Options - B, C, and D: Vitamin B12 is not available in plant sources and vegetarians present with vitamin B12 deficiency.

Solution for Question 55:

Correct Option D - Folate therapy using the B12 stores

- Vitamin B12 deficiency can lead to peripheral neuropathy, among other neurological manifestations. Folate supplementation alone can correct the anemia caused by folate deficiency, but it does not address the underlying vitamin B12 deficiency.
- When folate is administered, it can mask the anemia associated with vitamin B12 deficiency, leading to improvement in the blood picture. However, the neurological symptoms can worsen or progress because the underlying vitamin B12 deficiency remains untreated.

Incorrect Options:

Option A - Malabsorption of folate: Malabsorption of folate could contribute to folate deficiency, but it would not explain the aggravation of neurologic manifestations with folate therapy.

Option B - Treatment with folate unmasking pyridoxine deficiency: Pyridoxine deficiency (vitamin B6 deficiency) is not directly related to the scenario described. It would not explain the aggravation of neurologic manifestations with folate therapy.

Option C - Deficiency of folate reductase in CNS: Folate reductase deficiency in the central nervous system (CNS) is a rare genetic disorder and would not be the most likely cause in this case. It does not explain the aggravation of neurologic manifestations with folate therapy.

Solution for Question 56:

Correct Option A - Niacin

Pellagra is a nutritional disorder characterized by the "3 Ds" - dermatitis, diarrhea, and dementia. The skin lesions typically appear in sun-exposed areas and may present as a reddish rash, scaling, and thickening of the skin. The patient's history of a maize-based diet suggests a lack of niacin intake, as maize is deficient in niacin unless treated with alkaline substances.

Incorrect Options:

Option B - Zinc: Zinc deficiency does not typically cause the specific symptoms and skin lesions seen in pellagra. Zinc deficiency can lead to other manifestations such as delayed wound healing, hair loss, and immune system dysfunction.

Option C - Thiamine: Thiamine deficiency can lead to a condition called beriberi, characterized by neurological and cardiovascular symptoms. It does not cause the specific skin lesions seen in pellagra.

Option D - Iron: Iron deficiency can cause anemia and other symptoms such as fatigue, pale skin, and weakness. It does not result in the characteristic skin lesions associated with pellagra.

Solution for Question 57:

Correct Option: D - Folate therapy caused rapid use of vitamin B12 stores aggravating symptoms:

- This option is the most probable reason for the patient's condition. Folate therapy can improve anemia caused by folate deficiency, but it can also exacerbate neurological symptoms in individuals with vitamin B12 deficiency. This phenomenon is known as "masking" or "aggravation" of vitamin B12 deficiency symptoms.

Incorrect Options:

Option A - Folate is not well absorbed: This option suggests that the patient's condition is due to poor absorption of folate. However, in the scenario provided, the patient was treated with folate therapy, which implies that folate was successfully absorbed and administered. Therefore, this option is not the most probable reason for her condition.

Option B - Folic acid therapy unmasked pyridoxine deficiency: Pyridoxine refers to vitamin B6, and this option suggests that the patient's symptoms worsened due to the unmasking of a pyridoxine deficiency caused by folic acid therapy.

Option C - Deficiency of folate reductase in CNS: Folate reductase is an enzyme involved in the conversion of folic acid to its active form, which is necessary for various cellular processes.

Solution for Question 58:

Correct Option B - 7-dehydrocholesterol to cholecalciferol

- Vitamin D is produced endogenously in the body in a process that involves the skin, liver and kidneys.
- 7-dehydrocholesterol is present in the epidermis of the skin. The UVB radiation present in sunlight reacts with 7-dehydrocholesterol converting it into cholecalciferol. Sunlight plays a crucial role in this conversion without which the Vitamin D synthesis does not progress further.

Incorrect Options :

Option A - 1,25 dihydroxy cholecalciferol to 7-dehydrocholesterol: This choice is incorrect as there is no reaction in the Vitamin D synthesis which converts 1,25 dihydroxy cholecalciferol back into 7-dehydrocholesterol.

Option C - Cholecalciferol to 25-hydroxycholecalciferol: This step occurs in the liver and it does not require sunlight. The enzyme involved in the hydroxylation of cholecalciferol is 2,5 alpha-hydroxylase.

Option D - 25-hydroxycholecalciferol to 1,25 dihydroxycholecalciferol: This is the final step and it takes place in the kidney with the help of the enzyme 1 alpha-hydroxylase. This step does not require sunlight.

Solution for Question 59:

Correct Option A - Pellagra

- Pellagra is a condition caused by the deficiency of Niacin (Vitamin B3) in the body. The symptoms seen in the patient including pain in the abdomen, loss of appetite and diarrhoea along with glossitis and pigmented rash in sun-exposed areas are symptoms commonly associated with Pellagra.
- Since the patient has a staple diet of Maize which does not contain Vitamin B3 and does not have any other source of niacin, it is conclusive that the diagnosis, in this case, is Pellagra.

Incorrect Options:

Option B - Scurvy: Caused by a deficiency of Vitamin C. Common symptoms of scurvy include bleeding gums, loosening of teeth, joint pain, and delayed wound healing. These symptoms were not presented in the above scenario.

Option C - Megaloblastic anaemia : Megaloblastic anaemia is caused by a deficiency of vitamin B12. Common symptoms include fatigue, weakness and pale skin but do not include pain in the abdomen or any other skin manifestations.

Option D - Beriberi: Beriberi is caused by a deficiency of Vitamin B1. It causes fatigue, muscular weakness, and peripheral neuropathy.

Solution for Question 60:

Correct Option D - Vitamin C: Serum methylmalonate level

- Vitamin C functions as a cofactor for prolyl-3-hydroxylases, prolyl-4-hydroxylases, and lysyl hydroxylases. These enzymes are required for the hydroxylation of proline and lysine for synthesis of collagen.
- Vitamin B-12 is required for methylmalonic acid (MMA) to be converted to succinyl-CoA, and in combination with folic acid, for homocysteine (HC) to be converted to methionine. So Vitamin B-12 is the enzyme associated with MMA and not Vitamin C.

Incorrect options:

Option A - Vitamin B1 (Thiamine): RBC Trans-ketolase : Thiamine and Magnesium acts as coenzymes for RBC Transketolase in the Pentose Phosphate Pathway which produces ribose-5-phosphate and NADPH for regulating cancer cell growth.

Option B - Vitamin B2 (Riboflavin): RBC glutathione reductase: Vitamin B2 is found in the the coenzymes flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN) and converts oxidized glutathione to reduced glutathione and maintains the level of reduced glutathione in the body.

Option C - Vitamin B9 (Folic acid): FIGLU excretion/Histidine load test : A test for folate deficiency involves measurement of urinary FIGLU excretion after a histidine load. It was observed that folate-deficient individuals receiving the histidine for the FIGLU test made a hematological response that alleviated the anemia associated with this deficiency.

Solution for Question 61:

Correct Option C - Glutathione reductase

- Glutathione reductase (GR) catalyzes the reduction of oxidized glutathione (GSSG) to reduced glutathione (GSH) using NADPH and FAD, a derivative of the water-soluble vitamin riboflavin. In its reduced form, glutathione plays key roles in the cellular control of reactive oxygen species.
- Reduced glutathione removes H₂O₂ and accumulation of H₂O₂ may decrease the life span of the erythrocyte by causing oxidative damage to the cell membrane, leading to hemolysis.

Incorrect Options:

Option A - Transketolase : Thiamin or B1 is the coenzyme for Transketolase. Deficiency of B1 leads to beri beri and wernicke's encephalopathy.

Option B - Pyruvate dehydrogenase : Thiamin diphosphate(B1) is the coenzyme for three multi-enzyme complexes that catalyze oxidative decarboxylation reactions: pyruvate dehydrogenase in carbohydrate metabolism; α -ketoglutarate dehydrogenase in the citric acid cycle; and the branched-chain keto-acid dehydrogenase involved in the metabolism of leucine, isoleucine, and valine.

Option D - Isocitrate dehydrogenase : Niacin (Vitamin B3) , in the form of nicotinamide adenine dinucleotide (NAD), the coenzyme for three dehydrogenases in the cycle— isocitrate dehydrogenase, α -ketoglutarate dehydrogenase, and malate dehydrogenase.

Solution for Question 62:

Correct option C - Vitamin C

- Vitamin C deficiency develops after weeks to months of vitamin C depletion.
- Signs of vitamin C deficiency in scurvy include skin changes, fragility of blood capillaries, gum decay, tooth loss, and bone fracture, many of which can be attributed to deficient collagen synthesis.

- Vitamin C is a water-soluble antioxidant that maintains vitamin E and many metal cofactors in the reduced state.
- Vitamin C enhances the absorption of iron, and this depends on the presence of the vitamin in the gut. Therefore, increased intakes may be beneficial.

Incorrect Options:

Option A - Glutathione: In erythrocytes and other tissues, the enzyme glutathione peroxidase, containing selenium as a prosthetic group, catalyzes the destruction of H₂O₂ and lipid hydroperoxides by reduced glutathione, protecting membrane lipids and hemoglobin against oxidation by peroxides.

Option B - β -carotene: The most common provitamin A carotenoid in foods and dietary supplements is beta-carotene. In the body, beta-carotene converts into vitamin A (retinol).

Option D - Zinc: Zinc deficiency leads to hair loss, changes in their nails, diarrhea, more infections, feeling irritable, loss of appetite, impotence, eye problems.

Solution for Question 63:

Correct Option B - B12 If vit B12 is deficient in the body, the active form of folate {THF} is not formed, and so methyl THF { inactive form } is trapped and gets accumulated. It is also called a functional deficiency of folate. B12 deficiency is diagnosed by a Schillings test.

Incorrect Options:

Option A - B2: Vit B2 (riboflavin) deficiency causes corneal vascularization: pink eye due to increasing vascularization cornea, cheilosis: angular stomatitis due to cracks at the corner of mouth, seborrheic dermatitis, and glossitis: magenta tongue.

Option C - B1: Vit B1(thiamine) deficiency causes dry beriberi: which affects the CNS, and wet beriberi: which affects the CVS. In chronic alcoholics, severe vitamin B1 deficiency causes Wernicke Korsakoff psychosis syndrome: patient experiences encephalopathy and confabulations. Option D - B6: Vit B6 (pyridoxine) deficiency causes increased homocysteine levels in blood and urine and excretion of xanthurenic acid in urine.

Solution for Question 64:

Correct Option A - Scurvy:

Vitamin C plays an important role in collagen synthesis which helps in wound repair and healing. Lack of vitamin C causes Scurvy. Thus, scurvy is associated with defects in collagen maturity.

Incorrect Options:

Option B - Rickets: It is caused due to the deficiency of vitamin D in children. In adults deficiency of vitamin D causes osteomalacia.

Option C - Beriberi is caused due to deficiency of vitamin B1. The most common cause of B1 deficiency is alcoholism.

Option D - Night blindness: It occurs due to a deficiency of vitamin A. Earliest manifestation of vitamin A deficiency is night blindness.

Solution for Question 65:

Correct Option B - Pyruvate dehydrogenase:

- Thiamine (vitamin B1) is an essential coenzyme involved in several metabolic pathways, including the conversion of pyruvate to acetyl-CoA through the pyruvate dehydrogenase complex (PDC).
- When thiamine is deficient, as in thiamine deficiency, the activity of the pyruvate dehydrogenase complex is impaired. As a result, pyruvate cannot be efficiently converted into acetyl-CoA, leading to a buildup of pyruvate.

Incorrect Options:

Option A - Pyruvate carboxylase: Pyruvate carboxylase is responsible for the conversion of pyruvate to oxaloacetate in an anaplerotic reaction that replenishes TCA cycle intermediates.

Option C - Phosphoenolpyruvate carboxykinase: This enzyme is responsible for the conversion of oxaloacetate to phosphoenolpyruvate (PEP), an important step in gluconeogenesis. It is not directly involved in the conversion of pyruvate to acetyl-CoA and is not typically affected by thiamine deficiency.

Option D - Transketolase: This enzyme is involved in the non-oxidative phase of the pentose phosphate pathway (PPP), which generates ribose-5-phosphate and other sugars. Transketolase requires thiamine pyrophosphate as a cofactor, and thiamine deficiency can impair its activity. However, transketolase deficiency is not directly associated with lactic acidosis.

Solution for Question 66:

Correct Choice. C

Explanation: Thiamine deficiency is the first seen micronutrient deficiency in chronic alcoholism, as alcohol interferes with thiamine absorption.

Reasons:

- High energy status.
- Missed mixed balance diet.
- Essential micronutrient deficiencies.

Even thiamine supplements are taken without a stop on alcohol consumption there will be no change in the symptoms as the alcohol interferes with the thiamine absorption.

TRIAD:

- Global confusion
- Ophthalmoplegia
- Ataxia

Incorrect Choices:

Option A. A deficiency of cobalamin (Vitamin B12) manifests as macrocytic anaemia. Alcohol use disorder may result in macrocytic anaemia, but the common symptoms are mood disturbances, loss of balance, paresthesia, peripheral neuropathy, and gastrointestinal upset from malabsorption.

Option B . Folate (Vitamin B9) deficiency occurs due to a lack of folic acid in the blood or alcohol dependency. Symptoms include pale skin, decreased appetite, fatigue, weakness, a smooth and tender tongue, and mouth sores.

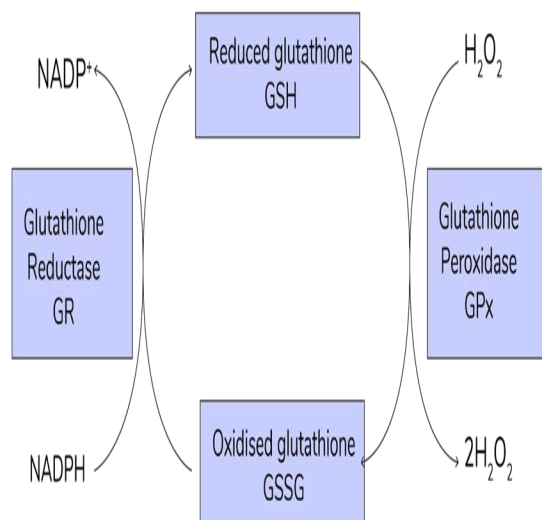
Option D . Alcoholism may induce or aggravate niacin deficiency (Vitamin B3), leading to alcoholic pellagra encephalopathy. The common symptoms are confusion, hallucinations,

oppositional hypertonus, ataxia, and myoclonus.

Solution for Question 67:

Correct Option B - Glutathione reductase

- Flavin adenine dinucleotide (FAD), a coenzyme generated from vitamin B2, is needed as a cofactor for the enzyme glutathione reductase in order for it to work properly. Low FAD levels are caused by a vitamin B2 deficiency, and this lowers glutathione reductase activity.
- In order to measure vitamin B2 levels, glutathione reductase activity in red blood cells can be monitored.



Incorrect Options: Option A - Transketolase

- An example of an enzyme that works in the pentose phosphate pathway to produce NADPH and ribose sugars for biosynthesis is transketolase.

Option C - Kynureninase:

- The tryptophan metabolism enzyme kynurenine is not directly correlated with the presence of vitamin B2.

Option D - Pyruvate dehydrogenase:

- The cofactors for the enzyme complex pyruvate dehydrogenase include thiamine (vitamin B1) and lipoic acid but not vitamin B2.
-

Solution for Question 68:

Correct Option B - Competitive inhibition

- Dicoumarol is structurally similar to vitamin K, and it competes with vitamin K for binding to the enzyme called vitamin K epoxide reductase (VKOR).
- Competitive inhibition occurs when a molecule competes with the substrate for the active site of an enzyme. In this case, dicoumarol competes with vitamin K for binding to the active site of VKOR. When dicoumarol binds to VKOR, it prevents vitamin K from binding and inhibits its action.

Incorrect Options:

Option A - Suicide inhibition: Suicide inhibition occurs when a molecule irreversibly binds to the active site of an enzyme, resulting in the inactivation of the enzyme.

Option C - Non-competitive inhibition: Non-competitive inhibition occurs when a molecule binds to a site on the enzyme other than the active site, leading to a conformational change in the enzyme that affects its activity.

Option D - Uncompetitive inhibition: Uncompetitive inhibition occurs when an inhibitor binds to the enzyme-substrate complex, preventing the release of the product.

Previous Year Questions

1. Which test should be done for detecting metabolite levels in inborn errors of metabolism in neonates?

- A. Tandem mass spectroscopy
 - B. Complete blood count (CBC)
 - C. Urinalysis
 - D. Genetic testing
-

2. Which of the following substances is used in a tube for collecting blood samples specifically for measuring alcohol concentration?

- A. Heparin + EDTA
 - B. Sodium fluoride + Potassium dichromate
 - C. Heparin
 - D. Sodium citrate
-

3. A patient presents with a history of vomiting. Arterial blood gas analysis reveals the following: pH: 7.5 pCO₂: 48 mm Hg HCO₃⁻: 30 mm Hg What is the most likely acid-base abnormality?

- A. Metabolic alkalosis
 - B. Metabolic acidosis
 - C. Respiratory acidosis
 - D. Respiratory alkalosis
-

4. Which of the following enzymes is typically elevated in prostate cancer?

- A. Acid phosphatase
 - B. Alkaline phosphatase
 - C. Lactate dehydrogenase
 - D. Alpha-fetoprotein
-

5. The development of cataracts in patients with Diabetes Mellitus is primarily due to the accumulation of which of the following substances in the lens?

- A. Sorbitol
 - B. Dulcitol
 - C. Mannitol
 - D. Galactitol
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	1
Question 5	1

Solution for Question 1:

Correct Option A - Tandem mass spectroscopy:

- Tandem mass spectroscopy (MS/MS) is widely used for newborn screening for metabolic disorders.
- It allows for the detection of multiple metabolic diseases from a single blood sample.
- It is highly sensitive and can identify various metabolic abnormalities by analyzing specific metabolites and their precursors.

Incorrect Options:

Option B - Complete blood count (CBC):

- CBC is a common blood test used to evaluate overall health and detect a wide range of disorders, including anemia, infection, and many other diseases. However, it is not specifically designed for metabolic disease screening.

Option C - Urinalysis:

- Useful for evaluating kidney function and urinary tract issues but not comprehensive enough for metabolic disease screening

Option D - Genetic testing:

- Identifies specific genetic mutations, often used when there is a high suspicion of a particular disorder, but not a first-line screening tool due to cost and specificity.
-

Solution for Question 2:

Correct Option B - Sodium fluoride + Potassium dichromate:

- Sodium fluoride is an antiglycolytic agent that inhibits the breakdown of glucose, and Potassium dichromate is a reagent that stabilizes the sample for accurate alcohol concentration measurement. This combination is specifically used for alcohol testing because it prevents the metabolism of alcohol in the sample.

Incorrect Options:

Option A - Heparin + EDTA: Heparin and EDTA are anticoagulants used for different types of blood tests, but they are not specifically used for alcohol concentration testing.

Option C - Heparin: Heparin is an anticoagulant used to prevent blood clotting, but it is not suitable for alcohol concentration tests as it does not preserve the sample for this specific purpose.

Option D - Sodium citrate : Sodium citrate is an anticoagulant used mainly for coagulation studies, but it is not used for measuring blood alcohol concentration.

Solution for Question 3:

Answer: A) Metabolic alkalosis

Explanation:

The arterial blood gas (ABG) values indicate:

- pH: 7.5 (Alkalosis)
- pCO₂: 48 mmHg (Elevated)
- HCO₃⁻: 30 mEq/L (Elevated)

These findings suggest a primary metabolic alkalosis with respiratory compensation .

- Primary Mechanism: Vomiting leads to loss of gastric HCl , causing an increase in bicarbonate (HCO_3^-) and resulting in metabolic alkalosis.
- Compensatory Mechanism: The body attempts to retain CO_2 via hypoventilation (elevated pCO_2) to counteract the alkalosis.
- Metabolic Acidosis (Option B) is incorrect because metabolic acidosis would present with a low pH (<7.35) and a low HCO_3^- , which is not the case here.
- Respiratory Acidosis (Option C) is incorrect because respiratory acidosis is characterized by low pH and high pCO_2 , but in this case, the pH is elevated.
- Respiratory Alkalosis (Option D) is incorrect because respiratory alkalosis involves low pCO_2 due to hyperventilation, whereas this patient has elevated pCO_2 as a compensatory response.

Solution for Question 4:

Answer: A) Acid phosphatase

Explanation:

Acid phosphatase , specifically prostatic acid phosphatase (PAP) , is an enzyme that is elevated in prostate cancer , particularly in advanced or metastatic cases . PAP was historically used as a tumour marker for prostate cancer , though prostate-specific antigen (PSA) is now the preferred marker.

- Significance: Elevated in prostate cancer with bone metastases Can be used to monitor disease progression Less commonly used now due to the superior sensitivity of PSA
- Elevated in prostate cancer with bone metastases
- Can be used to monitor disease progression
- Less commonly used now due to the superior sensitivity of PSA
- Elevated in prostate cancer with bone metastases
- Can be used to monitor disease progression
- Less commonly used now due to the superior sensitivity of PSA
- Alkaline Phosphatase (Option B) is incorrect because while alkaline phosphatase (ALP) is elevated in bone metastases , it is not specific for prostate cancer and can also be seen in liver disease and bone disorders.

- Lactate Dehydrogenase (Option C) is incorrect because LDH is a non-specific marker of cell turnover and is elevated in various malignancies, not just prostate cancer.
 - Alpha-Fetoprotein (Option D) is incorrect because AFP is primarily a marker for hepatocellular carcinoma and germ cell tumours , not prostate cancer.
-

Solution for Question 5:

Answer: A) Sorbitol

Explanation:

Sorbitol accumulation in the lens is the primary cause of diabetic cataracts . In diabetes mellitus , chronic hyperglycaemia leads to the activation of the polyol pathway , where:

- Osmotic stress → water influx → lens swelling
- Oxidative stress → protein damage → cataract formation

Diabetic cataracts typically present as snowflake cataracts in younger individuals or as accelerated senile cataracts .

- Dulcitol (Option B) is incorrect because dulcitol (galactitol) accumulates in galactosemia , not diabetes. It causes galactose-induced cataracts due to galactokinase or GALT deficiency .
 - Mannitol (Option C) is incorrect because mannitol is an exogenous osmotic diuretic and does not accumulate in the lens.
 - Galactitol (Option D) is incorrect because it is another name for dulcitol , and its accumulation is linked to galactosemia , not diabetes.
-